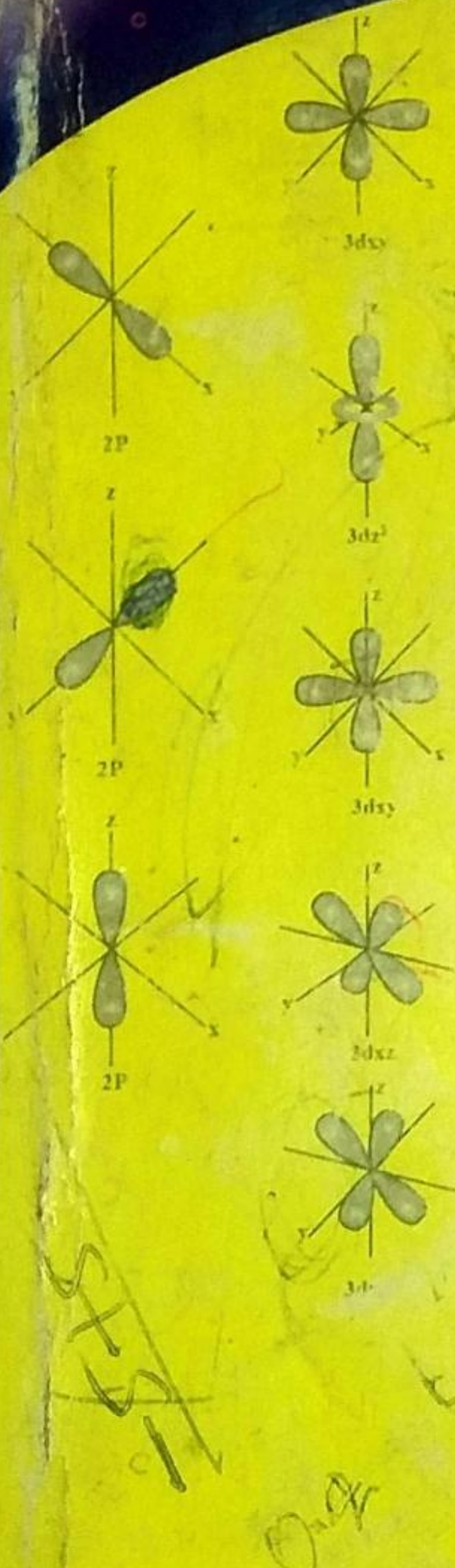
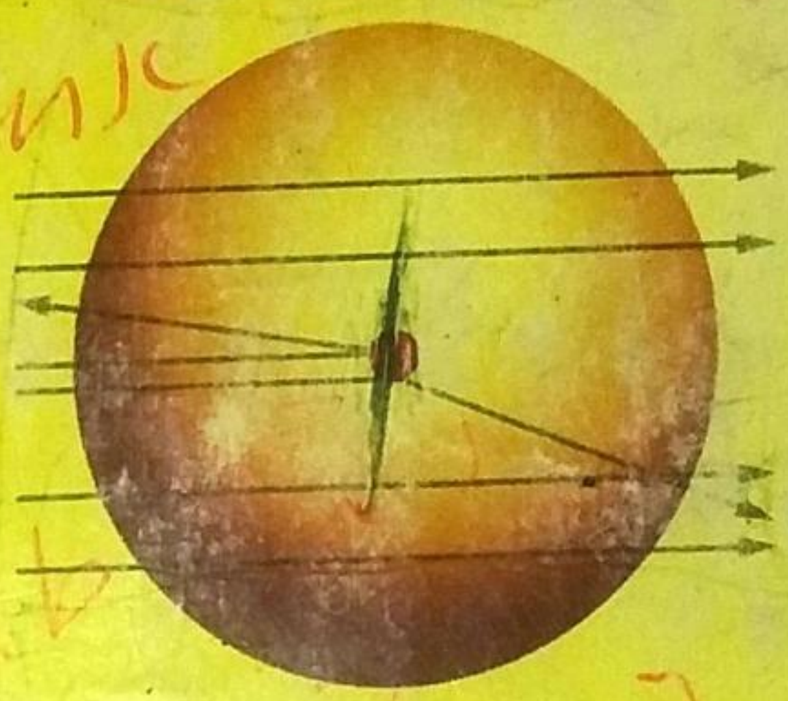
Extreme Series

# CHEMISTRY

## GRADE 11 - 12



Based on Common Currently Used National and International Curriculum

Extreme Series

2011 Edition

Mengesha Moges

# Ethiopian Digital library

public channel

## Description

✔ መጽሀፍት በPDF ያገኛሉ
✓Top books
✓Extreme books
✅ ለትምህርታዊ መፅሀፍቶች
✅ ቴክኖሎጂ መፃሀፍቶች
✅ የሳይንስ መጽሀፍቶ
✅ የመደበኛ ተማሪዎችን መጽሀፍትን፣
✅ የሶሻልና የናቹራል መጽሀፎች
✅ የጥያቄና መልስ
የሚያገኙበት
@ethiolibrary1
@ethiolibrary1
jemal6454@gmail.com

t.me/ethiolibrary1

Invite Link

Notifications

On

# Extreme CHEMISTRY

## For Grade 11-12

**Page layout and Cover Design**

- Biruk Gohatsibah

**Secreterial Work**

- Mahider Almaw
- Medhanit Begashaw

Based on Common Currently Used

National and International Curriculum

# Table of Content

## Unit 1: Fundamental Concepts of Chemistry


| | | |
|---|---|---|
| 11.1 | Definition of Chemistry | 2 |
| 11.2 | Measurements and Units in Chemistry | 4 |
| 11.3 | Uncertainty In Measurements | 11. |
| 11.4 | Decimal place and Rounding off | 13 |
| 11.5 | Scientific Notation | 14 |
| 11.6 | The Scientific Method | 17 |
| 11.7 | Some Experimental Skills in Chemistry | 19 |


## Unit 2: Atomic Structure and The Periodic Table


| | | |
|---|---|---|
| 2.1 | Dalton's Atomic Theory | 34 |
| 2.2 | Modern Atomic Theory | 35 |
| 2.3 | Electromagnetic Radiation (EMR) and Atomic Spectra | 49 |
| 2.4 | The Bohr Model of the hydrogen atom | 60 |
| 2.5 | The Quantum model of the atom | 66 |
| 2.6 | Electron Configuration and Orbital Diagrams | 76 |
| 2.7 | Periodic Classification of Elements | 82 |

# Unit 1

## Fundamental Concepts of Chemistry

### Unit Outline:


1.1 Definition of Chemistry
1.2 Measurements and units in chemistry
1.3 Uncertainty in measurements
1.4 Decimal place and rounding off
1.5 Significant figures
1.6 The scientific method
1.7 Some experimental skills in chemistry

## Introduction

Chemistry is the science of molecules and their transformations. It is the science not so much of the one hundred and ten elements, but of the infinite variety of molecules that may built from them.

**Chemistry is useful to almost every one by**

- ➦ Increasing crop production
- ➦ Manufacturing drugs and medicines
- ➦ Making building materials such as cement, paints, plastics.
- ➦ Converting natural resources into new substances.
- ➦ Preserving food and chemicals.
- ➦ Protecting the environment
- ➦ Treating drinking water with chlorine.
- ➦ Manufacturing cosmetics, soaps, detergents, clothes, shoes, ...

In general, chemistry is important and interesting science.

## 1.1 Definition of Chemistry

Chemistry is the science of matter and its transformation.

Matter is anything that has mass and occupies space. It undergoes two changes. These are

### Chemical change

- ➦ It is a change in which new substance with new property is formed.
- ➦ It is difficult to reverse chemical change.
- ➦ It is accompanied by heat change

**Example**

All chemical reactions.

### Physical Change

- ➦ It is a change in which no new substance is formed.
- ➦ It is easily reversed.
- ➦ It is accompanied by small or no heat change.

**Example**

- ➦ Melting of ice
- ➦ Powdering of chalk.
- ➦ Formation of Cloud.

All measurable properties of matter fall into two categories: **Extensive and intensive properties.**

**Extensive property** depends on how much matter is being considered.

**Example**

Mass, length, Volume, etc

**Intensive Property** does not depend on the amount of matter being considered.

**Example**

Temperature, density, melting point etc

**The Scope of chemistry** is the extent of its applications in production of new substances, discovering new substances, understanding matter and improving the living conditions for the society.

## Illustrative Example

1. Which of the following is not chemical change?

   A. Souring of milk
   B. Decaying of food
   C. Rusting of iron
   D. tearing of paper

**Explanation:** Tearing of paper is change in size or shape, thus it is physical change.

**Answer: D**

2. All of the following are chemical properties. EXCEPT

   A. Fermentation of sugar.
   B. Reactivity of Sodium with water.
   C. Water boils at 100°C.
   D. Gold does not react with water.

**Answer: C**

3. Classify the following as intensive or extensive properties.

a. Volume
b. Color
c. Area
d. Thickness
e. Density
f. Freezing point
g. Length
h. Conductivity
i. Specific Gravity

**Answer**

- a, c d, and g are extensive properties.
- b, e, f, h and i are intensive properties.

4. Write the main uses of chemical changes.

**Explanation:** To produce new substances and energy.

# Major Fields of Chemistry

### Organic Chemistry

- Is the study of carbon compounds except carbonates, hydrogen carbonates, cyanides, cyanate, thiocyanate, and oxides of carbon.

### Inorganic Chemistry

- Is the chemistry of all elements and inorganic compounds.

### Analytical Chemistry

- Is concerned with the quantitative determination and qualitative identification of a sample of matter.

### Physical Chemistry

- Is the study of the relationship between matter and energy.

### Bio Chemistry

- Deals with processes and the structure of substances in living things.

**Other branches of chemistry include:**

- ➥ Textile chemistry
- ➥ Polymer Chemistry
- ➥ Water Chemistry
- ➥ Agricultural Chemistry
- ➥ Environmental Chemistry
- ➥ Food Chemistry.
- ➥ Geochemistry
- ➥ Astrochemistry

## 1.2 Measurements and Units in Chemistry

The study of chemistry depends heavily on measurement.

Measurement is the act of finding the magnitude of a given quantity interms of units.

Measured quantities are classified into two: **Basic and derived quantities.**

Table 1.1. Basic quantities and their basic SI units

| Basic quantity | Name of unit | Symbol of unit |
|---|---|---|
| Length | Meter | m |
| Mass | Kilogram | Kg |
| Time | Second | S |
| Electrical current | Ampere | A |
| Temperature | Kelvin | K |
| Amount of substance | Mole | mol |
| Luminous intensity | Candela | Cd |

**Mass** is the measure of the quantity of matter in an object.

Its SI unit is kilogram. Other Units of mass are gram (g), milligram (mg).

$$1\ kg = 1000g = 10^6 mg$$

**Temperature** is the measure of the average kinetic energy of molecules.

Its SI unit is Kelvin. Other units of temperature are degree Celsius (°C), degree Fahrenheit (°F).

Lowest attainable temperature is -273.15°C or 0K.

$K - 0 = °C - (-273.15)$

$K = °C + 273.15$

$$K = °C + 273$$
$$°C = K - 273$$

The relationship between degree Celcius and degree Fahrenheit is obtained from boiling and freezing point of water.

Boiling point of water is 100°C or 212°F.

Freezing point of water is 0°C or 32°F.

FOR COMMENT

Subtract freezing point from both sides:

$$\frac{°C - 0}{°F - 32} = \frac{100 - 0}{212 - 32}$$

$$\Rightarrow \frac{°C}{°F - 32} = \frac{100}{180} = \frac{5}{9}$$

$$\therefore \boxed{°C = \frac{5}{9}(°F - 32)} \qquad \boxed{°F = \frac{9}{5}°C + 32}$$

**Example**

Convert 27°C into °F and K.

**Solution:**

➦ $°F = \frac{9}{5}°C + 32 = \frac{9}{5} \times 27 + 32 = 80.6°F$

➦ $K = °C + 273 = 27 + 273 = 300K$

**Mole** is the amount of substance that contains as many elementary particles as there are atoms in 12g of carbon isotope.

$$1\ mole = 6.02 \times 10^{23}\ particles\ (atoms, molecules, ions, formula\ units)$$

$$\text{Number of mole} = \frac{\text{given mass}}{\text{molar mass}} \qquad \boxed{n = \frac{m}{M}}$$

- Number of atoms = $n \times 6.02 \times 10^{23}$
- Number of atoms in a molecule = $n \times$ sum of subscripts $\times 6.02 \times 10^{23}$

**Example**

Calculate the number of atoms in 16g of $CH_4$.

**Solution:**

$$n = \frac{m}{M} = \frac{16g}{32g/mol} = 0.5mol$$

Number of atoms = n × sum of subscripts × $6.02 \times 10^{23}$

$= 0.5 \text{ mol} \times (1 + 4) \times 6.02 \times 10^{23}$ atoms/mol

$= 1.505 \times 10^{24}$ atoms

## Illustrative Examples

5. Which of the following is not basic SI unit?

A. Second B. Kelvin C. Gram D. Meter

**Answer: C**

6. -40°F is _____ K

A. -40 B. 233 C. 273 D. 300g

**Solution:**

-40°F = -40°C

$\Rightarrow K = °C + 273 = -40 + 273 = 233K$

**Answer: B**

7. How many atoms are present in 22 grams of carbon dioxide, $CO_2$? (EUEE)

A. $3.01 \times 10^{23}$

B. $6.02 \times 10^{23}$

C. $2 \times 6.02 \times 10^{23}$

D. $1.5 \times 6.02 \times 10^{23}$

**Solution:**

$$n = \frac{m}{M} = \frac{22g}{44g/mol} = 0.5mol$$

Number of atoms = n × sum of subscripts × $6.02 \times 10^{23}$

$$= 0.5 \times (2 + 1) \times 6.02 \times 10^{23}$$

$$= 1.5 \times 6.02 \times 10^{23} \text{ atoms}$$

**Answer: B**

8. One mole of $(NH_4)_2HPO_4$ contains _____ moles of hydrogen atoms.

   A. One B. Five C. Six D. Nine

**Solution:**

Number of moles of atoms in a compound = number of mole of compound × subscript of the atom

$\Rightarrow (NH_4)_2PO_4 = N_2H_9PO_4$

number of H - atom = $n(NH_4)_2HPO_4 \times 9$

$= 1 \text{ mol} \times 9 = 9 \text{ moles}$

**Answer: D**

9. How many moles are there is 159g of alanine, $C_3H_7NO_2$? (EUEE)

**Solution:**

$M, C_3H_{11}NO_2 = 3 \times 12 + 7 \times 1 + 1 \times 14 + 2 \times 16 = 89 g/mol$

$$n = \frac{m}{M} = \frac{159g}{89g/mol} = 1.8mol$$

10. What is the lowest attainable temperature in °C and K?

**Answer: -273.15°C or 0K**

Table 1.2. Some derived quantities and their SI Units

| Derived quantity | Name of SI unit | Symbol of Unit |
|---|---|---|
| Area | Square meter | $m^2$ |
| Volume | Cubic meter | $m^3$ |
| Density | Kilogram per cubic meter | $kg/m^3$ |
| Pressure | Pascal | Pa |
| Concentration | Mole per liter | mol/L |
| Weight | Newton | N |
| Force | Newton | N |
| Work | Joule | J |
| Speed | Meter Per second | m/s |

**Volume** is the space occupied by an object. Its SI unit is cubic meter ($m^3$). Other units of volume are liter (L), milliliter (mL), cubic centimeter ($cm^3$) and cubic decimeter ($dm^3$).

$1m^3 = 10^3dm^3 = 10^3L = 10^6cm^3 = 10^6mL$

**Note**

Volumetric flask, graduated cylinder, pipet and buret are laboratory equipments used to measure volume.

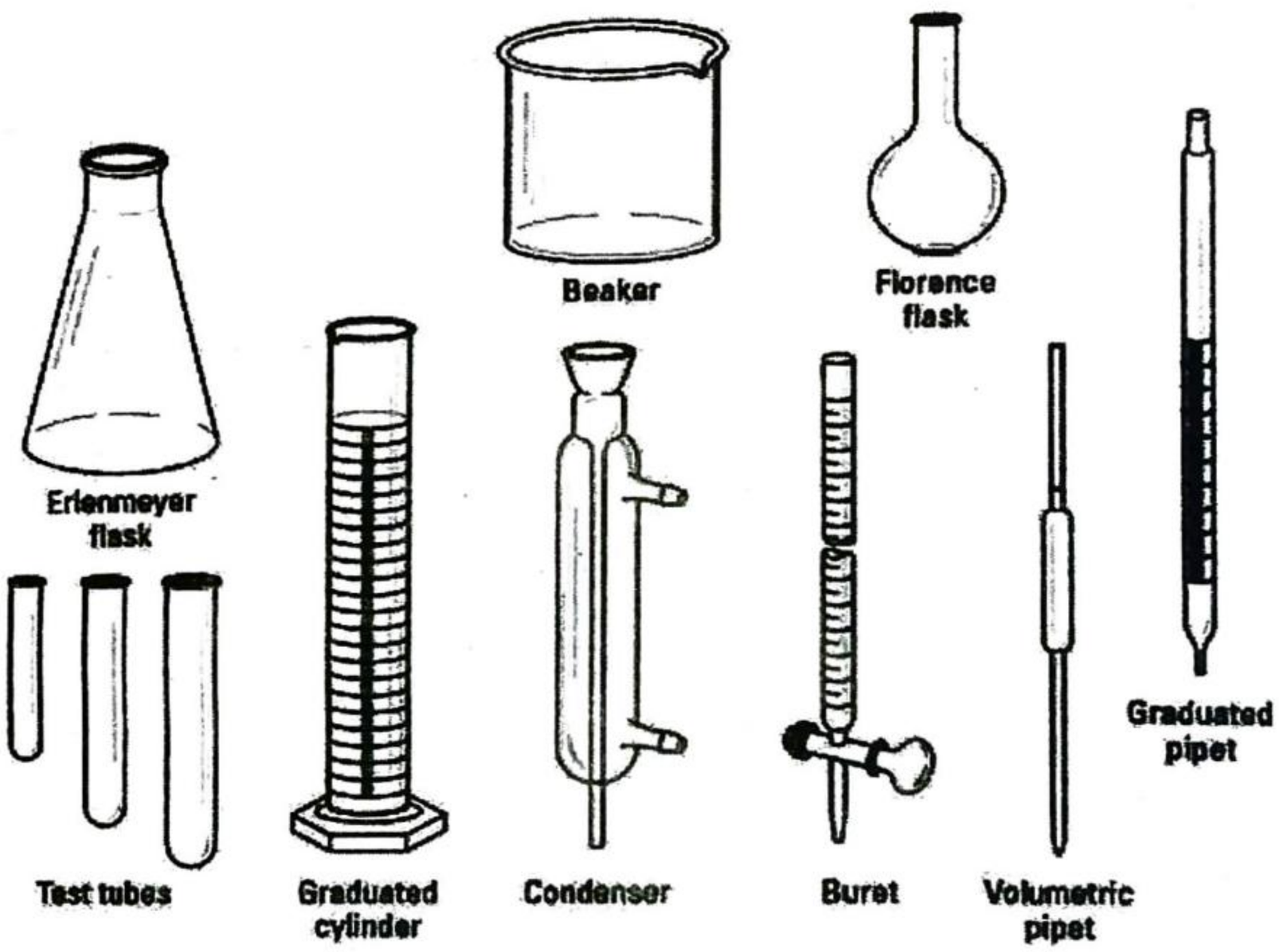



Figure 1.1 These pieces of glass ware are used to measure volume of liquids

**Pressure** is force per area.

$$P = \frac{F}{A}$$

The SI unit of pressure is Pascal (Pa). Other units of pressure are atmosphere (atm), millimeter of mercury (mmHg), centimeter of mercury (cmHg) and torr.

1 atm = 76cmHg = 760 mmHg = 760 torr = 101325 Pa

→ 1 mmHg = 1 torr

1 kPa = 1000Pa

**Density (ρ)**: is mass divided by volume.

$$\rho = \frac{m}{V}$$

The SI unit of density is kilogram per cubic meter ($kg/m^3$). Other units are gram per mililiter (g/mL), gram per liter (g/L).

$$kg/m^3 = 1 g/L = 0.001 g/mL$$

## Common Prefixes used in SI units and Conversion of units

Table 1.3. Some common Prefixes used in SI units

| Prefix | Symbol | Factor |
|---|---|---|
| Exa | E | $10^{18}$ |
| Peta | P | $10^{15}$ |
| Tera | T | $10^{12}$ |
| Giga | G | $10^{9}$ |
| Mega | M | $10^{6}$ |
| Kilo | K | $10^{3}$ |
| Hecto | H | $10^{2}$ |
| Deca | D | $10^{1}$ |
| Deci | d | $10^{-1}$ |
| Centi | c | $10^{-2}$ |
| Milli | m | $10^{-3}$ |
| Micro | μ | $10^{-6}$ |
| nano | n | $10^{-9}$ |
| Pico | P | $10^{-12}$ |
| Femto | F | $10^{-15}$ |
| Atto | a | $10^{-18}$ |

## Conversion of Units

➡ To convert one unit into another, first find the conversion factor.

**Example**

Convert 160cm into dm.

**Solution:**

$$\frac{1\text{ cm}}{1\text{ dm}} = \frac{10^{-2}\text{ m}}{10^{-1}\text{ m}}$$

$$\Rightarrow \frac{1\text{ cm}}{1\text{ dm}} = \frac{10^{-2}}{10^{-1}}$$

$$1\text{ cm} = 10^{-1}\text{ dm}$$

$\Rightarrow$ The conversion factor is $10^{-1}$.

$$\therefore 160\text{cm} = 160\,\cancel{\text{cm}} \times 10^{-1}\,\frac{\text{dm}}{\cancel{\text{cm}}} = 16\text{dm}$$

## Illustrative Examples

11. How many milligrams are in 0.2 decigrams? (EUEE)

A. 20 B. 2000 C. 0.002 D. 0.00002

**Solution:**

$$\frac{1\text{ dg}}{1\text{ mg}} = \frac{10^{-1}\text{ g}}{10^{-3}\text{ g}}$$

$$\frac{1\text{ dg}}{1\text{ mg}} = 10^{2}$$

$$\Rightarrow 1\text{ dg} = 10^{2}\text{ mg}$$

$$\therefore 0.2\text{dg} = 0.2 \times 10^{2}\text{mg} = 20\text{mg}$$

**Answer: A**

12. What fraction of a second is a picosecond, ps? (EUEE)

A. $10^{-15}$ B. $10^{-12}$ C. $10^{-10}$ D. $10^{-2}$

**Answer: B**

13. Convert the following

a. $100\text{cm}^3$ into $\text{dm}^3$

b. $6.54\text{m}^3$ to $\mu\text{L}$

c. 0.65 ng to $\mu$g

**Solution:**

a. $$\frac{1\text{cm}^3}{1\text{dm}^3} = \frac{10^{-6}\text{m}^3}{10^{-3}\text{m}^3} \Rightarrow 1\text{cm}^3 = 10^{-3}\text{dm}^3$$

$$\therefore 100\text{cm}^3 = 100 \times 10^{-3}\text{ dm}^3 = 0.1\text{dm}^3$$

b. $$\frac{1\text{m}^3}{1\mu\text{L}} = \frac{10^{3}\text{L}}{10^{-6}\text{L}} \Rightarrow 1\text{m}^3 = 10^{9}\,\mu\text{L}$$

$$\therefore 6.54\text{m}^3 = 6.54 \times 10^{9}\mu\text{L}$$

c. $\frac{1ng}{1\mu g} = \frac{10^{-9}g}{10^{-6}g} \Rightarrow 1ng = 10^{-3}\mu g$

$\therefore 0.65\ ng = 0.65 \times 10^{-3}\mu g = 6.5 \times 10^{-4}\ \mu g$

14. From the seven basic SI units, which one has a prefix?

**Explanation:** The kilogram has a prefix kilo ($10^3$). The rest have no prefixes.

15. Given that the density of gold is 19.3 g/cm$^3$. Calculate the volume of a gold hugget with a mass of 5.98g.

**Solution:**

$$\rho = \frac{m}{V} \Rightarrow V = \frac{m}{\rho} = \frac{5.98g}{19.3g/cm^3} = 0.310cm^3$$

## 1.3 Uncertainty In Measurements

- Uncertainity in measurement can be caused by
  1. Observational error
  2. Instrumental error
  3. Variability in the item measured.
  4. Environment where the measurement is being made.
- Chemistry uses two types of number: **exact and inexact**.

**Exact numbers** include numbers with defined values and those that are obtained by counting.

**Example**

There are 1000m in 1km.

**Inexact numbers** are measured numbers.

**Example**

The length of a table.

**Precision and Accuracy in measurements**

**Precision** tells us how closely multiple measurements of the same thing are to one another.

**Accuracy** tells us how close a measurement is to the **true value**.

**Example**

The true mass of an aspirin tablet is 0.370g. Three students are asked to determine the mass of the tablet. Each student weighs the aspirin tablet three times. The results (in grams) are:

| Trial | Student A | Student B | Student C |
|---|---|---|---|
| 1 | 0.335 | 0.357 | 0.369 |
| 2 | 0.331 | 0.375 | 0.373 |
| 3 | 0.333 | 0.338 | 0.371 |
| Average | 0.333 | 0.357 | 0.371 |

- Student A's results are precise but not accurate.
- Student B's results are neither precise nor accurate.
- Student C's results are both precise and accurate.

**Errors = |True value - measured value|**

There are two types of errors in measurement. These are:

**Random errors** - vary in direction and magnitude.

- mainly affect the precision of measurement

**Systematic errors** - lie in one direction of the true value, that is all above or all below the true value.

- caused by instruments and personal errors.

## Illustrative Example

16. The difference between an accepted value and an experimental value is called a (an).
    A. Error
    B. Measured value
    C. Percent error
    D. Precision

**Answer: A**

17. The true value for the density of water at 4°C and one atmosphere is 1g/mL. Student A and B determined the density of water at 4°C and 1 atm and obtained the following values in g/mL.

| Student | Trials | | | | | | Average |
|---|---|---|---|---|---|---|---|
| A | 1 | 2 | 3 | 4 | 5 | 6 | |
| | 0.91 | 0.92 | 0.90 | 0.88 | 0.91 | 0.90 | 0.90 |
| B | 1.10 | 0.99 | 1.01 | 0.97 | 0.91 | 1.03 | 1.00 |

a. Which set of data is the most precise?
b. Which set of data is the most accurate?
c. Which student has made random and which one has made systematic error.

**Explanation:**

a. Student A has made precise but less accurate measurement.

b. Student B has made accurate but less precise measurement.

c. Student A has made systematic error and student B has made random error.

18. Indicate which of the following allow you to give exact numbers when you measure them.

a. The mass of a bar copper.

b. The number of centimeters in a meter.

c. The area of a rectangle.

d. The number of hours in a day.

e. The thickness of a black board.

**Explanation:**

➦ b and d can be expressed by exact numbers.

➦ a, c and e can be expressed by inexact numbers.

19. Four students measured the mass of a piece of gold whose true mass is 18.0g. Their results are: 18.19, 17.4g, 19.0g and 17.8g.

a. What is the best estimate for the mass of a piece of gold.

b. Explain whether the results are precise or accurate.

**Explanation:**

a. The best estimate value is the average value of the four students.

$$= \frac{18.1+17.4+19.0+17.8}{4} = 18.075 \approx 18.1\text{g}$$

b. The average value is close to the true value. Thus, they have made accurate measurement.

20. Write two causes for uncertainty in measurement.

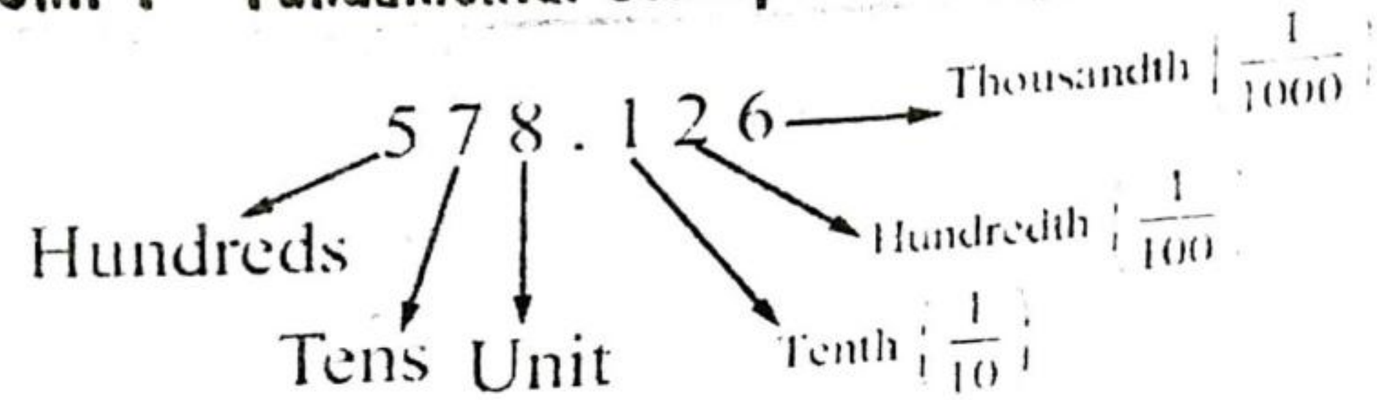
**Answer:**

Observational and personal errors.

## 1.4 Decimal place and Rounding off

**A decimal place** is the number of digital to the right of the decimal point.

**A decimal point (.)** is used to separate the fraction part from the whole number part.




**Rounding off** is finding a number that is closest to a given number but with fewer digits.

**Rules for rounding of a number**

1. If the digit to be removed is from 0 to 4, the preceding digit remains same.

Example

Round off 6.7314 to two decimal place.

**Answer: 6.73**

2. If the digit to be removed is from 6 to 9, add one to the preceding digit.

Example

Round of 7.8374 to two decimal place.

**Answer: 7.84**

3. If the digit to be removed is 5, the preceding digit is unchanged if it is even and increases by one if it is odd.

Example

Round of 8.7635 to 3 decimal place.

$\Rightarrow$ 8.764

Round of 8.7625 to 3 decimal place.

$\Rightarrow$ 8.762

Note

1. If 5 is followed by zeros only, the left most digit is unchanged.
2. If 5 is followed by non - zero digits, the left most digit is increased by one.

Example

21.6500 $\Rightarrow$ 21.6 but 21.6534 $\Rightarrow$ 21.7

## 1.5 Scientific Notation

- It is a way of expressing large or small numbers as a factor of the powers of 10.
- It can be expressed by the form $y \times 10^x$, where, y is 1 to 10

excluding 10 and x is any integer.

- When we shift decimal point from left to right, the power is negative but from right to left the power becomes positive.

**Example**

$0.000000132 = 1.32 \times 10^{-7}$

$6240000000000 = 6.24 \times 10^{12}$

## Significant figures

- Significant figures are the meaningful digits in a reported number. The last digit in a measured number is referred to as **uncertain digits** and the uncertainty associated with a measured number is generally considered as ±1 in the place of the last digit.
- When we report the width of a card to be 2.5cm, we are implying that its width is 2.5 ±0.1cm.
- If the width of the card is 2.45cm, this implies its width is 2.45±0.01cm.

## Rules of determine number of significant figures (sf)

1. Any digit that is not zero is significant. **Eg.** 156.3 has four sf
2. Zeros located between non - zero digits are significant. **Eg.**406 has three sf
3. Zeros to the left of non - zero digit are not significant. **Eg.** 0.00231 has three sf
4. Zeros to the right of the last non - zero digit are significant if the number contains a decimal point. **Eg.** 3.100 has four sf
5. Zeros to the right of the last non - zero digit in a number without decimal point may or may not be significant. To avoid ambiguity, use scientific notation.

**Example**

500 may have different significant figures.

$\Rightarrow 5 \times 10^2 = 1$ sf

$\Rightarrow 5.0 \times 10^2 = 2$ sf

$\Rightarrow 5.00 \times 10^2 = 3$ sf

## Calculations with significant figures

1. **In addition and subtraction**, the answer cannot have more digits to the right of the decimal point than any of the original numbers.

**Example**

103.50 ←—— two digits after the decimal place

<u>+0.316</u> ←—— three digits after the decimal place

102.816 ←—— round to 102.82

2. **In multiplication and division**, the number of significant figures in the product or quotient is determined by the original number that has the smallest number of significant figures.

**Example**

1.4 × 8.011 = 11.2154 ← round to 11

3. **Exact numbers** can be considered to have an infinite number of significant numbers and do not limit the number of significant figures in a calculated result.

## Illustrative Example

21. How many significant figures are there in 0.0500? (EUEE)

A. One B. Three C. Two D. Four

**Explanation:** Zeros before 5 are not significant, but after 5 are significant due to the presence of decimal point.

**Answer: B**

22. A measuring cylinder has a precision of ±0.01mL. A liquid sample that has a volume of 10mL is measured by measuring cylinder. How many significant figures should be reported for the measurement? (EUEE)

A. 4 B. 5 C. 6 D. 7

**Explanation:** A precision ±0.01mL implies there are two decimal places after the decimal place of the measured volume.

⇒10.00mL

**Answer: A**

23. Determine the number of significant figures.

a. 1.056
b. 0.000691
c. 62415
d. 600

**Solution:**

a. 4 b. 3 c. 5 d. ambiguous

24. Solve the following and write your answer to the correct significant figures.

a. 317.5mL + 0.675mL
b. 47.80L - 2.075L
c. 13.5g ÷ 45.18L
d. 6.25cm × 1.175cm

**Solution:**

a. 318.2mL
b. 45.73L
c. 0.2999 g/L
d. $7.34cm^2$

25. Write the following in the given number of significant figures by using scientific notation.

a. 3467893412 (2 sf)
b. 46789.8763 (3 sf)
c. 0.00000000069998 (4 sf)
d. 613423 (1 sf)

**Solution:**

a. $3.5 \times 10^9$
b. $4.68 \times 10^4$
c. $7.000 \times 10^{-10}$
d. $6 \times 10^5$

## 1.6 The Scientific Method

**Scientific method** is logical, careful and systematic approach to solve scientific problems.

### Steps in Scientific Method

1. **Observation** is looking through carefully the nature of an event.
   - ➦ There are two types of observations, qualitative and quantitative observations.

**Qualitative observations** have no numbers with them.

Example

The sky is blue. Honey is sweet.

Water is liquid.

**Quantitative Observations** have numbers with them and are more useful.

Example

A book weighs 10 grams. Water boils at 100°C.

2. **Hypothesis** is tentative explanation for an observation.
   - ➦ Hypothesis is useful if it can be used to make predictions.
3. **Experiment or testing hypothesis** is examining facts about hypothesis.

4. **Theory** is tested hypothesis.
   - ➦ It always serves as aid to new experiments and is constantly being tested.
   - ➦ It can be rejected or modified when it is proven incorrect.
5. **Scientific law** is a general statement that summarizes facts that come from many experiments.

**Example**

The law of gravitational force.

The law of conservation of mass.

## Illustrative Examples

26. What is the first step of the scientific method? (EUEE ... 2008 E.C)
   A. Making observations
   B. Forming a hypothesis
   C. Performing an experiment
   D. Predicting the result of an experiment.

**Answer: A**

27. Which of the following represents a tentative explanation of a certain scientific law?
   A. Hypothesis
   B. Observation
   C. Experiment
   D. Theory

**Answer: D**

28. In order to advance to the level of a theory, a hypothesis should (EUEE... 2008)
   A. be obviously accepted by most people.
   B. be repeatedly confirmed by experimentation.
   C. be a fully functional experiment.
   D. report the past experience.

**Answer: B**

29. Write the difference between quantitative and qualitative observation.

**Explanation:** Quantitative observations are numerical data while qualitative observations means non - numerical data.

30. Classify the following as quantitative or qualitative observations.
   a. The leaves of most trees are green in summer.
   b. The apple consists of over 95% water.

c. Chemistry is an easy subspect.

d. I got 90% in my last chemistry examination.

e. My hair is black.

**Solution:**

- **a, c and e** are qualitative observations.
- **b and d** are quantitative observations.

31. State the limitations of the scientific method.

**Explanations:** Scientific method is too rigid and doesn't permit conclusions to be reached from limited informations.

32. Write the steps in the scientific method.

**Solution:**

Observation ⟶ Hypothesis ⟶ Experiment ⟶ Theory ⟶ Law

## 1.7 Some Experimental Skills in Chemistry

**Chemistry** is an experimental science that relates theoretical study with practical activities.

### Chemistry laboratory apparatus

- Laboratory equipments are classified into three based on their uses. These are:

1. **Reaction vessels** are used for the occurrence of a reaction.

Example

Beakers, flasks, test tubes.

2. **Measuring equipments** are used to measure the quantities of substances.

Example

Measuring cylinder, pipet, buret, balance, thermometer, volumetric falsks.

3. **Supporting and heating devices**

Example

Stand, clamp, test tube holder, tong, Bunsen burner, splint lamp.

### Chemistry Laboratory Safety rules

- Follow the following laboratory safety rules when you perform an activity in the laboratory.

1. Wear gloves, goggles and laboratory coat.

2. Wear shoes that cover your feet.
3. Tie back lose hair.
4. Do not taste any chemical.
5. Do not suck the solutions in the pipet by mouth, rather use a rubber suction bulb or other device to fill a pipe.
6. To dilute concentrated acids, add acid to water (AW!) but not water to acid (WA!).
7. Never look directly into a test tube.
8. Point test tubes that are being heated away from you and others.
9. Carry out only allowed experiments.
10. Read the labels on the reagent bottles carefully to make sure that you are using the right chemicals.
11. When lighting Bunsen burner, first light the match stick then turn on the gas.
12. If corrosive bases are swallowed, weak acids such as acetic acid or lemon juice is administered.
13. If corrosive acids are swallowed, administer weak bases such as magnesium hydroxide ($Mg(OH)_2$), aluminum hydroxide ($Al(OH)_3$).
14. If an acid or a base enters your eye, wash with water repeatedly and then consult doctor.
15. If a concentrated acid is spilled or splashed on your body, first wash the affected part with running water and then wash with 10% of $Na_2CO_3$ solution.
16. Before leaving the laboratory, clean your apparatus, work surface and your hands well.

**Writing a laboratory Report**

➦ A laboratory report is a written composition of the results of an experiment.

➦ To write a laboratory report include the following.

1. Title _____
2. Objective _____
3. Materials and Chemicals _____
   _____
   _____
4. Procedure _____
   _____

5. Observation _____

6. Result and discussion _____

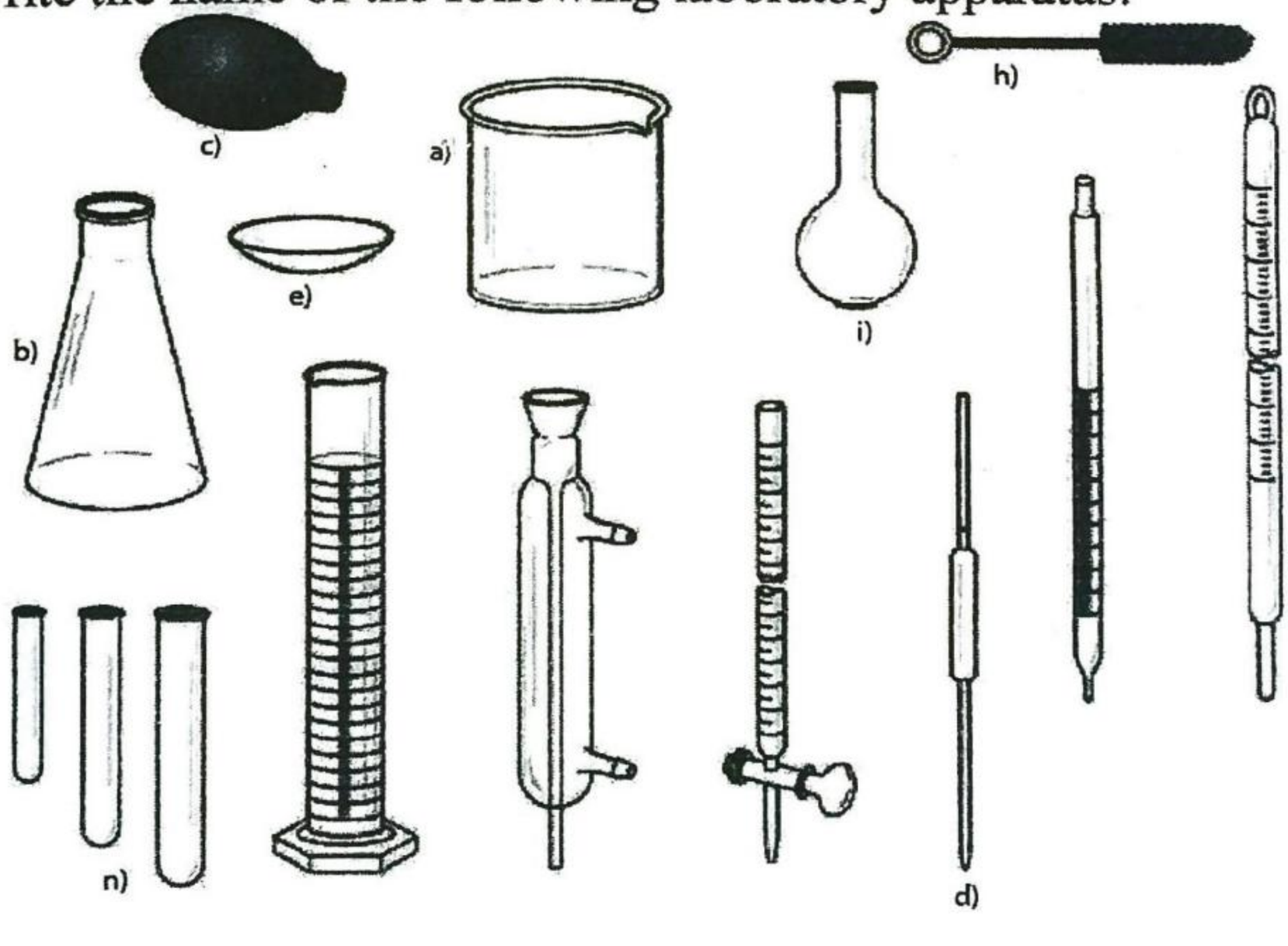
7. Conclusion _____

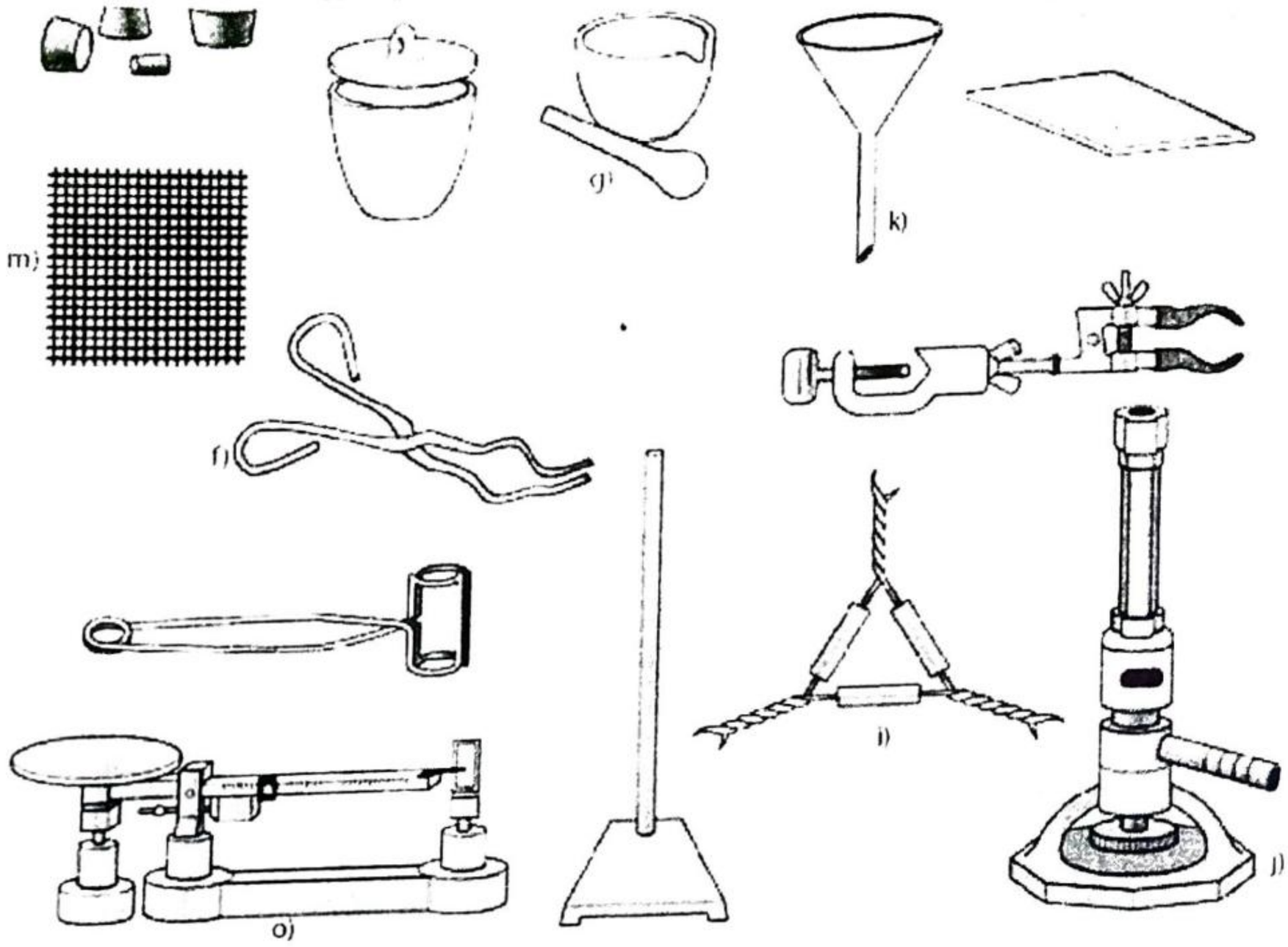
**Laboratory report answers the following questions.**

- ➦ What did you do?
- ➦ How did you do it?
- ➦ Why did you do it?
- ➦ What happened?

## Illustrative Examples

33. Write the name of the following laboratory apparatus.






**Solution:**

a. Beaker
b. Conical flasks
c. Glass or plastic funnel
d. Dropping pipette or teat pipette
e. Evaporating dish
f. Tong
g. Crucible and lid
h. Plastic or nickel spatuala
i. Tripod
j. Bunsen burner
k. Funnel
l. Round botton flask
m. Wire gauze
n. Test tubes
o. Balance

34. State some laboratory rules that are not stated in this book.

**Solution:**

- Never remove chemicals from the laboratory.
- Never return extra chemicals into the reagent bottles.
- Add boiling chips to liquids to be boiled.
- Never place chemicals directly on the pan balances, use filter paper of other equipments.
- Eating and drinking in the laboratory are not allowed.

35. Write the name of laboratory apparatus used to

a. Melt solids.

b. Measure volume of liquids.

c. Measure mass of solid substances.

d. Told hot test tubes.

e. Burn solid substances.

**Solution:**

a. Crucible

b. Pipette, volumetric flask, measuring cylinder, buret.

c. Top pan balance, double beam balance, digital balance.

d. Test tube holder.

e. Tongs

36. Write a short laboratory report of the experiment of burning magnesiun ribbon to produce basic oxide.

**Explanation:**

**Title:** Burning of magnesium ribbon.

**Objective**: To prepare basic oxide from burning of magnesium ribbon.

**Materials and chemicals**: Magnesium ribbon, Bunsen - burner, test tube or crucible, water, litmus paper, tong.

**Procedure**: Turning on the Bunsen burner, holding the magnesium ribbo with tong and burning it.

**Observation**: A bright light was observed when magnesium burnt in air. The white ash formed was collected in the crucible.

**Result and discussion:** At the end of burning, white ash was formed and dissolved in water. When the solution was checked with indicators it turned the litmus paper to blue.

**Conclusion**: The white ash formed was known to be basic oxide since it changed red litmus paper to blue.

## Solved Problems

37. Relate the different units of length.

**Solution:**

$1km = 10^3m = 10^5cm = 10^6mm = 0.62$ mile

$1m = 1.094$ yard

1 inch $= 2.54cm = 0.0254m = 25.4mm$

1 foot $= 12$ inch

1 yard $= 3$ feet

1 mile $= 5280$ feet

$1 cm = 10mm$

38. Relate the different units of mass and volume.

**Solution:**

- $1 kg = 1000g = 10^6mg$
- 1 pound $= 0.4536kg$
- 1 ounce $= 0.02835$ kg
- 1 metric tone $= 1000kg$

**Volume Units**

- $1m^3 = 10^3dm^3 = 10^3L = 10^6cm^3 = 10^9mm^3$
- $1dm^3 = 1L$, $1cm^3 = 1mL$
- $1L = 1.06$ quarts.

39. Show that kg/L and g/mL are equivalent units of volume.

**Explanation:** $1 kg = 1000g$ and $1L = 1000mL$

$$\Rightarrow \frac{1kg}{1L} = \frac{1000g}{100mL} \Rightarrow 1kg/L = 1g/mL$$

40. Bromine is a reddish - brown liquid. Calculate its density if 586g of the substance occupies 188mL.

**Solution:**

$$\rho = \frac{m}{v} = \frac{586g}{188mL} = 3.12g/mL$$

41. Determine the number of significant figures in each of the following measurements.

a. 0.005L
b. 60,156 tons
c. 6 kg
d. 60m
e. $9.50 \times 10^{4}$cm
f. $10.001 m^3$

**Solution:**

a. 1
b. 5
c. 1
d. ambigous
e. 3
f. 5

42. Carryout the following operations in the correct number of significant figures.

a. 7.310 km ÷ 5.70km
b. $(3.26 \times 10^{-3}mg) - (7.88 \times 10^{-5}mg)$
c. $(4.02 \times 10^{6}\ dm) + (7.74 \times 10^{7}\ dm)$
d. $(1.0 \times 10^{4}\ m) \times (9.9 \times 10^{6}m)$
e. $(7.95 \times 10^{4}L) \div (2.5 \times 10^{2}L)$

**Solution:**

a. 1.28
b. $3.18 \times 10^{-3}mg$
c. $8.1 \times 10^{7}dm$
d. $9.9 \times 10^{10}\ m^2$
e. $3.2 \times 10^{2}$

43. At what temperature does the numerical reading on a Celcius thermometer equal that on Fahrenheit thermometer?

**Solution:**

$$°C = °F$$

$$\frac{5}{9}(°F - 32) = \frac{9}{5}°C + 32 \text{ but } °F = °C$$

$$\frac{5}{9}°C - \frac{5}{9} \times 32 = \frac{9}{5}°C + 32$$

$$\frac{5}{9}°C - \frac{9}{5}°C = 32 + \frac{5}{9} \times 32$$

$$-\frac{56}{45}°C = 32 \times \frac{14}{9}$$

$$°C = -40$$

∴ At -40, °C and °F are equal.

44. Chalcopyrite, the principal ore of copper contains 35% copper by mass. How many grams of copper can be obtained from $5.11 \times 10^3$kg of the ore?

**Solution:**

Percentage means parts per hundred $\left(\frac{1}{100}\right)$

$\Rightarrow$ Mass of copper = 35% × mass of ore

$$= \frac{35}{100} \times 5.11 \times 10^3 \text{ kg}$$

$$= 1.8 \times 10^3 \text{kg}$$

45. How many seconds are in a solar year (365.24 days)?

**Solution:**

1 day = 24 hours = 1440 minutes = 86,400 seconds

$\Rightarrow$365.24 days = 86,400 sec × 365.24 = $3.156 \times 10^7$s

46. Classify the following as chemical or physical property.
   a. The normal color of bromine is red - orange.
   b. Iron rusts in the presence of air and moisture.
   c. Dynamite can explode when it interacts with oxygen.
   d. The density of uranium metal is 19.07 g/mL.
   e. Aluminum metal melts at 660°C.

**Solution:**

- a, d and e are physical properties.
- b and c are chemical properties.

47. Oxygen freezes to a solid at -218°C. What is this temperature in °F? in K?

**Solution:**

$$°F = \frac{9}{5}.°C + 32 = \frac{9}{5} \times -218 + 32 = -360.4°F$$

$$K = °C + 273 = -218 + 273 = 55K$$

48. Calculate the number of moles of $1.2 \times 10^{24}$ HCl molecules.

**Solution:**

$$n = \frac{\text{given number of molecules}}{6.02 \times 10^{23} \text{molecule/mol}}$$

$$n = \frac{1.2 \times 10^{24} \text{molecules}}{6.02 \times 10^{23} \text{molecules/mol}} = 2\text{mol}$$

49. Calculate the number of hydrogen atoms in 8.5g of ammonia. ($M_{NH_3} = 17g/mol$)

**Solution:**

$$n_{NH_3} = \frac{m_{NH_3}}{M_{NH_3}} = \frac{8.5g}{17g/mol} = 0.5mol$$

$$\Rightarrow n_H = n_{NH_3} \times \text{subscript of H} = 0.5mol \times 3 = 1.5mol$$

$\therefore$ Number of atoms of hydrogen $= n \times 6.02 \times 10^{23}$ atoms/mol

$= 1.5 \text{ mol} \times 6.02 \times 10^{23}$ atoms/mol

$= 9.03 \times 10^{23}$ atoms

50. Convert the following

a. $0.0712 \text{ cm}^3$ to $\text{dm}^3$

b. $6A^\circ$ to $\mu m$

c. $6.72 \times 10^{-7}$m to nm

**Solution:**

a. $71.2\text{dm}^3$ b. $6 \times 10^{-4}\mu m$ c. 672 nm

51. Hexane, a solvent used for rubber cement, has a specific gravity of 0.668. What is the density of hexane in $g/cm^3$ and in $kg/m^3$? Water has a density of $1 g/cm^3$.

**Solution:**

$$\text{Specific gravity} = \text{specific density} = \text{sp.gr} = \frac{\rho_{\text{substancees}}}{\rho_{\text{water}}}$$

$$\text{sp.gr} = \frac{\rho_{\text{hexane}}}{\rho_{\text{water}}}$$

$$\rho_{\text{hexane}} = \text{sp. gr.} \rho_{\text{water}}$$

$$= 0.668 \times 1 \text{ g/cm}^3 = 0.668 \text{ g/cm}^3$$

$$1 \text{ g/cm}^3 = 1000 \text{ kg/m}^3$$

$$\therefore 0.668 \text{ g/cm}^3 = 668 \text{ kg/m}^3$$

52. What is scientific method?

**Explanation:** Scientific method is a set of guidelines for the practice of science to ensure a certain scientific method.

# Supplementary Exercise

I. Match the quantities stated on the left with their SI unit stated on the right.

| | A | B |
|---|---|---|
| 1 | Length | A. square meter ($m^2$) |
| 2 | mass | B. Newton (N) |
| 3 | pressure | C. Kelvin (K) |
| 4 | Force | D. Pascal (Pa) |
| 5 | Temperature | E. Kilogram (kg) |
| 6 | Time | F. Ampere (A) |
| 7 | Current | G. Cubic meter ($m^3$) |
| 8 | Volume | H. Second (s) |
| 9 | Density | I. Kilogram per cubic meter ($kg/m^3$) |
| 10 | Area | J. Meter (m) |
| | | K. Mole (mol) |
| | | L. Unit less |

II. Choose the correct answer

11. Which of the following has no unit?
    A. Amount of substance
    B. Luminuous intensity
    C. specific gravity
    D. weight

12. All of the following are chemical changes. EXCEPT
    A. Heating of sugar
    B. decaying of food
    C. reaction of potassium with water
    D. sublimation of iodine to vapour

13. Which of the following is not intensive properties?
    A. colour
    B. Area
    C. Odour
    D. Conductivity

14. Which of the following branches of chemistry involves separating, identifying and determing the relative amount of a sample of matter?
    A. Analytical chemistry
    B. Inorganic chemistry
    C. Organic chemistry
    D. Physical chemistry
15. Which one is used as a unit of time for long time interval?
    A. second
    B. microsecond
    C. millisecond
    D. hour
16. A fraction $10^{12}$ represents a prexix
    A. giga
    B. mega
    C. tera
    D. pico
17. Which one of the following is a cause for uncertainty?
    A. The person doing the experiment
    B. The measuring equipment
    C. The environment where the measurement is being made.
    D. All of the above
18. Which of the following measurement can be expressed by exact numbers?
    A. The volume of water in a cylinder
    B. The area of a rectangle
    C. The mass of a piece of calcium metal
    D. The number of microseconds in a minute
19. A concise verbal statement or mathematical equation that summarizes a broad variety of observation is
    A. theory
    B. law
    C. hypothesis
    D. observation
20. Which one of the following is measuring equipment?
    A. Spirit burner
    B. test tube
    C. thermometer
    D. boiling tube
21. Which of the following is not allowed in the chemistry laboratory?
    A. Wearing any type of cloth and shoe
    B. Knowing the hazards of the materials being used.
    C. Reading the labels on the reagent bottles carefully
    D. Using glycerin when inserting glass tubing into rubber stoppers.

22. Which of the following has three significant figures?

A. 530
C. 0.003
B. 0.7130
D. 4.01

23. A blood sample analysis showed a total of 0.00467g lead. What is the result if it is expressed in micrograms?

A. 0.467μg
C. $4.67 \times 10^6$μg
B. $4.67 \times 10^3$μg
D. $4.67 \times 10^4$μg

24. Which of the following is true?

A. A scientific notation for 0.00021 is $2.1 \times 10^{-4}$
B. A substance with density 2.7g/ml floats on water
C. one meter has about the same length as a yard.
D. The prefix kilo means multiplied by 0.001.

25. The number $5.68 \pm 0.0001$ has _____ significant figures

A. 4
C. 6
B. 5
D. 3

26. Which of the following is derived quantity?

A. force
C. temperature
B. mass
D. time

27. Which of the following has the lowest uncertainty?

A. 6.0
C. 6.000
B. 6.00
D. 6.0000

28. The knowledge of chemistry is not important in one of the following?

A. in agriculture
C. in medicine
B. in geology
D. none

29. Which of the following is not correct?

A. Density has no units.
B. Every measurement has a unit tied to it.
C. Thee Kelvin degree is larger than the Celsius degree
D. Changes of state are physical properties.

30. The density of carbon in the form of diamond is 3.51g/cm$^3$. If you have a small diamond with a volume of 0.027cm$^3$, What is its mass in gram?
    A. 0.095
    B. $2.1 \times 10^{23}$
    C. 0.0078
    D. 0.027

31. One atmosphere is not equivalent with
    A. 760mmHg
    B. 760torr
    C. 760cmHg
    D. 101.3kpa

32. Which of the following is caused by poor calibration of instruments?
    A. Random error
    B. Relative average deviation
    C. Average deviation
    D. systematic error

33. The number of significant figures in $2.35 \times 10^{-5}$A is
    A. 5
    B. 4
    C. 3
    D. 6

34. One mole is not equal to
    A. $6.02 \times 10^{23}$ particles
    B. A gram atom
    C. A gram molecule
    D. A formula unit

35. Which one of the following has the largest number of atoms?
    A. 1g$H_2$
    B. 1g$O_2$
    C. 1g$N_2$
    D. 1g$F_2$

III. Give short answer

36. Calculate the number of mole of hydrogen in 171g of sucrose ($C_{12}H_{22}O_{11}$).

37. State the number of significant figures in the following quantities.
    a. 6.00g of gold
    b. 0.004sec
    c. 100 test tubes
    d. $6.001 \times 10^6$mL of water

38. Round off the following numbers to the indicated number of significant figures (sf.)
    a. 8.348 (2sf)
    b. 8.348 (3sf)
    c. 8.345 (3sf)
    d. 8.375 (3sf)
    e. 2.99298 (2 sf)

39. A volumetric flask has a volume of 250mL and weighs 100g. If it weighs 227g when filled with pure ether, what is the density of ether?

40. Perform the following calculations and express the result to the proper number of significant figures

    a. $21.235 + 3.24$  b. $135.24 - 48.1$  c. $(3.60 \times 273)/1372$

41. Convert the following temperatures as indicated

    a. -127°C to K  b. 3K to °C  c. 212°F to K

42. Find the mass of 1.5mol of calciumhydroxide ($Ca(OH)_2$)

43. The ____ of measurement describes how well repeated measurements of a value agree with one another while ____ is determined by the agreement between the measured quantity and the correct value.

44. A student determined his body temperature to be 38°C. What is the error involved in his measurement, if the normal body temperature is 37°C?

45. Calculate the number of seconds in 10min.

አጋዥ መጽሐፍቶችን ለማግኘት በ
Telegram @ethiolibrary1 ላይ
ይቀላቀሉን
+251927273995
+251909282436
jemal6454@gmail.com

ኣጋዥ መጻሕፍቲ ብ Telegram ንምርካብ ናብዚ ይቀላቀሉን
@ethiolibrary1
+251927273995
+251909282436
jemal6454@gmail.com
Ahmed Ahmed

# Unit 2

## Atomic Structure and The Periodic Table

## Unit Content Outline:

2.1 Dalton's atomic theory
2.2 Modern atomic theory
2.3 Electromagnetic radiation
2.4 The Bohr model of the hydrogen atom
2.5 The quantum model of the atom
2.6 The electron configuration and orbital diagram
2.7 The periodic classification of elements

**Introduction**

The concept of 'atom' was originated by the Greeks, especially **Democritus and Leucippus.**

**Democritus**: believed that matter is discrete meaning division has limit.

According to Democritus, the world was made up of two things, **atoms** and **empty space**.

**Aristotle**: was the opponent of Democritus.

He believed that matter is continuous meaning division has no limit.

## 2.1 Dalton's Atomic Theory

**Dalton** formulated a precise definition for the indivisible building blocks of matter that we call atoms.

➦ His work marked the beginning of the modern era of chemistry.

**Postulates of Dalton**

1. Elements are composed of extremely small particles called atoms.
2. Atoms are indivisible and indestructible.
3. All atoms of a given element are identical, having the same size, mass and chemical properties.
4. The atoms of one element are different from the atoms of other elements.
5. Compounds are composed of atoms of more than one element. In any compound, the ratio of the number of atoms of any two of the elements present is either an integer or a simple fraction.
6. A chemical reaction involves only the separation, combination or rearrangement of atoms, it does not result in their creation or destruction.

➦ Dalton made no attempt to describe the structure or composition of atoms.

➦ Dalton's sixth postulate is another way of stating the law of **conservation of mass**.

Law of conservation mass (expressed by Lavoisier in 1774) states that matter is neither created not destroyed in ordinary chemical reaction.

➦ **Dalton's fifth** postulate supports the laws of definite and multiple proportions.

**Law of definite proportion** (expressed by Proust in 1799) states that all pure samples of the same compound contain the same elements combined in fixed proportion by mass.

**Law of multiple proportion** (Dalton, 1803) states that, if two elements can combined to form more than one compound, the masses of one element that combine with a fixed mass of the other element are in ratios of small whole numbers.

**Note**: Modified postulates of Dalton are

1. Atoms are indivisible.
2. Atoms of the same element have same mass and other properties.

## 2.2 Modern Atomic Theory

**Postulates of Modern Atomic Theory**:

1. Elements are made up of small particles called atoms.
2. Atoms are indestructible during ordinary chemical reactions.
3. Atoms are divisible into sub-atomic particles such as electrons, protons and neutrons.
4. Atoms of the same element have different mass and physical properties due to isotopes.
5. Atoms form compounds by combining with small whole number ratios.
6. Atoms of different element are different.

### Illustrative Examples

1. The Greek Philosoper that proposed the first atomic theory was

   A. Plato
   B. Parmenides
   C. Dmocritus
   D. Aristotle

**Explanation: Plato** - believed that the earth, air, fire and water are comprised of unchanging geometric solids.

**Parmenides** - believed that all changes are illusion.

**Democritus** - proposed the first atomic theory.

**Aristotle** - was the opponent of Democritus.

**Answer: C**

2. Which of the following is not true about Dalton's atomic theory?
   A. Atoms of different element are different.
   B. Atoms are indivisible.
   C. Atoms of the same element have different mass.
   D. Atoms combine in small whole number ratio to form compounds.

**Answer: C**

3. Write the main draw backs of Dalton's atomic theory.

**Explanation:**

- ➦ Dalton failed to explain why atom is divisible.
- ➦ He failed to explain why atoms of the same element have different mass.

4. What were the bases for Dalton's atomic theory?

**Explanation:**

- ➦ Law of conservation of mass.
- ➦ Law of definite composition.
- ➦ Law of multiple proprtion.

5. Show how the five oxides of nitrogen illustrate the law of multiple proportion.

**Explanation:** Atomic mass of N = 14, O = 16

The five oxides of nitrogen are $N_2O$, NO, $N_2O_3$, $NO_2$ and $N_2O_5$.

| Oxides | Mass of ratio of N to O | Ratio of Oxygen dividing each by the smallest |
|---|---|---|
| $N_2O$ | $28:16 \Rightarrow \frac{28}{28}:\frac{16}{28}$ | $\frac{16}{28} \div \frac{16}{28} = 1$ |
| NO | $14:16 \Rightarrow \frac{14}{14}:\frac{16}{14}$ | $\frac{16}{14} \div \frac{16}{28} = 2$ |
| $N_2O_3$ | $28:48 \Rightarrow \frac{28}{28}:\frac{48}{28}$ | $\frac{48}{28} \div \frac{16}{28} = 3$ |
| $NO_2$ | $14:32 \Rightarrow \frac{14}{14}:\frac{32}{14}$ | $\frac{32}{14} \div \frac{16}{28} = 4$ |
| $N_2O_5$ | $28:80 \Rightarrow \frac{28}{28}:\frac{80}{28}$ | $\frac{80}{28} \div \frac{16}{28} = 5$ |

∴ The ratio oxygen in the fives oxides of nitrogen by fixing the mass of nitrogen is 1: 2: 3: 4: 5.

6. A 17.52g sample of element M reacts with 46.52g of a sample of element N, to form the compound $M_2N_3$. What is the mass of reactants and product? What can you conclude from the result?

**Solution**:

Based on the law of conservation of mass:

Mass of reactants = Mass of products

➦ Mass of reactants = mass of product = 17.52 + 46.52 = 64.04g

7. Write two modified postulates of Dalton.

**Explanation**:

- ➦ Atom is indivisible.
- ➦ Atoms of the same element have same mass and other properties.

## Thomson's Experiment, the discovery of electron

- ➦ Thomson used a discharge tube to study the electron.
- ➦ A high potential difference was applied across the two electrodes (anode and cathod).
- ➦ When electricity was switched on a beam of rays emitted from the cathode. These rays were called **cathode rays** because they came out from the cathode.

## Properties of Cathode rays

- ➦ They move in straight lines.
- ➦ They are deflected by both magnetic and electric fields.
- ➦ They are deflected toward positive plate.
- ➦ They are found in all elements.

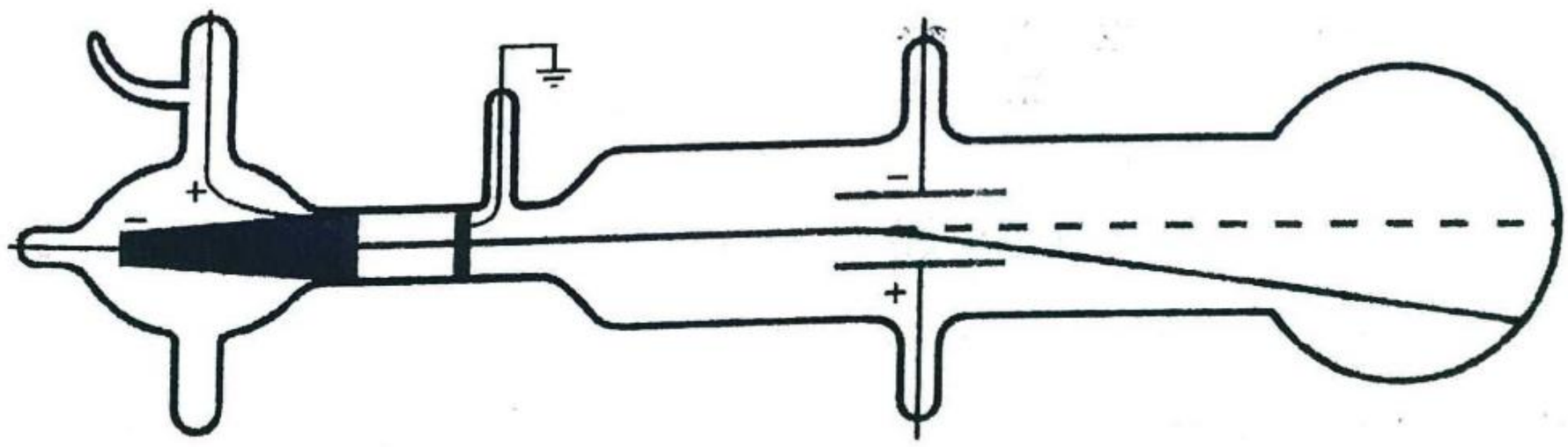



Figure 2.1 Cathode ray tube used by Thomson

- The charge of an electron was determined by **R. Millkan** using oil **drop experiment**.
- The charge of an electron is $-1.602 \times 10^{-19}$C.
- The charge to mass ratio of an electron was determined by **J.J. Thomson**.
- The charge to mass ratio ($e^-/m$) of an electron is $-1.76 \times 10^8$C/g.
- From charge to mass ratio and charge of an electron, it is possible to calculate its mass.

$$\Rightarrow m_{e^-} = \frac{e^-}{e^-/m}$$

$$m_{e^-} = \frac{-1.602 \times 10^{-19}\,C}{-1.76 \times 10^{8}\,C/g}$$

$$\boxed{\begin{aligned} m_{e^-} &= 9.10 \times 10^{-28}\,g \\ m_{e^-} &= 9.10 \times 10^{-31}\,kg \end{aligned}}$$

$m_e$- = mass of an electron

$e^-$ = charge of electron

$\frac{e^-}{m}$ = charge to mass ratio of an electron.

## Illustrative Examples

8. Which of the following was determined by the Millikan's oil drop experiment? (EUEE)

   A. Charge on the electron.
   B. e/m ratio for an electron.
   C. Mass of an electron
   D. Velocity of an electron.

**Answer: A**

9. Which of the following is not true about an electron?

   A. It has the smallest mass.
   B. It has negative charge.
   C. It has the highest charge to mass ratio.
   D. It was discovered by J. Dalton.

**Answer: D**

10. Calculate the mass of Avogadro's number of electrons.

**Solution:**

mass of one electron $= 9.10 \times 10^{-28}$g

mass of $6.02 \times 10^{23}$ electrons = ?

$\Rightarrow 6.02 \times 10^{23} \times 9.10 \times 10^{-28}\text{g} = 4.48 \times 10^{-4}\text{g}$

11. Calculate the charge of Avogadro's number of electrons.

**Solution**:

Charge of one electron = $-1.602 \times 10^{-19}C$

Charge of $6.02 \times 10^{23}$ electrons = ?

$\Rightarrow 6.02 \times 10^{23} \times (-1.602 \times 10^{-19}C) = -9.64 \times 10^{4}C$

12. Why an electron has the highest charge to mass ratio?

**Explanation**: Because it has the smallest mass.

## Radioactivity and Discovery of the Nucleus

**Radioactivity** is spontaneous emission of particles and/or radiation by radioactive elements.

➦ Elements with unstable nucleus that emit radiation and/or particles are called **radioactive elements**.

**Example**

Uranium (U-238), Carbon (C - 14)

Chromium (Cr - 55), Radium (Ra - 226) etc

In general elements whose atomic number is greater than or equal to 83 are radio active elements.

➦ There are three types of rays emitted by radioactive elements. These are

1. Alpha ($\alpha$) rays
2. Beta ($\beta$) rays
3. Gamma ($\gamma$) rays

Table 2.1. The properties of the three types of rays

| Rays | Symbol | Charge | Mass in gram | Deflection | Penetration |
|---|---|---|---|---|---|
| Alpha | ${}_{2}^{4}\alpha, {}_{2}^{4}He^{2+}$ | +2 | $6.65\times10^{-24}$ | Deflected toward negative plate | Stopped by paper |
| Beta | ${}_{-1}^{0}\beta, {}^{0}_{-1}e$ | -1 | $9.11\times10^{-28}$ | Deflects toward positive plate | Stopped by aluminum |
| Gamma | ${}_{0}^{0}\gamma$ | 0 | 0 | Undeflected | Stopped by thick lead sheet |

## Discovery of Nucleus

**Nucleus** was discovered by **E. Rutherford** using alpha scattering experiment. He alloted alpha particles to pass through gold foil. Three main observations were made

1. Most of the alpha particles went through the thin gold foil without any change in path. This indicated most of the space in an atom was **empty**.
2. A few alpha particles were deflected through small angles. This showed that the nucleus has positive charge.
3. Few alpha particles reflected with large angles because of the **heavy nucleus**.

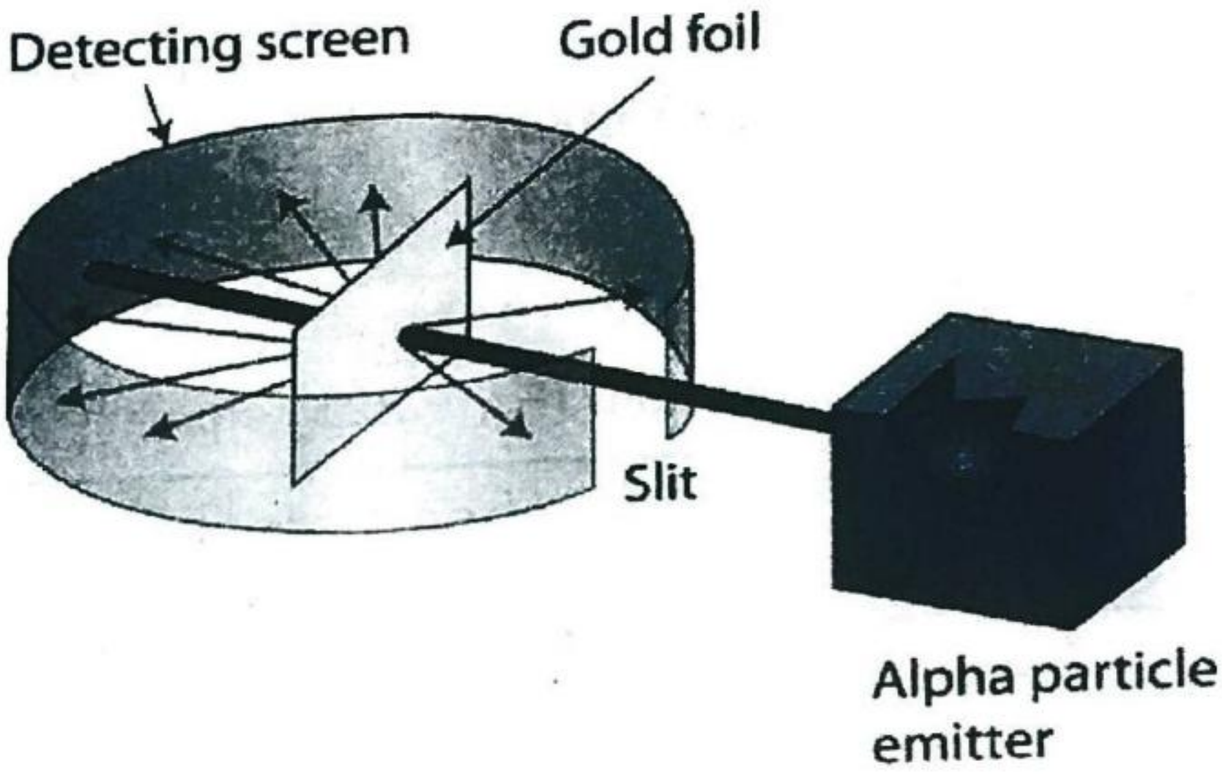



Figure 2.2 Rutherford's gold foil experiment

## The Neutron

It was discovered by **James chadcuick**. When he bombarded a thin sheet of beryllium with alpha particles, a very high energy radiation similar to gamma rays was emitted by the metal. Later experiments showed that the rays actually consisted of electrical neutral particles called **neutrons**.

Table 2.2. The three fundamental sub – atomic particles

| Particle | Symbol | Mass in gram | Mass in amu | Charge in Coulomb | Charge unit |
|---|---|---|---|---|---|
| Electron | e | $9.10 \times 10^{-28}$ | 0 | $-1.6022\times10^{-19}$ | -1 |
| Proton | p | $1.673\times10^{-24}$ | 1 | $+1.6022\times10^{-19}$ | -1 |
| Neutron | n | $1.675\times10^{-24}$ | 1 | 0 | 0 |

- Proton and neutron are collectively known as **nucleons.**
- Proton and neutron are found in the nucleus.
- Electron is found outside the nucleus.
- When a radioactive element emits an alpha particle, its mass decreases by four and its charge decreases by two.

**Example**

$$^{238}_{92}U \longrightarrow ^{4}_{2}\alpha + ^{234}_{90}Th$$

- When a radioactive element emits a beta particle, its mass remains same but its charge increases by one.

**Example**

$$^{1}_{0}n \longrightarrow ^{0}_{-1}\beta + ^{1}_{1}H, ^{235}_{92}U \longrightarrow ^{0}_{-1}\beta + ^{235}_{93}Np$$

## Illustrative Examples

13. Which of the following types of rays combine to form atoms of helium? (EUEE ... 2008 E.C)

A. gamma rays
B. beta rays
C. alpha rays
D. x - rays

**Explanation:** Experiments showed that alpha rays were nuclei of helium.

**Answer: C**

14. What did Rutherford's alpha particle experiment show?

A. Electrons have a negative charge.
B. A proton is a hydrogen atom without electron.
C. Electrons circle the nucleus of an atom in orbits.
D. Most of the mass and all of the positive charge of an atom is found in the tiny nucleus.

**Explanation:** Rutherford discovered that an atom has a tiny center with positive charge and heavy mass.

**Answer: D**

15. The mass and charge of a proton are $1.673 \times 10^{-24}$g and $1.602 \times 10^{-19}$C. Find its charge to mass ratio.

**Solution:**

$$\frac{\text{charge}}{\text{mass}} = \frac{e}{m} = \frac{1.602\times10^{-19}\text{C}}{1.673\times10^{-24}\text{g}} = 9.58\times10^{4}\text{C/g}$$

16. What are the two regions of an atom?

**Solution:**

Nucleus and atomic shell.

17. Write the two important uses of neutrons.

**Solution:**

- It reduces the strong electrostatic repulsion forces between protons.
- Half of the mass of an atom is almost the mass of a neutron.

18. Is there an atom without a neutron?

**Explanation**: Ordinary hydrogen (protinium) has no neutron.

19. Why an atom has equal number of protons and electrons?

**Explanation**:

An atom is neutral if it has equal number of protons and electrons.

20. Compare the mass of a proton and an electron.

**Solution:**

$$\frac{m_P}{m_e} = \frac{1.673\times10^{-24}\,g}{9.10\times10^{-28}\,g} = 1838$$

$\Rightarrow m_P = 1838\ m_e$ $m_P$ = mass of proton, $m_e$ = mass of electron

21. Sulfur - 35, emits an alpha particle. What is the mass of the new isotope formed?

**Solution:**

Let the new isotope formed be x

$${}^{35}_{16}S \longrightarrow {}^{4}_{2}He + {}^{a}_{b}X$$

Since mass is conserved, $a + 4 = 35$ and charge is conserved, $b + 2 = 16$

$\Rightarrow a = 35 - 4 = 31$ and $b = 16 - 2 = 14$

$\therefore X = {}^{31}_{28}Si$ and its mass becomes 31.

22. When thorium - 234, ${}^{234}_{90}Th$ emits a beta particle, what will be the charge of the resulting isotope?

**Solution:**

$${}^{234}_{90}Th \longrightarrow {}^{0}_{-1}B + {}^{x}_{y}Pa$$

$\Rightarrow x + 0 = 234$ $\quad y + (-1) = 90$

$x = 234$ $\quad y = 91$

$\therefore$ The charge of Pa is +91

## Atomic number and Mass number

**Atomic number (Z)**: is the number of proton (s) in an atom. For a neutral atom:

$$Z = P = e$$

**Mass number (A)**: is the sum of atomic number and neutron number.

$$A = Z + n = P + n$$

Mass number is the same as number of nucleons.

## Isotopes, Isotones, Isobars and Isoelectronics

**Isotopes** are atoms of the same elements with same atomic number but different mass number.

Example

$^{1}_{1}H, ^{2}_{1}H, ^{3}_{1}H$ or $H-1, H-2, H-3$

**Isotones** are atoms of different element with the same **neutron number**.

Example

$^{23}_{11}Na$ and $^{24}_{12}Mg$ have same neutron number.

**Isobars** are atoms of different element with the same mass number.

Example

$^{14}_{6}C$ and $^{4}_{7}N$

**Isoelectronic** are different species with the same **electron number**.

Example

$_{10}Ne, \ _{11}Na^{+1}, \ _{9}F^{-1}$

## Atomic Mass

The actual mass of an atom is very small and determined by using **mass spectrometer**.

Example

The actual mass of C - 12 is $2 \times 10^{-23}$g.

**Atomic mass unit (amu)** is the unit of a scale of relative atomic masses of the elements.

$$1\text{amu} = \frac{1}{12} \times \text{mass of C - 12}$$

$$= \frac{1}{12} \times 2 \times 10^{-23}\,\text{g} = 1.67 \times 10^{-24}\,\text{g}$$

$$\text{Atomic mass of an atom} = \frac{\text{Absolute mass of an atom}}{1.67 \times 10^{-24}\ \text{g/ amu}}$$

The absolute mass of magnesium atom is $4 \times 10^{-23}$g. Calculate its atomic mass.

**Solution**

$$\text{Atomic mass of magnesium} = \frac{\text{Absolute mass of magnesium}}{1.67 \times 10^{-24}\text{g/amu}}$$

$$= \frac{4 \times 10^{-23}\,\text{g}}{1.67 \times 10^{-24}\,\text{g/amu}} = 24\text{amu}$$

**Average or relative atomic mass** is the average mass of an atom in a representative sample atoms of an element.

$$\text{Average atomic mass} = \frac{A_1f_1 + A_2f_2 + \ldots A_nf_n}{100}$$

- $A_1, A_2, \ldots A_n$ atomic mass of first, second, $n^{th}$ isotopes, respectively.
- $f_1, f_2, \ldots f_n$, percentage abundance of first, second, nth isotopes respectively.

$$f_1 + f_2 + \ldots f_n = 100$$

$$\text{Average atomic mass} = A_1x_1 + A_2x_2 + \ldots A_nx_n$$

- $x_1, x_2, \ldots x_n$ fraction abundance of first, second, $n^{th}$ isotopes respectively.

$$x_1 + x_2 + \ldots x_n = 1$$

## Example

Copper has two naturally occurring isotopes with the given mass and percentage abundance.

| Isotope | mass (amu) | relative abundance |
|---|---|---|
| Cu - 63 | 62.93 | 69.08% |
| Cu - 65 | 64.93 | 30.91% |

Calculate the average atomic mass of copper.

**Solution**

$A_1 = 62.93$ $f_1 = 69.09\%$

$A_2 = 64.93$ $f_2 = 30.91\%$

$$\text{Average atomic mass} = \frac{A_1 f_1 + A_2 f_2}{100}$$

$$= \frac{62.93 \times 69.09 + 64.93 \times 30.91}{100} = 63.55\,\text{amu}$$

## Illustrative Examples

23. What is the difference between Chlorine - 35 and chlorine -37? (EUEE ...2004 E.C)

A. Chlorine - 37 has two more protons than Chlorine - 35

B. Chlorine - 37 has two more neutrons than Chlorine - 35

C. Chlorine - 37 has two more electrons than Chlorine - 35

D. Chlorine - 37 has two more protons and once more neutron than Chlorine - 35.

**Explanation:** Isotopes have the same proton, electron and atomic number but different neutron and mass number.

$\Rightarrow$ Cl - 37 has two more neutrons than Cl - 35.

**Answer: B**

24. A monoatomic ion has 20 protons and a +2 charge: (EUEE)

A. has 16 protons

B. has the symbol $Ar^{2+}$

C. has 18 neutrons

D. is isoelectronic with Ar

**Explanation:** For mono atomic ion, e = p - charge = 20 - 2 = 18 Argon (Ar) has also 18 electrons. Thus, they are isoelectronic.

**Answer: D**

25. Which of the following has 45 neutrons, 35 protons and 36 electrons?

A. $^{80}_{35}Hg$ B. $^{103}_{45}Rh^-$ C. $^{80}_{35}Br^-$ D. $^{71}_{35}Lu^-$

**Solution:**

Let the species be $^{4}_{2}X$

$\Rightarrow A = P + n = 35 + 45 = 80, z = p = 35$

$Charge = p - e = 35 - 36 = -1$

$\therefore {}^{80}_{35}X^{-1}$, thus X = Br

**Answer: C**

26. Natural copper is a mixture of only two isotopes one of which has 29 protons, 34 neutrons and a relative abundance of 69.17%. The relative atomic mass of Copper is 63.6 amu. How many neutrons must be there in the second isotope?

A. 29 B. 36 C. 32 D. 35

**Solution**

$A_1 = p + n_1 = 29 + 34 = 63, f_1 = 69.17\%$

$A_2 = p + n_2 = 29 + n_2, f_2 = 100 - f_1 = 30.83\%$

$$\text{Average atomic mass} = \frac{A_1f_1 + A_2f_2}{100}$$

$$63.6 = \frac{63 \times 69.17 + (29 + n_2)30.83}{100}$$

Solving for $n_2$, $n_2 = 36$

**Answer: B**

27. In which pair of the elements do the nuclei of the atoms contain the same number of neutrons?

A. $^{7}_{3}Li$ and $^{9}_{4}Be$

B. $^{23}_{11}Na$ and $^{24}_{12}Mg$

C. $^{14}_{7}N$ and $^{16}_{8}O$

D. $^{32}_{16}S$ and $^{35}_{17}C$

**Answer: B**

28. Determine the number of protons, electrons and neutrons in:

a. $^{127}_{53}I$ b. $^{59}_{27}Co^{2+}$ c. $^{52}_{24}Cr^{2+}$ d. $^{79}_{34}Se^{2-}$

**Solution:**

a. For neutral atom, P = z = e = 53

$n = A - p = 127 - 53 = 74$

b. For charged atoms, e = p - charge = 27 - 2 = 25

$n = A - p = 59 - 27 = 32$

c. e = p - charge = 24 - 2 = 22

$n = A - p = 52 - 24 = 28$

d. e = p - charge = 34 - (-2) = 34 + 2 = 36

$n = A - p = 79 - 34 = 45$

29. Why do isotopes of an element have similar chemical properties but a different physical properties?

**Explanation:**

- Isotopes have similar chemical properties because they have same electron number.
- Isotopes have different physical properties because they have different mass number.

30. Lithium has two naturally occurring isotopes, Li - 6 and Li - 7. The average atomic mass of lithium is 6.94amu. Which isotope is the most abundant in nature?

**Explanation:** The most abundant isotope of an element has atomic mass nearer to the average mass. Thus, Li - 7 is the most abundant isotope of lithium in nature.

31. The element gallium has two stable naturally occurring isotopes. $^{69}Ga$ and $^{71}Ga$. Given that

| Isotope | Mass in amu |
|---|---|
| Ga - 69 | 68.9 |
| Ga - 71 | 70.9 |

The average atomic mass of gallium is 69.7amu. Calculate the relative percentage abundance of each isotope.

**Solution:**

$A_1 = 68.9, f_1 = 100 - f_2$

$A_2 = 70.9, f_2 = 100 - f_1$

$$\text{Average atomic mass} = \frac{A_1f_1 + A_2f_2}{100}$$

$$69.7 = \frac{68.9f_1 + 70.9(100 - f_1)}{100}$$

Solving for $f_1$, $f_1 = 60\%$ and $f_2 = 100 - f_1 = 100 - 60 = 40\%$

32. Natural copper consists of Cu - 63 and Cu - 65. What is the fraction abundances of each isotope if the average mass of copper is 63.5?

**Solution:**

$A_1 = 63$, $X_1 = 1 - X_2$

$A_2 = 65$, $X_2 = 1 - X_1$

Average atomic mass $= A_1X_1 + A_2X_2$

$63.5 = 63X_1 + 65\ (1 - X_1)$

$63.5 = 63X + 65 - 65X_1$

Solving for $X_1$, $X_1 = 0.75$

$\Rightarrow X_2 = 1 - X_1 = 1 - 0.75 = 0.25$

If the required is percentage abundance, $f_1 = X_1 . 100 = 75\%$ and $X_2 = 0.25 \times 100 = 25\%$

33. The atomic mass of the two stable isotopes of boron $^{10}_{5}B(0.1978)$ and $^{11}_{5}B(0.8022)$ are 10.0129 amu and 11.0093 amu, respectively. Calculate the relative atomic mass of boron?

**Solution:**

$A_1 = 10.0129$ amu, $X_1 = 0.1978$

$A_2 = 11.0093$ amu, $X_2 = 0.8022$

Average atomic mass $= A_1X_1 + A_2X_2$

$= 10.0129 \times 0.1878 + 11.0093 \times 0.8022 =$ 10.81 amu

34. The absolute mass of sodium atom is $3.81 \times 10^{-23}$ g. What is its atomic mass?

**Solution:**

$$\text{Atomic mass of Na} = \frac{\text{Absolute mass of Na}}{1.67 \times 10^{-24}\ \text{g/amu}}$$

$$= \frac{3.81 \times 10^{-23}\ \text{g}}{1.67 \times 10^{-24}\ \text{g/amu}} = 22.8\text{amu}$$

35. The atomic mass of aluminum is 27 amu. What is the absolute mass of aluminum atom?

**Solution:**

$$\text{Absolute mass} = \text{Atomic mass} \times 1.67 \times 10^{-24}\text{g/amu}$$
$$= 27 \text{ amu} \times 1.67 \times 10^{-24} \text{ g/amu} = 4.5 \times 10^{-23}\text{g}$$

36. Write the symbol of silver atom (Z = 47) that has 61 neutrons.

**Solution:**

$A = p + n = 47 + 61 = 108$

Thus, the symbol of silver is $^{108}_{47}Ag$.

## 2.3 Electromagnetic Radiation (EMR) and Atomic Spectra

**Electromagnetic radiation** is radiation that can be described interms of oscillating or wave - like electric and magnetic field in space.

**Some characteristics of EMR**

1. They are produced by the oscillation of electric charge and magnetic field residing on the atom.
2. They are characterized by their wave lengths and frequencies or wave numbers.
3. The energy carried by an electromagnetic radiation is directly proportional to its frequency.
   - ➦ The emission or absorption of radiation is quantized and each quantum of radiation is called **photon**.
4. All types of radiation travel with the same velocity and no medium is required for propagation.
5. When visible light is passed through a prism, it split up into seven colors which correspond to definite wavelengths.

### Units of light wave

i. **Wave - length (λ)** is the distance between two adjacent crests or troughs in a particular wave.

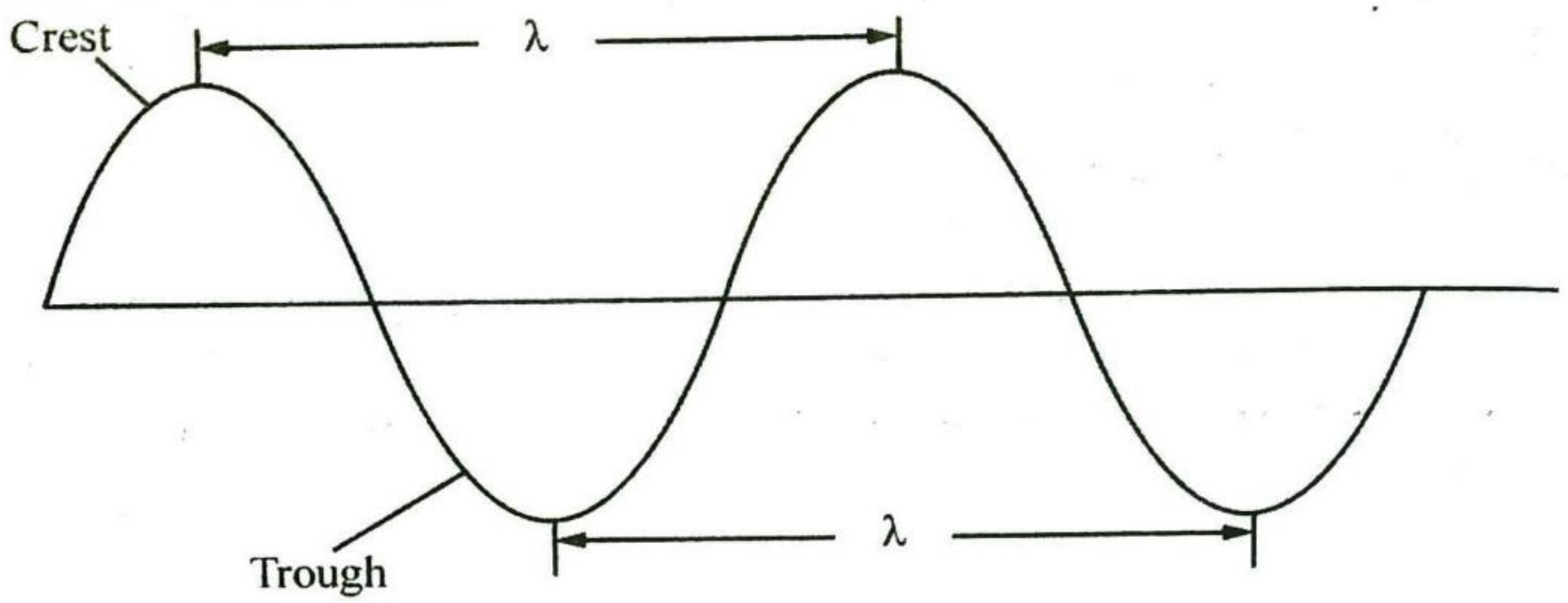



Figure 2.3

ii **Wave - number** $(\bar{v})$ is the reciprocal of wave - length expressed in centimeters, inverse ($cm^{-1}$)

$$\boxed{\bar{v} = \frac{1}{\lambda}}$$

iii **Frequency (v)** is the number of waves which can pass through a point in one second.

➦ Its unit is second inverse ($s^{-1}$) or Hertz (Hz).

iv **Speed (c)** is the product of wave - length and frequency.

$$\boxed{c = \lambda . v}$$

All electromagnetic radiations travel through vacuum with a constant speed of $3 \times 10^8 m/s$.

v. **Energy** of a wave of the particular radiation can be calculated by applying the relation:

$$\boxed{E = hv}$$

But $v = \frac{c}{\lambda}$

$$\Rightarrow \boxed{E = \frac{hc}{\lambda}}$$

But $\bar{v} = \frac{1}{\lambda}$

$$\Rightarrow \boxed{E = hc\bar{v}}$$

h - plank's constant

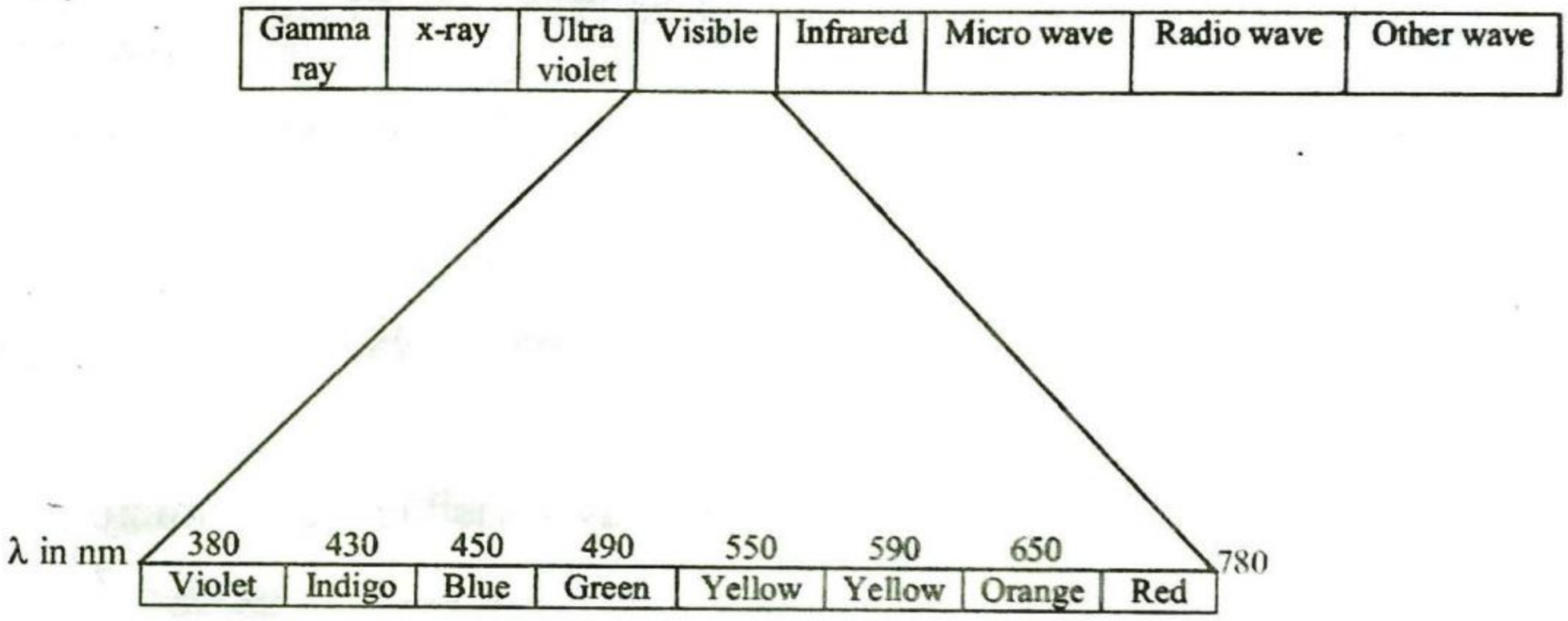
$h = 6.626 \times 10^{-34}$ J.s

c = speed of light

➦ Energy is directly proportion to wave number but inversely proportional to wave - length.

## Electromagnetic Spectrum

➦ The arrangement of all types of electromagnetic radiations in order of increasing wave lengths or decreasing frequencies is known as **electromagnetic spectrum.**




**Amplitude (A)** is the height of the crest or the depth of the trough of the electromagnetic wave.

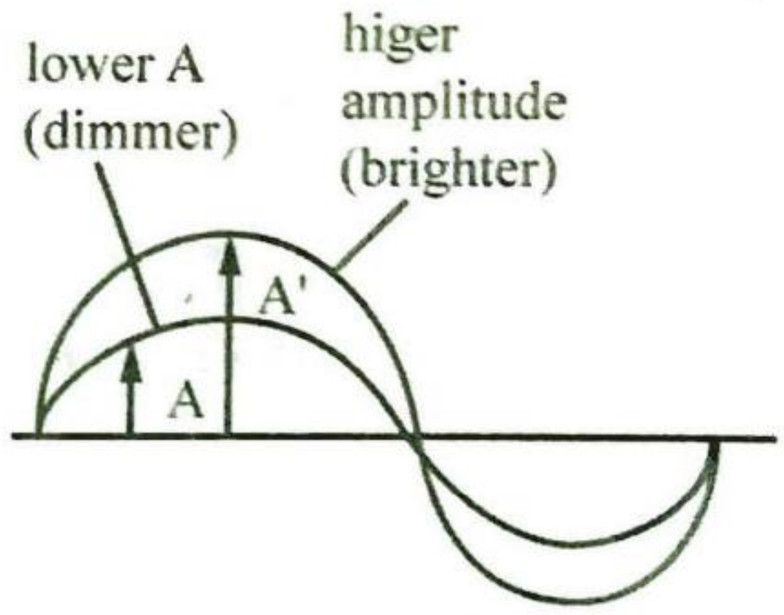



Figure 2.4

➦ It is related to the intensity of radiation. In the visible region of electromagnetic spectrum, the higher the amplitude, the higher is the intensity of the light.

## Illustrative Examples

37. Which of the following are not electromagnetic waves? (EUEE .. 2008 E.C.)

A. Infrared waves
C. Radio waves
B. Gamma waves
D. Sound waves

**Answer: D**

38. What is the distance that a radio wave travel in 0.250s? (EUEE)

A. $1.2 \times 10^7$m
B. $12 \times 10^7$m
C. $7.5 \times 10^7$m
D. $5.6 \times 10^7$m

**Solution:**

$$c = \lambda v = \frac{\lambda}{T} \text{ because } v = \frac{1}{T}$$

$$\lambda = c.T = 3 \times 10^8 \text{ m/s} \times 0.250\text{s} = 7.5 \times 10^7 \text{ m}$$

**Answer: C**

39. Which of the following electromagnetic radiations has the shortest wavelength? (EUEE ... 2004 E.C)

A. x-ray
B. UV ray
C. gamma ray
D. macro waves

**Explanation:**

**Wave length**: gamma ray < x - rya < UV - ray < visible < IR < micro wave

**Energy and frequency**: gamma ray > x - ray > UV - ray > visible > IR > micro wave

40. What is the frequency in MHz of a wave with a wave length of 600nm?

**Solution:**

$$c = \lambda v$$

$$v = \frac{c}{\lambda} = \frac{3 \times 10^8 \text{ m/s}}{6 \times 10^{-7} \text{ m}} = 5 \times 10^{14} \text{ s}^{-1} = 5 \times 10^8 \text{ MHz}$$

$\boxed{1 \text{ MHz} = 10^6 \text{Hz}}$

41. Some diamonds appear yellow because they contain nitrogenous compounds that absorb purple light of frequency $7 \times 10^{14}$Hz. Calculate the wave length in

a. nm
b. pm
c. A°
d. μm

**Solution:**

a. $\lambda = \frac{c}{v} = \frac{3 \times 10^8 \text{ m/s}}{7 \times 10^{14} \text{ s}^{-1}} = 4.29 \times 10^{-7} \text{ m} = 429\text{nm}$

b. $1\text{m} = 10^{12}\text{Pm} \Rightarrow 4.29 \times 10^{-7}\text{m} = 4.29 \times 10^5\text{pm}$

c. $1\text{m} = 10^{10}\text{A}° \Rightarrow 4.29 \times 10^{-7}\text{m} = 4.29 \times 10^3\text{A}°$

d. $1\text{m} = 10^6\mu\text{m} \Rightarrow 4.29 \times 10^{-7}\text{m} = 0.429\mu\text{m}$

42. The frequency of the green light from a traffic signal is centered at $5.75 \times 10^{14}$ Hz. Calculate the wave number of the green light.

**Solution:**

First find wave length in centimeters.

$$\Rightarrow \lambda = \frac{c}{v} = \frac{3\times10^{10}\,cm/s}{5.75\times10^{14}\,s^{1}} = 5.22\times10^{-5}\,cm$$

$$\bar{v} = \frac{1}{\lambda} = \frac{1}{5.22\times10^{-5}\,cm} = 1.92\times10^{4}\,cm^{-1}$$

43. Explain the difference and similarity between electro - magnetic radiations.

**Explanation:**

**Similarity**: All electromagnetic radiations travel through vacuum with the same speed.

**Difference**: Different electro magnetic radiations have different wave length, frequency and energy.

44. Give some examples of electromagnetic radiations.

**Solution:**

- The light from the sun.
- The x-ray used by dentists.
- The energy used to cook food in microwave oven.
- The radiant heat from a fire place.

## The quantum Theory and Photon

- Planck said that atoms and molecules could emit or absorb energy in **discrete quantities**, like small packages or bundles.
- He gave **quantum or photon** to the smallest quantity of energy that can be emitted or absorbed in the form of electromagnetic radiation.

$$E_{Photon} = h\upsilon$$

- According to plank's quantum theory, energy is always emitted in multiplies of hv, for example, hv, 2hv, 3hv, ...

## The Photoelectric Effect

- The ejection of electrons by a metal bomarded with light at least a minimum frequency (threshold frequency) is called **Photoelectric effect**.

➦ The minimum amount of energy required to eject electron from metal surface is called **work function (Φ).**

$$\Phi = hv_o = \frac{hc}{\lambda_o}.$$

$v_o$ - Threshold frequency

$\lambda_o$ - maximum wave length (threshold wavelength)

h - Planck's constant

➦ If the frequency is below threshold frequency there will be no photoelectric effect.

➦ If energy more than work functions supplied by using radiation, the excess energy goes into kinetic energy of the ejected electrons.

$\Rightarrow E = hu = \Phi + kE$

$\Rightarrow hu = hu_o + kE$

$$kE = hu - hu$$

$$kE = h(u - uo)$$

$$kE = \frac{1}{2}mv^2.$$

m - mass

v - velocity

$$v = \sqrt{\frac{2h}{m}(v - v_o)}$$

kE - kinetic energy

v - frequency of light

➦ Electromagnetic radiations exhibit **wave** and **particle** properties.

$E = mc^2$, Einstein's famous equation

$E = hv = \frac{hc}{\lambda}$, Max Planck's equation

➦ Relating these equations results:

$$mc^2 = \frac{hc}{\lambda}$$

$$\lambda = \frac{h}{mc}$$

This equation is used if the wave properties is dominant.

➦ For a particle like electron with velocity V, the equation becomes:

$$\lambda = \frac{h}{mV}$$

This equation is known as **de Broglie's equation**.

## Illustrative Examples

45. What is the magnitude of quantum energy and the frequency for an object whose wavelength is $0.6 \times 10^{-6}$m? (UEE .. 2008 E.C)

A. $3.31 \times 10^{-19}$J, $5 \times 10^{14}$Hz
C. $1.99 \times 10^{-25}$J, $3.98 \times 10^{-40}$Hz
B. $3.98 \times 10^{-40}$J, $2 \times 10^{15}$Hz
D. $9.94 \times 10^{-25}$ Hz

**Solution:**

$$E = hu = \frac{hc}{\lambda} = \frac{6.626\times10^{-34}\,J.s \times 3\times10^{8}\,m/s}{0.6\times10^{-6}\,m}$$

$$E = 3.31 \times 10^{-19}J$$

$$u = \frac{E}{\lambda} = \frac{3.31\times10^{-19}\,J}{6.626\times10^{-34}\,J.s} = 5\times10^{14}\,Hz$$

**Answer: A**

46. The energy required to remove an electron from an iron atom amounts to $7.21 \times 10^{-19}$J. What is the maximum wave length of light that can perform this?

A. 250nm
B. 276nm
C. 289nm
D. 300nm

**Solution:**

$$\Phi = \frac{hc}{\lambda_{max}} = \lambda_{max} = \frac{hc}{\Phi}$$

$$\lambda_{max} = \frac{6.626\times10^{-34}\,J.s\times3\times10^{8}\,m/s}{7.21\times10^{-19}} = 2.76\times10^{-7}\,m = 276nm$$

**Answer: B**

47. Calculate the energy of a photon of wave length $5 \times 10^{4}$nm (infrared region) and a photon of wave length $5 \times 10^{-2}$nm (x - ray region).

**Solution:**

i For IR region:

$$E = \frac{hc}{\lambda} = \frac{6.626\times10^{-34}\,J.s\times3\times10^{8}\,m/s}{5\times10^{4}\times10^{-9}\,m} = 3.98\times10^{-21}\,J$$

ii For x - ray region

$$E = \frac{hc}{\lambda} = \frac{6.626\times10^{-34}\,J.s\times3\times10^{8}\,m/s}{5\times10^{-2}\times10^{-9}\,m} = 3.98\times10^{-15}\,J$$

48. Explain the difference between Plank's theory and the classical perceptions of the nature of energy.

**Explanation**: Plank's theory suggested that energy is not continuous, rather expressed in discrete packets. However, prior this it was believed that energy is continuous without limit.

49. The threshold frequency for metallic potassium is $5.46 \times 10^{14}$Hz. Calculate

a. the work function

b. kinetic energy

c. velocity of the emitted electron, if the wave length of light shining on the potassium surface is 350 nm, and mass of an electron is $9.1 \times 10^{-31}$kg.

**Solution**:

a. $\Phi = hu_o = 6.626 \times 10^{-34}J.s \times 5.46 \times 10^{14}s^{-1} = 3.6 \times 10^{-19}J$

b. $kE = h(v - v_o) = 6.626 \times 10^{-34}J.s \left(\frac{c}{\lambda} - 5.46\times10^{14}\right)s^{-1}$

$kE = 2.08 \times 10^{-19}J$

c. $v^2 = \frac{2kE}{m}$

$$V = \sqrt{\frac{2kE}{m}} = \sqrt{\frac{2\times2.08\times10^{-19}\,kg.m^2/s^2}{9.1\times10^{-31}\,kg}}$$

$V = 6.7 \times 10^5 m/s$

50. What is the wave length of a proton having a velocity of $1.38 \times 10^7$cm/s?

**Solution**:

Mass of a proton = $1.673 \times 10^{-27}$ kg

$V = 1.38 \times 10^7 cm/s = 1.38 \times 10^5$ m/s

$$\lambda = \frac{h}{mV} = \frac{6.626\times10^{-34}\,J.s}{1.673\times10^{-27}\,kg.1.38\times10^{5}\,m/s}$$

$\lambda = 2.88 \times 10^{-12}m = 2.88$ pm

51. Calculate the wave length associated with an electron traveling at 40% of the velocity of light.

**Solution:**

$$\text{V of electron} = 40\%\,c = \frac{40}{100} \times 3 \times 10^8\ \text{m/s}$$

$$\text{V of electron} = 1.2 \times 10^8\ \text{m/s}$$

$$\lambda = \frac{h}{mV} = \frac{6.626 \times 10^{-34}\ \text{kg/m}^2/\text{s}}{9.1 \times 10^{-31}\ \text{kg} \times 1.2 \times 10^8\ \text{m/s}}$$

$$\lambda = 6.07 \times 10^{-12}\text{m} = 6.07\ \text{pm}$$

52. A microwave oven operates with a radiation having a wave length of 1.20cm. Calculate the energy of this radiation in joules per photon and in kilojoules per mole of photons.

**Solution:**

$$E = \frac{hc}{\lambda} = \frac{6.626 \times 10^{-34}\ \text{J.s} \times 3 \times 10^{10}\ \text{cm/s}}{1.20\text{cm}}$$

$$E = 1.66 \times 10^{-23}\text{J/photon}$$

1 mole of photons $= 6.02 \times 10^{23}$ photons

$\Rightarrow$1 photon $= 1.66 \times 10^{-23}$J

$6.02 \times 10^{23}$ photons = ?

$\Rightarrow$ E = 9.99 J/mol = 0.00999 kJ/mol

53. Let the wave length of 400nm strikes the surface of a metallic cesium in a photo cell. The maximum kinetic energy of the electrons ejected is $1.54 \times 10^{-19}$J. Calculate the work function of cesium and the longest wave length of the light that is capable of ejecting the electrons from the metal.

Solve by your own!

**Answer: $\Phi = 3.43 \times 10^{-19}$J, $\lambda_{max} = 579$ nm**

54. Place the following types of radiations in order of increasing energy per photon:

I. green light from a mercury lamp
II. x - rays from an instrument in a dentist's office
III. micro waves in a micro wave oven
IV. an FM music station at 96.3 MHz.

**Solution:**

$\Rightarrow$ IV < III < I < II

## Atomic Line Spectra

- The emission spectra of atoms in the gas phase do not show a continuous spread of wave lengths from red to violet.
- These atoms produce bright lines in distinct parts of the visible spectrum.
- Those spectra are the emission of light only at specific wave lengths.
- Every element has a unique emission spectrum.
- The characteristic lines in atomic spectra can be used in chemical analysis to identify elements, much as finger prints are used to identify people.
- The spectrum produced by sunlight is called **a continuous spectrum**, because one color merges into the next without any gaps or missing colors.

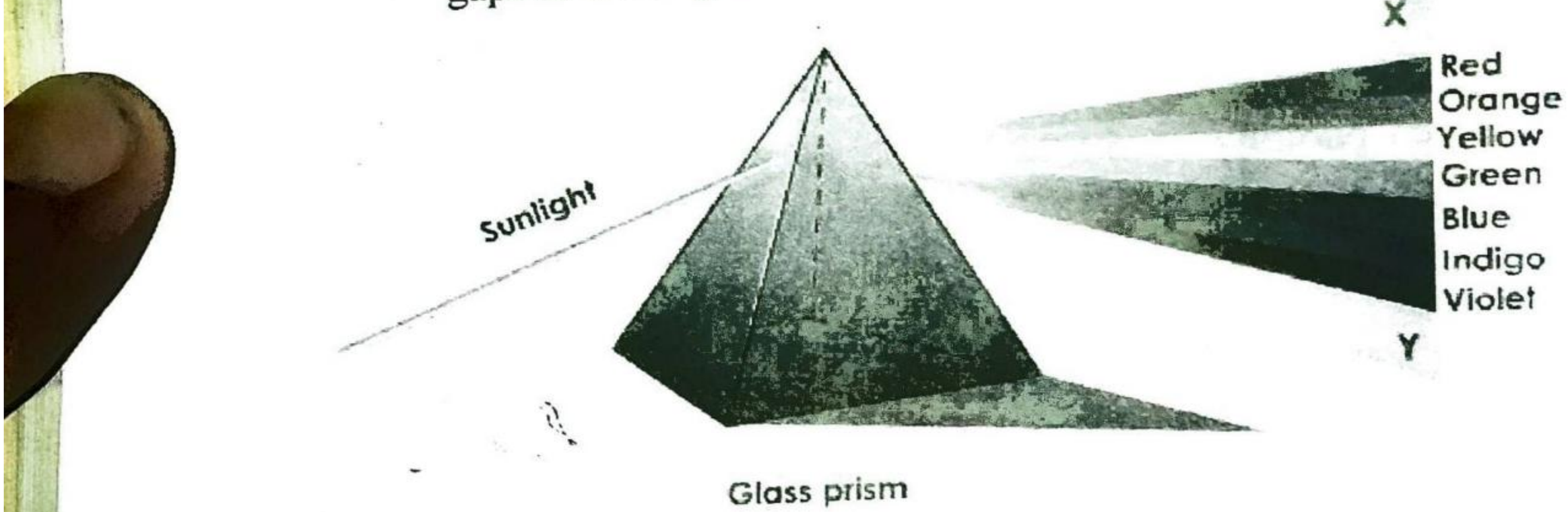



Figure 2.5 Sunlight passing through a prism

## Illustrative Examples

55. Which of the following is true about atomic spectra?
    A. Atomic spectra are also referred to as dot spectra
    B. Atomic spectra are a series of fine lines of individual colors separated by colored spaces.
    C. There are fewer energy states in each atom than there are electrons.
    D. The intensity of the lines in an atomic spectrum vary with temperature.

**Explanation:**

- Atomic spectra are also called line spectra.
- Atomic spectra are fine lines separated by colorless or black spaces.
- There are more energy states in each atom than there are electrons.
- The intensity of the lines in an atomic spectrum vary with temperature and pressure affects their width.

**Answer: D**

56. Which of the following produces atomic or line spectra?

A. Sunlight
B. Red - hot iron bar
C. Heated solid
D. Excited atoms

**Explanation:** Excited atoms produce line spectra.

**Answer: D**

57. Write the application of Atomic spectra.

**Solution**

Line spectra serve as sensitive and useful tools for identifying elements.

58. How can atomic spectra be used to advance astronomy?

**Solution**

- Analyses of line spectra enabled scientists to identify the elements that compose the sun and stars.
- Atomic spectra can be used to inform scientists about the physical conditions of places that are far away in our galaxy. They can be used to estimate the temperature of the surface of stars and planets.
- They can be also used to estimate the gas pressure and magnetic field strength on and around planets and stars.

59. Describe how rainbow is formed in the sky.

**Explanation**

- After it rains, the sky is filled with moisture in the form of water droplets. When the sun comes out, the white light it produces reflects off the water droplets. The water droplets act like a prism and separate the white light into spectrum of colors. All of the colored lights, or light frequencies bend in different ways to form a rainbow in the sky.

60. Who passed sunlight through a glass prism to show that the colors separates into a rainbow spectrum?

**Solution**

Sir Isac - Newton

## 2.4 The Bohr Model of the hydrogen atom

➦ **Bohr** used the quantum theory developed by **Planck** and the photoelectric theory developed by **Einstein** to formulate his model of the hydrogen atom.

1. The electron in a hydrogen atom travels around the nucleus in circular path called **orbit**.
2. The energy of an electron in an orbit is directly proportional to its distance from the nucleus.
3. Orbits are quantized.
4. As long as an electron stays in a given orbit, it neither gains nor loses energy. A definite amount of energy is absorbed when an electron jumps from lower orbit to higher orbit and emitted in the form of light when it falls into a lower energy orbit.

$$\Delta E = E_f - E_i = h\upsilon$$

$\Delta E$ - change in energy

$E_f$ - energy in the final state (orbit)

$E_i$ - energy in the initial state

➦ **Bohr** derived an equation for the energy of an electron in the $n^{th}$ orbit.

$$E_n = -R_H \cdot \frac{Z^2}{n^2}$$

➦ $R_H$ - Rydberg constant, which has a value of $2.18 \times 10^{-18}$J

➦ n - positive integer (1, 2, 3, ...∞)

➦ Z - atomic number, for hydrogen, it is 1.

$$E_n = \frac{-2.18 \times 10^{-18}\,J}{n^2}$$

➦ The negative sign (-) implies as n increases energy also increases.

## Example

Calculate the energy of the first, second and third orbits in the hydrogen atom.

### Solution

➦ First orbit (n = 1)

$$E_1 = \frac{-2.18\times10^{18}\,J}{n_1^2} = \frac{-2.18\times10^{-18}\,J}{(1)^2} = -2.18\times10^{-18}\,J$$

➦ Second orbit (n = 2)

$$E_2 = \frac{-2.18\times10^{-18}\,J}{n_2^2} = \frac{-2.18\times10^{-18}\,J}{(2)^2} = -5.45\times10^{-19}\,J$$

➦ For the third orbit (n = 3)

$$E_3 = \frac{-2.18\times10^{-18}\,J}{n_3^2} = \frac{-2.18\times10^{-18}\,J}{3^2} = -2.45\times10^{-19}\,J$$

$\Rightarrow E_1 < E_2 < E_3$

➦ The energy change ($\Delta E$) for hydrogen atom can be derived as follows:

$\Delta E = E_f - E_i$ but we know that

$$E_f = \frac{-2.18\times10^{-18}\,J}{n_f^2} \text{ and } E_i = \frac{-2.18\times10^{-18}\,J}{n_i^2}$$

Substituting these two equations in the initial equation:

$$\Rightarrow \Delta E = \left(\frac{-2.18\times10^{-18}\,J}{n_f^2}\right) - \left(\frac{-2.18\times10^{-18}\,J}{n_i^2}\right)$$

$$\boxed{\Delta E = 2.18\times10^{-18}\,J\left(\frac{1}{n_i^2} - \frac{1}{n_f^2}\right)}$$

➦ The radius of an orbit can be determined by the formula:

$$\boxed{r = \frac{h^2\varepsilon_o n^2}{8\pi^2 m e^2 z}}$$

$\Rightarrow$ h, $\pi$, m, $\varepsilon_o$ and e are constants, when we replace by another constant, $a_o$, the formula becomes:

$$r = \frac{n^2, a_o}{z} = \frac{0.53n^2A^\circ}{z}$$

$$\boxed{r = 0.53n^2A^\circ}$$ , z = 1 for H

## Example

The radius of the first and second orbits become:

- First orbit (n = 1)
  $r = 0.53\ (1)^2A^\circ = 0.53A^\circ$
- Second orbit (n = 2)
  $r = 0.53\ (2)^2A^\circ = 2.12A^\circ$
- The emission spectrum of hydrogen covers a wide range of wave lengths from the infrared to the ultraviolet. They are named after their discovers.

Table 2.3. The various series in atomic hydrogen emission spectrum

| Series | $n_f$ | $n_i$ | Spectrum region |
|---|---|---|---|
| Lyman | 1 | 2, 3, 4, ... | Ultraviolet |
| Balmer | 2 | 3, 4, 5, ... | Visible and ultraviolet |
| Paschen | 3 | 4, 5, 6, ... | Infrared |
| Brackett | 4 | 5, 6, 7, ... | Infrared |
| Pfund | 5 | 6, 7, 8, ... | Infrared |

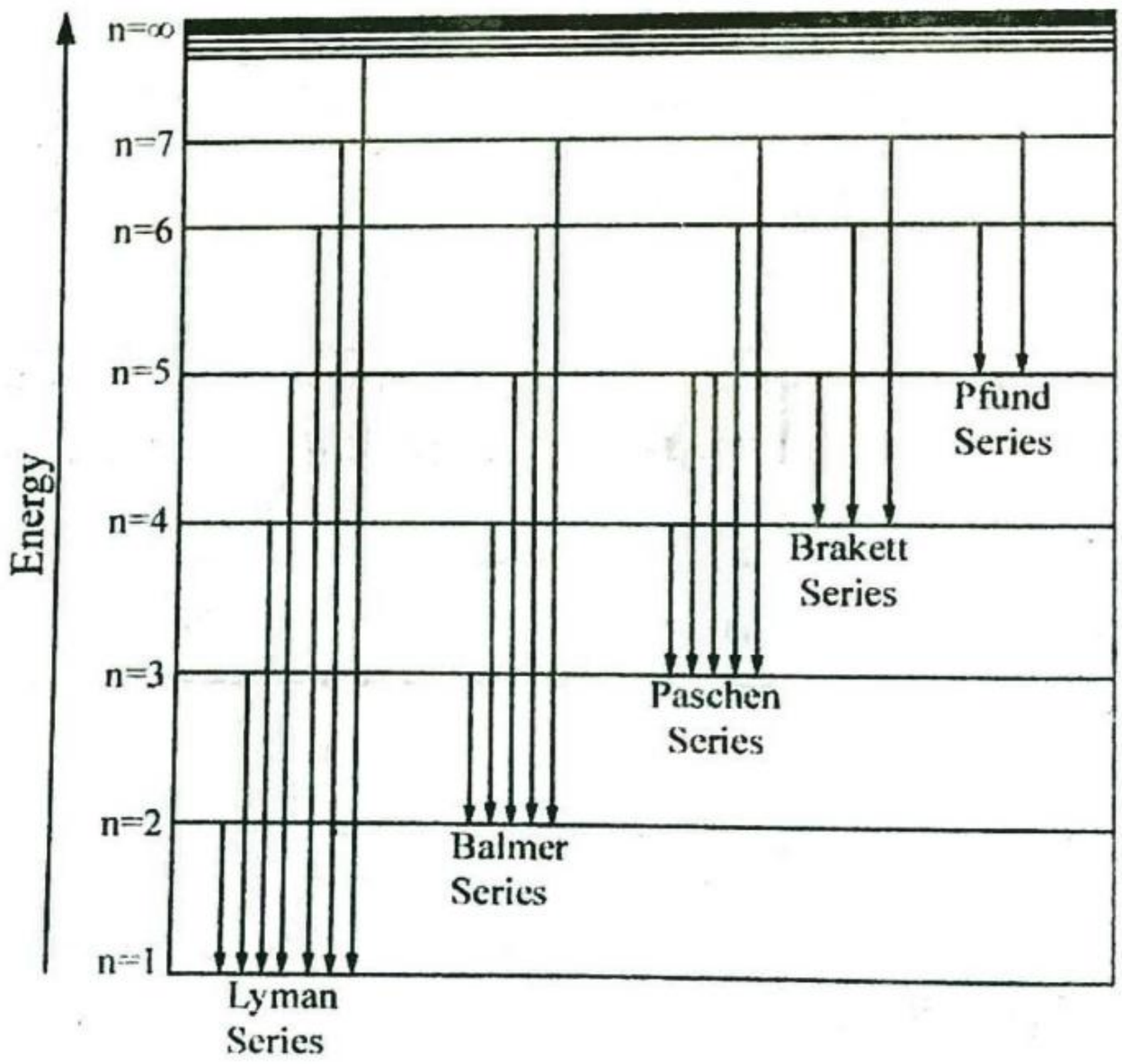



Figure 2.6 The energy levels in the hydrogen atom and the various emission series

## Limitation of the Bohr Theory

- Bohr model failed to predict the wave lengths of spectral lines of atoms more complicated than hydrogen.
- It is applied only to species that have one electron number such as hydrogen, (H), Lithium ion ($Li^{+2}$), Beryllium ($Be^{3+}$).

## Illustrative Examples

61. What new concept did Bohr adapt and use to formulate his model of the atom? (EUEE ... 2008 E.C)

    A. Electromagnetic theory developed by Maxwell.

    B. The quantum concept developed by Planck.

    C. The photoelectric theory developed by Thomson.

    D. Neutron theory developed by Chadwick.

**Explanation:** Bohr used the quantum concept developed by Planck and the photo electric effect developed by Einstein to formulate his model of the hydrogen atom.

**Answer: B**

62. Hydrogen atoms are excited to the stationary state designated by the principal quantum number, n = 4. What will be the number of lines (frequencies) emitted by the hydrogen atoms? (EUEE, 2003 E.C)

    A. 1 B. 4 C. 6 D. 16

**Solution**

Number of spectral lines are directly proportional to principal quantum number.

**Answer: B**

63. Which of the following electron transitions requires the smallest energy to be absorbed by the hydrogen atom? (EUEE)

    A. From n = 1 to n = 2

    B. From n = 2 to n = 3

    C. From n = 3 to n = 4

    D. From n = 4 to n = 5

**Solution**

$$\Delta E = 2.18\times10^{-18}\,J\left(\frac{1}{n_i^2}-\frac{1}{n_f^2}\right)$$

For n = 1 to n = 2

for n = 1 to n = 2

$$\Delta E = 2.18\times10^{-18}\,J\left(\frac{1}{1^2}-\frac{1}{2^2}\right)=1.64\times10^{-18}\,J$$

For n = 2 to n = 3

$$\Delta E = 2.18\times10^{-18}\,J\left(\frac{1}{2^2}-\frac{1}{3^2}\right)=3.03\times10^{-19}\,J$$

For n = 3 to n = 4

$$\Delta E = 2.18\times10^{-18}\,J\left(\frac{1}{3^2}-\frac{1}{4^2}\right)=1.06\times10^{-19}\,J$$

For n = 4 to n = 5

$$\Delta E = 2.18\times10^{-18}\,J\left(\frac{1}{4^2}-\frac{1}{5^2}\right)=4.90\times10^{-20}\,J$$

**Answer: D**

64. According to Bohr, what type of motions does an electron have in an atom and what happens when energy is applied to the atom.

**Explanation:** According to Bohr, an electron moves in discrete, fixed circular path called orbits around the nucleus. If photons of the applied energy equals the difference in energy between the two orbits, the atom absorbs a photon and the electron moves to larger orbit.

65. Why was Bohr's theory for the hydrogen atom initially accepted and why was it ultimately discarded?

**Explanation:** Bohr's theory explained the experimentally observed line spectrum of hydrogen but failed to explain more complicated atoms that is why it was ultimately discarded.

66. What is the frequency of a photon emitted during a transition from n = 5 state to n = 2 states in the hydrogen atom?

**Solution**

$$\Delta E = 2.18\times10^{-18}\,J\left(\frac{1}{n_i^2-n_f^2}\right)$$

$$=2.18\times10^{-18}\,J\left(\frac{1}{5^2}-\frac{1}{2^2}\right)$$

$$\Delta E = -4.578\times10^{-19}J$$

But we know that, $\Delta E = hv$

$$\Rightarrow v = \frac{\Delta E}{h} = \frac{4.578\times10^{-19}\,J}{6.626\times10^{-34}\,J.s} = 6.91\times10^{14}\,s^{-1}$$

67. The first line of the Balmer series occurs at a wave length of 656.3nm. What is the energy difference between the two energy levels involved in the emission that results in this spectral line?

**Solution**

$$\Delta E = hv = \frac{hc}{\lambda} = \frac{6.626\times10^{-34}\,J.s\times3\times10^{8}\,m/s}{6.563\times10^{-7}\,m}$$

$$\Delta E = 3.03\times10^{-19}J$$

For Balmer series, $n_f = 2$ and for the first line, $n_i = 3$

$$\Rightarrow E = 2.18\times10^{-18}\,J\left(\frac{1}{3^2}-\frac{1}{2^2}\right) = 3.03\times10^{-19}\,J$$

68. Calculate the frequency and wave length of the emitted photon when an electron drops from the n = 4 to n = 1 level in a hydrogen atom.

**Solve by your own.**

**Answer:** $\lambda = 9.7\times10^{-8}m$, $V = 3.1\times10^{15}$ Hz

69. An electron in an orbit of principal quantum number $n_i$ in a hydrogen atom makes a transition to an orbit of principal quantum number 2. The photon emitted has a wave length of 434 nm. Calculate $n_i$.

**Solution**

$$\Delta E = \frac{hc}{\lambda} = \frac{6.626\times10^{-34}\,J.s\times3\times10^{8}\,m/s}{4.34\times10^{-7}\,m}$$

$$\Delta E = 4.58\times10^{-19}J$$

$$\Delta E = 2.18\times10^{-18}\,J\left(\frac{1}{n_i^2}-\frac{1}{n_f^2}\right)$$

$$\left(\frac{1}{n_i^2}-\frac{1}{n_f^2}\right) = \frac{\Delta E}{2.18\times10^{-18}\,J} \Rightarrow n_i = 5$$

70. How much energy, in kilojoules per mole is released when an electron makes a transition from n = 4 to n = 3 in a hydrogen atom? Is this energy sufficient to back the H - H bond (436 kJ/mol is needed to break this bond)?

**Solution**

$$\Delta E = 2.18\times10^{-18}\,J\left(\frac{1}{4^2}-\frac{1}{3^2}\right)=1.06\times10^{-19}\,J/Photon$$

1 mole of photon = $6.02 \times 10^{23}$ photons

$\Rightarrow \Delta E = 1.06 \times 10^{-19} J/mol \times 6.02 \times 10^{23}$ photon/mol

$\Delta E = 6.38 \times 10^{4} J/mol = 63.8\ kJ/mol$

$\Rightarrow$ This energy is not sufficient to break the H - H bond.

71. Calculate the energy required to remove an electron from the lowest level of the hydrogen atom to produce hydrogen ion.

**Solution**

$n_i = 1, n_f = \infty$

$$\Rightarrow \Delta E = 2.18\times10^{-18}\,J\left(\frac{1}{1^2}-\frac{1}{\infty^2}\right)=2.18\times10^{-18}\,J$$

72. Calculate the radius of the fourth orbit of the hydrogen atom.

**Solution**

$r = 0.53n^2A° = 0.53(4)^2A° = 8.48A°$

## 2.5 The Quantum model of the atom

- It is also called **wave mechanical model or Erawin Schrodinger model** of an atom.
- Macroscopic objects do exhibit a wave like motion but the wave length is too small for humans to perceive.
- A wave is associated with any object, even a moving elephant but is dominated by the particle nature.
- Electron is microscopic particle which moves with a very high speed. As a result, it is very difficult to trace the exact location and speed of an electron at the same time around the nucleus.

**Heisenberg uncertainty principle** states that it is impossible to know simultaneously both the momentum (mass times velocity) and the position of a particle with certainty.

$$\Delta x.\Delta p \geq \frac{h}{4\pi}$$

$\Delta x$ – change in position

- The concept of **electron density** gives the probability that an electron will be found in a particular region of an atom. This region is called **orbital**.

➦ **Orbital** is a region in space around the nucleus where high probability of getting an electron.

## Quantum Numbers

➦ In quantum mechanics, three **quantum numbers** are required to describe the distribution of electrons in atoms. They are called **the principal quantum number, angular momentum quantum number and magnetic quantum number.**

➦ A fourth quantum - number, **the spin quantum number** describes the behavior of a specific electron and completes the description.

### 1. The principal quantum number (n)

➦ It has integral values, 1, 2, 3, ...

➦ It describes the main energy levels.

➦ It is used to determine:

**i Size of orbital**

➦ As n increases, size of orbital also increases.

**ii Energy of orbital**

➦ As n increases, energy of orbital also increases.

**iii Number of orbitals**

| Number of orbitals = $n^2$ |
|---|

**Example**

Calculate the number of orbitals in the fourth shell.

Number of orbitals = $n^2 = 4^2 = 16$

| The maximum capacity of an orbital is 2 electrons. |
|---|

**iv Number of electrons in main energy levels.**

| Number of electrons in main shell = $2n^2$ |
|---|

**Example**

Determine the number of electrons in the fourth shell (n = 4).

**Solution**

Number of electrons = $2n^2 = 2(4)^2 = 32$

$2n^2$ $2\times4^2$ 32

v **Number of sub - shells in any main shell.**

Number of sub shell = n

**Example**

Calculate the number of sub - shells in the fourth shell.

**Solution**

Number of sub shell = n = 4, these are s, p, d and f.

vi **The average distance of the electron from the nucleus in a particular orbital.**

➦ The larger n is, the greater the average distance of an electron in the orbital from the nucleus.

**2. The Azimutal quantum number ($\ell$)**

➦ It is also called angular momentum or subsidiary or orbital shape quantum number.

➦ It has value from 0 to n - 1.

The maximum value of $\ell$ is n - 1

**Example**

If n = 1, $\ell = 0$

n = 2, $\ell = 0, 1$

n = 3, $\ell = 0, 1, 2$

n = 4, $\ell = 0, 1, 2, 3$

➦ The number of possible values of $\ell$ is equal to n.

➦ It gives

i **Shape of orbitals**

ii **The type of sub - shells**

| Value of l | Sub - shell |
|---|---|
| 0 | s |
| 1 | p |
| 2 | d |
| 3 | f |
| 4 | g |

iii **the number of electrons in sub - shell**

Number of electrons in sub shell = $4\ell + 2$

**Example**

Calculate the number of electrons in s, p, d and f sub – shells.

**Solution**

For s - sub - shell, $\ell = 0$

$\Rightarrow$ Number of electrons $= 4\ell + 2 = 4 \times 0 + 2 = 2$

For p - sub shells, $\ell = 1$

$\Rightarrow$Number of electrons $= 4\ell + 2 = 4 \times 1 + 2 = 6$

For d - sub shell, $\ell = 2$

$\Rightarrow$Number of electrons $= 4\ell + 2 = 4 \times 2 + 2 = 10$

For f - sub shell, $\ell = 3$

$\Rightarrow$ Number of electrons $= 4\ell + 2 = 4 \times 3 + 2 = 14$

iv **The number of orbitals in sub - shell**

Number of orbitals in sub - shell $= 2\ell + 1$

| Sub - shell | Value of l | Number of orbitals |
|---|---|---|
| s | 0 | 1 |
| p | 1 | 3 |
| d | 2 | 5 |
| f | 3 | 7 |
| g | 4 | 9 |

v **The energy of sub - shells**

- As the value of $\ell$ increases, energy of sub shell also increases.
- With the same value of n, there are different values of $\ell$. For example; n = 2, $\ell = 0, 1$

$\Rightarrow \ell = 0$ means 2s and $\ell = 1$ means 2p. Thus, the energy of 2p is greater than the energy of 2s.

3. **The magnetic quantum number ($m_\ell$)**
   - It is also called **orbital orientation quantum number.**
   - It is an integer having values from $-\ell$ to $+\ell$

**Example**

If the value of $\ell$ is 0, $m_\ell = 0$.

$\ell = 2$, $m\ell = -2, -1, 0, 1, 2$

$\ell = 1$, $m\ell = -1, 0, 1$

The possible values of $m\ell = 2\ell + 1$

- It prescribes the orientation of the orbital in three dimensional space about the nucleus.
- s - orbitals has one orientation in space.
- p - orbitals have three orientations in space.
- d - orbitals have five orientations in space.
- f - orbitals have seven orientations in space.

**4. The spin quantum number (ms)**

- It has two possible values $+\frac{1}{2}$ (represented by ↑) and $-\frac{1}{2}$ (represented by ↓).
- It describes the direction of spin of an electron.

## Illustrative Examples

73. What values of l are permitted for an electron with n = 4? (EUEE, 2008 E.C)

A. 1, 2, 3

B. 1, 2, 3, 4

C. 0, 1, 2, 3, 4

D. 0, 1, 2, 3

**Solution**

The possible values of l are from 0 to n - 1 $\Rightarrow$ l = 0, 1, 2, 3

**Answer: D**

74. Which of the following electrons, identified only by their n and l quantum numbers have the highest energy? (EUEE)

A. $n = 2, \ell = 0$

B. $n = 4, \ell = 1$

C. $n = 3, \ell = 2$

D. $n = 4, \ell = 2$

**Solution**

As n increases energy also increases. With the same value of n, as $\ell$ increases energy also increases.

**Answer: D**

75. For an electron that has quantum numbers n = 4 and ml = 0, which of the following is true? (EUEE)

A. It must have quantum number n = 0.

B. It must have quantum number $\ell = 0$.

C. It must have quantum number ms = ±1/2.

D. It may have quantum number $\ell = 0, 1, 2, 3$

**Explanation:** For n = 4 and ml = 0, $\ell$ may be 0, 1, 2 or 3 and ms may be $+\frac{1}{2}$ or $-\frac{1}{2}$

**Answer: D**

76. Which of the following quantum numbers is not allowed in the hydrogen atom?

A. $n = 2, \ell = 1, m\ell = 1$

B. $n = 1, \ell = 0, m\ell = 0$

C. $n = 8, \ell = 7, m\ell = -6$

D. $n = 2, \ell = 0, m\ell = 2$

**Explanation:** The possible values of $m\ell$ are from $-\ell$ to $+\ell$.

**Answer: D**

77. What are the possible values of azimutal quantum number for an electron in the 4f sub - shell?(EUEE, 2003 E.C.)

A. 0, 1, 2, 3 B. 1, 2, 3 C. 3 D. 4

**Solution**

For any sub - shell, there is only single vlaues of l. Thus, for 4f it is 3.

**Answer: C**

78. How many orbitals are in an atomic sub - level with l = 3?

A. 3 B. 5 C. 7 D. 9

**Solution**

Number of orbitals in sub - shell = $2\ell + 1 = 2 \times 3 + 1 = 7$

**Answer: C**

79. What is the maximum number of electrons in an atom that can have can have the quantum numbers n = 5, $m\ell = +1$?

A. 8 B. 16 C. 32 D. 64

**Solution**

For $n = 5, \ell = 0, m\ell = 0$

$\ell = 1, m\ell = -1, 0, 1$

$\ell = 2, m\ell = -2, -1, 0, 1, 2$

$\ell = 3, m\ell = -3, -2, -1, 0, 1, 2, 3$

$\ell = 4, m\ell = -4, -3, -2, -1, 0, 1, 2, 3, 4$

Thus, there are four orbitals with $m\ell = +1$ orientations. As a result, the maximum number of electrons $= 4 \times 2 = 8$.

80. Which set of quantum numbers (n, l, ml, ms) is not permitted by the rules of quantum mechanics?

A. $1, 0, 0, \frac{1}{2}$

B. $2, 1, -1, -\frac{1}{2}$

C. $3, 3, 1, -\frac{1}{2}$

D. $4, 3, 2, \frac{1}{2}$

**Explanation:** The value of n is not equal to the value of $\ell$.

**Answer: C**

81. Identify the types of sub - shells represented by the quantum numbers n and $\ell$.

a. $n = 3, \ell = 2$

b. $n = 4, \ell = 3$

c. $n = 5, \ell = 0$

d. $n = 2, \ell = 1$

**Explanation:** n identifies main shell and $\ell$ identifies sub - shell.

a. 3d b. 4f c. 5s d. 2p

82. Determine the maximum number of electrons in an atom that can have the following quantum numbers.

a. $n = 3$

b. $n = 4, \ell = 2$

c. $n = 2, ms = +\frac{1}{2}$

d. $n = 4, m_l = +1$

e. $n = 2, \ell = 0, ms = -\frac{1}{2}$

f. $n = 4, \ell = 3, m\ell = 2$

**Solution**

a. Number of electrons $= 2n^2 = 2(3)^2 = 18$

b. It is 4d and contains 10 electrons.

c. $n = 2$ means $\ell = 0, 1$ that is 2s and 2p sub - shells.

$\Rightarrow$In 2s there is 1 electron with $+\frac{1}{2}$ spin

$\Rightarrow$ In 2p there are 3 electrons with $+\frac{1}{2}$ spins.

Therefore, there are 4 electrons.

d. 6 electrons (2 in 4p, 2 in 4d and 2 in 4f)

e. 1 electron in 2s with $-\frac{1}{2}$ spin

f. 2 electrons in 4f.

83. Calculate the number of orbitals in an atom with the following quantum numbers.

a. $n = 2$

b. $\ell = 2$

c. $n = 3, \ell = 0$

d. $n = 4, \ell = 2, m\ell = 2$

**Solution**

a. Number of orbitals $= n^2 = 2^2 = 4$

b. Number of orbitals $= 2\ell + 1 = 2 \times 4 + 1 = 9$

c. Number of orbitals $= 2\ell + 1 = 2 \times 0 + 1 = 1$

d. Number of orbitals = possible values of $m\ell = 1$

84. Give the values of the quantum numbers associated with the following orbitals.

a. 3s b. 4p c. 6d d. 5f

**Solution**

a. $n = 3, \ell = 0, m\ell = 0$

b. $n = 4, \ell = 1, m\ell = -1, 0, 1$

c. $n = 6, \ell = 2, m\ell = -2, -1, 0, 1, 2$

d. $n = 5, \ell = 3, m\ell = -3, -2, -1, 0, 1, 2$

85. What is the total number of electrons that can be held in all orbitals having the same principal quantum number, n?

**Answer: $2n^2$**

86. Arrange the following orbitals in increasing order of energy.

a. 3s, 3p, 3d

b. 4s, 4f, 4d, 4p

c. 2s, 3s, 4s

**Answer**

a. $3s < 3p < 3d$ because as the value of $\ell$ increases energy also increases.

b. $4s < 4p < 4d < 4f$ because as the value of $\ell$ increases energy also increases.

c. $2s < 3s < 4s$, because as the value of n increases energy also increases.

87. List the possible sub - shells and orbitals associated with the principal quantum number, n = 5.

**Solution**

➦ 5s, one orbital, 5p, three orbitals, 5d, ten orbitals, 5f, seven orbitals and 5g nine orbitals.

88. How many sub - shells are there in the shell designated by n = 6? List them

**Solution**

Number of sub - shells = n = 6, these are 6s, 6p, 6d, 6f, 6g and 6h.

## Shapes of Atomic Orbitals

**1. s - orbitals**

➦ For any value of n, the value of $\ell$ is zero for s - orbital.

➦ It has spherical shape and one orientation in space.

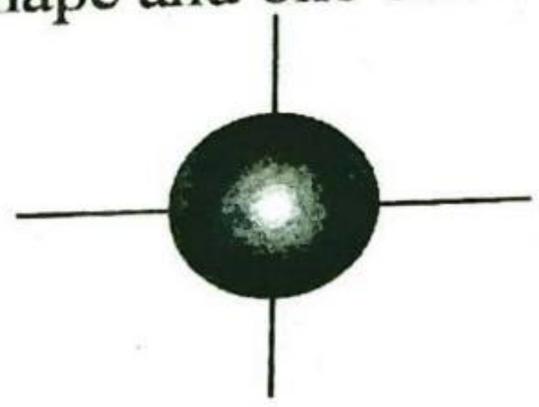


Figure 2.7 Shape of s - orbitals

➦ All s - orbitals have the same shape and orientation but different size.

**2. p - orbitals**

➦ For any value of n, the value $\ell$ is 1 for p - orbitals.

➦ There are three different orientations ($m\ell = -1, 0, 1$) in space for the three p - orbitals.

➦ There is no 1p - orbital because of the value of $\ell$

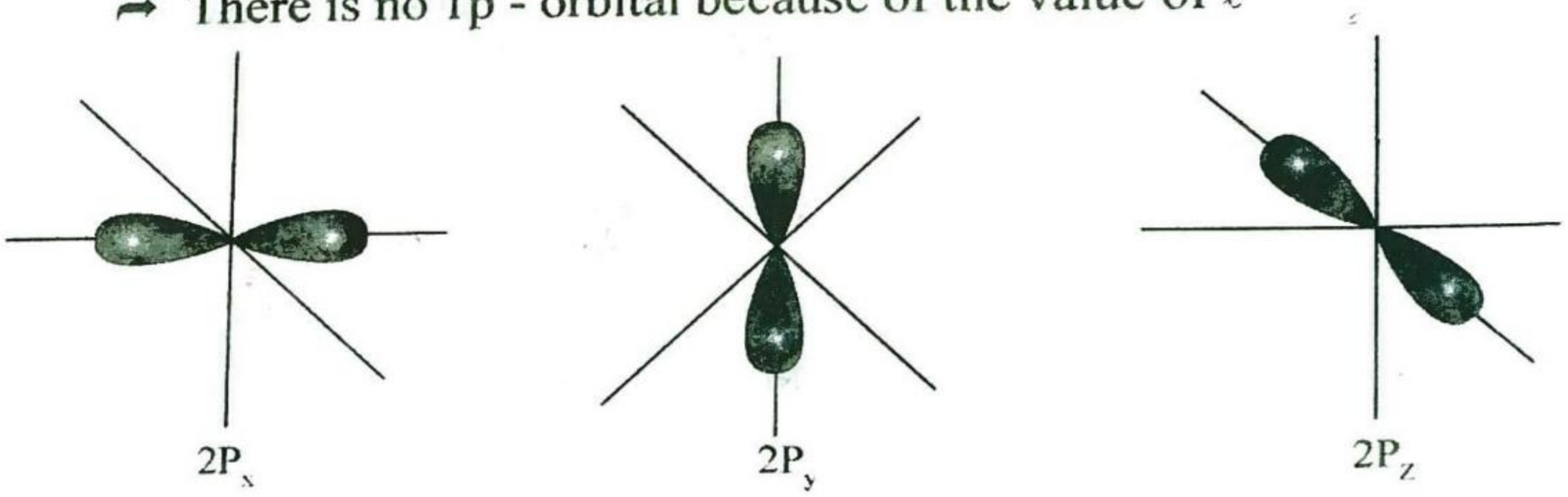



Figure 2.8 Shape and Orientations of p - orbitals

- As the value of n increases, the size of p - orbitals also increases.
- p - orbitals with the same value of n have the same energy and are called **degenerate - orbitals.**

**Example**

2px, 2py and 2pz are degenerate orbitals.

**3. d - orbitals**

- For any value of n, the value of $\ell$ is 2 for d - orbitals.
- There are five d - orbitals with five different orientations ($m\ell = -2, -1, 0, 1, 2$) in space.

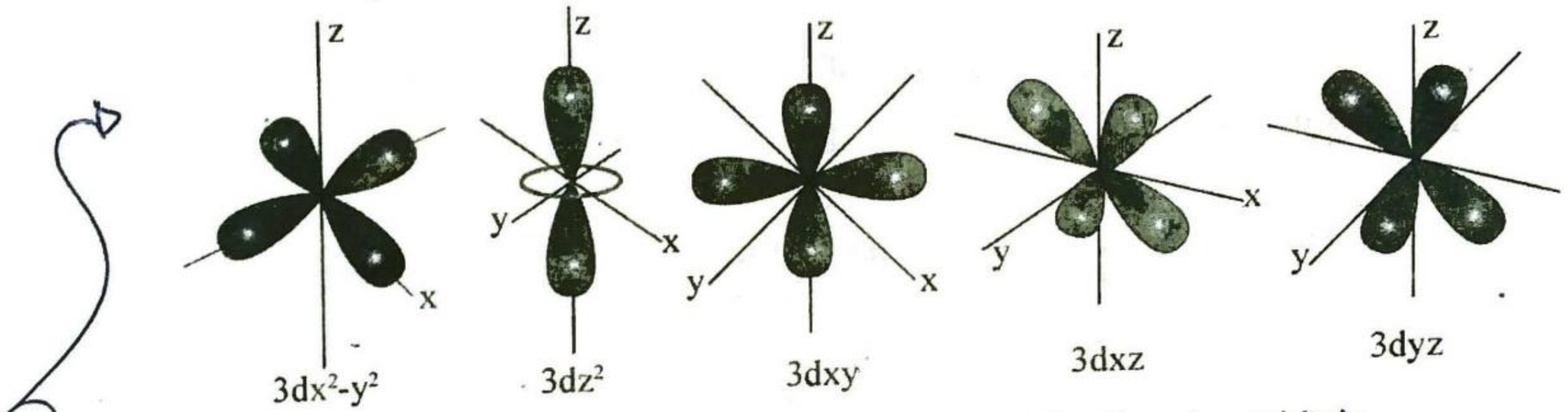



Figure 2.9 Shape and orientations of the five d – orbitals

- d - orbitals with the same n values are also **degenerate orbitals.**

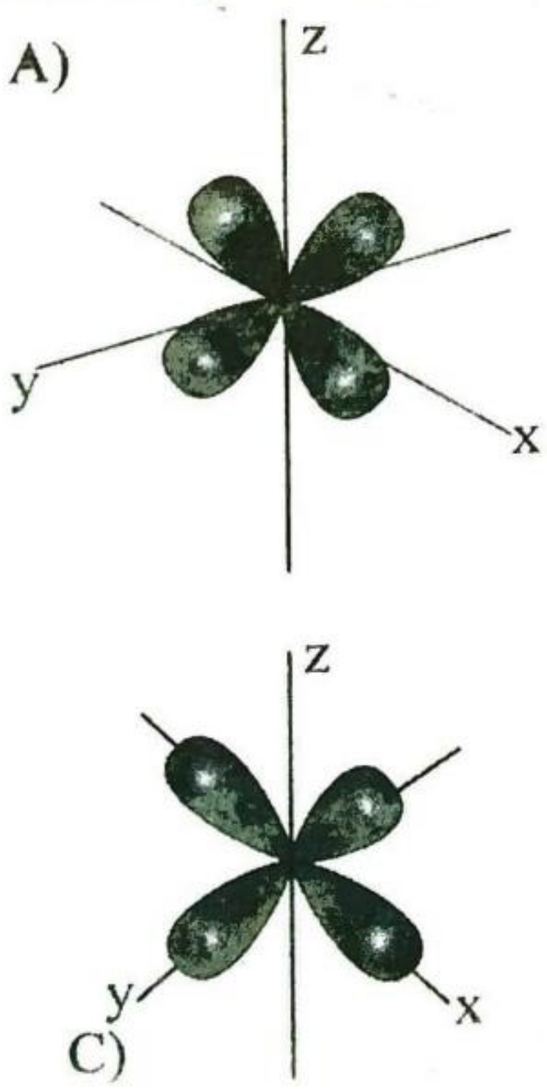
**Example**

- $3dx^2y^2$, $3dz^2$, 3dxy, 3dxz and 3dyz are degenerate orbitals.

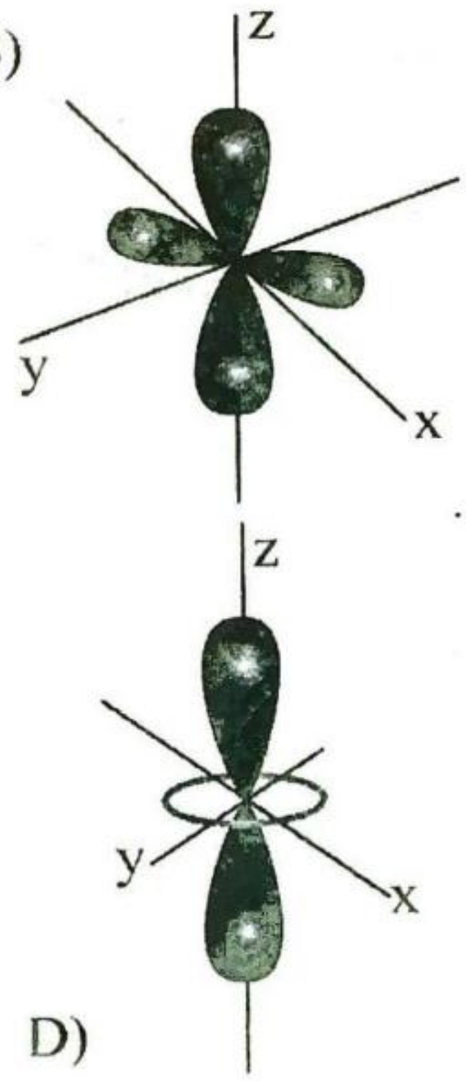
## Illustrative Examples

89. Which of the following diagrams describes the electron density in the $dx^2 - y^2$ orbital? (EUEE, 1999 E.C)

A)




B)




C)




D)




**Answer: C**

90. Why degenerate orbitals have the same energy?

**Explanation**: Because they have the same value of principal and angular momentum quantum numbers.

**Example**

4px, 4py and 4pz, n = 4 and l = 1

## 2.6 Electron Configuration and Orbital Diagrams

**Electron configuration** is distribution of electrons in atomic shells, sub - shells or orbitals. There are different types of electron configuration. These are

1. Main shell electron configuration
2. Sub - shell electron configuration
3. Orbital diagram configuration
4. Molecular orbital configuration

➦ The electron configuration of atoms or ions or molecules is based on the following three principles.

1. **The Aufbua (building up) principle**: states that atomic shells or orbitals with the lowest energy filled first.

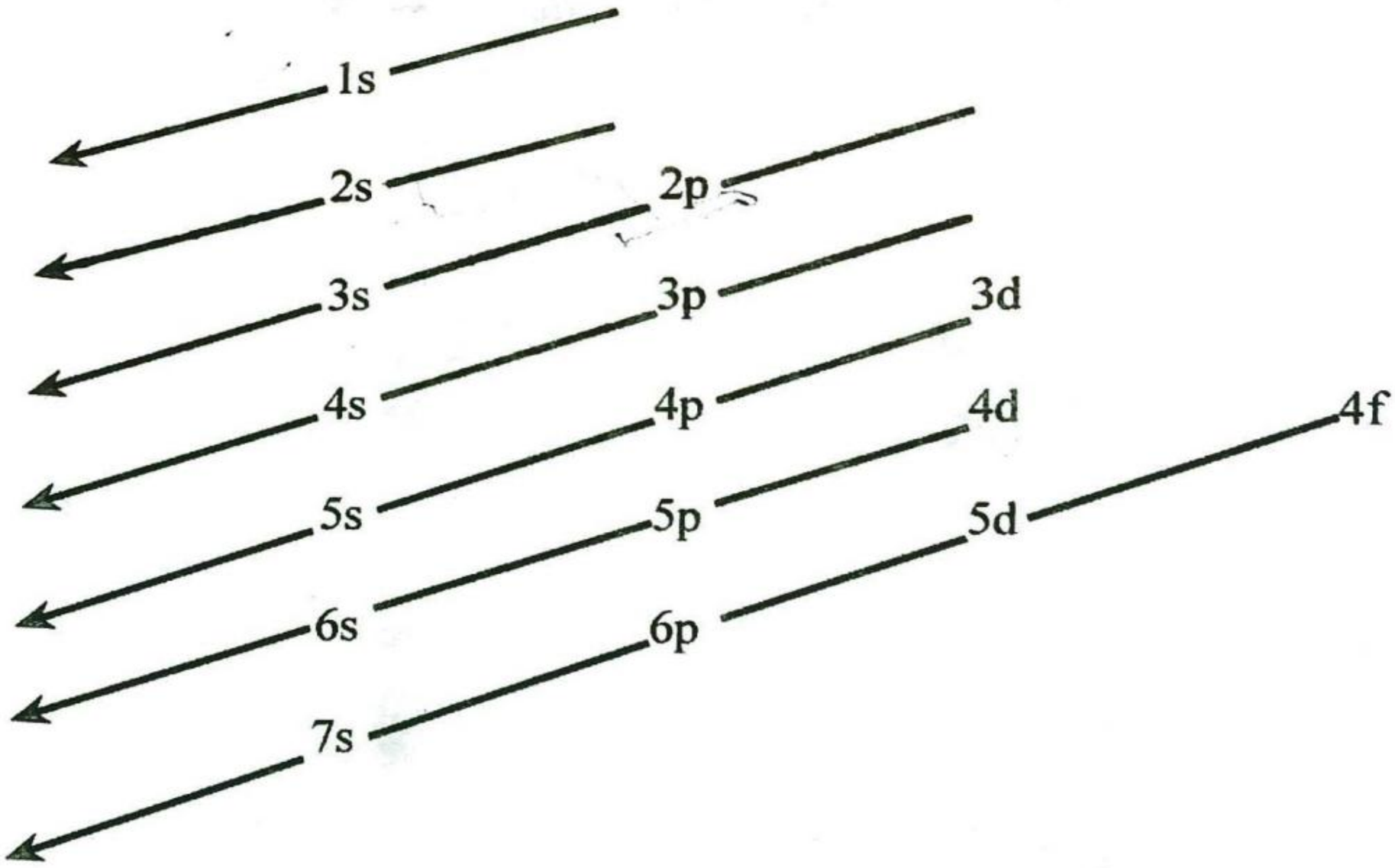



Figure 2.10 Diagonal rule of filling atomic orbitals

**Example**

Write electron configuration for sodium (z = 11) using sub = shells.

**Solution**

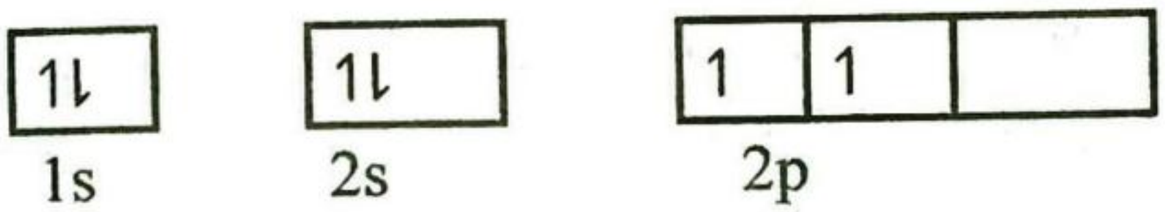
$\Rightarrow 1s^2 2s^2 2p^6 3s^1$

2. **Hund's rule** states that the most stable arrangement of electrons in orbitals of equal energy is the one in which the number of electrons with the same spin is maximized.

   ➦ It is applied for elements or atoms that have degenerate orbitals which are not fulfilled.

**Example**

$C: 1s^2 2s^2 2p_x^1 2p_y^1 2p_z^0$ or by using orbitals diagram




3. **The Pauli Exclusion Principle** states that no two electrons in an atom can have the same four quantum numbers. At least they differ in spin quantum number. For example, consider helium (z = 2), He: $1s^2$

   For the first electron: n =1, $\ell = 0$, $m\ell = 0$, ms = +1/2

   For the second electron: n = 1, $\ell = 0$, $m\ell = 0$, ms = - 1/2

**Note**

Orbital diagram configuration is very important to determine the number of unpaired electrons.

## Illustrative Examples

91. What is the maximum number of unpaired electrons in a d-sub shell? (EUEE ... 2008 E.C)

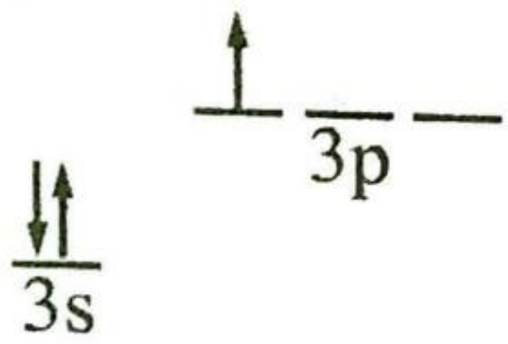
A. 2 B. 5 C. 3 D. 4

**Explanation:** The maximum number of unpaired electrons is a sub - shell is equal to the number of orbitals it contains.

- ➦ s - sub shell, 1 - orbital — 1 - unpaired
- ➦ p - sub shell, 3 - orbitals — 3 - unpaired electrons
- ➦ d - sub shell, 5 - orbitals — 5 - unpaired electrons
- ➦ f - sub shell, 7 - orbitals — 7 - unpaired electrons

**Answer: B**

92. The following energy level diagram represents outermost shell of what ground state element?




A. B B. He C. Al D. Be

**Solution**

➦ It is that of aluminum, Al (Z = 13) Al: $1s^2 2s^2 2p^6 3s^2 3p^1$

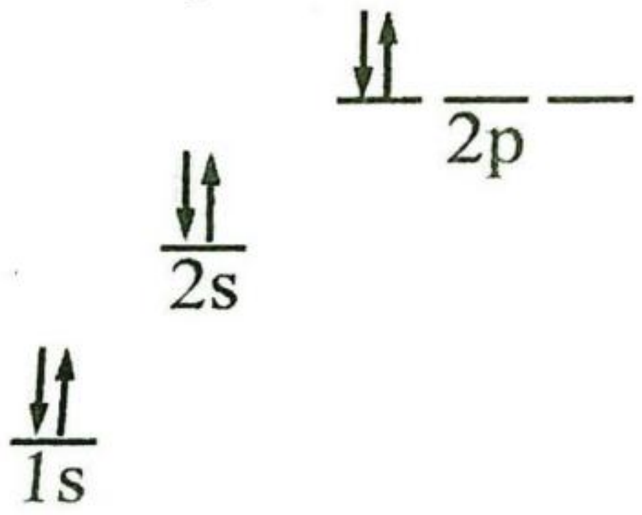
**Answer: C**

93. For which of the following elements is Hund's rule used in writing the electron configuration?

A. C B. B C. Be D. Li

**Answer: A**

94. What can you conclude from the figure below? (EUEE)




A. Hund's rule has been violated.
B. The Pauli exclusion principle has been violated.
C. The Aufbua principle has been violated.
D. This is valid diagram.

**Answer: A**

95. The Pauli exclusion principle _____

A. Tells the energy of an electron.
B. Tells the shape of an electron.
C. Describes the position in space of an electron.
D. Limits the number of electrons in an orbital to be two.

**Explanation**

➦ Pauli limited the number of electrons in an orbital to be two.

**Answer: D**

96. Which one of the following is the correct electronic configuration of Ni? (EUEE)

A. $1s^2 2s^2 2p^6 3s^2 3p^6 3d^8 4s^2$
B. $1s^2 2s^2 2p^6 3s^2 3p^6 3d^9 4s^1$
C. $1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 4p^6 5s^2$
D. $1s^2 2s^2 2p^6 3s^2 3p^6 3d^6 4s^2 4p^2$

**Solution**

Ni (Z = 28): $1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d^8$

**Answer: A**

97. Which element corresponds to the electron configuration $[Ne]3s^2 3p^1$?

A. Mg B. Al C. Si D. Ga

**Solution**

Ne (Z = 10), thus, the atomic number of the element becomes 10 + 2 + 1 = 13. It is aluminum (Al).

**Answer: B**

98. A neutral atom X at its ground state has half - filled 4f orbitals. What is the atomic number of X?

**Solution**

Write electron configuration and determine the atomic number of X.

$\Rightarrow 1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d^{10} 4p^6 5s^2 4d^{10} 5p^6 6s^2 4f^7$

$\therefore Z = 2 + 2 + 6 + 2 + 6 + 2 + 10 + 6 + 2 + 10 + 7 + 2 + 7 = 63$

99. Write the valence electron of Potassium using the four quantum numbers.

**Solution**

➦ Write noble gas core configuration and identify the outer most shell electron.

➦ K(Z = 19): $[Ar]4s^1$, thus the valence shell configuration is $4s^1$.

$\Rightarrow n = 4, l = 0, ml = 0, ms = +\frac{1}{2}$

100. Write electron configuration for the following atoms and ions.

a. Cr
b. Cu
c. $N^{3-}$
d. $Fe^{2+}$
e. Br
f. Zn

**Solution**

➦ Cr and Cu violates the building up principle

a. Cr (Z = 24): $1s^2 2s^2 2p^6 3s^2 3p^6 4s^1 3d^5 = [Ar]4s^1 3d^5$

b. Cu (Z = 29): $1s^2 2s^2 2p^6 3s^2 3p^6 4s^1 3d^{10}$

c. For ions find the number of electrons.

$e = p - charge = 7 - (-3) = 10$

$\Rightarrow N^{3-}$ $(Z = 7)$: $1s^2 2s^2 2p^6 = [Ne]$

d. $e = p - charge = 26 - (+2) = 24$

For transition elements, electrons are lost from s - orbital first and next from d - orbital.

$Fe^{2+}$: $1s^2 2s^2 2p^6 3s^2 3p^6 3d^6 = [Ar]3d^6$

e. Br $(Z = 35)$: $1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d^{10} 4p^5 = [Ar]4s^2 3d^{10} 4p^5$

f. Zn $(Z = 30)$: $1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d^{10} = [Ar]4s^2 3d^{10}$

101. Two electrons in an orbital always have opposite spins. Why?

**Explanation:** The spin of an electron is magnetic property not an actual rotation like a magnet with its south and north poles. Electrons which are paired have opposite magnetic properties, which lead to opposing spins.

102. Why there are no more than two electrons in an orbital?

**Explanation:** Because an atomic orbital is a mathematical function that describes the wave like behavior of either an electron or a pair of electrons.

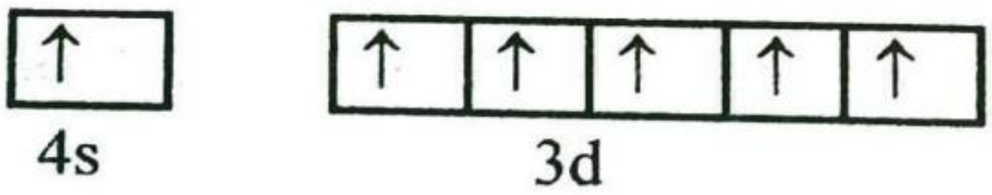
103. Determine the number of unpaired electrons in

a. Cr b. Ge c. $Zr^{2+}$ d. $Se^{2-}$

**Solution**

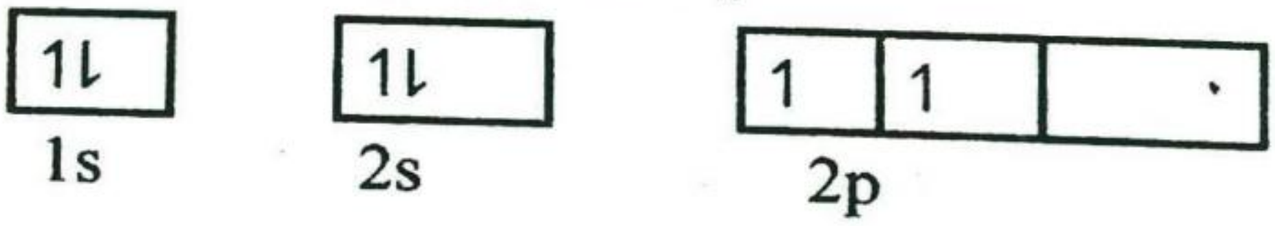
➦ Unpaired electrons are found in valence shells. Represent inner shells by the nearest noble gas. Use orbital diagram to find the number of unpaired electrons.

a. Cr $(Z = 24)$: $[Ar]4s^1 3d^5$




$\Rightarrow$ Number of unpaired electrons = 6

b. Ge $(Z = 32)$: $[Ar]4s^2 3d^{10} 4p^2$




$\Rightarrow$ Number of unpaired electrons = 2

c. $Zr^{2+}(Z = 40)$: $[Kr]4d^2$

⇒Number of unpaired electrons = 2

d. $Se^{2-}$ $(Z = 34)$: [Kr]

⇒ No unpaired electrons.

104. Portions of orbital diagrams representing the ground state electron configurations of certain elements are shown here. Which of them violate the Pauli exclusion principle? Which violate Hund's rule?

**Solution**

**Explanation**

- I and III violate Pauli exclusion principle.
- II and IV violate the Hund's rule.

105. Which of the following species has the greatest number of unpaired electrons: $S^+$, S or $S^-$?

**Explanation**

- $S^+(Z = 16)$, $e = 15$: $[Ne]3s^23p^3$, 3 unparied electrons
- $S(Z = 16)$, $e = 16$: $[Ne]3s^23p^4$, 2 unpaired electrons.
- $S(Z = 16)$: $e = 17$: $[Ne]\ 3s^23p^5$, 1 unpaired electron.

∴ $S^+$ has the greatest number of unpaired electrons.

106. An atom may absorb a quantum of energy and promote one of its electrons to a higher energy orbital. When it happens, we say that the atom is in an excited atoms are given. Identify these atoms and write their ground state configurations:

a. $1s^12s^1$

b. $1s^22s^22p^23d^1$

c. $[Ar]4s^13d^{10}4p^4$

d. $1s^22s^22p^64s^1$

e. $[Ne]3s^23p^43d^1$

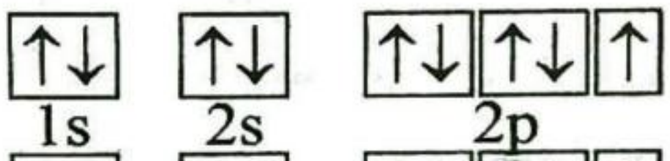
**Solution**

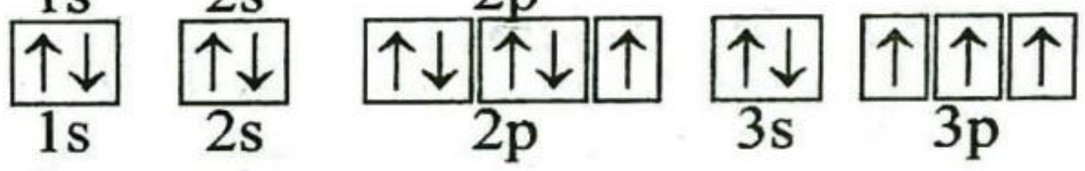
| | Atom | Ground state electron configuration |
|---|---|---|
| a) | He (Z = 2) | $1s^2$ |
| b) | N (Z = 7) | $1s^2 2s^2 2p^3$ |
| c) | As (Z = 33) | [Ar] $4s^2 3d^{10} 4p^3$ |
| d) | Na (Z = 11) | $1s^2 2s^2 2p^6 3s^1$ |
| e) | Cl (Z = 17) | [Ne] $3s^2 3p^5$ |

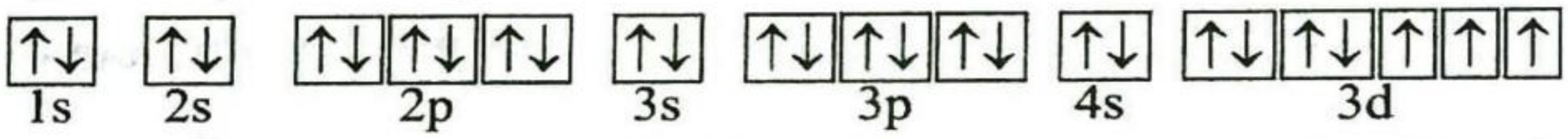
107. Write orbital diagram configuration for

a. O b. P c. Co

**Solution**

a. 


b. 


c. 


a. O (Z = 8) b. P (Z = 15) c. Co (Z = 27)

## 2.7 Periodic Classification of Elements

### Historical development of the Periodic table

#### 1. Dobernier's triad

- It was the first attempt for classification of elements.
- He arranged three elements in increasing order of atomic masses and said that the atomic mass of the middle element is the average of the other two elements.

**Example**

$^{7}Li^{23}$, $Na^{39}$, K is one triad.

$$\Rightarrow \text{Atomic mass of Na} = \frac{\text{Atomic mass K} + \text{Atomic mass of Li}}{2}$$

$$= \frac{39 + 7}{2} = 23$$

**Note:**

The law of triad was not applied even for all elements discovered at that time.

2. **Newland's law of octaves**: states that every eighth element has similar properties with the first element if they are arranged in increasing order of their atomic masses.

| | | | | | | |
|---|---|---|---|---|---|---|
| Li | Be | B | C | N | O | F |
| Na | Mg | Al | Si | P | S | Cl |
| K | Ca | | | | | |

Example

N and P have similar properties.

Note

- Newlands' law of octaves is not applied after calcium.
- When noble gases were discovered, they became the nineth elemes.

## 3. Mendaleev's Periodic law

- Russian chemist Dmitri Mendeleev and the German chemist Luthar Meyer independently proposed periodic law, which states: "The properties of elements is the periodic function of their atomic masses."
- Mendeleev's periodic table include 66 known elements.
- Mendeleev introduced periodicity.

## Advantage of Mendeleev's Periodic Table

- Leaving gap for undiscovered elements initiated other scientists to discover them.

## Limitations of Mendeleev's Periodic table

- Wrong order of the atomic masses of some elements.
- No space for isotopes of an element.

## The Modern Periodic Table

- The English physicist, Henry Mosely discovered **atomic number**.
- Every element has its own atomic number.
- The modern periodic law sates that, the properties of elements is a periodic function of their atomic number.
- In the modern periodic table there are 16 groups (8 'A' groups and 8 'B' groups) and seven periods.

Table 2.4. Common name of 'A' group elements

| Group | Common name |
|---|---|
| IA | Alkali metals except hydrogen |
| IIA | Alkaline earth metals |
| IIIA | Boron family |
| IVA | Carbon family |
| VA | Nitrogen family |
| VIA | Oxygen family (Chalogens) |
| VIIA | Halogens (Salt makers) |
| VIIIA | Noble gas or 'O' groups |

Table 2.5. The number of element in each period

| Period | Number of elements |
|---|---|
| 1 | 2 (H and He) |
| 2 | 8 (Li - Ne) |
| 3 | 8 (Na - Ar) |
| 4 | 18 (K - Kr) |
| 5 | 18 (Rb - Xe) |
| 6 | 32 (Cs - Rn) |
| 7 | Incomplete (Fr - ?) |

## Classification of Elements

### 1. Representative elements

- They are 'A' group elements.
- They are s - block and p - block elements.
- They include metals, non - metals and metalloids.

Group = Number of valence electrons.
Period = Number of shells.

### 2. Transition elements

- They are 'B' group elements.
- They are d - block elements.
- They are found between IIA and IIIA.
- They are all metals.

- They form colored and complex compounds.
- They have variable valence number.
- They are four series of elements (3d, 4d, 5d and 6d)

| Period = Number of shells |
|---|

- To determine the group of transition elements use the simple formula given below.

| Valence configuration | Group |
|---|---|
| $ns^1(n-1)d^{10}$ | IB |
| $ns^2(n-1)d^{10}$ | IIB |
| $ns^2(n-1)d^1$ | 2 + 1 = IIIB |
| $ns^2(n-1)d^2$ | 2 + 2 = IVB |
| $ns^2(n-1)d^3$ | 2 + 3 = VB |
| $ns^1(n-1)d^5$ | 1 + 5 = VIB |
| $ns^2(n-1)d^5$ | 2 + 5 = VIIB |
| $ns^2(n-1)d^6$ | VIIIB |
| $ns^2(n-1)d^7$ | |
| $ns^2(n-1)d^8$ | |

### 3. Inner transition (rare earth) metals

- They are f - block elements.
- They are all metals.
- They are two series elements (4f and 5f) called **lanthanides and actinides** respectively.

## Illustrative Examples

107. Which of the following types of elements are good oxidizing agents? (EUEE, 2008 E.C)

A. Alkali metals
B. Halogens
C. Lanthanides
D. Transition elements

**Explanation**

- Halogens are powerful oxidizing agents while alkali metals are best reducing agents.

**Answer: B**

108. Which group of elements is characterized by $ns^2np^2$ outer - electron configuration? (EUEE)

A. group 2A
B. group 4A
C. group 4B
D. group 3B

**Explanation**

| Outer - electron configuration | Group |
|---|---|
| $ns^1$ | IA |
| $ns^2$ | IIA |
| $ns^2np^1$ | IIIA |
| $ns^2np^2$ | IVA |
| $ns^2np^3$ | VA |
| $ns^2np^4$ | VIA |
| $ns^2np^5$ | VIIA |
| $ns^2np^6$ | VIIIA |

**Answer: B**

109. In which of the following period of the periodic table is an element with atomic number 20 is placed?

A. 1 B. 2 C. 3 D. 4

**Solution**

$Z = 20$, $1s^2 2s^2 2p^6 3s^2 3p^6 4s^2$ Period = valence shell = 4

**Answer: D**

110. Determine the group, period and block of the following elements.

a. Rb b. Mn c. S d. Kr

**Solution**

a. Rb(Z = 37)

$1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d^{10} 4p^6 5s^1$

⇒Valence shell configuration: $5s^1$

⇒ Group = IA

- Period = 5
- Block = s

b. Mn(Z = 25): $1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d^5$
   - ➦ Valence shell configuration: $4s^2 3d^5$
   - ➦ Group = 2 + 5 = VIIB
   - ➦ Period = 4
   - ➦ Block = d

c. S (Z = 16): $1s^2 2s^2 2p^6 3s^2 3p^4$
   - ➦ Valence shell configuration: $3s^2 3p^4$
   - ➦ Group = VIA
   - ➦ Period = 3
   - ➦ Block = p

d. Kr (Z = 36): $1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d^{10} 4p^6$
   - ➦ Valence shell configuration: $4s^2 4p^6$
   - ➦ Group = VIIIA
   - ➦ Period = 4
   - ➦ Block = P

111. Two elements X and Y are found in group IIA and VA respectively. Write the formula of the compound of X and Y.

**Explanation:**

Valence number of X = 2

Valance number of Y = 8 - 5 = 3

Formula: $\overset{2}{X}\!\!\times\!\!\overset{3}{Y} = X_3Y_2$

112. What is the most important relationship among elements in the same group in the periodic table?

**Explanation:** Elements in the same group have the same valence electrons. As a result, they have similar chemical properties.

113. Briefly describe the significance of Mendeleev's periodic table.

**Explanation:** He left gaps for undiscovered elements. This initiated other scientists to discover them.

114. What is the similarity among Dobernier's triad, Newland's octaves and Mendeleev's periodic law?

**Explanation:** Three of them arranged elements based on their atomic masses.

# Periodic Properties

1. **Number of valence electrons** remains same down a group but increases across a period.
2. **Number of shells** increases down a group but remains same across a period.
3. **Atomic size** (radius) is half the distance between the nuclei of any two closest atoms in a substance.
   - ➦ Atomic size increases down a group but decreases across a period from left to right.

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| H 0.037 | | | | | | | He 0.08 |
| Li 0.134 | Be 0.125 | B 0.090 | C 0.077 | N 0.075 | O 0.073 | F 0.071 | Ne 0.160 |
| Na 0.154 | Mg 0.145 | Al 0.130 | Si 0.118 | P 0.110 | S 0.102 | Cl 0.099 | Ar 0.190 |

Figure 2.11 Atomic radii in nm of the first three periods elements

- ➦ The more positive the cation, the smaller is its ionic radius.

**Example**

$Al^{3+} < Mg^{2+} < Na^{+}$

- ➦ The more negative the anion, the larger is its ionic radius.

**Example**

$F^{-} < O^{2-} < N^{3-} < C^{4-}$

4. **Ionization energy (IE)** is the amount of energy required to remove an electron from isolated gaseous atom.
   - ➦ First ionization energy (IE) removes the first valence electron.

$Al\ (g) + IE_1 \rightarrow Al^{+}(g) + e^{-}$

   - ➦ Second ionization energy ($IE_2$) removes the second valence electron.

$Al^{+}(g) + IE_2 \rightarrow Al^{2+}(g) + e^{-}$

➦ Third ionization energy ($IE_3$) removes the third valence electron and so on.

$Al^{2+}(g) + IE_3 \rightarrow Al^{3+}(g) + e^-$

$$IE_1 < IE_2 < IE_3 < IE_4$$

➦ Ionization energy (IE) decreased down a group and increases across a period.

## Exception in periods due to stability of orbitals

**Period 2:** IE of Be > B, N > O

**Period 3**: IE of Mg > Al, P > S

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| Li | Be | B | C | N | O | F | Ne |
| 520 | 900 | 801 | 1086 | 1402 | 1314 | 1681 | 2081 |
| Na | Mg | Al | Si | P | S | Cl | Ar |
| 496 | 738 | 578 | 789 | 1012 | 1000 | 1251 | 1521 |

Figure 2.12 First ionization energy is kJ/mol of Period 2 and 3 elements

5. **Electron affinity (EA)** is the energy released when one mole of electrons added to gaseous atom.

**Example**

$Cl(g) + e^- \rightarrow Cl^-(g) + EA$, EA = -348 kJ/mol

➦ Electron affinity decreases down a group and increases across a period.

6. **Electron negativity** is the power of an atom to pull electrons to itself when it is in a molecule.

➦ Electron negativity decreases down a group and increases across a period.

The three most electron negativity elements are fluorine, oxygen and nitrogen respectively.

**Metallic character (Electro positively)** is the property of metals.

➦ Metals: are malleable, ductile, luster and sonorous.

- are good conductors of heat and electricity.

- are all solids except mercury which is liquid.
- have high melting and boiling points
- are reducing agents

➦ Metallic character increases down a group and decreases across a period from left to right.

**Example**

Consider period 3 and group VA

**Period 3:**

| Na | Mg | Al | Si | P | S | Cl | Ar |
|---|---|---|---|---|---|---|---|

➦ Na, Mg an dAl are metals.

➦ Si is metalloid and P, S, Cl and Ar are non - metals.

**Group VA**

| |
|---|
| N |
| P |
| As |
| Sb |
| Bi |

➦ N and P are non - metals.

➦ As and Sb are metalloids.

➦ Bi is metal.

## Importance of the Periodic table

1. It is useful in predicting properties of elements.
2. It is also used to predic properties of compounds.
3. It is useful in predicting formula of compounds.
4. It is used to refer symbols, atomic masses and atomic numbers.
5. It is simple and easy way to remember properties of elements by writing electron configuration.

## Illustrative Examples

115. With in a specific period, an increase in atomic number is usually accompanied by (EUEE 2003 E.C.)

A. An increase in atomic radius and an increases in electron negativity.

B. A decrease in atomic radius and an increase in electro-negativity.

C. An increase in atomic radius and a decrease in electron negativity.

D. A decrease in atomic radius and a decrease in electro negativity.

**Explanation**

➦ As atomic size increases, electro negativity decreases, and vice - versal.

➦ As atomic number increases, atomic size usually decreases.

**Answer: B**

116. Which set is arranged in order of decreasing radii (EUEE ... 1995 E.C)

A. $Na > Na^+ > Mg^{2+} > Al^{3+}$

B. $Na < Mg^{2+} < Na^+ < Al^{3+}$

C. $Al^{3+} > Na^+ < Mg^{2+} > Na$

D. $Na^+ > Na > Mg^{2+} > Al^{3+}$

**Explanation**

➦ Cation has smaller atomic radius than its neutral form. Thus, $Na > Na^+$.

**Answer:A**

117. In which case the metals are arranged in order of increasing metallic character?

A. Mg, Ca, K, Cs

B. Mg, Ca, Cs, K

C. Ca, Mg, K, Cs

D. K, Mg, Ca, Cs

**Explanation**

➦ Mg and Ca are in group IIA and K and Cs are in groups IA. Metallic character increases down a group and decreases across a period.

$\Rightarrow Mg < Ca < K < Cs$

**Answer: A**

118. A periodic property

A. Changes according to periods

B. Repeats itself in the same manner with increase in atomic number.

C. Changes according to ionization energy.

D. Changes with temperature and mass number.

**Answer: B**

119. One of the properties of metal is, they are all

A. Liquid at room temperature
B. Soft
C. Good conductors of heat and electricity.
D. Insulators

**Answer:** C

120. Arrange the following in decreasing order of ionization energy.

a. $Fe, Fe^{3+}, Fe^{2+}, Fe^{+}$
b. B, Be, O, N
c. Al, S, P, Mg
d. F, I, Cl, Br

**Solution**

a. $Fe^{3+} > Fe^{2+} > Fe^{+} > Fe$
b. $N > O > Be > B$
c. $P > S > Mg > Al$
d. $F > Cl > Br > I$

121. Arrange the following in increasing order of atomic radius.

a. $Na^{+}, F^{-}, Al^{3+}, Mg^{2+}$
b. Li, Be, Na, Mg
c. C, B, F, Ne
d. $A_s^{3-}, Se^{2-}, P^{3-}, S^{2-}$

**Solution**

a. $Al^{3+} < Mg^{2+} < Na^{+} < F^{-}$
b. $Be < Li < Mg < Na$
c. $Ne < F < C < B$
d. $S^{2-} < P^{3-} < Se^{2-} < As^{3-}$

122. Write the four most electronegative elements in descending order.

**Solution**

$F > O > N > Cl$

123. Ionization energy is always positive but electron affinity is positive and negative. Explain

**Explanation**

- Ionization energy is always positive because it is the energy required for removal of electrons.
- A negative value of electron affinity indicates the process is exothermic while positive electron affinity indicates, the reaction is energetically favorable.

124. Arrange the elements in each of the following groups in order of increasing electron affinity.

a. Li, Na, K, Rb
b. F, Cl, Br, I

**Solution**

a. $Rb < K < Na < Li$

b. $I < Br < F < Cl$, Chlorine exceptional has the highest negative electron affinity

## Solved Problems

125. Bromine has two stable isotopes. For bromine the isotopic masses are 78.9183 amu (relative abundance, 50.69%) and 80.9163 amu (relative abundance 49.31%). Calculate the average atomic weight of bromine.

**Solution**

$$\text{Average atomci mass} = \frac{78.9183 \times 50.69 + 80.9163 \times 49.31}{100}$$

$$= 79.90 \text{ amu}$$

126. The metallic element chromium has four stable isotopes.

| Exact mass | Fraction abundance |
|---|---|
| 49.9461 | 0.0435 |
| 51.9405 | 0.8379 |
| 52.9407 | 0.0950 |
| 53.9389 | 0.0236 |

Calculate the average atomic mass of chromium.

Av. atomci mass = $49.9461 \times 0.0435 + 51.94.05 \times 0.8379 + 52.9407 \times 0.0950 + 53.9389 \times 0.0236 = 51.9$ amu

127. A retina of a human eye can detect light when radiant energy incident on it is at least $4.0 \times 10^{-17}$J. For light of 585 nm wave length, how many photons does this energy correspond to?

**Solution**

$$E_{photon} = \frac{hc}{\lambda} = \frac{6.626 \times 10^{-34} \text{ J.s} \times 3 \times 10^{8} \text{ m/s}}{5.85 \times 10^{-7} \text{ m}} = 3.4 \times 10^{-19} \text{ J/photon}$$

$\Rightarrow$ 1 photon = $3.4 \times 10^{-19}$J

? = $4.0 \times 10^{-17}$J

$$\text{Number of photons} = \frac{4.0\times10^{-17}\,\text{J}\times 1\text{ photon}}{3.4\times10^{-19}\,\text{J}} = 118\text{photons}$$

128. What is the wave length in nanometer of radiation that has an energy content of $1.0 \times 10^3$ kJ/mol?

**Solution**

$1 \times 10^3$ kJ/mol $= 1 \times 10^6$ J/mol

1 mole of photons $= 6.02 \times 10^{23}$ photons $= 1 \times 10^6$J

1 photon = ?

$$\Rightarrow E_{\text{Photon}} = \frac{1\text{ Photon}\times 1\times10^{6}\,\text{J}}{6.02\times10^{23}\,\text{Photons}} = 1.66\times10^{-18}\,\text{J}$$

$$E_{\text{Photon}} = \frac{hc}{\lambda}$$

$$\lambda = \frac{hc}{E_{\text{Photon}}} = \frac{6.626\times10^{-34}\,\text{J.s}\times 3\times10^{8}\,\text{m/s}}{1.66\times10^{-18}\,\text{J}} = 1.20\times10^{-7}\,\text{m}$$

$\lambda = 120$ nm

129. Thermal neutrons are neutrons that move at speeds comparable to those of air molecules at room temperature. These neutrons are most effective in initiating a nuclear chain reaction among $^{235}U$ isotopes. Calculate the wave length associated with a beam of neutrons moving at $7 \times 10^2$m/s. (Mass of a neutron=$1.675\times10^{-27}$kg)

**Solution**

$$\lambda = \frac{h}{mv} = \frac{6.626\times10^{-34}\,\text{J.s}}{1.675\times10^{-27}\,\text{kg}\times 7\times10^{2}\,\text{m/s}} = 5.65\times10^{-10}\,\text{m}$$

$\lambda$=5.65A°

130. Light with a wave length of 400nm strikes the surfaces of metallic cesium in a photo cell and the maximum kinetic energy of the electrons ejected is $1.54 \times 10^{-19}$J. Calculate the work function and the longest wavelength of light that is capable of ejecting electrons from that metal. Determine the velocity of an electron. (Mass of an electron = $9.1 \times 10^{-31}$ kg)

**Solution**

First find frequency of light

$$v = \frac{c}{\lambda} = \frac{3\times10^{8}\,m/s}{4\times10^{-7}\,m} = 7.5\times10^{14}\,Hz$$

$$kE = h\,(v - v_o)$$

$$\frac{kE}{h} = v - v_o$$

$$u_o = u - \frac{kE}{h} = 7.5\times10^{14}\,Hz - \frac{1.54\times10^{-19}\,J}{6.626\times10^{-34}\,J.s}$$

$$u_o = 5.18\times10^{14}\,Hz$$

$$\Phi = hv_o = 6.626\times10^{-34}J.s\times5.18\times10^{14}s^{-1}$$

$$\Phi = 3.43\times10^{-19}J$$

$$\lambda_{max} = \frac{c}{v_o} = \frac{3\times10^{8}\,m/s}{5.18\times10^{14}\,m/s} = 5.79\times10^{-7}\,m = 579nm$$

131. Compare the wave length for an electron (mass = $9.1\times10^{-31}$kg) traveling at a speed of $1\times10^{7}$m/s with that for a ball (mass 0.10 kg) traveling at 35 m/s.

**Solution**

- For electron: $\lambda = 7.3\times10^{-11}$m
- For the ball: $\lambda = 1.9\times10^{-34}$ m

$\Rightarrow$ As mass increases, wave length decreases.

132. Calculate the wave length of the third line in the Paschen series for hydrogen atom.

**Solution**

For Paschen series: $n_f = 3$ and ni for 3rd line = 6

$$\Delta E = 2.18\times10^{-18}\left(\frac{1}{6^2}-\frac{1}{3^2}\right) = 1.82\times10^{-19}J$$

$$\lambda = 1.09\times10^{-6}m = 1.09\mu m$$

133. What is the number of nucleons in $^{234}_{90}Th$ ?

**Solution**

Number of nucleons means mass number = 234

134. What sub-shells (s) is/are being filled in each of the following regions of the periodic table.

a. Group IA and IIA

b. Group IIIA - VIIIA

c. Transition elements

d. The lanthanides and actinides.

**Solution**

a. ns b. np c. (n - 1)d d. (n - 2) f

135. Arrange the atoms (ions) in the following groups in order of increasing size.

a. Be, C, B, F, Li

b. $Al^{3+}$, F, $Na^{+}$, $Mg^{2+}$,$O^{2-}$

c. $C^{4-}$, $F^{-}$, Ne, $N^{3-}$, $Na^{+}$

d. $Ca^{2+}$, $Cl^{-}$, $As^{3-}$, $K^{+}$, $S^{2-}$

**Solution**

a. $F < C < B < Be < Li$

b. $Al^{3+} < Mg^{2+} < Na^{+} < F^{-} < O^{2-}$

c. $Na^{+} < Ne < F^{-} < N^{3-} < C^{4-}$

d. $Ca^{2+} < K^{+} < Cl^{-} < S^{2-} < As^{3-}$

136. How many photons of light of $7 \times 10^{3}$ A° are equivalent to 1.00J of energy?

**Solution**

$$E_{Photon} = \frac{hc}{\lambda} = \frac{6.626\times10^{-34}\,J.s\times3\times10^{8}\,m/s}{7\times10^{-7}\,m}$$

$$E_{Photon} = 2.84\times10^{-19}\,J$$

$2.84 \times 10^{-19}J = 1$ photon

$1J = ?$

$$\text{Number of photons} = \frac{1J\times1\ Photon}{2.84\times10^{-19}\,J} = 3.52\times10^{18}\,\text{Photons}$$

137. When a certain metal was irradiated with light of frequency $3.2 \times 10^{16}$Hz, the photo electrons emitted had twice the kinetic energy as did photo electrons emitted when the same metal was irradiated with light of frequency $2.0 \times 10^{16}$Hz. Calculate the threshold frequency of the metal.

**Solution**

Given $kE_1 = 2kE_2$

$$\frac{kE_2}{kE_1} = \frac{h(u_2 - u_o)}{h(u_1 - u_o)}$$

$$\frac{2kE_2}{kE_2} = \frac{u_1 - u_o}{u_2 - u_o}$$

$$2u_2 - 2u_o = u_2 - u_o$$

$$2u_2 - u_1 = u_o$$

$$u_o = (2 \times 2 \times 10^{16} - 3.2 \times 10^{16}) \text{ Hz}$$

$$u_o = 8 \times 10^{15} \text{Hz}$$

138. Calculate the kinetic energy of the electron emitted when light of frequency $3 \times 10^{15}$ Hz is shone on a metal surface which has a threshold frequency $1 \times 10^{15}$Hz.

**Answer: kE = $1.3 \times 10^{-18}$J**

139. The wave number of the first line in the Balmer series of hydrogen is 15200 $cm^{-1}$. What is the wave number of the first line in the Balmer series of $Be^{3+}$?

**Solution**

$$\Delta E_{Be} = Z^2 \times \Delta E_H$$

$$\bar{v}_{Be} = Z^2 . \bar{v}_H = (4^2)(15200\,cm^{-1}) = 2.43 \times 10^5\,cm^{-1}$$

140. Determine the number of unparied electrons in

a. Eu
b. $pd^{3+}$
c. $Tl^+$
d. $Fe^+$
e. Au

**Answer**

a. 7
b. 1
c. 0
d. 5
e. 1

141. Arrange the following in decreasing order of ionic radius. Co, $Co^+$, $Co^{2+}$, $Co^{3+}$

**Answer:** $Co > Co^+ > Co^{2+} > Co^{3+}$

142. Arrange the following in increasing order of electro-negativity. Be, Mg, Ca, Sr.

**Answer:** Sr < Ca < Mg < Be

# Supplementary Exercise

1. Every atom consists of electrons, protons and neutrons except

   A. helium atom
   B. Lithium atom
   C. ordinary hydrogen atom
   D. boron atom

2. The number of neutrons in Cd are

   A. 109 B. 48 C. 61 D. 96

3. Non-metals when compared to metals:

   A. are less electronegative and have smaller radii
   B. have greater ionization energies and larger radii
   C. are more electronegative and have smaller radii
   D. are less electro positive and have smaller ionization energy

4. An atom containing two electrons which n = 3, l = 1, ml = 1, ms = -½ and n = 3, l = 1, ml = 1, ms = -½ may not exist based on

   A. Hund's rule
   B. Pauli's exclusing principle
   C. Heisenberg's uncertainty principle
   D. Buiding up principle

5. The charge on an electron was determined by measurement of the effect of an electric field on the rate at which oil droplets fell under the influence of gravity. This discovery was made by

   A. R.Millikan
   B. J.J. Thomson
   C. E. Rutherford
   D. H. Becquerel

6. An element with high ionization energy, very high electronegativity and high electronaffinity most likely classified as

   A. metal
   B. non-metal
   C. metalloid
   D. noble gas

7. All of the following increases down a group except

   A. metallic character
   B. Atomic size
   C. electropositivity
   D. Ionization energy

8. The classification of elements in the modern periodic table is mainly based on

   A. mass number
   B. Natural abundance
   C. physical property
   D. electron arrangement

9. A Particular shaped volume of space in an atom where the probability of finding an electron is maximum is known as _____

   A. an orbital
   B. an orbit
   C. a sub shell
   D. a shell

10. Thomson, Dalton, Rutherford and Bohr, all support which of the following statements?

    A. All matter is composed of atoms
    B. Atoms are mostly composed of empty space
    C. Atoms are composed of an electron and a proton
    D. Electron orbits the nucleus of an atom at distinct energy levels

11. The atomic number of an element with valence electron configuration 3s23p5 is

    A. 7  B. 17  C. 15  D. 18

12. The sub shell that is found in every main shell is

    A. s  B. p  C. d  D. f

13. The elements $^{14}C$ and $^{14}N$ have the same

    A. chemical property
    B. mass number
    C. neutron number
    D. proton number

14. The earliest of the attempts to classify elements was that of

A. Newlands octaves
C. Moseley's classification
B. Mendeleev's classification
D. Dobreiner's triads

15. Which group in the modern periodic table has elements in the three states of matter?

A. Alkali metals
C. Boron family
B. Chalcogens
D. halogens

16. Arrange the various regions of the electromagnetic spectrum is the correct order, in terms of wavelengths

A. Infrared > ultraviolet > visible > microwave > radio wave
B. Microwave > infrared > visible > ultraviolet > radio wave
C. radio wave > microwave > infrared > visible> ultraviolet
D. visible > infrared> ultraviolet > microwave > radio wave

17. The quantum numbers for a given orbital are n = 2, l = 1, m$\ell$ = 0. We would usually represent this as

A. 1s  B. 2s  C. 2pz  D. 3dxy

18. Which of the following has the smallest ionic radius?

A. $Li^+$  B. $Na^+$  C. $K^+$  D. $Rb^+$

19. How many different sub shell exist with principal quantum number, n = 3?

A. 9  B. 3  C. 6  D. 18

20. Which of the following is the most electronegative element?

A. Li  B. Na  C. Be  D. K

21. Which electronic sub shell do the lanthanides have incompletely filled?

A. 4f  B. 5f  C. 6f  D. 5g

22. In the reaction $^{223}_{87}Fr \rightarrow {}^{223}_{88}Ra + X$ , particle x is

A. a neutron
C. a beta particle
B. an α- particle
D. a proton

23. How many different values can m1 assume in the electron sub shell designated by quantum numbers n = 5, l = 4?

A. 3 B. 4 C. 8 D. 9

24. The energy of a photon having a frequency of $1\times 10^{9}$ MHz is

A. $6.626 \times 10^{-25}$J
B. $6.626 \times 10^{-19}$J
C. $6.626 \times 10^{-17}$J
D. $6.626 \times 10^{-15}$J

25. All of the following are isoelectronic except

A. $K^{+}$ B. $s^{2-}$ C. Ar D. $Al^{3+}$

26. The total number of s-electrons in $Mg^{2+}$ are

A. 4 B. 6 C. 10 D. 12

27. Which of the following is invalid electronic configuration?

A. 2, 8, 8, 2
B. 2, 8, 9, 1
C. 2, 8, 6
D. 2, 8, 18, 2

28. How many 3d electrons are present in the ground state of copper atom?

A. 9 B. 10 C. 8 D. 5

29. Which of the following is correct increasing order of size?

A. $O^{2-} > F^{-} > Na^{+} > Mg^{2+} > Al^{3+}$
B. $Al^{3+} > Mg^{2+} > Na^{+} > F^{-} > O^{2-}$
C. $Na^{+} > Mg^{2+} > Al^{3+} > F^{-} > O^{2-}$
D. $Mg^{2+} > Al^{3+} > Na^{+} > F^{-} > O^{2-}$

30. The element whose atoms have the electron configuration of [Kr] $4d^{10}5s^{2}5p^{2}$

A. is in group IIA
B. is in period 4
C. is a transition element
D. has 50 protons

**II. Answer the following questions**

31. Which one has large size?

a. Al or $Al^{3+}$
b. S or $S^{2-}$
c. Cu+ or $Cu^{+2}$
d. Be or B

32. The particle that has no electrical charge is _____

33. The particle with smallest mass is _____

34. Arrange the following in decreasing order of electronegativity

a. F, Be, C, O, N

b. Li, K, Na, Rb

c. Be, Mg, Ca, Sr

35. Find the number of un paired electrons in

a. $Cr^{3+}$ 3 b. $Mn^{4+}$ 3 c. Ag

36. When n = 3, the value of l can be 0, 1 and 2

37. When l = 1, the values of $m_\ell$ can be -1, 0 and 1

38. The number of electrons in the third shell before the 4s starts to fill is 8

39. When l = 3, the sub shell is f

40. Give the quantum number associated with each of the following

a. Orbital shape ሠ

b. Orbital energy ∩

c. Orbital orientation ሠ

አጋዥ መጽሐፍቶችን ለማግኘት በ
Telegram @ethiolibrary1 ላይ
ይቀላቀሉን
+251927273995
+251909282436
jemal6454@gmail.com

# Unit 3

## Chmeical Bonding and Structure

## Unit Content Outline:

3.1 Ionic bonding

3.2 Formation of ionic bonding

3.3 Covalent bonding

3.4 Coordinate covalent bonding

3.5 Molecular geometry

3.6 Intermolecular forces in covalent compounds

3.7 Metallic bonding

3.8 Chemical bonding theories

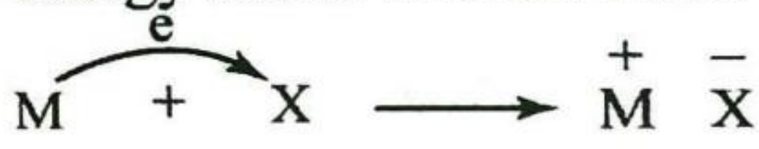
3.9 Types of solids

## Introduction

- Most of the substance around us are compounds and molecules. These substance are formed from atoms.
- The force that holds atoms together is called **chemical bond.** The types of electrons that participate in bond formation are known as **valance electrons**
- There are three types of chemical bonds. These are:
  1. Ionic bond
  2. Covalent bond
  3. Metallic bond

# 3.1 Ionic bonding

- Ionic bonding is also called **electro valent bond**
- It is formed by complete transfer of electrons from metals to non-metals.
- It is the electrostatic attraction force between positive and negative ions.
- The formation of ionic bond is generally, favored when an atom of low ionization energy reacts with an atom of high electron affinity.




- Compounds formed by ionic bond are termed as **ionic compounds.**

**Example**

$NaCl$, $KF$, $Al_2O_3$, $MgCl_2$ etc

## Properties Ionic compounds

- They are aggregates of positive and negative ions.
- They are all crystalline solids.
- They have high melting and boiling points.
- They are hard and brittle.
- The have high density
- They are resistant to heat but many will break by heat.
- They are usually soluble in polar solvent, water but insoluble in non- polar solvents such as benzene, carbon tetra chloride, etc.
- They are non- volatile.
- They transmit electricity in molten and aqueous states but not in solid state.

## Illustrative Examples

1. Which of the following is Not a form of chemical bonding? (EUEE...2008 E.C)

   A. Covalent bonding
   B. Metallic bonding
   C. Ionic bonding
   D. Hydrogen bonding

**Explanation:** Hydrogen bonding is intermolecular force. It is not a type of bonding.

**Answer: D**

2. Four elements A, B, C, and D have atomic numbers Z-1, Z+1 and Z+2 respectively. If Z is 9, then the bond between which pair of elements will be ionic? (EUEE)

   A. A and C
   B. C and D
   C. B and D
   D. B and C

**Explanation:** The valance electrons of metals is usually from 1 to 3 and non- metals from 5-8. ionic bond is formed between metals and non-metals except noble gases.

| Element | Atomic number | Electron configuration |
|---|---|---|
| A | Z - 1 = 9 - 1 = 8 | 2, 6 ... non - metal |
| B | Z = 9 | 2, 7 ... non - metal |
| C | Z + 1 = 9 + 10 | 2, 8 ... noble gas |
| D | Z + 2 = 9 + 2 = 11 | between D and A |

Thus, ionic bond is formed between D and A or D and B

**Answer: C**

3. **Which of the following compounds is unlikely to contain ionic bonds?**

   A. CO  B. NaF  C. Li Cl  D. $MgBr_2$

**Answer: A**

4. Which of the statements below best explains why atoms react chemically with each other?

   A. When atoms react, they gain proton and are more stable.
   B. When atoms react, they lose all their electrons and they become more stable.
   C. When atoms react, they lose, gain or share electrons and are then less stable.
   D. When atoms react, they lose, gain or share electrons to attain full outer energy level and are than more stable.

**Explanation:** All atoms except noble gases react chemically to lose, gain or share electrons and become more stable by lowering their energy.

**Answer: D**

5. All of the following their are charactersties of most ionic compounds in the solid phase. EXCEPT

A. Hight melting point
B. Solubility in water
C. High electrical conductivity
D. Insolubility in organic solvents

**Explanation:** Ionic compounds in solid phase do not transmit electricity because the ions are not free to carry charge .

**Answer: C**

6. What is the difference between cations and anions?

**Solution**

- Cations are positive ions formed when atoms lose electrons.
  Example: $Na^+$, $Ca^{2+}$, $Mg^{2+}$, etc
- Anions are negative ions formed when atoms gain electrons.
  Example: $Cl^-$, $O^{2-}$, $N^{3-}$, etc.

7. Which of the following forms ionic bond with bromine? why?

Boron, Carbon, Oxygen, Potassium

**Explanation:** Potassium forms ionic bond with bromine because it is metal.

8. How many ionic bonds result for the formation of:

a. NaF
b. $Mg Cl_2$
c. $Li_3N$

**Explanation:**

a. one ionic bond because valance electron is transfered from Na to F
b. two ionic bonds because two valance electrons of magnesium are transfered to chlorine
c. three ionic bonds.

9. Write the formula and name of ionic compounds formed between the following elements.

a. Calcium and nitrogen
b. Aluminum and flourine
c. Magnesium and oxygen
d. Barium and sulphur

**Solution**

a. $Ca_3N_2$-calciumnitride
b. $AlF_3$ - Aluminum flouride
c. MgO - Magnesium oxide
d. BaS - Barium Sulphide

10. Arrange the following in increasing order of ionic character. NaF, $Na_2O$, $Na_3N$, NaCl

**Explanation:** As the electro-negativity difference between metal and non-metal increases, ionic character also increases

$\Rightarrow NaCl < Na_3N < Na_2O < NaF$

11. Ionic bond is also known as electrovalent bond. Why?

**Explanation:** Because an ionic bond is attraction force between cation and anion

12. Draw the Lewis structure for the following ionic compounds.

a. NaCl b. $Mg_3N_2$ c. $Al_2O_3$ d. KBr

**Solution:**

a. $Na^+ \left[:\ddot{C}l:\right]^-$

b. $Mg_3^{2+}\left[:\ddot{N}:\right]_2^{3-}$

c. $Al_2^{3+}\left[:\ddot{O}:\right]_3^{2-}$

d. $K^+\left[:\ddot{B}r:\right]^{-1}$

13. Use sub-shell motion and Lewis electron dot symbol to represent the electron configuration of the following ions.

a. $Li^+$ b. $P^{3-}$ c. $Al^{3+}$ d. $O^{2-}$

**Solution:**

| Sub-shell notation | Lewis symbol |
|---|---|
| a. $1s^2$ or [He] | $[Li]^+$ |
| b. $1s^2\ 2s^2\ 2p^6\ 35^2\ 3p^6$ or [Ar] | $\left[:\ddot{P}:\right]^{3-}$ |
| c. $1s^2\ 2s^2\ 2p^6$ or [Ne] | $[Al]^{3+}$ |
| d. $1s^2\ 2s^2\ 2p^6$ or [Ne] | $\left[:\ddot{O}:\right]^{2-}$ |

## 3.2 Formation of Ionic bonding

➥ It is affected by three factors.

1. Ionization energy
2. Electron affinity
3. Lattice energy

**1. Ionization energy**

➦ Elements with low ionization energy favours formation of ionic bonding

**2. Electron affinity**

➦ It is an exothermic process and plays a role in making the ion more stable than the separated atom in the formation of ionic bond.

➦ Elements with high electron affinity favor formation of ionic bond.

**3. Lattice energy (U)** is the energy released when one mole of ionic solid is formed from its ions.

➦ It is also the energy required to break one mole of ionic solid into isolated gaseous ions.

**Example**

$$Na^+(g) + Cl^-(g) \underset{+U}{\overset{-U}{\rightleftharpoons}} NaCl(S)$$

➦ Lattice energy affects strength, melting point and solubility of ionic compounds.

$$U \alpha \frac{\text{Strength, melting point}}{\text{solubility of ionic compound}}$$

Factors affecting lattice energy.

➦ There are two factors that affect lattice energy.

1. Atomic size
2. The magnitude of the charge.

$$\text{Lattice energy } \alpha \frac{\text{Magnitude of charge}}{\text{Atomic size } |\text{radii}|}$$

**Example**

Lattice energy of: LiF > LiCl > Li Br > Li I because flourine has the smallest size.

Lattice energy of: $Na_2O < MgO < Al_2O_3$, because $Al^{3+}$ has the largest magnitude of charge and smallest size.

Lattice energy is calculated by using Hess's law through the Born-Haber cycle

- **Hess's law** states that enthalpy change, $\Delta H$ is independent on the paths the reactants proceed to form products.

  Mathematically: $\Delta H = \Delta H_1 + \Delta H_2 + \Delta H_3 + \ldots$

- Consider the formation of lithium fluoride, LiF

**Step 1**: Conversion of lithium solid into lithrism gas

$$Li\,(s) \longrightarrow Li\,(g),\ \Delta H_{sub} = 155KJ$$

**Step 2:** Decomposition of $F_2(g)$

$F_2\,(g) \longrightarrow 2F^-\,(g)$, $\Delta H_{diss} = 158$ KJ, to form one mole of F (g), $\Delta H$ diss $= \frac{1}{2} \times 158KJ = 79KJ$ because enthalpy change is extensive property.

**Step 3**: Ionization of lithium atom

$$Li\,(g) \longrightarrow Li^+\,(g),\ \Delta HIE = 520\ KJ$$

**Step 4:** Ionization of fluorine atom

$$F\,(g) + e \longrightarrow F^-(g)$$

**Step 5:** Bringing together one mole of Li (g) and $F^-$ (g) to form one mole of Li F (s).

$$Li^+\,(g) + F^-\,(g) \longrightarrow LiF\,(s),\ U = -1349\ KJ$$

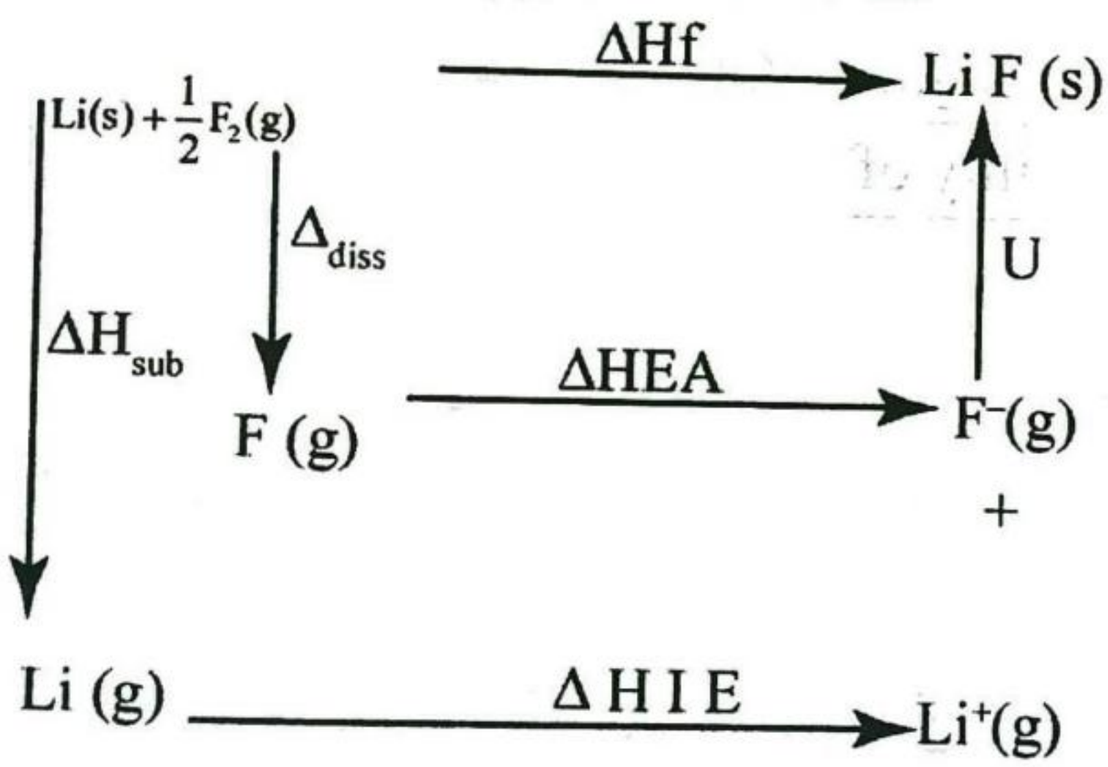



Figure 3.1 The Born- Haber cycle for the formation of lithium fluoride

Based on **Hess's law:**

$$\Delta Hf = \Delta H_{sub} + \Delta_{diss} + \Delta H_{IE} + \Delta H_{EA} + U$$

$$\Delta HF = (152 + 79 + 520 - 328 - 1349)\ KJ$$

$$\Delta Hf = -923\ KJ$$

$\Delta H_f$ – ent halpy of formation

$\Delta H_{sub}$ – enthalpy of sublimation

$\Delta H_{diss}$ – enthalpy of dissociation

$\Delta H_{IE}$– enthalpy of ionization energy,

$\Delta H_{EA}$–enthalpy of electron affinity

## Illustrative Examples

14. Which of the following compounds would be expected to have the highest melting point? (EUEE)

A. $BaF_2$ B. $BaCl_2$ C. $BaBr_2$ D. $BaI_2$

**Explanation:** As size decrease lattice energy and melting point increases

**Answer:** A

15. Which of the following ionic compounds has the smallest lattice energy?

A. LiF B. CsI C. NaF D. LiI

**Answer: B**

16. Arrange the following ionic compounds in decreasing order of solubility in water.

NaCl, LiCl, RbCl, KCl, CsCl

**Explanation:** As lattice energy increases, solubility of ionic compounds in water decreases.

Cs Cl > Rb Cl > KCl > NaCl > Li Cl

17. Arrange the compounds NaF, MgO and AlN in order of increasing lattice energy.

**Solution:**

NaF < MgO < AlN

18. Calculate the lattice energy of cesium chloride, Cs Cl and draw the Born-Haber cycle.

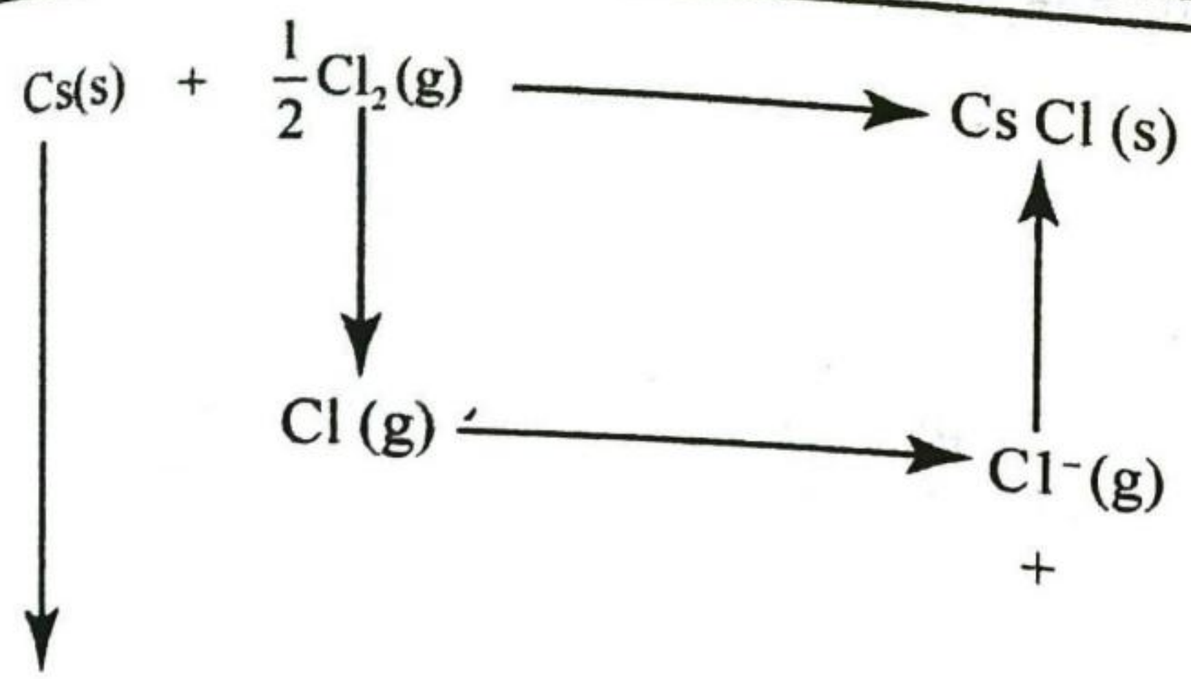
**Given:** $\Delta H_{sub} = 44$ KJ/ mol

$\Delta H_{EA}$ (C1) = − 349 KJ/mol

$\Delta H_{IE}$ (Cs) = 376 KJ/mol

$\Delta H_{diss}$ ($Cl_2$) = 243 KJ/mol

$\Delta Hf$ (CsCl (s)) = −423KJ/mol



$\Delta H_{diss}(Cl_2) = \frac{1}{2} \times 243 = 121.5 KJ/mol$

$\Delta Hf = \Delta H_{sub} + \Delta H_{diss} + \Delta H_{IE} + \Delta H_{EA} + U$

$U = \Delta H_f - (\Delta H_{sub} + \Delta H_{diss} + \Delta H_{IE} + \Delta H_{EA})$

$U = -423 - (77 + 376 + 121.5 - 349)$

$U = -649.5$ KJ/mol.

19. Using the following data, calculate enthalpy of formation of KF (s).

$\Delta H_{sub}(K) = 90KJ/mol$

$\Delta H_{diss}(F) = 160$ KJ/mol

$\Delta H_{IE}(K) = 419KJ/mol$

$\Delta H_{EA}(F) = -328KJ/mol$

$U = -808$ KJ/mol

**Solution:**

$\Delta Hf = \Delta H_{sub} + \Delta H_{diss} + \Delta H_{IE} + \Delta H_{EA} + U$

$= 90 + 80 + 419 - 328 - 808 = -547$ KJ/mol

## Ionic compounds that violate the octet rule

- **The octet rule** states that, during the formation of ion its valence shell has eight electrons by gaining, losing or sharing of valence electrons. For ionic compounds to obey the octet rule, there must be eight electrons in the valence shell of the cation.

## Ionic compounds below octet.

- Compounds of lithium and berylium have less than eight electrons in the valence shall of cations.

**Example**

$LiF$, $Li_2O$, $BeF_2$, $BeO$ etc.

**Ionic compounds above octet.**

➦ The cations of transition and post transition metals have above eight electrons in their valence shells. Thus, their compounds violate the octet rule.

**Example**

$FeCl_3$, $CuO$, $GaF_3$, $AgCl$ etc

$Fe^{3+}$: $1s^2 2s^2\ 2p^6\ 3s^2\ 3p^6\ 3d^5$

Valence shell configuration: $3S^2\ 3P^6\ 3d^5$

Therefore, 2 + 6 + 5 = 13 electrons in its valence shell.

➦ Ions with completely filled sub shells that have **$ns^2\ np^6\ nd^{10}$** valence shell configuration show pseudo **noble gas configuration.** Such type of ions follow the **18 electrons rule.**

**Example**

$Cu^+$, $Zn^{2+}$, $Ga^{3+}$, $Ag^+$, $Sn^{4+}$, etc.

$Cu^+ : 1s^2 2s^2 2p^6 3s^2 3p^6 3d^{10}$

$zn^{2+} : 1s^2 2s^2 2p^6 3s^4 3p^6 3d^{10}$

$Ag^+ : 1s^2 2s^2 2p^6 3s^2 3p^6 4d^2 3d^{10} 4p^6 4d^{10}$

$Sn^{4+} : 1s^2 2s^2 2p^6 3S^2 3p^6 3d^{10} 4524p^6 4d^{10}$

## Illustrative Examples

20. Which of the following ionic compounds does not obey the octet rule?

A. NaI B. MgO C. $Li_3N$ D. $Al_2O$

**Explanation:** $Li_3N$ does not obey the octet rule because $Li^+$ has $1s^2$ valence shell electron configuration. Thus, it has 2 electrons.

**Answer: C**

21. Which of the following cations has pseudo noble gas configuration?

A. $V^{3+}$ B. $Mn^{4+}$ C. $Cd^{2+}$ D. $Ca^{2+}$

**Solution:**

➦ $V^{3+}$ $(Z = 23)$ : $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 3d^2$ does not show.

- $Mn^{4+}(Z=25): 1s^2\ 2s^2\ 2p^6\ 3s^2\ 3d^6\ 3d^3$ does not show,
- $Cd^{2+}(Z=48): 1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 3d^{10}\ 4s^2\ 4p^6\ 4d^{10}$ show.
- $Ca^{2+}(Z=20): 1s^2\ 2s^2\ 2p^6 3s^2\ 3p^6$ does not show.

**Answer: C**

22. Check weather the following ionic compounds obey the octet rule or not.

a. CoO
b. SnO
c. $AlF_3$
d. $Ni\ F_2$
e. Kcl

**Solution:**

a. $Co^{2+}(z=27): ls^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 3d^7$ thus Co O does not obey.

b. $Sn^{2+}(Z=50): 1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 4s^2\ 3d^{10}\ 4p^6\ 4d^{10}\ 5p^2$ thus Sn O does not obey the octet rule.

c. $Al^{3+}(Z=13): 1s^2\ 2s^2\ 2p^6$, thus $A1F_3$ obeys the octet rule.

d. $Ni^{2+}\ (Z=28): 1s^22\ s^22\ p^6\ 3s^2\ 3p^6\ 3d^8$, thus $NiF_2$ does not obey.

e. $K^+\ (Z=19): 1s^22\ s^2p^6\ 3s^2\ 3p^6$, thus KCl obeys be octet rule.

## 3.3 Covalent Bonding

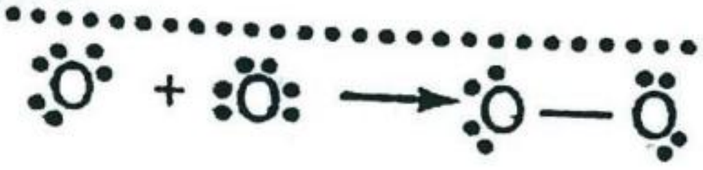
- **Covalent bonding** is inter atomic attraction resulting from sharing of electrons between the atoms.
- The shared electrons are called **bonding pairs (bp)** while the unshared electrons are known as **lone pairs (Lp)**.
- If one pair of electrons is shared, the resulting bond is **single covalent bond.**

Example

$$H\cdot + \cdot H \longrightarrow H-H$$

- If two pairs of electrons are shared, then the bond formed will be double **covalent bond**.

Example



- If three pairs of electrons are shared , the resulting bond is known as **triple covalent bond**.

**Example**

$$:\underset{\bullet}{N}: + :\underset{\bullet}{N}: \longrightarrow :N \equiv N:$$

## Lewis structure for covalent compounds

Follow the following rules in order to draw the Lewis structure for covalent compounds.

1. Determine the total number of valence electrons and pair them.

**Example**

Consider $SF_4$

S: 2, 8, 6, valence electrons = 6

F: 2, 7, valence electrons = 7

S: 1 × 6 = 6

F: 4 × 7 = 28

Total number of valence electrons = 34

$$\text{Pairs} = \frac{34}{2} = 17$$

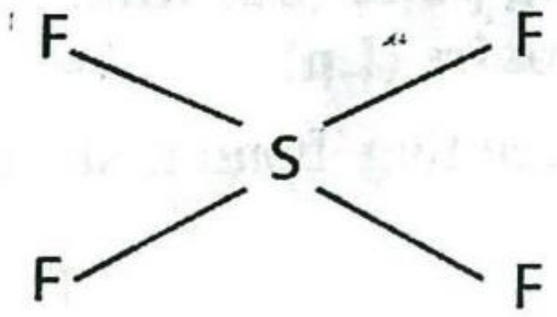
2. Identify the central atom (more positive or larger in size)

**Note:**

Hydrogen and flourine cannot be central atom.

In the above example, S is central atom.

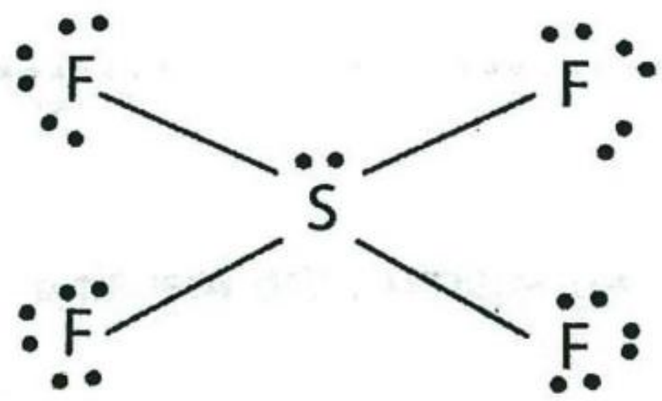
3. Draw the skeletal structure. For $SF_4$:




4. Fulfill the octet of the terminal atoms (atoms that surround the central atom) and place the raiming electrons on the central atom if any.

If the teminal atoms is hydrogen no need to fulfill its octet.

For $SF_4$




bp = 4

Lp = 13

5. Make multiple bonds (double or triple) by using lone pair of electrons on the terminal atoms to fulfill the octet of the central atom if necessary.

**Example**

For $SO_3$

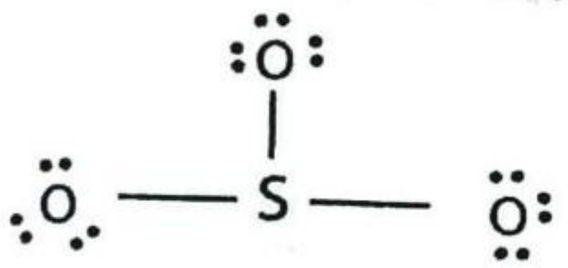
S: 2, 8, 6

O: 2, 6

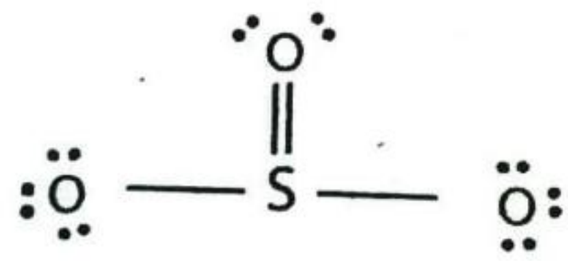
Total valence electrons: $1 \times 6 + 3 \times 6 = 24$

$$\text{Pairs} = \frac{24}{2} = 12$$

Centeral atom: S



Make O – S double bond to fulfill the octet of S.



bp = 4
Lp = 8

## Illustrative Example

22. How many bonding pairs and lone pairs, respectively does the ion $ICl_4^-$ have? (EUEE... 2008 E.C.)

A. 3, 2 B. 4,2 C. 5,1 D. 4,1

**Solution:**

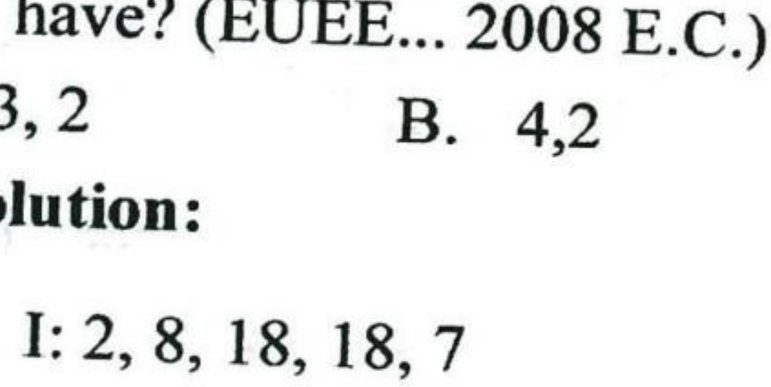
I: 2, 8, 18, 18, 7 Cl: 2, 8, 7

Charge = +1 because the ion gains one electron.

Total number of valence electrons: $1 \times 7 + 4 \times 7 + 1 = 36$

$$\text{Pairs} = \frac{36}{2} = 18$$

Centeral atom: I



bp = 4
Lp = 2 on the centeral atom.

**Answer: B**

23. What does the correct Lewis structure for $CCl_4$ molecules show? (EUEE)

A. 5 bonds

B. 24 Unshared electrons.

C. no unshared electrons.

D. 10 shared electrons.

**Solution:**

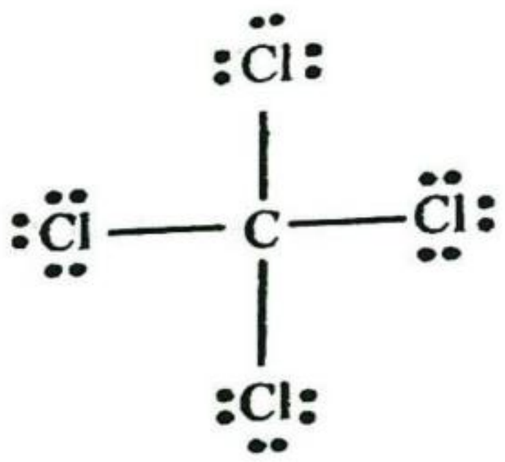
CC14: C: $1 \times 4 = 4$

C1: $4 \times 7 = 28$

Total valence electorns $= 4 + 28 = 32$

Pairs $= \frac{32}{2} = 16$

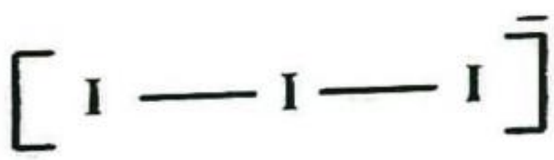
Centeral atom: C



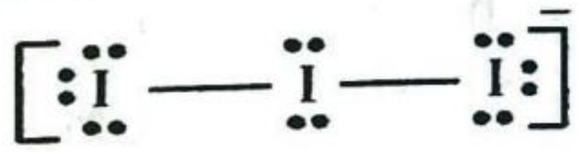
Lp = 12
bp = 4

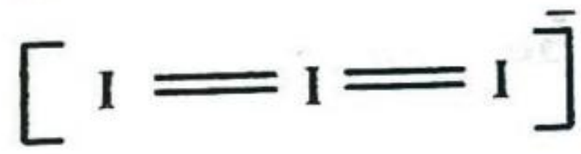
**Answer: B**

24. Which of the following Lewis structure TS correct for tri iodide ion $I_3^-$?

A. 

B.

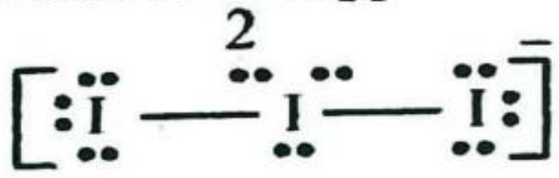
C. 

D. 

**Solution:**

Total valance electrons $= 3 \times 7 + 1 = 22$

Pairs $= \frac{22}{2} = 11$



**Answer: B**

25. Draw the Lewis structure and determine the number of bonding and lone pairs.

a. $N\overset{+}{O}$

b. $Cl\overset{-}{O_4}$

c. $SCl_2$

d. $PH^+_4$

e. $BrF_3$

f. $C\overset{2-}{O_3}$

g. $CH_3COOH$

h. $COCl_2$

**Solution**

For positive ion subtract the charge from total valence electrons

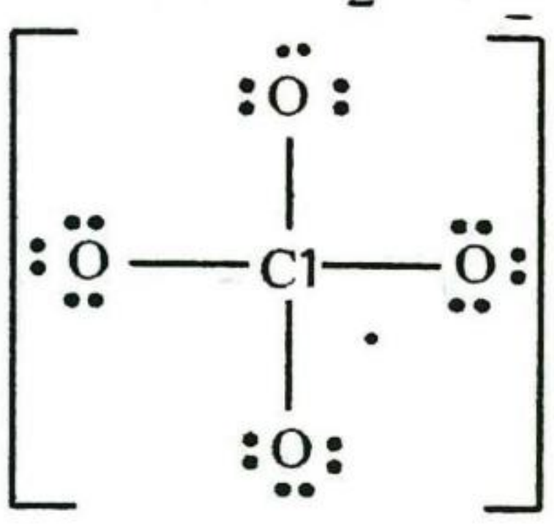
a. Total valence electrons: $1\times 5 + 1\times 6 - 1 = 0$

$$\text{Paris} = \frac{10}{2} = 5$$

$[:N \equiv O:]^+$ Lp = 2, bp = 3

b. Total valence electrons: $1 \times 7 + 4\times 6 + 1 = 32$

$$\text{Pairs} = \frac{32}{2} = 16$$



Lp = 12
bp = 4

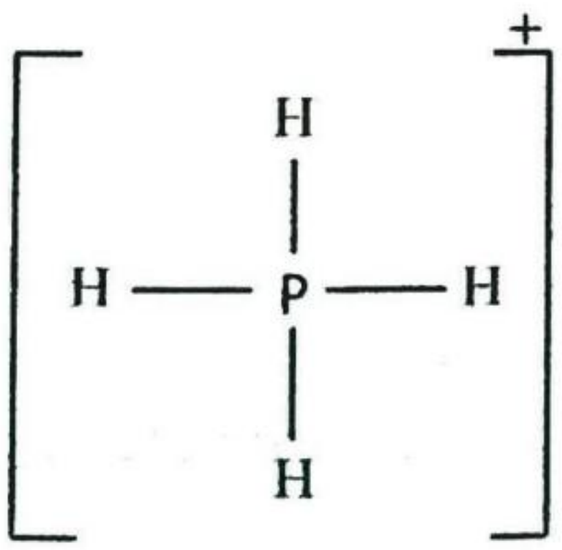
c. Total valence electrons: $1\times 6 \times 2 \times 7 = 20$

$$\text{Pairs} = \frac{20}{2} = 10$$

bp = 4
Lp = 0

d. Total valence electrons: $1\times5 + 4 \times 1 - 1 = 8$

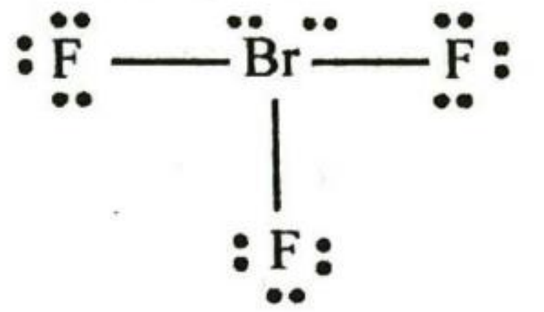
$$\text{Pairs} = \frac{8}{2} = 4$$



bp = 4
Lp = 0

e. Total valence electrons: $1\times 7 + 3 \times 7 = 28$

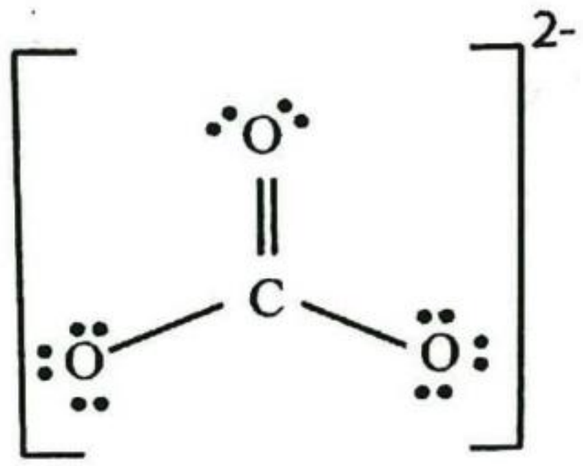
$$\text{Pairs} = \frac{28}{2} = 14$$



Lp = 11
bp = 3

f. Total number of valence electrons: $1 \times 4 + 3 \times 6 + 2 = 24$

$$\text{Pairs} = \frac{24}{2} = 12$$

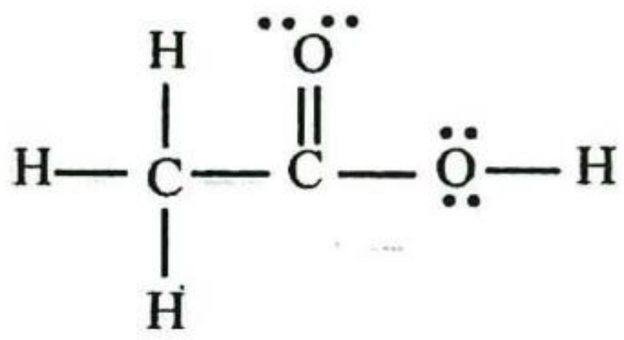


Lp = 8
bp = 4

g. Total number of valence electrons: $2 \times 4 + 4 \times 1 + 2 \times 6 = 24$

$$\text{Pairs} = \frac{24}{2} = 12$$

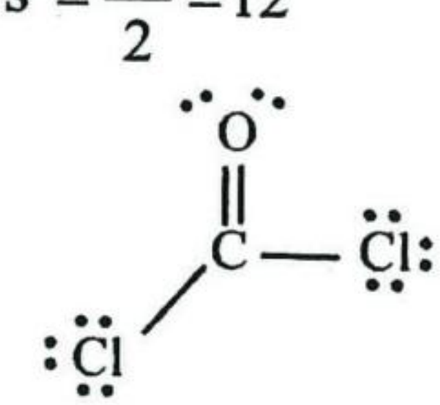
In $CH_3COOH$, there are three central atoms, the two carbon atoms and one oxygen atom.



bp = 8
Lp = 4

h. Total number of valence electrons: $1 \times 4 + 1 \times 6 + 2 \times 7 = 24$

$$\text{Pairs} = \frac{24}{2} = 12$$



bp = 4
Lp = 8

26. Arrange single, double and triple bonds in decreasing order of bond strength.

**Solution**

➥ Triple bond > double bond > single bond

27. What is the attraction force in covalent bonding

**Solution**

➦ The attraction force in covalent bonding is the force between the shared electrons and the nuclei of the bonded atoms.

28. Covalent bonding is formed between what type of elements?

**Solution**

➦ Covalent bonding is mainly formed between non-metals and between metalloids and non-metals.

## 3.4 Coordinate covalent bond

➦ It is also called **dative or donor-acceptor bond**.

➦ It is inter-atomic attraction resulting from the sharing of a lone pair of electrons from one atom with another atom.

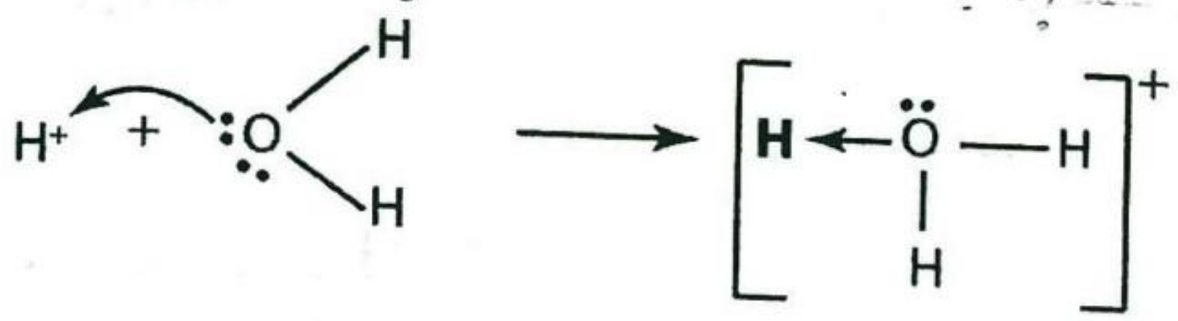
➦ The shared electrons are obtained from one atom and the other atom has vacant or unfulfilled orbital. $A: + B \rightarrow A: B$

**Example**

Consider the following species

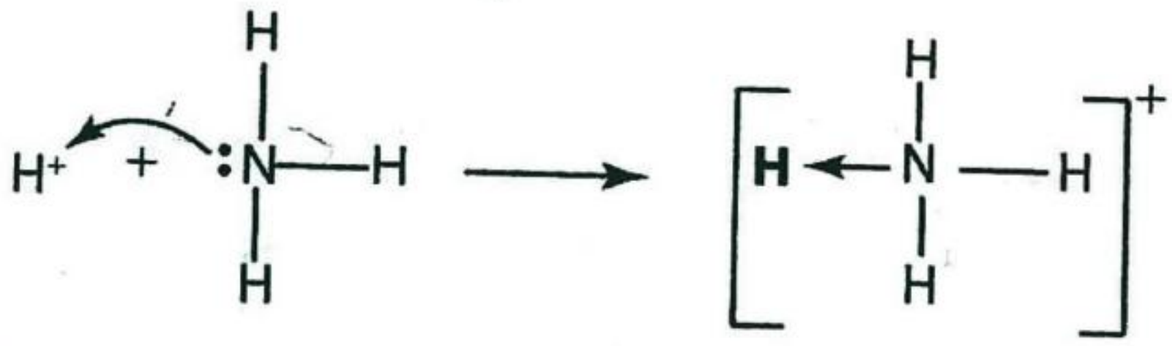
a. $H_3O^+$ b. $NH_4^+$ c. $F_3BNH_3$

a. $H^+ + H_2O \rightarrow H_3O^+$

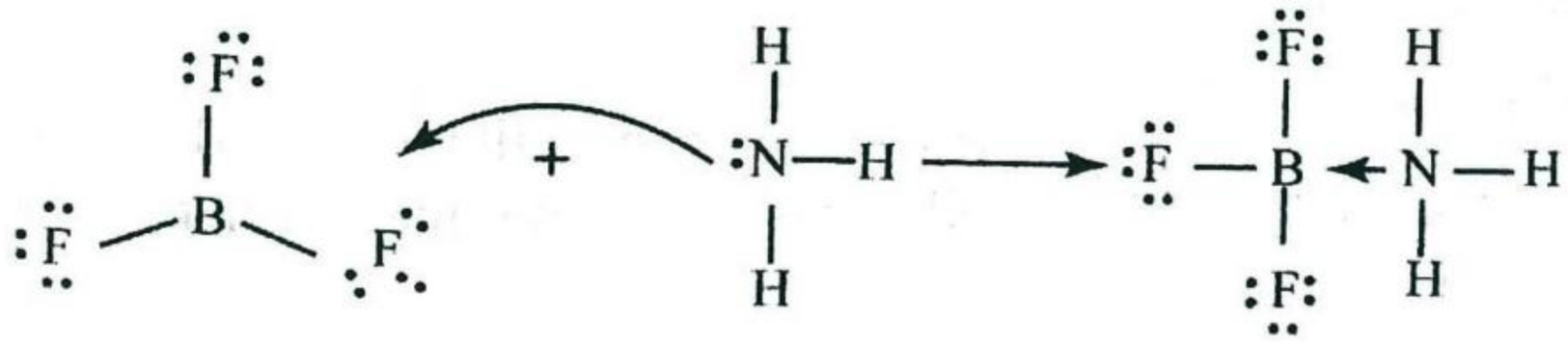


The bold bond is dative bond

b. $H^+ + NH_3 \rightarrow NH_4^+$



c. $BF_3 + NH_3 \rightarrow F_3BNH_3$



## Illustrative Example

29. Which of the following shows the formation of coordinate covalent bond?

A. $A\cdot + \cdot B \longrightarrow A:B$

B. $A\cdot + \cdot\underset{\cdot\cdot}{\overset{\cdot\cdot}{B}}: \longrightarrow A^{+}\,[:\underset{\cdot\cdot}{\overset{\cdot\cdot}{B}}:]^{-1}$

C. $A + :B \longrightarrow [A:B]^{+}$

D. $A: + :B \longrightarrow A: :B$

**Explanation:**

- The first ideal reaction show formation of single covalent bond.
- The second reaction shows formation of ionic bond.
- The fourth reaction shows formation of double covalent bond.

**Answer: C**

30. Which of the following molecules doesn't have a coordinate covalent bonds? (EUEE. 2000 E.C)

A. $H_3O^+$

B. $NH_4^+$

C. $BF_3\ NH_3$

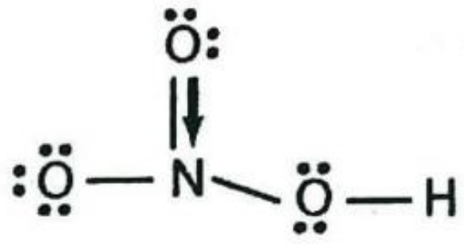
D. $CO_2$

**Answer: D**

31. Identify the coordinate covalent bond in the following species by drawing Lewis structure

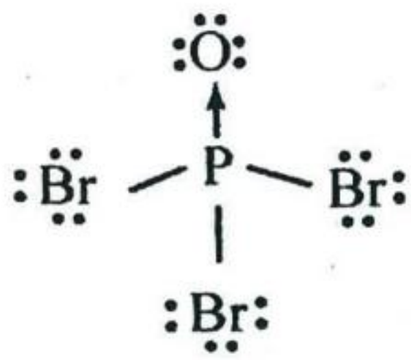
a. $HNO_3$ b. CO c. $PO\ Br_3$

**Solution:**

a. 

the bold arrow is coordinate covalent bond.

b.

c. 

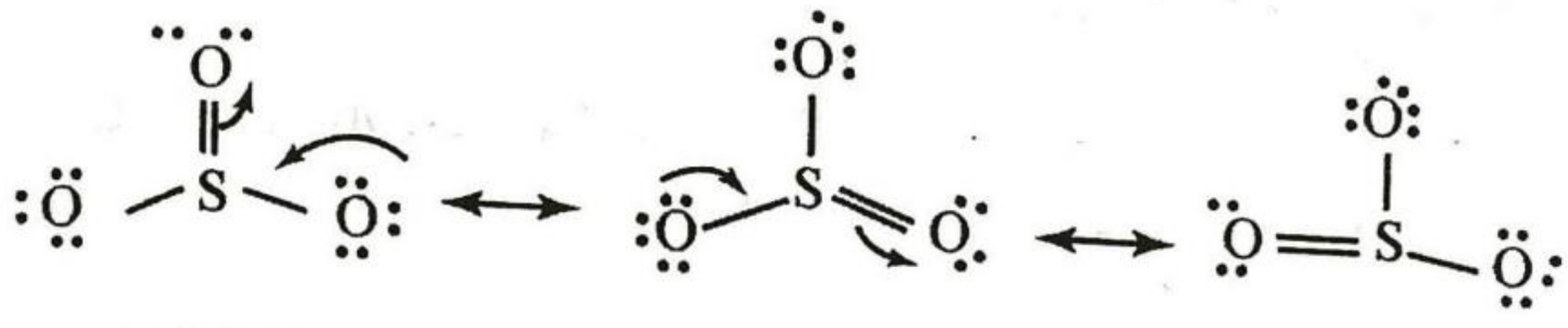
## Resonance Structure

**Resonance** is a condition occurring when more than one valid Lewis structures can be written for a particular molecule due to the presence of **multiple bonds**.

➦ The various structures are called **resonance structures** and the actual species is called **resonance hybrid.**

**Example**

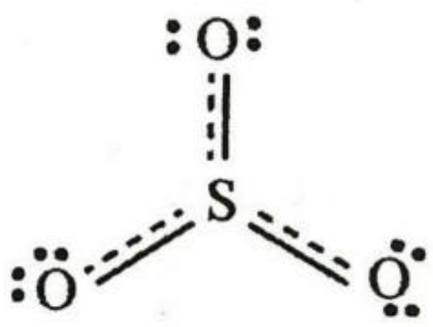
Consider $SO_3$



resonanace structure I

resonance structure II

resonance structure III



Resonance hybrid

**Covalent compounds that violate the octet rule.**

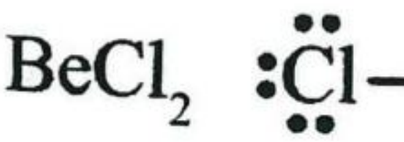
➦ All covalent compounds that have no eight electrons around the centeral atoms violate the octet rule. This can be discussed in three different ways.

**1. Covalent compounds below octet.**

➦ These covalent compounds have less than eight electrons around the centeral atom.

**Example**

$BeCl_2$, $BF_3$, $AlCl_3$, etc

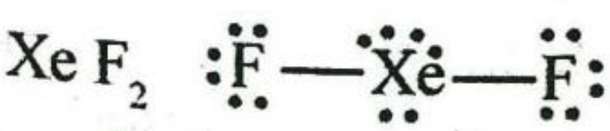
$BeCl_2$  2 electrons on Be

**2. Covalent compounds above octet.**

➦ Such that of covalent compounds have greater than eight electrons around the centeral atom.

**Example**

$SF_4$, $SF_6$, $PCl_4$, $Xe\ F_4$, $Xe\ F_2$, $Sb\ F_5$ etc

$Xe\ F_2$ 

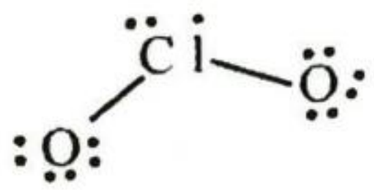
10 electrons on Xe

### 3. Covalent compounds with odd number of electrons.

➦ All covalent compounds with odd number of electrons violet the octet rule.

**Example**

NO, $NO_2$, $ClO_2$, $ClO_3$, etc.

$ClO_2$  there are 7 electrons around Cl.

## Illustrative Example

33. Which of the following compounds has no resonance structure

A. $HNO_3$ B. $CO_3^{2-}$ C. $SO_2$ D. $NH_3$

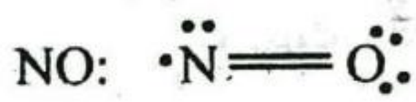
**Explanation:** $NH_3$ has no resonance structure because all the bonds are nitrogen-hydrogen single bonds.

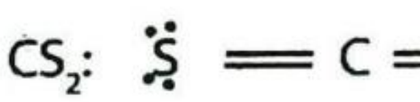
**Answer: D**

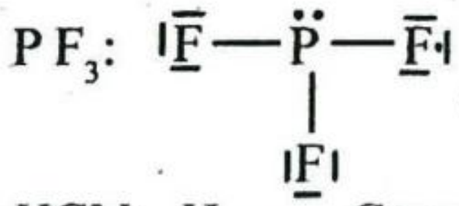
34. Which molecule has Lewis structure that does not obey the octet rule? (EUEE ... 2004 E.C)

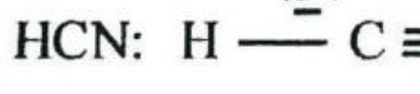
A. NO B. $CS_2$ C. $PF_3$ D. HCN

**Explanation:** Draw Lewis structure to know weather a given compound obey or not the octet rule

NO:  does not obey octet rule

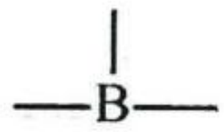
$CS_2$:  obey the octet rule.

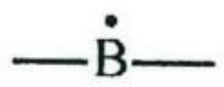
$PF_3$:  obeys the octet rule.

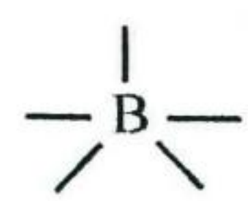
HCN:  obeys the octet rule.

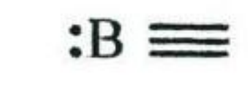
**Answer: A**

35. In which pattern does B obey the octet rule?

A. 

B. 

C. 

D. 

**Answer: D**

36. Draw the resonance structures of

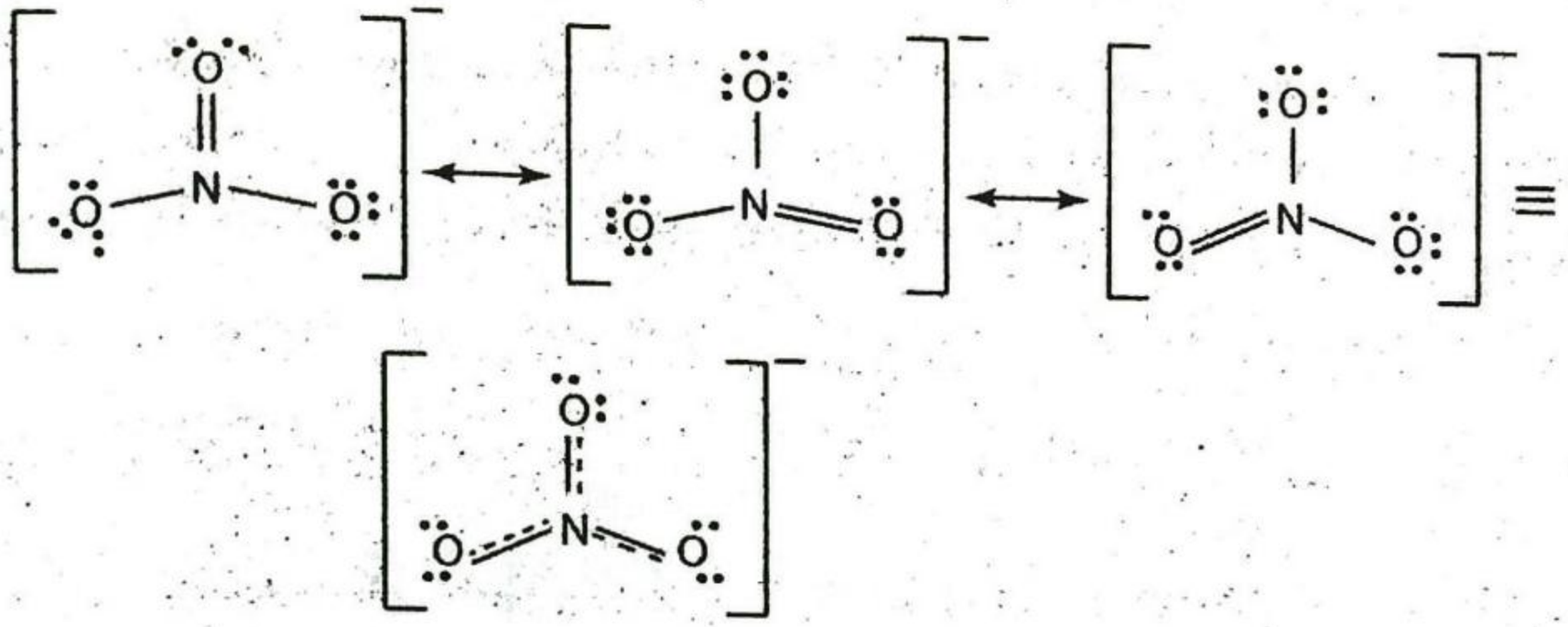
a. $NO_3^-$ c. $HNO_3$ e. $NO_2F$

b. $CO_3^{2-}$ d. N2 04 f. $NO_2^-$

g. $CH_2 = CH - CH = CH_2$ h. $C_6H_6$

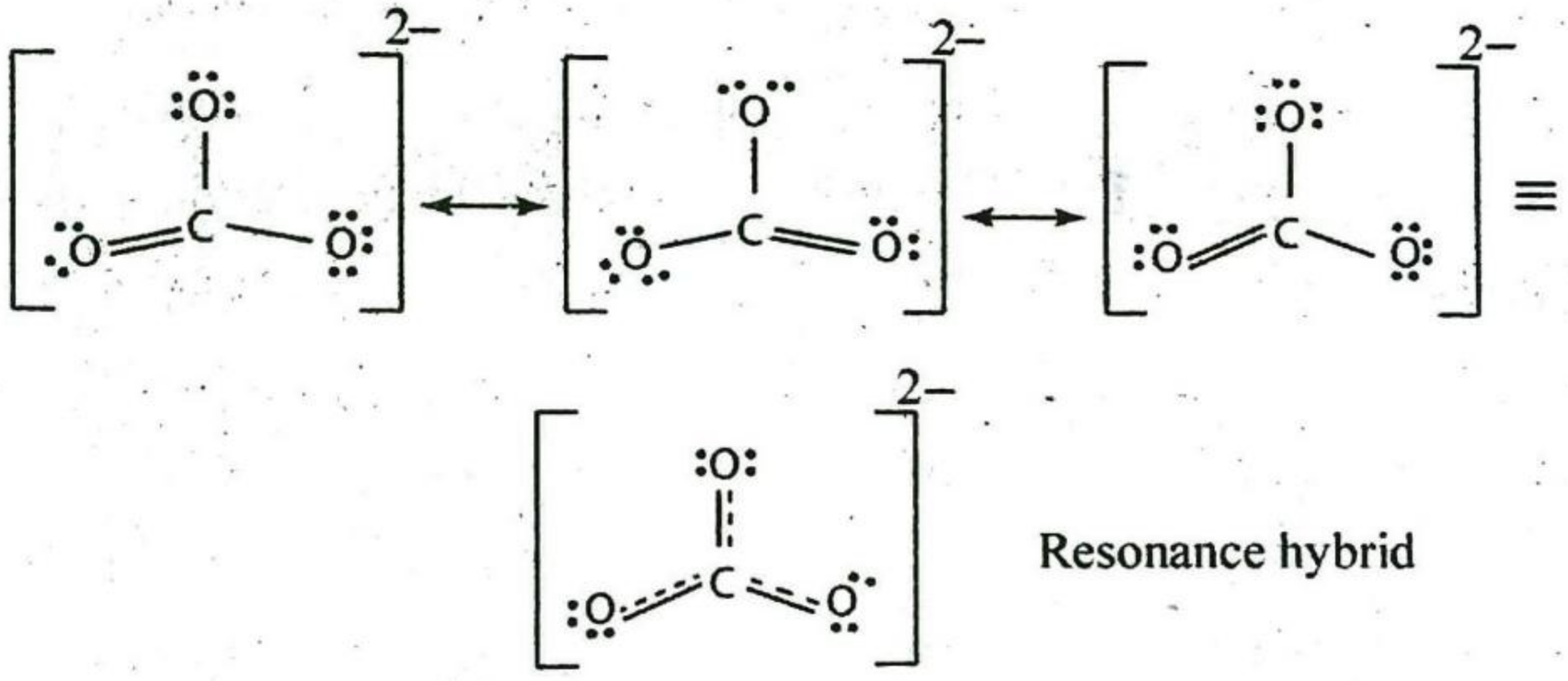
**Solution:**

a)



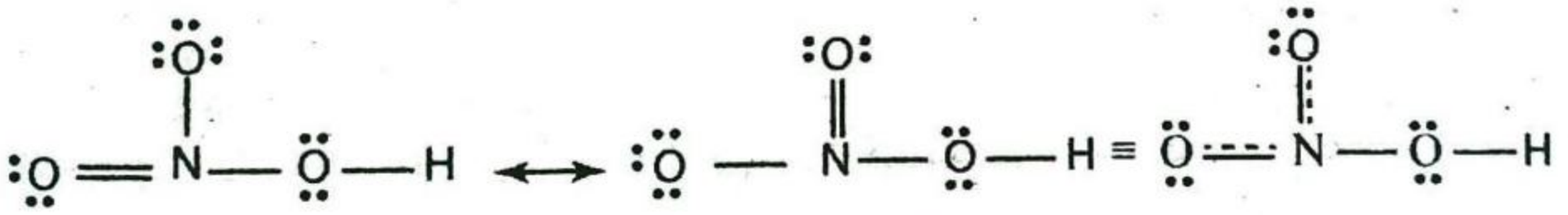
Resonance hybrid

b)

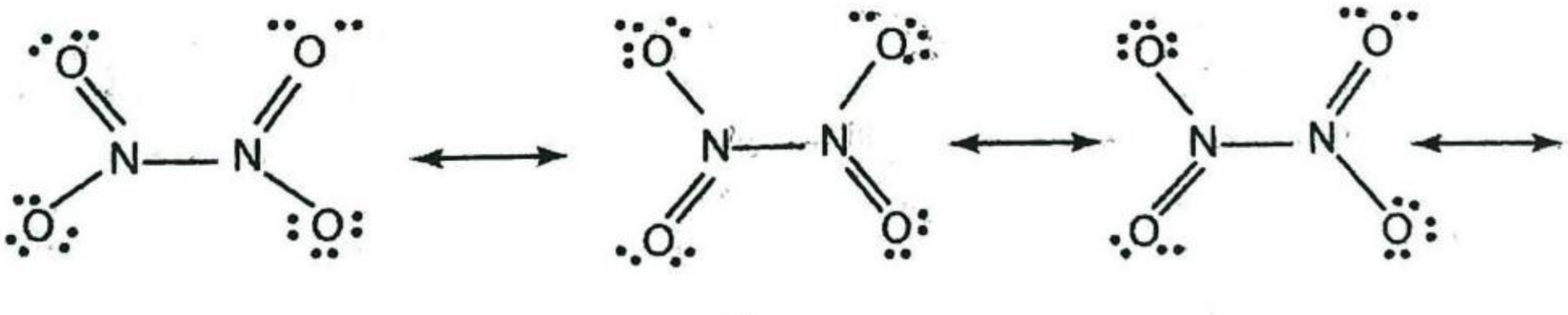


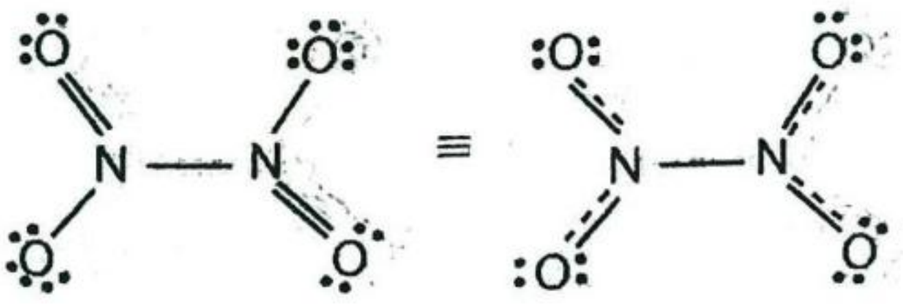
Resonance hybrid

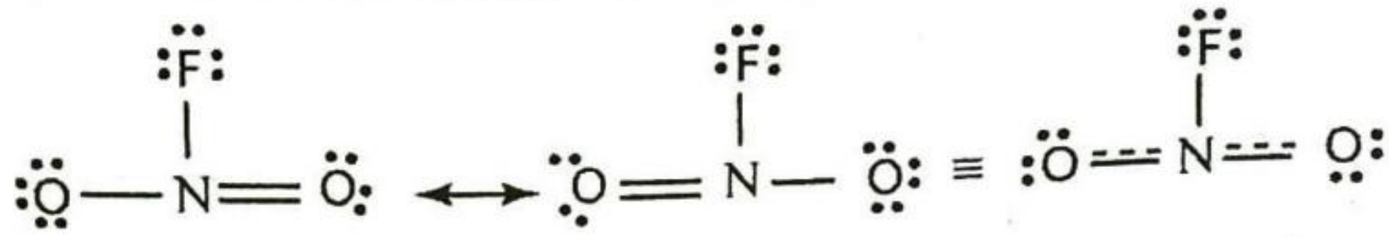
c)

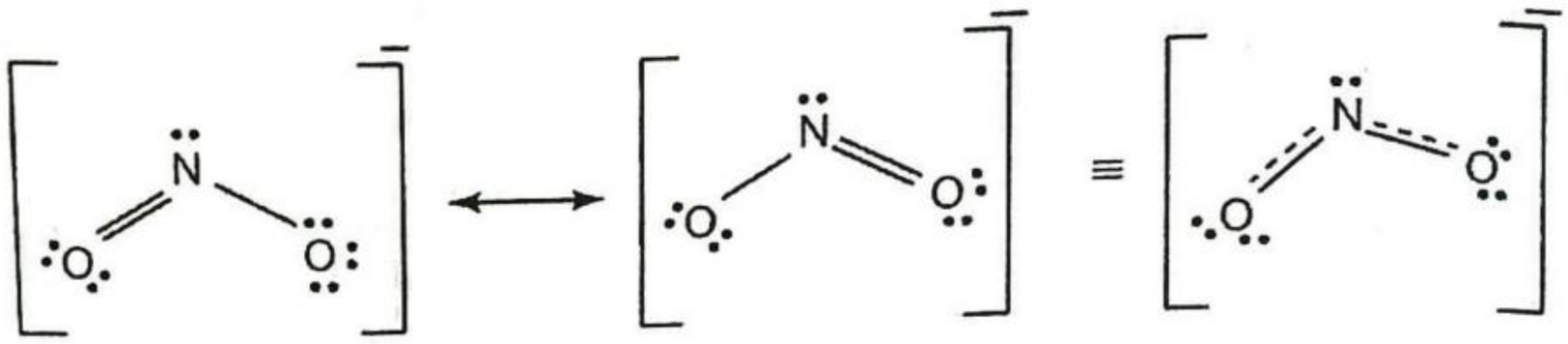


d)



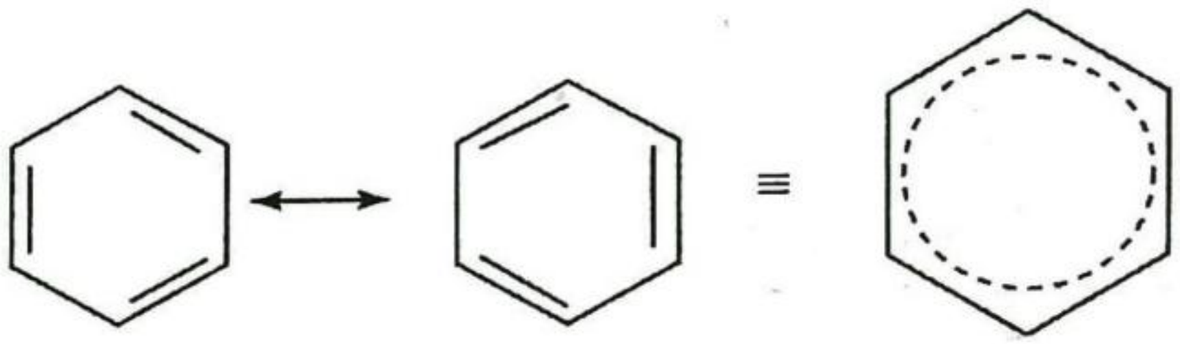


e) 

f) 

g) $\overset{\oplus}{C}H_2 - CH = CH - \overset{\ominus}{\ddot{C}}H_2 \longleftrightarrow \overset{\ominus}{\ddot{C}}H_2 - CH = CH - \overset{+}{C}H_2 \equiv$

$$CH_2 \overset{- - -}{=} CH = CH \overset{- - -}{=} CH_2$$

h) 

37. All of the following species violate the octet rule classify these species as below octet, above octet or with odd number of electrons.

a. $BH_3$ c. $IO_3$ e. $IC\bar{l}_4$ g. $IC\bar{l}_2$

b. $NO_2$ d. $XeF_2$ f. $IO_2$ h. $AlH_3$

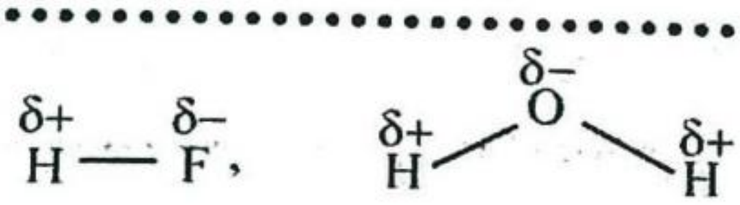
**Solution:**

- a and h are below octet
- b, c and f are odd number of electrons.
- d, e and g are above octet.

## Polar and Non-polar covalent bonds

- **Polar covalent bond** is a bond formed between atoms with different electro-negativities.
- It is a covalent bond formed when electrons are shared unequally. Theorem negative atom carries a partial negative charge ($\delta -$) and the other atoms carries a partial positive charge ($\delta +$)

### Example

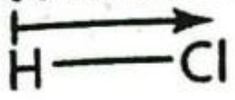


- Molecules formed by polar covalent bond are called **polar molecules.**

- The measure of the polarity of a molecule is known as **dipole-moment (μ).**

- Polar-molecules have dipole - moment greater than **zero**. This dipolar character of a molecule is often represented by an arrow. This arrow points toward the negative charge center.

**Example**




- **Dipole moment** (μ) is the product of the magnitude of separated charges and the distance between them.

$$\mu = q.d$$

- The SI unit of μ is coulomb meter (c.m)
- Other unit of μ is Debye (D)

$$1D = 3.33 \times 10^{-30} \text{ C.m}$$

As the electronegativity difference between the bonded atoms increases, polarity and magnitude of dipole-moment also increases

**Example**

Polarity of HF > HCl > HBr > HI

**Non - polar covalent bond** is formed between atoms with similar electronegativity.

- It is formed when electrons are shared equally.
- Molecules formed by non-polar covalent bond are called non-polar covalent molecules.
- These molecules have no dipole-moment.

**Example**

There are non-polar covalent molecules with polar bond. Example: $CCl_4$, $CO_2$, $CS_2$, etc

## Illustrative Examples

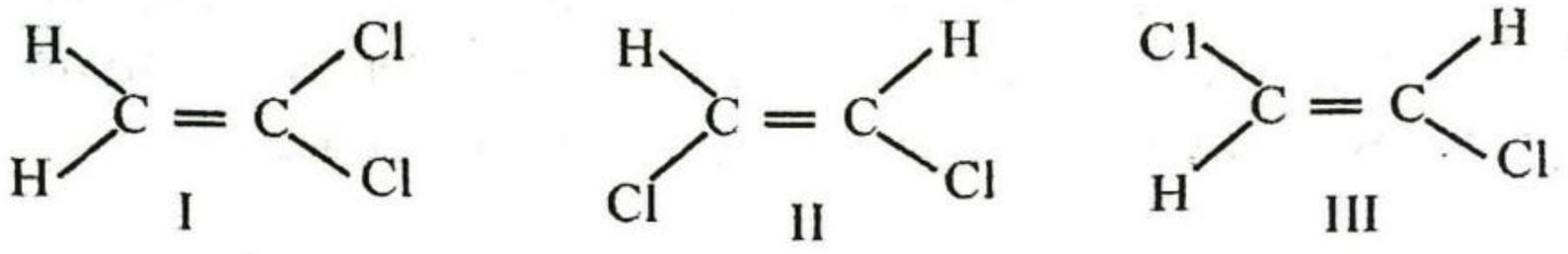
37. Which bond between the following groups of elements will be the most polar? (EUEE... 2003 E.C)

A. S–F B. S–Cl C. S–Br D. S–I

**Explanation**: As the electro-negativity difference increases between the bonded atoms bond polarity also increases.

**Answer: A**

38. Dichloro ethylene $C_2H_2Cl_2$, has the following Lewis strictures. Which one has a dipole-moment of zero? (EUEE)



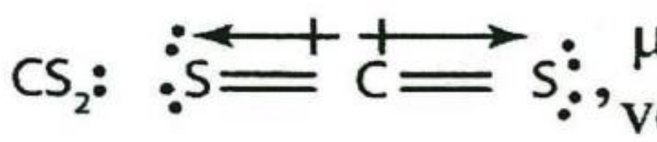
A. I B. II C. II and III D. III

**Answer: D**

39. Which of the following molecuels is non-polar?

A. $CS_2$ B. $SO_2$ C. $CHCl_3$ D. $SF_4$

**Explanation:** Draw the Lewis structure and predict their dipole-moment using vector method.



$\mu = 0$ because the sum of the two vectors if equal to zero. Thus, $CS_2$ is non - polar.

**Answer: A**

40. Arrange the following molecules in decreasing order of polarity based on the given dipole moment.

| Molecule | dipole moment (D) |
|---|---|
| HCl | 1.07 |
| $H_2O$ | 1.85 |
| $NF_3$ | 0.23 |
| CO | 0.12 |
| $NH_3$ | 1.47 |

**Explanation:** The higher the dipole moment, the more polar is the molecule.

$\rightarrow H_2O > NH_3 > HCl > NF_3 > CO$

41. Classify the following molecules as polar or non-polar.

a. $CH\,Cl_3$ d. $CCl_4$ g. $Te\,Cl_2$

b. $CH_2\,Cl_2$ e. $PH_3$ h. $HB\,F_2$

c. $CH_3\,Cl$ f. $B\,Cl_3$ i. $SO_2$

**Solution:**

➦ a, c, e, g, h and i are polar molecules

➦ b, d and f are non-polar molecules.

42. Which one has higher boiling point? Explain.

a. $I_2$ or ICl  b. $H_2O$ or $H_2S$  c. HF or HCl

**Explanation:**

a. ICl because it is polar

b. $H_2O$ because it is more polar than $H_2S$

c. HF because it is more polar than HCl

43. Write three non-polar molecules with polar bonds.

**Solution:**

➦ $CO_2$, $CCl_4$, $BCl_3$,

44. Is there a polar covalent molecule with non-polar bond?

**Explanation:** There is no polar molecule with non-polar bond.

## Properties of covalent compounds.

➦ Covalent compounds exist in the three states of matter at room temperature. For example

- $H_2$
- $O_2$
- $CH_4$
- $SO_2$

} are gases

- $H_2O$
- $C_5H_{12}$
- $C_2H_{50}OH$

} are liquids

- $I_2$
- $C_{20}H_{42}$

} are solids

➦ They have lower boiling and melting points

➦ They are volatile.

➦ Non - polar compounds are non - electrolytes.

➦ Some polar covalent compounds are electrolyte.

➦ Polar covalent compounds have higher boiling point than non-polar covalent compounds.

➦ Polar covalent compounds are soluble in polar solvent but non-polar covalent compounds are soluble in non-polar solvent.

# 3.5 Molecular Geometry

## 1. Electron pair geometry (Electron Pair arrangement)

➦ Electron set geometry is the set of electrons around the central atom.

➦ A set of electron means an electron domain.

> One set of electrons may be one single bond, one double bond, one triple bond, one lone pair or one electron

## Types of electron pair Arrangement

### 1. Linear electron pair arrangement

➦ Two sets of electrons around the centeral atom

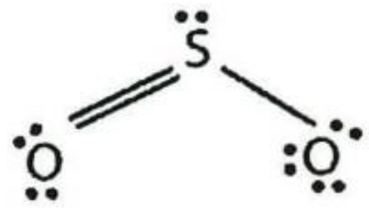
**Example**

$HCN$, $BeCl_2$, $CO_2$, $C_2H_2$

HCN: H – C ≡ N: , there are two sets of electrons around carbon (C)

### 2. Trigonal planar electron pair arrangement

➦ Three sets of electrons around the centeral atom.

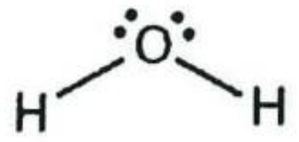
$SO_2$:  there are three sets of electrons around sulphur (S).

### 3. Tetrahedral electron pair arrangement

➦ Four sets of electrons around the centeral atom.

**Example**

$H_2O$, $NH_3$. $CH_4$, $Si\ Cl_4$ etc

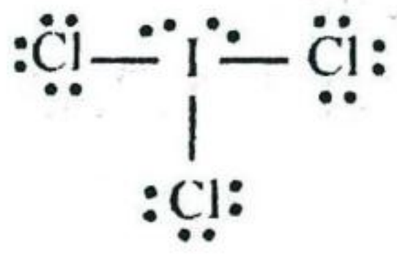
$H_2O$  there are four sets of electrons around the centeral atom oxygen (O)

### 4. Trigonal bi pyramidal electron pair arrangement

➦ Five sets of electrons around the centeral atom.

**Example**

$PF_5$, $ICl_3$, $SF_4$ etc

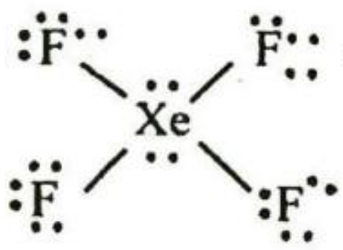
$ICl_3$  Five sets of electrons around iodine (I)

**5. Octahedral electron pair arrangement.**

➦ Six set of electrons around the centeral atom.

**Example**

$SF_6$, $XeF_4$, $IC1^-_4$ etc

$XeF_4$  There are six sets of electrons around xenone (Xe)

**Bond angle** is the angle between two atoms bonded to a central atom in a molecule.

## Illustrative Examples

45. Which of the following is not true about covalent compound?
    A. Polar covalent molecules have higher boiling point than non-polar molecules.
    B. Covalent compounds are molecules
    C. Dipole moment is the measure of a polar bond
    D. Covalent compounds have higher boiling point than ionic compounds

**Explanation:** Ionic compounds have higher boiling point than covalent compounds

**Answer: D**

46. All of the following when added to water will produce electric solution. EXCEPT (EUEE-2004 E.C)
    A. $N_2$ (g) B. NaI (s) C. HCl (g) D. KOH (s)

**Explanation:** $N_2$(g) is non-polar covalent molecule. It does not produce electrolytic solution due to lack of ions.

**Answer: A**

47. Why some polar covalent compounds are electrolytes?

**Explanation:** Polar covalent compounds such as $NH_3$, HF, HCl, HI, HBr dissolve in water and produce ions that carry charge.

48. Write some examples of non-polar covalent compounds that have high tendency to evaporate at room temperature.

**Explanation:** Non-polar covalent compounds have high tendency to evaporate are volatile covalent compounds.

**Example**

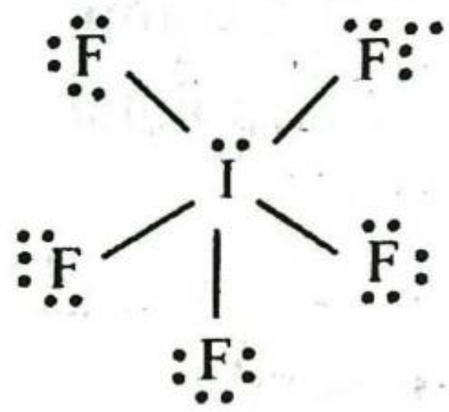
$C_6H_6$, $C_7H_8$, $C_5H_{12}$ etc.

## Illustrative Examples

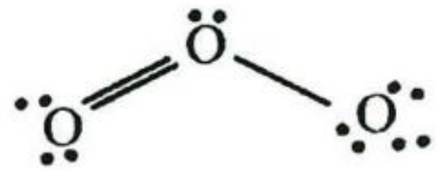
49. Determine the number of sets of electrons and identify the types of electron pair arrangements in the special below.

a. $IF_5$
b. $O_3$
c. $SO_3^{2-}$
d. $CS_2$
e. $SbF_5$
f. $TiCl_6^{3-}$

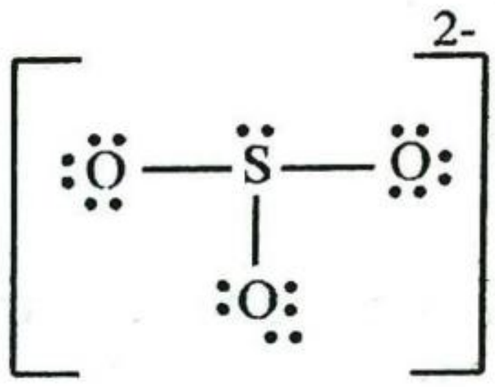
**Explanation:** Draw the Lewis structure for each species to determine the number of sets of electrons.

a. 

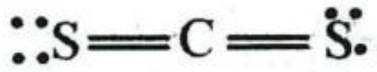
➦ six sets of electrons.
➦ octahedral electron arrangement

b. 

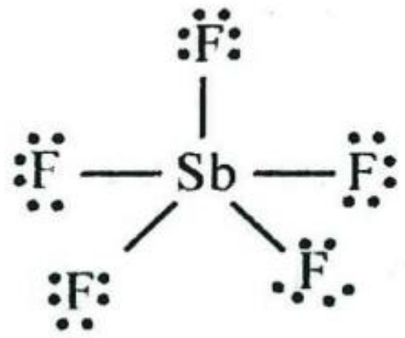
➦ three sets of electrons
➦ trigonal planar electron arrangement

c. 

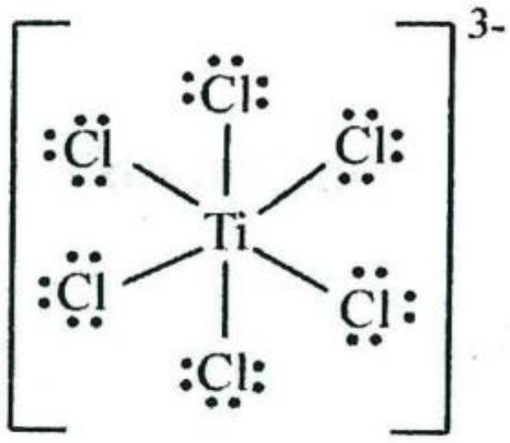
➦ four sets of electrons
➦ tetrahedral electron arrangement

d. 

➦ two sets of electrons
➦ linear electron arrangement.

e. 

➦ Five sets of electrons
➦ Trigonal bi-pyramidal electron arrangement.

f. 

➦ six sets of electrons
➦ octahedral electron arrangement.

# Molecular Shape

- Molecules and molecular ions have three- dimentional shapes
- Lewis can predict shapes of molecules using **Lewis structures and valence shell electron-pair repulsion (VSEPR) model.**
- The basis of the **VSEPR** model is that electron pairs in the valence shell of an atom **repel** one another.
- The repulsion force is in the order of:

**Lp – Lp > Lp – bp > bp – bp**

- Lp – Lp ⇒ lone pair – lone pair repulsion.
- Lp –bp ⇒ lone pair –bonding pair repulsion.
- bp –bp ⇒ bonding pair –bonding pair repulsion.

Molecular shape is the same as electron arrangement if there is no lone pair on the central atom.

## 1. Molecular shape with two sets of electrons

### $AX_2$ type

- Electron arrangement –linear
- A - central atom
- X - bonding set electron
- Molecular shape – **linear**
- Bond angle – 180°

**Example**

Be $Cl_2$, $CO_2$, $CS_2$, HCN, $C_2H_2$ etc

Be $Cl_2$ :Cl — Be — Cl: (180°)

## 2. Molecular shape with three sets of electrons

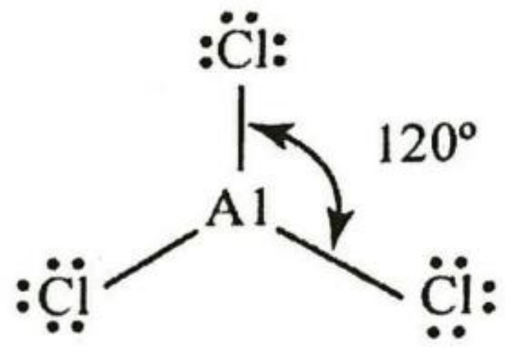
### i. $AX_3$ type

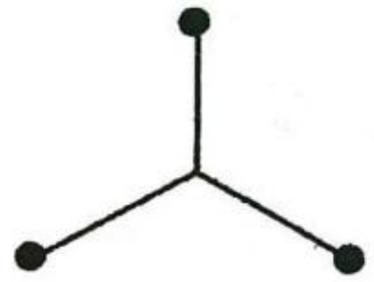
- Electron arrangement - trigonal planar
- Molecular shape = **trigonal planar**
- Bond angle =120°

**Example**

$BC1_3$, $A1C1_3$, $CO_3$, etc

$AlCl_3$



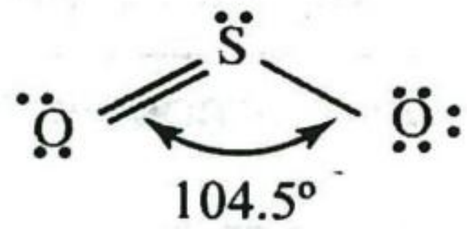
### ii. $AX_2E$ type

**X** – represents single, double or triple bond.

**E** – represents lone pair or single electron.

- Electron arrangement - trigonal planar
- Shape = bent, angular or V- shape.
- Bond angle = 104.5°

$SO_2$



## 3. Molecular shape with four sets of electrons

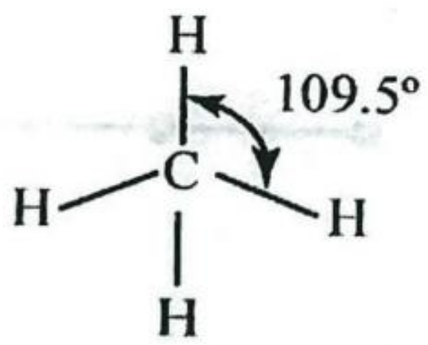
### i. $AX_4$ type

- Bonding pains = 4, lone pair = 0
- Electron arrangement - tetrahedral
- Molecular shape = tetrahedral
- Bond angle = 109.5°

**Example**

$CH_4$, $NH^+_4$, $SiC_{l4}$, $PH^+_4$ etc

$CH_4$



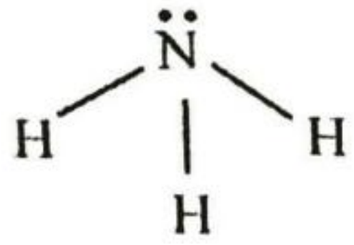
### ii. $AX_3E$ type

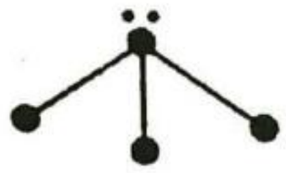
- Bonding pairs = 3, lone pairs = 1
- Electron arrangement - tetrahedral
- Molecular shape - trigonal pyramidal
- Bond angle = 107°

**Example**

$NH_3$, $PH_3$, $H_3O^+$, etc

$NH_3$





### iii. $AX_2E_2$ type

- Bonding pairs = 2, lone pairs = 2
- Electron arrangement - tetrahedral
- Molecular shape - Bent
- Bond angle - 104.5°

**Example**

$H_2O$, $NH_2^-$, $OF_2$, etc

## 4. Molecular shape with five sets of electrons.

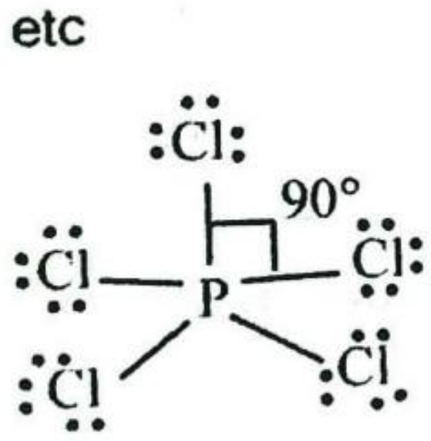
### i. $AX_5$ type

- Bonding pairs = 5, lone pairs = 0
- Electron arrangement – trigonal bipyramidal
- Shape - trigonal bipyramidal
- Bond angle = 90°, 120°

**Example**

$PCl_5$, $SiF_5^-$, etc

$PCl_5$



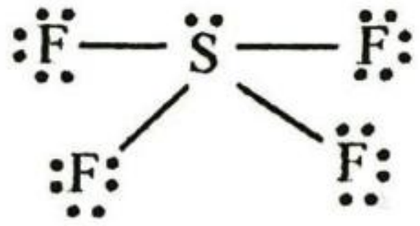
### ii. $AX_4E$ type

- Bonding Pairs = 4, lone pairs = 1
- Electron arrangement –trigonal bi pyramidal
- Molecular –shape = **see –saw or distorted tetrahedron**
- Bond angle = 90°

**Example**

$SF_4$, $IC1^+_4$, $IF_4^+$, $Xe\ O_2\ F_2$

$SF_4$

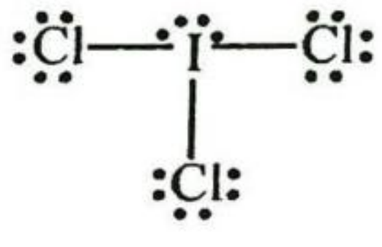


iii. **$AX_3E_2$ type**

- Bonding pairs = 3, lone pairs = 2
- Electron pairs arrangement - trigonal bi pyramidal
- Molecular shape = **T –shaped**
- Bond angle = 90°

**Example**

$ICl_3$, $BrF_3$, $ClF_3$ etc

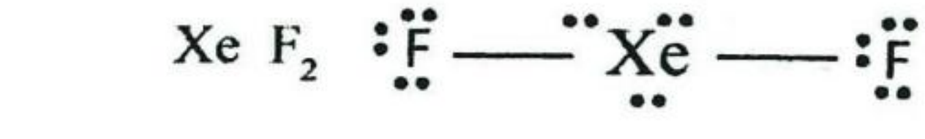


iv. **$AX_2E_3$ type**

- Bonding pairs = 2, lone pairs = 3
- Electron arrangement trigonal bi pyramidal
- Molecular shape linear
- Bond angle - 180°

**Example**

$XeF_2$, $IC1^-_2$, $BeF^-_2$ etc

$Xe\ F_2$ 

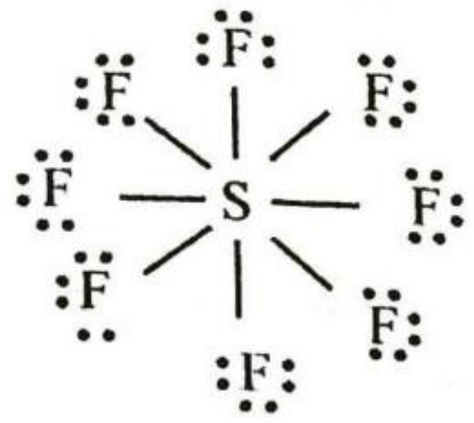
**5. Molecular shape with six sets of electrons.**

i. **$AX_6$ type**

- Bonding pairs = 6, lone pair = 0
- Molecular shape – octahedral
- Electron pair arrangement - octahedral
- Bond angle = 90°

Example

$SF_6$



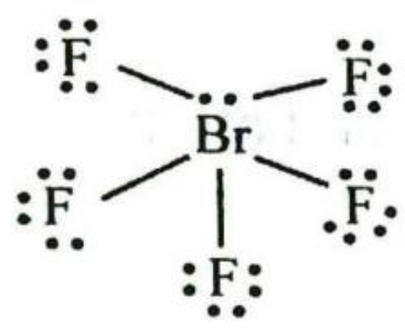
### ii. $AX_5$ E type

- Bonding pairs = 5, lone pair = 1
- Molecular shape - **square pyramidal**
- Electron arrangement - octahedral

Example

$BrF_5$, $ICl_5$, $TeF_5^-$, $SeCl_5^-$ etc

Br $F_5$



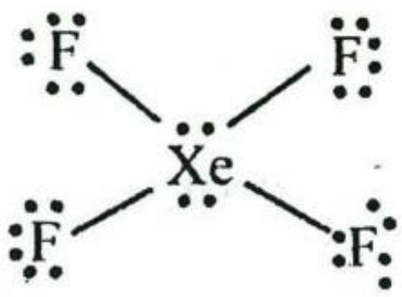
### iii. $AX_4$ $E_2$ type

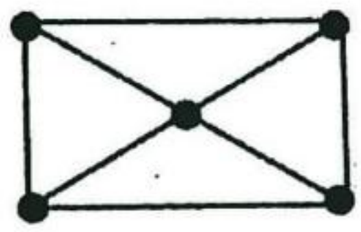
- Bonding pairs = 4, lone pairs = 2
- Electron arrangement - octahedral
- Molecular shape - **square Planar**

Example

Xe $F_4$, $ICl_4^-$ Br $F_4^-$ etc

Xe $F_4$





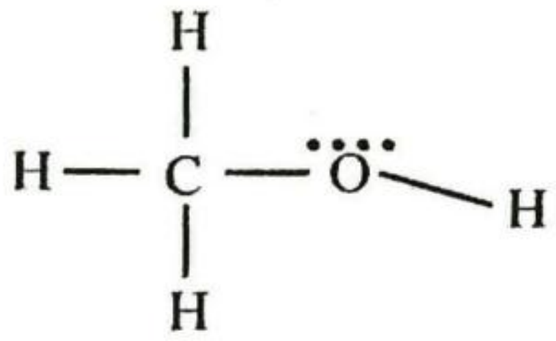
## Molecular shape of molecules with more than one central atoms.

- The shapes of molecules with more than one centeral atoms is predicted with respect to each centeral atom.

For example, for methanol ($CH_3OH$), there are two centeral atoms C and O.

- Draw the Lewis structure and predict the shape with respect to carbon and oxygen.

$CH_3OH$



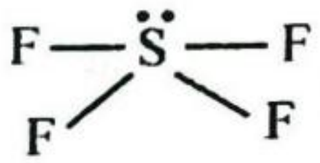
The shape with respect to carbon is tetrahedral and with respect to oxygen is angular.

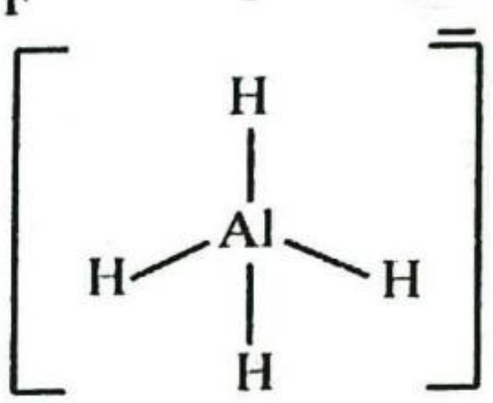
## Illustrative Examples

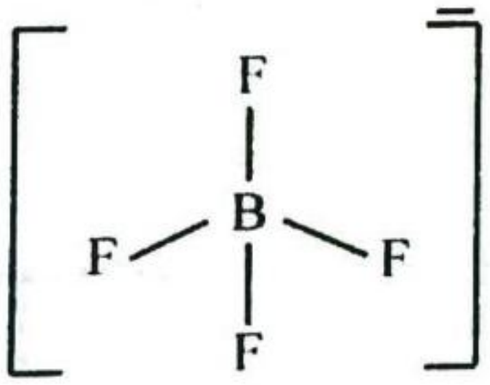
51. Which of the following molecules does NOT have a tetrahedral atom? (EUEE ... 2008 E.C)

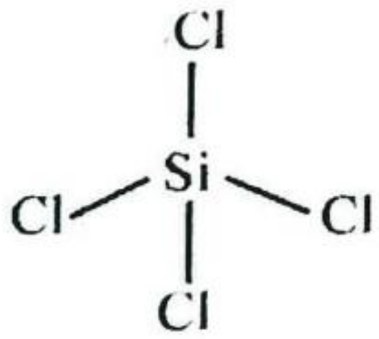
A. $SF_4$ B. $AH_4^-$ C. $BF^-_4$ D. $Si\ Cl_4$

**Explanation:** Draw Lewis structure and observe the sets of electrons around the centeral atom.

$SF_4$  trigonal bi pyramidal centeral atom

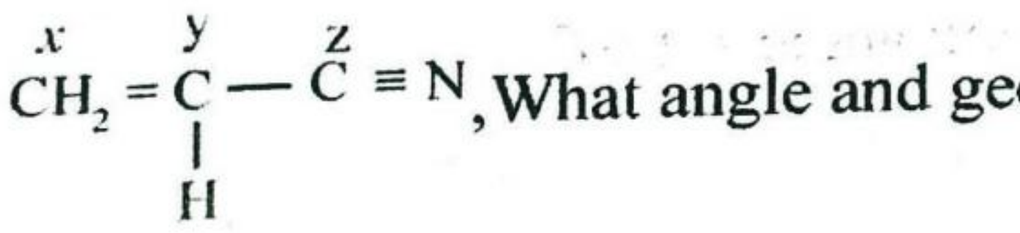
$Al\ \bar{H}_4$  tetrahedral centeral atom.

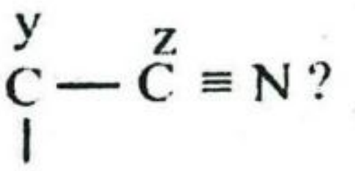
$BF^-_4$  tetrahedral centeral atom

$Bi\ Cl_4$  Tetrahedral centeral atom.

**Answer: A**

52. Acrylonitrile has the following Lewis structure with designation of x, y and Z for each carbon atom:



What angle and geometry of  ?

A. 109°, tetrahedral
C. 180°, linear
B. 120°, trigonal pyramidal
D. 90°, T- shaped.

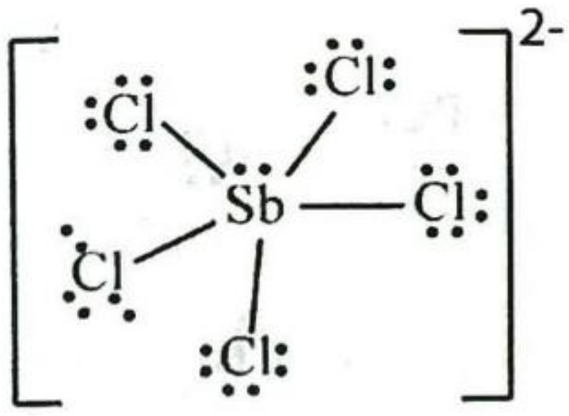
**Answer: C**

53. Antimony (Sb) is a group V element. What will be the molecular geometry and number of lone pair electrons respectively that exist in the ion ($SbCl_5^{2-}$)? (EUEE...2008 E.C.)

A. See saw, 1
C. Square planar, 2
B. See saw, 2
D. Linear, 3

**Explanation:** Draw Lewis structure and predict the shape.

$[Sb\ Cl_5]^{2-}$



Shape = square pyramidal

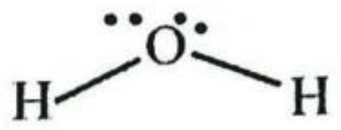
Lone pair electrons one central atom = 2

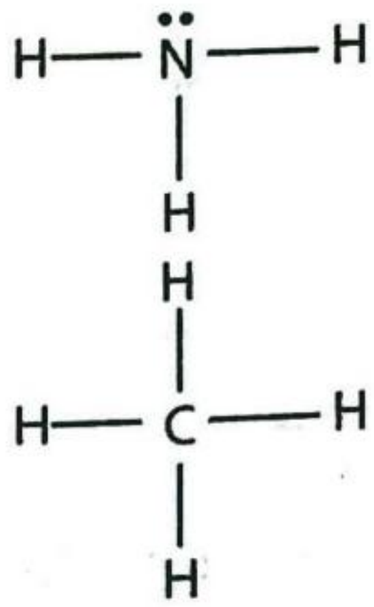
**Answer:**

54. Which of the following species has the smallest H — X — H bond angle where X is centeral atom?

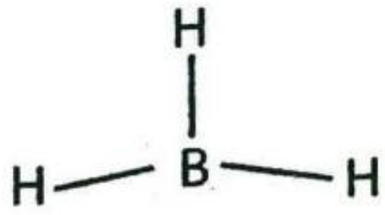
A. $OH_2$
B. $NH_3$
C. $CH_4$
D. $BH_3$

**Explanation:**

$OH_2$  Bond angle = 104.5°

$NH_3$  Bond angle = 107°

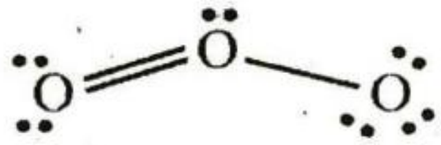
$CH_4$  Bond angle = 109.5°

$BH_3$  Bond angle = 120°

**Answer: D**

55. What is the molecular geometry of $O_3$? (EUEE)

A. Trigonal planar
B. Bent
C. Trigonal pyramidal
D. Tetrahedral

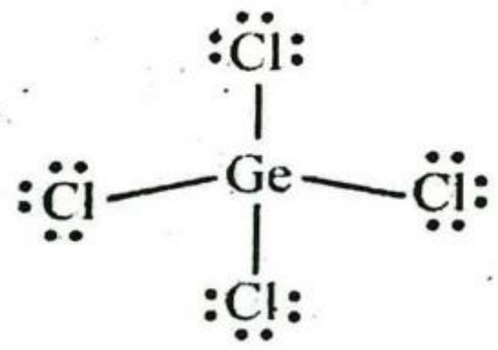
**Explanation:** $O_3$  bent

**Answer: B**

56. Which of the following is NOT probable shape of a molecule? (EUEE)

A. A tetrahedral, Ge $Cl_4$
B. A bent, HCN
C. A trigonal bi pyramidal, $PCl_5$
D. A T-shape, Br $F_3$

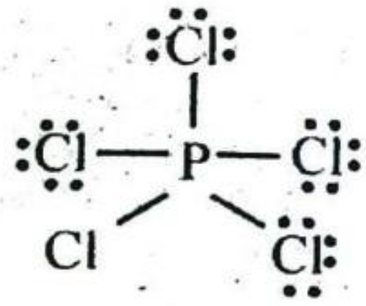
**Explanation:**

Ge $Cl_4$: 

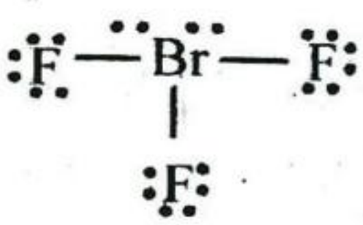
tetrahedral

HCN: H— C ≡ N:

linear

$PCl_5$: 

trigonal bipyramidal

Br $F_3$ 

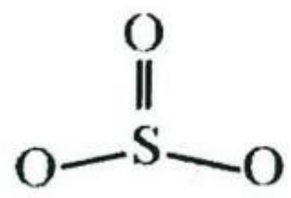
T- shape

**Answer: B**

57. Which reaction involves a change in the electron pair geometry for the underlined atom? (EUEE)

A. $\underline{B}F_3 + \bar{F} \longrightarrow \underline{B}F_4^-$
B. $\underline{N}H_3 + H^+ \longrightarrow \underline{N}H_4^+$
C. $2\underline{S}O_2 + O_2 \longrightarrow 2\underline{S}O_3$
D. $H_2\underline{O} + H^+ \longrightarrow H_3\underline{O}^+$

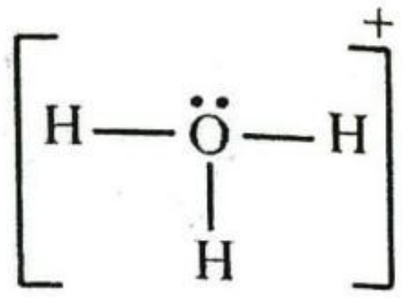
**Explanation:** Electron pair geometry means electrons arrangement around the centeral atom.

$SO_3$ 

trigonal planar

$H_2O$

tetrahedral

$H_3O^+$ 

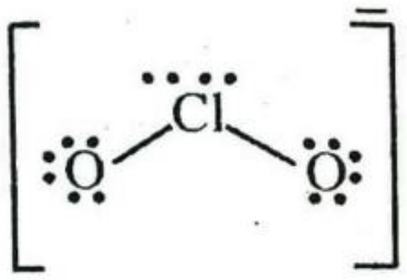
tetrahedral

**Answer: A**

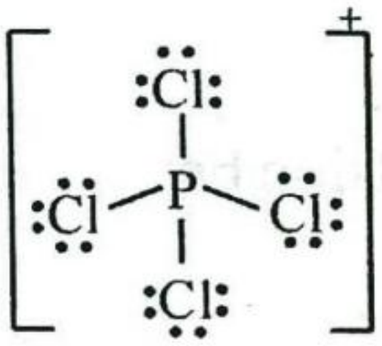
58. Use the VSEPR model to predict the electron pair arrangement and molecular shape of

a. $Cl\overline{O}_2$
b. $PCl^+_4$
c. $ICl^+_4$
d. $CH^-_3$
e. $AsH_3$
f. $SbCl^-_6$
g. $C_2H_2$
h. $BeF^-_4$

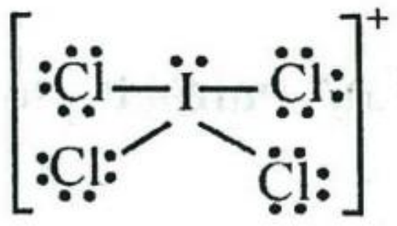
**Explanation:**

a.  shape - bent

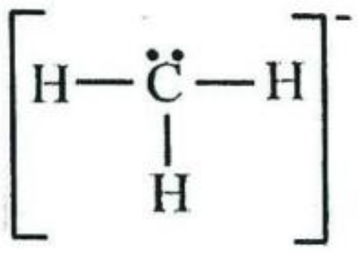
electron pair arrangement = tetrahedral

b.  shape = tetrahedral

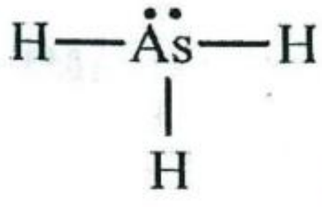
electron pair arrangement - tetrahedral

c.  shape = see saw

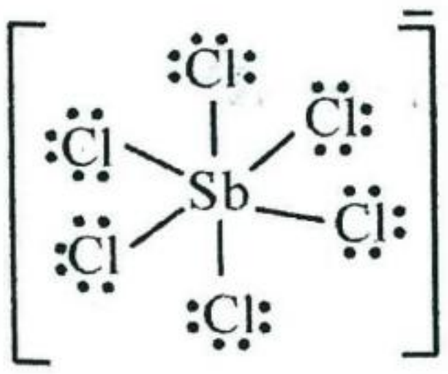
electron pair arrangement = trigonal bipyramidal

d.  shape = trigonal pyramidal

electron pair arrangement - tetrahedral

e.  shape = trigonal pyramid

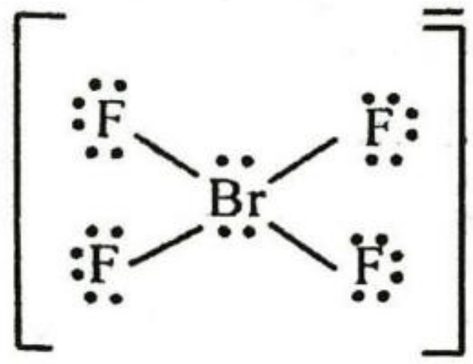
electron pair geometry = tetrahedral

f.  shape = octahedral

electron pair arrangement = octahedral

g. H — C ≡ C — H shape = linear

electron pair arrangement = linear

h. 

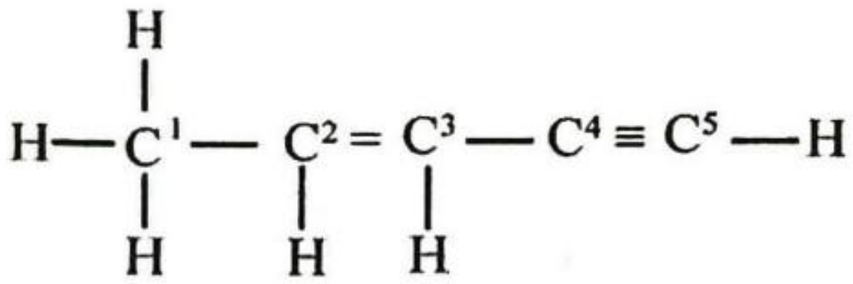
shape = square planar

electron pair arrangement = octahedral

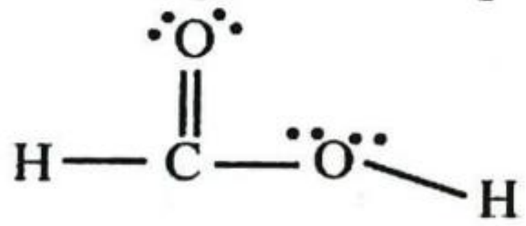
59. Predict the shape of the following with respect to each centeral atom.

a. $CH_3 — CH = CH — C \equiv CH$ b. H COOH

**Explanation:** Draw the Lewis structure and predict the shape

a. 

- Shape with respect to C – 1 = tetrahedral
- Shape with respect to C– 2 and C – 3 = trigonal planar
- Shape with respect to C – 4 and C – 5 = linear

b. 

- Shape with respect to C = trigonal planar
- Shape with respect to O = angular

60. What are the three aspects that minimize the repulsion between valence shall electron pairs?

**Explanation:**

- Lone pair - Lone pair
- Lone pair - bonding pair
- Bonding pair - bonding pair

61. What is molecular geometry?

**Solution:**

Molecular geometry is the relative positions of the atomic nuclei.

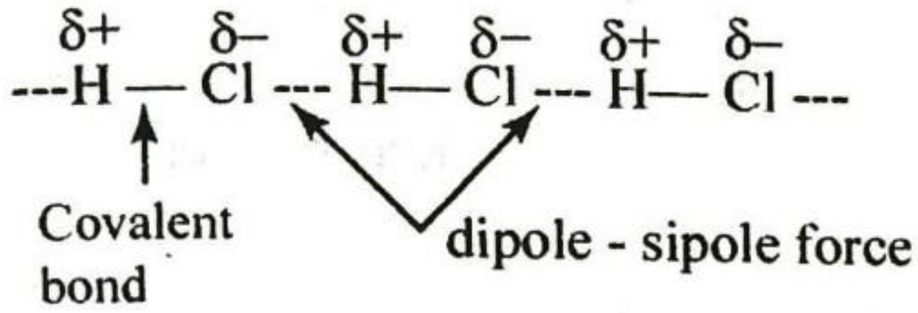
## 3.6 Intermolecular forces in covalent compound

- Attractive forces between molecules are called **intermolecular forces.**
- There are three types of intermolecular forces in covalent compounds.
  1. Dipole - dipole force
  2. Hydrogen bond
  3. London dispersion force

1. **Dipole-dipole forces** are -forces that exist between polar molecules, that is between molecules that possess dipole moments.
   - ➦ The higher the dipole moments, the stronger is the dipole- dipole force strength.

**Example**

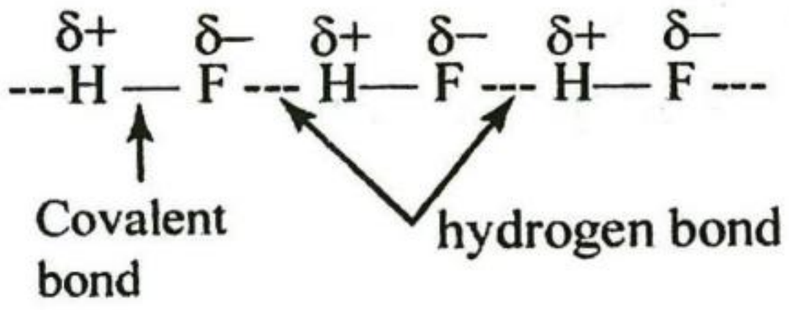
HCl



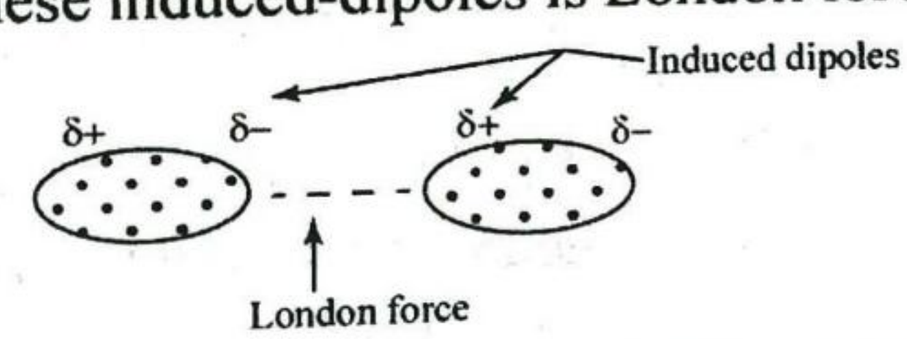

2. **Hydrogen bond**
   - ➦ It is a particularly strong type of dipole-dipole force.
   - ➦ It is formed between molecules that have hydrogen- fluorine, hydrogen-oxygen and hydrogen nitrogen bonds.

**Example**

HF, $H_2O$, $NH_2$ $CH_2$ COOH, $CH_3$ OH, $CH_3COOH$ etc are hydrogen bond.




3. **London dispersion forces** are forces that exist between non-polar molecules.
   - ➦ London force is the weakest intermolecular force.
   - ➦ Non-polar molecules form induced dipoles the attraction force between these induced-dipoles is London force.




The strength of London force is mainly affected by size. As size of non-polar molecules increases strength of London forces also increase

**Example**

$I_2 > Br_2 > Cl_2 > F_2$

$C_5H_{12} > C_4H_{10} > C_3H_8 > C_2H_6$ Strength of London force

**Ion-dipole forces** are intermolecular forces which occur between ion (cation or anion) and a polar molecule. These forces are responsible for the solubility of ionic compounds in water.

Table 3.1. The difference between intermolecular and intermolecular forces

| Intra molecular forces | Intermolecular forces |
|---|---|
| ➦ hold atoms together in a molecule<br>➦ stabilize individual molecule<br>➦ affect chemical property<br>➦ stronger than intermolecular forces | ➦ hold molecules together<br>➦ responsible for the properties of the bulk matter<br>➦ affect physical property<br>➦ much weaker than intramolecular forces. |

## Illustrative Examples

62. Which of the following explains, why at room temperature, $I_2$ is a solid, $Br_2$ is liquid and $C1_2$ is a gas? (EUEE.... 2004 E.C)

A. Ionic bonding
C. Hydrogen bonding
B. Hybridization
D. London dis perssion forces

**Explanation:** As size increases, strength of London dispersion force also increases. That is why, $Cl_2$ is gas, $Br_2$ is liquid and $I_2$ is solid at room temperature.

**Answer: D**

63. There is a strong covalent bond between the N-atoms in nirtogen gas, $N_2$. Why, then does nitrogen have such a low boiling point of –196°C?

A. The bond between the nitrogen -atoms is triple.
B. N is very electronegative, only next to F and O.
C. The strong bond an intermolecular one determines the boiling point of the substance
D. Boiling point is determined by intermolecular force, which in this case is weak as the molecule is non-polar

**Explanation:** $N_2$ is non-polar molecule tied together with the weekest intermolecular force.

**Answer: D**

64. Of $Br_2$ Ne, HCl and $N_2$, which is likely to have the largest London dispersion forces?

A. $Br_2$
B. Ne
C. HCl
D. $N_2$

**Explanation:** London disperssion forces occur in non-polar molecules

**Answer: A**

65. Arrange the following in decreasing order of boiling point.

a. $CF_4$, $CH_4$, $CCl_4$, $ICl_4$

b. $H_2O$, HF, $NH_3$, $H_2S$

c. $CH_4$, $C_2H_6$, $C_5H_{12}$, $C_{10}H_{22}$

**Solution:**

a. $ICl_4 > CCl_4 > CF_4 > CH_4$

b. $HF > H_2O > NH_3 > H_2S$

c. $C_{10}H_{22} > C_5H_{12} > C_2H_6 > CH_4$

66. Which molecule in each pair has stronger London dispersion forces? Explain.

a. Ar or Kr

b. $C_{15}H_{30}$ or $C_{50}H_{100}$

**Explanation:**

a. Kr because it has larger size.

b. $C_{50}H_{100}$ because it has larger size.

67. $N_2$ and CO are is electronic which substance would you expect to have higher boiling point?

**Explanation:** CO has higher boiling point than $N_2$ because it is polar?

68. Identify the types of intermolecular forces that are dominant in each substance below.

a. IF

b. $CHCl_3$

c. $CH_3NH_2$

d. HCOOH

e. $PCl_5$

f. $AgNO_3$

**Solution:**

a. dipole - dipole

b. dipole - dipole force

c. hydrogen bonding

d. hydrogen bonding

e. London dispersion force

f. Ion-dipole force

69. Why ice floats over liquid water?

**Explanation:** The hydrogen bond in ice hold the water molecules in a rigid but open structure and more volume is occupied. This means the molecules of water are far apart in ice than liquid water. That is why solid water (ice) is less dense than liquid water at O°.

70. Arrange the following in order of increasing boiling point. RbF, $CO_2$, $CH_3OH$, $CH_3Br$

**Solution:**

$$CO_2 < CH_3Br < CH_3OH < RbF$$

71. Arrange the following in decreasing order of strength.
London dis perssion force, hydrogen bond,
dipole - dipole force, covalent bond.

**Solution:**

➦ Covalent > hydrogen bond > dipole - dipole force > London disperssion force.

## 3.7 Metallic bonding

➦ Metallic bond is bonding in solid metals.
➦ It is attraction force between free electrons of metals and their nuclie.
➦ The electron-sea model is a very simple model which pictures the metals as an array of metal cations in a "sea" of electrons.

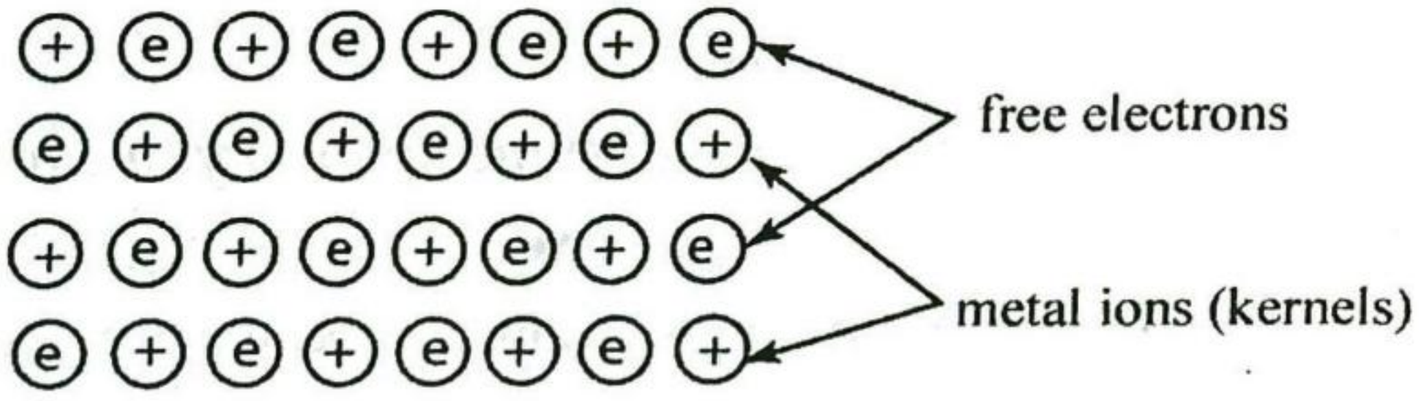



Figure 3.2 The electron - sea model

**Factors affecting metallic bond strength.**

There are two factors that affect the strength of metallic bond strength. These are

1. Size
2. Number of valence electrons

**1. Size**

➦ As size of the metal increases, strength of metallic bond decreases

**Example**

Li < Na < K < Rb, size

Li > Na > K > Rb, metallic bond strength.

**2. Number of valence electrons.**

➦ As the number of valence electrons increases strength of metallic bond strength also in creases.

Example

Al>Mg>Na metallic bond strength

Al, 2, 8, 3 Mg, 2, 8, 2 Na, 2,8, 1

- **Metallic bond is responsible for the following physical properties metals.**
  - Metals are malleable (hammered into sheet)
  - Metals are ductile (drawn into wire)
  - Metals are conductors
  - Metals have high melting and boiling point.
  - They are strong and durable

## Illustrative Examples

72. Which of the following transmit electricity in solid state? (EUEE)

A. Water
B. Copper
C. Sodium chloride
D. Sulphur

**Solution:**

Metals transmit electricity in the three states of matter.

**Answer: B**

73. Which of the following has the highest melting point?
(EUEE ... 2004 E.C)

A. Iodine
B. Tungsten
C. Mercury
D. Bromine

**Solution:**

Metals in general have higher melting and boiling points than non-metals. The melting point of metals ranges from – 39°C (Hg) to 3410°c (W)

**Answer: B**

74. Which of the following is Not true of metallic bonding?

A. It gives rise to excellent electrical conductivity.
B. Electrons are free to move throughout the structure.
C. The strength of metallic bond increase down a group.
D. The strength metallic bond affects the boiling point of metals.

**Solution:**

➥ Down a group metallic bond strength decreases because of increase in size.

**Answer: C**

75. How is metallic bonding different from ionic and covalent bonding?

**Explanation:** In metallic bonding no transfer of electrons like ionic bonding. Metallic bonding does not involve sharing of electrons like covalent bonding.

76. Which one has higher melting point? Explain

a. Li or Be b. Na or K

**Explanation:**

a. Be because it has smaller size and one more valence electron than Li.

b. Na because it has smaller size than K

77. Which one has greater metallic character?

a. Li or Be c. K or Rb

b. Mg or Ca d. Mg or Al

**Explanation:** Down a group metallic character increases but across a period decreases.

a. Li b. Ca c. Rb d. Mg

## 3.8 Chemical Bonding Theories

➥ There are two commonly used theories of chemical bonding based on the concept of quantum mechanics.

These are:

1. The valence Bond Theory (VBT)
2. The molecular orbital theory (MOT)

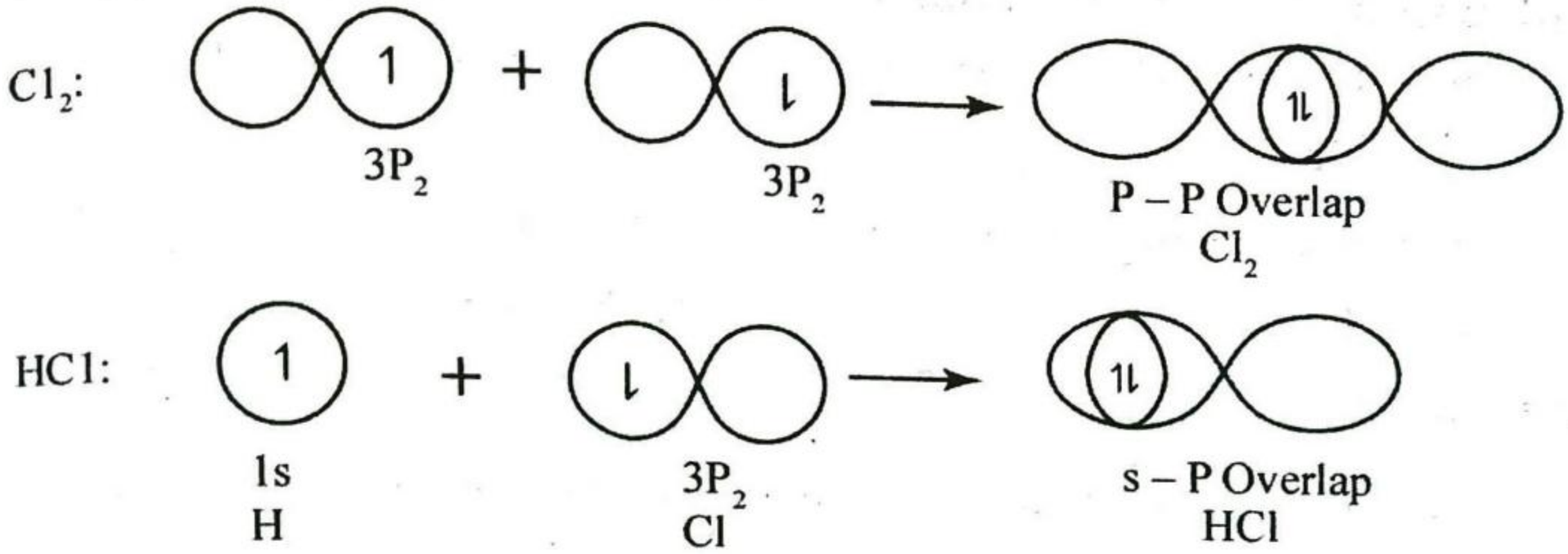
### 1. The Valence Bond Theory (VBT)

➥ According to VBT a covalent bond is formed when half - filled orbitals overlap in head to head or side to side ways.

When half-filled orbitals overlap in end to end, head to head or linear overlap **sigma bond (δ - bond)** is formed.

For example, consider the formation of $H_2$, $Cl_2$ and HCl

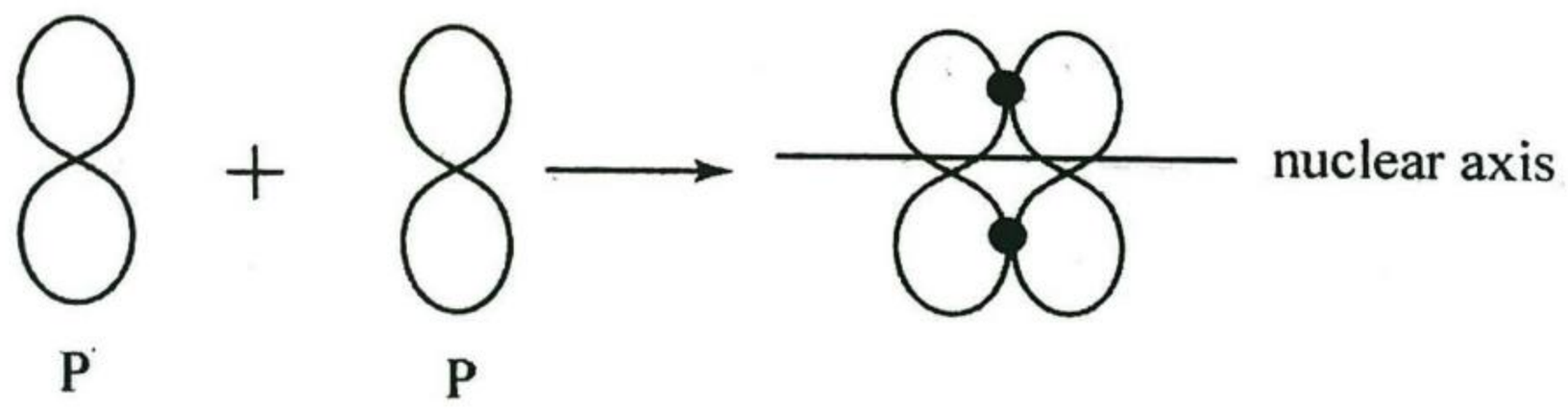
➦ During sigma bond formation the electron density lie along the nuclear or bond axis.




➦ When half-filled atomic orbitals overlap in side to side or parallel way, **Pi (π) bond is formed**.

Half - filled s - orbitals do not form Pi bond due to their spherical shape.

➦ During Pi bond formation, the electron density lies above and below the nuclear axis.




➦ According to VBT, the higher the overlap, the stronger is the bond.

➦ A sigma bond is stronger than a Pi bond.

➦ All single bonds are sigma bonds.

➦ A double bond contains one sigma and one Pi bonds

➦ A triple bond contains one sigma and two Pi bonds.

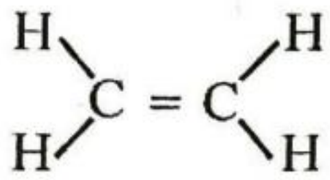
## Illustrative Examples

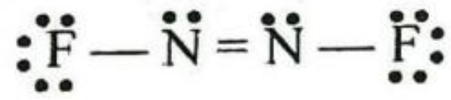
78. Which molecule listed below has two sigma (δ) and two Pi bonds? (EUEE)

A. $N_2$ B. $C_2H_4$ C. $N_2F_2$ D. HCN

**Explanation:** Draw the Lewis structure to determine the number of sigma and Pi bonds.

$N_2$: $:N \equiv N:$ 1 triple bond = $1\delta + 2\pi$

$C_2H_4$:  4 single bonds = 4δ bonds, 1 double bond = $1\delta + 1\pi$ Thus, $C_2H_4$ has 5δ and 1π bonds.

$N_2F_2$  two single bonds = 2δ, one double bond = $1\delta + 1\pi$ thus $N_2F_2$ has 3δ and 1π bonds.

HCN: $H—C \equiv N:$ 1 single bond = 1δ, 1 triple bond = $1\delta + 2\pi$ thus HCN has 2δ and 2π bonds.

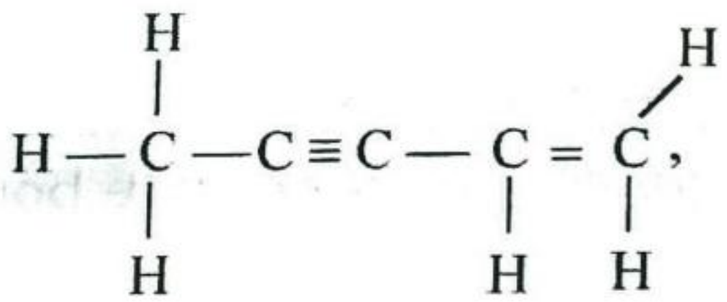
**Answer: D**

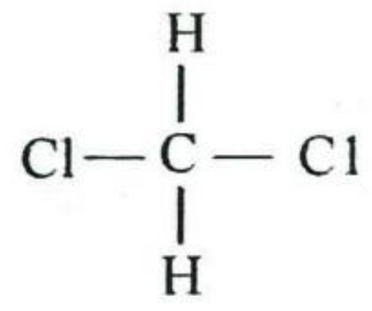
79. Determine the number of sigma and Pi bonds in

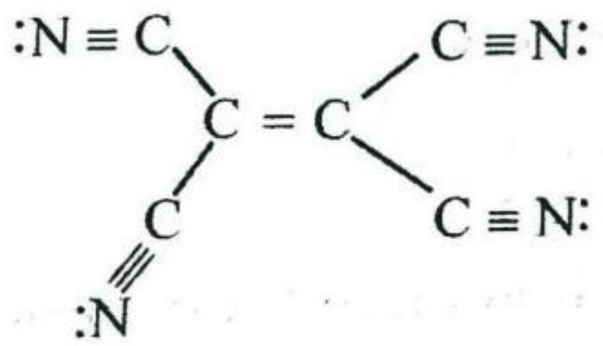
a $CH_3C \equiv C\,CH = CH_2$ b. $CH_2Cl_2$ c. $C_6N_4$

**Solution:**

Draw detail structure formula to observe the single, double and triple bonds.

a.  $3\pi$ bonds and $10\delta$ bonds.

b.  $4\delta$ bonds but no Pi bond.

c.  $9\pi$ bonds and $9\delta$ bonds.

80. What is bond axis or nuclear axis?

**Solution:**

Bond axis is an imaginary line that joins the nuclie of the bonding atoms.

81. Bonds formed from atomic s- orbitals are always sigma bonds. Why?

**Explanations:** s- orbitals have spherical shape which is not ready for side to side over laps.

## Hybridization of Orbitals

**Need for modification of valence bond theory.**

- The old version of valence bond theory is limited to diatomic molecules only. It could not explain the structures and bond angles of molecules with more than three atoms such as $BCl_3$, $CH_4$, $H_2O$, $NH_3$ etc.
- **Linus Pauling** modified the valence bond theory using hybridization concept to explain the structure and bond angle of molecules.

**Hybridization** is the intermixing of two or more pure atomic orbitals of the **same atom** with almost the same energy to give same number of identical and degenerate new type of orbitals called **hybrid orbitals**.

- For atomic orbitals to undergo hybridization:
  1. The atomic orbitals of same atom with almost the same energy.
  2. The fulfilled, half- filled or even empty orbitals can undergo hybridization.
- The hybrid orbitals have the same **energy** and **shape**.
- The hybrid orbitals have one big lobe associated with a small lobe on the other side as shown below.

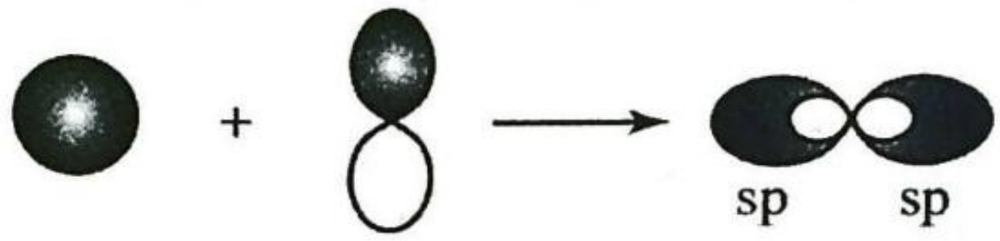
Figure 3.3 Shape of hybrid orbital

- The number of hybrid orbitals formed is equal to the number of pure atomic orbitals under going hybridization.
- The hybrid orbitals participate in the **sigma bond** formation with other atoms.

# Types of hybridization

## 1. sp - hybridization

- Intermixing of one 's' and one 'P' orbitals
- sp - hybrid orbitals are arranged linearly by making 180° of angle
- They each possess 50% 's' and 50% 'p' character

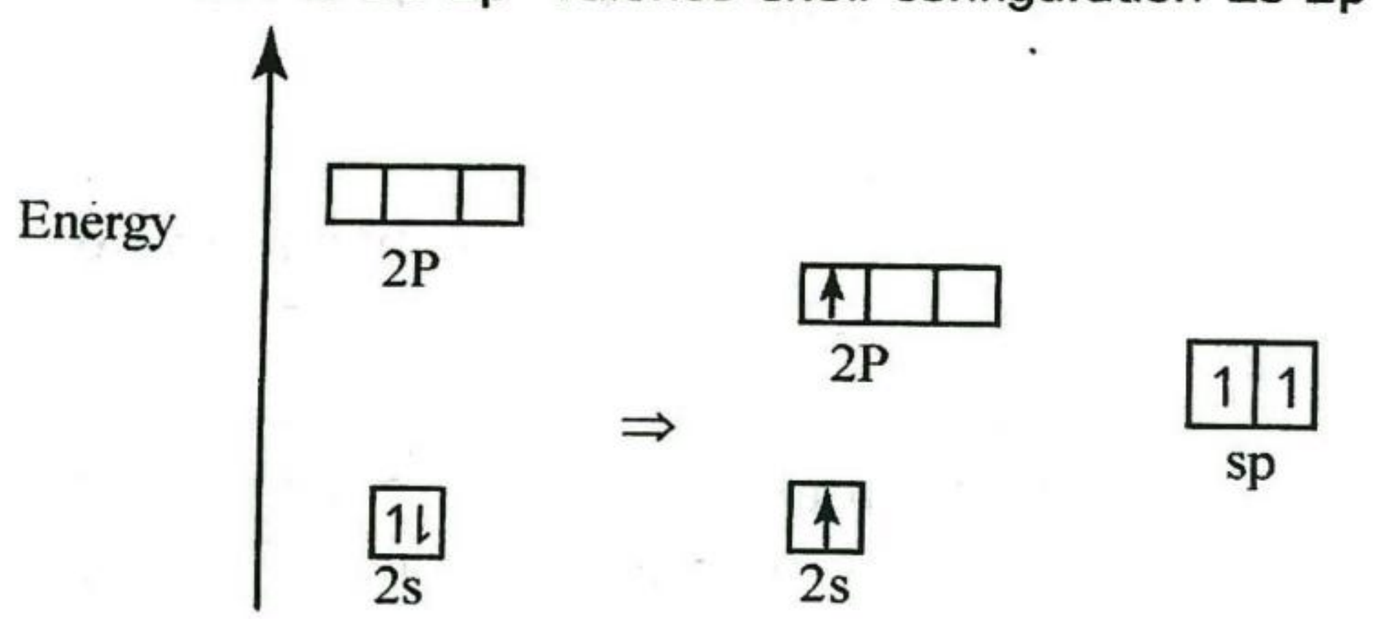



s- orbital p- orbital Two sp hybrid obitals

**Example**

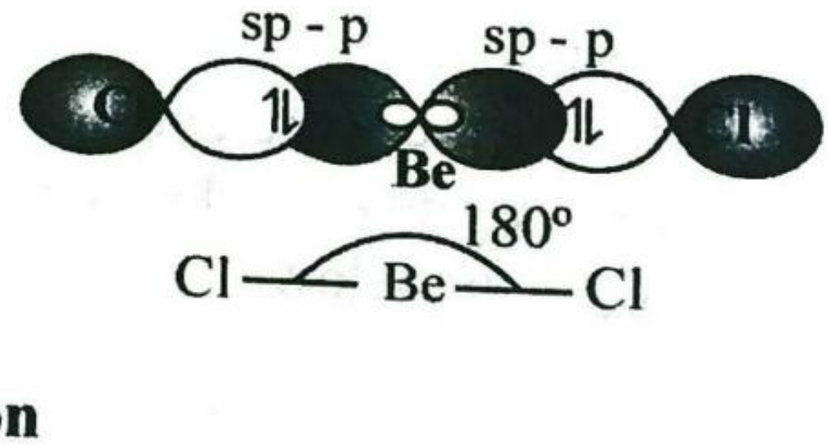
Beryllium chloride ($BeCl_2$)

Be: $1s^2 2S^2 2p^\circ$ valence shell configuration $2s^2 2p^\circ$




Be in ground state | Be in excited state | Be in hybridization state

⇒ Here, beryllium atom is sp hybridized and forms a sigma bond with the P- orbitals of the two chlorine atoms.



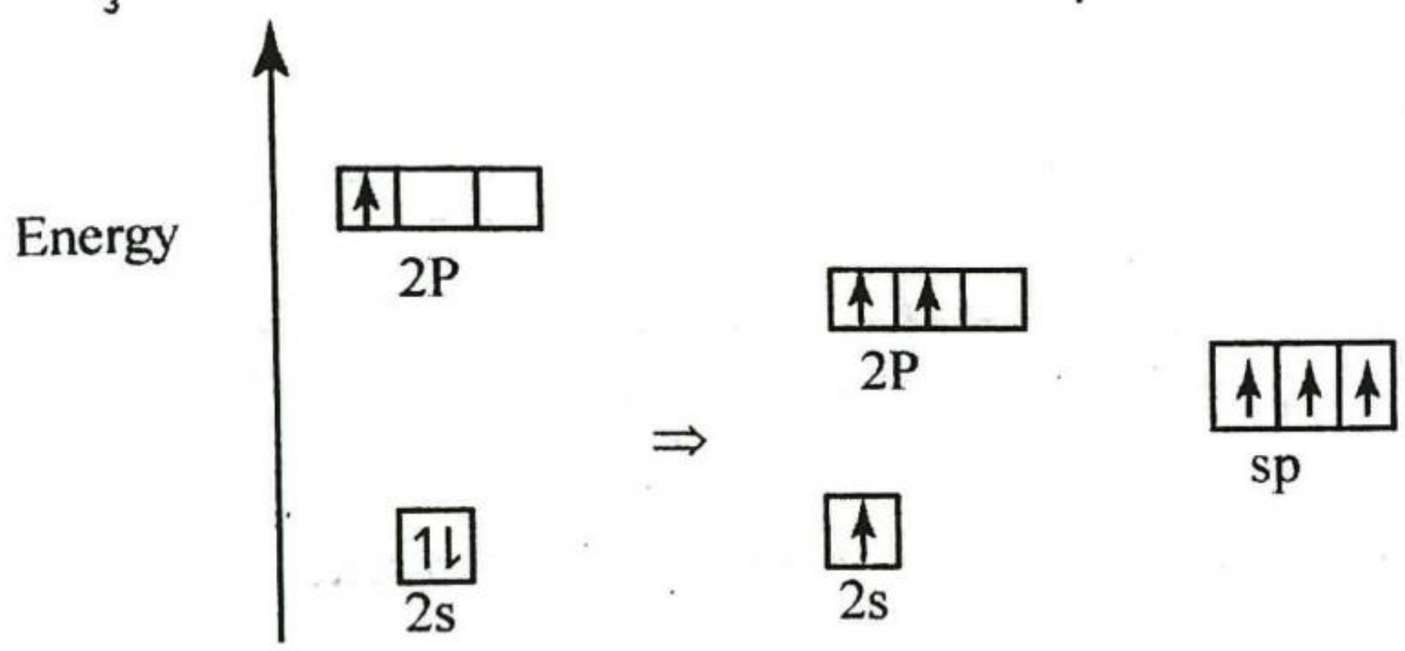

## 2. $sp^2$ - hybridization

- Intermixing of one **'s'** and two **'P'** orbitals almost equal energy.
- The three $sp^2$ hybrid orbitals are oriented in trigonal planar symmetry at angles of **120°** to each other.
- The $sp^2$ hybrid orbital have 33.3% 's' character and 66.7% p character.

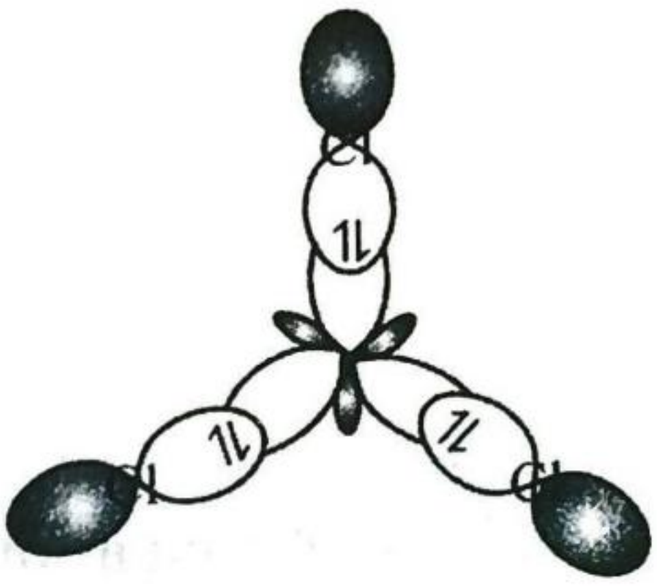
s - orbital  Px - orbital  Py - orbital  $sp^2$ hybrid orbitals

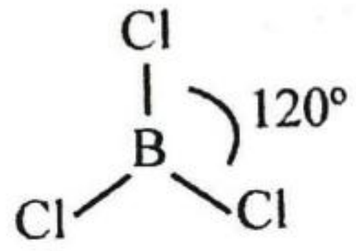
**Example**

$BC1_3$  B: $2s^2 2p^1$




B in ground state  B in excited state  B in hybridization state

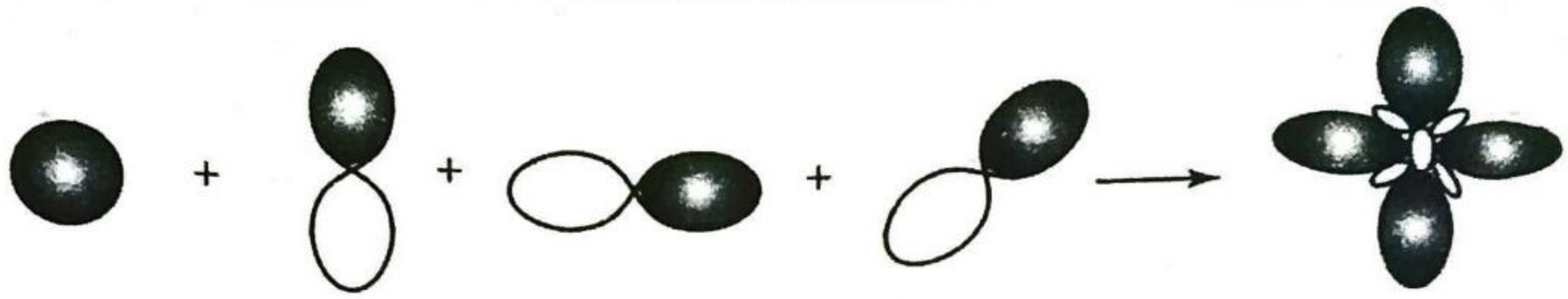





Trigonal planar structure of $BC1_3$

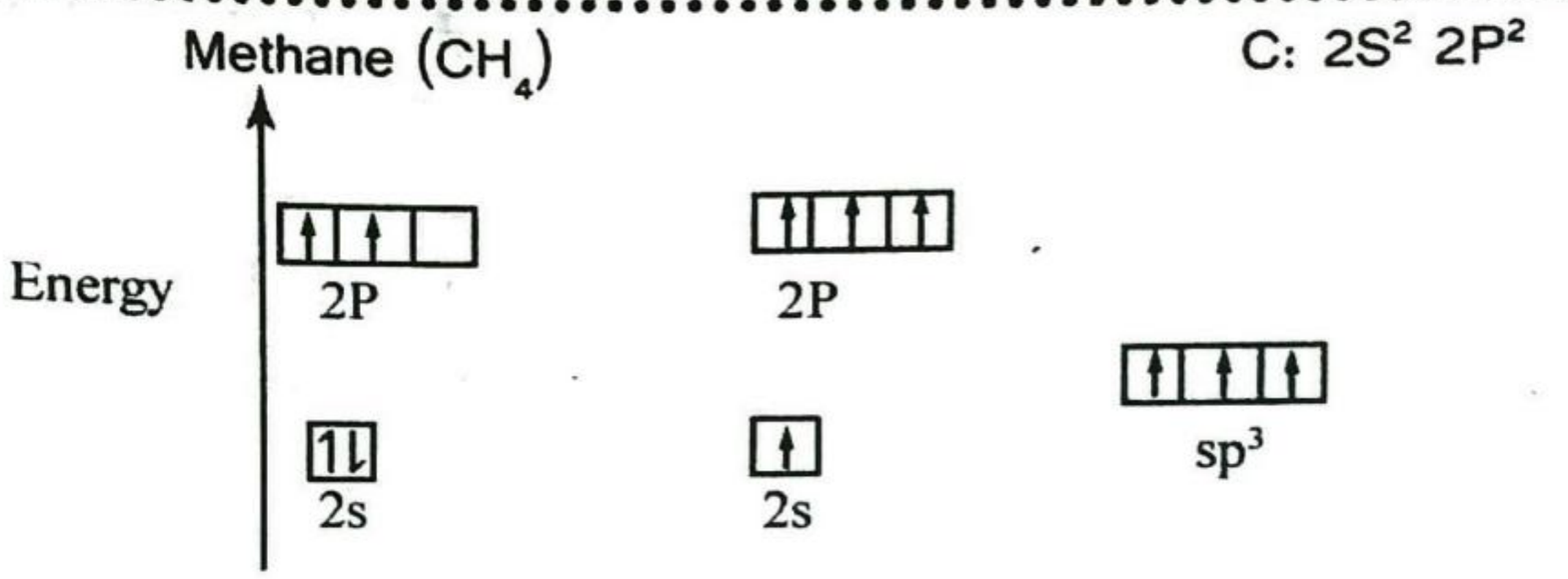
3. **$sp^3$ - hybridization**

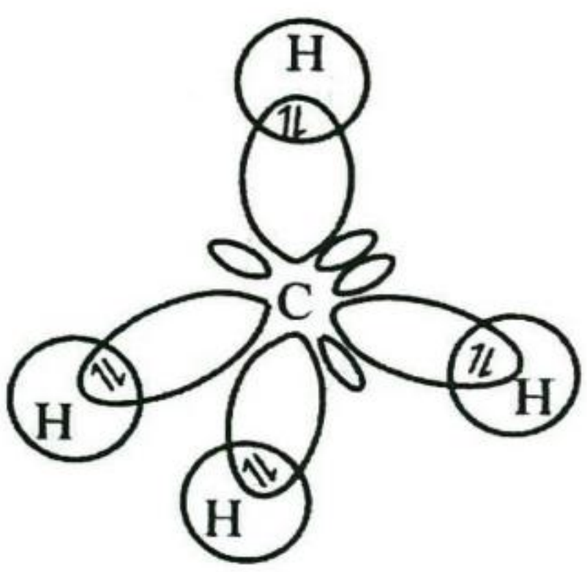
   - In $sp^3$ hybridization one **'s'** and three **'p' orbitals** of almost equal energy intermix to give four identical and degenerated hybrid orbitals
   - These four **$sp^3$** hybrid orbitals are oriented in tetrahedral summery with **109.5° angle** with each other.
   - The **$sp^3$** hybrid orbitals have 25% 's' character and 75% 'P' character

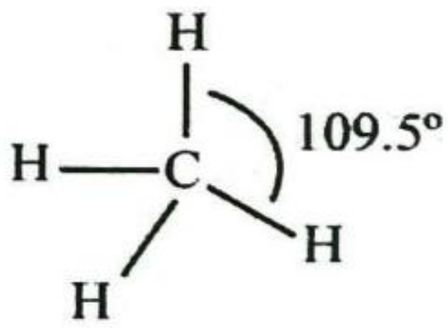


s - orbital Px - orbital Py - orbital $P_z$- orbital $sp^3$ hybrid orbitals

**Example**




## 4. $sp^3d$ - hybridization

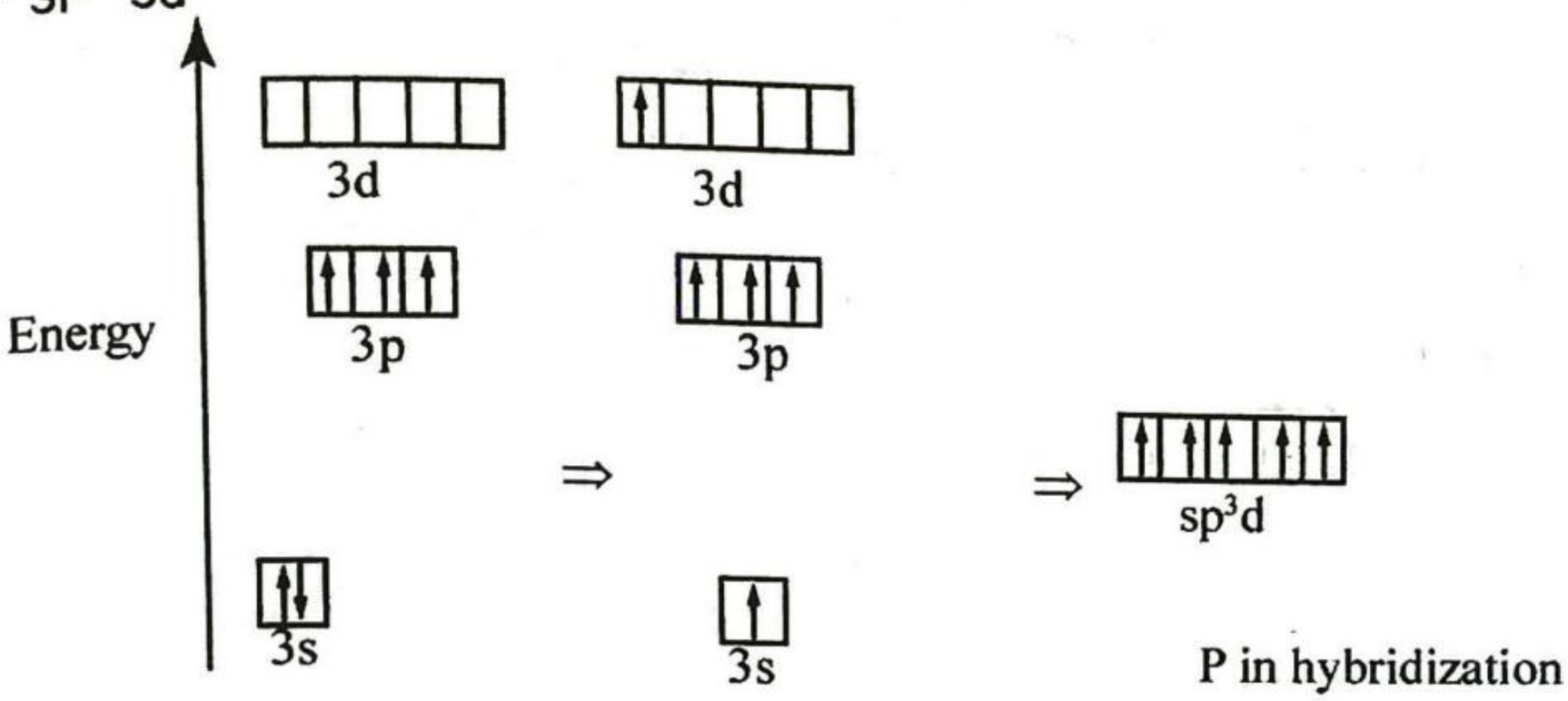
- In **$sp^3d$** hybridization one 's', three 'p' and one 'd' orbitals of almost equal energy intermix to give five identical and degenerate hybrid orbitals, which are arranged in **trigonal bi pyramidal** symmetry.
- Three are arranged in **trigonal plane** and the remaining two orbitals are present above and below the trigonal plane at right angles.
- The **$sp^3d$** hybrid orbitals have 20% 'S' 60% and 20% 'd' characters.

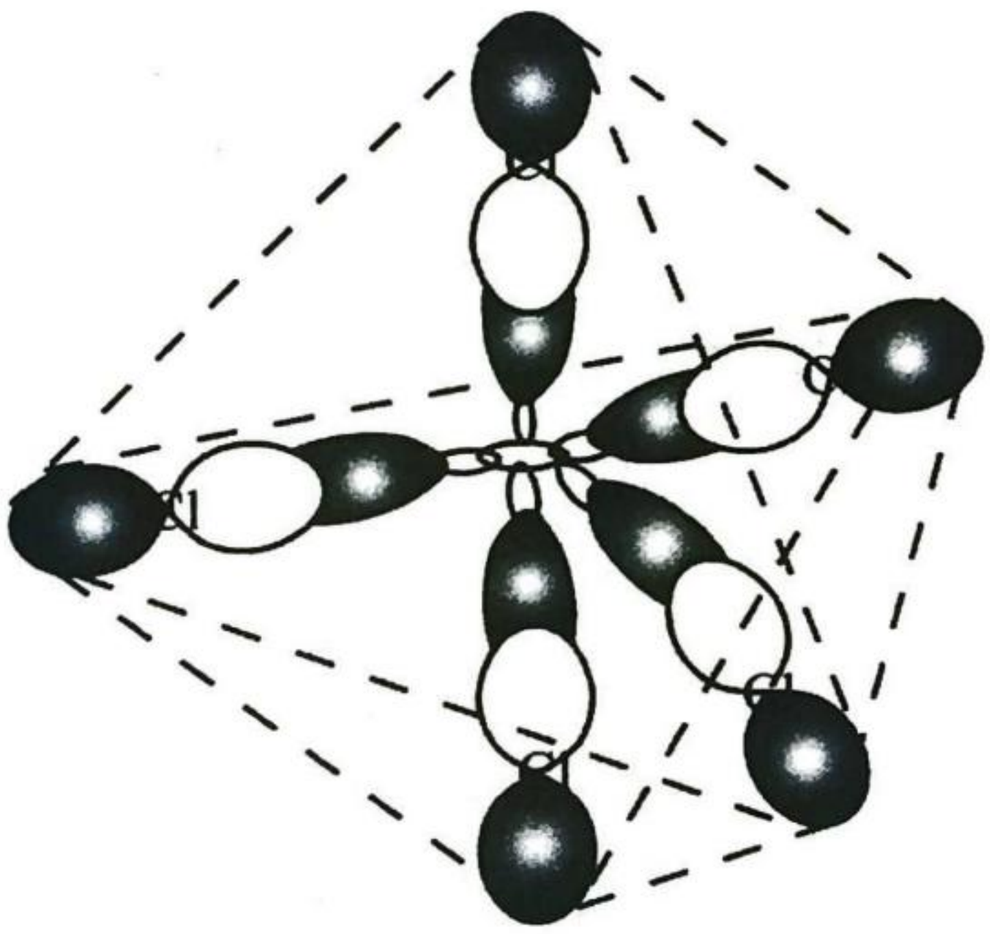
$sp^3d$ hybrid orbitals

**Example**

Phosphorus pent a chloride ($PCl_5$)

$_{15}P$: $3S^2\ 3P^3\ 3d^0$



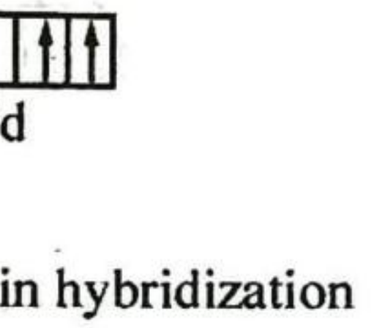


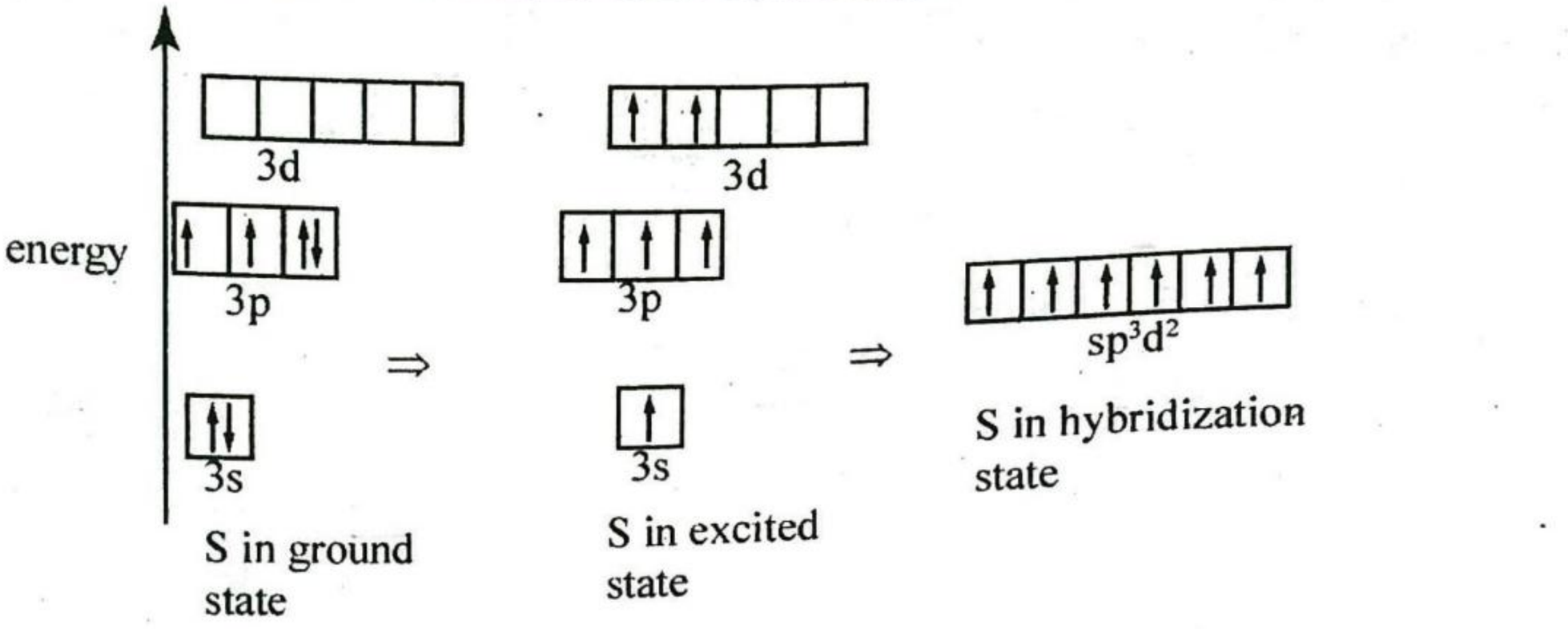

Figure 3.4 Trigonal bi pyramidal structure of $PCl_5$

5. **$sp^3d^2$ hybridization**
   - ➦ Intermixing of one **'s'**, three **'p'** and two **'d'** orbitals of same energy by giving six identical and degenerate hybrid orbitals called **$sp^3d^2$** hybridization
   - ➦ These six hybrid orbitals are arranged in octahedral symmetry by making 90° angles to each other.

**Example**

Sulphur hexa fluoride ($SF_6$)

$_{16}S$: $3S^2\ 3P^4\ 3d^0$




## Hybridization and multiple bonds.

### 1. Double bond and hybridization ($sp^2$).

During double bond formation one of the P-orbital is not hybridized for Pi bond formation.

**Example**

$C_2H_4$

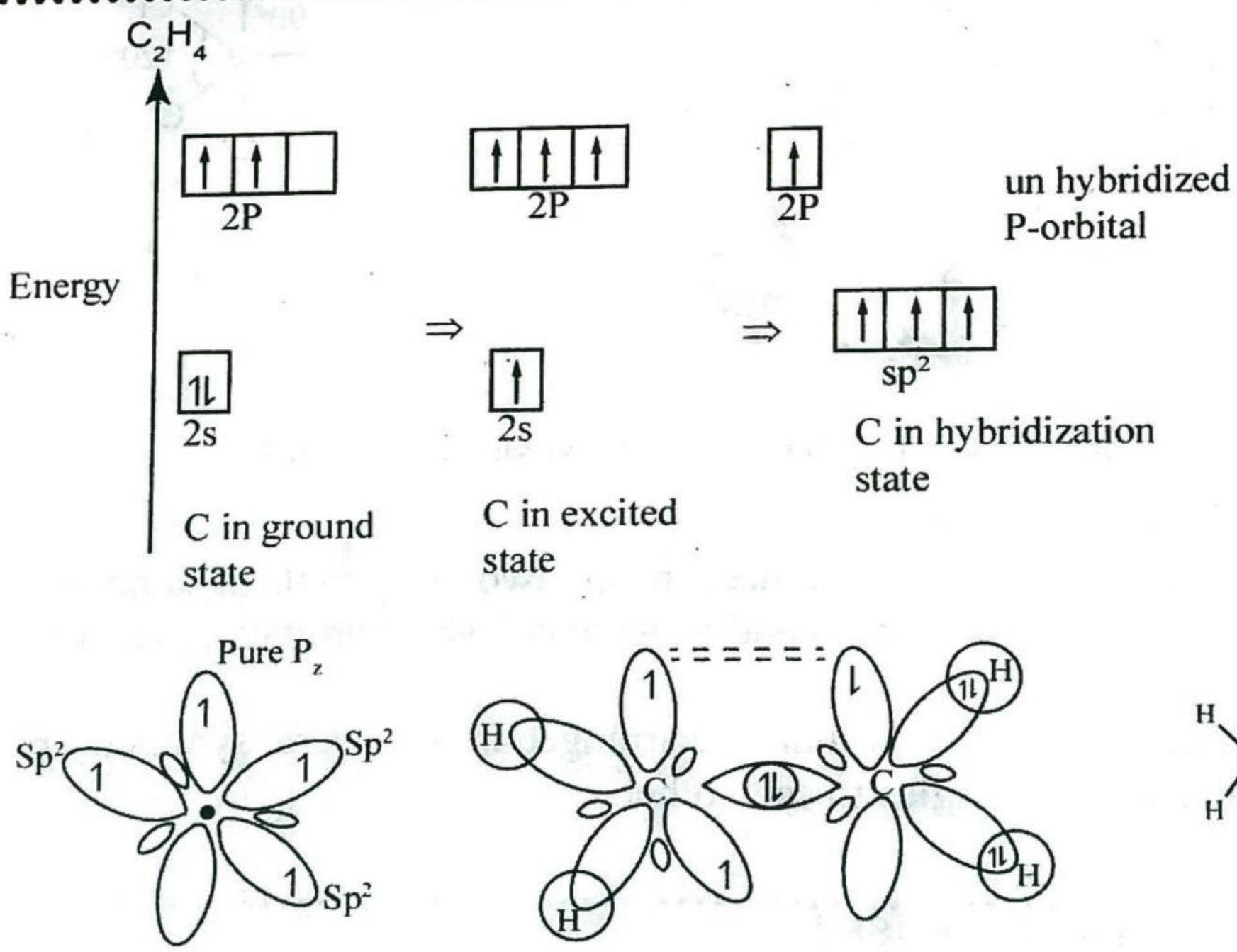



Figure 3.5 Planar structure of ethene ($C_2H_4$)

## 2. Triple bond and hybridization (sp).

➦ During triple bond formation two P-orbitals are not hybridized for the two Pi-bonds formation. **Eg.** Ethyne ($C_2H_2$)

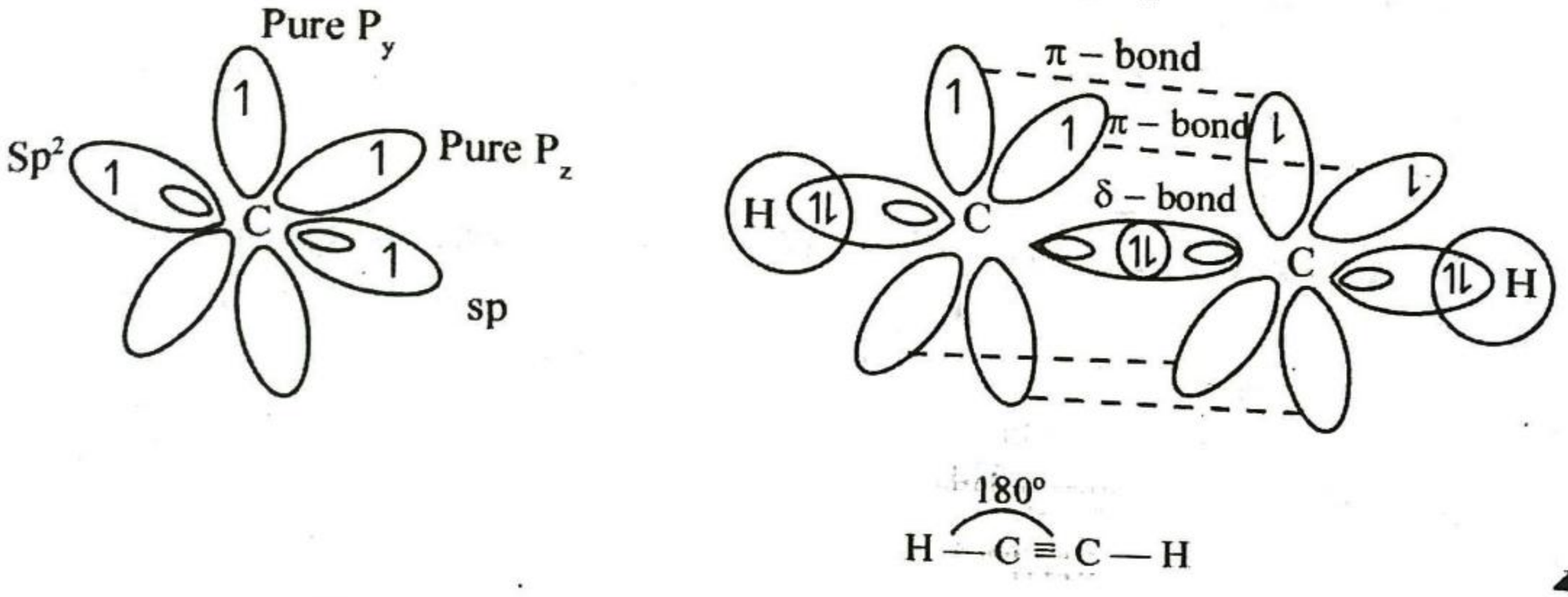



Figure 3.6 Linear structure of ethyne molecule.

**Note:**

Number of set of electrons is equal to number of hybrid orbitals.

| Number of sets of electrons | hybridization |
|---|---|
| 2 | sp |
| 3 | $sp^2$ |
| 4 | $sp^3$ |
| 5 | $sp^3d$ |
| 6 | $sp^3d^2$ |

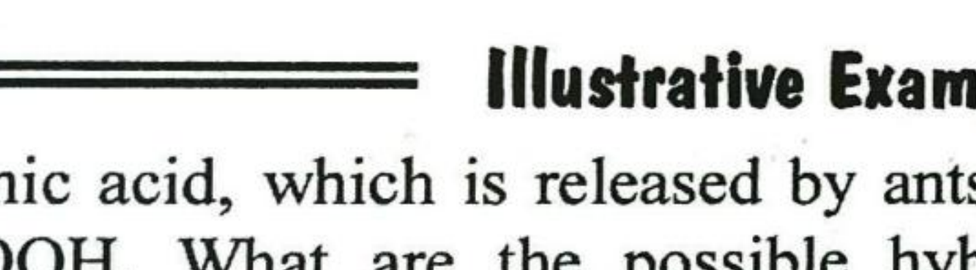
## Illustrative Examples

82. Formic acid, which is released by ants, has a molecular formula of HCOOH. What are the possible hybridizations that exist in the molecule? (EUEE... 2008 E.C)

A. $sp^2$ and $sp^3$
B. sp and $sp^3$
C. sp, $sp^2$ and $sp^3$
D. sp and $sp^2$

**Solution:**

Draw the Lewis structure and determine the number of sets of electrons around C and O




➦ Around C, three sets of electrons ⇒ $sp^2$ hybridized

➦ Around O, four sets of electrons ⇒ $sp^3$ hybridized

**Answer: A**

83. How many atomic orbital are required for an $sp^3$ hybridization? (EUEE... 2008 E.C.)

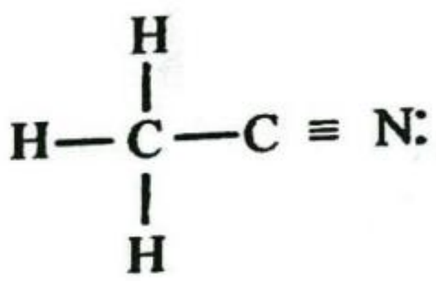
A. 2 B. 6 C. 4 D. 8

**Solution:**

Number of atomic orbital combined = number of hybrid orbitals formed. Four hybrid orbitals are obtained from the combination of four atomic orbitals, one 's' and three 'p' orbitals.

**Answer: C**

84. What is the hybridization of the carbon atom attached to nitrogen in acetonitrile shown below?



A. sp
B. $sp^2$
C. $sp^3$
D. $sp^4$

**Solution:**

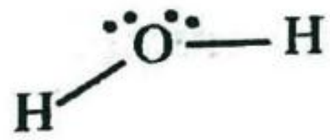
The carbon attacted to nitrogen is surrounded by two sets of electrons. Thus, its hybridization is sp.

**Answer: A**

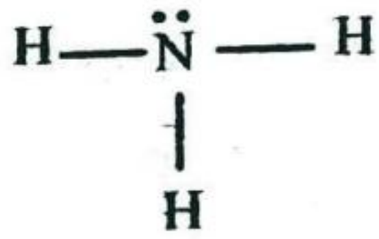
85. Identify the hybridization of the centeral atom of

a. $H_2O$
b. $NH_3$
c. $CO_2$
d. $SO_2$
e. $BrF_5$
f. $ICl_3$

**Solution:**

a. 

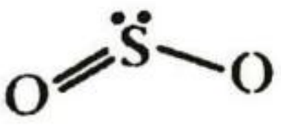
➦ Four sets of electrons
➦ Oxygen is $sp^3$ hybridized

b. 

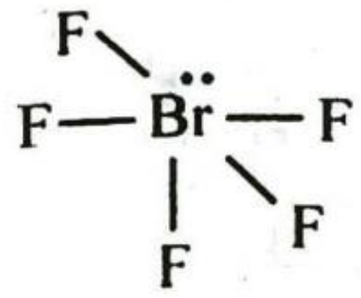
➦ Four sets of electrons
➦ Nitrogen is $sp^3$ hybridized

c. $CO_2$ O = C = O

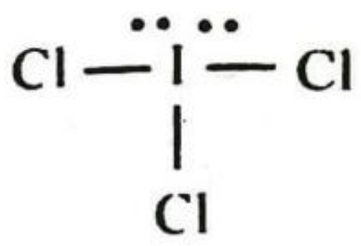
- two sets of electrons
- Carbon is sp hybridized

d. 

- three sets of electrons
- S is $sp^2$ hybridized.

e. 

- Six sets of electrons
- Br is $sp^3d^2$ hybridized

f. 

- Fire sets of electrons
- I is $sp^3d$ hybridized

86. The following compounds have more than one centeral atom. Identify the type of hybridization with respect to each centeral atom.

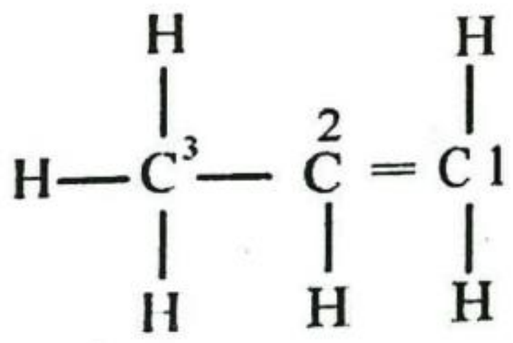
a. $CH_3 — CH = CH_2$

b. $H_2 C = C = CH_2$

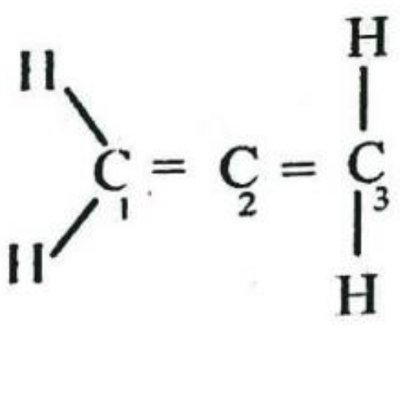
c. $HNO_3$

d. $CH_3 COOH$

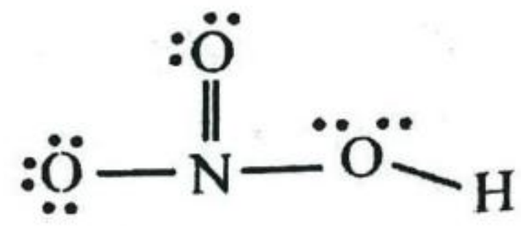
**Solution:**

a. 

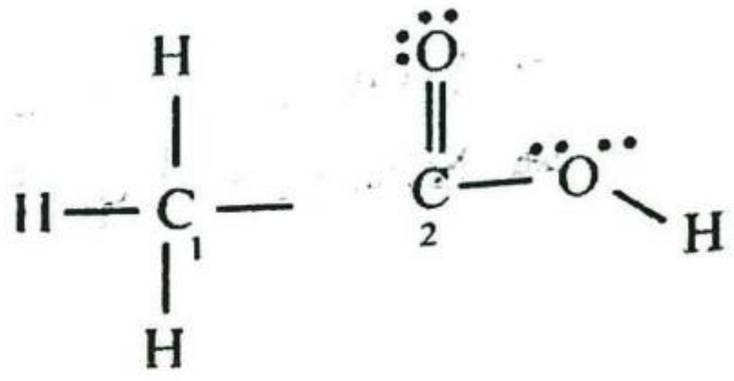
- C – 1 and C – 2 are $sp^2$ hybridized
- C - 3 is $sp^3$ hybridized

b. 

- C – 1 and C – 3 are $sp^2$ hybridized
- C – 2 sp hybridized

c. 

- N is $sp^2$ hybridized
- O is $sp^3$ hybridized

d. 

- C – 1 is $sp^3$ hybridized
- C – 2 is $sp^2$ hybridized
- O is $sp^3$ hybridized

87. Identify the type of orbital over lap for the formation the compounds below.

a. $BH_3$
b. $Be\,Cl_2$
c. $C_2H_2$
d. $SF_6$
e. $PCl_5$
f. $CCl_4$

**Solution:**

a. $sp^2$ – s overlap
b. sp – p overlap
c. sp – s overlap
d. $sp^3d^2$ – P overlap
e. $sp^3d$ – P overlap
f. $sp^3$ – P overlap

88. Explain the relation ship between s - character and bond length.

**Explanation:** As s- character increases, bond length decreases because s- orbital is nearer to the nucleus.

89. Draw the condensed structure of a compound that contains only carbon and hydrogen atoms that has
    a. three $sp^3$ hybridized carbons
    b. one $sp^3$ and two $sp^2$ hybridized carbon atoms
    c. Two $sp^3$ and two sp hybridized carbon atoms

**Solution:**

a. $CH_3\,CH_2\,CH_3$ or cyclo propane
b. $CH_3 — CH = CH_2$
c. $CH_3 — C \equiv C — CH_3$ or $HC \equiv C — CH_2 — CH_3$

## Molecular Orbital theory (MOT)

- When any number of atomic orbital overlaps, they combine to form an equal number of new orbitals called **molecular orbital**.
- Molecular orbitals obtained by addition of atomic orbital are called **bonding molecular orbitals** represented by $\delta$ **or** $\pi$.
- Molecular orbitals formed by subtraction of atomic orbitals are known as **anti-bonding molecular orbitals** and denoted by $\delta^*$ **or** $\pi^*$.
- Bonding molecular orbitals have lower energy than the original atomic orbitals.
- Anti bonding molecular orbitals have higher energy than the original atomic orbitals.

## Molecular orbital theory principles

1. Orbitals are conserved. The total number of molecular orbitals equals the total number of atomic orbitals that contributed.

2. Conservation of energy is obeyed. Bonding molecular orbitals have lower energy than the average of the contributing atomic orbitals. Anti bonding molecular orbitals have higher energy.

3. When assigning electrons to molecular orbitals, the **Pauli Exclusion principle and Hund's rule still apply**.

   **Rule 1**: For species that have 14 or below 14 electrons the order of filling is δ1s δ1s* δ2s δ2s* π2py π2pz δ2px

   **Rule 2**: For species that have above 14 electrons

   δ1s δ1s* δ2s δ2s* δ2px π2py π2pz π2*py π2*pz π2px*

   ➦ π2py and π2pz } are degenerate molecular orbitals.
   ➦ π2*py and π2*pz }

4. The maximum number of electrons in a given molecular orbital is two
5. In ground state configurations, electrons enter molecular orbitals of same energies single before they pair up.
6. When two atomic orbitals overlap end to end they form sigma molecular orbital.
7. When two atomic orbitals overlap in side to side way they form pi molecular orbital.

**A molecular orbital diagram** shows the relative energy and number of electrons in each MO and the atomic orbitals from which they form.

### Example

$H_2$

Number of electrons = 2 × 1 = 2

Electron configuration: $(\delta 1s)^2$

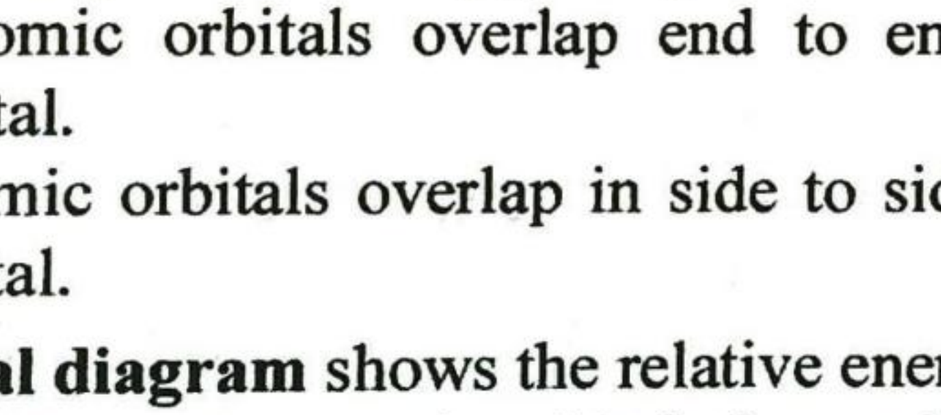



Figure 3.7 Molecular orbital diagram and bonding in the molecule of $H_2$

**Bond order (BO)** is one half the difference between the number of bonding ($n_b$) and anti bonding ($n_a$) electrons.

$$\boxed{Bo = \frac{1}{2}(n_b - n_a)}$$

- As the value of bonder order increases, the stability of the species also increases.
- Bond order equals to zero means the given species does not exist (unstable).
- A species with unpaired electrons exhibits **paramagnetic property** (attracted by an external magnetic field)
- A species with all the electrons are paired exhibits **diamagnetic species** (not attracted by magnetic field).

## Illustrative Examples

90. What would happen to the $O_2$ molecule upon ionization to $O_2^+$? (EUEE ... 2008 E.C)

A. The bond length will increase and the bond energy will increases.
B. The bond length will increase and the bond energy will decrease.
C. The bond length will decrease and the bond energy will increase.
D. The bond length will decrease and the bond energy will decrease.

**Solution:**

Determine bond order first. As bond order increases bond length decreases and bond energy increases.

$O_2$: number of electrons $= 8 \times 2 = 16$

Electron configuration: $(\delta_{1S})^2 (\delta^*_{1S})^2 (\delta 2s)^2 (\delta^*_{2S})^2 (\delta_2 px)^2 (\pi_2 p_y)^2 (\pi_2 p_z)^2 (\pi_2 {}^*py)^1 (\pi_2 {}^*p_2)^1$

$$BO = \frac{1}{2}(n_b - n_a) = \frac{1}{2}(10-6) = 2$$

- $O^+_2$: number of electrons $= 8 \times 2 - 1 = 15$

$\Rightarrow (\delta_{1S})^2 (\delta^*_{1S})^2 (\delta_{2S})^2 (\delta 2\sigma^*)^2 (\delta 2Px)^2 (\pi 2Py)^2 = (\pi 2P_z)^2 (\pi 2P^*y)^1$

$$BO = \frac{1}{2}(n_b - n_a) = \frac{1}{2}(10-5) = 2.5$$

$\Rightarrow$ BO of $O_2^+$ is higher than $O_2$. $O_2^+$ more stable than $O_2$.

$\Rightarrow O_2 \longrightarrow O^+_2$, bond length decreases and bond energy increases

**Answer: C**

91. According to the molecular orbital theory which one of the following is paramagnetic?

A. $F_2$ B. $H_2$ C. $NO^+$ D. NO

**Solution:**

➦ All species with odd number of electrons are paramagnetic.

**Answer: D**

92. Which of the following is a characteristics of anti bonding molecular orbital? (EUEE)

A. It is a molecular orbital with high probability of finding electrons parallel of finding electrons parallel to the region between the bonded atoms

B. It has no electrons

C. It is a molecular orbital with high probability of finding electrons away from the region between the bonded atoms.

D. It is a molecular orbital with a high probability of finding electrons in the region between bonded atoms.

**Explanation:** In anti bonding molecular orbital, the electron density is concentrated away from the region between the nuclei while in a bonding molecular orbital the electron density is concentrated between the two nuclei

**Answer: C**

93. Why does the peroxide ion $\left(O_2^{2-}\right)$ have longer bond than does super oxide $O^-_2$? (EUEE)

A. $O_2^-$ has a greater bond order than $O_2^{2-}$

B. $O_2^-$ has a greater bond order than $O_2^-$

C. $O_2^-$ has a smaller bond order than $O_2^-$

D. $O_2^-$ has a smaller bond order than $O_2^-$

**Solution:**

$\text{BO of } O_2^{2-} = 1$

$\text{BO of } O_2^- = 1.5$

➦ Bond order and bond length are inversely related

**Answer: D**

94. Which of the following molecules has an unpaired electron (S)?

A. $F_2$ B. $N_2$ C. $CN^-$ D. $O_2$

**Answer: D**

95. Which of the following is the most stable?

A. CN B. $CN^-$ C. $CN^+$ D. $C_2^+$

**Solution:**

As BO increases, stability also increases

BO for CN = 2.5

BO for $CN^-$ = 3

BO for $CN^+$ = 2

BO for $C_2^+$ =1.5

**Answer: B**

96. Draw molecular orbital diagram for

a. $C_2$ b. $F_2$

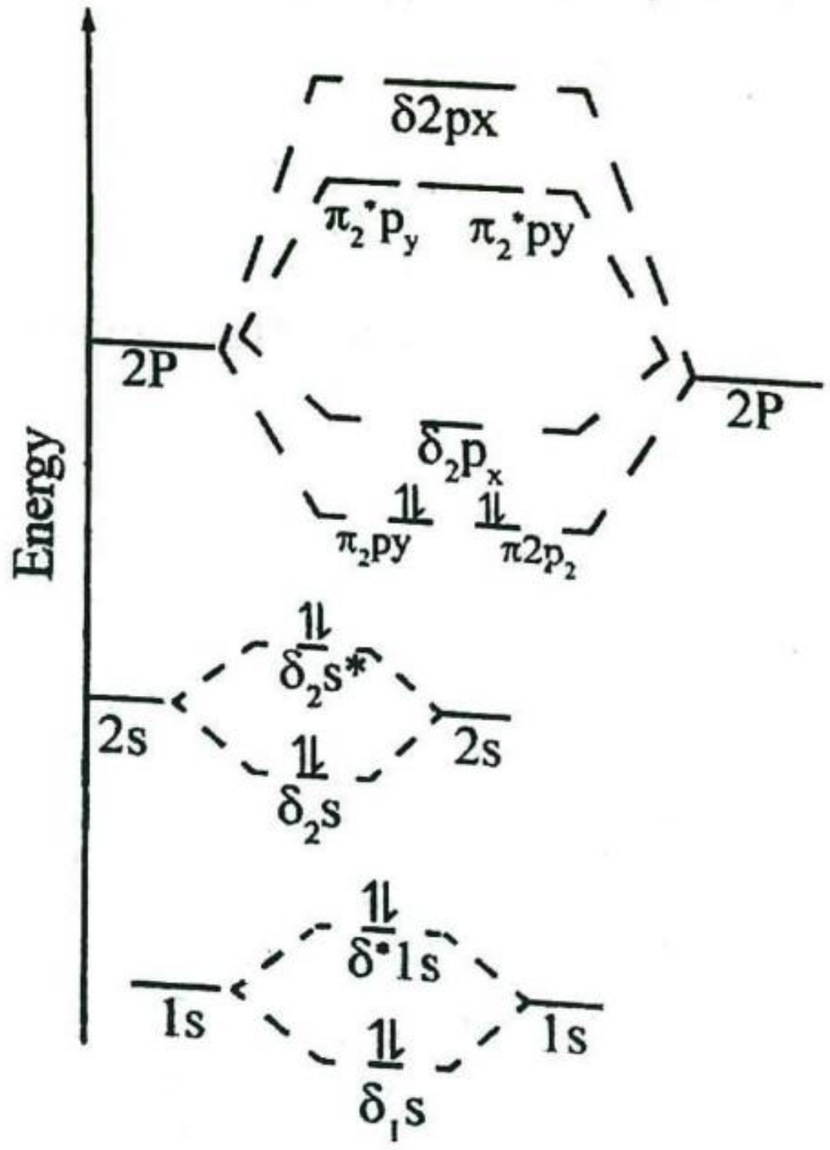
**Solution:**

First write electron configuration bases on MOT.

a. $C_2$: number of electrons = $2 \times 6 = 12$

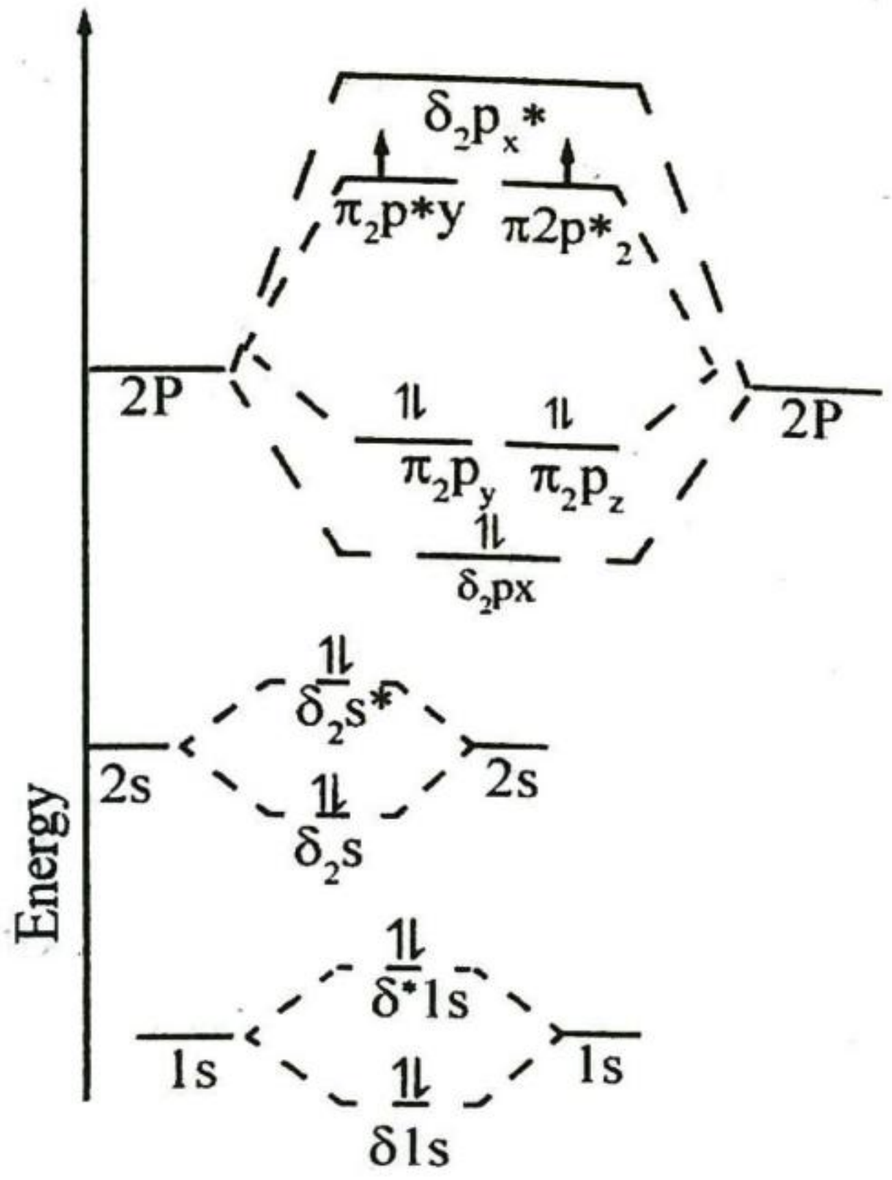
➡ Use rule 1 to write electron configuration

$(\delta 1s)^2 (\delta^*_{1s})^2 (\delta 2s)^2 (\delta^*_{2s})^2 (\pi 2py)^2 = (\pi 2pz)^2$




b. $O_2$: electrons = $2 \times 8 = 16$, use rule 2

$$(\delta 1s)^2 (\delta^*_{1s})^2 (\delta 2s)^2 (\delta^*_{2s})^2 (\delta 2px)^2 (\pi 2py)^2 = (\pi 2p_z)^2 (\pi_2 p^*_y)^1 = (\pi_2 p^*_z)^1$$




97. Identify those which exist or do not exist by determining their bond order.

a. $He_2$
b. $H_2^+$
c. $Li_2$
d. $B_2$
e. NO
f. $CN^+$

**Solution:**

a. $(\delta 1s)^2 (\delta^*_{1s})^2$, $BO = \frac{1}{2}(2-2) = 0$

$\Rightarrow He_2$ does not exist

b. electron $= 2\times1 - 1 = 1$

$(\delta 1s)^1$, $BO = \frac{1}{2}(1-0) = 0.5$

$\Rightarrow H_2^+$ does exist

c. electron $= 2\times3 = 6$

$(\delta_{1s})^2 (\delta^*_{1s})^2 (\delta_{2s})^2$, $BO = \frac{1}{2}(4-2) = 1$

$\Rightarrow Li_2$ does exist

d. electron $= 2\times5 = 10$

$$(\delta_{1s})^2 (\delta^*_{1s})^2 (\delta_{2s})^2 (\delta^*_{2s})^2 (\pi_2 py)^1 = (\pi_2 P_z)^1$$

BO = 1

$\Rightarrow B_2$ is stable

e. electron = 7 + 8 = 15

$$(\delta 1s)^2 (\delta 1s*)^2 (\delta 2s)^2 (\delta 2*s)^2 (\delta 2P*)^2 (\pi 2py)^2 = (\pi 2Pz)^2 (\pi 2Py*)^1$$

BO = 2.5

$\Rightarrow$ NO is stable.

f. electron = (6 + 7) – 1 = 12

$$(\delta_{1s})^2 (\delta_{1s}*)^2 (\delta_{2s})^2 (\delta^*_{2s})^2 (\pi_2 Py)^2 = (\pi_2 P_2)^2$$

$$BO = \frac{1}{2}(8-4) = 2$$

$\Rightarrow CN^+$ is stable

98. Write the centeral ideas of Molecular orbital theory.

**Solution**

- Atomic orbitals of all atoms mathematically combine to from molecular orbitals.
- Electrons then populate the molecular orbitals and are delocalized across entire molecule.

## Types of solids

- Depending on the arrangement of atoms, there are two types of solids. These are

1. Amorphous solid
2. Crystallime solid

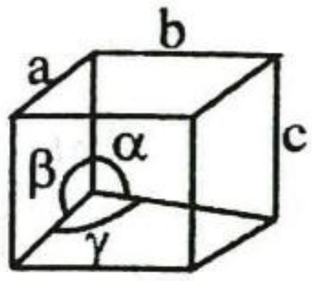
Table 3.2. properties of amorphous and crystalline solids

| Type of solid | Properties | Example |
|---|---|---|
| Crystalline solids | Consist of individual crystal each with regular shape.<br>Cleave to give flat, well defined faces<br>Melt at definite temperature | metals<br>ionic compounds<br>sugar |

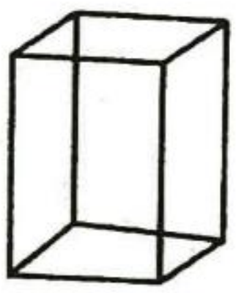
| | | |
|---|---|---|
| Amorphous solid | have no well defined shape<br>break to give curved or irregular faces<br>soften then melt over a range of temperature | Paraffin<br>asphalt<br>window glass<br>rubber |

- **A crystal** is a piece of solid with regular shape.
- The basic repeating structural unit of crystalline solid is a **unit cell.**

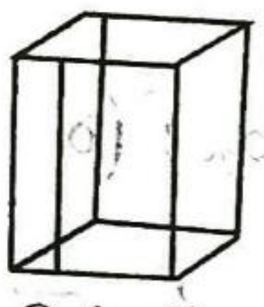
There are seven types of unit cells



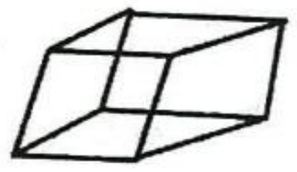

simple cube
$a = b = c$
$\alpha = \beta = \gamma = 90°$



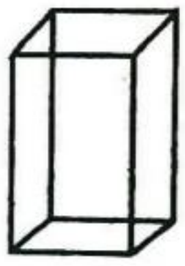
Tetragonal
$a = b \neq c$
$\alpha = \beta = \gamma = 90°$



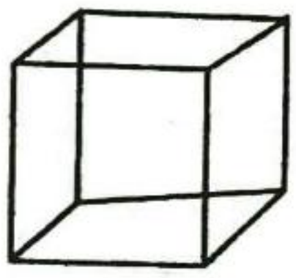
Orthorhombic
$a \neq b \neq c$
$\alpha = \beta = \gamma = 90°$



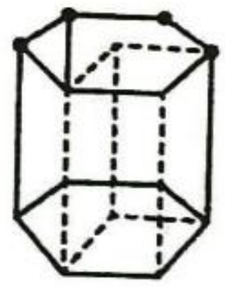
Rhombohedral
$a = b = c$
$\alpha = \beta = \gamma \neq 90°$



Monoclinic
$a \neq b \neq c$
$\alpha = \gamma = 90°, \beta \neq 90°$



Triclinic
$a \neq b \neq c$
$\alpha \neq \beta \neq \gamma = 90°$

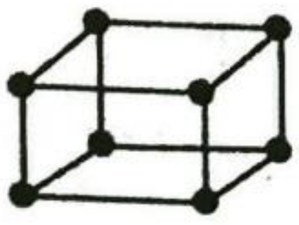


Hexagonal
$a = b \neq c$
$\alpha = \beta = 90°, \gamma = 120°$

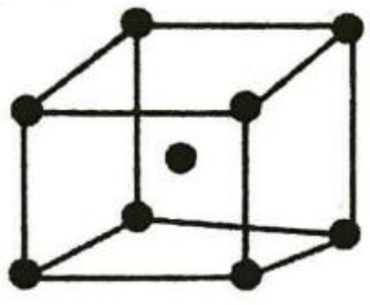
Figure 3.8 The seven types of unit cells. Angle $\alpha$ is defined by edges b and c, angle $\beta$ by ages a and c and angle $\gamma$ by adges a and b.

- There are three types of cubic cells. These are:

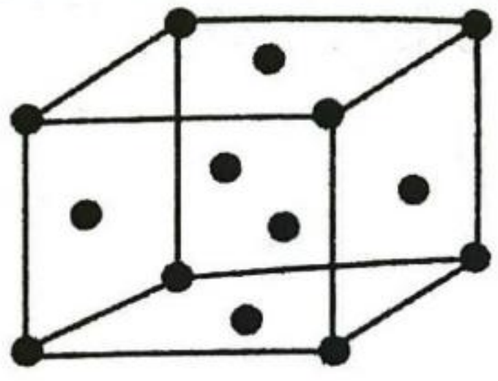
1. Simple cubic cell
   - Each sphere is found at each corner.
2. Body centered cubic cell
   - Each sphere at each corner and one sphere in the center.
3. Face centered cubic cell
   - There are spheres at the center of each of the six faces of the cube, in addition to the eight corner spheres.



Simple cube



Body - centered cubic



Face centered cubic

➦ The crystalline solid further classified in to four different types. These are

1. Ionic crystal
2. Covalent network crystal
3. Metallic crystal
4. Molecular crystal.

Table 3.3.

| Type of crystal | Interaction force | Properties | Example |
|---|---|---|---|
| Ionic crystals | Ionic bond | ➦ are hard and brittle<br>➦ have high melting point<br>➦ contain ions<br>➦ less volatile<br>➦ have high heat of fusion and vaporization<br>➦ transmit electricity in molten and aqueous states | NaCl<br>Cao<br>KBr |
| Covalent network crystals | Covalent bond | ➦ contain atoms<br>➦ are hard and brittle<br>➦ very high melting point<br>➦ very low volatility<br>➦ poor conductors | $Si O_2$<br>Si C<br>diamond |

| | | | |
|---|---|---|---|
| Molecular crystals | Intermolecular force | ➦ contain molecules<br>➦ soft to hard and brittle<br>➦ low melting point<br>➦ high volatility<br>➦ low heat of fusion and vaporization.<br>➦ Poor conductor of heat and electricity | Solid $H_2O$<br>Solid $CO_2$<br>Sugar<br>$C_{12}H_{22}O_{11}$ |
| Metallic crystal | metallic bond | ➦ contains atoms<br>➦ very soft to hard<br>➦ malleable and ductile<br>➦ very low to high melting point (-39°c- Hg to 3410°c - W) | Cu, Fe, Ag |

## Illustrative Examples

99. Which of the following is covalent network crystal?

A. Na Cl
C. $Si O_2$
B. Cu
D. $C_{12} H_{22} O_{11}$

**Answer: C**

100. All of the following are crystalline solids. EXCEPT

A. potassium chloride
C. Rubber
B. dry ice
D. Iron

**Answer: C**

101. Which of the following is true about crystalline solids?

A. They melt over a range of temperature
B. They have no well defined shape.
C. They break and give regular faces.
D. None of the above.

**Answer: C**

102. Identify the attractive forces in the four types of crystals.

**Solution:**

- Ionic crystal - ionic bond
- Covalent network crystal -covalent bond
- Molecular crystal - intermolecular force
- Metallic crystal - metallic bond

## Solved Problems

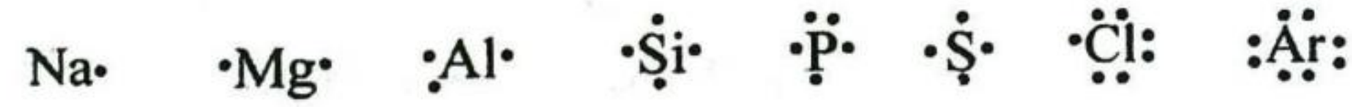
103. What types of electrons participate in bond formation?

**Solution:**

- Valence electrons- electrons in the valence shell

104. Write the Lewis electron dot symbol for period three elements.

**Solution:**

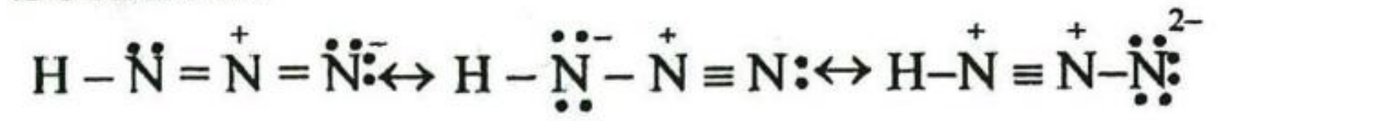


105. Arrange the ionic compounds below in increasing order of melting point. KF, KI, KBR, Kcl

**Solution:** KI < KBr < KCl < KF

106. Write three resonance structure for hydrazoic acid $HN_3$.

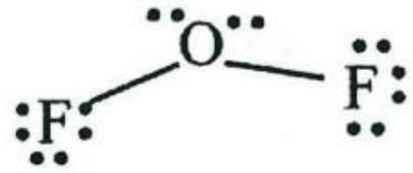
**Solution:**

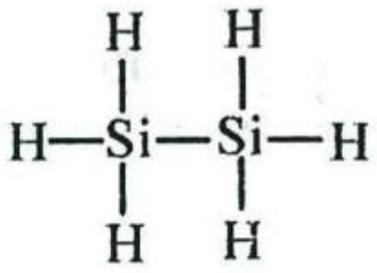


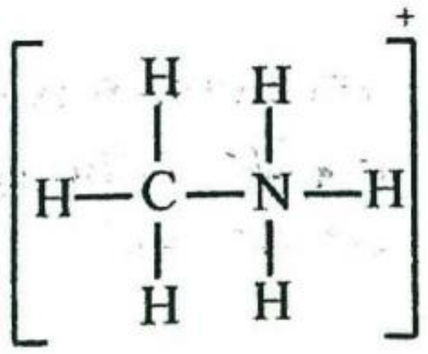
107. Draw the Lewis structures for the following molecules and ions and determine the number of lone

a. $OF_2$ b. $Si_2H_6$ c. $CH_3\overset{+}{N}H_3$

**Solution:**

a.  2 lone pairs

b.  no lone pair

c.  no lone pair

108. Four atoms are arbitrarily labelled A, B, C and D. Their electro-negativities are as follows:

A = 3.8 B = 3.3 C = 2.8 and D = 1.2. If the atoms of these elements form the molecules AB, AC, AD and BD, how would your arrange these molecules in increasing order of polarity?

**Explanation:** As the electro-negativity gap between the bonded atoms increases, polarity also increases.

$\Rightarrow$ AB = 3.8 – 3.3 = 0.5
AC = 3.8 – 2.8 = 1.0
AD = 3.8 –1.2 = 1.6
BD = 3.3 – 1.2 = 1.1
$\therefore$ AB < AC < BD < AD

109. Predict the geometries of the following species

a. $AlCl_3$
b. $ZnCl_2$
c. $TeCl_4$
d. $AlH_4^+$
e. $ICl_2^-$
f. $BeF_4^{2-}$
g. $SeF_4$
h. $CO_3^{2-}$
i. $IF_3$

**Solution:**

a. Trigonal planar
b. Angular
c. See saw
d. tetrahedral
e. linear
f. tetrahedral
g. see saw
h. trigonal planar
i. T-shaped

110. Describe the change in hybridization (if any) of the Al atom in the following reaction

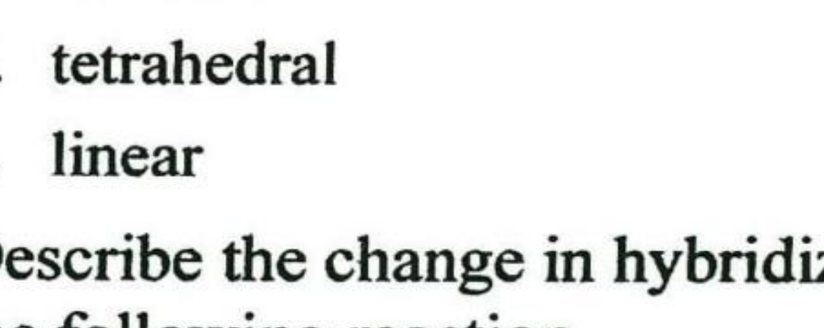
$$AlCl_3 + Cl^- \longrightarrow AlCl_4^-$$

**Solution**

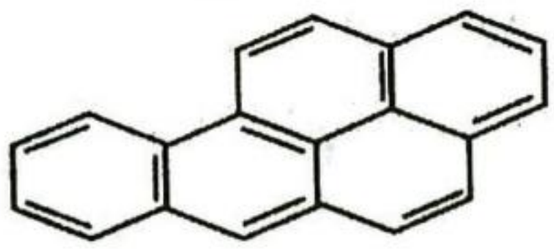
➦ Changed from $sp^2$ to $sp^3$

111. Determine the number of sigma and Pi bonds in the compounds below.

a.




Tryptophan is one of the 20 amino acid in human bod

b. 

Benzo ($\alpha$) pyrene - is a potent carcinogen found in coal and cigarette smoke.

**Solution:**

a. $5\pi$ bonds and 27 $\delta$ bonds

b. $10\pi$ bonds and 35 $\delta$ bonds

112. Arrange the following specias inorder of increasing stability.

$Li_2, Li_2^+, Li_2^-$

**Solution:**

$Li_2^+ = Li_2^- < Li_2$

113. Determine the bond order of C – N, C = N and C ≡ N bonds and arrange these bonds in increasing order of bond length.

**Solution:**

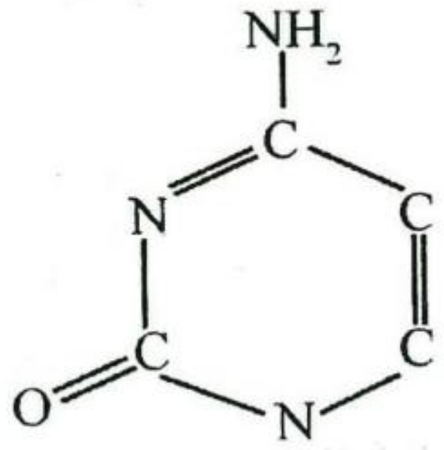
BO = 1, C – N

BO = 2, C = N

BO = 3, C ≡ N

C ≡ N < C = N < C – N

114. What is the hybridization of C and of N in this molecule?




**Solution:**

➦ All the carbon atoms are $sp^2$ hybridized

➦ The nitrogen with double bond is $sp^2$ hybridized. The other two N are $sp^3$ hybridized.

115. Arrange the following in decreasing order polarity. $SbH_3$, $AsH_3$, $NH_3$, $PH_3$

**Solution:**

$NH_3 > Sb\,H_3 > AsH_3 > PH_3$

116. List properties of water that results from hydrogen bonding

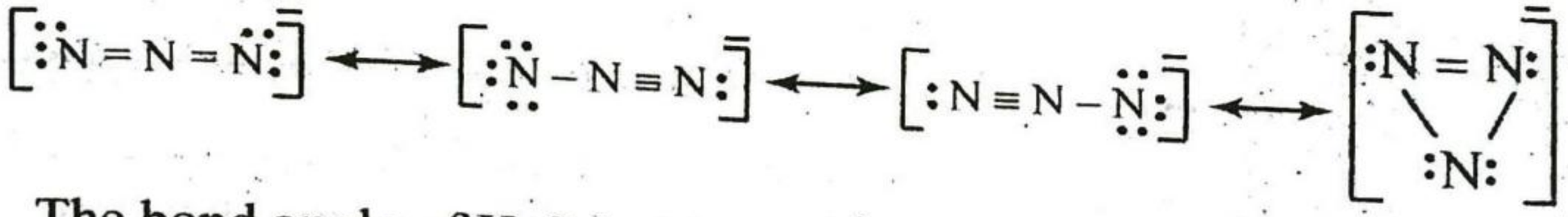
**Solution:**

➦ high melting and boiling point

- high heat of vaporization
- low density of ice compared to water
- high specifie heat, high ionic conductance of hydronium and hydroxide ions.

117. Draw all possible structures of $N_3^-$.

**Solution:**



118. The bond angle of $H_2O$ is 105° and $H_2S$ is 92°. Explain this difference.

**Explanation:** The larger size in S atom as compared to O minimized electron repulsions and allow the bond in $H_2S$ to be more purely P-type.

119. Write two non-polar compounds with polar bonds

**Solution:**

- $CO_2$, $BF_3$, $CCl_4$, $CF_4$

120. Write the formula of

a. Calcium nitride
b. Aluminum sulphate
c. Iron (III) carbonate
d. Ammonium phosphate

**Solution:**

a. $Ca_3N_2$
b. $Al_2(SO_4)_3$
c. $Fe_2(CO_3)_3$
d. $(NH_4)_3PO_4$

121. Give the chemical name for

a. $Cr_2O_3$
b. $Li_3P$
c. $HgCl_2$
d. $PbSO_4$

**Solution:**

a. Chromium (III) oxide
b. Lithium phosphide
c. Mercury (II) chloride
d. Lead (II) sulphate

122. The bond energies for H –A, H –B, H –C and H –D are 436, 565, 297 and 427 respectively in kJ/mol. Arrange these molecules in increasing order of boiling point.

**Answer:** H – C < H – D < H – A < H – B

123. Compare the concepts of hybrid and molecular orbitals with respect to:

i energy of the resulting orbitals with respect to one another

ii number of atoms involved from a given number of ground state orbitals.

iii the number of orbital produced.

**Solution:**

| hybrid orbitals | molecular orbitals |
|---|---|
| all have same energy | bonding MO has lower energy than anti bonding MO. |
| one centeral atom | more than one atom |
| same number | same number |

124. Sulphur forms both $SF_2$ and $SF_4$ molecules but oxygen forms only $OF_2$. Explain.

**Explanation:** Sulphur may form more than two bonds with F by promoting its 3p electrons to the empty 3d orbital to form an expanded octet but oxygen cannot do this due to absence of '2d' orbitals.

125. What are the preconditions for the formulation of covalent bond according to valence bond theory?

**Solution**

- When a half - filled orbital of one atom overlaps with a half - filled orbital of another atom.
- The total number of electrons in both orbitals is no more than two.

126. What is hybridization?

**Solution**

It is an imaginary mixing process in which the orbitals of an atom rearrange themselves to form new atomic orbitals called hybrid orbitals.

127. Explain the difference between bonding and anti bonding molecular orbitals.

**Solution**

- Bonding molecular orbitals results from the addition of atomic orbitals but anti bonding molecular orbitals result from subtraction of atomic orbitals.
- Bonding molecular orbitals have lower energy than atomic orbitals but anti bonding molecular orbitals have higher energy than atomic orbitals.

128. For each of the following pairs of ionic compounds, specify which compound has higher lattice energy.

a. KCl or CaO
b. LiF or LiI
c. $Mg_3N_2$ or NaCl
d. $AlF_3$ or $MgF_2$

**Solution**

a. CaO
b. LiF
c. $Mg_3N_2$
d. $AlF_3$

129. Arrange the following bonds in increasing order of polarity.

a. P – H, H – O, C – Cl
b. O – F, C – S, H – Br

**Solution**

a. P – H < C – Cl < H – O
b. O – F < C – S < H – Br

130. What is the electron sea model for metals?

**Solution**

The electron – sea model is a very simple model, which pictures the metal as an array of metal cations in a "sea" of electrons.

131. Arrange the following molecules in an increasing order of boiling point: HF, HCl, HBr and HI.

**Solution**

HI < HBr < HCl < HF

132. $N_2$ and CO are isoelectronic, which substance would you expect to have lower boiling point?

**Solution**

$N_2$ has lower boiling point because it is non – polar.

133. What type of atom or molecule has induced dipole?

**Solution**

An atom or molecule that is temporary polarized is said to be have an induced dipole.

134. $PH_3$ and $H_2S$ are isoelectric and their dipole moments are 0.56D and 0.95D respectively. Which substance have a higher boiling point?

**Solution**

$H_2S$ because it has higher dipole moment.

135. Explain the difference between amorphous and crystalline solids.

**Solution**

Crystaline solids
- Consist of individuals crystals, each with well defined shape.
- Cleave to give flat, well defined faces.
- Melt at definite temperature.

Amorphous solids
- are non - crystaline
- have no well defined shape
- break to give curved or irregular faces
- soften, then melt over a temperature range

136. What will happen to the energy of the system when a chemical bond is formed?

**Solution**

Energy of the system decreases.

137. List the types of bonds present in CO.

**Solution**

Sigma, Pi and dative bonds.

138. The hybrid states of carbon in diamond, graphite and acetyle respectively are

**Solution**

$sp^2$, $sp^3$, sp

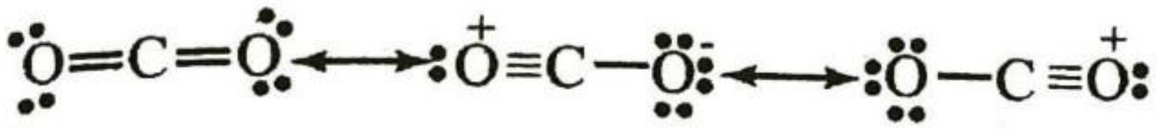
139. The order of O – O bond length in $O_2$, $H_2O_2$, $O_3$ is

**Solution**

$H_2O_2 > O_3 > O_2$

140. Write the resonance structures of $CO_2$.

**Solution**



c. Al(1)

d. NaCl(aq)

e. NaCl(s)

f. graphite

141. Given the dipole moment of the molecules.

| Molecules | Dipole moment |
|---|---|
| HF | 1.82 |
| HCl | 1.08 |
| HBr | 0.82 |
| HI | 0.44 |

Arrange these molecules in increasing order of:

a. Polarity

b. Bond length

**Explanation:**

a. As the value of dipole moment increases, polarity of covalent molecules also increases. Therefore, increasing order of polarity will be:
HI < HBr < HCl < HF

b. The higher the value of the dipole moment, the shorter is the bond length and the higher the bond strength. As a result, increasing order of bond length will be: HF < HCl < HBr < HI

142. Draw the Lewis structure for:

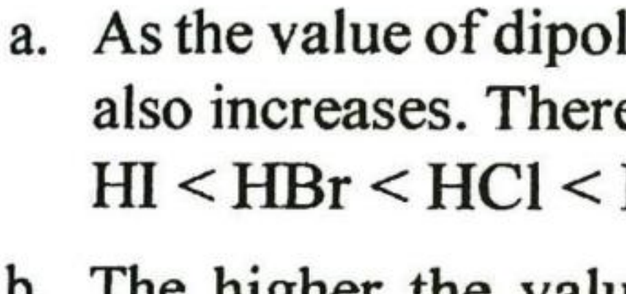
a. Formic acid, HCOOH

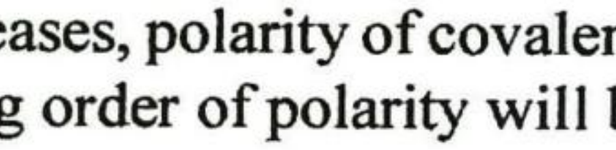
b. Hydrogen peroxide, $H_2O_2$

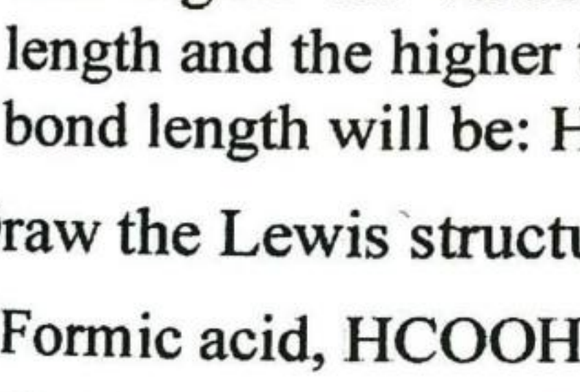
c. Chlorate ion, $ClO_3^-$

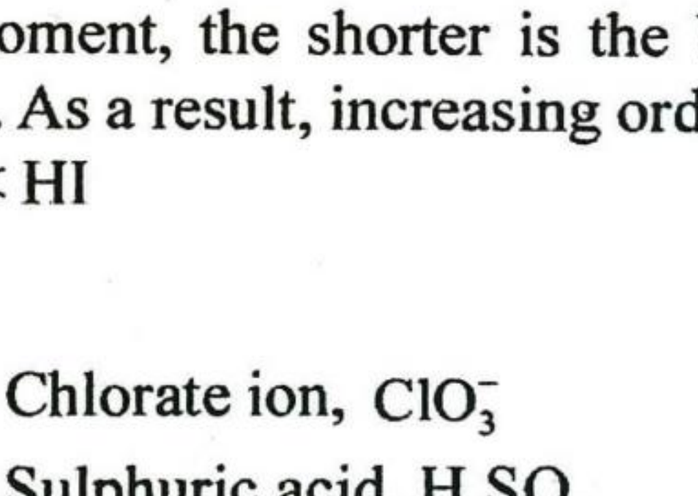
d. Sulphuric acid, $H_2SO_4$

**Explanation:**

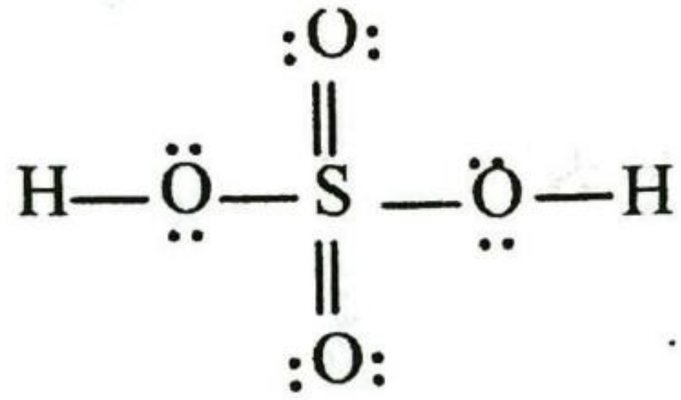
a. 

b. 

c. 

d. 

But the observed (actual) structure of $H_2SO_4$ is:



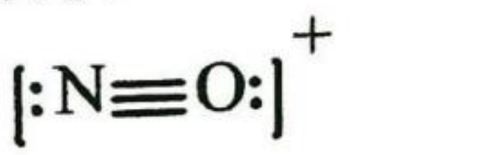
143. Draw the Lewis structure for the follow species and show formal charges.

a. $NO^+$ b. $NO^-_2$ c. $SCN^-$ d. $ClF^+_2$

**Explanation:**

Draw the Lewis structure by applying the rules.

a. NO+

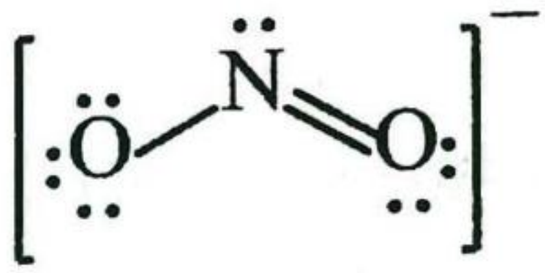


Formal charges for

$$N:\left[5-\left(2+\frac{1}{2}\times 6\right)\right]=0$$

$$O:\left[6-\left(2+\frac{1}{2}\times 6\right)\right]=+1$$

b. $NO_2$



Formal charges for:

$$\text{O atom (singly bonded)} =\left[6-\left(6+\frac{1}{2}\times 2\right)\right]=-1$$

$$\text{O atom (doubly bonded)} =\left[6-\left(4+\frac{1}{2}\times 4\right)\right]=0$$

$$\text{N atom} =\left[5-\left(2+\frac{1}{2}\times 6\right)\right]=0$$

c. $SCN^-$

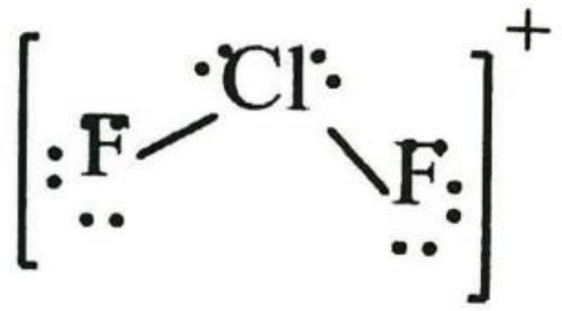
$$\left[:C{=}S{=}\ddot{N}:\right]^-$$

$$C:\left[4-\left(4+\frac{1}{2}\times 4\right)\right]=-2$$

$$S:\left[6-\left(0+\frac{1}{2}\times 8\right)\right]=+2$$

$$N:\left[5-\left(4+\frac{1}{2}\times 4\right)\right]=-1$$

d. $ClF^+_2$



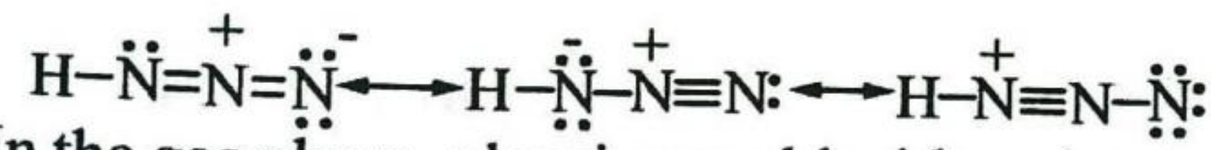
$$F:\left[7-\left(6+\frac{1}{2}\times 2\right)\right]=0$$

$$Cl:\left[7-\left(4+\frac{1}{2}\times 4\right)\right]=+1$$

144. Draw the resonance structures for

a. $HN_3$, hydrazoic acid

**Explanation:**

a. 

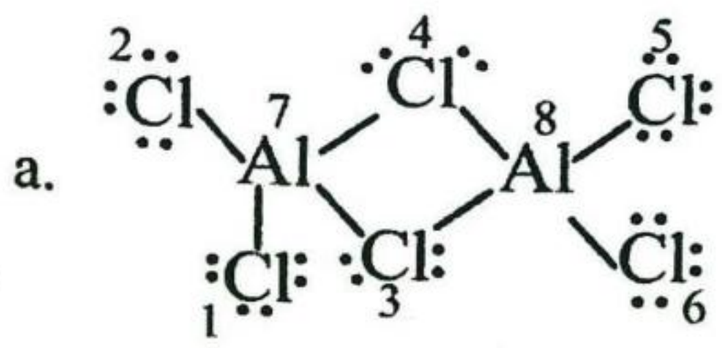
145. In the gas phase, aluminum chloride exists as a dimmer (a unit of two) with the formula $Al_2Cl_6$.

a. Draw the Lewis structure.

b. Determine the formal charge of each atom.

c. Indicate the coordinate covalent bond in the molecule.

**Explanation:**

a. 

b. Formal charges for:

$$Cl^1: \left[7-\left(6+\frac{1}{2}\times 2\right)\right]=0$$

$$Cl^2: \left[7-\left(6+\frac{1}{2}\times 2\right)\right]=0$$

$$Cl^3: \left[7-\left(4+\frac{1}{2}\times 4\right)\right]=+1$$

$$Cl^4: \left[7-\left(4+\frac{1}{2}\times 4\right)\right]=+1$$

$$Cl^5: \left[7-\left(6+\frac{1}{2}\times 2\right)\right]=0$$

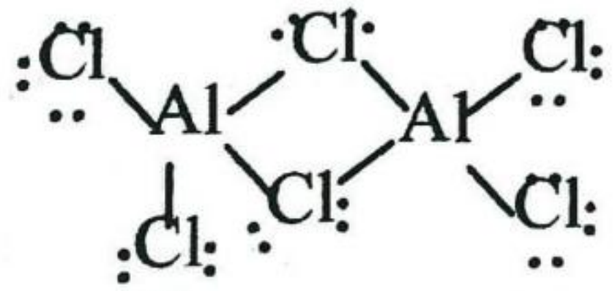
$$Cl^6: \left[7-\left(6+\frac{1}{2}\times 2\right)\right]=0$$

$$Al^7: \left[3-\left(0+\frac{1}{2}\times 8\right)\right]=-1$$

$$Al^8: \left[3-\left(0+\frac{1}{2}\times 8\right)\right]=-1$$

The sum of formal charges is zero

c.




146. How many orbitals does the following hybrid orbitals contain?

a. $sp^2$

b. $sp^3d$

c. $sp^3d^2$

**Explanation:**

a. Three

b. five

c. six

147. How many P atomic orbitals are required to generate a set of the following hybrid orbitals?

a. Sp  b. $sp^2$  c. $sp^3d$

**Explanation:**

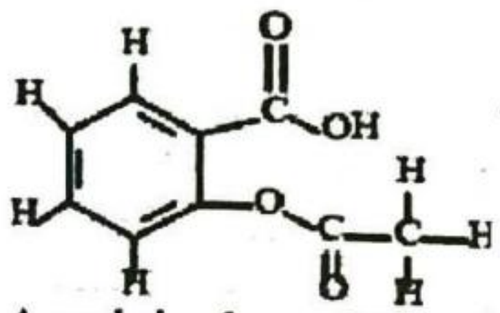
a. One  b. two  c. three

148. Determine the number of sigma and pi bonds in

a. Aspirin  b. Acetaminophen

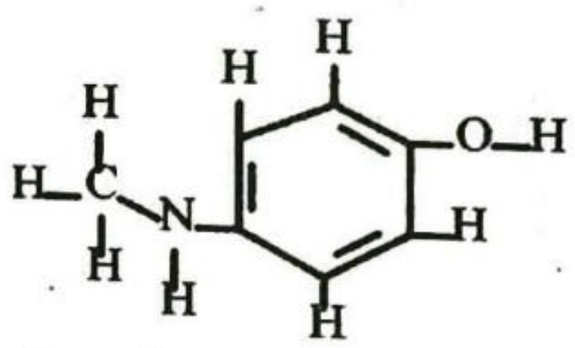
**Explanation:** Draw the Lewis structure for each.

a.



Aspirin has 21 σ bonds and 5π bonds.

b.



Acetaminophen has 20 σ bonds and 3π bonds.

149. Write the formula and name of the ionic compound formed between the reaction of:

a. Magnesium and nitrogen (EUEE, 2005 E.C)

b. Calcium and sulphur

c. Potassium and phosphorus

**Explanation:** Determine the valence of magnesium and nitrogen. Write electron configuration for group IA and IIA metals, their valency are 1 and 2 respectively.

Valence of nitrogen = 8 – valence electrons = 8 – 5 = 3

Valence of sulphur = 8 – valence electrons = 8 – 6 = 2

Valence of phosphorus = 8 – valence electrons = 8 – 5 = 3

Therefore, use criss – cross method, to write the formula of the ionic compounds.

a. $Mg^{2} \times N^{3} = Mg_3N_2$ – magnesium nitride

b. $Ca_2 \times S_2 = CaS$ – calcium sulphide

c. $K^{1} \times P = K_3P$ – potassium phosphide

150. What is the type of interaction force in ionic compounds responsible for their solubility?

**Solution**

Ion - dipole force

## Supplementary exercise

### I. Multiple choice type

1. Which of the following substances has the least ionic character in its bond?

   A. NaF B. $CF_4$ C. $MgF_2$ D. KF

2. The orbitals providing the most efficient overlap are

   A. s – s B. s – p C. sp – sp D. p – p

3. What is the bond order of $H_2^-$ according to the morcular orbital theory?

   A. 1 C. 2

   B. 1.5 D. 0.5

4. The structure of $ClO_3F$ is

   A. tetrahedral C. square – planar

   B. trigonal-planar D. trigonal bipysmidal

5. Which of the following compounds has the highest melting point?

   A. $MgF_2$ B. $CaF_2$ C. $SrF_2$ D. $BaF_2$

6. The bond formed when ammonia reacts with aluminum chloride is

   A. Ionic bond C. metallic bond

   B. covalent bond D. dative bond

7. All of the following have the same molecular shape except one. Identity it

   A. $CHCl_3$ B. $CH_4$ C. $NH_3$ D. $PH_4^+$

8. Identify the molecule that does not have double or triple bond.

   A. $CO_2$ B. $Cl_2$ C. CO D. $N_2$

9. Only one has linear geometry, which one?

   A. $OF_2$ B. $CS_2$ C. $H_2O$ D. $CH_4$

10. Which of the following statements is NOT true?

   A. The reason why water is liquid at room temperature while hydrogen sulphide is gas is due to hydrogen bond in the former

   B. The nitrogen in ammonia has a lone pair of electrons which can be used to form coordinate covalent bond

   C. A bond formed between two elements of differing electronegativity is referred to as a polarized bond.

   D. Electronegative element such as sulphur and chlorine are involved in hydrogen bonding.

11. In which series are compounds arranged in order of decreasing ionic character?

   A. NaI > NaBr > NaCl > NaF C. NaCl > NaI > NaCl > NaF

   B. NaBr > NaI > NaCl >NaF D. NaF > NaCl > NaBr > NaI

12. How many pairs of unshared valence electrons are on the centeral atom of $PCl_3$?

   A. 1 B. 0 C. 2 D. 3

13. Which of the following molecules is non-polar?

   A. $CS_2$ C. $CHCl_3$

   B. $SO_2$ D. $SF_4$

14. Which one of the following represents the correct electron-pair geometry and molecular geometry of $Cl_2O$?

   A. Linear, Linear

   B. Tetrahedral, bent

   C. Trigonal planar, T-shaped

   D. Square planar, linear

15. Which of the following molecules is the most stable molecular species?

   A. $O_2^+$ B. $O_2^-$ C. $CN^-$ D. $O_2^{2-}$

16. Which of the following has the shortest bond length?

   A. $O_2$ B. $N_2$ C. $H_2$ D. $F_2$

17. A neutral molecule having the general formula $AB_3$ has two unshared pair of electrons on A. what is the hybridization of A?

A. sp B. $sp^2$ C. $sp^3$ D. $sp^3d$

18. Given $CF_4$, $CS_2$, $PH_3$, $PCl_5$ and $SF_6$. Which of the following does not describe any of the molecules above?

A. Linear
B. Octahedral
C. Square planar
D. Tetrahedral

19. Which of the following molecules has the greater number of lone pair of electrons?

A. $CCl_4$ B. $NCl_3$ C. $BCl_3$ D. $BeCl_2$

20. The strongest of the intermolecular forces is

A. dipole-dipole
B. dipole – induced diole
C. London force
D. Hydrogen bond

21. Which of the following contains both ionic and dative bond?

A. HCl B. $NH_4Cl$ C. NaCl D. $CCl_4$

22. Which of the following has the largest bond angle?

A. $OF_2$ B. $BF_3$ C. NaF D. $NF_3$

23. Which of the following has the largest bond angle?

A. $OF_2$ B. $BeCl_2$ C. $NH_4^+$ D. $NO_3^-$

24. Which of the following is the strongest?

A. covalent bond
B. dipole-dipole force
C. Hydrogen bond
D. London force

25. The most important reason for bond formation is

A. noble gas electronic structure
B. electro static attraction
C. lowering of energy
D. electron pair sharing

26. Which of the following has the highest boiling point?

A. Xe B. Ar C. Kr D. Ne

27. Which of the following compounds has most likely been formed by covalent bonding of atoms?

A. CaO B. $SiO_2$ C. NaCl D. $MgCl_2$

28. Which of the following violates the octet rule?
    A. $XeF_2$ B. $OF_2$ C. $PF_3$ D. $SF_2$
29. Identify the compound formed by an overlap of $sp^3$ and s orbitals?
    A. $BeH_2$ B. $BCl_3$ C. $CH_4$ D. $PCl_2$
30. Which of the following has the highest melting point?
    A. Ca B. Na C. K D. Mg

II. Answer the following questions

31. Which one in each pair is expected to exhibit hydrogen bonding
    a. $CH_3OH$ or $CH_3OCH3$
    b. $CH_3SH$ or $CH_3NH_2$
    c. $(CH_3)_2NH$ or $(CH_3)3N$
32. Deduce the type of hybridization on the central atom
    a. $XeF_2$
    b. $XeF_4$
    c. $BrF_6^-$
    d. $NO_2^-$
33. Which one is more stable
    a. $NO^+$ or NO
    b. CO or $CO^+$
    c. CN or $CN^-$
    d. $N_2$ or $N_2^-$
34. Write two diatomic molecules in period 2 that are paramagnetic.
35. State homonuclear diatomic molecules of the second period elements which are paramagnetic.

Write the type of hybrid orbitals for molecules that possess the following molecular shape

    a. Linear
    b. Trigonal planar
    c. tetrahedral
    d. Trigonal bipyraidal
    e. octahedral

36. Which type of intermolular force is dominant in the following substances?
    a. $IF_3$ b. $N_2$ c. $NH_3$ d. HCl
37. Name the following compounds
    a. $AlBr_3$
    b. RbI
    c. BeO
    d. $Ba_3N_2$

38. Show the direction of the bond moment (dipole moment) in each of the following bonds

H – F, N – H, N – O, C – Cl

39. Compare the following in increasing order of boiling point

$CH_4$, $C_2H_6$, $C_3H_8$, $C_4H_{10}$

40. Classify the following as conductors or non-conductors.

a. NaCl(1)
b. Al(s)
c. Nacl(aq)
d. graphite
e. $C_2H_5OH$
f. HCl(aq)

አጋዥ መጽሐፍቶችን ለማግኘት በ
Telegram @ethiolibrary1 ላይ
ይቀላቀሉን
+251927273995
+251909282436
jemal6454@gmail.com

አጋዥ መጽሐፍቶችን ለማግኘት በ Telegram
@ethiolibrary1 ላይ ይቀላቀሉን
+251927273995
+251909282436
jemal6454@gmail.com

# Unit 4

## Chemical Kinetics

## Unit Content Outline:

4.1 Rate of chemical reaction

4.2 Determination of rate of reaction

4.3 Factors affecting rate of reaction

4.4 Theories of reaction rates

4.5 Rate equation and order of reaction

4.6 Concentration - time equation

4.7 Reaction mechanism

## Introduction

Some reactions are fast others are slow.

### Example

Explosion of gun powder, reaction of sodium with water are fast reactions.

➦ Fermentation, rusting of iron and weathering are slow reactions.

**Chemical kinetics** is the study of the rate of reaction under various conditions and how reaction - rate data are used to deduce probable reaction mechanisms.

## 4.1 Rate of Chemical reaction

**Rate of reaction** is the change in concentration of reactant or product per second.

$$\text{Rate} = \frac{\text{change in concentration of reactant}}{\text{change in time}}$$

or

$$\text{Rate} = \frac{\text{change in concentration of product}}{\text{change in time}}$$

For a general reaction: $B \rightarrow C$

$$\text{Rate} = \frac{-\Delta[B]}{\Delta t} = \frac{\Delta[C]}{\Delta t}$$

$$\Delta[B] = [B]_f - [B]_i$$
$$\Delta[C] = [C]_f - [C]_i$$

- The subscript f and i are final and initial states.
- [ ] is concentration usually expressed in mole per liter.
- Rate is scalar quantity, that is why there is a negative sign infront of the change in concentration of reactant.
- As time goes, concentration of reactant decreases and concentration of product increases.

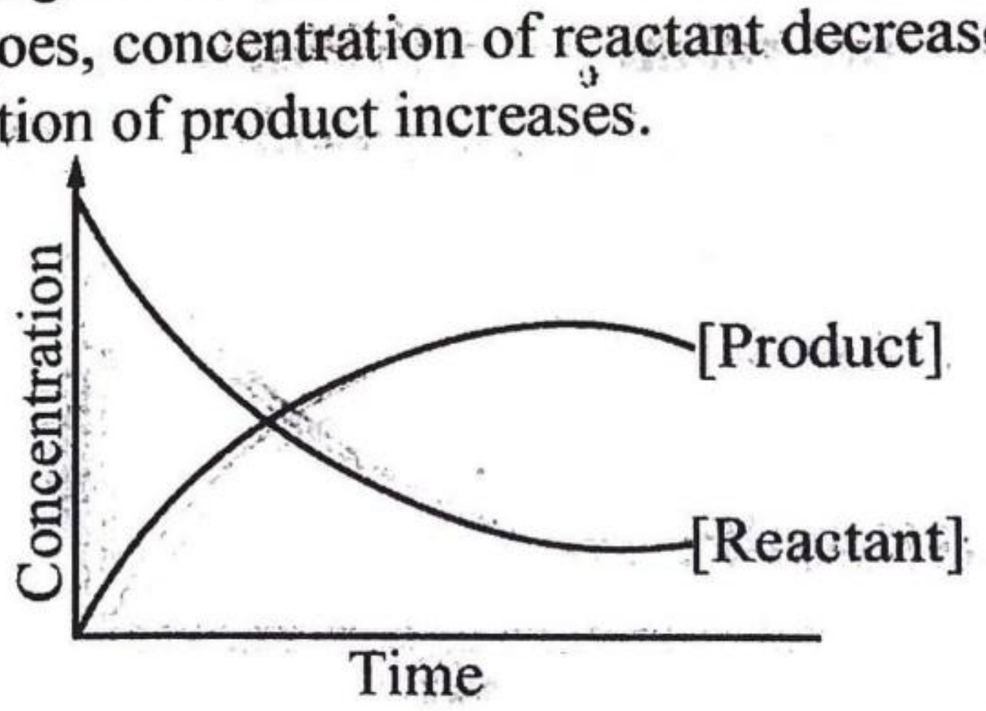



Figure 4.1 A plot showing the concentration of reactant and product as a function of time

# Rate and Stoichiometry

For a hypotetical reaction: $aA + bB \rightarrow cC + dD$

$$\text{Rate} = \frac{-\Delta[A]}{a\Delta t} = \frac{-\Delta[B]}{b\Delta t} = \frac{\Delta[C]}{c\Delta t} = \frac{\Delta[D]}{d\Delta t}$$

$$\Rightarrow \frac{\Delta[A]}{\Delta t} = \frac{a}{b}\frac{\Delta[B]}{\Delta t} = -\frac{a}{c}\frac{\Delta[C]}{\Delta t} = -\frac{a}{d}\frac{\Delta[C]}{\Delta t}$$

$$\frac{\Delta[B]}{\Delta t} = \frac{b}{a}\frac{\Delta[A]}{\Delta t} = -\frac{b}{c}\frac{\Delta[C]}{\Delta t} = -\frac{b}{d}\frac{\Delta[D]}{\Delta t}$$

$$\frac{\Delta[C]}{\Delta t} = -\frac{c}{a}\frac{\Delta[A]}{\Delta t} = -\frac{c}{b}\frac{\Delta[B]}{\Delta t} = \frac{c}{d}\frac{\Delta[D]}{\Delta t}$$

$$\frac{\Delta[D]}{\Delta t} = -\frac{d}{a}\frac{\Delta[A]}{\Delta t} = -\frac{d}{b}\frac{\Delta[B]}{\Delta t} = \frac{d}{c}\frac{\Delta[C]}{\Delta t}$$

## Illustrative Example

1. Which of the following expressions is correct for the rate of a reaction given below? (EUEE ... 2008 E.C.)

$$6H^+ + BrO_3^- + 5Br^- \rightarrow 3Br_2 + 3H_2O$$

A. $\frac{\Delta[Br^-]}{\Delta t} = 5\frac{\Delta[H^+]}{\Delta t}$

B. $\frac{\Delta[Br^-]}{\Delta t} = \frac{5}{6}\frac{\Delta[H^+]}{\Delta t}$

C. $\frac{\Delta[Br^-]}{\Delta t} = \frac{6}{5}\frac{\Delta[H^+]}{\Delta t}$

D. $\frac{\Delta[Br^-]}{\Delta t} = 6\frac{\Delta[H^+]}{\Delta t}$

**Solution**

$$\frac{\Delta[Br^-]}{\Delta t} = 5\frac{\Delta[BrO_3^-]}{\Delta t} = \frac{5}{6}\frac{\Delta[H^+]}{\Delta t} = \frac{-5}{3}\frac{\Delta[Br_2]}{\Delta t}$$

**Answer: B**

2. Potassium chlorate, $KClO_3$ decomposes according to the reaction: $2KClO_3 \rightarrow 2KCl + 3O_2$. If the rate of decomposition of $KClO_3$ at a certain time is determined to be $2.2 \times 10^{-2}$mol/L.s, what is the rate of formation of $O_2$ (EUEE)

A. $1.1 \times 10^{-2}$

B. $2.2 \times 10^{-2}$

C. $3.3 \times 10^{-2}$

D. $4.4 \times 10^{-2}$

**Solution**

$$\text{Rate of } O_2 = \frac{\text{coefficient of } O_2}{\text{Coefficient } KClO_3} \times \text{rate } KClO_3$$

$$= \frac{3}{2} \times 2.2 \times 10^{-2}\,\text{mol.L}^{-1}.\text{s}^{-1}$$

$$= 3.3 \times 10^{-2}\,\text{mol.L}^{-1}.\text{s}^{-1}$$

**Answer: C**

3. Given the reaction: $C_3H_8(g) + 5O_2(g) + \rightarrow 3CO_2(g) + 4H_2O(g)$

   a. Write rate expression interms of reactants and products.

   b. If the rate of formation of $CO_2$ is 0.30 $mol^{-1}L^{-1}s^{-1}$, calculate the rate of rest.

   **Solution**

   a. $$\text{rate} = \frac{-\Delta[C_3H_8]}{\Delta t} = \frac{-\Delta[O_2]}{5\Delta t} = \frac{\Delta[CO_2]}{3\Delta t} = \frac{\Delta[H_2O]}{4\Delta t}$$

   b. $$\text{rate } C_3H_8 = \frac{1}{3} \times 0.3 = 0.1 mol.L^{-1}.s^{-1}$$

   $$\text{rate } O_2 = \frac{5}{3} \times 0.3 = 0.5 mol.L^{-1}.s^{-1}$$

   $$\text{rate } O_2 = \frac{5}{3} \times 0.3 = 0.5\, mol.L^{-1}.s^{-1}$$

   $$\text{rate } H_2O = \frac{4}{3} \times 0.3 = 0.4\, mol.L^{-1}.s^{-1}$$

4. What is the unit of rate of a reaction if the unit of time is second?

   **Solution**

   $$\text{unit of rate} = \frac{\text{unit of concentration}}{\text{unit of time}} = mol/L.s = mol.L^{-1}.s^{-1}$$

5. At a given set of conditions, the average rate of appearance of oxygen gas in the reaction: $2O_3(g) \rightarrow 3O_2(g)$ is $3.6 \times 10^{-3}$ atm/s. What is the average rate of the disappearance of $O_3$ expressed in units of atm/s.

   **Solution**

   $$\text{Rate } O_3 = \frac{2}{3} \times \text{rate of } O_2$$

   $$= \frac{2}{3} \times 3.6 \times 10^{-3} \text{ atm/s}$$

   $$= 2.4 \times 10^{-3} \text{atm/s}$$

6. For the synthesis of ammonia (Haber process), $N_2 + 3H_2 \rightleftharpoons 2NH_3$, the rate of the reaction was measured as:

   $$\frac{\Delta[NH_3]}{\Delta t} = 2 \times 10^{-4} mol.L^{-1}s^{-1}$$

   what is the rate of the reaction expressed in terms of $H_2$?

   **Solve by your own: Answer: $3 \times 10^{-4} mol.L^{-1}s^{-1}$**

7. Write rate expression for the following reactions.

a. $4NH_3 + 7O_2 \rightarrow 4NO_2 + 6H_2O$

b. $2N_2O_5 \rightarrow 4NO_2 + O_2$

**Solution**

a. $r = \frac{-\Delta[NH_3]}{4\Delta t} = \frac{-\Delta[O_2]}{7\Delta t} = \frac{\Delta[NO_2]}{4\Delta t} = \frac{-\Delta[H_2O]}{6\Delta t}$

b. $r = \frac{-\Delta[N_2O_5]}{2\Delta t} = \frac{\Delta[NO_2]}{4\Delta t} = \frac{\Delta[O_2]}{\Delta t}$

8. At some point in the reaction:

$4A + B \rightarrow 2C + D$, $[A] = 0.5M$, at a time 10 min later $[A] = 0.4M$. What is the rate of the reaction during this time interval, expressed in $mol.L^{-1}s^{-1}$?

**Solution**

$[A]_i = 0.5M \qquad [A]_f = 0.4M, t = 10\ min = 600s$

$$\text{Rate} = \frac{-\Delta[A]}{4\Delta t} = \frac{-([A]_f - [A]_i)}{4 \times (600s - 0)} = 4.17 \times 10^{-5}\ mol.L^{-1}.s^{-1}$$

9. Thiosulfate ion is oxidized by iodine according to the following reaction:

$$2S_2O_3^{2-}(aq) + I_2(aq) \longrightarrow S_4O_6^{2-}(aq) + 2I^-(aq)$$

If, in a certain experiment, 0.006mol of $S_2O_3^{2-}$ is consumed in 1.0L of each second, at what rates are $S_4O_6^{2-}$ and $I^-$ produced in this solution?

**Solution**

$$[S_2O_3^{2-}] = \frac{n}{v} = \frac{0.006mol}{1L} = 0.006mol/L$$

$$\text{Rate of } S_2O_3^{2-} = \frac{-\Delta[S_2O_3^{2-}]}{2\Delta t} = \frac{1}{2}\left(\frac{0.006\ mol/L}{1sec}\right)$$

$$= 0.003\ mol.\ L^{-1}.s^{-1}$$

$$\Rightarrow \text{Rate of } S_4O_6^{2-} = \frac{1}{2} \times \text{rate of } S_2O_3^{2-}$$

$$= \frac{1}{2} \times 0.003 mol.L^{-1}.s^{-1}$$

$$= 1.5 \times 10^{-3} mol.L^{-1}.s^{-1}$$

$$\text{Rate of } I^{-1} = \frac{2}{2} \times \text{rate of } S_2O_3^{2-} = 0.003 \text{mol.L}^{-1}.\text{s}^{-1}$$

10. For the reaction: $A + B \rightarrow 2C$, the rate of the reaction may be expressed as: $r = \frac{-\Delta[A]}{\Delta t}$. What does the negative sign indicate?

**Explanation**

➦ Concentration of A decreases with time.

11. Define chemical kinetic and write its two main concerns.

**Solution**

➦ Chemical kinetic is the branch of chemistry that deals with the study of reaction rates and their mechanisms.

➦ It is mainly concerned on the rate and mechanism of reactions.

12. What is the rate of reaction?

**Solution**

Rate of a reaction is expressed in terms of decrease of concentration of one of the reactant or rate of increase of concentration of one of the product.

## 4.2 Determination of rate of reaction

➦ To determine a rate of a reaction, we need to measure changes in concentration over time.

➦ A change in time can be measured with a stop watch or other limiting device.

➦ An accurate, quantitative estimate of a rate of reaction at any given time can be obtained from the slope of the tangent to the concentration time graph at that particular instant. **Eg.** $A \rightarrow B$

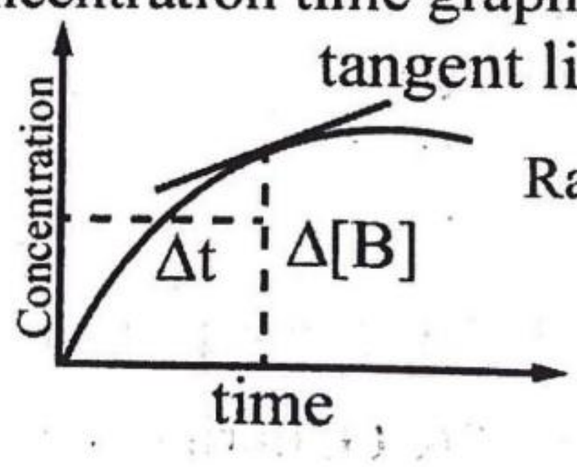



Figure 4.2 Estimation of the rate of a reaction based on $\Delta[B]$ with time

➦ The rate of the reaction is greatest at the beginning and slows down with time.

➦ Some of the methods used to determine the concentration of reactants or products are:

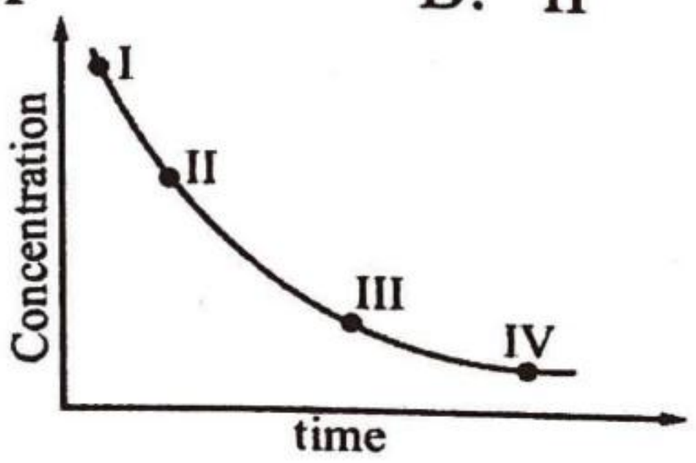
- Amount of precipitate formed
- Color change

- Gain or loss of weight
- Increase or decreases in volume or pressure for gases.

## Illustrative Examples

13. Given the concentration of reactant versus time graph. At which point the rate of a reaction is maximum.

A. I B. II C. III D. IV




**Solution**

The rate of a reaction is maximum at the beginning of a reaction.

**Answer: A**

14. How can the rate of a reaction be measured?

**Explanation:**

By following the loss of reactant or formation of product.

15. Given the reaction: $N_2O_5(g) \rightarrow 2NO_2(g) + \frac{1}{2}O_2(g)$. Based on the given data below, answer the following questions.

| time in min | $[N_2O_5]$, M | $[NO_2]$, M | $[O_2]$, M |
|---|---|---|---|
| 0 | $1.24\times10^{-2}$ | 0 | 0 |
| 10 | $0.92\times10^{-2}$ | $0.64\times10^{-2}$ | $0.16\times10^{-2}$ |
| 20 | $0.68\times10^{-2}$ | $1.12\times10^{-2}$ | $0.28\times10^{-2}$ |
| 30 | $0.50\times10^{-2}$ | $1.48\times10^{-2}$ | $0.37\times10^{-2}$ |
| 40 | $0.37\times10^{-2}$ | $1.74\times10^{-2}$ | $0.44\times10^{-2}$ |
| 50 | $0.28\times10^{-2}$ | $1.92\times10^{-2}$ | $0.48\times10^{-2}$ |

a. Calculate the rate of formation of $NO_2$ over the first 10 min.

b. Calculate the rate of decomposition of $N_2O_5$ during the time interval t = 30 min to t = 50 min.

c. Calculate the rate of formation of $O_2$ over the first 20 min.

**Solution**

a. $r = \frac{\Delta[NO_2]}{\Delta t} = \frac{[NO_2]_f - [NO_2]_i}{t_f - t_i}$

$$r = \frac{0.64\times10^{-12} - 0}{(10-0)\,\text{min}} = 6.4\times10^{-4}\,\text{mol.L}^{-1}.\text{min}^{-1}$$

b. $$r = \frac{-\Delta[N_2O_5]}{\Delta t} = \frac{-([N_2O_5]_f - [N_2O_5]_0)}{t_f - t_i}$$

$$r = \frac{-(0.28\times10^{-2} - 0.50\times10^{-2})}{50-30} = 1.10\times10^{-4}\,\text{mol.L}^{-1}.\text{min}^{-1}$$

c. $$r = \frac{2\Delta[O_2]}{\Delta t} = \frac{2(0.28\times10^{-2} - 0.50\times10^{-2} - 0)}{20-0} = 2.8\times10^{-4}\,\text{mol.L}^{-1}.\text{s}^{-1}$$

16. What is instantaneous rate of reaction?

**Solution**

Instantaneous rate is the rate of the reaction determined from the slope of tangent line to a concentration time curve.

## 4.3 Factors Affecting the rate of reactions

➦ There are different factors that affect the rate of reactions. These are:

### 1. Concentration of Reactants

➦ As concentration of reactants increases rate of reaction also increases except for **zero - order reaction**.

➦ For gaseous reactants, as partial pressure of gas increases, rate of the reaction also increases.

**Example**

$Mg + 2HCl\ (1M) \rightarrow MgCl_2 + H_2$ ... slow

$Mg + 2HCl\ (3M) \rightarrow MgCl_2 + H_2$ ... fast

### 2. Nature of reactants

➦ The more reactive the reactant, the faster is the reaction.

**Example**

$2Na + 2H_2O \rightarrow 2NaOH + H_2$ ... fast

$Ca + 2H_2O \rightarrow Ca(OH)_2 + H_2$ ... slow

$\Rightarrow$ Na is more reactive than Ca.

### 3. Surface area

➦ It affects the rate of reactions involving solid reactants.

- As surface area increases, the rate of a reaction also increases.
- Powdering or grinding increases surface area.

**Example**

$CaCO_3(chips) + 2HCl \rightarrow CaCl_2 + CO_{.2} + H_2O$ ... slow

$CaCO_3(powder) + 2HCl \rightarrow CaCl_2 + CO_2 + H_2O$ ... fast

- **A homogeneous reaction** is a reaction occurring in one phase.

**Example**

$N_2(g) + 3H_2(g) \rightarrow 2NH_3(g)$

$HI(aq) + H_2O(l) \rightarrow H_3O^+(aq) + I^-(aq)$

- **Heterogeneous reaction** is a reaction that involves different phases.

**Example**

$Mg(s) + 2HCl(aq) \rightarrow MgCl(aq) + H_2(g)$

$CaCO_3(s) + H_2SO_4(aq) \rightarrow CaSO_4(aq) + CO_2(g) + H_2O(l)$

**4. Temperature of the reactants**

- As temperature of reactants increases the rate of the reactions also increases.

**5. Catalysts** are substances that raise the rate of a reaction by lowering activation energy.

- **Activation energy (Ea)** is the minimum extra energy above and over the average potential energy of the reactants which must be supplied to the reactants to enable them to cross over the energy barrier between reactants and products.
- It depends on the chemical nature of reactants.
- It doesn't change with temperature.
- It is always positive.
- Reactants with smaller activation energy are fast whereas with high activation energy are slow.

⇒ Some reactions are stable in the absence of catalyst.

**Example**

$2H_2O_2(l) \rightarrow 2H_2O(l) + O_2(g)$

- An insect called **a bombardier beetle** uses this reaction as its defense mechanism. The beetle uses biological catalyst called an enzyme to produce a stream of super heated steam to spray on its enemies.

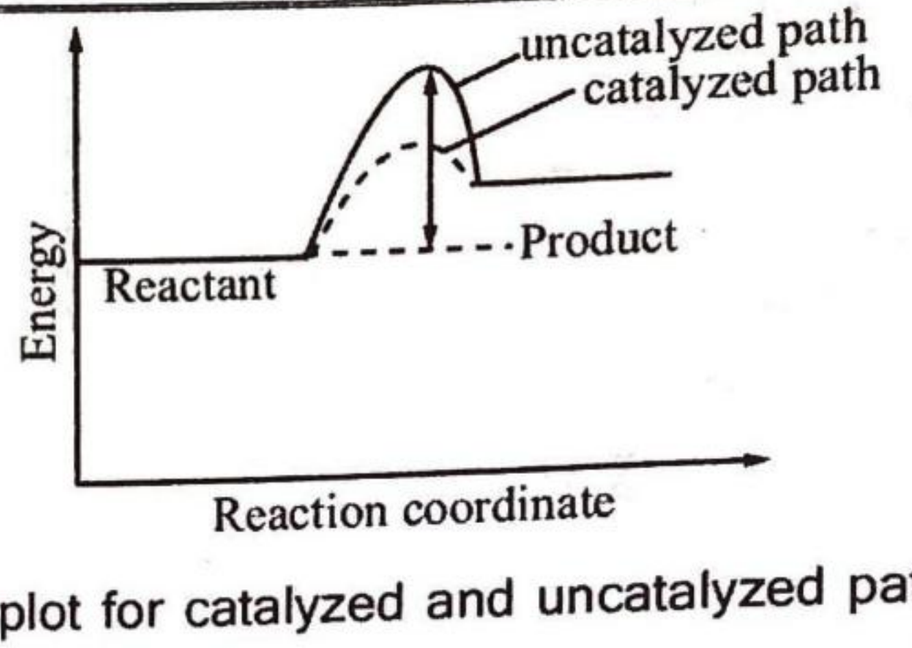



Figure 4.3 Energy plot for catalyzed and uncatalyzed path way for a reaction

## Types of Catalysts

There are three types of catalysts.

1. **Homogeneous catalyst** has similar phase with reactants and products.

**Example**

$$CH_3OH(aq) + CH_3COOH(aq) \xrightarrow{H^+(aq)} CH_3COOCH_3(aq) + H_2O(l)$$

2. **Heterogeneous catalyst** presents in different phases from reactants and products.

**Example**

$$2SO_2(g) + O_2(g) \xrightarrow{V_2O_5(s)} 2SO_3(g)$$

➦ Heterogeneous catalysts provide surface area for the reactants to collide.

3. **Enzymes** are biological catalysts that speed up biochemical reactions in living things.

➦ Enzymes are effective and specific.

Inhibitors are negative catalysts that slow down the rate of reactions.

## Illustrative Examples

17. How does a catalyst increases the rate of a chemical reaction? (EUEE)
    A. by shifting the equilibrium to the right
    B. by lowering activation energy
    C. by shifting the equilibrium to the left
    D. by increasing the activation energy

**Solution**

➦ Catalysts increase the rate of a reaction by lowering activation energy.

**Answer: B**

18. Which of the following does not affect the rate of a chemical reaction?
    A. Enthalpy of reaction
    B. Surface area
    C. Concentration of reactants
    D. Temperature

**Answer: A**

19. A homogeneous liquid reaction mixture is often heated to increase the rate of reaction. This is best explained by the fact that raising the temperature (EUEE)
    A. increase the heat of reaction
    B. decrease the energy of activation
    C. increase the vapor pressure of the liquid
    D. increase the average kinetic energy of the reactants

**Solution**

Temperature is the measure of the average kinetic energy of particles.

**Answer: D**

20. The minimum amount of energy that the reacting molecules must posses at the time of collisions in order to produce effective collisions is
    A. Activation energy
    B. Potential energy
    C. Free energy
    D. Internal energy

**Answer: A**

21. The value of activation energy is primary determined by _____

**Answer:** Chemical nature of reactants and products.

22. The catalyst remains unchanged at the end of the reaction in terms of _____ and _____.

**Solution**

➦ Chemical composition and quantity or amount.

23. What is the difference between positive and negative catalysts.

**Solution**

➦ Positive catalysts speed up a reaction but negative catalysts slow down the rate of a reaction.

24. Mention some effects of temperature that can be experienced in your daily life.

**Solution**

- ➦ Food is cooked by raising temperature
- ➦ Food is preserved by lowering temperature.
- ➦ During surgery, the body of the patient should be at low temperature to slow down the rates of biochemical reactions.

25. Give five ways used to increase reaction rates.

**Solution**

- ➦ Increase concentration of reactants.
- ➦ Raising temperature of reactants.
- ➦ Increasing the pressure of gaseous reactants.
- ➦ Increasing surface area of solid reactants.
- ➦ Using positive catalysts.

## 4.4 Theories of Reaction Rates

- ➦ There are two theories used to explain reaction rates.

### Collision Theory

- ➦ It is a theory of reaction rates that assumes molecules must collide in order to react.
- ➦ For a reaction to occur:
  - the particles of the reactants must collide
  - the particles of the reactants must have proper orientation
  - the particles of the reactanting species must collide with energy greater than or equal to activation energy
- ➦ There are two types of collisions. These are

1. **Effective collisions** - results a reaction to form desired products.
   - ➦ Reactants collide with sufficient energy to break or rearrange bonds and proper orientation.
2. **Ineffective collision** - do not result in a reaction to form desired products and reactants have no proper orientations.

## Concentration and Collisions Theory

- Concentration is directly proportional to number of collisions.
- As the number of collisions increases the rate of a reaction also increases.

## Temperature and Collision Theory

- As temperature increases, average kinetic energy of particles also increases. This increases the number of collisions.
- The value of rate constant, k is affected by temperature. This was explained by **Arrhenius equation**.

$$\boxed{k = Ae^{-\frac{E_a}{RT}}}$$

k - rate constant

A - frequency factor expressed in L. $mol^{-1}.s^{1}$

R - universal gas constant ($8.314 J.mol^{-1}.K^{-1}$)

T - temperature in kelvin

$E_a$ - activation energy

- Take natural logarithm of each side of the equation:

$$\ln k = \ln Ae^{-\frac{E_a}{RT}}$$

$$\ln k = \ln A + \ln e^{-\frac{E_a}{RT}}$$

$$\ln k = -\frac{E_a}{RT} + \ln A$$

This is an equation of a straight line of form: $y = mx + b$.

- The slope of the line is $-\frac{E_a}{R}$
- If we plot ln k versus $\frac{1}{T}$, we will get decreasing graph.

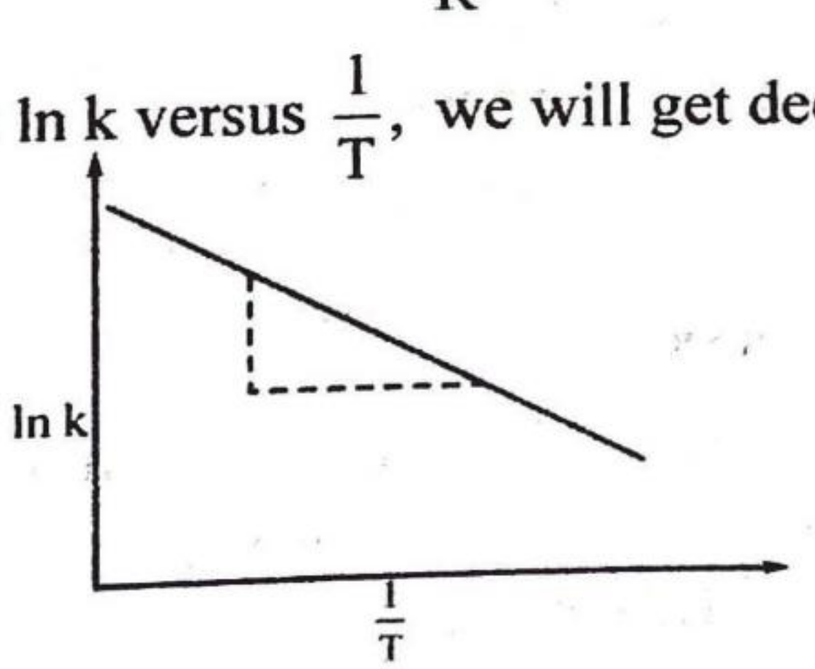



Figure 4.4 ln k vs $\frac{1}{T}$

**Note**

From the slope of the line it is possible determine the activation energy.

**Example**

Derive a formula for the above Arrhenius equation at two temperature, $T_1$ and $T_2$.

**Solution**

$$\ln k_1 = -\frac{E_a}{RT_1} + \ln A \ldots(1)$$

$$\ln k_2 = -\frac{E_a}{RT_2} + \ln A \ldots(2)$$

Subtract (1) from (2)

$$\Rightarrow \ln k_2 - \ln k_1 = \left(\frac{-Ea}{RT_2} + \ln A\right) - \left(\frac{-Ea}{RT_1} + \ln A\right)$$

$$\boxed{\ln \frac{k_2}{k_1} = \frac{Ea}{R}\left(\frac{1}{T_1} - \frac{1}{T_2}\right)}$$

$$\Rightarrow \ln \frac{k_2}{k_1} = 2.303109 \frac{k_2}{k_1}$$

$$2.30310g \frac{k_2}{k_1} = \frac{Ea}{R}\left(\frac{1}{T_1} - \frac{1}{T_2}\right)$$

$$\boxed{10g \frac{k_2}{k_1} = \frac{Ea}{2.303R}\left(\frac{1}{T_1} - \frac{1}{T_2}\right)}$$

**Example**

Calculate the activation energy for the decomposition of hydrogen iodide at two different temperatures.

$2HI\,(g) \rightarrow H_2(g) + I_2(g)$

$k_1 = 2.15 \times 10^{-8} L.mol^{-1}.s^{-1}$ at 377°C and

$k_2 = 2.39 \times 10^{-7} L.mol^{-1}.s^{1}$ at 427°k

**Solution**

$T_1 = 377°C = 377 + 273 = 650\ K$

$T_2 = 427°C = 427 + 273 = 700\ K$

$$Ea = \frac{\left(2.303\,R \log \frac{k_2}{k_1}\right).T_1T_2}{T_2 - T_1}$$

$$Ea = \frac{\left(2.303 \times 8.314 J.mol^{-1}.k^{-1} \log \frac{2.39 \times 10^{-7}}{2.15 \times 10^{-8}}\right) \times 650K \times 700K}{700K - 650K}$$

$$Ea = 182000 J/mol = 182\ kJ/mol$$

## 2. The Transition state Theory

➦ According to this theory, collision between reactants yields a short - lived, high energy species called **activated complex or transition state** before the formation of products.

**Example**

$$A_2 + B_2 \longrightarrow A_2B_2^{\neq} \longrightarrow 2AB$$

Reactants Activated compelx Product

$$A — A + B — B \rightarrow A \cdots B \rightarrow 2A — B$$

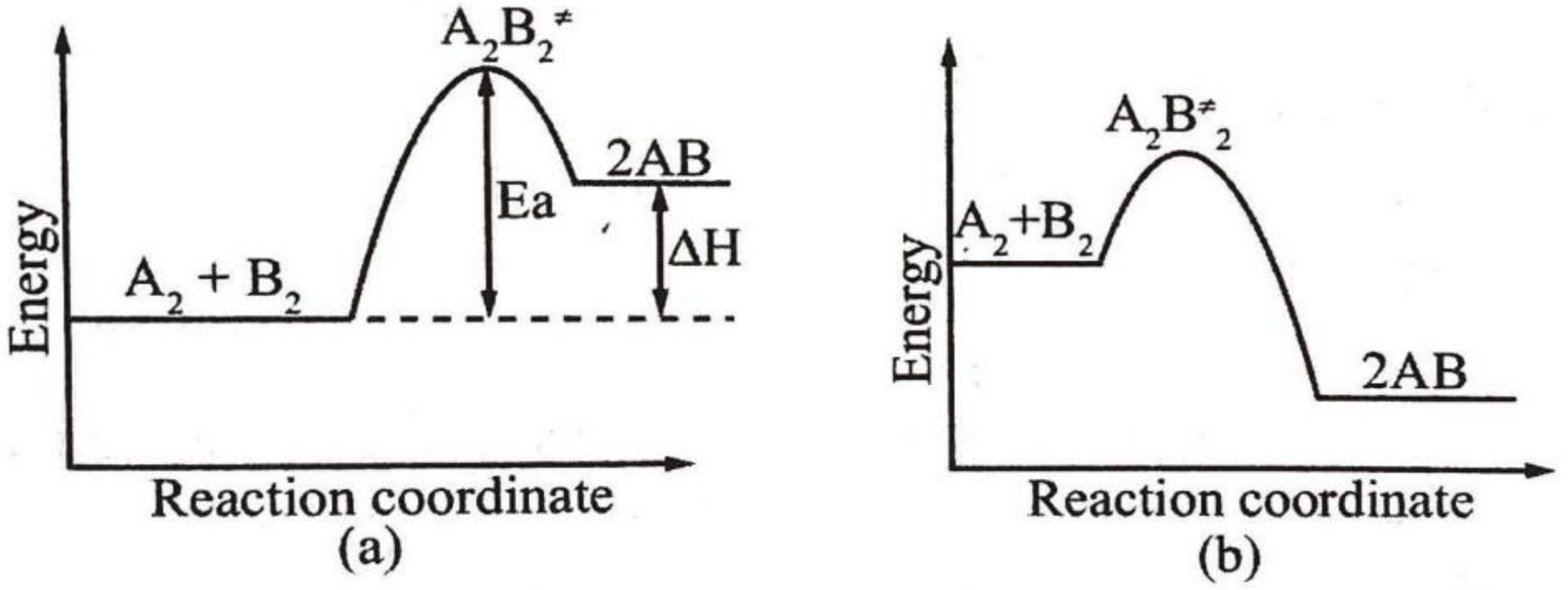



Figure 4.5 Energy diagram (a) for endothermic reaction and (b) exothermic reaction

## Endothermic reaction

➦ It is a reaction that absorbs heat.
➦ It has positive enthalpy change ($\Delta H > 0$).
➦ Heat content of product ($H_p$) is greater than heat content of reactant (Hr).

$$\Delta H = H_p - H_r$$

## Exothermic reaction

➦ It releases heat.
➦ It has negative enthalpy change ($\Delta H < 0$).
➦ Heat content of product is less than heat content of reactant.

## Illustrative Examples

26. If we increase the concentration of a reactant, what happens to the collisions between particles? (EUEE)

A. There are more collisions.
B. There are fewer collisions.
C. There are the same number of collisions, but they have less energy.
D. There are the same number of collisions, but they have more energy.

**Solution**

Concentration is directly proportional to number of collisions.

**Answer: A**

27. Rate law for the reaction A + 2B → C is found to be: Rate = k [A][B]. If the concentration of reactant 'B' is doubled, keeping concentration of 'A' constant, what will be the value of rate constant? (EUEE ... 2008 E.C.)

A. the same
B. halved
C. quadrupled
D. doubled

**Solution**

➦ Rate constant is independent on concentration. It is affected by temperature.

**Answer: A**

28. What is the activation energy of a reaction whose rate constant doubles when the temperature is increased from 25°C to 35°C?

A. 52.9J B. 5.29J C. 52.9kJ D. 5.29kJ

**Solution**

$T_1 = 25°C = 298\ k$

$T_2 = 35°C = 308k$

$K_2 = 2K_1$

$$Ea = \frac{2.303R \log \frac{k_2}{k_1} . T_1 . T_2}{T_2 - T_1}$$

$$Ea = \frac{2.303 \times 8.314 J/mol . \log \frac{2k_1}{k_1} . 298 \times 308k}{(308 - 298)k}$$

$Ea = 52897.8J/mol = 52.9kJ/mol$

**Answer: C**

29. The rate constant at 327°C is $2.7 \times 10^{-4} M^{-1} s^{1}$. What is the temperature at which the rate constant increases to $3.2 \times 10^{-2} M^{-1}.s^{-1}$? (Energy of activation for the reaction is 166.7kJ/mol)(EYEE ... 2000 E.C)

A. 165°C
B. 350°C
C. 427°C
D. 700°C

**Solution**

$T_1 = 327 + 273 = 600k$

$$\frac{2.303 \times 8.314}{1.667 \times 10^{5}} \log \frac{3.2 \times 10^{-2}}{2.7 \times 10^{-4}} = \frac{1}{T_1} - \frac{1}{T_2}$$

Solving for $T_2$, $T_2 = 700k$

$T_2 = 700 - 273 = 427°C$

**Answer: C**

30. Explain why all reactions require activation energy?

**Explanation:** All reactions have activation energy (energy required to combine reactants) which is used for the occurrence of reactions.

31. For a collision to be effective, what are the requirements?

**Solution**

- The colliding molecules must possess certain minimum amount of energy called threshold energy.
- The colliding species should have proper orientations.

32. What is the slope of ln k vs $\frac{1}{T}$ graph?

**Solution**

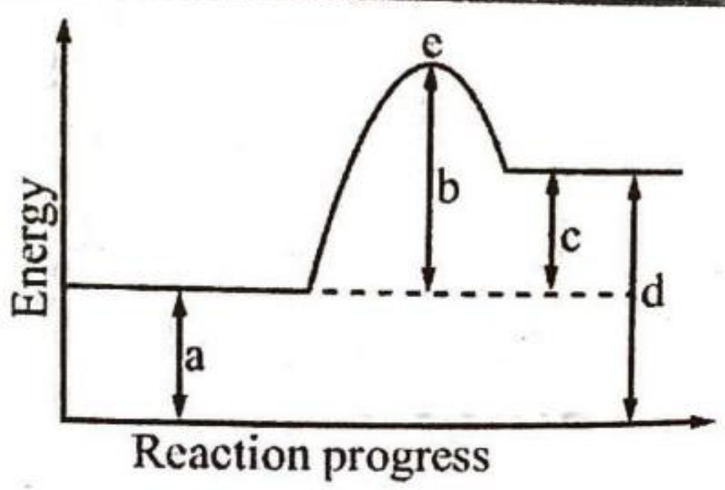
A graph of lnk vs $\frac{1}{T}$ has slope equal to $-\frac{Ea}{R}$

33. Based on the Arrhenius equation, when temperature decreases, what will happen to the value of rate constant?

**Solution**

As the value of temperature decreases, the rate constant also decreases.

34. In the energy diagram below, write the terms represented by letters a to e.




**Solution**

a. Heat content of reactant (Hr)
b. Activation energy (Ea)
c. Heat of reaction (ΔH)
d. Heat content of product (Hp)
e. Activated complex or transition state.

## 4.5 Rate Equation or Rate law and Order of reaction

**Rate law** - relates concentration of reactants, rate and order of reaction.

**Example**

$aA + bB \rightarrow cC + dD$

$Rate = k[A]^x[B]^y$

- k - rate constant
- [A] - concentration of reactant A
- [B] - concentration of reactant B
- x and y are order of reaction with respect to reactant A and B respectively.

**Order of a reaction** is obtained experimentally.

- It is not the same as the coefficients of the reactants.
- It can be negative, zero, positive or fraction.
- The overall order of the reaction is the sum of the order of the reaction with respect to each reactant.

**Example**

$R = k[A]^2[B]$

Over all order = 2 + 1 = 3

# Determination of Rate law

**The initial method** is an easy method used to determine the order of the reaction and rate constant by doing a series of experiments in which the initial amounts are varied.

- For a general reaction: $aA + bB + cC \rightarrow dD + eE + fF$

Initial rate, $(rate)_0 = k[A]_o^x[B]_o^y[C]_o^z$

$[A]_o$ - initial concentration of A

$[B]_o$ - initial concentration of B

$[C]_o$ - initial concentration of C

- X is determined by varying concentration of A while keeping constant concentration of B and C.
- Y is determined by varying concentration of B but keeping constant concentration of A and C.
- Z is determined by varying concentration of C but keeping constant concentration of A and B.

## Example

The reaction: $2NO\ (g) + Cl_2(g) \rightarrow 2NOCl\ (g)$ was studied at 263K and the following data were obtained.

| Experiment | $[NO]_O$, M | $[Cl_2]_o$, M | initial rate, $M.s^{-1}$ |
|---|---|---|---|
| 1 | 0.10 | 0.10 | 0.18 |
| 2 | 0.10 | 0.20 | 0.36 |
| 3 | 0.20 | 0.20 | 1.44 |

Determine

a. rate law

b. overall order

c. the value of rate constant, k

**Solution**

a. $Rate = k[NO]^x[Cl_2]^y$
Use experiment 2 and 3 to find the value of x.

$$\frac{0.36}{1.44} = \frac{(0.10)^x (0.20)^y}{(0.20)^x (0.20)^y}$$

$$\frac{1}{4} = \left(\frac{1}{2}\right)^x$$

$$\left(\frac{1}{2}\right)^2 = \left(\frac{1}{2}\right)^x$$

$$\Rightarrow x = 2$$

➦ Use experiment 1 and 2 to find y.

$$\frac{\text{Rate}_1}{\text{Rate}_1} = \frac{k[NO]_1^x [Cl_2]_1^y}{k[NO]_2^x [Cl_2]_2^y}$$

$$\frac{0.18}{0.36} = \frac{(0.10)^x (0.10)^y}{(0.10)^x (0.20)^y}$$

$$\frac{1}{2} = \left(\frac{1}{2}\right)^y$$

$$\Rightarrow y = 1$$

$$\Rightarrow \text{Rate} = k[NO]^2[Cl_2]$$

b. Overall order $= 2 + 1 = 3$

c. Use the simplest data to find the value of k.

$$\text{Rate}_1 = k[NO]_1^2 [Cl_2]_1$$

$$k = \frac{\text{Rate}_1}{[NO]_1^2 [Cl_2]_1}$$

$$= \frac{0.18 M.s^{-1}}{(0.10 M)^2 (0.10 M)}$$

$$= 180 M^{-2}.s^{-1} = 180 L^2.mol^{-2}.s^{-1}$$

Table 4.1. For zero, First, second, and third order reactions use the table below to determine order of reaction

| Reaction | Doubling concentration of a reactant, A | Tripling concentration of a reactant A |
|---|---|---|
| Zero - order $r = k[A]^\circ$ | Rate is unaffected | Rate is unaffected |
| First - order $r = k[A]$ | Rate doubles | Rate triples |
| Second - order $r = k[A]^2$ | Rate quadruples | Rate increases by a factor of 9 |
| Third order $r = k[A]^3$ | Rate increases by a factor of 8 | Rate increases by a factor of 27 |

**Note**

- The rate of zero - order reaction does not depend on concentration of reactant.
- The unit of rate constant for zero - order reaction is the same as the unit of rate of reaction.
- Rate constant has different units for reactions with different orders.

## Illustrative Examples

35. The rate law of the overall reaction: $A + B \rightarrow C$ is: rate = $k[A]^2$ which of the following will NOT increase the rate of the reaction? (EUEE ...2008 E.C)

A. Increasing concentration of reactant A.

B. Increasing concentration of reactant B.

C. Increasing the temperature of teh reaction.

D. Adding a catalyst for th reaction.

**Solution**

- The reaction is zero - order in reactant B. The rate of zero - order reaction is independent on concentration.

**Answer: B**

36. The decomposition of nitrosyl chloride was studied as:

$2NOCl(g) \rightleftharpoons 2NO(g) + Cl_2(g)$ and the following data were obtained.

| $[NOCl]_0$ (molecules/$cm^3$) | Initial rate in molecules/$cm^3$.s |
|---|---|
| $3.0 \times 10^{16}$ | $5.98 \times 10^4$ |
| $2.0 \times 10^{16}$ | $2.66 \times 10^4$ |
| $1.0 \times 10^{16}$ | $6.64 \times 10^4$ |
| $4.0 \times 10^{16}$ | $1.06 \times 10^5$ |

What is the rate law for the above decomposition?

A. $r = k\,[NOCl]^2$

B. $r = k[NOCl]$

C. $r = k\,[NOCl][NO]$

D. $r = k[NOCl][Cl_2]$

**Solution**

Use only two simple data to find the order of the reaction in NOCl. Rate = $k[NOCl]^x$

$$\frac{r_2}{r_4} = \frac{k[NOCl]_2^x}{k[NOCl]_4^x}$$

$$\frac{2.66\times10^4}{10.6\times10^4} = \left(\frac{2\times10^6}{4\times10^{16}}\right)^x$$

$$\frac{1}{4} = \left(\frac{1}{2}\right)^x$$

$\Rightarrow x = 2$

$\therefore$ Rate $= k[NOCl]^2$

**Answer: A**

37. In a reaction, A + B → product, the rate is doubled when concentration of B is doubled and the rate increases by a factor of 8 when the concentration of both reactants A and B are doubled, the rate law for the relation can be written as:

A. Rate $= k[A][B]$
B. Rate $= k[A][B]^2$
C. Rate $= [A]^2[B]$
D. Rate $= k[A]^2[B]^2$

**Solution**

➦ The reaction is first order with respect to reactant B because doubling concentration of B doubles the rate.

➦ The overall order is 3 because when concentration of both reactants double the rate increases by 8.

$\Rightarrow$Rate $= k[A]^2[B]$

**Answer: C**

38. The overall order of the reaction:

$2NO + O_2 \rightarrow 2NO_2$ with rate law, rate $= k[NO]^2[O_2]$ the overall order is (EUEE)

A. 0 B. 1 C. 2 D. 3

**Solution**

Over all order $= 2 + 1 = 3$

**Answer: D**

39. Initial rates and concentrations are measured for the following reaction:
$2NO(g) + 2H_2(g) \rightarrow N_2(g) + 2H_2O\ (g)$

| $[NO]_0$, M | $[H_2]_0$, M | initial rate, $mol.L^{-1}.s^{-1}$ |
|---|---|---|
| 0.005 | 0.0005 | 0.000025 |
| 0.005 | 0.0010 | 0.000050 |
| 0.0010 | 0.0010 | 0.00020 |

The rate equation for the reaction is

A. Rate = $k[NO]$
B. Rate = $k[NO]^2$
C. Rate = $k[NO], [H_2]$
D. Rate = $k[NO]^2[H_2]$

**Solution**

- Using the first two data, the order is one in $H_2$.
- Using the last two data, the order is two in NO.

$\Rightarrow$ Rate = $k[NO]^2[H_2]$

**Answer: D**

40. The reaction between NO and $I_2$ is second order in NO and first order in $I_2$. What change occurs in the rate of the reaction if the concentration of each reactant is tripled? (EUEE)

A. 3 - fold increase
B. 6 - fold increase
C. 18 - fold increase
D. 27 - fold increase

**Solution**

$Rate_1 = k[NO]^2 [I_2]$, $Rate_2 = k[NO]_2^2[I_2]_2$

$[NO]_2 = 3[NO]$, and $(3\ [I_2])_1$

Rate 2 = $k(3[NO]_1)^2 (3[I_2]_1)$

Rate 2 = $k9\ [NO]_1^2, 3[I_2]_1$

Rate 2 = $27\ k[NO]_1^2[I_2]_1$

Rate 2 = 27 Rate 1

**Answer: D**

41. Which of the following rate law is third order overall?

A. Rate = $k[A][B]^3$
B. Rate = $k[A][B]^2$
C. Rate = $k[A]^3[B]^3$
D. Rate = $k[A]^5[B]^2$

**Answer: B**

42. For the reaction: $2A + B \rightarrow 2C$, the following data were obtained.

| Experiment | $[A]_0$, M | $[B]_0$, M | $(rate)_0$, mol/L.s |
|---|---|---|---|
| 1 | 0.5 | 0.5 | 10 |
| 2 | 0.5 | 1.0 | 20 |
| 3 | 0.5 | 1.5 | 30 |
| 4 | 1.0 | 0.5 | 40 |

**Determine**

a. rate law

b. the value of rate constant

**Solution**

a. Use experiment 1 and 2 to find the order in A and B and 4 to find the order with respect to B.
$\Rightarrow Rate = k[A]^2[B]$

b. $$k = \frac{Rate}{[A]^2[B]} = \frac{10\ mol/L.s}{(0.5M)^2(0.5H)} = 80L^2.mol^{-2}.s^{-1}$$

43. Given the reaction

$$H_2O_2(aq) + 3I^-(aq) + 2H^+(aq) \rightarrow I^-_3(aq) + 2H_2O(l)$$

| Experiment | Initial concentration in mol/L | | | Initial rate in mol/L.s |
|---|---|---|---|---|
| | $H_2O_2$ | $I^-$ | $H^+$ | |
| 1 | 0.010 | 0.010 | 0.00050 | $1.15\times10^{-6}$ |
| 2 | 0.020 | 0.010 | 0.00050 | $2.30\times10^{-6}$ |
| 3 | 0.010 | 0.020 | 0.00050 | $2.30\times10^{-6}$ |
| 4 | 0.010 | 0.010 | 0.00100 | $1.15\times10$ |

Determine the order of the reaction with respect to each reactant.

**Solution**

$Rate = k[H_2O_2]^x[I^-]^y[H^+]^z$

To find x use experiment 1 and 2.

$$\frac{Rate\ 1}{Rate\ 2} = \frac{[H_2O_2]_1^x}{[H_2O_2]_2^x}$$

$$\frac{1.15\times10^{-6}}{2.30\times10^{-6}} = \left(\frac{0.010}{0.020}\right)^x$$

$$\left(\frac{1}{2}\right)^1 = \left(\frac{1}{2}\right)^x$$

$x = 1$

➦ To find y use experiment 1 and 3

$$\frac{\text{Rate 3}}{\text{Rate 1}} = \frac{[I^-]_3^y}{[I^-]_1^y}$$

$$\frac{2.30\times10^{-6}}{1.15\times10^{-6}} = \left(\frac{0.020}{0.010}\right)^x$$

$$(2)^1 = 2^y$$

$$y = 1$$

➦ To find z use experiment 1 and 4.

$$\frac{\text{Rate 4}}{\text{Rate 1}} = \left(\frac{0.00100}{0.00050}\right)^z$$

$$\frac{1.15\times10^{-6}}{1.15\times10^{-6}} = (2)^z$$

$$1 = (2)^z \Rightarrow z = 0$$

$\Rightarrow$Rate = $k[H_2O_2][I^-]$, thus the order of the reaction with respect to $H_2O_2$ is 1, $I^-$ is 1 and $H^+$ is zero.

44. State rate law

**Solution**

Rate law states that the rate is equal to rate constant times the concentration of reactants raised to a simple integer or fractions.

45. Consider the reaction is which nitric oxide is oxidized to nitrogen dioxide: $2NO(g) + O_2(g) \rightarrow 2NO_2(g)$

For which rate law is: rate = $kPNO^2.\ PO_2$. If this reaction takes place in a sealed vessel and the partial pressure of nitric oxide is doubled, what effect would this have on the rate of the reaction?

**Solution**

The order of the reaction with respect to NO is 2. Doubling partial pressure of NO increases the rate of the reaction by a factor of 4.

46. A particular reaction was found to depend on the concentration of the hydrogen ion, $H^+$. The initial rate changed as a function of $[H^+]$ as follows:

| $[H^+]$, M | 0.005 | 0.01 | 0.02 |
|---|---|---|---|
| Initial rate in M/s | $3.4\times10^{-7}$ | $1.7\times10^{-7}$ | $0.85\times10^{-7}$ |

What is the order of the reaction in $[H^+]$?

**Solution**

$$\text{Rate} = k\,[H^+]^x$$

$$\frac{\text{Rate 1}}{\text{Rate 2}} = \frac{k[H^+]_1^x}{k[H^+]_2^x}$$

$$\frac{3.4\times10^{-7}}{1.7\times10^{-7}} = \left(\frac{0.005}{0.01}\right)^x$$

$$2 = \left(\frac{1}{2}\right)^x$$

$$(2)^1 = (2)^{-x} = -x = 1 = x = -1$$

The order of the reaction in $[H^+]$ is -1.

47. Write the factor that does not affect the rate of zero - order reaction.

**Solution**

➦ Concentration or partial pressure of gas.

48. The reaction: $2X + Y \rightarrow Z$, was studied and the following data were obtained.

| Experiment | [X] | [Y] | Rate [$mol.L^{-1}.s^{-1}$] |
|---|---|---|---|
| 1 | 3.0 | 3.0 | 1.8 |
| 2 | 3.0 | 1.5 | 0.45 |
| 3 | 1.5 | 1.5 | 0.45 |

What is the rate equation?

**Solution**

$$\text{Rate} = k[x]_1[y]^b$$

$$\frac{\text{Rate 1}}{\text{Rate 2}} = \left(\frac{3.0}{1.5}\right)^b \qquad \frac{\text{Rate 1}}{\text{Rate 2}} = \left(\frac{3.0}{1.5}\right)^a$$

$$\left(\frac{1.8}{0.45}\right) = (2)^b \qquad \left(\frac{0.45}{0.45}\right) = 2^a$$

$$2^2 = (2)^b \qquad a = 0$$

$$b = 2$$

$$\therefore \text{Rate} = k[Y]^2$$

49. A reaction is third order with respect to a given reactant. By what factor does the reaction rate decreases if the concentration of the reactant is reduced to $\frac{1}{2}$ of its initial concentration?

**Solution**

Let the reactant be A

$$\text{Rate } 2 = k[A]_2^3$$

$$[A]_2 = k\frac{1}{2}[A]_1$$

$$\text{Rate } 2 = k[A]_2^3$$

$$= k\left(\frac{1}{2}[A]_1\right)^3$$

$$= \frac{1}{8}k[A]_1^3$$

$$\Rightarrow \text{Rate } 2 = \frac{1}{8}.\text{Rate } 1$$

$\therefore$ The reaction rate decreases by a factor of $\frac{1}{8}$.

50. Determine the overall order and the unit of rate constant. For

a. $\text{Rate} = k[A][B]$

b. $\text{Rate} = k[A]^2[B]^{1/2}$

c. $\text{Rate} = k[A]^2[B][C]^{-1/2}$

**Solution**

a. Overall order $1 + 1 = 2$

$$\text{Unit of } k = \frac{M.s^{-1}}{M.M} = M^{-1}.s^{-1} = L.mol^{-1}.s^{-1}$$

b. Overall order $= 2 + \frac{1}{2} = \frac{5}{2}$

$$\text{Unit of } k = \frac{M.s^{-1}}{M^2.M^{\frac{1}{2}}} = M^{-3/2}.s^{-1} = L^{3/2}.mol^{-3/2}.s^{-1}$$

c. Overall order $= 2 + 1 + \left(-\frac{1}{2}\right) = \frac{5}{2}$

$$\text{unit of } k = M^{-3/2}.s^{-1}$$

51. For a hypothetical reaction: $2A + B \rightarrow$ products, the rate constant is equal to $6.5 \times 10^{-5} L^2.mol^{-2}.s^{-1}$. What is the order of the reaction?

**Solution**

From the unit f rate constant it is possible to determine the order of the reaction. The order of the reaction is three.

52. What is the maximum value for order of a reaction?

**Solution**

Generally the order of a reaction cannot be greater than three.

## 4.6 Concentration - time Equation (Integrated rate law)

➦ It expresses the concentration of a reactant as a function of time.

### 1. Integrated rate law for zero - order reaction

$A \rightarrow Product$

$Rate = k[A]^0 = k$

$$Rate = \frac{-\Delta[A]}{\Delta t}$$

$$\Rightarrow k = \frac{-\Delta[A]}{\Delta t}$$

$$-\Delta[A] = k\Delta t$$

$$\Rightarrow [A] = -kt$$

Using calculus

$$\int_{[A]_0}^{[A]_t} d[A] = -k\int_0^t dt$$

$$[A]_t - [A]_0 = -kt$$

$$\boxed{[A]t = -kt + [A]_0}$$

This is an equation of a straight line with slope -k.

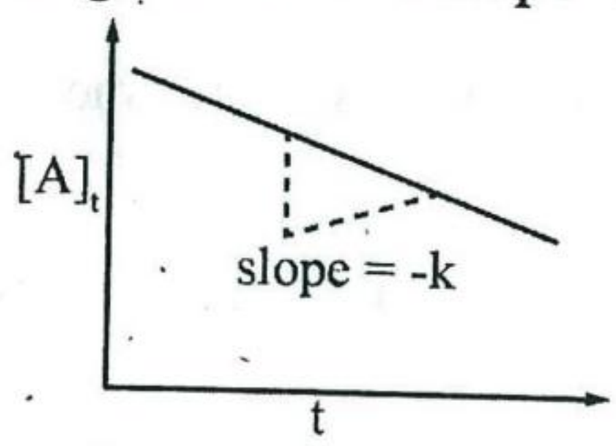



Figure 4.6 Concentration time graph for zero order reaction

➦ Most important examples of zero - order reactions are found in the action of enzymes.

**Example**

The decomposition reaction:

$2N_2O(g) \rightarrow 2N_2(g) + O_2(g)$ is a zero order reaction that occurs on a hot platinum surface. The concentration of $N_2O$ reduced from 0.4M to 0.1M in 20 seconds. Calculate the value of rate constant of the reaction.

**Solution**

$$[N_2O]_0 = 0.4M,\ [N_2O] = 0.1M,\ t = 20\ sec$$
$$[N_2O]_t = -kt + [N_2O]_0$$
$$[N_2O]_t - [N_2O] = -kt$$
$$k = \frac{[N_2O]_0 - [N_2O]_t}{t}$$
$$k = \frac{0.4M - 0.1M}{20sec}$$
$$k = 0.015M.s^{-1}$$

## Half - life of zero - order reaction

**Half - life ($t_{1/2}$)** is the time required for the concentration of a reactant to be half of its initial concentration.

$$\Rightarrow [A]_t = \frac{1}{2}[A]_0, t = t_{½}$$
$$[A]_t - [A]_0 = -kt$$
$$[A]_0 - [A]_t = kt$$
$$[A]_0 - \frac{1}{2}[A]_0 = kt_{½}$$
$$\frac{[A]_0}{2} = kt_{½}$$
$$\boxed{t_{½} = \frac{[A]_0}{2k}}$$

Half - life of zero - order reaction is directly proportional to initial concentration of reactant.

### Illustrative Examples

53. A drug decomposes by zero - order kinetics with rate constant of 2mg. $mL^{-1}$ $month^{-1}$. If the initial concentration is 100mg. $mL^{-1}$, how long will it take for the drug to decompose by 10% (EUEE ... 2008)

A. 2 months
B. 3 months
C. 5 months
D. 4 months

**Solution**

$$[A]_0 - [A]_t = kt$$

$$t = \frac{[A]_0 - [A]_t}{k}$$

10% decomposed means 90% left.

$$\Rightarrow [A]_t = 90\%\ [A]_0$$

$$[A]_t = \frac{9}{10}[A]_0$$

$$\Rightarrow t = \frac{[A]_0 - \frac{9}{10}[A]_0}{k}$$

$$= \frac{100 - \frac{9}{10}\times 100}{2} = \frac{10}{2} = 5 \text{ months}$$

**Answer: C**

54. For a zero - order reaction, a plot of _____ versus _____ is linear.

A. $\frac{1}{[A]_t}, t$ B. $\ln\frac{1}{[A]_t}.t$ C. $[A]_t, t$ D. $\ln [A]_t, t$

**Answer: C**

55. What is the half - time, $t_{1/2}$ for a zero order reaction: A→B? (k rate constant)

A. ln 2x B. ln k$[A]_0$ C. $\frac{[A]_0}{2k}$ D. ln 2|$[A]_0$

**Answer: C**

56. The rate constant of a zero - order reaction is $2.5\times10^{-4}.mol.L^{-1}.s^{-1}$. Find its half - life if the initial concentration of the reactants 0.6M.

**Solution**

$$t_{½} = \frac{[A]_0}{2k} = \frac{0.6 mol.L^{-1}}{2\times 2.5\times10^{-4} mol.L^{-1}.s^{-1}}$$

$$t_{½} = 1.2\times10^{3} s$$

## 2. Integrated rate law for first order reaction

A→Product

$$\text{Rate} = k[A] \text{ or } \text{Rate} = \frac{-\Delta[A]}{\Delta t}$$

$$\Rightarrow \frac{-\Delta[A]}{\Delta t} = k[A]$$

$$-\Delta[A] = k[A]\Delta t$$

$$[A] = -k[A]_t$$

$$\int_{[A]_0}^{[A]_t} d\frac{[A]}{[A]} = -k\int_{t_0}^{t} dt$$

$$\ln [A]_t - \ln [A]_0 = -kt$$

$$\boxed{\ln [A]_t = -kt + \ln [A]_0}$$

➦ This equation is an equation of a straight line with slope - k.

$$\boxed{\log[A]_+ = -\frac{kt}{2.303} + \log[A]_0}$$

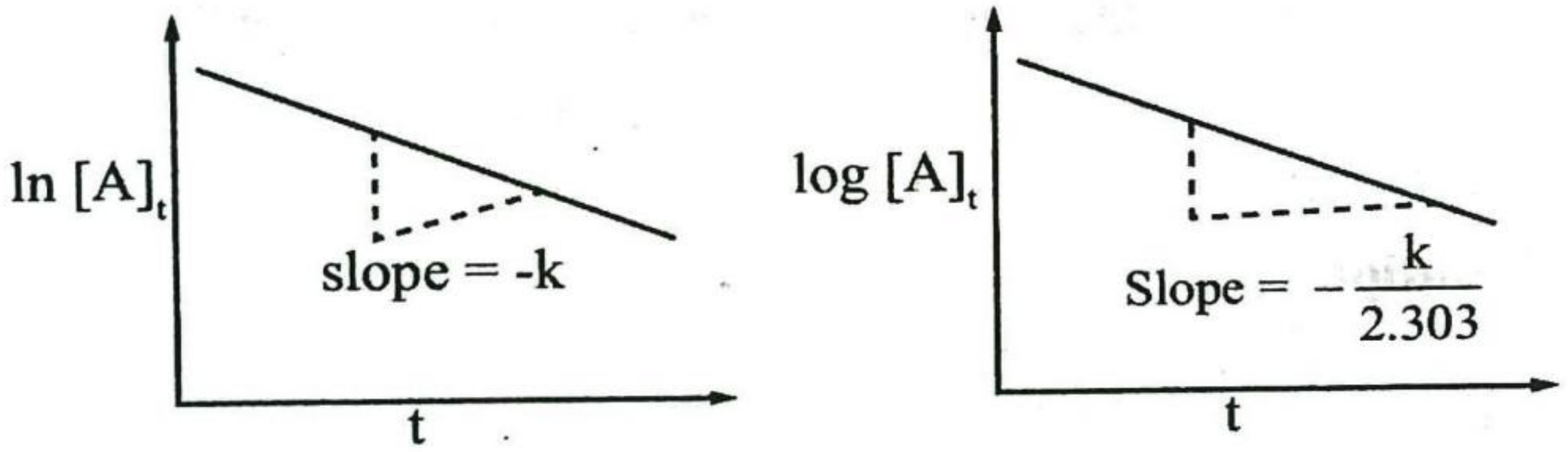



Figure 4.7 Concentration time graph for first order reaction

## Half - life for first order reaction

$$[A]_t = \frac{1}{2}[A]_0, t = t_{½}$$

$$\ln [A]_t = -kt + \ln [A]_0$$

$$\ln [A]_t - \ln [A]_0 = -kt$$

$$\ln \frac{[A]_t}{[A]_0} = -kt$$

$$\ln \frac{[A]_0}{[A]_t} = kt$$

$$\ln \frac{[A]_0}{\frac{1}{2}[A]_0} = kt_{½}$$

$$t_{½} = \frac{\ln 2}{k}$$

$$\boxed{t_{½} = \frac{0.693}{k}}$$

Half - life of first order reaction is independent on initial concentration of reactant.

## Illustrative Examples

57. For a first order reaction, a plot of _____ versus _____ is linear (EUEE ... 2008 E.C)

A. $\frac{1}{[A]_t}, t$ B. $\ln\frac{1}{[A]_t}, t$ C. $[A]_t, t$ D. $\ln [A]_t, t$

**Solution**

➦ For first order reaction the graph of $\ln [A]_t$ vs t or $\log [A]_t$ vs t is linear.

**Answer: D**

58. The decomposition of carbondisulphide, $CS_2$, to carbonmonosulfide, CS and Sulphur, S is first order with $k = 2.8 \times 10^{-7}s^{-1}$ at 1000°C. What is the half - life of the reaction at 1000°C? (EUEE ... 2007 E.C)

A. $5 \times 10^7 s$ C. $3.8 \times 10^5 s$

B. $4.7 \times 10^{-6} s$ D. $2.5 \times 10^6 s$

**Solution**

$$t_{1/2} = \frac{0.693}{k} = \frac{0.693}{2.8\times10^{-7}s^{-1}} = 2.5\times10^6 s$$

**Answer: D**

59. The half - life for the first order decomposition of nitromethane, $CH_3NO_2$ at 500K is 650 second. If the initial concentration of $CH_3NO_2$ is 0.5M, what will its concentration be in (M) after 1300 seconds have elapsed? (EUEE)

A. 0.125 B. 0.140 C. 0.250 D. 0.425

**Solution**

$$t_{1/2} = \frac{0.693}{k}$$

$$k = \frac{0.693}{650s} = 1.07\times10^{-3}s^{-1}$$

$$\log\frac{[A]_0}{[A]_t} = \frac{kt}{2.303}$$

$$\frac{[A]_0}{[A]_t} = 10^{\frac{kt}{2.303}}$$

$$\frac{[A]_0}{[A]_t} = 10^{0.6}$$

$$[A]_t = \frac{[A]_0}{10^{0.6}} = \frac{0.5}{10^{0.6}} = 0.25M$$

$$\Rightarrow [CH_3NO_2]_t = 0.25M$$

**Answer: C**

60. Dinitrogen pentoxide, $N_2O_5$, dissolved in $CCl_4$ decomposes by a first - order reaction into $N_2O_4$ and $O_2$. How many seconds will it take for the concentration to drop from 0.1M to 0.05M if the rate constant at the reaction temperature is $6.93 \times 10^{-4}s^{-1}$?

**Solution**

$$t = \frac{-2.303 \log \frac{[N_2O_5]_t}{[N_2O_5]_0}}{k}$$

$$t = \frac{-2.303 \log \frac{0.05}{0.10}}{6.93 \times 10^{-4} s^{-1}} = \frac{0.693}{6.93 \times 10^{-4} s^{-1}}$$

$$t = 1000s$$

61. What is the half - life of the irreversible first order reaction, A→B, if 75% of A is converted to B in 60 minutes?

**Solution**

75% of A converted means 25% left unreacted

$$[A]_t = 25\%\ [A]_0$$

$$[A]_0 = 4[A]_t$$

$$\log \frac{[A]_0}{[A]_t} = \frac{kt}{2.303}$$

$$k = \frac{2.303 \log \frac{[A]_0}{[A]_t}}{t}$$

$$k = \frac{2.303 \log 4 \frac{[A]_t}{[A]_t}}{60\,min} = 0.0231\,min^{-1}$$

$$t_{½} = \frac{0.693}{k} = \frac{0.693}{0.023\,min^{-1}} = 30\,min$$

62. In the fermentation of sugar in an enzymatic solution, which is initially 0.2M, the concentration of the sugar is reduced to 0.1M in 10 hours and to 0.05M in 20 hours. What is the order of the reaction and the rate constant?

**Solution**

[Sugar] is reduced from:

➦ 0.2M to 0.1M in 10 hours

➦ 0.1M to 0.05M in 20 hours

This is true for first order reaction.

$$t_{1/2} = \frac{0.693}{k}$$

$$k = \frac{0.693}{t_{1/2}} = \frac{0.693}{10 \times 3600s} = 1.92 \times 10^{-5}\,s$$

63. What is the slope of a first order reaction with a graph of log [A] Vs time?

**Solution**

$$\text{Slope} = \frac{-k}{2.303}$$

64. The half-life of a first order reaction: X→Y is 100min. The concentration of X would be reduced to 10% of the original concentration in _____ ____ minutes.

**Solution**

$$t_{y_2} = \frac{0.693}{k}$$

$$k = \frac{0.693}{t_{y_2}} =$$

$$K = \frac{0.693}{100\,\text{min}} = 6.93 \times 10^{-3}\,\text{min}^{-1}$$

$$[X]_t = 10\%[X]_0, t = ?$$

$$t = \frac{2.303 \log \frac{[x]_0}{[x]_t}}{k}$$

$$t = \frac{2.303 \log \frac{10[x]_t}{[x]_t}}{k} = \frac{2.303}{6.93 \times 10^{-3}\,\text{min}^{-1}}$$

$$t = 332\,\text{min}$$

65. A first order - reaction is 50% complete in 100 min. The time required for 100% completion would be _____.

**Solution**

$$[A]_t = 50\%\ [A]_0,\ t = 100\text{min}$$

$$k = \frac{2.303\log\frac{[A]_0}{[A]_t}}{t} = \frac{2.303\log\frac{2[A]_t}{[A]_t}}{100\,\text{min}}$$

$$k = 6.93\times10^{-3}\ \text{min}^{-1}$$

$$[A]_t = 100\%[A]_0 = t = ?$$

$$t = 200\,\text{min}$$

### 3. Integrated rate law for second order reaction

$$A \rightarrow \text{Product}$$

$$\text{Rate} = k[A]^2 \text{ or } \text{Rate} = \frac{-\Delta[A]}{\Delta t}$$

$$k[A]^2 = \frac{-\Delta[A]}{\Delta t}$$

$$k[A]^2\Delta T = -\Delta[A]$$

$$\int_{[A]_0}^{[A]_b} d\frac{[A]}{[A]^2} = -k\int_{t_0}^{t} dt$$

$$\boxed{\frac{1}{[A]_t} = kt + \frac{1}{[A]_0}}$$

➦ This is an equation of a straight line with slope k.

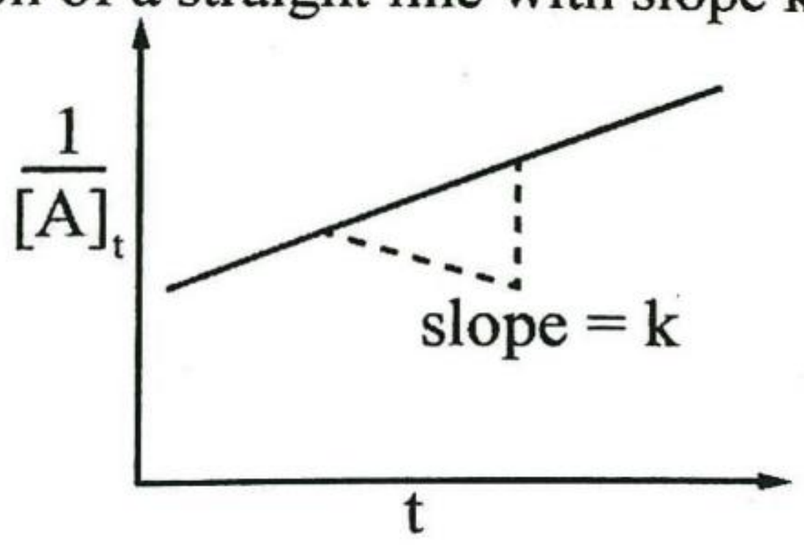



Figure 4.8 Concentration - time graph of second order

## Half - life of a second order reaction

$$[A]_t = \frac{1}{2}[A]_0, t = t_{1/2}$$

$$\frac{1}{[A]_t} - \frac{1}{[A]_0} = kt$$

$$\frac{1}{\frac{1}{2}[A]_0} - \frac{1}{[A]_0} = kt_{1/2}$$

$$\boxed{t_{1/2} = \frac{1}{k[A]_0}}$$

**Half - life of second order reaction is inversely proportional to the initial concentration of the reactant.**

## Illustrative Examples

66. For a second - order reaction, a plot of _____ versus _____is linear.

A. $\frac{1}{[A]_t}$, t

B. $\ln[A]_t$, t

C. $[A]_t$, t

D. $\log [A]_t$, t

**Answer: A**

67. The decomposition of hydrogen iodide is second order reaction with rate constant, $k = 30L.mol^{-1}.s^{1}$ at 443°C. Calculate

a. the half - life

b. the time required for concentration of HI to drop from 0.4M to 0.1M.

**Solution**

a. $t_{1/2} = \frac{1}{k[A]_0} = \frac{1}{30L.mol^{-1}.s^{-1} \times 0.4mol.L^{-1}}$

$t_{1/2} = 0.0835$

b. $\frac{1}{[A]_t} - \frac{1}{[A]_0} = kt$

$\frac{1}{0.1} - \frac{1}{0.4} = kt$

$\frac{10 - 2.5}{k} = t$

$t = 0.25$ sec

# 4.7 Reaction Mechanism

- **Reaction mechanism** is the sequence of bond making and bond breaking steps that occurs during the conversion of reactants into products.
- Some reactions occur in a single step.

**Example**

$NO(g) + O_3(g) \rightarrow NO_2(g) + O_2(g)$

➦ Most chemical reactions, however, involve a sequence of steps.

**Example**

The reaction given below occurs in three steps reaction mechanism.

$$2NH_3(aq) + OCl^-(aq) \xrightarrow[\text{solution}]{\text{basic}} N_2H_4(aq) + Cl^-(aq) + H_2O(l)$$

**Step 1:** $NH_3(aq) + OCl^-(aq) \xrightarrow{\text{fast}} NH_2Cl(aq) + OH^-(aq)$

**Step 2:** $NH_2Cl(aq) + NH_3(aq) \xrightarrow{\text{slow}} N_2H_5^+(aq) + Cl^-(aq)$

**Step 3:** $N_2H_5^+(aq) + OH^-(aq) \xrightarrow{\text{fast}} N_2H_4(aq) + H_2O(l)$

➦ Each step in the reaction mechanism is called **an elementary reaction.**

➦ The **slowest step** in the reaction mechanism is called **the rate determining step.**

⇒Step 2 in the above example is the rate determining step.

➦ Species that are found in the reaction mechanism but not in the overall reaction are called **intermediates.**

⇒ $NH_2Cl$, $OH^-$ and $N_2H_5^+$ are intermediates in the above example.

➦ There are two requirements for a **plausible reaction mechanism.**

1. It must be consistent with the stoichiometry of the overall reaction.
2. It must agree with the experimentally determined rate law.

## Elementary Process

1. Elementary processes are either unimoleucalr, biomolecualr, or termolecular.

**Unimolecualr** - a process in which a single molecule decomposes.

**Example**

$NH_4NO_2 \rightarrow N_2 + 2H_2O$

**Biomolecular** - a process involving the collision of two molecules.

**Example**

$$O_3 + NO \rightarrow NO_2 + O_2$$

**Termoleuclar** - a process in which three molecules collide at the same time.

**Example**

$$2CO + O_2 \rightarrow 2CO_2$$

2. The exponents of the concentration terms in the rate law for an elementary process are the same as the stoichiometric coefficients in the balanced equation for the process.

**Example**

$$2NO\ (g) + O_2\ (g) \rightarrow 2NO_2(g)$$

$$\text{Rate} = k[NO_2]^2[O_2]$$

3. Elementary process are reversible and some may reach condition of equilibrium in which the rates of the forward and reverse processes are equal.

4. Certain species are produced and consumed in the elementary process.

5. One elementary process may occur much more slowly than all others. This step is rate determining step.

Overall rate = the rate of the slowest step

**Example**

The reaction: $NO_2(g) + CO(g) \rightarrow NO(g) + CO_2(g)$ can be thought of as occurring in two elementary steps.

1. $NO_2 + NO_2 \rightarrow NO + NO_3$ (slow step)

2. $NO_3 + CO \rightarrow NO_2 + CO_2$ (fast step)

   a. Identify the rate determining step and intermediate (s).

   b. Determine the molecularity of this reaction.

   **Solution**

   a. step 1, $NO_3$ b. Bimolecular

Table 4.2. Elementary step, molecularity and rate law

| Elementary Step | Molecularity | Rate law |
|---|---|---|
| $X \rightarrow$product | Unimolecular | Rate = k[X] |
| $2X \rightarrow$product<br>$X + Y \rightarrow$product | Biomolecular | Rate = $k[X]^2$<br>Rate = k[X][Y] |
| $3X \rightarrow$product<br>$2X + Y \rightarrow$product<br>$X+Y+Z \rightarrow$product | Termolecular | Rate = $k[X]^3$<br>Rate = $k[X]^2[Y]$<br>Rate = k[X][Y][Z] |

Molecularity cannot be negative, zero or fraction. It is 1, 2 or 3.

## Reaction Mechanism Involving fast equilibrium

- A reaction mechanism may involve a first step in which both the forward and reverse reactions are very fast compared with the reaction in the second step.
- The intermediate in the rate determining step should be replaced by using the fast equilibrium.

### Example

The rate law for the reaction:

$2NO(g) + O_2(g) \rightarrow 2NO_2(g)$ is found to be Rate = $k[NO]^2[O_2]$, show that the mechanism below is consistent with the rate law.

i $\quad 2NO(g) \underset{k_2}{\overset{k_1}{\rightleftharpoons}} N_2O_2(g)$ . . . fast

ii $\quad N_2O_2(g) + O_2(g) \longrightarrow 2NO_2(g)$ . . . slow

**Solution**

⇒It must be consistent with the stoichiometry of the overall reaction.

$$2NO(g) \rightleftharpoons N_2O_2(g)$$

$$\underline{N_2O_2(g) + O_2(g) \longrightarrow 2NO_2(g)}$$

over all reaction: $2NO(g) + O_2(g) \rightarrow 2NO_2(g)$

⇒ The rate law must agree with the experimentally determined rate law.

Step (ii) is the rate determining step

$\text{Rate} = k_3[N_2O_2][O_2]$ ...(1)

But, $N_2O_2$ is an intermediate. Use step (i) to replace it.

$r_f = k_1[NO]^2$ $\qquad$ $r_b = k_2[N_2O_2]$

At equilibrium $r_f = r_b$

$\Rightarrow k_1[NO]^2 = k_2[N_2O_2]$

$[N_2O_2] = \frac{k_1[NO]^2}{k}$ ...(2)

Substitute (2) in (1)

$\Rightarrow \text{Rate} = k_3 \frac{k_1}{k_2}[NO]^2[O_2]$

$\text{Rate} = k[NO]^2[O_2]$, $k = \frac{k_3k_1}{k_2}$

$r_f$ - rate of forward reaction

$r_b$ - rate of backward reaction

## Illustrative Example

68. Which of the following statement (s) is (are) applicable to balanced chemical equation of an elementary reaction? (EUEE ... 2008 E.C)

i Order is same as molecularity

ii Order is less than the molecularity

iii Order is greater than the molecularity

iv Molecularity can never be zero.

A. i $\qquad$ B. i, ii $\qquad$ C. i, iv $\qquad$ D. i, iii

**Solution**

➦ Order and molecularity are same in an elementary reaction.

➦ Molecularity can be 1, 2 or 3 but not zero.

**Answer: C**

69. Considering the mechanism for a reaction below, which of the following statements is correct?(EUEE ... 2007 E.C)

**Step 1**: $HBr + O_2 \rightarrow HOOBr$

**Step 2**: $HBr + HOOBr \rightarrow 2HOBr$

**Step 3**: $2HOBr + 2HBr \rightarrow 2Br_2 + 2H_2O$

A. $Br_2$ is a reactant

B. HBr is a product

C. HOBr is a catalyst

D. HOOBr is a reaction intermediate

**Solution**

The overall reaction is: $4HBr + O_2 \rightarrow 2Br_2 + 2H_2O$. Thus HOOBr and HOBr are intermediates.

**Answer: D**

70. The mechanism of a reaction is shown below:

$HCOOH + I^- \rightarrow HOI + OH^-$ (slow)

$HOI + I^- \rightarrow I_2 + OH^-$ (fast)

$2OH^- + 2H_3O^+ \rightarrow 4H_2O$ (fast)

What is the rate law based on this mechanism?

A. Rate = $k[HCOOH][I^-]$

B. Rate = $k[HCCOH]^2[I^-]$

C. Rate = $k[HCOOH][I^-]^2$

D. Rate = $k[HCOOH]$

**Solution**

The slowest step is the rate determining step and order is the same as coefficients of the reactants.

$\Rightarrow$Rate = $k[HCOOH][I^-]$

**Answer: A**

71. Ozone ($O_3$) is depleted from tropospheric atmosphere by the following cycle.

$NO + O_3 \rightarrow NO_2 + O_2$

$NO_2 + O \rightarrow 2O_2$

Net $O_3 + O \rightarrow 2O_2$

What observation could you make about the cycle?

A. $NO_2$ is a catalyst

B. $NO_2$ is an intermediate

C. O is a catalyst.

D. O is an intermediates

**Answer: B**

72. Each of the following choices gives the reaction and the corresponding rate law. Which individual step in a chemical reactions elementary reaction?

A. $2A \rightarrow P$, Rate = $k[A]$

B. $A + B \rightarrow P$, Rate = $k[A][B]$

C. $A + 2B \rightarrow P$, Rate $k[A]^2$

D. $A + B + C \rightarrow P$, Rate $k[A][C]$

**Solution**

For an elementary reaction, order is the same as coefficients of reactants.

**Answer: B**

73. The reaction: $2O_3 \rightleftarrows 3O_2$ proceeds through the mechanism below.

$O_3 \underset{k_2}{\overset{k_1}{\rightleftharpoons}} O_2 + O$ . . . fast

$O_3 + O \xrightarrow{k_3} 2O_2$. . . slow

a. Determine the rate law.

b. Identify the intermediates.

**Solution**

a. The second step is the rate determining step.

$\Rightarrow$Rate $= k_3[O_3][O]$ but O is an intermediate, use the first step to replace it $r_f = k_1[O_3]$ $r_b = k_2[O_2][O]$

At equilibrium $r_f = r_b$

$$\Rightarrow k_1[O_3] = k_2[O_2][O]$$

$$[O] = \frac{k_1}{k_2}\frac{[O_3]}{[O_2]}$$

$$\therefore \text{Rate} = k_3\frac{k_1}{k_2}[O_3]\frac{[O_3]}{[O_2]} = k[O_3]^2[O_2]^{-1}$$

b. O

74. A possible mechanism for the reaction: $2A + B \rightarrow C + D$ is

1. $A + A \underset{k_2}{\overset{k_1}{\rightleftharpoons}} A_2$ . . . fast equilibirum
2. $A_2 + A \rightleftharpoons A_3$ . . . slow
3. $A_3 + B \longrightarrow A + C + D$ . . . fast

Determine the rate law and identify the intermediates.

Solve by your own: Answer: Rate $= k[A]^3$, intermediate: $A_2$, $A_3$

75. What is the rate law for a single - step reaction: $A + B \rightarrow 2C$?

**Answer: Rate = k[A][B]**

76. Identify the number of molecular species and the type of molecularity for the following reactions.

a. $2HI \rightarrow H_2 + I_2$

b. $2CO + O_2 \rightarrow 2CO_2$

c. $NH_4NO_2 \rightarrow N_2 + 2H_2O$

**Solution**

a. two, biomolecular

b. three, termoleuclar

c. one, unimolecular

77. What are the two requirements for a given reaction mechanism to be consistent with the reaction?

**Solution**

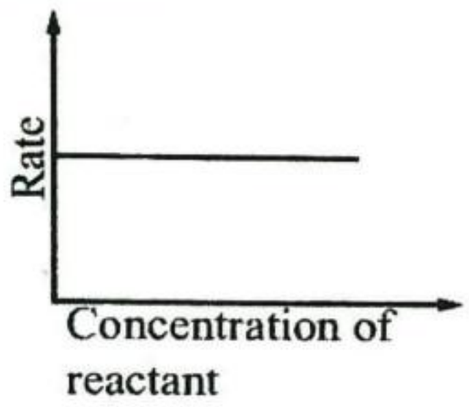
➦ The overall reaction must be the same as the reaction.

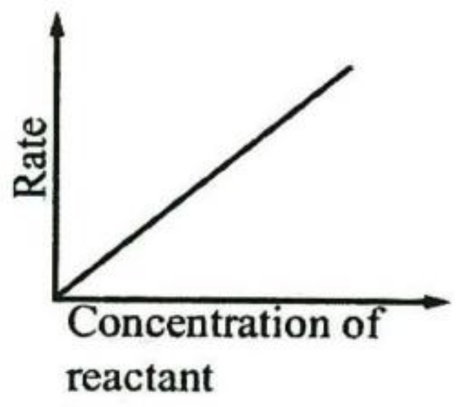
➦ The rate law must agree with the experimentally determined rate law.

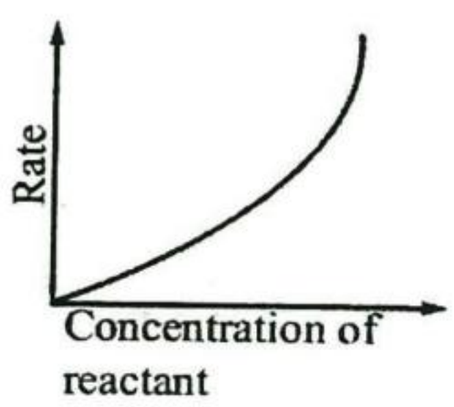
## Solved Problems

78. Explain the difference between zero, first and second order reactions by using graphs of reaction rate against reactant concentration.

**Solution**

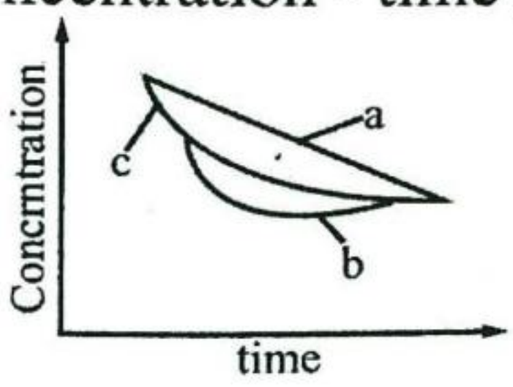



⇒zero-order reaction ⇒first-order reaction ⇒second -order reaction

79. Identify the order of reactions represented with letters a, b and c in the concentration - time graph below.




**Solution**

a. Zero - order

b. Second - order

c. First - order

80. What is the difference between rate and rate constant?

**Explanation**

➦ Rate is a variable proportional to the concentration of reactants but rate constant is a number which depends on the temperature only but not on concentration.

81. Write some fast and slow reactions.

**Solution**

- Fast reactions combustion of fuel, reaction of active metals with acids, precipitation reactions, firing of gun and neutralization reaction.
- Slow reactions: fermentation, souring of milk, geographical reaction, oxidation of copper and rusting of iron.

82. A second order reaction: A→B, has $[A]_0 = 0.8M$ and $[A]_t = 0.5M$ after 30 min.

a. What is the value of rate constant?

b. What is the half - life of the reaction?

c. At what time will [A] = 0.2M be?

**Solution**

a. $$\frac{1}{[A]_t} - \frac{1}{[A]_0} = kt$$

$$\frac{[A]_0 - [A]_t}{[A]_t[A]_0} = kt$$

$$k = \frac{[A]_0 - [A]_t}{[A]_t - [A]_0} = \frac{(0.8 - 0.5)M}{30\,min \times 0.8 \times 0.5M^2}$$

$$k = 0.025M^{-1}.min = 0.025L.mol^{-1}min^{-1}$$

b. $$t_{1/2} = \frac{1}{k[A]_0} = \frac{1}{0.025M^{-1}.min^{-1} \times 0.8M} = 50\,min$$

c. $$t = \frac{[A]_0 - [A]_t}{k[A]_t[A]_0} = \frac{(0.8 - 0.2)M}{0.025M^{-1}.min^{-1} \times 0.8M \times 0.2M}$$

$$t = 150\ min$$

83. In the reaction $2X \rightarrow Y$, concentration of X is found to be 0.54M at t = 50 sec and 0.42M at t = 62 sec. What is the average rate of the reaction during this time interval?

**Solution**

$$Rate = \frac{-\Delta[x]}{2\Delta t} = -\frac{1}{2}\left(\frac{[x]_f - [x]_i}{t_f - t_i}\right)$$

$$Rate = -\frac{1}{2}\left(\frac{0.42 - 0.54}{62 - 50}\right)\frac{M}{s} = 0.005M.s^{-1}$$

84. The rate of a first order reaction, $A \rightarrow P$ is $8 \times 10^{-4} mol.L^{-1}.s^{-1}$ when concentration of A is $0.4 mol.L^{-1}$. Calculate the value of rate constant.

**Solution**

$$Rate = k\,[A]$$

$$k = \frac{Rate}{[A]} = \frac{8 \times 10^{-4}\, mol.L^{-1}.s^{-1}}{0.4 mol.L^{-1}} = 2 \times 10^{-3} s^{-1}$$

85. The rate constant for a hypothetical reaction $2A \rightarrow D$ is $3 \times 10^{-5} mol.L^{-1}.s^{-1}$. What is the order of the reaction?

**Solution**

The order of the reaction is zero because the unit of rate constant is the same as the unit of rate only for zero order reaction.

86. A hypothetical reaction: $A_2 + B_2 \rightarrow 2AB$ follows the mechanism below.

$$A_2 \underset{k_2}{\overset{k_1}{\rightleftharpoons}} A + A$$

$$A + B_2 \xrightarrow{k_3} AB + B$$

$$A + B \xrightarrow{k_4} AB$$

What is the overall order of the reaction?

$k = A e^{-E_a/RT}$

$\boxed{r = k[A]}$

**Solution**

$$Rate = k_3[A][B_2] \text{ but A is an intermediate}$$

$$[A]^2 = \frac{k_1[A_2]}{k_2}$$

$$[A] = \left(\frac{k_1}{k_2}\right)^{1/2} [A_2]^{1/2}$$

$$\Rightarrow Rate = k_3 \left(\frac{k_1}{k_2}\right)^{1/2} [A_2]^{1/2}$$

$$Rate = k[A_2]^{1/2}[B_2]$$

$$\therefore \text{Overall order} = \frac{1}{2} + 1 = \frac{3}{2}$$

87. For a third order reaction, if the concentration of a reactant is increased by a factor of 4, what will happen to the rate of the reaction?

**Solution**

$$\text{Rate}_1 = k[A]_1^3 \qquad [A]_2 = 4[A]_1$$

$$\text{Rate}_2 = k[A]_2^3$$

$$\text{Rate}_2 = k(4[A]_1)^3$$

$$\text{Rate}_2 = 4^3 k[A]_1^3$$

$\text{Rate}_2 = 64\ \text{Rate}_1$, increases by a factor of 64.

88. What happens to the rate of a reaction for gaseous reactants if you increase the pressure?

**Solution**

The rate of the reaction also increases.

89. A chemist was able to determine the rate constant of a particular reaction at 100°C was four times faster than at 30°C. Calculate the approximate energy of activation for the reaction is kJ/mol?

**Solution**

$$T_1 = 30°C = 303K$$

$$T_2 = 100°C = 373K$$

$$K_2 = 4K_1$$

$$\log\frac{k_2}{k_1} = \frac{Ea}{2.303R}\left(\frac{T_2 - T_1}{T_1.T_2}\right)$$

$$Ea = \frac{2.303R\log 4\dfrac{k_1}{k_2}}{T_2.T_1}.T_1.T_2$$

$$Ea = \frac{2.303 \times 8.314\dfrac{J}{mol.k}\log 4 \times 303K.373K}{(373-303)K}$$

$$Ea = 18.6kJ/mol$$

90. Write the difference between positive and negative catalysts.

**Solution**

- Positive catalysts increase the rate of a reaction by lowering activation energy.
- Negative catalysts slow down the rate of a reaction by increasing the activation energy.

91. For the reaction: $X(g) + 2Y(g) \rightarrow XY_2(g)$, the following data were obtained.

| Run | Initial partial pressure in torr | | Initial rate in torr/s |
|---|---|---|---|
| | Px | $Py_2$ | |
| 1 | 320 | 380 | $5 \times 10^{-4}$ |
| 2 | 320 | 760 | $1 \times 10^{-3}$ |
| 3 | 640 | 760 | $4 \times 10^{-3}$ |

a. Determine rate law

b. Calculate the values of k

**Solution**

a. $\text{Rate} = kP_x^a.P_{y_2}^b$

b. $$k = \frac{5\times10^{-4}\,\text{torr/s}}{(320\,\text{torr})^2.(380\,\text{torr})}$$

$$k = 1.28 \times 10^{-11}\ \text{torr}^{-2}.\text{s}^{-1}$$

92. A first order reaction has a half - life of 10μs. What is the value of k for this reaction?

**Solution**

$$t_{1/2} = \frac{0.693}{k}$$

$$k = \frac{0.693}{t_{1/2}} = \frac{0.693}{10\times10^{-6}\text{s}} = 6.93\times10^{4}\text{s}^{-1}$$

93. The decomposition of a certain compound follows second order kinetics with $k = Ae^{-32000k/T}$. Calculate the activation energy of this reaction, in kJ/mol.

**Solution**

From Arrhenius equation we know that:

$$k = Ae^{-\frac{Ea}{kT}} = Ae^{-32000k/T}$$

$$\Rightarrow -\frac{Ea}{RT} = -32000\frac{k}{T}$$

$$\Rightarrow Ea = 32000\cancel{K}.8.314\frac{J}{mol.\cancel{K}}$$

$$Ea = 266\text{kJ/mol}$$

94. A second order reaction whose rate constant at 600°C was found to be $2 \times 10^{-3} L.mol^{-1}.s^{-1}$ has an activation energy of $83 kJ.mol^{-1}$. What is the value of rate constant at 800°C?

**Solution**

$$\log\frac{k_2}{k_1} = \frac{Ea}{2.303R}\left(\frac{T_2 - T_1}{T_1.T_2}\right)$$

$$\log\frac{k_2}{k_1} = \frac{83,000 J.mol^{-1}}{2.303 \times 8.314 J.mol^{-1}.k^{-1}}\left(\frac{1073 - 873}{1073.873}\right)^{k^{-1}}$$

$$\log\frac{k_2}{k_1} = 0.93$$

$$k_2 = 10^{0.93}.k_1$$

$$k_2 = 1.68 \times 10^{-2} L.mol^{-1}.s^{-1}$$

95. A reaction is 50% complete in 20 min. How long after its start will the reaction be 75% if its is first order reaction.

**Solution**

$$[A]_t = \frac{1}{2}[A]_0, t = 20 min$$

$$\log\frac{[A]_t}{[A]_t} = \frac{kt}{2.303}$$

$$k = \frac{2.303 \log \frac{2[A]_t}{[A]_t}}{t}$$

$$k = \frac{2.303 \log 2}{20 min} = 0.035 min^{-1}$$

$\Rightarrow$75% complete means 25% left

$$\frac{[A]_t}{[A]_0} = 25\% = \frac{25}{100} = \frac{1}{4}$$

$$[A]_0 = 4[A]_t$$

$$t = \frac{2.303 \log \frac{[A]_0}{[A]_t}}{k}$$

$$t = \frac{2.303 \log \frac{4[A]_t}{[A]_t}}{0.035 min^{-1}}$$

$$t = 40 min$$

96. The following rate data, obtained at 250k describe the reaction:

$$F_2(g) + 2ClO_2(g) \rightarrow 2FClO_2(g)$$

| Experiment | $[F_2]_0$, M | $[ClO_2]_{0.1}$M | $(Rate)_0$(mol/L.s) |
|---|---|---|---|
| 1 | 0.1 | 0.01 | $1.2v\times 10^{-3}$ |
| 2 | 0.1 | 0.04 | $4.8 \times 10^{-3}$ |
| 3 | 0.2 | 0.01 | $2.4 \times 10^{-3}$ |

a. Determine the rate law for the reaction.

b. Calculate the value of the rate constant.

**Solution**

Solve by your own.

a. Rate $= k[F_2][ClO_2]$

b. $k = 1.2\ M^{-1}.s^{-1}$

97. Write three heterogeneous catalysts used in industrial production of some substances.

**Solution**

- Production of solid fat from vegetable oil by hydrogenation in presence of platinum or nickel metal as a catalyst.
- In contact process of $H_2SO_4$ production, solid Vanadium (V) oxide or Platinum or Nickel metal as a catalyst.
- In the Haber synthesis of ammonia from $H_2(g)$ and $N_2(g)$ iron is used as a catalyst.

98. The rate constant for the formation of hydrogen iodide from the elements.

$H_2(g) + I_2(g) \rightarrow 2HI(g)$, is $2.7 \times 10^{-4}M^{-1}.s^{-1}$ at 600k and $3.2 \times 10^{-2}M^{-1}.S^{-1}$ at 700K. Find the activation energy, Ea.

**Answer: Ea = 167 kJ/mol**

99. If the rate constant for the first – order reaction $SO_2Cl(g) \rightarrow SO_2(g) + Cl_2(g)$ is $2.2 \times 10^{-5}s^{-1}$.

a. What is the half – life? How long would it take for the concentration of $SO_2Cl_2$ to decrease to

b. 25% of its initial value?

c. 12.5% of its initial value?

**Solution**

a. $t_{1/2} = \frac{0.693}{k} = \frac{0.693}{2.2\times10^{-5}\,s^{-1}} = 3.15\times10^{4}\,s$

b. $[SO_2Cl_2] = 25\%[SO_2Cl_2]_0 = \frac{1}{4} = [SO_2Cl_2]_0$

$$\log\frac{[SO_2Cl_2]_0}{[SO_2Cl_2]_t} = \frac{kt}{2.303}$$

$t = 6.3 \times 10^4 s$

c. $t = 9.5 \times 10^4 s$

100. What are the two requirements for a reaction mechanism?

**Solution**

- The sum of the elementary steps must give the overall balanced equation for the reaction.
- The mechanism must agree with the experimentally determined rate law.

# Supplementary Exercise

I. Multiple choice

1. The reaction between NO and $I_2$ is second order in NO and first order in $I_2$. What change occurs in the rate of the reaction if the concentration of NO is doubled and $I_2$ left unchanged?

A. doubled
C. eight times
B. quadruple
D. three times

2. A reaction is 50% complete in 2 hours and 75% complete in 4 hours. What is the order of this reaction?

A. 0 B. 1 C. 2 D. 3

3. For the reaction, $2NO\ (g) + O_2(g) \rightarrow 2NO_2(g)$, the following rate data were obtained.

| | | | |
|---|---|---|---|
| $[NO]_o$, M: | 0.10 | 0.10 | 0.40 |
| $[O_2]_o$, M: | 0.10 | 0.20 | 0.20 |
| $(rate)_o$, M/S: | $2.5 \times 10^{-4}$ | $5.0\times 10^{-4}$ | $8.0\times 10^{-3}$ |

What is the rate law for the reaction?

A. rate = $k[NO][O_2]$
C. rate = $k[NO]^2[O_2]$
B. rate = $k[NO][O_2]^2$
D. rate = $k[NO]_2[O_2]_2$

4. One of the steps in the manufacture of nitric acid is the oxidation of ammonia shown in this equation:

$4NH_3(g) + 5O_2(g) \rightarrow 4NO(g) + 6H_2O(g)$.

If gaseous water appears at a rate of 0.025mol. $min^{-1}$, at what rate does ammonia disappear?

A. 0.0040 mol. $min^{-1}$
C. 0.038 mol. $min^{-1}$
B. 0.017 mol. $min^{-1}$
D. 0.150 mol. $min^{-1}$

5. What will be the order of the reaction with respect to $OH^-$ if the rate law is, rate = $k/[OH^-]$?

A. 0 B. 1 C. –1 D. 2

6. For the decomposition of acetaldehyde the rate expression is: rate = k[Acetaldehyde]$^{3/2}$. If the concentration of acetaldehyde is doubled, then the rate would increase by a factor of

A. 5.50
C. 3.0
B. 2.83
D. 4.83

7. At low temperature, the rate law for the reaction

$CO_2(g) + NO_2(g) \rightarrow CO_2(g) + NO(g)$ is rate = $k[NO_2]^2$. Which of the following mechanism is consistent with this rate law?

A. $CO + NO_2 \rightarrow CO_2 + NO$

B. $2NO_2 \rightarrow N_2O_4$ (fast)

$N_2O_4 + 2CO \rightarrow 2CO_2 + 2NO$ (slow)

C. $2NO_2 \rightarrow NO_3 + NO$ (slow)

$NO_3 + CO \rightarrow NO2 + CO2$ fast

D. $2NO_2 \rightarrow 2NO + O_2$ fast

$2CO + O_2 \rightarrow 2CO_2$ (slow)

8. All of the following are correct expression for third order reactions except.

A. rate = $k[A]^3$

B. rate = $k[A]^2[B]$

C. rate = $k[A][B][C]$

D. rate = $k[A]^{3/2}[B]^{1/2}$

9. Which of the following is true about the factors affecting rate of reaction?

A. As surface area of solid reactants increases, the rate of the reaction also increases.

B. As concentration of reactants increases, the rate of any order of reaction increase.

C. As temperature increases, the rate of gaseous reaction also increase.

D. A catalyst always speed up the rate of the reaction.

10. A typical pathway for a chemical reaction that includes several simple steps is

A. elementary reactions
B. activation energy
C. an intermediate reaction
D. reaction mechanism

11. If we doubled the concentration of a reactant, the rate increases by eight times, the reaction is

A. First – order
B. Zero – order
C. Third – order
D. second – order

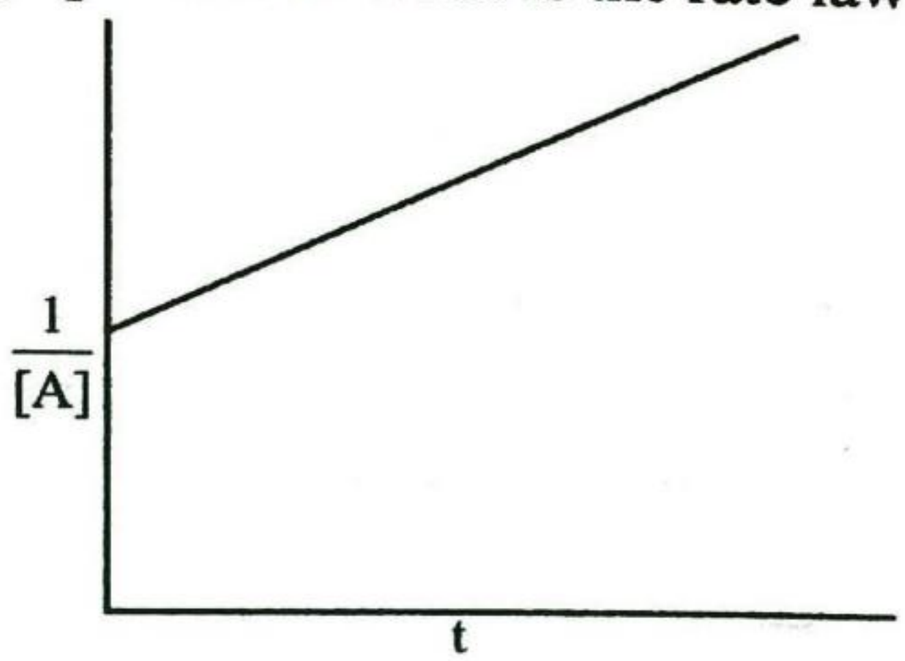
12. The activation energy of a reaction can be determined from the slope of which of the following graphs?

A. lnk versus T
B. $\frac{\ln k}{T}$ *versus* $\frac{1}{T}$
C. $\ln k$ *versus* $\frac{1}{T}$
D. $\frac{\ln k}{T}$ *versus* $T$

13. Consider the reaction $2N_2O_5 \rightarrow 4NO_2 + O_2$. What is the molecularity of the reaction?

A. third B. second C. first D. zero

14. For the reaction A + B → C the change in concentration of A with time is shown in the graph below. What is the rate law for this reaction?




A. rate = k[A]
B. rate = $k[A]^2$
C. rate = k[A] [B]
D. rate = $k [A]^2[B]$

15. The logarithm of the rate constant of a reaction is:

A. directly proportional to temperature changes
B. inversely proportional to temperature
C. not affected by temperature change
D. only affected by activation energy.

16. Consider the reaction X + Y →Z. The data below was obtained for the reaction after a kinetic study

| Run | [X],M | [Y],M | rate (mol/L.Sec) |
|---|---|---|---|
| 1 | 2.0 | 2.0 | 1.5 |
| 2 | 6.0 | 2.0 | 4.5 |
| 3 | 2.0 | 6.0 | 1.5 |

Which of the following is true about this reaction?

A. The reaction is second order in X

B. The rate of the reaction increases by a factor of four if the concentration of both X and Y is doubled

C. The reaction is zero order in y and first order in X

D. The overall order of the reaction is third.

17. For termolecular reaction, the order of the reaction is

A. 1 B. 2 C. 3 D. 0

18. Which of the following is not heterogeneous reaction?

A. $3Fe(s) + 4H_2O(g) \rightarrow Fe_3O_4(s) + 4H_2(g)$

B. $C(s) + O_2(g) \rightarrow CO_2(g))$

C. $CaCO_3(s) \rightarrow CaO(s) + CO_2(g)$

D. $N_2(g) + 3H_2(g) \rightarrow 2NH_3(g)$

19. All of the following are preconditions for a reaction to occur according to collision theory except

A. The particles of the reacting substances must collide

B. The particles of the reacting species must have proper orientation

C. The particles of the reactants must collide with energy greater than or equal to the activation energy

D. Ineffective collision results desired products.

20. Which of the following reaction is exothermic reaction?

A. $A + B \rightarrow C + D + heat$

B. $A+B + heat \rightarrow C+D$

C. $A + B \rightarrow C + D - heat$

D. $A+ B \rightarrow C+D, \Delta H=120KJ$

II. Give short answer

21. Give the molecularity of each of the following reactions and write rate expression for each.

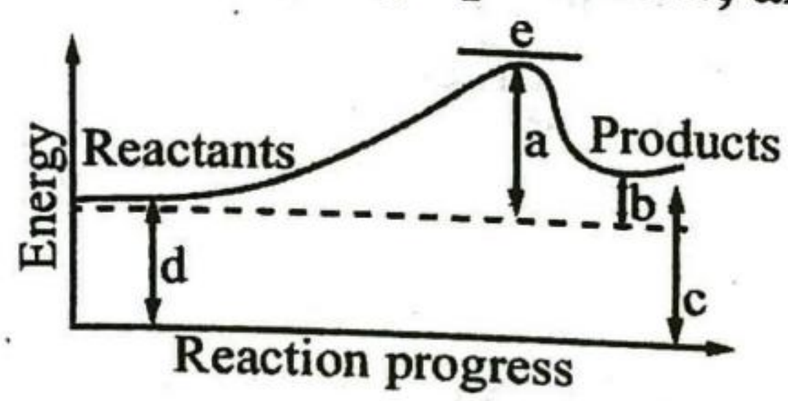
a. $2NO_2(g) \rightarrow NO(g) + NO_3(g)$

b. $CHCl_3(g) + Cl_2(g) \rightarrow CCl_4(g) + HCl(g)$

c. $2NO + O_2 \rightarrow 2NO_2$

d. $NH_4NO_2 \rightarrow N_2 + 2H_2O$

22. Based on the graph below, answer the following questions.




The letter that represents

i activation energy _____

ii enthalpy of the reaction

iii heat content of product _____

iv heat content of reactant _____

v activated complex _____

vi The graph represents _____ reaction

23. For the reaction $2NO + Br_2 \rightarrow 2NOBr$, the following mechanism has been suggested:

$NO + Br_2 \rightleftharpoons NOBr_2$ (fast)

$NOBr_2 + NO \rightleftharpoons 2NOBr$ (slow)

What is the rate law consistent with the mechanism?

24. Determine the overall order for:

a. rate = $k[A]^2[B]$

b. rate = $k[x]^{½}[y]^2$

c. rate = $\frac{k[A][B]}{[C]}$

25. Express the rate of the following reactions in terms of disappeance of reactants and appearance of products.

    a. $H_2O_2(g) + H_2(g) \rightarrow 2H_2O(g)$

    b. $4NH_3(g) + 3O_2(g) \rightarrow 2N_2(g) + 6H_2O(g)$

    c. $C_3H_8(g) + 5O_2(g) \rightarrow 3CO_2(g) + 4H_2O(g)$

26. _____ is the study of the rates of chemical reactions and mechanism by which reactions occur.

27. The short – lived combination of reacting atoms, molecules or ions that is intermediate between reactants and products is ____.

28. The rate at a given specific time is _____

29. Substances that slow down a reaction are _____

30. The kinetics of the reaction, $2x(g) + y_2(g) \rightarrow 2xy_2(g)$ was studied at 300K by varying the initial pressures of the reactants individually and finding the initial rates. The following data are typical of such a study.

| Run | $P_x^\circ$, mmHg | $P_{1/2}^\circ$, mmH | $(rate)_0$, mmHg/S |
|---|---|---|---|
| 1 | 100 | 100 | 20 |
| 2 | 100 | 200 | 40 |
| 3 | 200 | 100 | 80 |

Determine the order of the reaction with respect to each reactant and the rate law.

አጋዥ መጽሐፍቶችን ለማግኘት በ Telegram
@ethiolibrary1 ላይ ይቀላቀሉን
+251927273995
+251909282436
jemal6454@gmail.com

# Unit 5

## Chemical Equilibrium and Phase Equilibrium

## Unit Content Outline:

5.1 Attainment and Characteristics of a Chemical equilibrium

5.2 Equilibrium expression and equilibrium constant

5.3 Calculating the equilibrium Concentrations

5.4 Factors affecting chemical equilibrium

5.5 Phase Equilibrium

5.6 Phase Diagram

**Introduction**

Some reactions proceed to completion but some reactions do not proceed to completion.

- Reactions that go to completion are called **irreversible reactions**.
- Chemical reactions that do not go to completion are know as **reversible reactions.**

Table 5.1. The difference between irreversible and reversible reactions

| Reversible reactions | Irreversible reactions |
|---|---|
| ➦ are two way reactions<br>➦ do not go to completion<br>➦ are represented by double arrow (⇌) A ⇌ B<br>➦ reactants form products and products give back the original reactants.<br><br>Example<br>$N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$<br>$2SO_2(g) + O_2(g) \rightleftharpoons 2SO_3(g)$ | ➦ are one way reactions.<br>➦ go to completion.<br>➦ are denoted by single arrow (→) C→ D<br>➦ reactants form products and the amount of limiting reactant that left is negligible.<br><br>Example<br>$2Na + 2H_2O \rightarrow 2NaOH + H_2$<br>$NaOH + HCl \rightarrow NaCl + H_2O$ |

## 5.1 Attainment and Characteristics of a Chemical equilibrium

- **Equilibrium** is a condition in which the forward and reverse rates in a physical or chemical system are equal.
- A reaction that goes from left to right is called **forward reaction.**
- A reaction that proceeds from right to left is known as **reverse or backward reaction.**

Example

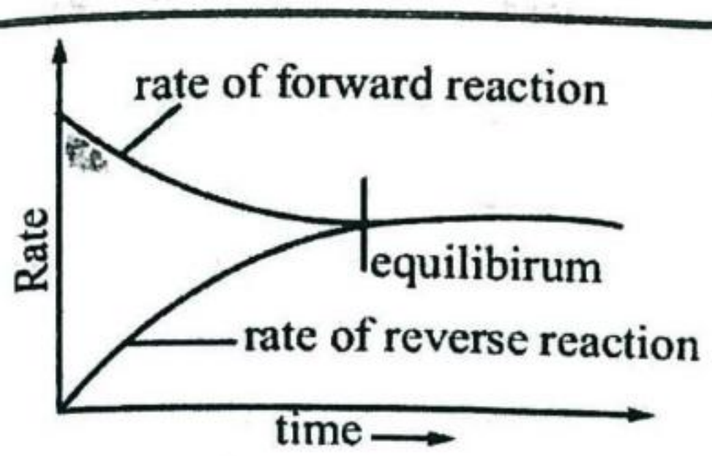
Consider the hypothetical reaction below:

$aA + bB \rightleftharpoons cC + dD$

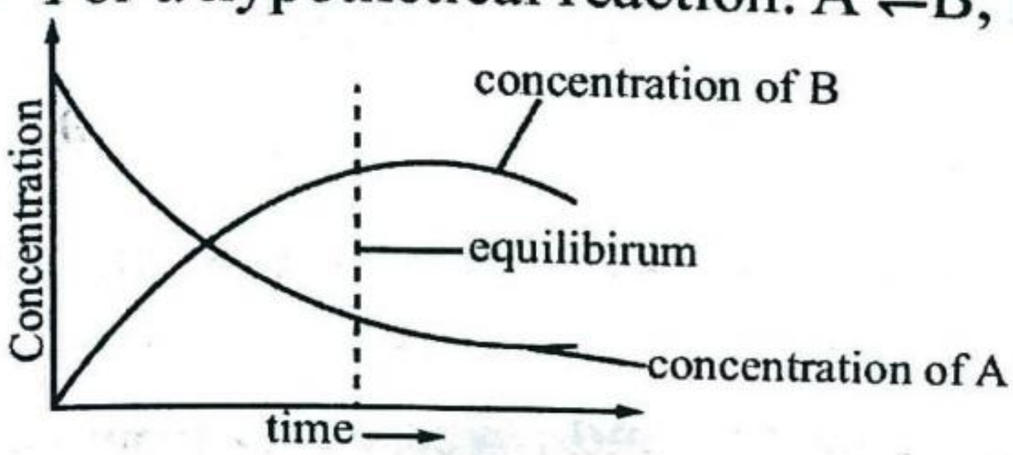
$\Rightarrow aA + bB \rightarrow cC + dD$ – forward reaction

$cC + dB \rightarrow aA + bB$ – backward reaction

- The rate versus time graph for a reversible reaction is given below:




➦ For a hypothetical reaction: A ⇌B, the concentration time graph is




**Equilibrium is attained when the rate of forward reaction is equal to the rate of backward reaction.**

## Characteristics of Chemical equilibrium

1. It is reached when forward rate is equal to backward rate.
2. It is dynamic meaning the reaction does not stop at equilibrium.
3. It is only applicable to reversible reactions.
4. The concentration of reactants is not necessarily equal to the concentration of products.
5. Entropy of the universe is zero at equilibrium.
6. Gibb's free energy change is zero.
7. The value of cell potential is zero.
8. The value of equilibrium constant is one.
9. Macro-properties such as temperature, pressure, volume and concentration should be constant.
10. The change in concentration of reactants and products is constant.
11. The rate constant of the forward and reverse reactions are not necessarily equal.

## Illustrative Examples

1. Which of the following correctly describes a chemical reaction at equilibrium? (EUEE ... 2004 E.C.)
   A. The concentration of reactants and products are equal.
   B. The change in concentration of reactants and products is constant.

C. The rate of forward reaction is less than the rate of reverse reaction.

D. The rate of forward reaction is greater than the rate of reverse reaction.

**Solution**

- ➦ At equilibrium, concentration of reactants is not necessary equal to concentration of products.
- ➦ Rate of forward and reverse reaction are equal.

**Answer: B**

2. Which of the following statements describes a system that is at equilibrium?

A. The rate constants for the forward and reverse reactions are equal.

B. The concentrations of reactants and products are equal.

C. The reaction cases.

D. The forward and reverse rates are equal.

**Solution**

- ➦ Forward rate constant and reverse rate constants are not equal and the reaction does not cease at equilibrium.

**Answer: D**

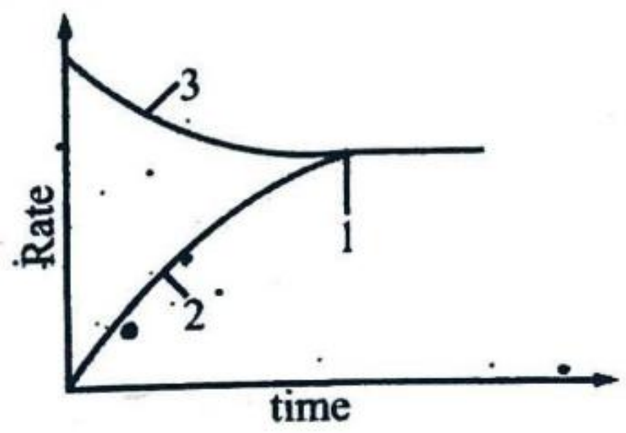
3. For the reversible reaction: $N_2 + 3H_2 \rightleftharpoons 2NH_3$, write the backward and forward reactions.

**Solution**

Forward reaction: $N_2 + 3H_2 \rightarrow 2NH_3$

Backward reaction: $2NH_3 \rightarrow N_2 + 3H_2$

4. Given the rate versus time graph below. Write the terms represented with numbers 1 to 3.



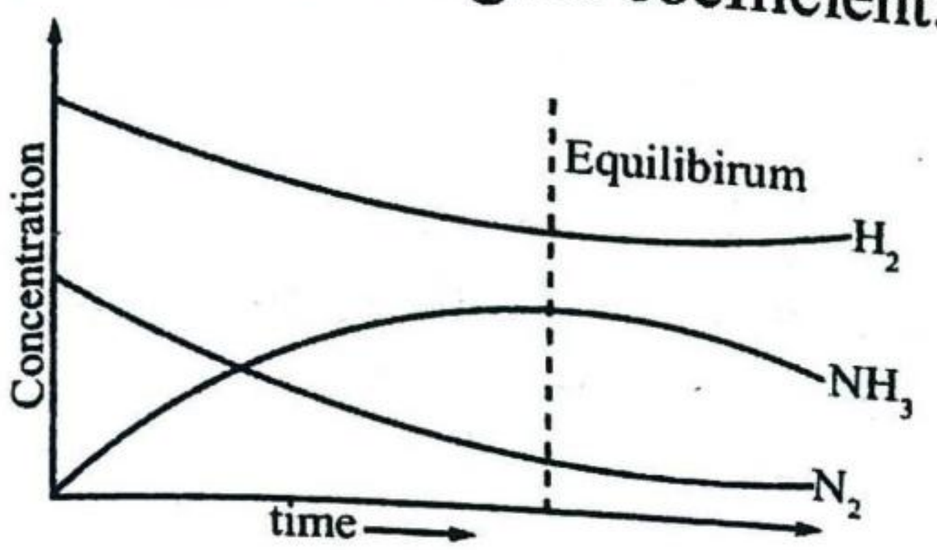

**Solution**

1. Equilibrium
2. Rate of reverse reaction.
3. Rate of Forward reaction.

5. Draw the concentration - time graph for the reaction $N_2 + 3H_2 \rightleftharpoons 2NH_3$

**Solution**

The concentration of $H_2$ should be above the concentration of $N_2$ because of its higher coefficient.




6. Write some physical process that are associated with equilibrium condition.

**Solution**

- Saturated solution (rate of dissolution is equal the rate of crystallization)
- The vapor pressure of liquid in a closed vessel (rate of evaporation is equal to rate of condensation).

## 5.2 Equilibrium expression and equilibrium constant

**Law of mass action or law of concentration action** is a law that characterized a dynamic equilibrium.

- It states that for a reversible reaction at equilibrium, at a given condition, equilibrium constant is equal to the product of the amount of substances formed divided by the amount of reactants each raised to the power of the corresponding coefficients.
- For a hypothetical reaction:

$aA + bB \rightleftharpoons cC + dD$

$r_f \alpha [A]^a[B]^b \Rightarrow r_f = k_f [A]^a[B]^b$

$r_b \alpha [C]^c[D]^d \Rightarrow r_b = k_b [C]^a[D]^d$

At equilibrium, $r_f = r_b$

$\Rightarrow k_f [A]^a[B]^b = k_b [C]^c[D]^d$

$$\boxed{K_c = \frac{k_f}{k_b} = \frac{[C]^c[D]^d}{[A]^a[B]^b}}$$

- $r_f$ - rate of forward reaction.
- $r_b$ - rate of backward reaction.
- $k_f$ - rate constant for the forward reaction.
- $k_b$ - rate constant for the backward reaction.
- $K_c$ - equilibrium constant in concentration.
- [ ] - concentration in mole per liter
- For gaseous reactants and products, the equilibrium constant may be represent by $K_p$ if partial pressures are given.

$$aA + bB \rightleftharpoons cC + dD$$

$$K_P = \frac{P_c^c . P_D^d}{P_A^a . P_B^b}$$

To write equilibrium constant expression follow the rules below.

- The equilibrium product concentrations appear in numerator.
- The equilibrium reactant concentrations appear in denominator.
- The equilibrium concentration of each species is raised to its coefficient.
- Pure solids (s) and liquids or solvents in dilute solution do not appear in the equilibrium constant expression.

## Example

Writ $K_c$ and $K_p$ expression for the following reactions at equilibrium.

a. $2SO_2(g) + O_2(g) \rightleftharpoons 2SO_3(g)$

b. $Fe_2O_3(s) + 3CO(g) \rightleftharpoons 2Fe(s) + 3CO_2(g)$

**Solution**

a. $K_c = \frac{[SO_3]^2}{[SO_2]^2[O_2]}, K_P = \frac{P_{SO_3}^2}{P_{SO_3}^2 . P_{O_2}}$

b. $K_c = \frac{[CO_2]^3}{[CO]^3}, K_P = \frac{P_{CO_2}^3}{P_{CO}^3}$

If all the reactants and products are in the same sate, the reaction is known as homogeneous reversible reaction.

If the reactants and products are in different states, the reaction is called heterogeneous reversible reaction.

## Illustrative Examples

7. The conventional equilibrium constant expression ($K_c$) for the system: $2ICl(s) \rightleftharpoons I_2(s) + Cl_2(g)$ is (EUEE ... 2007 E.C)

A. $\frac{[I_2][Cl_2]}{[ICl]^2}$

B. $\frac{[I_2][Cl_2]}{2[ICl]}$

C. $[Cl_2]$

D. $\frac{[I_2]+[Cl_2]}{2[ICl]}$

**Solution**

➦ Solids are not included the equilibrium constant expression.
$K_c = [Cl_2]$

**Answer: C**

8. What is the equilibrium constant expression for the following reaction? $2Hg[g] + O_2(g) \rightleftharpoons 2HgO(s)$?

A. $K = \frac{1}{[Hg]^2[O_2]}$

B. $K = [Hg]^2[O_2]$

C. $K = \frac{[HgO]^2}{[Hg]^2[O_2]}$

D. $K = \frac{[2HgO]}{[2Hg][O_2]}$

**Answer: A**

9. For the reactions A→B and B→A, the rate constant are $8\times10^2 s^{-1}$ and $5.2 \times 10^4 s^{-1}$ respectively. What is the value of the equilibrium constant for the equilibrium A ⇌B? (EUEE ... 1999 E.C)

A. $1.54 \times 10^{-3}$
B. $1.54 \times 10^{-2}$
C. $6.5 \times 10^{-3}$
D. $6.5 \times 10^{-2}$

**Solution**

$$K_c = \frac{k_f}{k_b} = \frac{[B]}{[A]}$$

$$K_c = \frac{8\times10^2 s^{-1}}{5.2\times10^4 s^{1}} = 1.54\times10^{-2}$$

**Answer: B**

10. If the equilibrium concentrations found in the reaction: $2A(g) + B(g) \rightleftharpoons C(g)$ are

$[A] = 2.4 \times 10^{-2}M$, $[B] = 4.6 \times 10^{-3}M$ and

$[C] = 6.2 \times 10^{-3}M$, then what is the value of Kc? (EUEE)

A. $1.3 \times 10^{-4}$

B. $2.3 \times 10^{3}$

C. 56

D. $5.8 \times 10^{2}$

**Solution**

$$K = \frac{[C]}{[A]^2[B]} = \frac{6.2\times10^{-3}}{(2.4\times10^{-2})^2(4.6\times10^{-3})}$$

$$K_c = 2.4\times10^{3}$$

**Answer: B**

11. At a certain temperature for the reaction: $2NO(g) + O_2(g) \rightleftharpoons 2NO_2(g)$, the equilibrium concentrations are, $[NO] = 0.52M$, $[O_2] = 0.24M$, $[NO_2] = 0.18M$. What is the value of kc at this temperature?

**Solution**

$$K_c = \frac{[NO_2]^2}{[NO]^2[O_2]} = \frac{(0.18)^2}{(0.52)^2(0.24)}$$

$$K_c = 0.50$$

12. At 295K, a 3L flask contains 7 mol HI, 4 mol $H_2$ and 0.6mol $I_2$ in equilibrium. What is the value of $K_C$ at this temperature?

**Solution**

First find the concentration of each species.

$$[HI] = \frac{n_{HI}}{v} = \frac{7mol}{3L} = 2.33M$$

$$[H_2] = \frac{n_{H_2}}{v} = \frac{4mol}{3L} = 1.3M$$

$$[I_2] = \frac{n_{I_2}}{v} = \frac{0.6mol}{3L} = 0.2M$$

$$K_c = \frac{[HI]^2}{[H_2][I_2]} = \frac{(2.33)^2}{(1.33)(0.2)}$$

$$K_c = 20.47$$

13. The value $K_c$ for the reaction: $COCl_2(g) \rightleftharpoons CO(g) + Cl_2(g)$ is $2.5 \times 10^{-10}$ at 300k. Starting with pure $COCl_2(g)$ in a closed vessel, if the concentration of CO(g) is $1 \times 10^{-5}M$ at equilibrium, what would be the concentration of $COCl_2(g)$? (EUEE)

**Solution**

Since the coefficient ratio is 1:1, at equilibrium $[CO] = [Cl_2]$

$= 1 \times 10^{-5}M$

$$K_c = \frac{[CO][Cl_2]}{[COCl_2]}$$

$$[COCl_2] = \frac{[CO][Cl_2]}{K_c} = \frac{(1\times10^{-5})^2}{2.5\times10^{-10}}$$

$[COCl_2] = 0.40M$

14. Write $K_c$ expression for

a. $AgCl(s) \rightleftharpoons Ag^+(aq) + Cl^-(aq)$

b. $NH_3(g) + H_2S(aq) \rightleftharpoons NH_4^+(aq) + HS^-(aq)$

c. $HF(g) + H_2O(l) \rightleftharpoons H_3O^+(aq) + F^-(aq)$

d. $Mg(s) + 2HCl(aq) \rightleftharpoons MgCl_2(aq) + H_2(g)$

e. $Ag^+(aq) + 2NH_3(aq) \rightleftharpoons Ag(NH_3)_2^+(aq)$

f. $PCl_5(s) \rightleftharpoons PCl_3(l) + Cl_2(g)$

**Solution**

a. $K_c = [Ag^+][Cl^-]$

b. $K_c = \frac{[NH_4^+][HS^-]}{[NH_3][H_2S]}$

c. $K_c = \frac{[H_3O^+][F^-]}{[HF]}$

d. $K_c = \frac{[MgCl_2][H_2]}{[HCl]^2}$

e. $K_c = \frac{[Ag(NH_3)_2^+]}{[Ag^+][NH_3]^2}$

f. $K_c = [Cl_2]$

15. Write $K_p$ expression for

a. $H_2(g) + F_2(g) \rightleftharpoons 2HF(g)$

b. $CO_2(g) + CaO(s) \rightleftharpoons CaCO_3(s)$

c. $4NH_3(g) + 5O_2 \rightleftharpoons 4NO(g) + 6H_2O(g)$

**Solution**

a. $K_P = \frac{P_{HF}^2}{P_{H_2}.P_{F_2}}$

b. $K_P = \frac{1}{P_{CO_2}}$

c. $K_P = \frac{P_{NO}^4.P_{H_2O}^6}{P_{NH_3}^4.PO_2^5}$

16. Classify the following as homogeneous or heterogeneous equilibrium.

a. $H_2CO_3(aq) + H_2O(l) \rightleftharpoons HCO_3^-(aq) + H_3O^+(aq)$

b. $NH_3(g) + H_2O(l) \rightleftharpoons NH_4^+(aq) + OH^-(aq)$

c. $Zn(s) + 2HCl(aq) \rightleftharpoons ZnCl_2(aq) + H_2(g)$

d. $H_2(g) + Br_2(g) \rightleftharpoons 2HBr(g)$

**Solution**

- a and d are homogeneous equilibria.
- b and c are heterogeneous equilibria.

17. Given the reaction at equilibrium: $A(s) + 2B(g) \rightleftharpoons AB_2(g)$. If the partial pressure of B and $AB_2$ are 4 atm and 3atm respectively at equilibrium, then find $K_p$.

**Solution**

$$K_P = \frac{P_{AB_2}}{P_B^2} = \frac{3\ atm}{(4\ atm)^2} = 0.1875 atm^{-1}$$

**Units of $K_c$**

For a hypothetical reaction: $aA + bB \rightleftharpoons cC + dD$

$$K_c = \frac{[C]^c[D]^d}{[A]^a[B]^b}$$

$$\text{unit of } k_c = \frac{M^c.M^d}{M^a.M^b} = M^{(c+d)-(a+b)}$$

$$\text{Unit of } K_c = (mol/L)^{\Delta n},\ \Delta n = (c + d) - (a + b)$$

**Example**

What is the unit of $k_c$ for $N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$?

**Solution**

$\Delta n = 2 - (3 + 1) = -2$

$\text{Unit of } K_c = (mol/L)^{\Delta n} = (mol/L)^{-2} = L^2.\ mol^{-2}$

**Units of $K_P$**

$aA + bB \rightleftharpoons cC + dD$

$K_P = \frac{P_C^c.P_D^d}{P_A^a.P_B^b}$, if the unit of pressure is atmosphere (atm)

$\text{unit of } K_P = (atm)^{(c+d)-(a+b)} = (atm)^{\Delta n}$

**Example**

What is the unit of $K_P$ for $2SO_3(g) \rightleftharpoons 2SO_2(g) + O_2(g)$?

**Solution**

$\Delta n = (2 + 1) - 2 = 1$

$\Rightarrow$unit of $K_P = (atm)^{\Delta n} = atm$

## Relationship between $k_c$ and $k_p$

Consider the hypothetical reaction:

$aA + bB \rightleftharpoons cC + dD$

$$K_C = \frac{[C]^c[D]^d}{[A]^a[B]^b} \ldots(1)$$

$$K_P = \frac{P_C^c . P_D^d}{P_A^a . P_B^b} \ldots(2)$$

From **ideal gas equation** we knwo that:

$$PV = nRT$$

$$P = \frac{n}{V}RT$$

$P = MRT$ because $M = \frac{n}{V}$

$\Rightarrow P_A = [A]RT$
$P_B = [B]RT$
$P_C = [C]RT$
$P_D = [D]RT$
} substitute in equation (2) above

$$K_P = \frac{([C]RT)^c([D]RT)^d}{([A]RT)^a([B]RT)^b}$$

$$K_P = \frac{[C]^c[D]^d}{[A]^a[B]^b} \cdot \frac{(RT)^{c+d}}{(RT)^{a+b}}$$

$$K_P = \frac{[C]^c[D]^d}{[A]^a[B]^b}(RT)^{(c+d)-(a+b)}$$

$$\boxed{K_P = K_c\,(RT)^{\Delta n}}$$

$R = 0.082$ atm.L/mol.k

$T$ = temperature in kelvin

**Note**

$K_p = K_c$ if $\Delta n = 0$

## Illustrative Examples

18. For the reaction: $CaCO_3(s) \leftrightharpoons CaO(s) + CO_2(g)$ $K_P$ is equal to (EUEE)

A. $K_c$ B. $K_cRT$ C. $K_c(RT)^2$ D. $K_c(RT)^{-1}$

**Solution**

$\Delta n = (1 + 0) - (0) = 1$

$K_P = K_c(RT)^{\Delta n} = K_c(RT)^1 = K_cRT$

**Answer: B**

19. Which of the following reversible reactions has the same $K_P$ and $K_c$ valeus?

A. $2NOCl\ (g) \leftrightharpoons 2NO(g) + Cl_2(g)$

B. $2CO_2(g) + H_2O(g) \leftrightharpoons 2O_2(g) + CH_2CO(g)$

C. $MgCO_3(s) \leftrightharpoons MgO(s) + CO_2(g)$

D. $2SO_3(g) \leftrightharpoons 2SO_2(g) + O_2(g)$

**Solution**

$K_c = K_p$ if $\Delta n = 0$

**Answer: B**

20. Determine the unit of $K_c$ for

a. $2HF(g) \leftrightharpoons H_2(g) + F_2(g)$

b. $CaCO_3(s) \leftrightharpoons Ca^{2+}(aq) + CO_3^{2-}(aq)$

**Solution**

a. $\Delta n = (1 + 1) - 2 = 0$

Unit of $K_c = (mol/L)^{\Delta n} = 1$

b. $\Delta n = (1 + 1) - 0 = 2$

Unit of $K_c\ (mol/L)^2 = mol^2.L^{-2}$

21. Find the unit of $K_P$ for the following reactions. Assume the unit of pressure is torr.

a. $2NO(g) + O_2(g) \leftrightharpoons 2NO_2(g)$

b. $H_2O\ (g) + C(s) \leftrightharpoons CO(g) + H_2(g)$

c. $2NO(g) + 7H_2(g) \leftrightharpoons 2NH_3(g) + 4H_2O(l)$

**Solution**

a. $\Delta n = 2 - (2 + 1) = -1$

Unit of $k_p = (torr)^{\Delta n} = (torr)^{-1}$

b. $\Delta n = (1 + 1) - 1 = 1$

Unit of $k_p = (torr)^1 = torr$

c. $\Delta n = 2 - (7 + 2) = -7$

Unit of $K_p = (torr)^{-7}$

22. The equilibrium constant, $K_c$, for the reaction $I_2(g) \leftrightharpoons 2I(g)$ is $4 \times 10^{-5} mol.L^{-1}$ at 727°C. Calculate $K_p$ at the given temperature.

**Solution**

$T = 727°C = 727 + 273 = 1000k$

$\Delta n = 2 - 1 = 1$

$K_p = K_c (RT)^{\Delta n} = K_c . RT$

$K_p = 4 \times 10^{-5} mol.L^{-1} \times 0.082 atm.L/mol.K \times 1000K$

$K_p = 3.28 \times 10^{-3} atm$

23. $K_p$ for the reaction below at 573K is 11.5atm. What is the value of $K_c$ at this temperature $PCl_5(g) \leftrightharpoons PCl_3(g) + Cl_2(g)$

**Solution**

$\Delta n = (1 + 1) - 1 = 1$

$K_p = K_c.(RT)^1$

$$K_c = \frac{K_p}{RT} = \frac{11.5atm}{0.082atm.L/mol.K \times 573K}$$

$K_c = 0.245 \text{ mol. } L^{-1}$

24. For the reaction: $A(g) \leftrightharpoons 2B(g)$ at equilibrium, 0.4mol of A was introduced in 2L vessel and allowed to decompose until it reached equilibrium with B. The concentration of A at equilibrium was found to be 0.02M at 127°C. Calculate $K_p$ for this reaction at the same temperature.

**Solution**

$$[A] := \frac{0.4mol}{2L} = 0.2M$$

$[A]_{converted} = [A]_0 - [A] = 0.2M - 0.02M$

Since the coefficient ratio of A to B is

1: 2, $[B]_{eq} = 2 \times 0.18M = 0.36M$

$$\Rightarrow K_c = \frac{[B]^2}{[A]} = \frac{(0.36M)^2}{(0.02M)} = 6.48M$$

$$K_p = K_c.RT = 6.48mol/L \times 0.082\ atm.L/mol.K \times 400K$$

$$K_p = 212.5atm$$

25. Determine $K_c$ from the given $K_p$ value

$Sb_2S_3(s) + 3H_2(g) \leftrightharpoons 2Sb(s) + 3H_2S(g)$ $K_p = 0.43$ at 713 K.

**Solution**

$$\Delta n = 3 - 3 = 0$$

$$\Rightarrow K_p = K_c = 0.43$$

## Solubility Product ($k_{sp}$)

➦ In general for sparingly soluble salt $M_aX_b$ salt that dissociates at equilibrium. $M_ax_b \leftrightharpoons aM^{b+} + bx^{a-}$

$$K_{sp} = [M^{b+}]^a[X^{a-}]^b$$

### Example

The solubility product for

a. AgCl: $AgCl(s) \leftrightharpoons Ag^+(aq) + Cl^-(aq)$

$$K_{sp} = [Ag^+][Cl^-]$$

b. $Ca_3(PO_4)_2$: $Ca_3(PO_4)_2(s) \leftrightharpoons 3Ca^{2+}(aq) + 2PO_4^{3-}(aq)$

$$K_{sp} = [Ca^{2+}]^3[PO_4^{3-}]^2$$

## Solubility Product for Weak Bases

$$Fe(OH)_3(s) \leftrightharpoons Fe^{3+}(aq) + 3OH^-(aq)$$

$$K_{sp} = [Fe^{3+}][OH^-]^3$$

## Factors Affecting Equilibrium Constant

There are two major factors that affect the value of equilibrium constant.

1. **Temperature**
   - ➦ As temperature increases the values of equilibrium constant also increases.
2. The form of balanced chemical equations which describe the equilibrium.

i When we reverse an equation, the new equilibrium constant is the reciprocal of the original.

$$A \leftrightharpoons B, K_1$$

$$B \leftrightharpoons A, K_2 = \left(\frac{1}{K_1}\right)$$

ii When we multiply the coefficients in a balanced equation by a common factor, n, we raise the equilibrium constant to the corresponding power n.

$$A \leftrightharpoons B, K_1$$

$$nA \leftrightharpoons nB, K_2 = (K_1)^n$$

iii When we reverse and multiply the equation by a constant n, by a constant n, the new equilibrium constant is the reciprocal of the original the power of n.

$$A \leftrightharpoons B, K_1$$

$$nB \leftrightharpoons nA, K_2 = \left(\frac{1}{K_1}\right)$$

iv When two or more equations are added to produce a net equation, the equilibrium constant for the net equation is the product of the equilibrium constants for the added equations.

**Example**

$A \leftrightharpoons B, k_1$ $\quad B \leftrightharpoons C, k_2$ $\quad C \leftrightharpoons D, k_3$

Net equation: $A \leftrightharpoons D, K = K_1 . K_2 . K_3$

## Illustrative Examples

26. The value of $K_{eq}$ for the following reaction is 0.25.

$SO_2(g) + NO_2(g) \leftrightharpoons SO_3(g) + NO(g)$ what is the value $K_{eq}$ at the same temperature for the reaction below? (EUEE ... 2008 E.C.)

$$2SO_2(g) + 2NO_2(g) \leftrightharpoons 2SO_3(g) + 2NO(g)$$

A. 0.25 B. 0.062 C. 0.50 D. 16

**Solution**

The original equation is multiplied by 2

$\Rightarrow K_{eq} = (K_{eq\ (original)})^2 = (0.25)^2 = 0.0625$

**Answer: B**

27. The following equilibrium constants were determined at 300°C.

$2N_2O(g) \leftrightharpoons 2N_2(g) + O_2(g)$, $K_c = 4.0 \times 10^{18}$

$N_2(g) + O_2(g) \leftrightharpoons 2NO(g)$, $K_c = 4.0 \times 10^{-31}$

What will be the equilibrium constant at 300°C for the gaseous reaction of: $2N_2O(g) + O_2(g) \leftrightharpoons 4NO(g)$? (EUEE)

A. $3.2 \times 10^{-12}$
B. $2 \times 10^{-13}$
C. $5 \times 10^{50}$
D. $1.6 \times 10^{-49}$

**Solution**

Multiply the second reaction by 2

$$2N_2O(g) \rightleftharpoons 2N_2(g) + O_2(g), K_1 = 4 \times 10^{18}$$

$$2N_2(g) + 2O_2(g) \rightleftharpoons 4NO(g), K_2^1 = (4 \times 10^{18}) = 1.6 \times 10^{-61}$$

$$2N_2O(g) + O_2(g) \rightleftharpoons 4NO(g), K = K_1 . K_2^1$$

$$K = (4 \times 10^{18})(1.6 \times 10^{-61}) = 6.4 \times 10^{-43}$$

**Answer:**

28. Which of the following mathematical relationships between K, $K_1$, and $K_2$ is correct?

$CO_2(g) + H_2(g) \leftrightharpoons CO(g) + H_2O(g)$, K

$Fe(s) + CO_2(g) \leftrightharpoons FeO(s) + CO(g)$, $K_1$

$Fe(s) + H_2O(g) \leftrightharpoons FeO(s) + H_2(g)$, $K_2$

A. $K = K_1 + K_2$
B. $K = K_1 \times K_2$
C. $K = K_1 | K_2$
D. $K = K_2 | K_1$

**Solution**

To obtain the first reverse the third.

$$Fe(s) + CO_2(g) \rightleftharpoons FeO(s) + CO(g), K_1$$

$$FeO(s) + H_2(g) \rightleftharpoons Fe(s) + H_2O(g), \frac{1}{K_2}$$

$$CO_2(g) + H_2(g) \rightleftharpoons CO(g) + H_2O(g), K = K_1 . \frac{1}{K_2}$$

**Answer: C**

29. At 298K the following two gaseous equilibrium involving $SO_2$, $SO_3$ are $O_2$ are established.

$$SO_2(g) + \frac{1}{2}O_2(g) \rightleftharpoons SO_3(g), K_1$$

$$2SO_3(g) \rightleftharpoons 2SO_2(g) + O_2(g), K_2$$

Which of the following equilibrium expression is correct?

A. $K_1 = K_2$ B. $K_2 = K_1$ C. $K_2 = \frac{1}{K_1^2}$ D. $K_2 = \frac{1}{K_1}$

**Solution**

The first reaction is reversed and multiplied by 2. Thus, $K_2 = \frac{1}{K_1^2}$

**Answer: C**

30. Given the equilibrium constant values for the following reactions:

i $N_2(g) + \frac{1}{2}O_2(g) \rightleftharpoons N_2O(g), K_1 = 3.4 \times 10^{-18}$

ii $N_2O_4(g) \rightleftharpoons 2NO_2(g), K_2 = 4.6 \times 10^{-3}$

iii $\frac{1}{2}N_2(g) + O_2(g) \rightleftharpoons NO_2(g), K_3 = 4.1 \times 10^{-9}$

Determine $K_c$ for the reaction

$2N_2O(g) + 3O_2(g) \rightleftharpoons 2N_2O_4(g)$

**Solution**

- Reverse (i) and multiply by 2.
- Reverse (ii) and multiply by 2.
- Multiply (iii) by 4.

$$2N_2O(g) \rightleftharpoons 2N_2(g) + O_2(g), K_1^1 = \left(\frac{1}{K_1}\right)^2 = 8.65 \times 10^{34}$$

$$4NO_2(g) \rightleftharpoons 2N_2O_4(g), K_2^1 = \left(\frac{1}{K_2}\right)^2 = 4.73 \times 10^4$$

$$\underline{2N_2(g) + 4O_2(g) \rightleftharpoons 4NO_2(g), K_3^1 = (K_3)^4 = 2.83 \times 10^{-34}}$$

$$2N_2O(g) + 3O_2(g) \rightleftharpoons 2N_2O_4(g), K = K_1^1 . K_2^1 . K_3^1$$

$K = (8.65 \times 10^{34})(4.37 \times 10^4)(2.83 \times 10^{-34})$

$K = 1.16 \times 10^6$

31. At a fixed, elevated temperature, the following equilibrium constants are determined.

$CoO(s) + H_2(g) \leftrightharpoons Co(s) + H_2O(g), K_1 = 67$

$CoO(s) + CO(g) \leftrightharpoons Co(s) + CO_2(g), K_2 = 490$

What is K at the same temperature for:

$CO_2(g) + H_2(g) \leftrightharpoons CO(g) + H_2O\ (g)$?

**Solution**

Reverse the second

$$C_oO(s) + H_2(g) \rightleftharpoons Co(s) + H_2O(g), K_1 = 67$$

$$\underline{CO_2(g) + Co(s) \rightleftharpoons C_oO(s) + CO(g), K_2^1 = \frac{1}{K_2} = \frac{1}{490}}$$

$$CO_2(g) + H_2(g) \rightleftharpoons H_2O(g) + CO(g), K = K_1.K_2^1$$

$$K = 67 \times \frac{1}{490} = 0.14$$

32. What is the major factor that affects the value of equilibrium constant, K?

**Solution**

Temperature

33. If the equilibrium constant for the reaction $2NO(g) \rightleftharpoons N_2(g) + O_2(g)$ is $3.6 \times 10^3$ at 275 K, then calculate the value of equilibrium constant at the same temperature for

a. $N_2(g) + O_2(g) \leftrightharpoons 2NO(g)$

b. $4NO(g) \leftrightharpoons 2N_2(g) + 2O_2(g)$

**Solution**

a. $K_{new} = \frac{1}{K_{original}} = \frac{1}{3.6 \times 10^3} = 2.8 \times 10^{-4}$

b. $K_{new} = (K_{original})^2 = (3.6 \times 10^3)^2 = 1.3 \times 10^7$

34. Determine $K_p$ for the reaction:

$\frac{1}{2}N_2(g) + \frac{1}{2}O_2(g) + \frac{1}{2}Br_2(g) \rightleftharpoons NOBr(g)$ from the following informations at 298K

i $2NO(g) \leftrightharpoons N_2(g) + O_2(g), Kp_1 = 2.4 \times 10^{30}$

ii $NO(g) + \frac{1}{2}Br_2(g) \rightleftharpoons NOBr(g), Kp_2 = 0.28$

**Solution**

➦ Reverse the first and multiply by $\frac{1}{2}$

$$\frac{1}{2}N_2(g) + \frac{1}{2}O_2(g) \rightleftharpoons NO(g), Kp_1^1 = \left(\frac{1}{Kp_1}\right)^{1/2} = 6.45 \times 10^{-16}$$

$$\underline{NO(g) + \frac{1}{2}Br_2(g) \rightleftharpoons NOBr(g), Kp_2 = 0.28}$$

$$\frac{1}{2}N_2(g) + \frac{1}{2}O_2(g) + \frac{1}{2}Br_2(g) \rightleftharpoons NOBr(g)$$

$$Kp = Kp_1^1.Kp_2 = 1.81 \times 10^{-16}$$

35. The solubility product $K_{sp}$ for the sparingly soluble salt $Ag_2CrO_4$ is $4 \times 10^{-12}$. What is the solubility of the salt?

**Solution**

$$Ag_2CrO_4(s) \leftrightharpoons 2Ag^+(aq) + CrO_4^{2-}(aq)$$

$K_{sp} = [Ag^+]^2[CrO_4^{2-}]$ but the coefficient ratio of $Ag^+$ to $CrO_4^{2-}$ = x

$\Rightarrow [Ag^+] = 2x, [CrO_4^{2-}] = x$

$K_{sp} = (2x)^2(x)$

$4x^3 = 4 \times 10^{-12}$

$x^3 = 1 \times 10^{-12}$

$x = \sqrt[3]{1\times10^{-12}} = 1\times10^{-4}\,mol/L$

Thus, the solubility pf $Ag_2CrO_4$ is $1 \times 10^{-4}$ mol/L to give a $Ag^+$ concentration of $2 \times 10^{-4}$mol/L and a $CrO_4^{2-}$ concentration of $1 \times 10^{-4}$ mol/L

As the value of $K_{sp}$ increases solubility of salt also increases.

36. What is the molar solubility of $Fe(OH)_3$ in a solution that is buffered at pH = 3.50 at 25°C? ($K_{sp}(Fe(OH)_3) = 4 \times 10^{-38}$)(EUEE ...2008 E.C)

**Solution**

$pOH = 14 - pH = 14 - 3.5 = 10.5$

$[OH^-] = 10^{-pOH} = 10^{-10.5} = 3.16 \times 10^{-11}M$

$K_{sp} = [Fe^{3+}][OH^-]^3$

$$[Fe^{3+}] = \frac{K_{sp}}{[OH^-]^3} = \frac{4\times10^{-38}}{(3.16\times10^{-11})^3}$$

$[Fe^{3+}] = 1.26 \times 10^{-6}M$

Thus, the molar solubility of $Fe(OH)_3$ is $1.26 \times 10^{-6}$M.

## Applications of equilibrium constant

➥ There are two important applications of equilibrium constant.

**1. Position of equilibrium**

a. $K_{eq} > 1$, formation of products is favored.

b. $K_{eq} < 1$, formation of reactants is favored.

c. $K_{eq} = 1$, both reactants and products formation is favored equally.

**Example**

$CO(g) + Cl_2(g) \leftrightharpoons COCl_2(g)$, K = 14

⇒Formation of $COCl_2$ is favored

$C(s) + H_2O(g) \leftrightharpoons CO(g) + H_2(g)$, $K = 1.6 \times 10^{-21}$

⇒The equilibrium shifts to the left.

$CaCO_3(s) \leftrightharpoons CaO(s) + CO_2(g)$, $K_p$ = 1 at 1200K

⇒Reactants and product formation is favored equally.

## 2. Extent of reaction

➦ A very large numerical value of $K_{eq}$ signifies that the forward reaction goes to completion or nearly so.

**Example**

$2H_2(g) + O_2(g) \leftrightharpoons 2H_2O(l)$, $K_p = 1.4 \times 10^{93}$ at 293k

➦ A very small numerical value of $K_{eq}$ shows that the extent of reaction is very small, that is forward reaction does not occur to any significant extent.

**Example**

$CaCO_3(s) \leftrightharpoons CaO(s) + CO_2(g)$, $K_p = 1.9 \times 10^{-23}$ at 298K.

# Reaction Quotient (Q)

➦ It is the ratio of the product concentration to the concentration of reactants, each raised to the power of its stoichiometric coefficients in the chemical equations.

$aA + bB \leftrightharpoons cC + dD$

$$Q_c = \frac{[C]^c[D]d}{[A]^a[B]^b}, Q_p = \frac{P_C^c.P_D^d}{P_A^a.P_B^b}$$

➦ $Q_C$ - reaction quotient in molarity.

➦ $Q_P$ - reaction quotient in terms of partial pressure.

# Application of reaction quotient

➦ Q > K, the equilibrium shifts to the left.

➦ Q < K, the equilibrium shifts to the right.

➦ Q = K, the reaction is at equilibrium.

## Illustrative Examples

37. At 445°C, $K_c$ for the following reaction is 0.020.

$2HI(g) \leftrightharpoons H_2(g) + I_2(g)$

A mixture of $H_2$, $I_2$ and HI in a vessel at 445°C has the following concentrations: [HI] = 2M, $[H_2]$ = 0.50M and $[I_2]$ = 0.10M. Which of the following statements concerning the reaction quotient, $Q_c$ is true for the above system? (EUEE ... 2007 E.C)

A. $Q_c$ is less than $K_c$, more HI will be produced
B. $Q_c$ is greater than $K_c$; more HI will be produced.
C. $Q_c$ is less than $K_c$; more $H_2$ and $I_2$ will be produced.
D. $Q_c$ is greater than $K_c$; more $H_2$ and $I_2$ will be produced.

**Solution**

$$Q_c = \frac{[H_2][I_2]}{[HI]^2} = \frac{(0.50)(0.10)}{(2)^2} = 0.0125$$

$\Rightarrow Q_c < K_c$, formation of products, $H_2$ and $I_2$ is favored.

**Answer: C**

38. The equilibrium constant is 0.09 at 295K for the reaction, $H_2O(g) + Cl_2O(g) \leftrightharpoons 2HOCl(g)$. For which of the following sets of conditions is the system at equilibrium? (EUEE ... 2003 E.C)

A. A 1L flask contains 1 mol of HOCl, 0.10mol of $Cl_2O$ and 0.10 mol of $Cl_2O$ and 0.10 mol of $H_2O$.
B. A 2L flask contains 0.084 mol of HOCl, 0.080mol of $Cl_2O$ and 0.98 mol of $H_2O$.
C. A 3L flask contains 0.25mol of HOCl, 0.0010mol of $Cl_2O$ and 0.56mol of $H_2O$.
D. A 4L flask contains 0.25mol of HOCl, 0.001mol of $Cl_2O$ and 0.56 mol of $H_2O$.

**Solution**

The system is at equilibrium if $Q_c = K_c$

A. $$[HOCl] = \frac{n_{HOCl}}{V} = \frac{1mol}{1L} = 1M$$

$$[Cl_2O] = \frac{n_{Cl_2O}}{V} = \frac{0.10mol}{1L} = 0.10M$$

$$[H_2O] = \frac{n_{H_2O}}{v} = \frac{0.10mol}{1L} = 0.10M$$

$$Q_c = \frac{[HOCl]^2}{[Cl_2O][H_2O]} = \frac{(1)^2}{(0.1)(0.1)} = 100$$

$\Rightarrow Q_c > K_c$, the reaction shits to the left.

B. $[HOCl] = \frac{0.084}{2} = 0.042M$

$[Cl_2O] = \frac{0.08}{2} = 0.04M$

$[H_2O] = \frac{0.98}{2} = 0.49M$

$$Q_c = \frac{[HOCl]^2}{[Cl_2O][H_2O]} = \frac{(0.042)^2}{(0.4)(0.49)} = 0.09$$

$\Rightarrow Q_c = K_c$, the system is at equilibrium.

**Answer: B**

39. At 373K, $K_p = 5 \times 10^{-5}$atm for the reaction $COCl_2(g) \leftrightharpoons CO(g) + Cl_2(g)$. For equal $P_{(CO)} = P_{(COCl_2)} = P_{(Cl_2)} = 1$atm placed in a flask. Which of the following is true at equilibrium?

A. The total pressure of the system must remains at 3 atm.

B. $P_{(CO)}$ and $P_{(Cl_2)}$ must decreases from their initial values.

C. P(CO) and $P_{(Cl_2)}$ must increase from their initial values.

D. $P_{(COCl_2)}$ will remain constant.

**Solution**

➦ $Q_P = \frac{P_{CO} \cdot P_{Cl_2}}{P_{COCl_2}} = \frac{1\ atm.1atm}{1atm} = 1atm$

➦ $Q_p > K_p$, the equilibrium will shift to the left. Thus, $P_{CO}$ and $P_{Cl_2}$ must decrease will $P_{COCl_2}$ must increase.

**Answer: B**

40. Which of the following reactions has the greatest extent based on the value of equilibrium constant?

A. $N_2(g) + 3H_2(g) \leftrightharpoons 2NH_3(g)$, $K_c = 3.6 \times 10^8$

B. $2CO(g) + O_2(g) \leftrightharpoons 2CO_2(g)$, $K_c = 2.25 \times 100^{23}$

C. $2NOCl(g) \leftrightharpoons 2NO(g) + Cl_2(g)$, $K_c = 4.7 \times 10^{-4}$

D. $N_2(g) + O_2(g) \leftrightharpoons 2NO(g)$, $K_c = 4.7 \times 10^{-31}$

**Solution**

The larger the value of equilibrium constant, the greater the extent of reaction.

**Answer: B**

41. For the reaction $PCl_5(g) \leftrightharpoons PCl_3(g) + Cl_2(g)$, $K_c = 0.0454$ at 261°C. If a vessel is filled with these gases such that the initial concentrations are $[PCl_3] = 0.50M$, $[Cl_2] = 0.20M$ and $[PCl_5] = 4.50M$. In which direction will a net charge occur?

**Solution**

$$Q_c = \frac{[PCl_3][Cl_2]}{[PCl_5]} = \frac{(0.50)(0.22)}{(4.50)} = 0.022$$

$Q_c < K_c$, the net change occurs from left to right.

42. The equilibrium constant for the reaction: $N_2(g) + O_2(g) \leftrightharpoons 2NO(g)$ is $4 \times 10^{-4}$ at 2000K. In the presence of a catalyst, equilibrium is attained ten times faster. What is the value of equilibrium constant, in the presence of catalyst at 2000K?

**Solution**

➦ $k = 4 \times 10^{-4}$ because catalysts do not affect the value of equilibrium constant.

43. Given the reaction: $2A + B \leftrightharpoons 3C$, $K_c = 0.01$. Is formation of C favored? Explain

**Solution**

➦ Formation of C is not favored because $K_c$ is less than one.

## 5.3 Calculating the Equilibrium Concentrations

➦ Rules to determine the equilibrium concentration.

1. Determine the direction of net change by comparing Q and $K_{eq}$.
2. State the initial concentrations and let x be change in concentration or partial pressure.
3. Express equilibrium concentration or partial pressure in terms of x.
4. Write equilibrium constant expression.
5. Substitute equilibrium concentration or partial pressure into $K_c$ or $K_p$ expression and solve for x.
6. Check the equilibrium concentrations or partial pressure by solving for $K_c$ and $K_p$.

## Example

Initially the concentration of $PCl_5$ is 0.02M. Find the equilibrium concentration of all species. $PCl_5(g) \leftrightharpoons PCl_3(g) + Cl_2(g)$, $K_c = 1 \times 10^{-2}$.

### Solution

$$PCl_5(g) \leftrightharpoons PCl_3(g) + Cl_2(g)$$

| | | | |
|---|---|---|---|
| Initial concentration | 0.02 | 0 | 0 |
| Change in concentration | -x | +x | +x |
| Equilibrium concentration | 0.02 - x | x | x |

$$K_c = \frac{[PCl_3][Cl_2]}{[PCl_5]}$$

$$1\times10^{-2} = \frac{x.x}{0.02 - x}$$

$x^2 = 2 \times 10^{-4} - 10^{-2}x$

$x^2 + 10^{-2}x - 2 \times 10^{-4} = 0$ solve for x by using quadratic formula.

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} \quad a = 1, b = 10^{-2}, c = -2 \times 10^{-4}$$

$$x = \frac{-10^2 \pm \sqrt{(10^{-2})^2 - (4\times1\times-2\times10^{-4})}}{2\times1}$$

$$x = \frac{-10^2 \pm 3\times10^{-2}}{2}$$

$x = 1 \times 10^{-2}$ or $x = -2 \times 10^{-2}$ but concentration cannot be negative. Thus the value of x is $1 \times 10^{-2}$.

At equilibrium

$[PCl_5] = 2 \times 10^{-2} - x = 2 \times 10^{-2} - 1 \times 10^2 = 1 \times 10^{-2}M$

$[PCl_3] = [Cl_2] = x = 1 \times 10^{-2}M = 0.01M$

Check

$$K_c = \frac{[PCl_3][Cl_2]}{[PCl_5]}$$

$$1\times10^{-2} = \frac{(1\times10^{-2})(1\times10^{-2})}{(1\times10^{-2})}$$

$1 \times 10^{-2} = 1 \times 10^{-2}$ True

## Illustrative Examples

44. When 0.50mol of $N_2O_4$ is placed in a 4.0 liter reaction vessel and heated to 400K, 80% of the $N_2O_4$ decomposes to $NO_2$ gas as follows: $N_2O_4(g) \leftrightharpoons 2NO_2(g)$ what will be the value of $K_p$, in units of pressure, at 400K for this reaction? (EUEE ... 2008 E.C.)

A. 2.62 B. 13.12 C. 50.48 D. 16.20

**Solution**

$\Rightarrow$80% $N_2O_4$ converted means 80% of 5 mol is changed.

$$\Rightarrow x = \frac{80}{100} \times 0.5\text{mol} = 4\text{mol}$$

$$[N_2O_4]_0 = \frac{n}{v} = \frac{0.5\text{mol}}{4\text{L}} = 0.125\text{M}$$

$$x = \frac{0.4\text{mol}}{4\text{L}} = 0.1\text{M}$$

| | $N_2O_4(g) \rightleftharpoons 2NO_2(g)$ | |
|---|---|---|
| Initial conc. | 0.125 | 0 |
| Change in conc. | -x | +2x |
| Equilibrium conc. | 0.125 - x | 2x |

But x = 0.1, thus, at equilibrium

$[N_2O_4] = 0.125 - x = 0.125 - 0.1 = 0.025\text{M}$

$[NO_2] = 2x = 2 \times 0.1 = 0.2\text{M}$

$$K_c = \frac{[NO_2]^2}{[N_2O_4]} = \frac{(0.2\text{M})^2}{(0.025\text{M})} = 1.6\ \text{mol/L}$$

$K_p = K_c\,(RT)^{\Delta n}$, $\Delta n = 2 - 1 = 1$

$K_p = 1.6\text{mol/L}\ (0.082\ \text{atm.L/mol.K} \times 400\text{K}) = 52.48\text{atm}$

**Answer: C**

45. In the reactiOn ($2SO_2 + O_2 \leftrightharpoons 2SO_3$, $K_{eq} = 100$). What will be the concentration of $O_2$, if the ratio of $SO_2$ is the same as the concentration of $SO_3$? (EUEE .. 2007 E.C.)

A. $[O_2] = [SO_2]$ C. $[O_2] = 100\text{M}$

B. $[O_2] = 0.01\text{M}$ D. $[O_2] = 0.1\text{M}$

**Solution**

$$K_{eq} = \frac{[SO_3]}{[SO_2][O_2]} \text{ but } [SO_3] = [SO_2]$$

$$\Rightarrow K_{eq} = \frac{1}{[O_2]}$$

$$[O_2] = \frac{1}{K_{eq}} = \frac{1}{100} = 0.01M$$

**Answer: B**

46. The equilibrium constant for the following reaction: $3H_2(g) + CO(g) \leftrightharpoons CH_4(g) + H_2O(g)$ is 3.393 at 1200K. A system at equilibrium has [CO] = 0.1522M, $[H_2]$ = 0.1566M and $[H_2O]$ = 0.0478M what is the $[CH_4]$?

**Solution**

$$K_c = \frac{[CH_4][H_2O]}{[H_2]^3[CO]}$$

$$[CH_4] = \frac{K_c[H_2]^3[CO]}{[H_2O]} = 0.0481M$$

47. The equilibrium constant, $K_c$ is $1.02 \times 10^{-19}$ at 298K for the reaction $C_2H_6(g) \leftrightharpoons H_2(g) + C_2H_4(g)$. If the initial concentration of $C_2H_6$ is 0.05 M at 298K, what will be the equilibrium concentrations of all species?

**Solution**

| | $C_2H_6(g) \leftrightharpoons H_2(g) + C_2H_4(g)$ | | |
|---|---|---|---|
| initial conc. | 0.05 | 0 | 0 |
| change inccnc | -x | +x | +x |
| equilibrium conc. | 0.05 - x | x | x |

$$K_c = \frac{[H_2][C_2H_4]}{[C_2H_6]}$$

$$1.02 \times 10^{-19} = \frac{x.x}{0.05 - x}$$

$$1.02 \times 10^{-19} = \frac{x^2}{0.05 - x}$$

➦ For very small value of $K_c$, the reaction will not proceed far to the right. So the value of x is too small and you can make an approximation.

$\Rightarrow 0.05 - x \approx 0.05$

$$1.02\times10^{-19}=\frac{x^2}{0.05}$$

$$x=\sqrt{0.05\times1.02\times10^{-19}}=7.1\times10^{-11}M$$

$\Rightarrow$At equilibrium, $[CH_2]=[C_2H_4]=7.1\times10^{-11}M$

$[C_2H_6]=0.05-7.1\times10^{-11}\approx 0.05M$

48. Given the reaction: $NH_4HS(s)\leftrightharpoons NH_3(g)+H_2S(g)$ $K_p=0.108\ atm^2$ at 25°C. What is the total gas pressure at equilibrium?

**Solution**

➦ At equilibrium $P_{NH_3}=P_{H_2S}$, because the coefficient ratio is one to one.

$$K_p=P_{NH_3}.P_{H_2S}$$

$$P_{NH_3}=P_{H_2S}=\sqrt{K_p}=\sqrt{0.108atm^2}=0.3atm$$

$$P_{total}=P_{NH_3}+P_{H_2S}=0.33atm+0.33atm=0.66atm$$

49. Starting with 1 atm of $SO_3(g)$, determine the partial pressure of each gas at equilibrium and the total pressure for the reaction: $2SO_3(g)\leftrightharpoons 2SO_2(g)+O_2(g)$, $K_p=3.2\times10^{-5}$

**Solution**

| | $2SO_3(g)\leftrightharpoons 2SO_2(g)+O_2(g)$ | | |
|---|---|---|---|
| Initial partial pressure | 1 | 0 | 0 |
| Change in pressure | -2x | +2x | +x |
| Equilibrium pressure | 1 - 2x | 2x | x |

$$K_p=\frac{P_{SO_2}^2.P_{O_2}}{P_{SO_3}^2}$$

$$3\times10^{-5}=\frac{(2x)^2(x)}{(1-2x)^2}$$

Since $K_p$ is small value use approximation method $1-2x\approx 1$

$$4x^3=3\times10^{-5}$$

$$x=\sqrt[3]{\frac{3.2}{4}\times10^{-5}}=0.02$$

$\Rightarrow$At equilibrium: $P_{SO_2}=2x=2\times0.02=0.04atm$

$$P_{O_2} = x = 0.02atm, P_{SO_3} = 1 - 0.02 = 0.98atm$$

$$P_{total} = P_{SO_3} + P_{SO_2} + P_{O_2}$$

$$= 0.98 + 0.02 + 0.04 = 1.04atm$$

50. An equilibrium mixture for the reaction: $2H_2S\ (g) \leftrightharpoons 2H_2(g) + S_2(g)$ had 1 mol of $H_2S$, 0.2mol of $H_2$ and 0.8mol of $S_2$ in a 2L flask. What is the value of $K_c$?

**Solution**

$$[H_2S] = \frac{1mol}{2L} = 0.5M$$

$$[H_2] = \frac{0.2mol}{2L} = 0.1M$$

$$[S_2] = \frac{0.8mol}{2L} = 0.4M$$

$$K_c = \frac{[H_2]^2(S_2)}{[H_2S]^2} = \frac{(0.1)^2(0.4)}{(0.5)^2} = 0.016mol/L$$

51. 4.5mol each of $H_2$ and $I_2$ are heated in a 10L sealed vessel. At equilibrium 3 mol of HI were found what is the value of equilibrium constant?

**Solution**

$$[H_2]_o = [I_2]_o = \frac{4.5mol}{10L} = 0.45M$$

$$[HI]_{eq} = \frac{3mol}{10L} = 0.3M$$

- O - initial concentration
- eq - equilibrium concentration

| | $H_2$ + | $I_2 \leftrightharpoons$ | 2HI |
|---|---|---|---|
| Initial | 0.45 | 0.45 | 0 |
| Change | -x | -x | +2x |
| Equilibrium | 0.45 - x | 0.45 - x | 2x |

But, at equilibrium, [HI] = 0.3M

$$\Rightarrow 2x = 0.3M$$

$$x = 0.15$$

Thus, $[H_2] = [I_2] = 0.45 - x = 0.45 - 0.5 = 0.30M$

$$K_c = \frac{[HI]^2}{[H_2][H_2]} = \frac{(0.3M)^2}{(0.3M)(0.3M)} = 1$$

52. For the reaction: $CaCO_3(g) \leftrightharpoons CaO(s) + CO_2(g)$ $K_p = 1.15$ atm at 1073K. If 40g of $CaCO_3$ was put into a 10L container and heated to 1073K, what percent of $CaCO_3$ would remain unreacted at equilibrium?

**Solution**

$$K_P = P_{(CO_2)} = 1.16atm$$

$$n_{CO_2} = \frac{PV}{RT} = \frac{1.16atm \times 10L}{0.082atm.L/mol.k \times 1073k}$$

$$n_{CO_2} = 0.132mol$$

$$n_{CaCO_3} \text{ initially} = \frac{M}{M} = \frac{40g}{100g/mol} = 0.40mol$$

$$\%CaCO_3 \text{ converted} = \frac{n_{CO_2}}{n_{CaCO_3}} \times 100 = \frac{0.132mol}{0.40mol} \times 100 = 33\%$$

Thus, 100 - 33 = 67% remains undecomposed.

## 5.4 Factors affecting chemical equilibrium

**Le Chatelier's principle** states that a change in any of the factors determining an equilibrium will cause the system to change in such a way that the effect of the change reduced or counteracted.

### 1. Concentration

- Increasing concentration of reactants or decreasing concentration of products shifts the equilibrium to the **right**.
- Decreasing concentration of reactants or increasing concentration of products shifts the equilibrium to the **left**.

Adding solid substances does not disturb the position of equilibrium because the concentration of solids is constant. However, removing solids shifts the equilibrium to the direction in which they are removed.

**Example**

$CaCO_3(s) \leftrightharpoons CaO(s) + CO_2(g)$

- Increasing $[CO_2]$ - shifts the equilibrium to the left.
- Removing $CaCO_3$ - shifts the equilibrium to the left.
- Adding CaO (s) - no effect.

### 2. Temperature

- Increasing temperature shifts the equilibrium to the **endothermic side (direction)**.
- Lowering temperature shifts the equilibrium to the **exothermic direction**.

If the forward reaction is endothermic reaction the reverse will be exothermic and vice versal.

**Example**

$2SO_2(g) + O_2(g) \leftrightharpoons 2SO_3(g)$, $\Delta H = -196kJ/mol$

- Forward reaction is exothermic while reverse reaction is endothermic.
- Lowering temperature shifts the equilibrium to the right.
- Increasing temperature shifts the equilibrium to the left.

### 3. Pressure and volume

- Pressure and volume affect the position of equilibrium for

i a reaction involving gases

ii a reaction in which the number of moles of reactants is not equal to number of moles of products

- Increasing pressure or lowering volume shifts the equilibrium to the **smaller number of moles.**
- Lowering pressure or increasing volume shifts the equilibrium to the **larger number of moles.**

Reason $V \alpha \frac{1}{P}$ (Boyle's law)

$V \alpha n$ (Avogadro's law)

**Example**

$$\underbrace{N_2(g) + 3H_2(g)}_{n=1+3=4mol} \rightleftharpoons \underbrace{2NH_3(g)}_{n=2mol}$$

- Lowering volume or increasing pressure shifts the equilibrium to the right.
- Increasing volume or lowering pressure shifts the equilibrium to the left.

### 4. Addition of inert gas

- It shifts the equilibrium to the larger number of gaseous species.

If an inert gas is added at constant volume there will be no change on the position of equilibrium.

**Note**

Adding catalyst to a reaction mixture speeds up both the forward and reverse reactions.

Equilibrium is achieved more rapidly but the equilibrium is not disturbed by catalyst.

## Illustrative Examples

53. In which of the following systems will the position of equilibrium shift to the left upon an increase in pressure, but to the right up on an increase in temperature? (EUEE ... 2008 E.C)

A. $CO_2(g) + H_2(g) \leftrightharpoons CO(g) + H_2O\ (g),\ \Delta H > 0$

B. $C_2H_4(g) + H_2O(g) \leftrightharpoons C_2H_5OH(g),\ \Delta H < 0$

C. $C_2H_6(g) \leftrightharpoons C_2H_4(g) + H_2(g),\ \Delta H > 0$

D. $2SO_2(g) + O_2(g) \leftrightharpoons 2SO_3(g),\ \Delta H < 0$

**Solution**

➦ Increase in pressure to the smaller number of molecules and increase in temperature to the endothermic side.

**Answer: C**

54. The hydrogen used in Haber process is made by the following reactions: $CH_4(g) + H_2O\ (g) \leftrightharpoons CO(g) + 3H_2(g),\ \Delta H° = +206kJ$, which of the following sets of conditions will favor the formation of $H_2$? (EUEE)

A. Low pressure and high temperature.

B. Low pressure and low temperature.

C. High pressure and low temperature.

D. High pressure and high temperature.

**Solution**

$$\underbrace{CH_4(g) + H_2O(g)}_{n=1+1=2} \rightleftharpoons \underbrace{CO(g) + 3H_2(g)}_{n=1+3=4},\ \Delta H > 0$$

$\Rightarrow$ Low pressure and high temperature favor formation of $H_2$.

**Answer: A**

55. Which change will increase the amount of $SO_3(g)$ at equilibrium?
$2SO_2(g) + O_2(g) \leftrightharpoons 2SO_3(g),\ \Delta H = -197kJ/mol$

i Increasing the temperature.

ii Decreasing Volume

iii Adding a catalyst.

A. I only

B. II only

C. I and II only

D. II and III

**Solution**

➥ The amount of $SO_3(g)$ is increased by:

- Lowering temperature
- Increasing pressure
- Lowering volume
- Removing $SO_3(g)$
- Increasing $[SO_2]$ and $[O_2]$

**Answer: B**

56. Consider the following equilibrium: $N_2O_4(g) \leftrightharpoons 2NO_2(g)$, what happens if the total pressure is increased by adding nitrogen gas?

A. $K_c$ increases

B. The equilibrium shifts to the left.

C. The equilibrium shifts to the right.

D. The position of equilibrium is not affected.

**Solution**

Addition of inert gas shifts the equilibrium to the larger number of moles.

**Answer: C**

57. If we add a catalyst to a chemical reaction that is at equilibrium then catalyst increases the rate at which equilibrium is achieved?

A. But it does not change the composition of the equilibrium.

B. And it will change the composition of the equilibrium.

C. And will change the equilibrium in the direction that produces heat.

D. And will change the equilibrium in the direction that occupies a large volume.

**Answer: A**

58. Consider the equilibrium: $H_2(g) + CO_2(g) \leftrightharpoons H_2O(g) + CO(g)$ for which $\Delta H$ for the reaction = +41kJ. How is the amount of $H_2$ affected in the system when it is subjected to an external stress? (EUEE)

A. Addition of $CO_2$ increases the amount of $H_2$

B. Increase in temperature decreases the amount of $H_2$.

C. Addition of a catalyst increases the amount of $H_2$.

D. Decrease in volume increases the amount of $H_2$.

**Solution**

- Due to equal number of moles of reactants and products, pressure and volume have no effect on the amount of $H_2$.
- Increasing temperature shifts the equilibrium to the right. As a result, the amount of $H_2$ decreases.

**Answer: B**

59. Consider the reaction, $PCl_5(g) \leftrightharpoons PCl_3(g) + Cl_2(g)$ in a closed container at equilibrium at a constant temperature. What will be the effect of adding more $PCl_5$ on the equilibrium concentration of $Cl_2(g)$? It

A. Decreases

B. Cannot be predicted without the value of $k_p$

C. Remains unaffected

D. Increases

**Solution**

- Addition of $PCl_5$ shifts the equilibrium to the right. More $Cl_2$ and $PCl_3$ are produced.

**Answer: D**

60. Given the reaction at equilibrium: $2NaHCO_3(s) \leftrightharpoons Na_2CO_3(s) + CO_2(g) + H_2O(l)$, $\Delta H > 0$. Explain the affect of the following on the position of equilibrium.

a. Adding $NaHCO_3(s)$

b. Removing $Na_2CO_3(s)$

c. Increasing temperature and volume.

d. Lowering temperature and pressure.

e. Increasing concentration of $CO_2(g)$

f. Adding inert gas

g. Adding catalyst

**Explanation:**

a. No effect because it is solid.

b. Shifts to the right because removing of substances in any state shifts the equilibrium in that direction.

c. If the combining effect of two or more factors is in the same direction, it will be predictable unless it becomes unpredictable.

$\Rightarrow$shifts to the right

d. Unpredictable because lowering temperature shifts to the left but lowering pressure shifts the equilibrium to the right.

e. Shifts to the left

f. Shifts to the right

g. No effect

61. Consider the reaction below at equilibrium. $2NO(g) + Cl_2(g) \leftrightharpoons 2NOCl(g)$, $\Delta H = -78kJ$. How would you increase the amount of product by using temperature and pressure?

**Explanation:**

➦ Formation of NOCl is favored by performing the reaction at low temperature and high pressure.

62. What are optimum conditions? Explain by using the factors that disturb position of equilibrium.

**Explanation:**

➦ Optimum conditions are conditions that increase the amount of product.

➦ Temperature:

- Endothermic reaction - increasing temperature.
- Exothermic reaction - lowering temperature.

➦ Concentration: $\Delta n > 0$ - lowering pressure

- $\Delta n < 0$ - increasing pressure

➦ Addition of inert gas, for $\Delta n > 0$.

## Application of Chemical equilibrium

➦ There are many applications of chemical equilibrium two of them are discussed below.

### 1. Production of Ammonia ($NH_3$)

➦ Ammonia is industrially manufactured by **Haber process.**

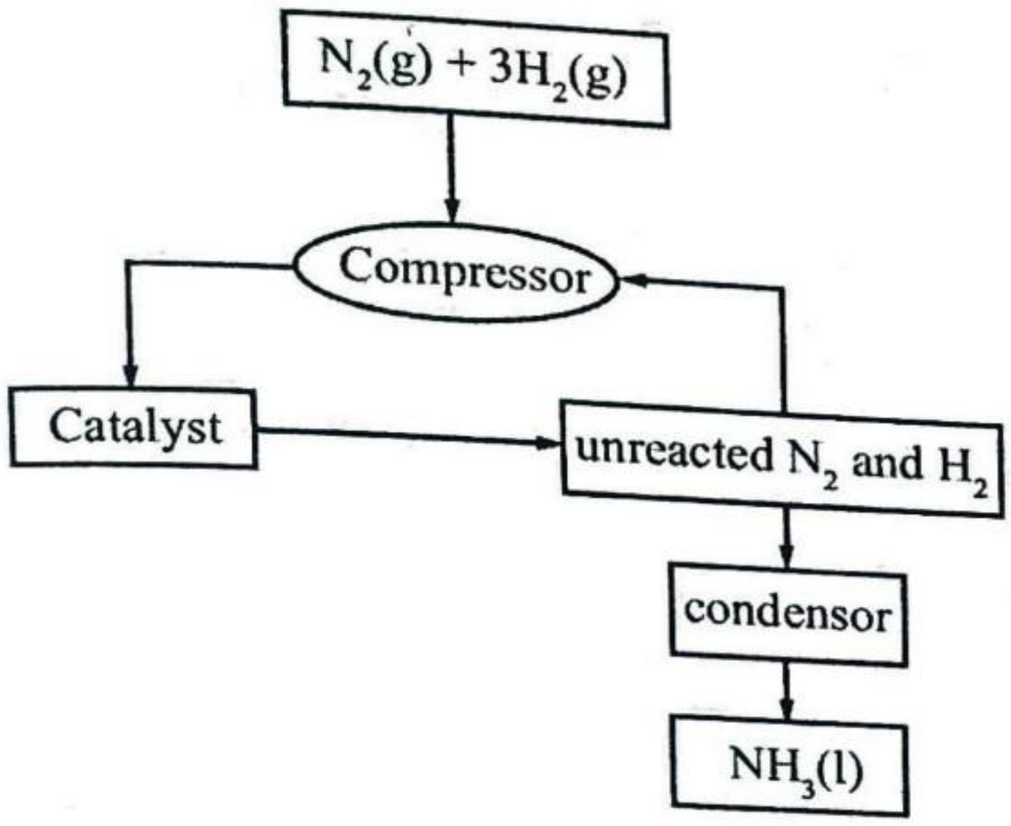
$N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g) + 92\ kJ.mol^{-1}$

➦ Formation of ammonia is favored by:

- Lowering temperature
- Increasing pressure
- Removing ammonia
- Increasing concentration of $N_2$ and $H_2$

Iron is used as a catalyst in the Haber process to reduce the time at which equilibrium is attained.

A flow chart for Haber process




## Physical Properties of Ammonia

- It is alkaline colorless gas.
- It has Pungent smell.
- It is lighter than air and extremely soluble in water.
- Liquefies on cooling under pressure.
- It causes tears in eyes.

## Uses of Ammonia

- It is used to make nitrogenous fertilizers such as urea, ammonium nitrate and ammonium sulphate.
- It is used to make nitric acid by **Ostwald process**.
- It is used as refrigerant.

### 2. Production of Sulfuric acid ($H_2SO_4$)

- Sulphuric acid is industrially manufactured by **contact process**.
- There are four steps in the contact process.

1. Burning of sulfur

$$S + O_2 \rightarrow SO_2$$

2. Burning of Sulfurdioxide

$$2SO_2(g) + O_2(g) \rightleftharpoons^{V_2O_5} 2SO_3(g) + 196\,kJ.mol^{-1}$$

➦ Formation of $SO_3(g)$ is favored by

- Lowering temperature
- Increasing pressure
- Removing $SO_3$
- Increasing concentration of $SO_2$ and $O_2$

3. Reaction of sulfurtrioxide with sulfuric acid

$$SO_3(g) + H_2SO_4(l) \rightarrow H_2S_2O_7(l)$$

Oleum (fumining sulphuric acid)

4. Dissolving oleum in water

$$H_2S_2O_7(l) + H_2O \rightarrow 2H_2SO_4(l)$$

➦ Sulfuric acid is also called **oil of vitriol.**

Flow chart for contact process.

Combustion chamber, burning of sulphur

↓

Combustion chamber, formation of $SO_3$

↓

Combustion tower, $SO_3$ is absorbed in $H_2SO_4$

↓

Hydration of Oleum, production of Sulphuric acid

The sulphur used in the contact process is industrially manufactured by Frasch process from the underground deposit.

## Uses of Sulphuric Acid

➦ It is used in large number of industries. As a result it is called the **"king of chemicals"**.

➦ It is used to make drugs, dyes, paints, sulphates, detergents, nitirc acid, hydrochloric acid and others.

➦ It is used in storage battries as electrolytes.

➦ It is dehydrating and oxidizing agent.

➦ It is used to clean metal surfaces before electroplating.

➦ It is used as catalyst in organic chemistry.

➦ It is used to make fertilizers.

## Illustrative Examples

63. If the following reaction is at equilibrium all of the following favor formation of $NH_3$. EXCEPT

$$N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g) + 92\ kJ.mol^{-1}$$

A. Increasing pressure

B. Decreasing temperature

C. Adding more $N_2$ and $H_2$

D. Increasing the volume of the reaction container.

**Solution**

Increasing volume shifts the equilibrium to the left. It does not favor formation of $NH_3$.

**Answer: D**

64. The catalyst that provides surface area for $SO_2$ and $O_2$ to collide in the contact process is

A. $MnO_2$

B. Fe

C. $V_2O_5$

D. Ni

**Solution**

Vanadium pentoxide ($V_2O_5$) provides surface area for $SO_2$ and $O_2$ to make contact. That is why is called contact process.

**Answer: C**

65. Why is $SO_3$ not directly combined with water to make sulphuric acid?

**Explanation**

Because the reaction can not be controlled easily and create a fog of sulphuric acid.

66. Write the industrial processes used to make the following substances.

a. Ammonia

b. Sulphur

c. Sulphuirc acid

d. Nitric acid

**Solution**

a. Haber process

b. Frasch process

c. Contact process

d. Ostwald process

67. Write the main uses of ammonia and sulphuric acid.

**Solution**

- ➦ Ammonia is mainly used to make fertilizers and nitric acid.
- ➦ Sulphuric acid is the most important chemical in industry as raw material, catalyst, oxidizing an dehydrating agents.

68. Which step in the contact process is directly related with the application of chemical equilibrium?

**Solution**

- ➦ Step 2: Combustion of sulphur dioxide

$$2SO_2(g) + O_2(g) \rightleftharpoons 2SO_3(g) + \text{heat}$$

## 5.5 Phase Equilibrium

**Equilibrium** occurs in physical processes under a certain temperature and pressure.

- ➦ Matter is anything that has mass and occupies space.
- ➦ Solid, liquid and gas are the three states of matter.

Table 5.1. The difference between the three states of matter

| Solids | Liquids | Gases |
|---|---|---|
| ➦ Occur at low temperature and high pressure | ➦ Occur at intermediate temperature and pressure | ➦ Occur at high temperature and low pressure |
| ➦ have definite shape and volume | ➦ have no definite shape but have definite volume | ➦ have no definite shape and volume |
| ➦ have high density, melting and boiling points | ➦ have lower density and boiling point than solids | ➦ have the lowest density and boiling point |
| ➦ have lower kinetic energy than attract ion force between particles. | ➦ have almost equal kinetic energy and attraction forces | ➦ have higher kinetic energy than attraction forces between molecules. |
| Example<br>Carbon, wood | Example<br>water, oil, alcohol | Example<br>Oxygen, Hydrogen, nitrogen |

## Common Terms

1. **Phase (P)** is part of a system that is physically and chemically homogeneous.

## Rules to determine the number of phases

➦ Homogeneous mixture has one phase.

**Example**

Sugar solution, P = 1

➦ Heterogeneous mixture has more than one phases.

**Example**

A mixture of oil and water, P = 2

➦ A pure substance in different states has different phases.

**Example**

Ice floats on water, P = 2.

➦ Allotropic forms of an element are different phases.

**Example**

White and red phosphorus, P = 2.

➦ A mixture of different solids has different phases.

**Example**

A mixture of sugar and salt, P = 2.

2. **Component (c)** is chemically distinct quantity present in a system.

| Number of components = Number of different chemical species (elements or compounds) in the given system. |
|---|

**Note**

A pure substance in different states has only one component.

3. **Degree of freedom (F)** is the minimum number of variables needed to describe each component in each phase of the system.

## Phase rule (Gibb's phase rule equation)

➦ Relates components and number of phases with degree of freedom.

$$F = C - P + 2$$

➥ The number 2 represents the two parameters temperature and pressure.

➥ For a pure substance with

a. one phase:
$F = C - P + 2 = 1 - 1 + 2 = 2$, such type of system is called **bivariant system.**

b. Two phases
$F = C - P + 2 = 1 - 2 + 2 = 1$, it is called **monovariant system.**

c. Three phases
$F = C - P + 2 = 1 - 3 + 2 = 0$, it is known as **invariant system.**

d. Four phases
$F = C - P + 2 = 1 - 4 + 2 = -1$, such type of system is not possible because defree of freedom is 0, 1 or 2.

## Illustrative Examples

69. Which of the following exists at low temperature and high pressure?

A. $H_2O(l)$ B. $H_2O(s)$ C. $H_2O(g)$ D. None

**Solution**

⇒solid exits at low temperature and high pressure.

**Answer: B**

70. The number of phase (s) in a mixture of $CH_4$, $C_2H_6$ and $C_3H_8$ are:

A. 1 B. 2 C. 3 D. 4

**Solution**

Three of them are non - polar gases. They form a gaseous solution. Thus, the number of phase is 1.

**Answer: A**

71. The direct change of solid to gas is

A. Deposition
B. Condensation
C. Evaporation
D. Sublimation

**Solution**

➥ The three states of matter are inter convertable by heating or cooling.

Solid → liquid, Melting
Liquid → Solid, Freezing
Liquid → gas, evaporation
Gas → liquid, condensation
Solid →gas, sublimation
Gas → solid, deposition

**Answer: D**

72. Arrange the three states of matter in decreasing order of
    a. Compressibility Gas Liquid. Solid
    b. Kinetic energy
    c. Attraction force between particles.
    d. Boiling point and density

**Solution**

a. Gas > liquid > solid
b. Gas > liquid > solid
c. Solid > liquid > gas
d. Solid > liquid > gas

73. Determine the number of phases and components in the following systems.
    a. 70% alcohol solution
    b. A mixture of benzene and water
    c. A mixture of hydrogen, carbondioxide and nitrogen.
    d. A mixture of powdered KCl, NaCl, $CaCO_3$ and $C_{12}H_{22}O_{11}$
    e. The allotropes of sulphur monoclinic and rhombic sulphur.
    f. Dry ice and carbon dixoide

**Solution**

a. P = 1, C = 2
b. P = 2, C = 2
c. P = 1, C = 3
d. P = 4, C = 4
e. P = 2, C = 1
f. P = 2, C = 1

74. Determine the degree of freedom in
    a. a saturated solution of salt with an excess of solid is present in equilibrium with its vapor in a closed vessel.
    b. Which ice, water and water vapor coexist at equilibrium.

**Solution**

a. C = 2, P = 2
   $F = C - P + 2 = 2 - 2 + 2 = 2$
b. C = 1, P = 3
   $F = C - P + 2 = 1 - 3 + 2 = 0$

75. Define the following terms.
    a. Boiling point
    b. Normal boiling point
    c. Vapor pressure

**Solution**

a. It is the temperature at which vapor pressure is equal to external pressure.
b. It is the temperature at which external pressure equal to one atmosphere.
c. It is the pressure exerted by vapor.

76. Volatile substances have higher vapor pressure than non - volatile substances. Why?

**Solution**

➦ Volatile substances have high tendency to evaporate at room temperature but non - volatile substances have little tendency to evaporate.

77. Classify the six changes of state as endothermic or exothermic process.

**Solution**

➦ Melting
➦ Evaporation
➦ Sublimation
} are endothermic processes

➦ Freezing
➦ Condensation
➦ Deposition
} are exothermic process

## 5.6 Phase Diagram

Phase diagram is a graph that shows the phases of substance existing at various temperatures and pressures.

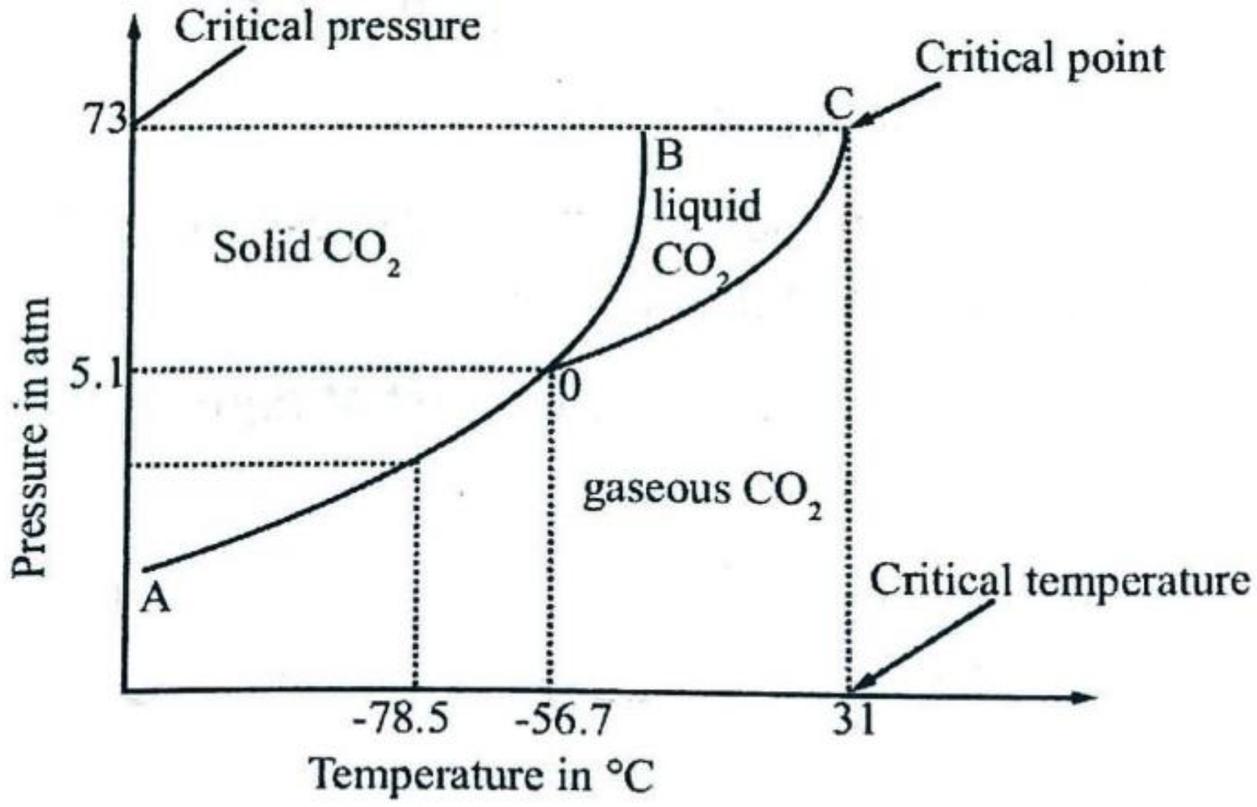



Figure 5.1 Phase diagram of $CO_2$

➦ Curve AO is **sublimation curve**, the solid and the gas present together at equilibrium.
➦ Curve OB is **fusion curve**, the solid and liquid are present together at equilibrium.

- Curve OC is **vaporization curve**, the liquid and gas phases are present together at equilibrium.
- Point O is **triple point**. At this point, the three states of matter coexist at equilibrium.
- The pressure at the critical point is called **critical pressure**.
- The temperature at the critical point is known as **critical temperature**.
- Above critical temperature, the vapor cannot be liquefied at any pressure.

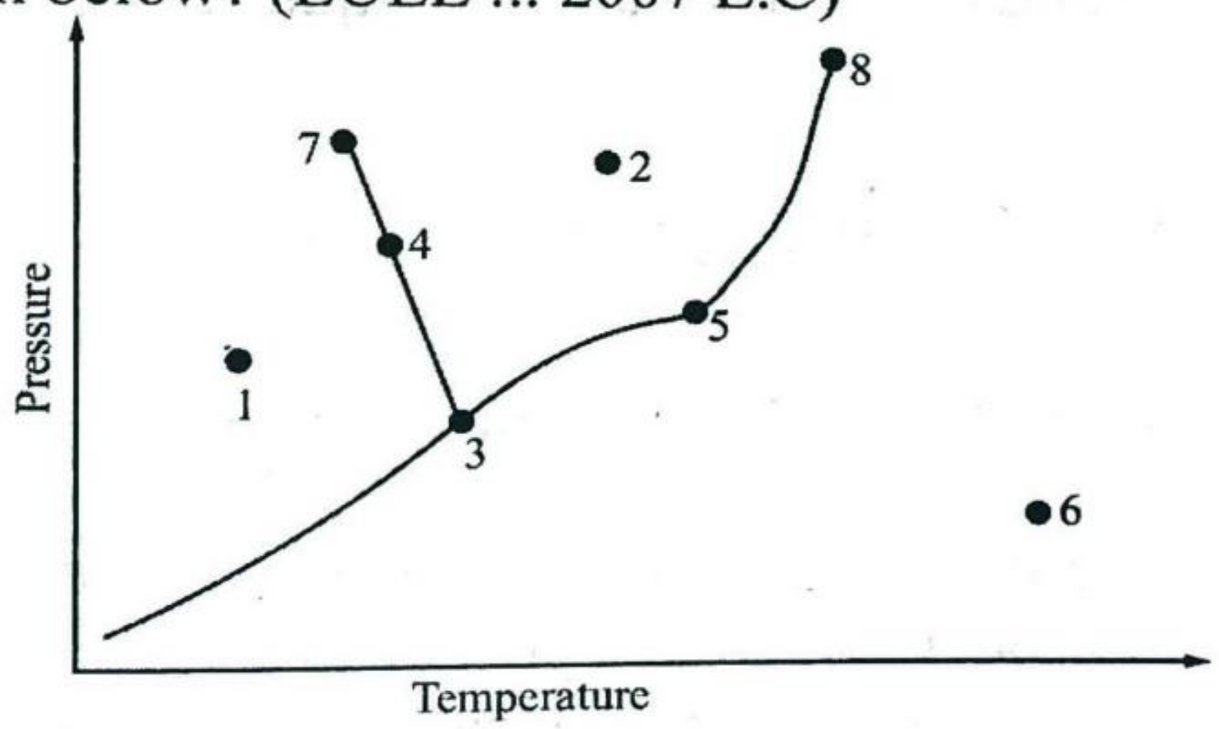
## Basic features of the Phase diagram of $CO_2$

- It is similar with phase diagram of water except that:
  - The fusion curve has positive slope.
  - Solid $CO_2$ is more dense than the liquid.
  - Solid $CO_2$ sublime directly without melting. **Dry ice (solid $CO_2$)** looks like water - ice but it doesn't melt.

Of all the thousands of substances for which phase diagrams are known, only water, bismuth and antimony have fusion curve with negative slope.

## Illustrative Examples

78. At which point can only the solid and liquid phases coexist in the phase diagram given below? (EUEE ... 2007 E.C)



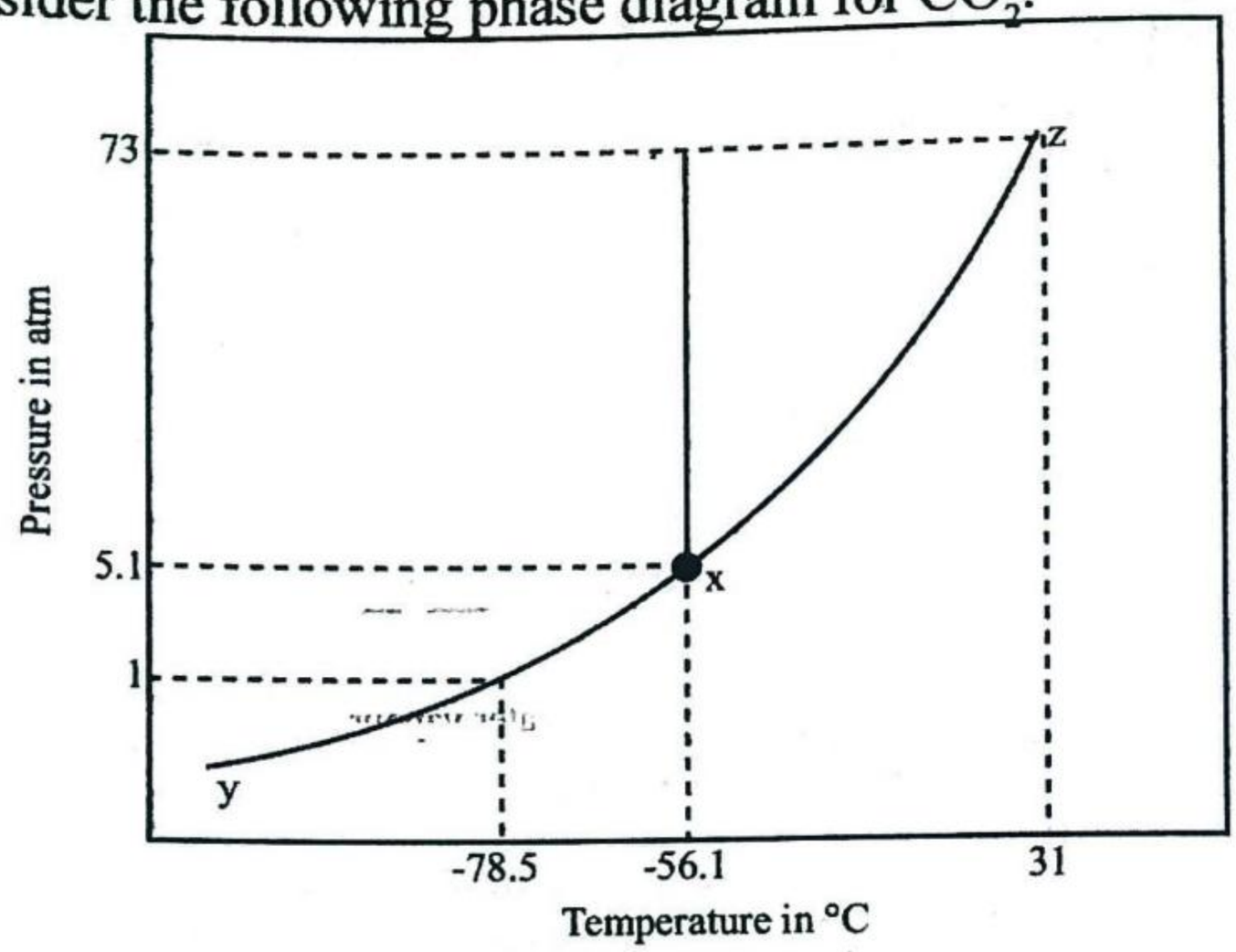

A. 3 B. 4 C. 5 D. 8

**Solution**

- Solid and liquid phases coexist at equilibrium at point 4.
- Liquid and gas phases coexist at equilibrium at point 5.
- Solid, liquid and gas phases coexist at point 3.

**Answer: B**

79. Consider the following phase diagram for $CO_2$.




What happen when a $CO_2$ sample initially at 1 atm and -70°C, the temperature increases from -70°C to -10°C at a constant pressure fo 60 atm? (EUEE, 2004 E.C)

A. $CO_2(g) \rightarrow CO_2(s)$
B. $CO_2 \rightarrow CO_2(l)$
C. $CO_2(g) \rightarrow CO_2(g)$
D. $CO_2(g) \rightarrow CO_2(l)$

**Solution**

At 60 atm, when the temperature is increased from -70°C to -10°C, solid $CO_2$ is converted to liquid $CO_2$.

**Answer: B**

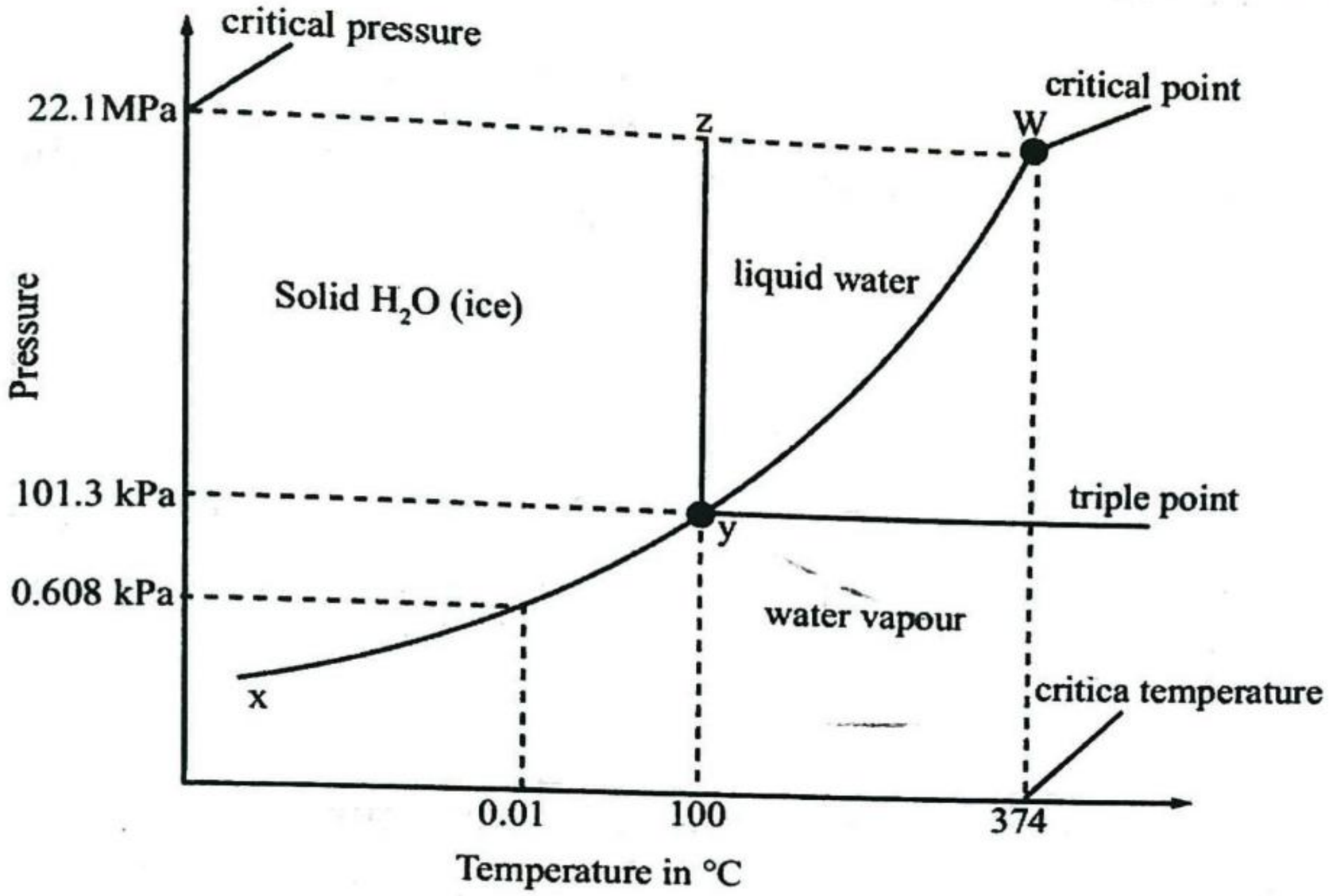
80. What is the use of phase diagram?

**Solution**

- ➦ It is used to find the phase of a system at a given temperature and pressure.
- ➦ It is used to determine the melting point, sublimation point and boiling point at different pressure.
- ➦ It is used to determine critical point, triple point and to predict the phase changes.

81. The triple point of water is 0.608 kPa at 0.01°C. Its critical point lies at 22.1MPa at 374°C. The sublimation curve and vaporization curve have positive slopes but the fusion curve has negative slope. Draw the phase diagram of water and label it.

**Solution**




## Solved Problems

82. A quantity of $AB_2$ was heated in a 10L vessel at 500k. $AB(g) \rightleftharpoons A(g) + 2B(g)$. At equilibrium the vessel contains, 0.10mol AB, 0.2 mol A and 0.3 MB. Calculate $K_c$ and $K_p$.

**Solution**

$$[AB_2] = \frac{n_{AB_2}}{v} = \frac{0.10\,mol}{10L} = 0.01M$$

$$[A] = \frac{n_A}{v} = \frac{0.2\,mol}{10L} = 0.02M$$

$$[B] = \frac{n_B}{v} = \frac{0.3mol}{10L} = 0.03M$$

$$K_c = \frac{[A][B]^2}{[AB_2]} = \frac{(0.02)(0.03)^2}{(0.01)} = 1.8\times10^{-3}\,mol/L$$

$$\Delta n = (1+2) - (1) = 1$$

$$\Rightarrow K_p = K_c\,(RT)^{\Delta n} = 1.8\ \ 10^{-3}mol/L \times 0.082\ atm.L/mol.K \times 500K$$

$$K_p = 0.0738\ atm$$

83. If the concentration of $CrO_4^{2-}$ ions in a saturated solution of silver chromate is $2 \times 10^{-4}$M, what is the solubility product of the salt?

**Solution**

$$Ag_2CrO_4(s) \rightleftharpoons 2Ag^+(aq) + CrO_4^{2-}(aq)$$

Since the coefficient ratio of $Ag^+$ to $CrO_4^{2-}$ is 2: 1, thus $[Ag^+] = 4 \times 10^{-4}M$.

$$\Rightarrow K_{sp} = [Ag^+]^2[CrO_4^{2-}]$$

$$K_{sp} = (4 \times 10^{-4})^2(2 \times 10^{-4}) = 1.6 \times 10^{-11}$$

84. One mole of $N_2O_4$ (g) at 27°C is kept in a closed vessel at one atmosphere pressure. If 20% of $N_2O_4$(g) is decomposed into $NO_2$, what will be the value of $K_p$?

**Solution**

20% 1 mol of $N_2O_4$ = 0.2 mol is decomposed

| | $N_2O_4(g) \rightleftharpoons 2NO_2(g)$ | |
|---|---|---|
| Initial mol | 1 mol | 0 |
| Change mol | -0.2mol | $+2 \times 0.2$mol |
| Equilibrium mol | 1-0.2 = 0.8mol | 0.4 mol |

$\Rightarrow$To find partial pressure of $N_2O_4$ and $NO_2$ find mole fraction.

$$X_{N_2O_4} = \frac{n_{N_2O_4}}{n_{N_2O_4} + n_{NO_2}} = \frac{0.8}{0.8+0.4} = 0.67$$

$$X_{NO_2} = \frac{n_{NO_2}}{n_{N_2O_4} + n_{NO_2}} = \frac{0.4}{0.8+0.4} = 0.33$$

$$P_{N_2O_4} = X_{N_2O_4}.P_T = 0.67 \times 1atm = 0.67atm$$

$$P_{NO_2} = X_{NO_2}.P_T = 0.33 \times 1atm = 0.33atm$$

$$\Rightarrow K_P = \frac{P_{NO_2^2}}{P_{N_2O_4}} = \frac{(0.33atm)^2}{(0.67atm)} = 0.16atm$$

85. Find $K_p/K_c$ for the reaction below $CO(g) + \frac{1}{2}O_2(g) \rightleftharpoons CO_2(g)$

**Solution**

$$\Delta n = 1 - \left(1 + \frac{1}{2}\right) = -\frac{1}{2}$$

$$K_p = K_c(RT)^{\Delta n}$$

$$\frac{K_p}{K_c} = (RT)^{\frac{1}{2}}$$

$$\frac{K_p}{K_c} = 1 : \sqrt{RT}$$

86. $A(g) + 2B(g) \rightleftharpoons 3C(g) + D(g)$, $K_p = 0.04$ atm at 1000K. Find $K_c$ in terms of R.

**Solution**

$$\Delta n = (3 + 1) - (2 + 1) = 1$$

$$K_c = \frac{K_p}{(RT)^{\Delta n}} = \frac{0.04atm}{1000K.R} = \frac{40atm}{RK}$$

87. The equilibrium constant, $K_c$ for the following reaction is 0.65 at 200°C.

$N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$

At the same temperature, calculate

a. $K_p$

b. $K_c$ for the reaction

i $2NH_3(g) \rightleftharpoons N_2(g) + 3H_2(g)$

ii $\frac{1}{2}N_2(g) + \frac{3}{2}H_2(g) \rightleftharpoons NH_3(g)$

**Solution**

a. $K_p = 0.65\ (0.032 \times 473)^{-2}$

$K_p = 4.32 \times 10^{-4}\ atm^{-2}$

b. i) $K_c = \frac{1}{0.65} = 1.54$

ii) $K_3 = (0.65)^{1/2} = 0.81$

88. Initially, 1 mol of A, 1.5mol of B and 0.4mol of C are placed in a 1.0 liter reaction vessel and allowed to come to equilibrium. If the final concentration of C is 0.80M, then find the value of equilibrium constant for the reaction: $A + B \rightleftharpoons 2C$.

**Solution**

$$[A]_0 = \frac{1mol}{1L} = 1M$$

$$[B]_0 = \frac{1.5mol}{1L} = 1.5M$$

$$[C]_0 = \frac{0.4mol}{1L} = 0.4M$$

| | A + | B ⇌ | 2C |
|---|---|---|---|
| Initial conc. | 1 | 1.5 | 0.4 |
| Change | -x | -x | +2x |
| Equilibrium | 1-x | 1.5-x | 0.4+2x |

But [C] = 0.80M at equilibrium

$\Rightarrow 0.4 + 2x = 0.80$

At equilibrium

[A] = 1 - x = 1 - 0.2 = 0.8M

[B] = 1.5 - x = 1.5 - 0.2 = 1.3M

[C] = 0.80M

$$K_c = \frac{[C]^2}{[A][B]} = \frac{(0.80\,M)^2}{(0.8\,M)(1.3\,M)} = 0.62$$

89. Given the equilibrium constant values of the following reactions at 25°C.

i $2H_2(g) + O_2(g) \rightleftharpoons 2H_2O(g)$, $k_c = 1.1 \times 10^{81}$

ii $H_2(g) + Cl_2(g) \rightleftharpoons 2HCl(g)$, $k_c = 4.4 \times 10^{32}$

Find the value of $K_p$ at 25°C for the reaction:

$2H_2O(g) + 2Cl_2(g) \rightleftharpoons 4HCl(g) + O_2(g)$

**Solution**

Reverse the first and multiply the second by 2 and find $k_c$.

$$2H_2O(g) \rightleftharpoons 2H_2(g) + O_2(g), \frac{1}{1.1\times10^{81}}$$

$$2H_2(g) + 2Cl_2(g) \rightleftharpoons 4HCl(g), (4.4\times10^{32})^2$$

$$2H_2O(g) + 2Cl_2(g) \rightleftharpoons 4HCl(g) + O_2(g)$$

$$K_c = (4.4\times10^{32})^2 . \frac{1}{1.1\times10^{81}}$$

$$K_c = 1.76\times10^{-16}$$

$\Rightarrow K_p = K_c (RT)^{\Delta n} = 1.76 \times 10^{-16} \times (0.082 \times 298)$

$K_p = 4.3 \times 10^{-15}$

90. Given the reaction at equilibrium. $2CO_2(g) \rightleftharpoons 2CO(g) + O_2(g)$, $\Delta H = +566kJ$ Explain the effect of the following on the position of equilibrium.

a. Lowering temperature
b. Removing $O_2$
c. Increasing volume and temperature.
d. Increasing pressure and decreasing temperature.
e. Increasing concentration of $CO_2$
f. Adding inert gas.

**Solution**

a. Shifts to the left.
b. Shifts to the right
c. Shifts to the right
d. Shifts to the left

e. Shifts to the right

f. Shifts to the right

91. Write $K_c$ and $K_p$ expression for the following reactions.

a. $4NH_3(g) + 3O_2(g) \rightleftharpoons 2N_2(g) + 6H_2O\,(g)$

b. $Ca(HCO_3)_2(s) \rightleftharpoons CaCO_3(s) + CO_2(g) + H_2O(s)$

c. $NO_2(g) \rightleftharpoons \frac{1}{2}N_2(g) + O_2(g)$

**Solution**

a. $K_c = \frac{[N_2]^2[H_2O]^6}{[NH_3]^4[O_2]^3}, K_P = \frac{P_{N_2}^2 . P_{H_2O}^6}{P_{NH_3}^4 . P_{O_2}^3}$

b. $K_c = [CO_2][H_2O], K_P = P_{CO_2} . P_{H_2O}$

c. $K_c = \frac{[N_2]^{\frac{1}{2}}[O_2]}{[NO_2]} = K_p = \frac{P_{N_2}^{1/2} . P_{O_2}}{P_{NO_2}}$

92. Is a mixture of 0.02 mol $NO_2$ and 0.75mol of $N_2O_4$ in a 5L flask at 298k at equilibrium? $N_2O_4(g) \rightleftharpoons 2NO_2(g)$, $K_c = 0.05$ at 298k.

**Solution**

Determine the value of $Q_c$ at 298K

$$[N_2O_4] = \frac{0.75}{5} = 0.15M$$

$$[Q_C = \frac{[NO_2]^2}{[N_2O_4]} = \frac{(0.004)^2}{0.55} = 1.1 \times 10^{-4}$$

$$[Q_C = \frac{[NO_2]^2}{[N_2O_4]} = \frac{(0.004)^2}{0.55} = 1.1 \times 10^{-4}$$

$\Rightarrow$ The reaction is not at equilibrium. It shifts to the right because $Q_c < K_c$.

93. What happens to water when the pressure remains constant at 1 atm but the temperature changes from -5°C to 85°C based on the phase diagram of water?

**Explanation:** Water remains liquid until the temperature reaches 100°C at 1 atm.

94. Equilibrium constant expressions do not contain concentration terms for solids or liquids of a single component. Explain why?

**Solution**

The concentration of the single component with a pure solid or liquid phase cannot change and the activities of pure solids and liquids are set equal to one. Due to this, we omit pure solid and liquids in the equilibrium constant expression.

95. For the reaction, $CaCO_3(l) \rightleftharpoons CaO(s) + CO_2(g)$, $K_p = 1.16atm$ at 800°C. If 40g of $CaCO_3$ was put into a 10L container and heated to 800°C. What percent of $CaCO_3$ would remain unreacted at equilibrium?

**Solution**

$$K_p = P_{CO_2} = 1.16atm$$

$$PV = nRT$$

$$n_{CO_2} = \frac{Pv}{RT} = \frac{1.16atm \times 10L}{0.082\frac{atm.L}{mol.k} \times 1073k}$$

$$n_{CO_2} = 0.13mol$$

$$n_{CaCO_3} = \frac{m}{M} = \frac{40g}{100g/mol} = 0.4mol$$

$$\%CaCO_3 decomposed = \frac{n_{CO_2}}{n_{CaCO_3}} \times 100 = \frac{0.13}{0.40} \times 100 = 32.95\%$$

$$\Rightarrow CaCO_3 \text{ undecomposed} = 100 - 32.95\% = 67.05\%$$

96. Arrange the following reactions in order of their increasing tendency to proceed toward completion.

a. $4NH_3(g) + 3O_2(g) \rightleftharpoons 2N_2(g) + 6H_2O(g)$, $K = 1 \times 10^{28}$

b. $N_2(g) + O_2(g) \rightleftharpoons 2NO(g)$, $K = 5 \times 10^{-31}$

c. $2HF(g) \rightleftharpoons H_2(g) + F_2(g)$, $K = 1 \times 10^{-13}$

d. $2NOCl(g) \rightleftharpoons 2NO(g) + Cl_2(g)$, $K = 4.7 \times 10^{-4}$

**Solution**

$b < c < d < a$

97. For the reaction: $H_2(g) + I_2(g) \rightleftharpoons 2HI(g)$ , an equilibrium mixture at 425°C contains $4.79 \times 10^{-4}$ mol/L $H_2$, $4.79 \times 10^{-4}$mol/L of $I_2$ and $3.53 \times 10^{-3}$mol/L of HI. Determine the equilibrium constant.

**Solution**

$$K_C = \frac{[HI]^2}{[H_2][I_2]}$$

$$K_C = \frac{(3.53 \times 10^{-3})^2}{(4.79 \times 10^{-4})^2} = 54.3$$

98. Hydrogen and nitrogen gases are pumped into an empty 5L vessel at 500°C. When equilibrium is reached, 3 mole of $N_2$, 2.1 moles of $H_2$ and 0.298 moles of $NH_3$ are present. Calculate $K_C$ and $K_P$.

$N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$

**Solution**

$$[N_2] = \frac{n_{N_2}}{V} = \frac{3mol}{5L} = 0.6mol/L$$

$$[H_2] = \frac{n_{H_2}}{V} = \frac{2.1mol}{5L} = 0.42mol/L$$

$$[NH_3] = \frac{n_{NH_3}}{V} = \frac{0.298mol/L}{5L} = 0.0596mol/L$$

$$K_C = \frac{[NH_3]^2}{[N_2][H_2]^3} = \frac{(0.0596)^2}{(0.6)(0.42)} = 0.08\ L/mol^2$$

$K_P = K_C (RT)^{\Delta n}$

$\Delta n = 2 - (1 + 3) = -2$

$T = 500°C = 500 + 273 = 773K$

$K_P = (0.08L^2/mol^2)(0.082\ atmL/mol.K.\ 773K)$

$K_P = 2 \times 10^{-5}\ atm^{-2}$

99. Consider the following heterogeneous equilibrium: $CaCO_3(s) \rightleftharpoons CaO(s) + CO_2(g)$ At 800, the pressure of $CO_2$ is 0.236atm. Calculate $K_P$ and $K_C$ at this temperature.

**Solution**

$K_P = P_{CO_2} = 0.236atm$

$\Delta n = 1, T = 800 + 273 = 1073K$

$K_P = K_C (RT)^{"n}$

$$K_C = \frac{K_P}{(RT)^{"n}}$$

$$K_C = \frac{0.236atm}{(0.082\,atm.L/\,mol.K \times 1073\,K)^1}$$

$K_C = 2.68 \times 10^{-3}\ mol/L$

100. Write the applications of equilibrium constant.

**Solution**

- To predict the composition of an equilibrium mixture.
- To predict the direction of the reaction.

➥ To calculate the equilibrium concentrations of reactants and products.

101. The equilibrium constant, $K_C$ is $1 \times 10^{-19}$ at 298K for the reaction:

$C_2H_6(g) \rightleftharpoons H_2(g) + C_2H_4(g)$

If the initial concentration of $C_2H_6$ is 0.1M at equilibrium, what will be the equilibrium concentrations of all species?

**Solution**

| | $C_2H_6(g) \rightleftharpoons H_2(g) + C_2H_4(g)$ | | |
|---|---|---|---|
| initial: | 0.1 | 0 | 0 |
| change: | -x | +x | +x |
| Equilibrium: | 0.1 - x | x | x |

$$K_C = \frac{[H_2][C_2H_4]}{[C_2H_6]}$$

$1\times10^{-19} = \frac{x.x}{0.1-x}$ use approximation method $0.1 - x \approx 0.1$

$$1\times10^{-19} = \frac{x^2}{0.1}$$

$$x = [C_2H_4] = [H_2] = 1\times10^{-10}\,M$$

$$[C_2H_6] = 0.1 - 1\times10^{-10} = 0.1M$$

102. Given the equilibrium constant values for the following reactions at 25°C.

$2H_2(g) + O_2(g) \rightleftharpoons 2H_2O(g)$, $K_C = 1 \times 10^{81}$

$H_2(g) + Cl_2(g) \rightleftharpoons 2HCl(g)$, $K_C = 4 \times 10^{32}$

What is the value of KC for the reaction:

$2H_2O(g) + 2Cl_2(g) \rightleftharpoons 4HCl\,(g) + O_2(g)$

**Solution**

Reverse the first and multiply the second by 2.

$$\Rightarrow 2H_2O(g) \rightleftharpoons O_2(g), K_{C_1^1} = 1\times10^{-8}$$

$$2H_2(g) + 2Cl_2(g) \rightleftharpoons 4HCl(g), K_{C_2^1} = 1.6\times10^{65}$$

$$2H_2O(g) + 2Cl_2(g) \rightleftharpoons 4HCl(g) + O_2(g)$$

$$K_C = K^1_{C_1} + K^1_{C_2} = (1\times10^{-81})(1.6\times10^{65}) = 1.6\times10^{-16}\,M$$

# Supplementary Exercise

## I. Multiple Choices

1. Which of the following reactions has equal values of $K_C$ and $K_P$?
   A. $CO(g) + 2H_2(g) \rightleftharpoons CH_3OH(g)$
   B. $PCl_5(g) \rightleftharpoons PCl_3(g) + Cl_2(g)$
   C. $H_2(g) + CO_2(g) \rightleftharpoons H_2O(g) + CO(g)$
   D. $Ni(s) + 4CO(g) \rightleftharpoons Ni(CO)_4(g)$
2. For the reaction $A(g) + 2B(g) \rightleftharpoons 2C(g)$ + heat, formation of C is favoured if:
   A. Temperature is increased
   B. Pressure is decreased
   C. Amount of A is increased
   D. Catalyst is added
3. For a reaction at equilibrium, which of the following is true?
   A. $Qp = Kp$ B. $Qp > Kp$ C. $Qp < Kp$ D. $Qp = 0$
4. For the reaction, $CO(g) + H_2O(g) \rightleftharpoons CO_2(g) + H_2(g)$, $Kc = 1.00$ at about 1100K. Which of the following must always be true concerning this reaction at equilibrium at 1100k?
   A. $2[CO] = [H_2O] = [CO_2] = [H_2]$
   B. $[CO] \times [H_2O] = [CO_2] \times [H_2]$
   C. $[CO] = [H_2O]$ and $[CO_2] = [H_2]$
   D. $[CO] \times [H_2O] = [CO_2] = [H_2] = 1.00$
5. In which of the following the yield of the product in the reaction, $A_2(g) + 2B(g) \rightleftharpoons V(g)$ + heat, would be higher?
   A. High temperature and high pressure
   B. High temperature and low pressure
   C. Low temperature and high pressure
   D. Low temperature and low pressure
6. Which one of the following is correct about a dynamic equilibrium?
   A. Is a form of static equilibrium
   B. Occurs when the rate constant of the forward process is equal to the rate constant of the reverse process.
   C. Only occurs in chemical equilibrium
   D. Exists when the rate of the forward reaction is equal to the rate of the reverse reaction.

7. For the reaction

$H^+(aq) + OH^-(aq) \rightleftharpoons H_2O(l)$, $kf = 1.4 \times 10^{11}$ and $Kr = 2.5 \times 10^{-5}$ at 250°C, then which one of the following is the value of $K_c$ for this process?

A. $5.6 \times 10^{15}$
C. $1.4 \times 10^{11}$
B. $1.8 \times 10^{-16}$
D. $3.5 \times 10^{6}$

8. Cobalt chloride is added to pure water. The $Co^{2+}$ and $Cl^-$ ions to set up the equilibrium shown below

$$Co(H_2O)_6^{2+} + 4Cl^- \rightleftharpoons COCl_4^{2-} + 6H_2O$$

Pink blue

Which statement below describes the change that the system undergo if water is added?

A. More water will be produced
B. more chloride ions will be produced
C. the equilibrium will shift to the right
D. the color will become more blue

9. For the equilibrium: $C(s) + CO_2(g) \rightleftharpoons 2CO(g)$ $K_p = 16.75$ atm at 500°C. What is the partial pressure of $CO_2$ that is in equilibrium with CO whose partial pressure is 0.50atm?

A. $2.99 \times 10^{-2}$atm
C. 0.25atm
B. $1.49 \times 10^{-2}$atm
D. 0.05atm

10. The gaseous reaction $A + 2B \rightleftharpoons C$ is exothermic.
A. An increase of temperature will drive the equilibrium towards the left
B. High pressure will favour the dissociation of C into A and B.
C. The systhesis of C from A and B is favoured by high temperatures and low pressure
D. Pressure has no effect on the equilibrium

11. Which of the following factors does not affect the position of chemical equilibrium?

A. temperature
C. pressure
B. concentration
D. catalysts

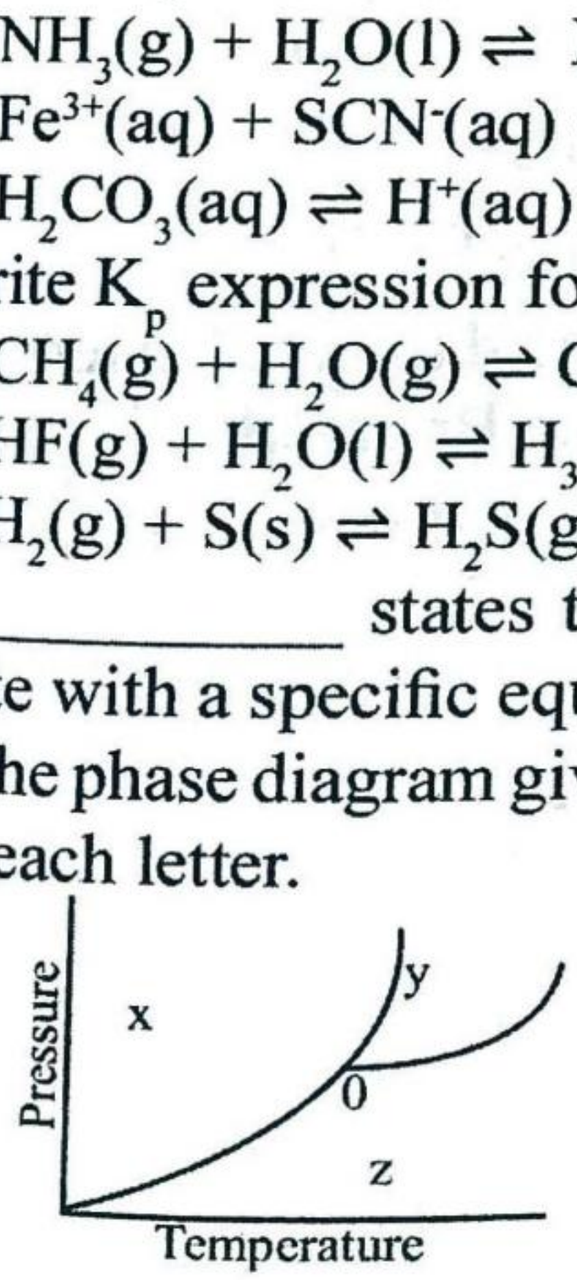
12. The unit of $K_c$ for the reaction, $H_3A(aq) \rightleftharpoons 3H^+(aq) + A^{3-}(aq)$ is

A. mol/L
C. $mol^2/L^2$
B. $mol^3/L^3$
D. $L^3/mol^3$

13. The system, $CaCO_3(s) \rightleftharpoons CaO(s) + CO_2(g)$ has
    A. 3 components, 1 phase and 2 degrees of freedom
    B. 2 components, 3 phase and 1 degrees of freedom
    C. 3 components, 3 phases and two degrees of freedom
    D. 2 components, 2 phase and 2 degrees of freedom
14. According to Lechatelier's principle, the addition of heat to the following reaction,
    $CO_2(g) + 2H_2O(g) \rightleftharpoons CH_4(g) + 2O_2(g)$, will cause it to shift to the right. This reaction can therefore described as
    A. Spontaneous
    B. unimolecular
    C. exothermic
    D. endothermic
15. Lechatelier's principle states that:
    A. Reaction equilibrium is obtained only in a certain conditions of temperature and pressure.
    B. increasing temperature while reducing pressure at same time, enhances equilibrium.
    C. stress applied to a system at equilibrium, shifts the system in such a way as to relieve the stress.
    D. Neither temperature nor pressure have a major effect on chemical equilibrium

## II. Give shot answer

16. Write $K_c$ expression for
    a. $NH_3(g) + H_2O(l) \rightleftharpoons NH_4^+ + OH^-(aq)$
    b. $Fe^{3+}(aq) + SCN^-(aq) \rightleftharpoons FeSCN^{+2}(aq)$
    c. $H_2CO_3(aq) \rightleftharpoons H^+(aq) + HCO^-_3(aq)$
17. Write $K_p$ expression for
    a. $CH_4(g) + H_2O(g) \rightleftharpoons CO(g) + 3H_2(g)$
    b. $HF(g) + H_2O(l) \rightleftharpoons H_3O{+}(aq) + F^-(aq)$
    c. $H_2(g) + S(s) \rightleftharpoons H_2S(g)$
18. _____ states that every reaction proceeds to an equilibrium state with a specific equilibrium constant.
19. In the phase diagram given below, write the phase of matter represented by each letter.




X. _____
Y. _____
Z. _____

20. The conversion of oxygen to ozone has a very small equilibrium constant

$\frac{3}{2}O_2(g) \rightleftharpoons O_3(g), K_c = 2.5 \times 10^{-29}$

Calculate the value of $K_C$ for

a. $3O_2(g) \rightleftharpoons 2O_3(g)$ b. $2O_3(g) \rightleftharpoons 3O_2(g)$

21. Lead sulfide can be converted to metallic lead in the following series of reactions:

$2PbS(s) + 3O_2(g) \rightleftharpoons 2PbO(s) + 2SO_2(g), K_1$

$PbO(s) + C(s) \rightleftharpoons Pb(l) + CO(g), K_2$

$PbO(s) + CO(g) \rightleftharpoons Pb(l) + CO_2(g), K_3$

If the over all reaction is

$2PbS(s) + 3O_2(g) + C(s) \rightleftharpoons 2Pb(l) + CO_2(g) + 2SO_2(g)$

What is the equilibrium constant for the overall reaction interms of $K_1$, $K_2$ and $K_3$?

22. The equilibrium constant kp for the following reaction is $4.00 \times 10^{-4}$ at 500°C.

$2HCN(g) \rightleftharpoons H_2(g) + C_2N_2(g)$, what is the equilibrium partial pressure of $H_2$ if pure HCN was originally present at 25atm?

23. $K_P = 0.11$ at 25°C for the reaction,

$NH_4HS(s) \rightleftharpoons H_2S(g) + NH_3(g)$. What is the total pressure in the flask at equilibrium?

24. For the reaction, $AgCl(s) \rightleftharpoons Ag^+(aq) + Cl^-(aq)$,

$Kc = 1.8 \times 10^{-10}$. If $[Ag^+] = 1 \times 10^{-5}M$, what is the $[Cl^-]$?

25. 1 mol of $Br_2$ is placed in a 1L vessel and heated to 1756l, a temperature at which the halogen molecules dissociate to Br atoms

$Br_2(g) \rightleftharpoons 2Br(g)$, if $Br_2(g)$ is 1% dissociated at this temperature, calculate $K_c$.

አጋዥ መጽሐፍቶችን ለማግኘት በ Telegram
@ethiolibrary1 ላይ ይቀላቀሉን
+251927273995
+251909282436
jemal6454@gmail.com

# Unit 6

## Carboxylic Acids, Esters, Fats and Oils

## Unit Content Outline:

6.1 Classification of Organic Compounds
6.2 Carboxylic acid
6.3 Nomenclature of Carboxylic acids
6.4 Esters
6.5 Aldehydes and Ketones
6.6 Sources and Structures of fats and Oils
6.7 Soaps and Detergents

## Introduction

- Carboxyic acids, esters, fats and oils are oxygen containing organic compounds.
- Organic chemistry is the chemistry of hydrocarbons and derivative hydrocarbons.
- Hydrocarbons are organic compound containing carbon and hydrogen only.
- Derivative hydrocarbons are obtained from hydrocarbons by replacing one or more hydrogen atoms with other atoms such as oxygen.

# 6.1 Classification of Organic Compounds

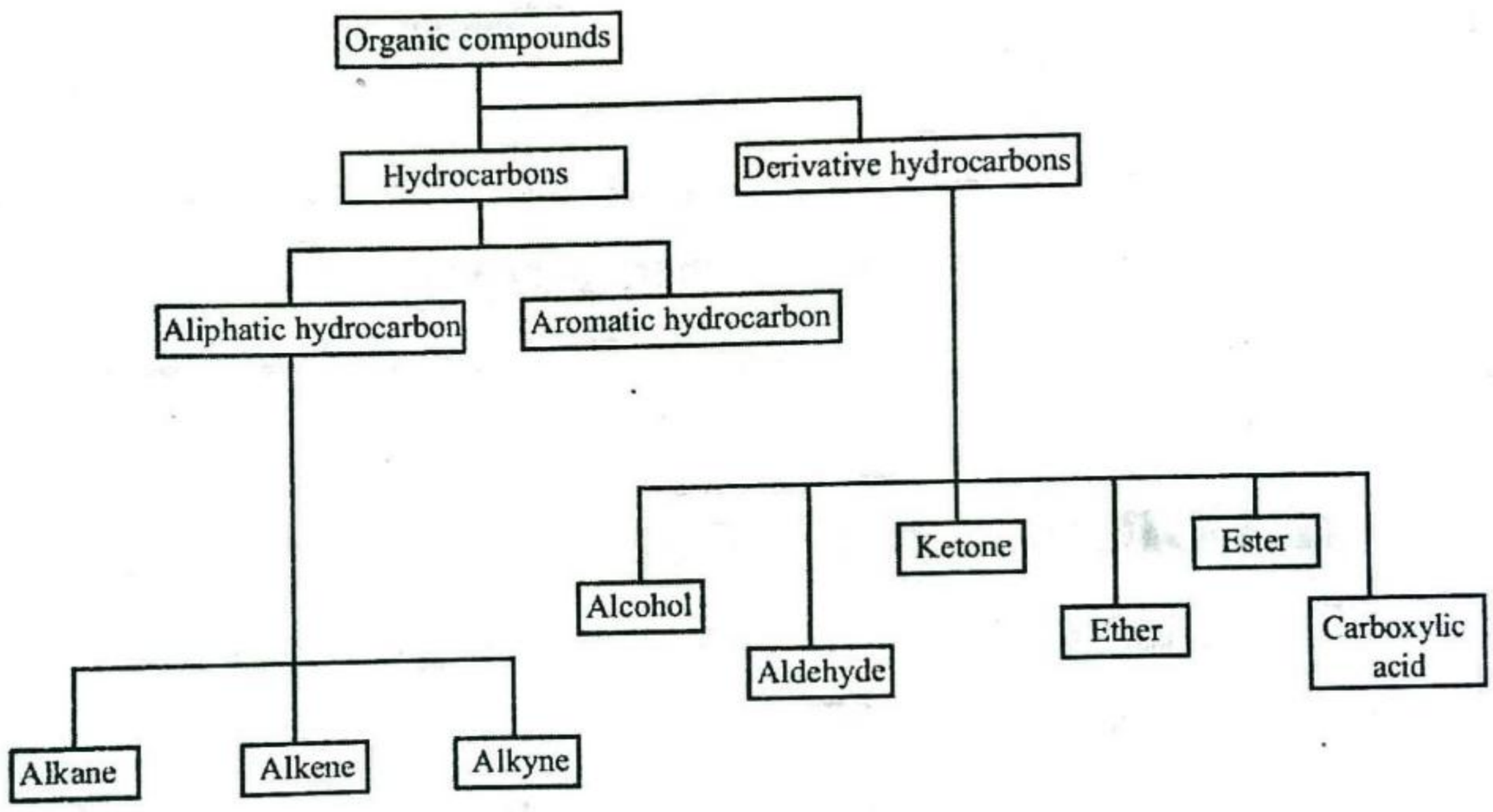
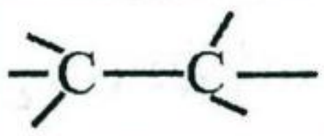
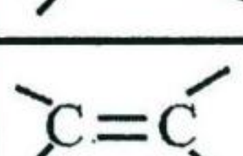
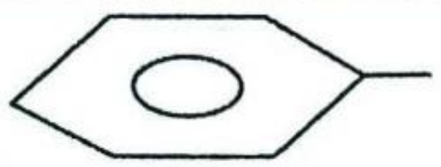



Table 6.1. Organic compounds, functional group, general formula and example

| Organic Compound | General formula | Functional group | Example |
|---|---|---|---|
| Alkane | RH |  | $CH_3CH_3$ |
| Alkene | $RCH = CH_2$ |  | $CH_2 = CH_2$ |
| Alkyne | $RC \equiv CH$ | $-C \equiv C-$ | $HC \equiv CH$ |
| Aromatic hydrocarbons | |  | |
| Alcohol | ROH | $-OH$ | $CH_3CH_2OH$ |

| | | | |
|---|---|---|---|
| Aldehyde | RHO | — CHO | $CH_3CHO$ |
| Ketone | RCOR | — CO | $CH_3COCH_3$ |
| Ether | ROR | — C — O — C — | $CH_3OCH_3$ |
| Carboxylic acid | RCOOH | —COOH | $CH_3COOH$ |
| Ester | RCOOR | — COO — | $CH_3COOCH_3$ |

## 6.2 Carboxylic acid

- They are acids that contain the carboxyl group, —COOH.
- Their general formula is RCOOH.
- Their functional group is —COOH
- They are organic acids.

## Classification of Carboxylic acids

### A. Based on number of carboxyl groups

1. **Monocarboxylic acids** - contain one carboxylic group.

Example

$CH_3COOH$

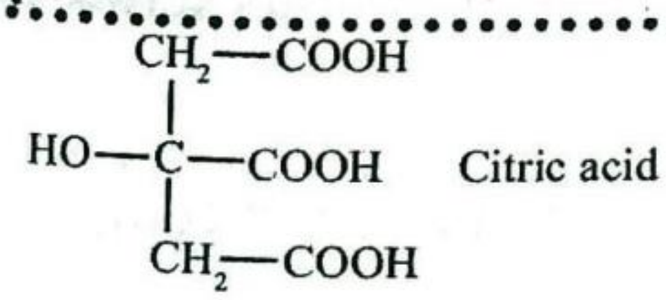
2. **Dicarboxylic acids** - contain two carboxyl groups.

Example

$HOOCCH_2COOH$

3. **Tricarboxylic acids** - posses three carboxyl groups.

Example




Citric acid

### B. Based on carbon - carbon bonds

1. **Saturated carboxylic acids** are acids with all carbon - carbon single bonds.

Example

$CH_3CH_2COOH$

2. **Unsaturated carboxylic acids** contain carbon - carbon double or triple bonds.

Example

$CH_3 - CH = CH - COOH$ $HC \equiv C - COOH$

Aromatic carboxylic acids contain benzene ring.

## Illustrative Examples

1. The organic compound $CH_3COCH_3$ is (EUEE ... 2004 E.C)

A. Aldehyde
B. Ester
C. Carbonyl
D. Ketone

**Solution**

If the functional group is $-\overset{\overset{O}{\|}}{C}-$, it is a ketone.

**Answer: D**

2. The general formula of carboxylic acid is

A. RCOOH
B. RCOOR'
C. RCOR'
D. ROH

**Solution**

Carboxylic acids contian - COOH group.

**Answer: A**

3. Part of the compound that determines its physical and chemical property is

A. Functional group
B. Structural formula
C. General formula
D. Homologous series

**Answer: A**

4. Identify the functional groups of the following organic compounds.

a. Carboxylic acids
b. Esters
c. Ketones
d. Aldehydes
e. Ethers
f. Alcohols

**Solution**

a. $-COOH$ or $-\overset{\overset{O}{\|}}{C}-OH$

b. $-COO-$ or $-\overset{\overset{O}{\|}}{C}-O-$

c. $-CO-$ or $-\overset{\overset{O}{\|}}{C}-$

d. $-CHO-$ or $-\overset{\overset{O}{\|}}{C}-H$

e. $-COC-$ or $-\overset{|}{\underset{|}{C}}-O-\overset{|}{\underset{|}{C}}-$ f. $-OH$ or $-O-H$

5. Write the molecular formula of mono carboxylic acids with carbon - carbon single bonds.

**Solution**: $C_nH_{2n}O_2$

6. Write the major organic acids found in the following living and non - living matter.

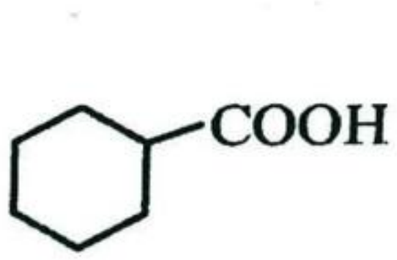
a. Lemon
b. Orange
c. Banana
d. Apple
e. Pineapple
f. Avocados
g. Salad cabbage
h. Grapes
i. Ant, bee
j. Rancid butter
k. Vinegar
l. Sour Milk
m. Aroma of goat

**Solution**

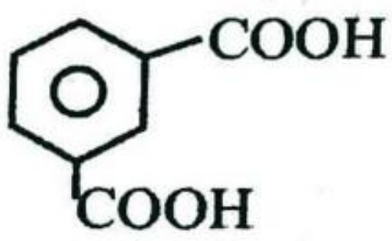
a. Citric acid, malic acid, tartaric acid, oxalic acid
b. Citric acid, malic acid, oxalic acid
c. Citric acid, tartaric acid, malic acid
d. Malic acid, quinic acid, citric acid
e. Citric and malic acids
f. Tartaric acid
g. Citric and malic acids
h. Citric acid, tartaric acid, malic and oxalic acids
i. Formic acid
j. Butryic acid
k. Acetic acid
l. Lactic acid
m. Caproic acid

7. Classify the following as monocarboxylic, dicarbxylci or tricarboxylic acids.

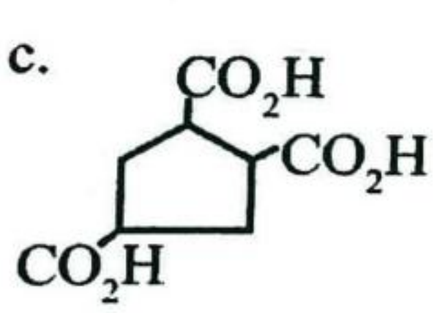
a.




b.




c.




**Solution**

a. Monocarboxylci acid
b. Dicrboxylic acid
c. Tricarboxylic acid

## 6.3 Nomenclature of Carboxylic Acids

➦ It is naming of carboxylic acids. There are two systems of naming carboxylic acids.

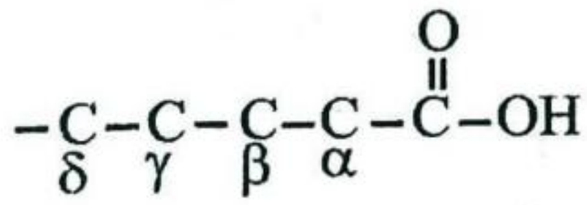
### 1. Common Name

i Common name of straight chain carboxylic acids.

➦ HCOOH - Formic acid
➦ $CH_3COOH$ - Acetic acid
➦ $CH_3CH_2COOH$ - Propionic acid
➦ $CH_3(CH_2)_2COOH$ - Butyric acid
➦ $CH_3(CH_2)_3COOH$ - Valeric acid
➦ $CH_3(CH_2)_4COOH$ - Caproic acid
➦ $CH_3(CH_2)_8COOH$ - Capric acid
➦ $CH_3(CH_2)_{10}COOH$ - Lauric acid
➦ $CH_3(CH_2)_{12}COOH$ - Myristic acid
➦ $CH_3(CH_2)_{14}COOH$ - Palmitic acid
➦ $CH_3(CH_2)_{16}COOH$ - Stearic acid

**ii Common name of branched chain caboxylic acids**

➦ Use the Greek letters α, β, γ, δ, ... on the 2nd, 3rd, 4th, 5th, ... carbon to state the position of the side chains or substituents.
➦ If the same side chain appears twice, three time, four times use prefixes, di -, tri, tetra respectively.
➦ Arrange the side chains in alphabetical order or size order.



**Example**

Write the common name for

a. $CH_3-\underset{\displaystyle CH_3}{\underset{|}{CH}}-COOH$, α – methylpropionic acid

b. $CH_3-\underset{\displaystyle Br}{\underset{|}{CH}}-\underset{\displaystyle Cl}{\underset{|}{CH}}-COOH$, β – bromo - α- chlorobutyric acid

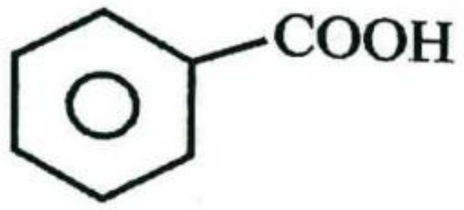
**iii Common name of dicarboxylic acids**

➦ $HO_2CCO_2H$ - Oxalic acid
➦ $HO_2CCH_2CO_2H$ - Malonic acid
➦ $HO_2C(CH_2)_2CO_2H$ - Succinic acid

- $HO_2C(CH_2)_3CO_2H$ - Glutaric acid
- $HO_2C(CH_2)_4CO_2H$ - Adipic acid
- $HO_2C(CH_2)_5CO_2H$ - Pimelic acid

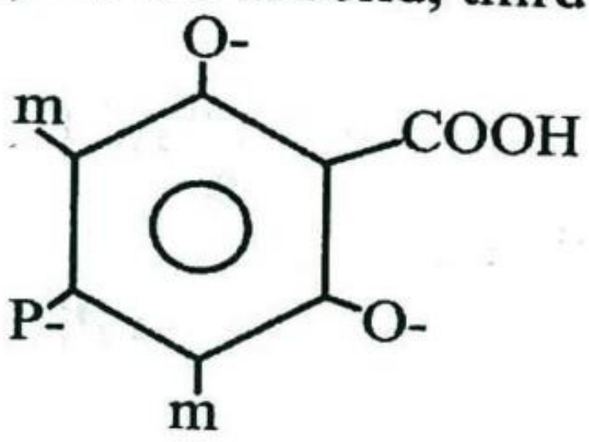
**iv Common name of aromatic carboxylic acids**

- The simplest aromatic carboxylic acid is commonly known as **benzoic acid**.

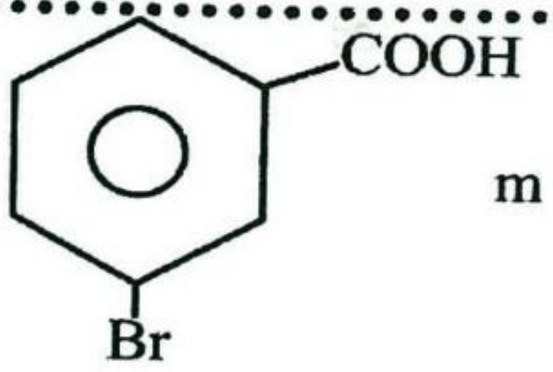



benzoic acid

- If side chains appear, we use the prefix ortho (O), meta (m) and Para (P) on the second, third and fourth carbon atoms respectively.




**Example**




m - bromobenzoic acid

## Illustrative Examples

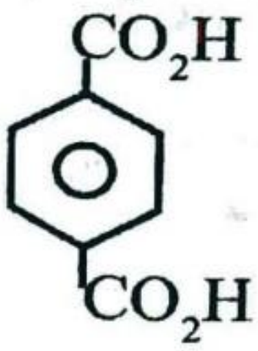
8. Write common name for the following carboxylic acids.

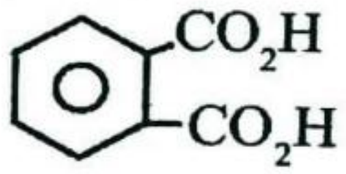
a. $CH_3COOH$

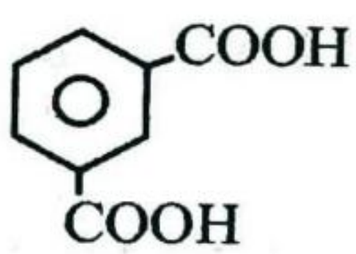
b. $CH_3\underset{\underset{Br}{|}}{C}HCH_2COOH$

c. $CH_2ClCOOH$

d. $CCl_3COOH$

e. 


f. 


g. 


**Solution**

a. Acetic acid

b. β - bromobutyric acid

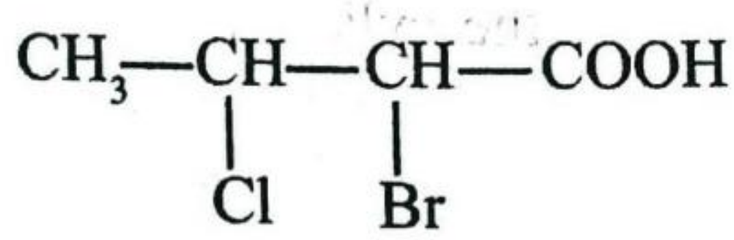
c. Chloro acetic acid

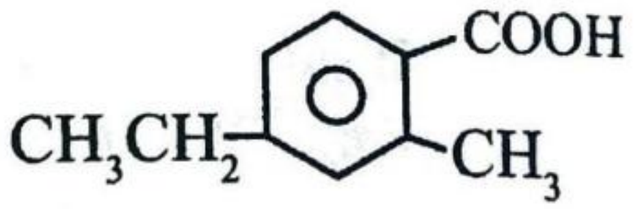
d. Tricholoro acetic acid

e. Terephthalic acid
f. Phthalic acid
g. Isophthalic acid

9. Draw the structural formula for the following organic acids.
   α. α - bromo - β - chlorobutyric acid
   b. Succinic acid
   c. P - ethyl - O - methyl benzoic acid
   d. Valeric acid

**Solution**

a. 

c. 

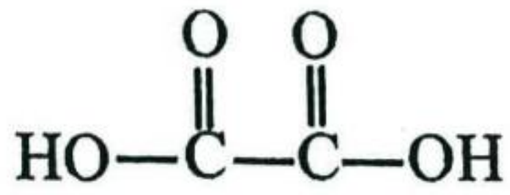
b. $HOOCCH_2CH_2COOH$

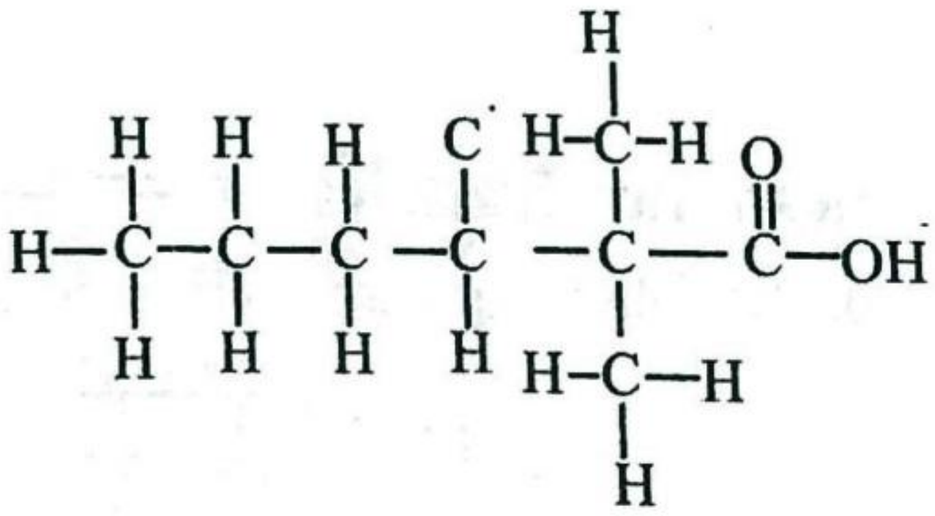
d. $CH_3CH_2CH_2CH_2COOH$

10. Draw the detail structural formula for
    a. Oxalic acid
    b. α, α- dimethyl caproic acid

**Solution**

Detail structural formula show all the bonds.

a. 

b. 

## 2. IUPAC name of carboxylic acids

**Rules**

1. The last letter of hydrocarbons "-e" is replaced by "oic acid".
2. Identify the longest chain and start with the carbonyl carbon $\left(-\overset{O}{\overset{\|}{C}}-\right)$ for open chain carboxylic acids.
3. Arrange the side chains in alphabetical or size order.
4. If the same side chain appears twice, three times, four times use the prefixes di - , tri and tetra respectively.

5. Number and number is separated by comma (,) and number and letter is separated by hyphen (-).
6. For dicarboxylic acids use the simple methode below.

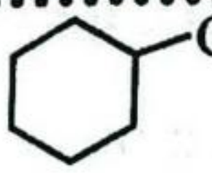
   Name of alkane + dioic acid.

**Example**

HOOCCOOH, Ethanedioic acid

7. For cyclic carboxylic acids, name of the cylic alkanes followed by carboxylcic acid.

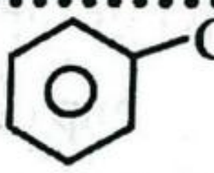
**Example**




,cyclohexane carboxylic acid

8. For aromatic hydrocarbon, benzene followed by the word carboxylic acid.

**Example**



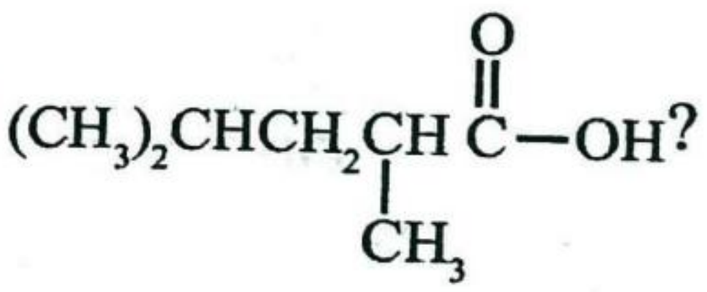

Benzene carboxylic acid

9. For unsaturated carboxylic acids, state the position of double or triple bond.

**Example**

$CH_3 - CH = CH - COOH$, 2 - butenoic acid

## Illustrative Examples

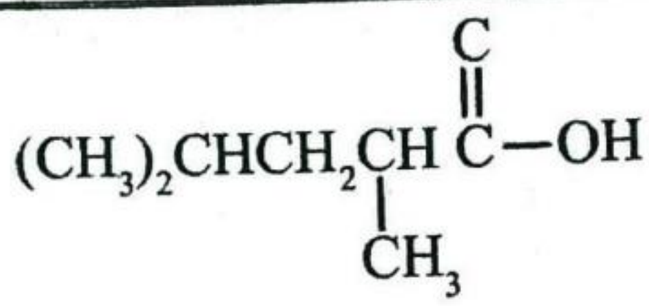
11. What is the IUPAC name for the compound  (EUEE ... 2007 E.C)



A. 2, 4 - dimethyl pentanoic acid
B. 1, 1, 3 - trimethyl butanoic acid
C. 1 - hydroxyl - 2, 4 - dimethylpentanone
D. 2 - carboxyisohexane.

**Solution**

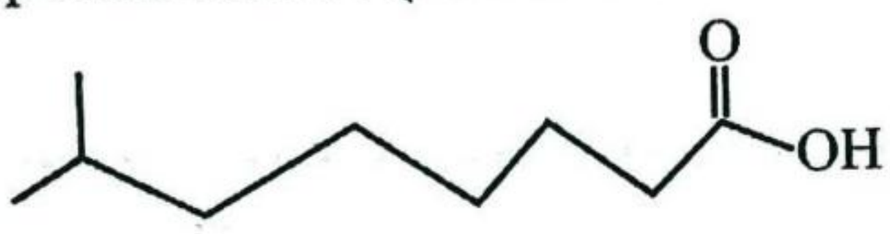
➦ The compound is carboxylic acid and count starting with the carbonyl carbon.



Its IUPAC name is: 2, 4 - dimethylpentanoic acid.

**Answer: A**

12. Which of the following is an acceptable IUPAC name for the organic compound below? (EUEE .. 2007 E.C)



A. Octanoic acid
B. 7 - methyleptanoic acid
C. 8 - methylheptanoic acid
D. 7 - methyl octanoic acid

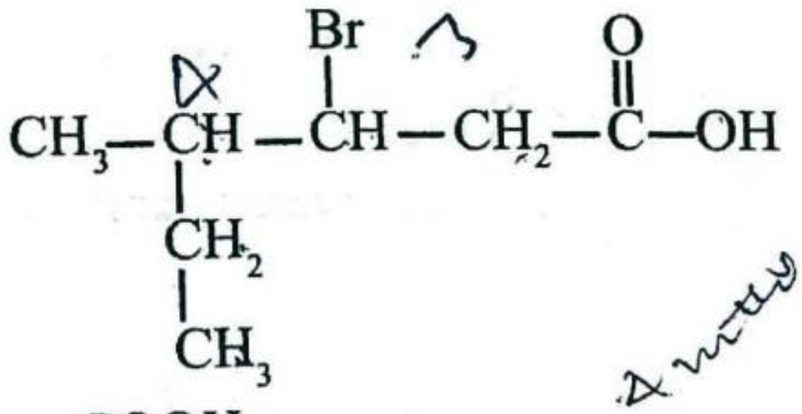
**Solution**

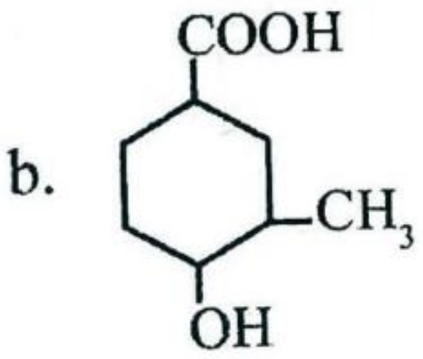
➦ It is line formula of carboxylic acid. Each angle represents a carbon atom and ____ is methyl group ⌒ is ethyl groups.

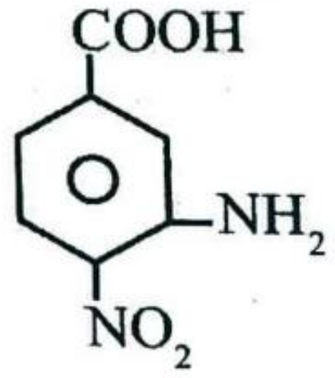
➦ The parent name is octanoic acid and there is methyl group on the seventh carbon atom. Thus, its name is 7 - methyl octanoic acid.

**Answer: D**

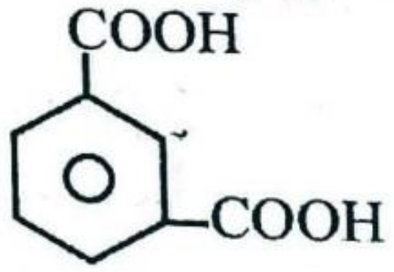
13. Give IUPAC name for the carboxylic acids given below.

a. 

b. 

c. 

d. $CH_3(CH_2)_{14}COOH$

e. 

**Solution**

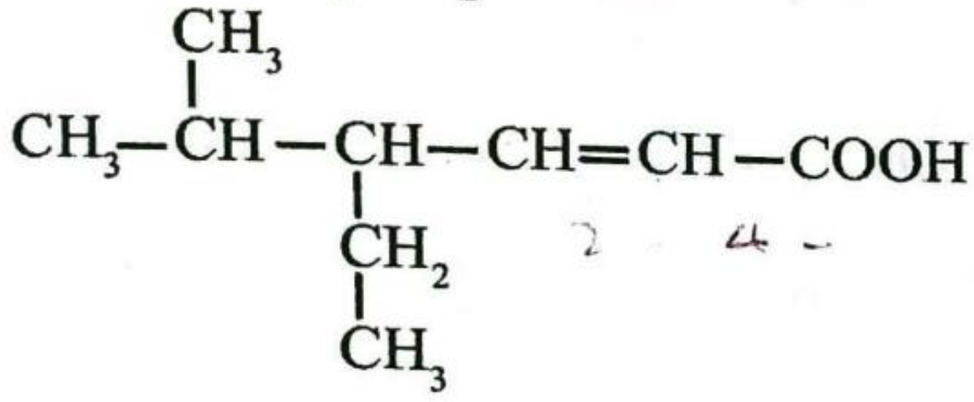
a. 3 - bromo - 4 - methyl hexanoic acid
b. 4 - hydroxyl - 3 - methyl cyclohexane carboxylic acid.
c. 3 - amino - 4 - nitro benzene carboxylic acid
d. Hexadecanoic acid
e. 1, 3 - benzene dicarboxylic acid

14. Write the IUPAC name for the following unsaturated and dicarboxylic acids.

a. $CH_3-CH_2-CH=CH-COOH$

b. $CH_3-C\equiv C-COOH$

c. $HOOC\,CH_2CH_2COOH$

d. 

**Solution**

a. 2 - pentenoic acid

b. 2 - butynoic acid

c. Butanedioic acid

d. 4 - ethyl - 5 - methyl - 2 - hexenoic acid

15. Draw the structures of the following carboxylic acids.

a. Tri fluoro ethanoic acid

b. 1, 2 - benzene dicarboxylic acid

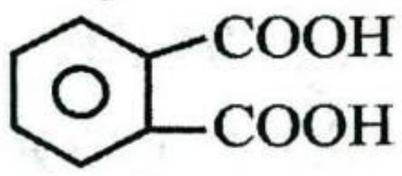
c. 2 - hydroxylpropanoic acid

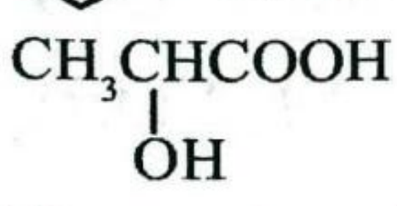
d. 5 - bromopentanoic acid

e. 3 - bromo - 2 - chloro - 2 - methyl heptanoic acid

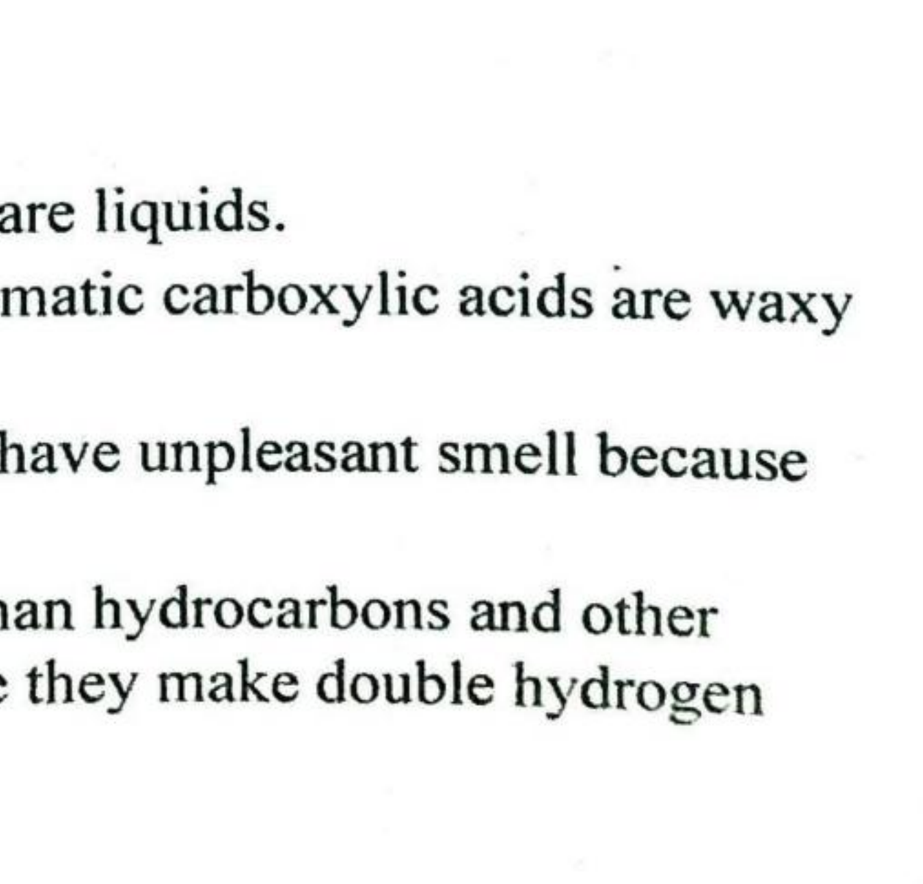
**Solution**

a. $CF_3COOH$

b. 

c. 

d. $BrCH_2CH_2CH_2CH_2COOH$

e. 

## Physical Properties of Carboxylic acids

- Lower member carboxylic acids are liquids.
- Higher member aliphatic and aromatic carboxylic acids are waxy solids.
- Lower member carboxylic acids have unpleasant smell because they are volatile.
- They have higher boiling point than hydrocarbons and other derivative hydro carbons because they make double hydrogen bonds with themselves.

**Example**

Boiling point of $CH_3COOH > CH_3CH_2OH > ... CH_3CH_3$

➦ As branching increases boiling point decreases with similar carbon atoms.

**Example**

$CH_3CH_2CH_2COOH > CH_3-\underset{\displaystyle CH_3}{\overset{\displaystyle |}{CH}}-COOH$

➦ Lower member carboxylic acids are completely miscible with water.

➦ As carbon number increases, solubility of carboxylic acids in water decreases.

➦ Solubility of dicarboxylic acids is higher than solubility of monocarboxylic acids.

**Example**

$HOOCCOOH > CH_3COOH$

➦ As carbon number increases, solubility of carboxylic acids in non - polar solvents increases.

## Illustrative Examples

16. When the following substances are arranged in the order of increasing melting point, the correct order is

A. $CH_3CH_2CH_3$, $CH_3COCH_3$, $CH_3CH_2CH_2OH$, $CH_3CH_2COOH$

B. $CH_3CH_2COOH$, $CH_3CH_2CH_2OH$, $CH_3COCH_3$, $CH_3CH_2CH_3$

C. $CH_3CH_2CH_2OH$, $CH_3CH_2COOH$, $CH_3CH_2CH_3$, $CH_3COCH_3$

D. $CH_3CH_2CH_3$, $CH_3COCH_3$, $CH_3CH_2COOH$, $CH_3CH_2CH_2OH$

**Solution**

Melting point: carboxylic acid > alcohol > Aldehyde/ketone > Alkane

**Answer: A**

17. Which of the following is the most soluble in water?

A. HCOOH

B. $CH_3COOH$

C. $CH_3CH_2COOH$

D. $CH_3CH_2CH_2COOH$

**Solution**

As carbon number increases, solubility in water decrease.

**Answer: A**

18. Which one of the following organic molecules has the highest water solubility? (EUEE ... 2004 E.C.)

A. $HOOCCH_2COOH$
B. $CH_3CH_2CH_2COOH$
C. $HOOCCH_2CH_2COOH$
D. $CH_3CH_2CH_2CH_2COOH$

**Solution**

Dicarboxylic acid with lower mass is the most soluble.

**Answer: A**

19. Which of the following is the most polar molecule?

A. $CH_4$ B. $CH_3OH$ C. HCOOH D. HCHO

**Solution**

Carboxylic acids are more polar than alcohols and aldehydes.

**Answer: C**

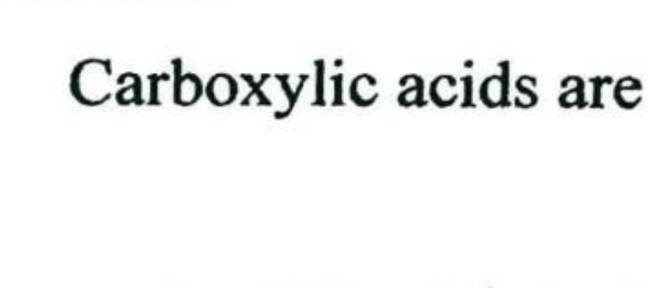
20. Arrange the following in decreasing order of volatility.

i HCOOH
ii $CH_3COOH$
iii $CH_3CH_2COOH$
iv $CH_3(CH_2)_2COOH$

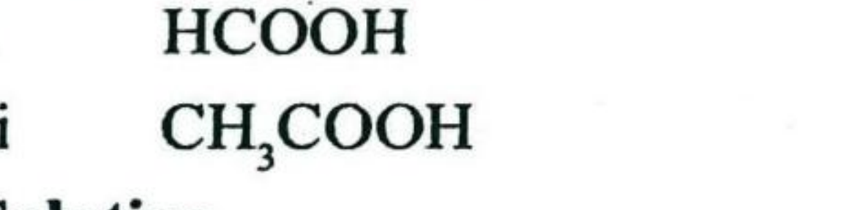
**Solution**

I > II > III > IV

21. Arrange the following in increasing order of boiling point.

a. 

b. $CH_3CH_2CH_2CH_2CH_2COOH$

c. 

**Solution**

a < c < b

22. Arrange the following aliphatic carboxylic acid in decreasing order of solubility in water?

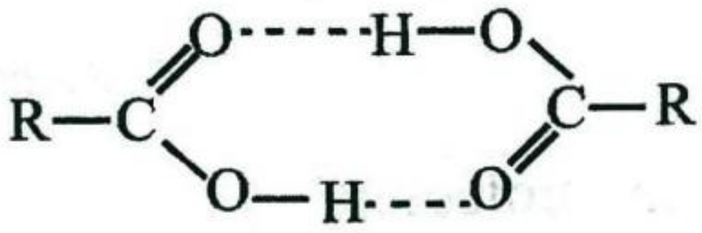
a. Methanoic acid
b. Octanoic acid
c. Decanoic acid
d. Hexanoic acid

**Solution**

a > d > b > c

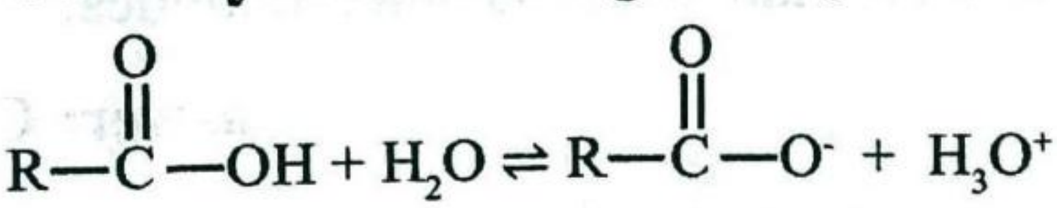
23. Why the boiling point of carboxylic acids is higher than alcohols of similar size?

**Explanation:** Because carboxylic acids form double hydrogen bonds but alcohols form single hydrogen bond with their molecules.



## Chemical Properties of carboxylic acids

**Carboxylic acids** are generally weak acids.



**Note**

The strength of carboxylic acids can be increased by replacing hydrogen with halogens.

➦ The more electronegative element replaced the stronger is the acid.

**Example**

Strength of

i $CH_2FCOOH > CH_2ClCOOH > CH_2BrCOOH > CH_3COOH$

ii $CF_3COOH > CCl_3COOH > CBr_3COOH > CH_3COOH$

The common reactions of carboxylic acids are:

**1. Reaction as acid**

i Carboxylic acids react with active metals.

Carboxylic acid + active metal → salt + hydrogen

$2RCOOH + Zn \rightarrow (RCOO)_2Zn + H_2$

Zinc carboxylate

Example

$2CH_3COOH + Mg \rightarrow (CH_3COO)_2Mg + H_2$

Magnesium acetate

ii Carboxylic acids react with bases.

**Carboxylic acid + Base → Salt + Water**

$RCOOH + NaOH \rightarrow RCOONa + H_2O$

Sodium carboxylate

Example

$CH_3COOH + KOH \rightarrow CH_3COOK + H_2O$

Potassium acetate

iii Carboxylic acids react with basic oxides.

**Carboxylic acid + basic oxide → Salt + water**

$2RCOOH + Na_2O \rightarrow 2RCOONa + H_2O$

Example

$2CH_3COOH + K_2O \rightarrow 2CH_3COOK + H_2O$

iv Carboxylic acids react with carbonates and hydrogen carbonates.

**Carboxylic acid + carbonate/hydrogen carbonate → salt + carbondioxide + water**

$2RCOOH + CaCO_3 \rightarrow (RCOO)_2Ca + CO_2 + H_2O$

Example

$2CH_3COOH + CaCO_3 \rightarrow (CH_3COO)_2Ca + CO_2 + H_2O$

## 2. Reduction

➦ Carboxylic acids undergo reduction reaction by lithium aluminum hydride ($LiAlH_4$) to give primary alcohols.

$RCOOH + LiAlH_4 \rightarrow RCH_2OH$

Carboxylic acid Primary alcohol

Example

$CH_3COOH + LiAlH_4 \rightarrow CH_3CH_2OH$

## 3. Esterification

➦ The reaction between carboxylic acids and alcohols to yield, esters is called **esterification reaction**.

$$\underset{\text{Carboxylic acid}}{RCOOH} + \underset{\text{alcohol}}{R'OH} \overset{H^+}{\rightleftharpoons} \underset{\text{Ester}}{RCOOR'} + H_2O$$

**Example**

$$CH_3COOH + CH_3OH \overset{H^+}{\rightleftharpoons} CH_3COOCH_3 + H_2O$$

## Illustrative Examples

24. Which of the following gives the correct order of decreasing acidity of carboxylic acids? (EUEE ... 2008)
    A. $Cl_3CCOOH, Cl_2CHCOOH, FCH_2COOH, CH_3COOH$
    B. $FCH_2COOH, CH_3COOH, Cl_2CHCOOH, Cl_2CCOOH$
    C. $CH_3COOH, FCH_2COOH, Cl_2CHCOOH, Cl_2CCOOH$
    D. $Cl_2CHCOOH, CH_3COOH, FCH_2COOH, Cl_3CCOOH$

    **Solution**

    ➦ The more substituted acid is the strongest acid
    $\Rightarrow Cl_3CCOOH > Cl_2CHCOOH > FCH_2COOH > CH_3COOH$

    **Answer: A**

25. Which of the description of the reaction between $C_3H_7OH$ and $CH_3COOH$?
    A. Addition
    B. Hydrolysis
    C. Esterification
    D. Elimination

    **Solution**

    The reaction between alcohol ($C_3H_7OH$) and carboxyic acid ($CH_3COOH$) is esterification.

    **Answer: C**

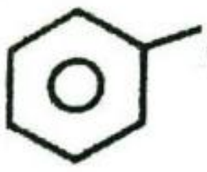
26. Which of these organic compounds bubbles when calcium is added to it? (EUEE)
    A. $CH_3CH_3$
    B. $CH_3COOH$
    C. $CH_3COCH_3$
    D. $CH_3CHO$

    **Solution**

    Calcium is active metal. It reacts with acid to form bubbles of gas.

    **Answer: B**

27. Complete and balance the following
    a. $CH_3CH_2COOH + LiAlH_4 \rightarrow$
    b. $CH_3CH_2COOH + Mg \rightarrow$
    c. $CH_3CH_2COOH + CaCO_3 \rightarrow$
    d. $CH_3COOH + NH_3 \rightarrow$

e.  $+ \; CH_3CH_2OH \rightleftharpoons^{H^+}$

f. $CH_3(CH_2)_2COOH + Ca(OH)_2 \rightarrow$

g. $CH_3(CH_2)_3COOH + Ba(HCO_8)_2 \rightarrow$

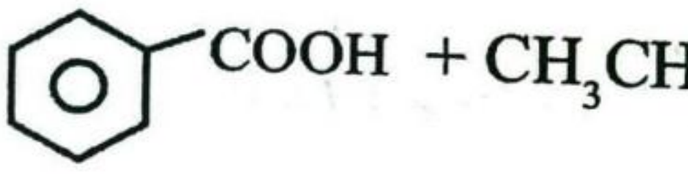
**Solution**

a. $CH_3CH_2COOH + LiAlH_4 \rightarrow CH_3CH_2CH_2OH$

b. $2CH_3CH_2COOH + Mg \rightarrow (CH_3CH_2COO)_2Mg + H_2$

c. $2CH_3CH_2COOH + CaCO_3 \rightarrow (CH_3CH_2COO)_2Ca + CO_2 + H_2O$

d. $CH_3COOH + NH_3 \rightarrow CH_3COONH_4$

e. 

f. $2CH_3(CH_2)_2COOH + Ca(OH)_2 \rightarrow (CH_3(CH_2)_2COO)_2Ca + 2H_2O$

g. $2CH_3(CH)_3COOH + Ba(HCO_3)_2 \rightarrow (CH_3(CH_2)_3COO)_2Ba + 2CO_2 + 2H_2O$

28. What are the two important reducing agents that reduce carboxylic acids into primary alcohols?

**Solution**

- Lithium aluminum hydride ($LiAlH_4$).
- Sodium boron hydride ($NaBH_4$).

## Preparation of Carboxylic acids

- There are different methods used to prepare carboxylic acids.

### 1. Oxidation of Primary alcohols

$$\underset{\text{Primary alcohol}}{RCH_2OH} \xrightarrow{KMnO_4} \underset{\text{Carboxylic acid}}{RCOOH}$$

**Example**

$$\underset{\text{Primary alcohol}}{RCH_2OH} \xrightarrow{KMnO_4} \underset{\text{Carboxylic acid}}{RCOOH}$$

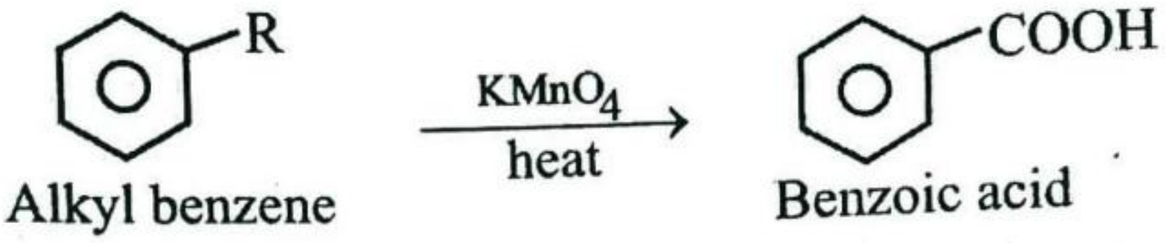
### 2. Oxidation of aldehydes

$$\underset{\text{Aldehyde}}{RCHO} \xrightarrow[\text{or } Na_2Cr_2O_7]{KMnO_4} \underset{\text{Carboxylic acid}}{RCOOH}$$

**Example**

$$CH_3CHO \xrightarrow{KMnO_4} CH_3COOH$$

### 3. Oxidation of alkyl benzene

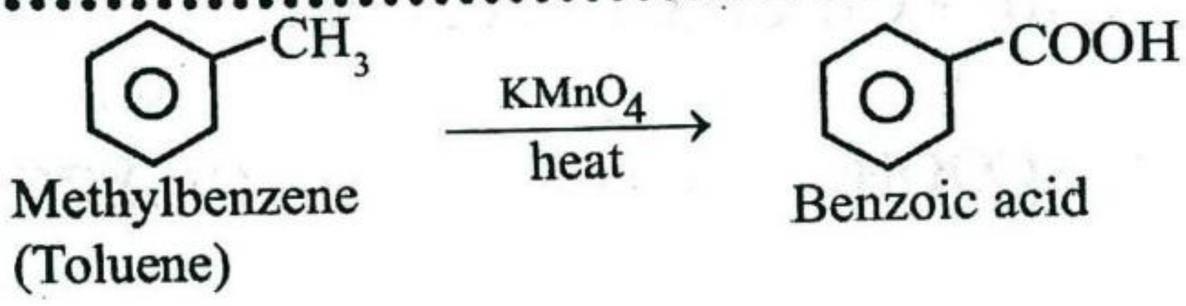



Alkyl benzene — Benzoic acid

R - alkyl group such as $-CH_3$, $-CH_2CH_3$, $-CH_2CH_2CH_3$ etc

> All alkyl benzenes yield benzoic acid upon oxidation with $KMnO_4$.

**Example**




Methylbenzene (Toluene) — Benzoic acid

### 4. Addition of Grignard reagents to carbondioxide

$$\underset{\text{Grignard reagent}}{RMgX} + CO_2 \longrightarrow RCO_2MgX \xrightarrow{H^+} RCO_2H$$

**Example**

$$CH_3CH_2CH_2Br \xrightarrow{Mg} CH_3CH_2CH_2BrMg \xrightarrow[2)\ H_3O^+]{1)\ CO_2}$$

$$CH_3CH_2CH_2CO_2H$$

### 5. Hydrolysis of cyano compounds

$$RC \equiv N \xrightarrow{H_3O^+} RCO_2H + NH_4^+$$

**Example**

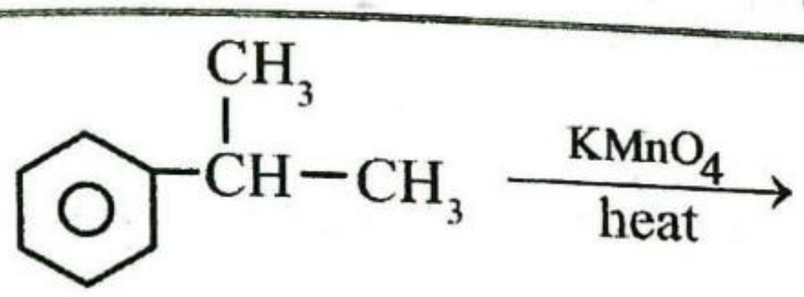
$$CH_3C \equiv N \xrightarrow{H_3O^+} CH_3COOH + NH_4^+$$

## Illustrative Examples

29. Complete and balance the following.

a. $CH_3CH_2CH_2OH \xrightarrow{KMnO_4}$

b. $CH_3CH_2CH_2CHO \xrightarrow{KMnO_4}$

c. 

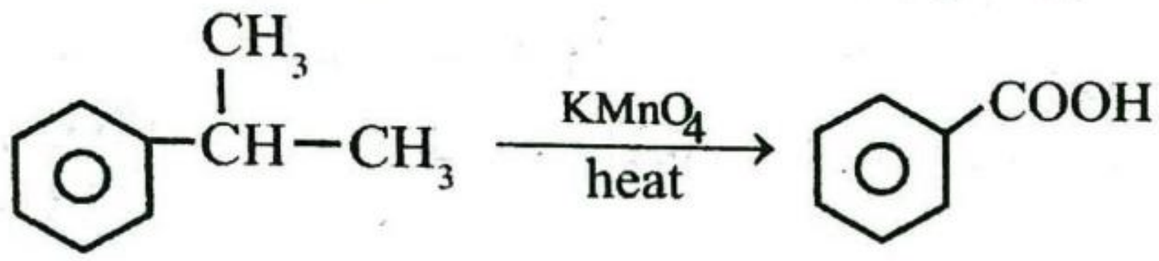
d. $CH_3CH_2Br \xrightarrow{Mg} CH_3CH_2BrMg \xrightarrow[2)\ H_3O^+]{1)\ CO_2}$

e. $CH_3CH_2C \equiv N \xrightarrow{H_3O^+}$

**Solution**

a. $CH_3CH_2CH_2OH \xrightarrow{KMnO_4} CH_3CH_2COOH$

b. $CH_3CH_2CH_2COH \xrightarrow{KMnO_4} CH_3CH_2CH_2COOH$

c. 

d. $CH_3CH_2Br \xrightarrow{Mg} CH_3CH_2BrMg \xrightarrow[2)\ H_3O^+]{1)\ CO_2} CH_3CH_2COOH$

e. $CH_3CH_2C \equiv N \xrightarrow{H_3O^+} CH_3CH_2COOH + NH_4^+$

## Fatty Acids

- They are long chain carboxylic acids.
- Fatty acids containing carbon - carbon double bond are called **unsaturated fatty acids.**

### Example

Oleic, $CH_3(CH_2)_7CH = CH(CH_2)_7COOH$

- Linolenic,
  $CH_3CH_2CH = CHCH_2CH = CHCH_2CH = CH(CH_2)_7COOH.$
- Fatty acids containing carbon - carbon single bond only are known as **saturated fatty acids.**

### Example

Lauric acid, $CH_3(CH_2)_{10}COOH$

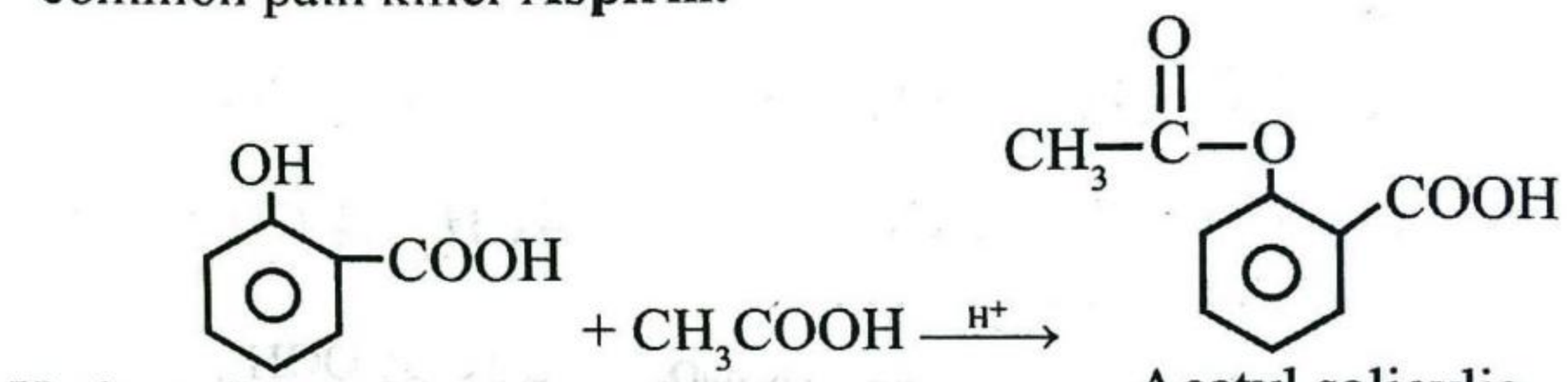
Myristic acid, $CH_3(CH_2)_{12}COOH$

Palmitic acid, $CH_3(CH_2)_{14}COOH$

Stearic acid, $CH_3(CH_2)_{16}COOH$

## Uses of carboxylic acids

- Fatty acids are used to make soaps, detergents and shampoos.
- Carboxylic acids are used to make greases, Crayons, plastics and synthetic odors and flavors.
- The most important carboxylic acid is acetic acid. It is used to make acetates, acetic anhydrides, vinyl acetate polymer and the common pain killer **Aspirin.**




Hydroxylbenzoic acid (Salicylic acid) + Acetic acid → Acetyl salicylic acid (Aspirin)

## Illustrative Examples

30. What is the chemical name for Aspirin (EUEE .. 2006 E.C.)

A. Acetyl salicyclic acid
B. Salicyclic acid
C. Methyl Salicylate
D. Sodium Salicylate

**Solution**

The chemical name of aspirin is acetyl salicylic acid.

**Answer: A**

31. Which of the following acid is used as chemical defense for ants?

A. HCOOH
B. $CH_3(CH_2)_2COOH$
C. $CH_3COOH$
D. $CH_3CH_2COOH$

**Solution**

Formic acid is the acid found in ants and bees.

**Answer: A**

32. State the name of the acids found in

a. Vinegar
b. Cheese
c. Rancid butter
d. Sour milk
e. Palm oil
f. Beef fat

**Solution**

a. Acetic acid
b. Propionic acid
c. Butanoic acid
d. Lactic acid
e. Palmitic acid
f. Stearic acid

33. Write the IUPAC name of the following fatty acids.

a. $CH_3(CH_2)_{10}COOH$
b. $CH_3(CH_2)_{12}COOH$
c. $CH_3(CH_2)_{14}COOH$
d. $CH_3(CH_2)_{16}COOH$

**Solution**

a. Dodecanoic acid
b. Tetradecanoic acid
c. Hexadecanoic acid
d. Octadecanoic acid

## 6.4 Esters

- **Esters** are pleasant smelling organic compounds.
- The general formula of esters is RCOOR'.
- The functional group of esters is —COO—.
- Esters are responsible for the aroma of many fruits.

**Example**

| Ester | Fruits |
|---|---|
| Methyl butanoate | Pineapple |
| Pentyl ethanoate | Banana |
| Octyl butanoate | Orange |
| Isoamyl ovalerate | Apples |
| Methyl Methanate | Rum essence |
| Methyl Anthranilate | Grapes |

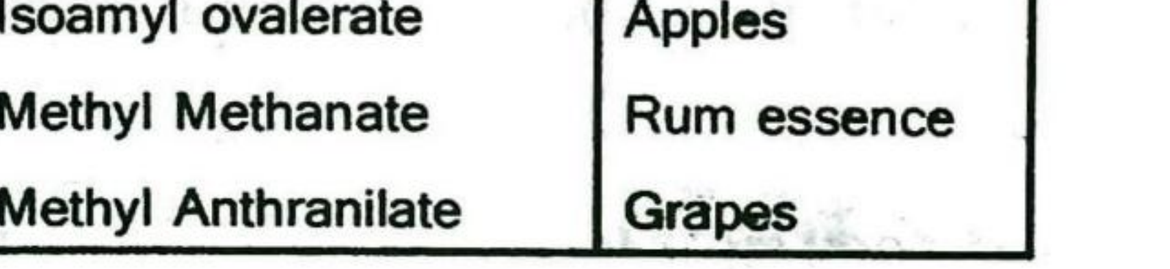
### Nomenclature of Esters

**1. Common name**

- The common name of the acid '-ic acid' is replaced with '-ate'.
- The alcohol portion of the ester is converted into alkyl or aryl group.

**Example**

Methanol → Methyl  Ethanol → Ethyl




**Example**

a. $HCOOCH_3$ - Methyl formate
b. $CH_3COOCH_2CH_3$ - Ethyl acetate

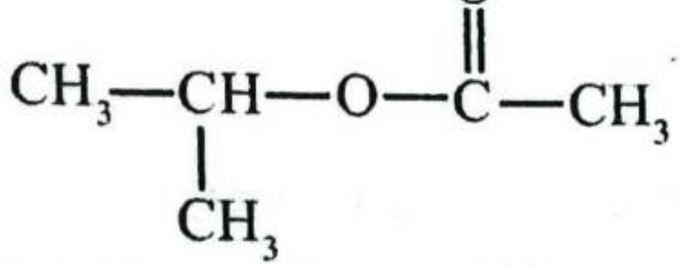
## 2. IUPAC name of Esters

➥ The IUPAC name of carboxylic acids "ic acid" is replaced with '-ate'.

**Example**

a. $HCOOCH_3$ - Methyl methanoate

b. $CH_3COOCH_2CH_3$ - Ethyl ethanoate

c.  - Isopropyl ethanoate

## Illustrative Examples

34. Which compound is an ester? (EUEE ... 2006 E.C)

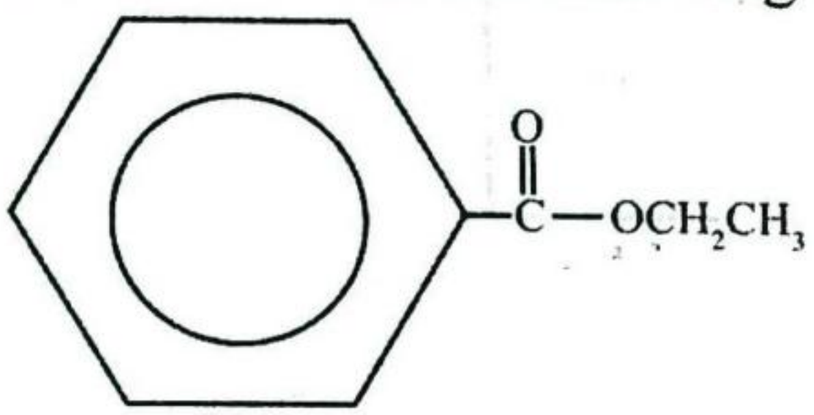
A. $CH_3COOH$
B. $CH_3OC_2H_5$
C. $C_2H_5CHO$
D. $HCOOCH_3$

**Solution**

Based on the general - formula of ester RCOOR', $HCOOCH_3$ is an ester.

**Answer: D**

35. What is the name of the following compound? (EUEE ... 2004 E.C)



A. Benzoate ester
B. Ethyl benzoate
C. Phenyl butyrate
D. Ethyl benzyl ketone

**Solution**

The acid portion is benzoic acid and the alcohol portion is ethanol. Thus, the name of the ester becomes ethyl benzoate.

**Answer: B**

36. Which of the following is responsible for the pleasant odour of many fruits?

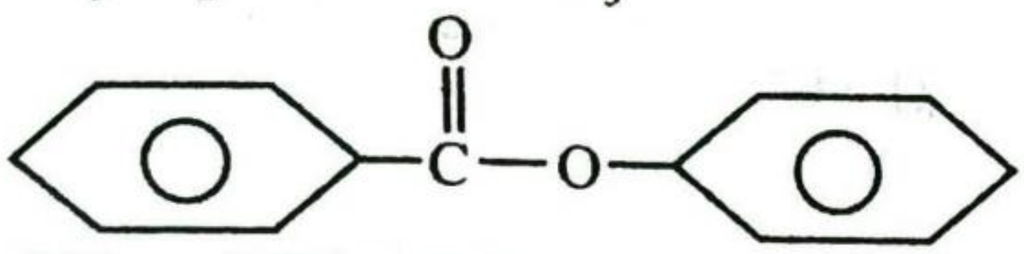
A. Ester
B. Carboyxlic acid
C. Ketone
D. Aldehyde

**Answer: A**

37. Write the common name of the following esters.

a. $CH_3CO_2CH_2CH_3$

b. $CH_3CH_2COOCH(CH_3)-CH_3$

c. 

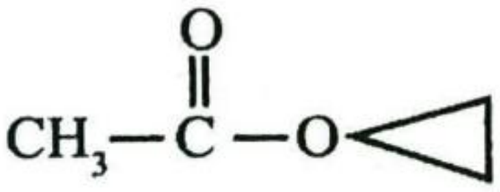
d. $CH_2=CHOOCCH_3$

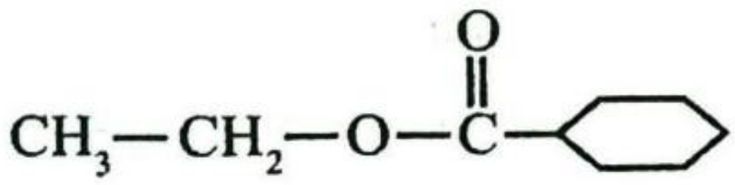
**Solution**

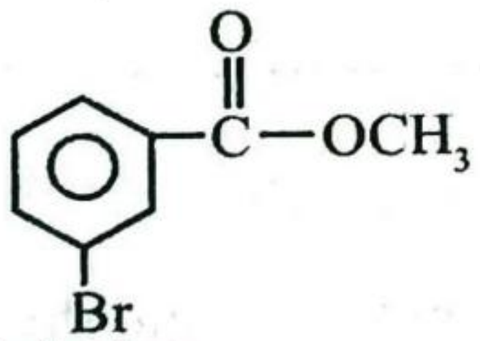
a. Ethyl acetate
b. Isopropyl propionate
c. Phenyl benzoate
d. Vinyl acetate

38. Give the IUPAC name for the following esters.

a. $CH_3-CH_2-C(=O)-O-CH_2-CH(CH_3)-CH_3$

.b. 

c. 

d. 


**Solution**

a. Isobutyl propanoate
b. Cyclopropyl ethanoate
c. Ethyl cyclohexane carboxylate
d. Methyl - 3 - bromo benzene carboyxlate

39. Draw the structure for

a. n - pentyl ethanoate
b. n - octylbutanoate
c. Isopropyl methanoate
d. t - butylpropanoate
e. Benzyl benzoate

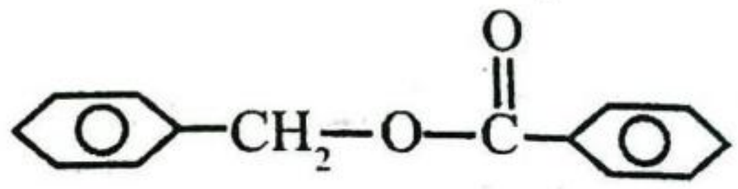
**Solution**

a. $CH_3CH_2CH_2CH_2CH_2OOCCH_3$

b. $CH_3(CH_2)_7OOCCH_2CH_2CH_3$

c. $HCOOCH(CH_3)-CH_3$

d. $CH_3-C(CH_3)_2-O-C(=O)-CH_2-CH_3$

e. 

## Physical Properties of Esters

- Esters are pleasant smelling organic compounds.
- They have lower boiling point than carboxylic acids.
- Low molecular mass esters are fairly soluble in water.

## Chemical Properties of Esters

1. **Hydrolysis** is the reaction of esters with water to produce carboxylic acid and alcohol

$$\underset{\text{Ester}}{R-C(=O)-OR'} + H_2O \overset{H^+}{\rightleftharpoons} \underset{\text{Carboxylic acid}}{RCOOH} + \underset{\text{alcohol}}{R'OH}$$

**Example**

$$CH_3-C(=O)-OCH_3 + H_2O \overset{H^+}{\rightleftharpoons} CH_3-C(=O)-OH + CH_3OH$$

- The reaction between ester and base is known as **saponfication**.

$$R-C(=O)-OR' + NaOH \longrightarrow \underset{\text{Sodium carboxylate}}{R-C(=O)-ONa} + R'OH$$

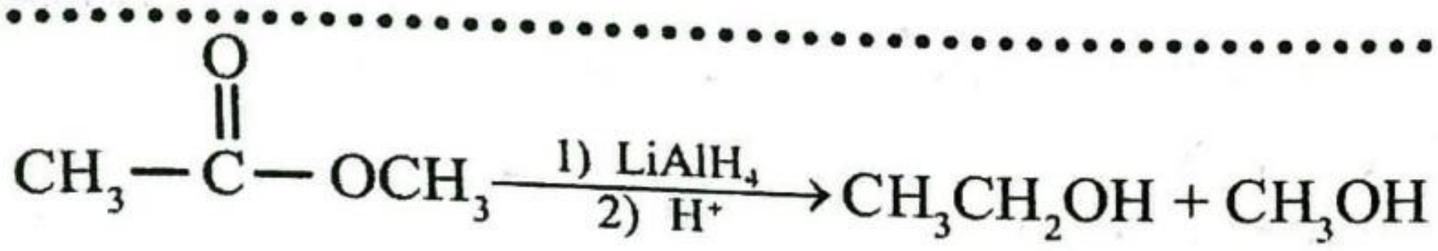
**Example**

$$\underset{\text{Methyl streate}}{CH_3(CH_2)_{16}C(=O)-OCH_3} + NaOH \longrightarrow \underset{\text{Sodium streate or soap}}{CH_3(CH_2)_{16}C(=O)ONa} + CH_3OH$$

2. **Reduction** - of ester gives two moles of alcohols.

$$R-C(=O)-OR' \xrightarrow[2)\ H^+]{1)\ LiAlH_4} RCH_2OH + R'OH$$

**Example**



## Illustrative Examples

40. Which is the name of a base - promoted ester hydrolysis reaction? (EUEE ... 2004 E.C)

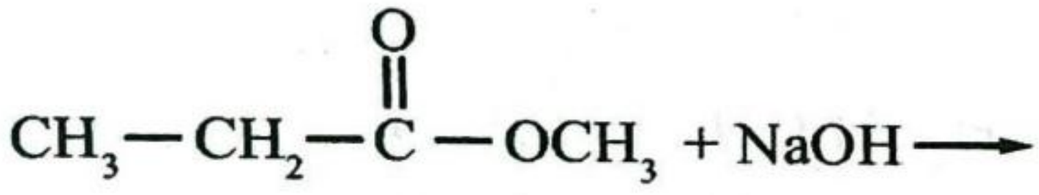
A. Acylation
B. Esterification
C. Condensation
D. Saponification

**Solution**

Base hydrolysis of ester is known as saponification.

**Answer: D**

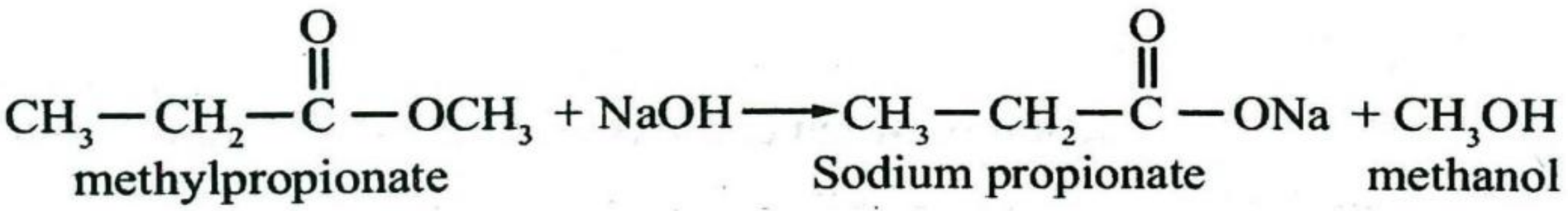
41. Consider the reaction:



What are the products of this reaction? (EUEE)

A. Sodium acetate and ethanol
B. Sodium acetate and methanol
C. Sodium Propionate and methanol
D. Methyl Propionic acid and methanol

**Solution**




**Answer: C**

42. Arrange the following esters in increasing order of boiling point.

a. $HCOOCH_3$
b. $HCOOCH_2CH_3$
c. $CH_3COOCH_2CH_3$
d. $CH_3COOCH_2CH_2CH_3$

**Solution**

As carbon number increases, boiling point also increases.

$a < b < c < d$

43. Arrange the following in decreasing order of solubility in water.

a. $CH_3COOCH_3$
b. $HCOOCH_3$
c. $CH_3COOH$
d. $CH_3CH_2OH$
e. $CH_3CH_2COOCH_3$

**Solution**

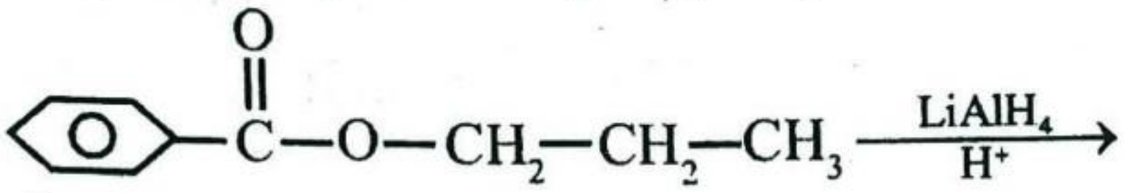
Solubility of carboxylic acid > alcohol > ester

$\Rightarrow c > d > b > a > e$

44. Complete and balance the following.

a. $CH_3-CH_2-\overset{\overset{O}{\|}}{C}-OCH_2CH_3 + H_2O \overset{H^+}{\rightleftharpoons}$

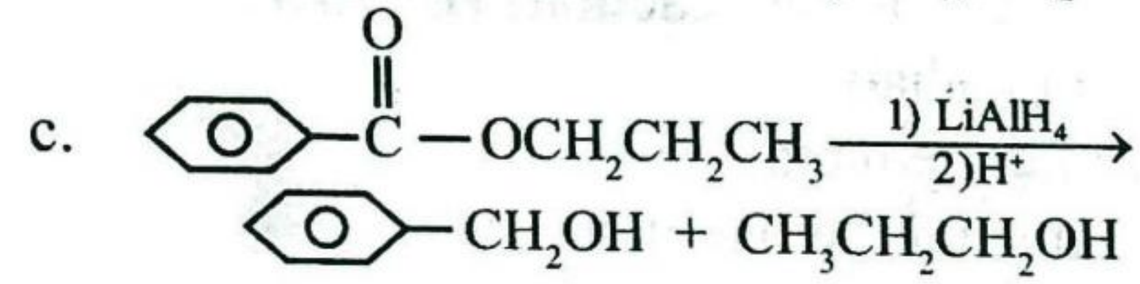
b. $CH_3CH_2CH_2OOCCH_2CH_2CH_3 + KOH \rightarrow$

c.  $\xrightarrow[H^+]{LiAlH_4}$

**Solution**

a. $CH_3-CH_2-\overset{\overset{O}{\|}}{C}-OCH_2CH_3 + H_2O \overset{H^+}{\rightleftharpoons} CH_3CH_2COOH + CH_3CH_2OH$

b. $CH_3CH_2CH_2OOCCH_2CH_2CH_3 + KOH \rightarrow CH_3CH_2CH_2OH + CH_3CH_2CH_2COOK$

c. 

## Preparation of Esters

### 1. Esterification

➦ It is the reaction between carboxylic acids and alcohols to form esters.

$$R-\overset{\overset{O}{\|}}{C}-OH + R'-OH \overset{H^+}{\rightleftharpoons} R-\overset{\overset{O}{\|}}{C}-OR' + H_2O$$

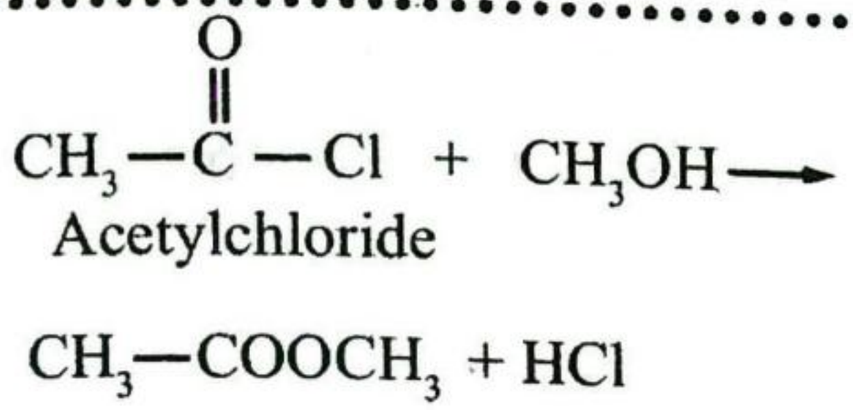
**Example**

$$\underset{\text{Ethanol}}{CH_3CH_2OH} + \underset{\text{Acetic acid}}{CH_3COOH} \overset{H^+}{\rightleftharpoons} \underset{\text{Ethylacetate}}{CH_3COOCH_2CH_3} + H_2O$$

### 2. The reaction between acylchloride and alcohol

$$\underset{\text{Acylchloride}}{R-\overset{\overset{O}{\|}}{C}-Cl} + \underset{\text{Alcohol}}{R'OH} \longrightarrow \underset{\text{Esters}}{R-\overset{\overset{O}{\|}}{C}-OR'} + HCl$$

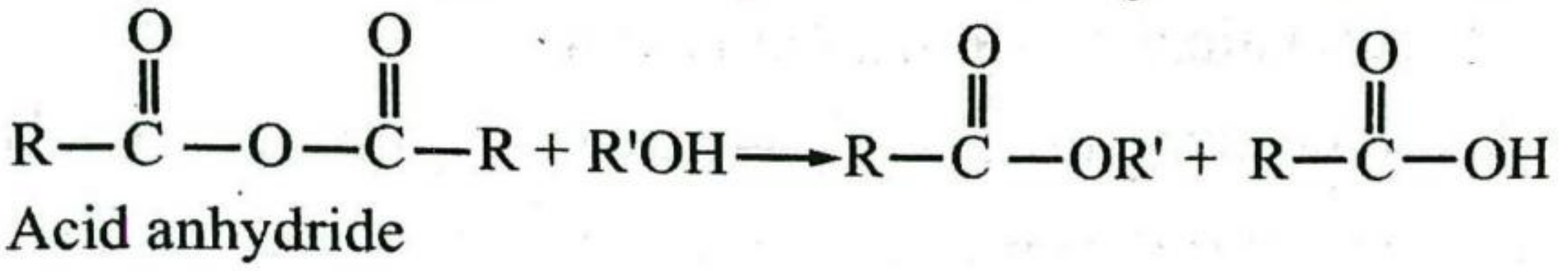
Example



Acetylchloride

$CH_3—COOCH_3 + HCl$

### 3. The reaction between carboxylic acid an hydride and alcohol



Acid anhydride

## Illustrative Examples

46. Which of these compounds is the ester formed from the reaction acetic acid and 1 - propanol? (EUEE ... 2006 E.C)

A. $CH_3COOHCH_2CH_2CH_3$

B. $CH_3CH_2\overset{\overset{O}{\|}}{C}—OH$

C. $CH_3CH_2CH_2OCH_2COOH$

D. $CH_3 — COOCH_2CH_2CH_3$

**Solution**

$$CH_3COOH + CH_3CH_2CH_2OH \rightarrow CH_3CH_2CH_2OOCCH_3 + H_2O$$

Acetic acid 1-propanol Propyl acetate

**Answer: D**

47. Which of the following compounds would react with the product of its oxidation product to form a sweet smelling liquid?

A. Butanol

B. Butanone

C. 1 - butanol

D. 2 - butanol

**Solution**

➦ Oxidation of primary alcohols yield carboxylic acids. The reaction between carboxylic acid and alcohol yield esters. But, oxidation of secondary alcohols yields ketone.

**Answer: C**

48. Write the name of the alcohol and acid used to prepare the following esters.

a. Benzyl benzoate
b. Isobutyl pentanoate
c. Cyclohexyal cyclohexane carboxylate

**Solution**

a. Benzyl alcohol and benzoic acid
b. Isobutyl alcohol and pentanoic acid
c. Cyclohexanol and cyclohexane carboxylic acid

49. Complete and balance the following.

a. $HCOOH + CH_3OH \overset{H^+}{\rightleftharpoons}$

b. $CH_3-CH_2-\overset{\overset{O}{\|}}{C}-Cl + CH_3CH_2OH \longrightarrow$

c. $CH_3-CH_2-\overset{\overset{O}{\|}}{C}-O-\overset{\overset{O}{\|}}{C}-CH_3 + CH_3OH \longrightarrow$

**Solution**

a. $HCOOH + CH_3OH \overset{H^+}{\rightleftharpoons} HCOOCH_3 + H_2O$

b. $CH_3-CH_2-\overset{\overset{O}{\|}}{C}-Cl + CH_3CH_2OH \longrightarrow CH_3-CH_2-\overset{\overset{O}{\|}}{C}-OCH_2CH_2 + HCl$

c. $CH_3-CH_2-\overset{\overset{O}{\|}}{C}-O-\overset{\overset{O}{\|}}{C}-CH_3 + CH_3OH \longrightarrow CH_3-CH_2-\overset{\overset{O}{\|}}{C}-OCH_3 + CH_3CH_2OH$

## 6.5 Aldehydes and Ketones

- Aldehyde has a general formula: $R-\overset{\overset{O}{\|}}{C}-H$
- Its functional groups: $-\overset{\overset{O}{\|}}{C}-H$
- The general formula of ketone is $R-\overset{\overset{O}{\|}}{C}-R'$

➦ The functional group of ketone is $-\overset{\overset{O}{\|}}{C}-$

## Nomenclature of aldehyde and ketones

➦ The name of aldehyde ends with '-al'.

➦ The name of ketone ends with '-one'.

## Nonenclature of aldehyde

HCHO - Methanal (formal dehyde)
$CH_3CHO$ - Ethanal (Acetaldehyde)
$CH_3CH_2CHO$ - propanal
$CH_3CH_2CH_2CHO$ - butanal

## Nomenclature of ketones

$CH_3-\overset{\overset{O}{\|}}{C}-CH_3$ - Propanone (acetone)

$CH_3-\overset{\overset{O}{\|}}{C}-CH_2-CH_3$ - butanone

$CH_3-CH_2-\overset{\overset{O}{\|}}{C}-CH_2CH_3$ 3 - pentanone

## Chemical Properties of aldehydes and Ketones

### 1. Oxidation

➦ Aldehydes give carboxylic acids upon oxidation.

$$R-\overset{\overset{O}{\|}}{C}-H \xrightarrow{KMnO_4} R-\overset{\overset{O}{\|}}{C}-OH$$

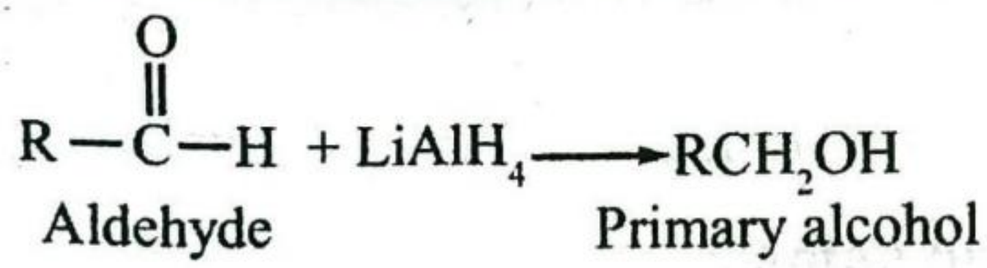
Aldehyde → Ketone

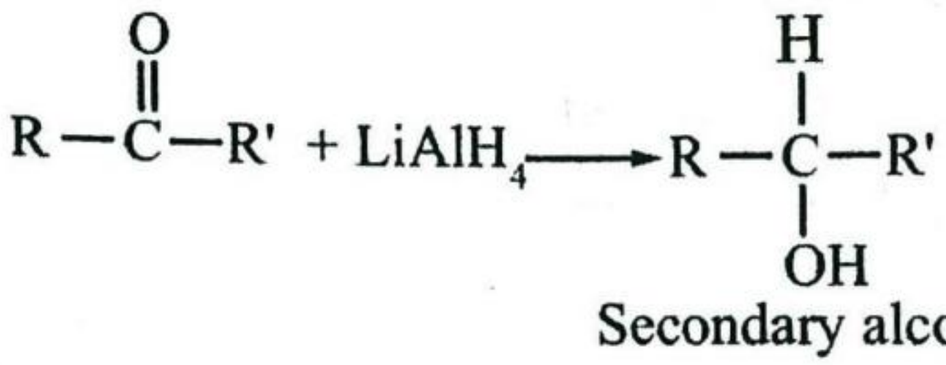
➦ Ketones do not oxidized at the given condition of temperature and pressure.

$$R-\overset{\overset{O}{\|}}{C}-R' \xrightarrow{KMnO_4} \text{No reaction}$$

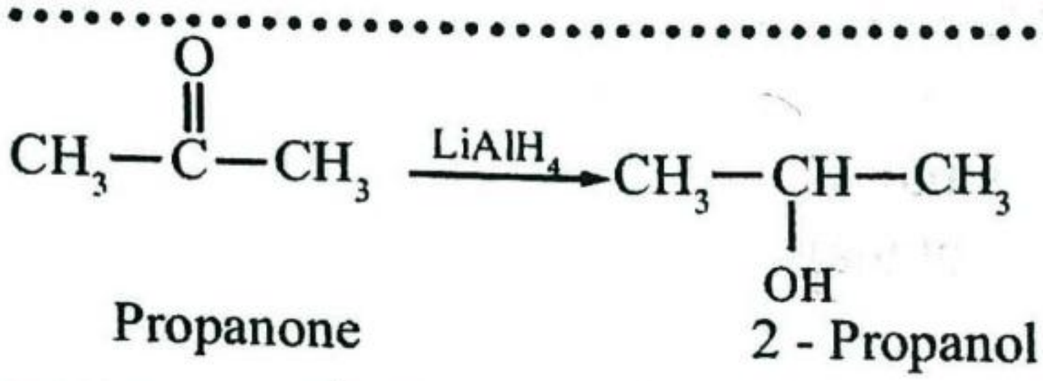
### 2. Reduction

➦ Reduction of aldehydes yield primary alcohols but ketones yield secondary alcohols.




**Example**




Acetone is the most important organic solvent and formaldehyde is used to form formalin.

## Illustrative Examples

50. Which of the following is NOT true about carbonyl compounds? (EUEE ... 2008 E.C)

A. Carbonyl compounds contain 3δ - bond and 1π – bond.

B. The carbon oxygen bond is both longer and weaker.

C. The bond angle is carbonyl is about 120°.

D. Carbonyl compounds may be hydrolyzed.

**Solution**

➦ Compounds containing $-\overset{O}{\overset{\|}{C}}-$ are carbonyl compounds. In $-\overset{O}{\overset{\|}{C}}-$, there are 3δ and 1π bonds. The carbon oxygen bond is shorter and stronger. The bond angle is 120°. From carbonyl compounds, esters can be hydroyzed.

**Answer: B**

51. Which of the following statements concerning the carbonyl group is aldehydes and Ketones is NOT true? (EUEE ... 2008 E.C)

A. The bond is polar, with a slight negative charge on the oxygen atom.

B. The bond angle about the central atom is 120°.

C. In condensed form, the carbonyl group can be written as CHO.

D. The bond is polar. Therefore, the carbonyl groups readily form hydrogen bonds with each other.

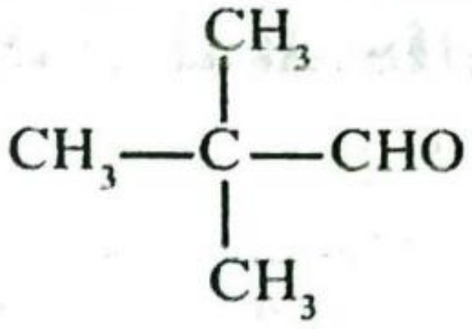
**Solution**

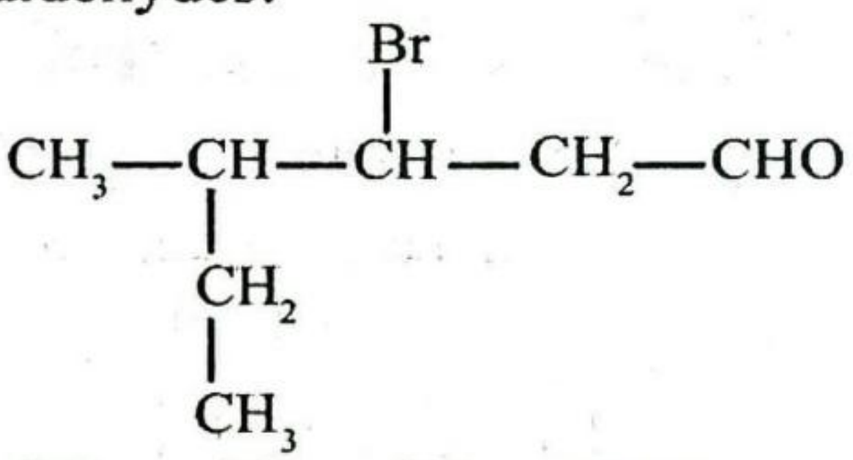
$\overset{\delta-}{C}=\overset{\delta+}{C}$ $\Rightarrow$ $\overset{120°}{\frown}$, for aldehyde, the condensed formula is CHO but for ketone it is CO.

$\Rightarrow$ Aldehydes and ketones do not form hydrogen bond with their molecules.

**Answer: D**

52. Write the IUPAC name for the following aldehydes.

a. 

b. 

c.

**Solution**

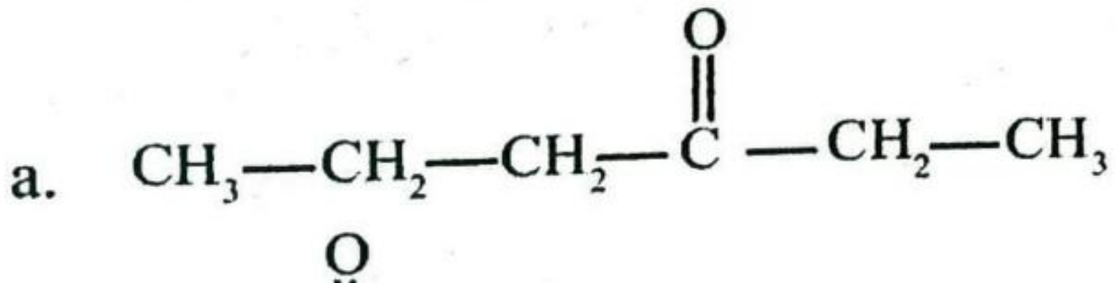
➦ Start from the carbonyl carbon to identify the longest chain.

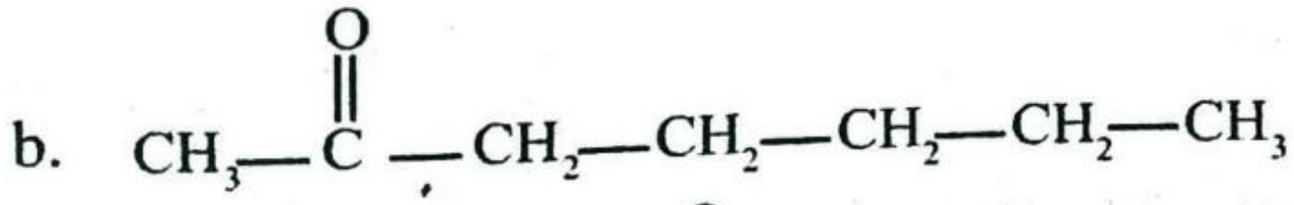
a. 2, 2 - dimethyl propanal

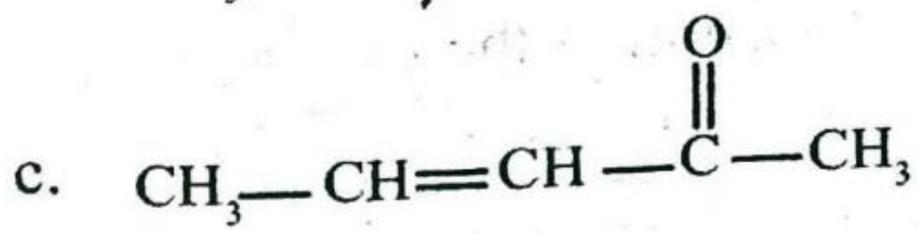
b. 3 - bromo - 4 - methyl hexanal

c. 2 - butenal

53. Give IUPAC name for the following ketones.

a. 

b. 

c. 

**Solution**

Count from the end nearer to the carbonyl carbon and state the position of carbonyl carbon.

a. 3 - hexanone
b. 2 - heptanone
c. 3 - penten - 2 - one

54. Draw the structure for the following aldehydes and ketones.

a. Acetaldehyde
b. Benzanal
c. Ethylmethyl ketone
d. Dimethyl ketone

**Solution**

a. $CH_3CHO$

b.

c. $CH_3CH_2-\overset{\overset{O}{\|}}{C}-CH_3$

d. $CH_3-\overset{\overset{O}{\|}}{C}-CH_3$

55. 40% formal dehyde is called _____

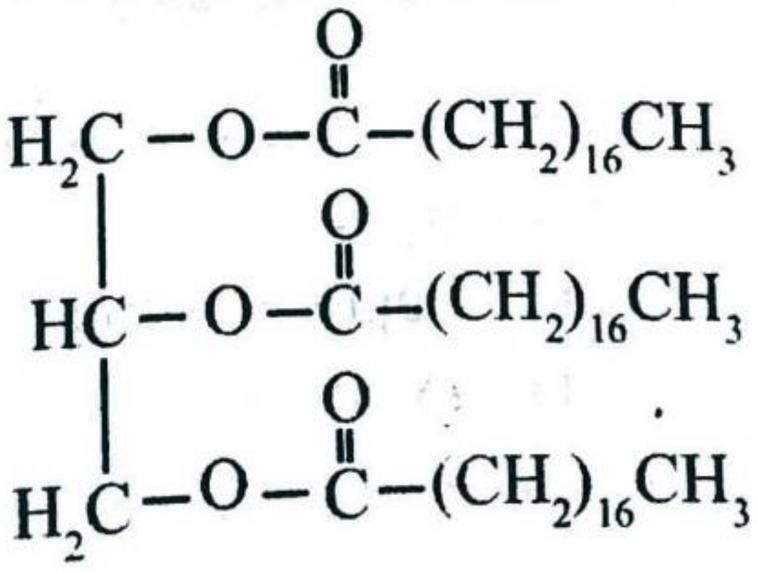
**Answer: Formalin**

## 6.6 Sources and Structures of fats and Oils

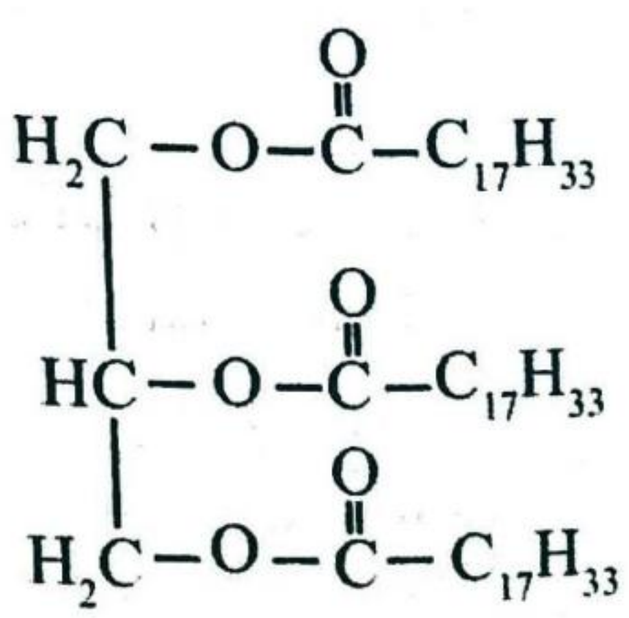
- Fats and oils are widely found in living things.
- Fats are a class of energy - rich organic compounds occurring in organic tissue especially in the adipose tissue of animals and in the seeds, nuts and fruits of plants.
- Fats are solids or semi - solids while oils are liquids at room temperature.
- Animals fats include butter, lard and tallow.
- Oils obtained from animals are liver oils, blubber oils and fish oils.
- Oils obtained from plants are peanut oil, soyabean oil and olive oil.

### Structure of Fats and Oils

- Fats and oils are triesters of glycerol, triacyl glycerols or triglycerides.
- Fats are esters of glycerol or saturated fatty acids but oils are unsaturated fatty acid.



Glyceryltristreate (stearin) fat



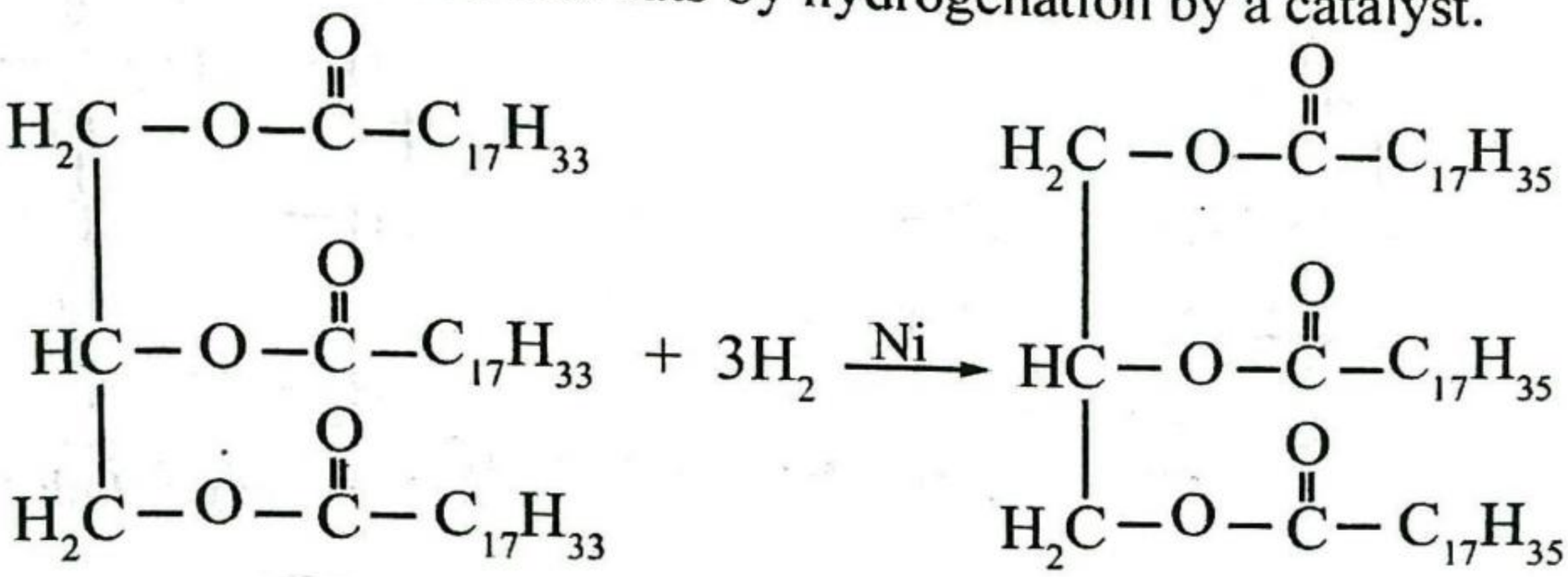
Glyceryltrioleate (Olein) oil

## Physical Properties of fats and oils

- Fat is solid while oil is liquid at room temperature.
- Fats and oils are insoluble in water but soluble in organic solvents.
- Fats and oils are greasy to touch and are non - volatile.
- They are odourless, colourless and neutral in pure state.

## Hardening of Oils

- Oils are converted into fats by hydrogenation by a catalyst.




Oil Fat

**Rancidity** is the spoilage of fats and oils due to unfavorable conditions.

## Illustrative Examples

56. Which of the following statements concerning fats and oils is incorrect? (EUEE ... 2008 E.C)

A. They are also called triacyl glycerols.
B. They are also called triglycerides.
C. They are fatty acid salts.
D. They are glycerol triesters.

**Solution**

Fats and oils are not fatty acid salts.

**Answer: C**

57. Commercially, liquid vegetable oils are converted to solid fats such as margarine by (EUEE)

A. Hydrogenation
B. Hydration
C. Saponification
D. Oxidation

**Solution**

$$\text{Oil} \xrightarrow{\text{hydrogenation}} \text{Fat}$$

**Answer: A**

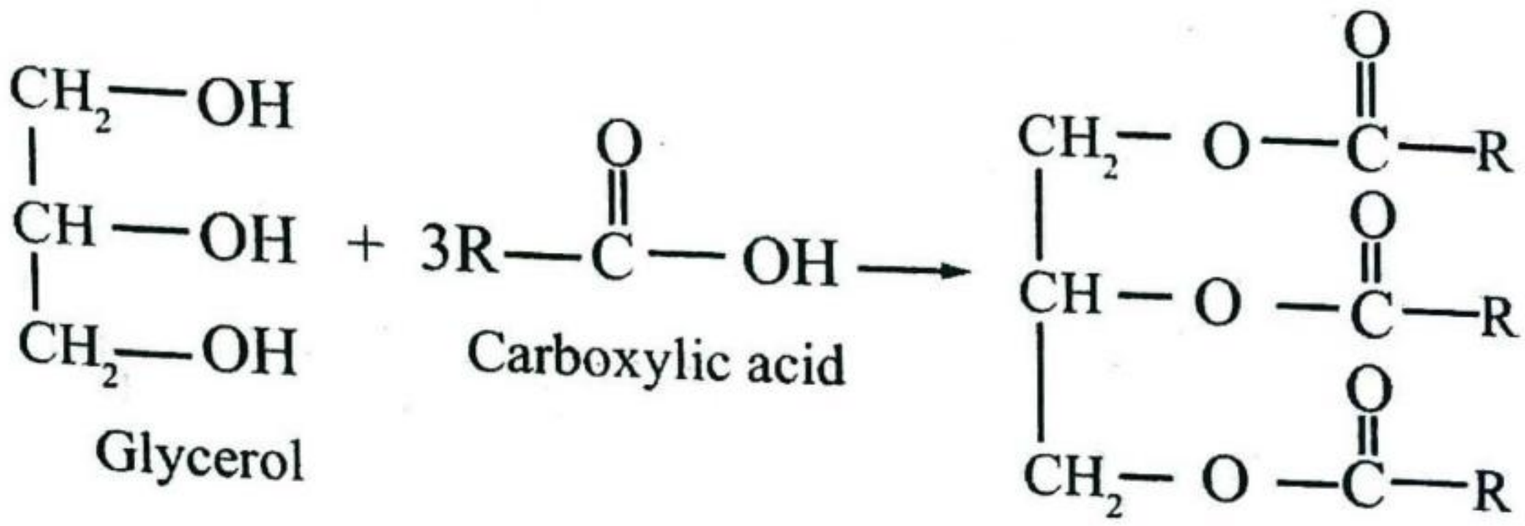
58. Which catalyst is used in the hydrogenation of vegetable oils?

A. Iron
B. Nickel
C. Platinum
D. Molybdenum

**Answer: B**

59. Write the general formula used to make fats and oils.

**Solution**




60. What is the main cause for rancidity of fats and oils?

**Solution**

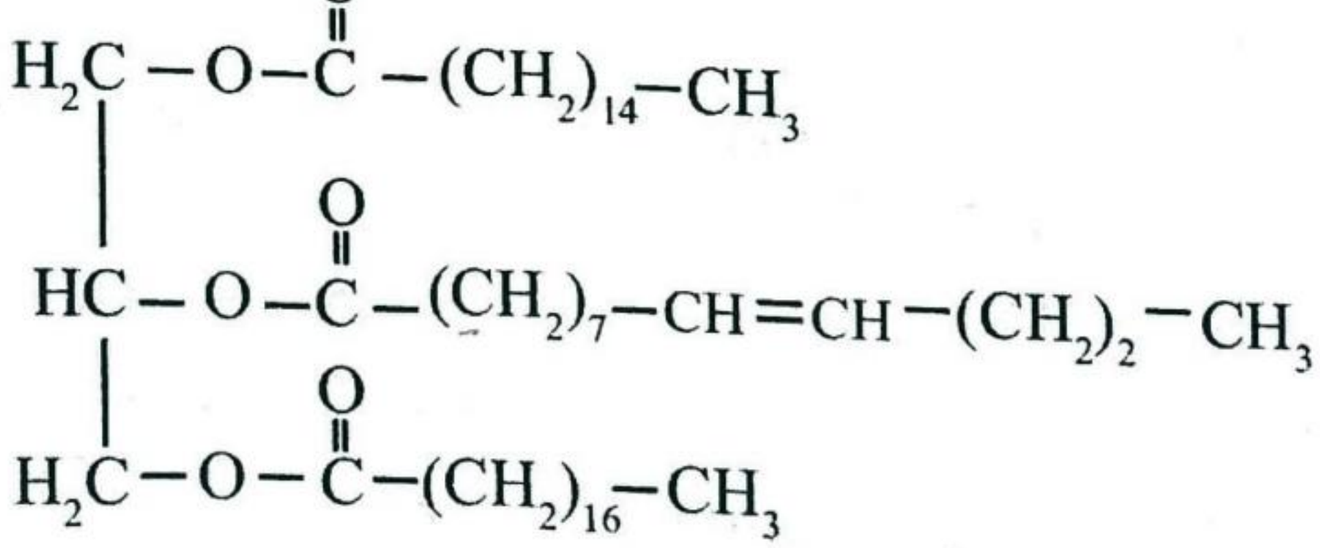
➡ It is caused mainly because of the hydrolysis of ester bond or linkage and oxidation across the double bonds. Other causes may be raw materials used to make fats and oils and storage under unfavorable conditions.

61. Write the structural formula for the following fats and oils.

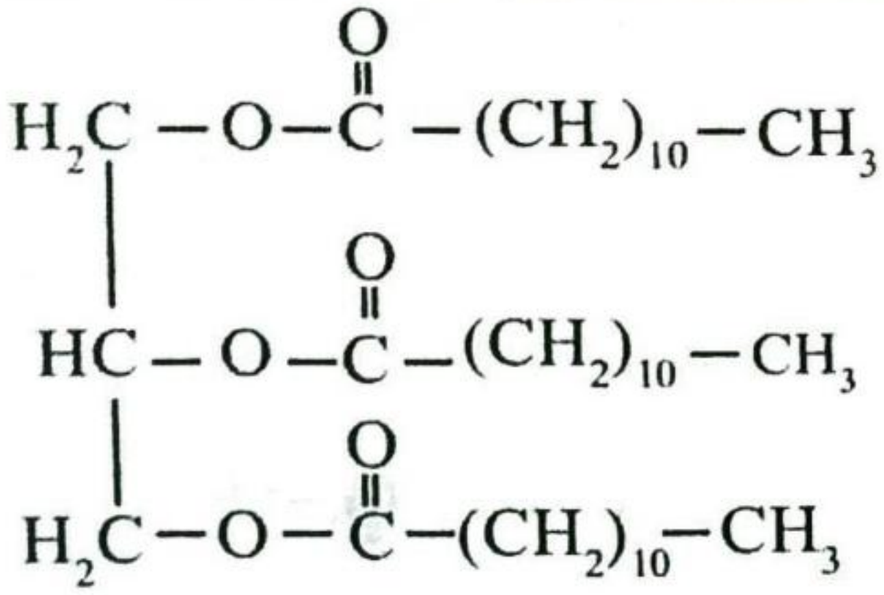
a. Glyceryl Palmitoolestreate
b. Glyceryltriluarate
c. Glyceryltriplamitate

**Solution**

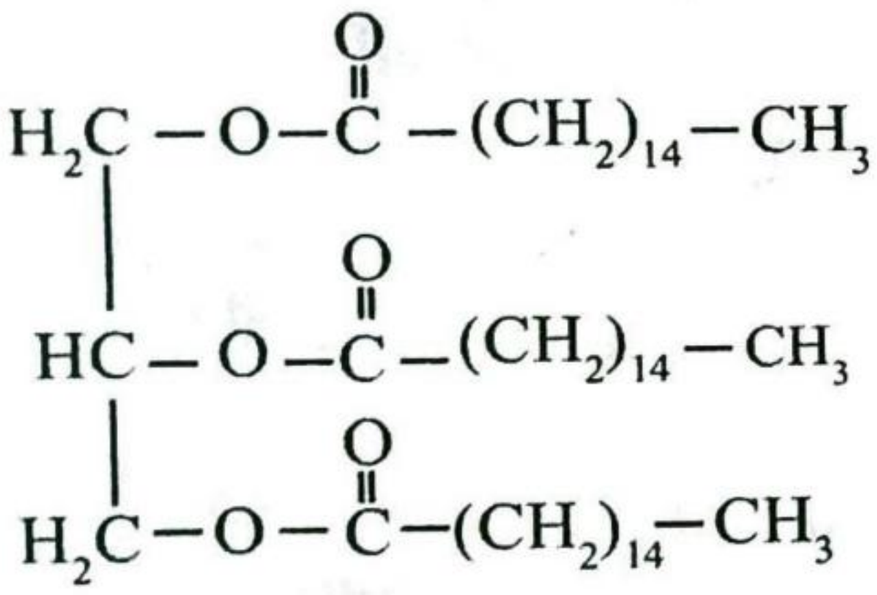
a.



b.



c.

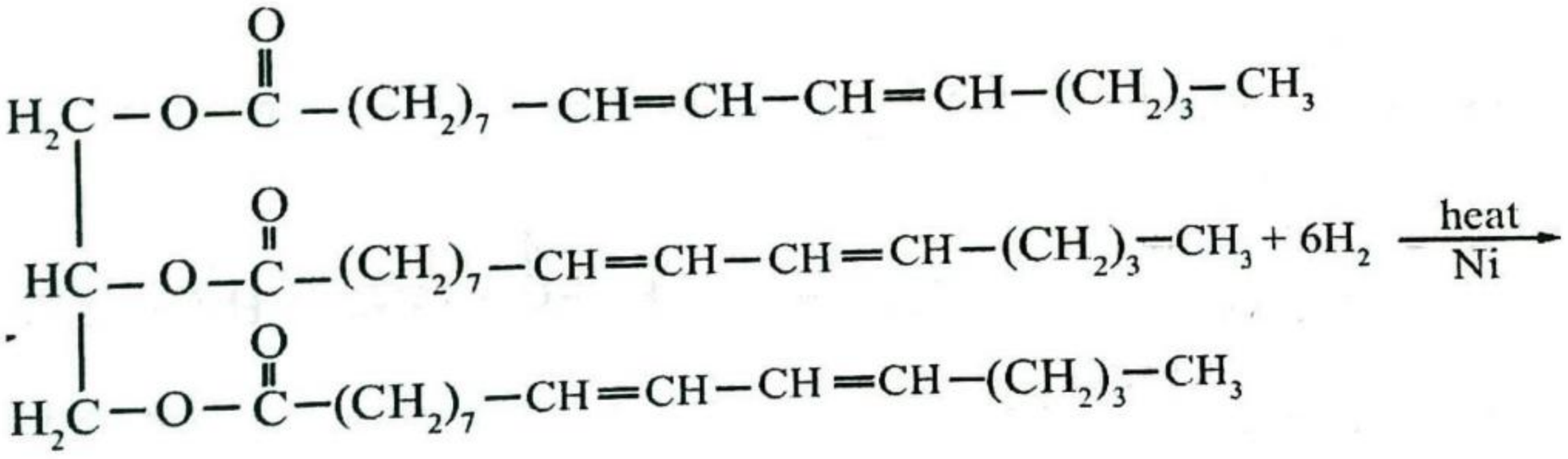


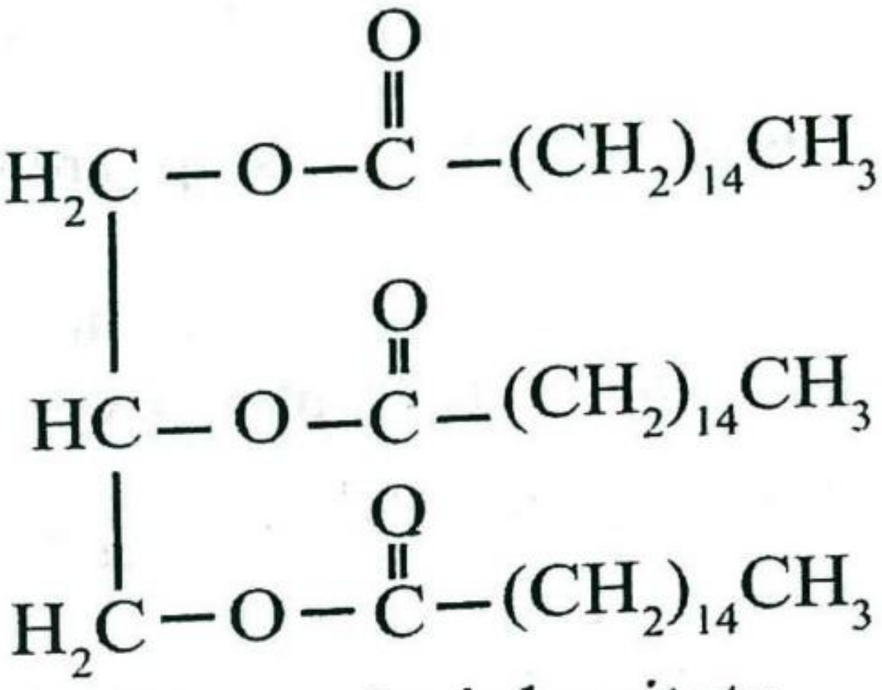
62. Write chemical equation for hydrogenation of:

a. Glyceryl trilinoleate b. Glyceryltrioleate

**Solution**

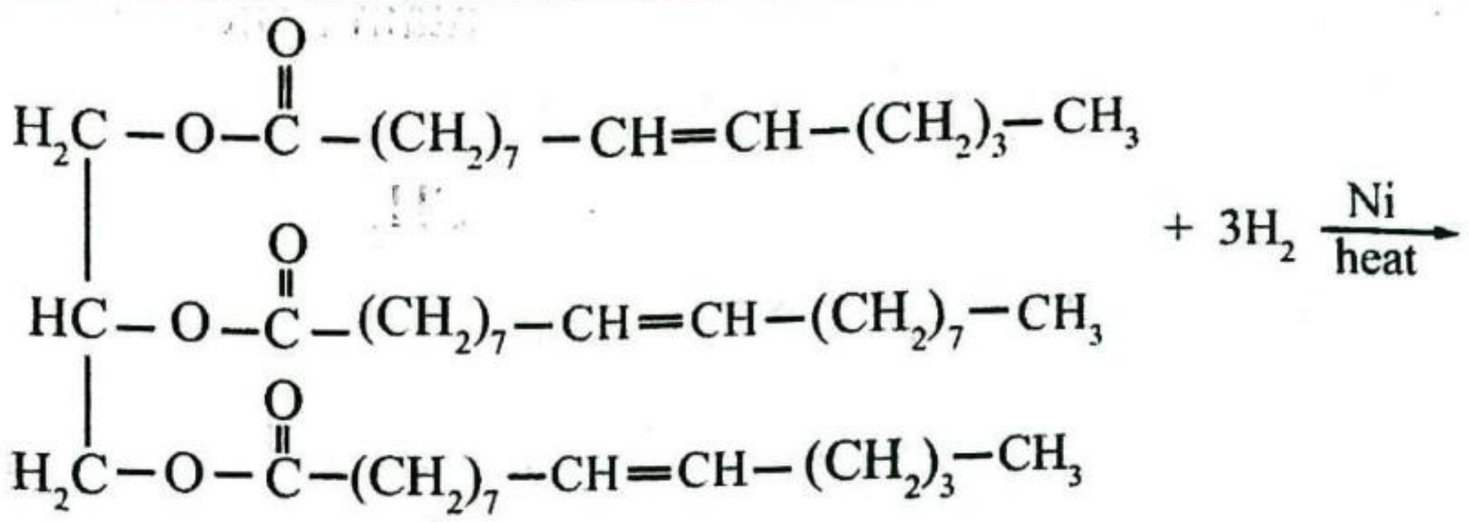
a.



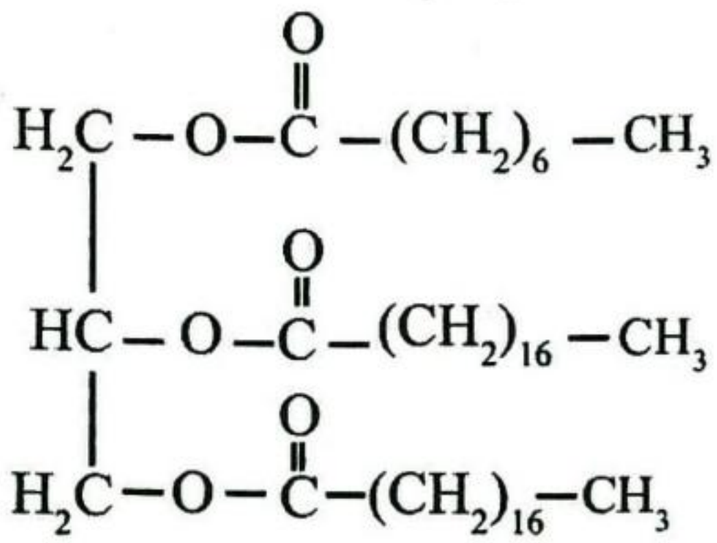


Glyceryl triplamitate
(Fat)

b.



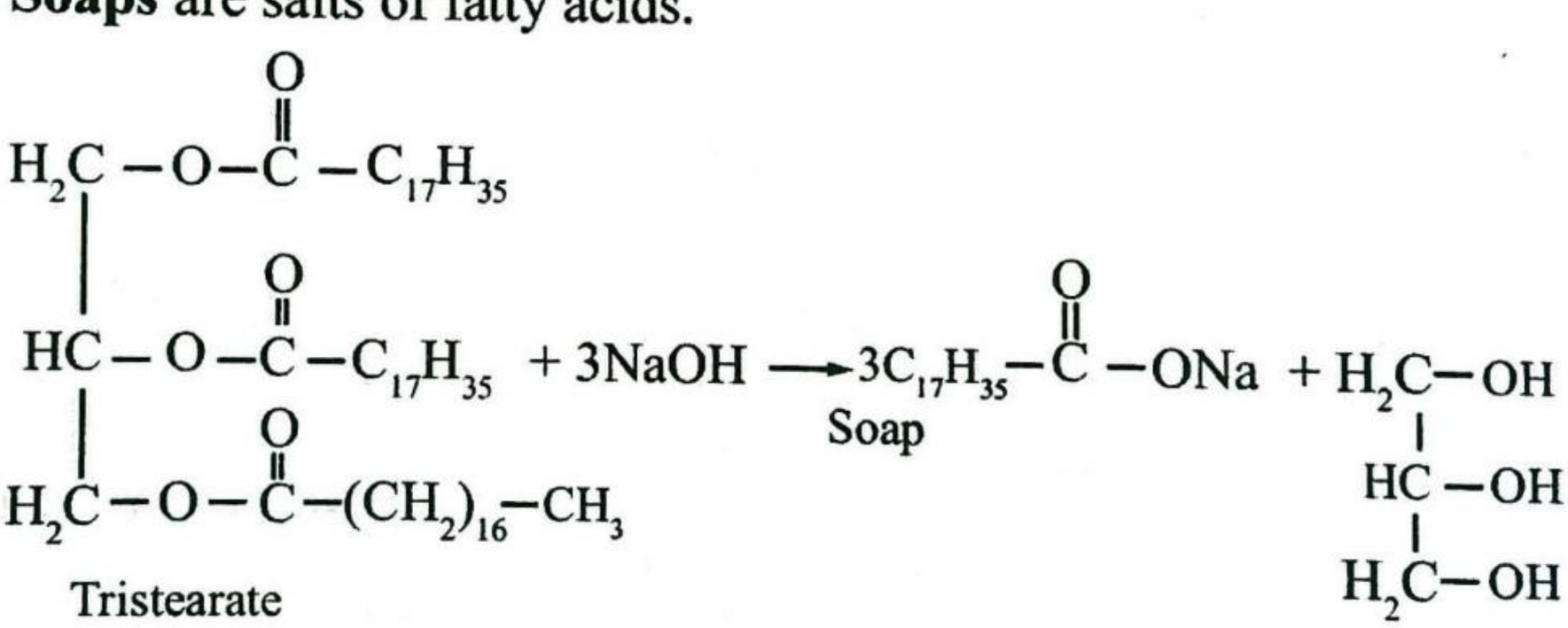
Glyceryl trioleate (Oil)



Glyceryl tristreate (Fat)

## 6.7 Soaps and Detergents

**Soaps** are salts of fatty acids.



Tristearate

Soap

- ➦ Sodium soaps are **hard soaps** while potassium soaps are **soft soaps**.
- ➦ The polar end of a soap molecule is said to be **hydrophilic or water loving** and the non - polar tail is **hydrophobic or water hating**.
- ➦ Polar head dissolves polar dirt and non - polar tail dissolves non - polar dirt.

A sodium soap is a solid at room temperature, whereas potassium soaps are usually liquid.

- ➦ Soaps and detergents affect the properties of surface of the dispersed material. Due to this, they are often called **surfactants**.

**Detergents** are synthetic substances that contain either sulphate ion or sulphonate groups.

$$CH_3(CH_2)_{12}CH_2OH + H_2SO_4 \rightarrow CH_3(CH_2)_{12}CH_2 - O - SO_2 - OH + H_2O$$

Myristly alcohol — Myristylhydrogen sulphate

$$CH_3(CH_2)_{12}CH_2 - O - SO_2 - OH + NaOH \rightarrow CH_3(CH_2)_{12}CH_2OSO_3Na$$

Sodium myristyl sulphate

## Illustrative Examples

63. Which of the following types of compounds are expected products from the saponification of a fat? (EUEE ... 2008 E.C)

A. Glycerol and fatty acid salts
B. Glycerol and fatty acids
C. Fatty acid salts and fatty acids.
D. Glycerol, fatty acids and fatty acids salts.

**Solution**

Fat + NaOH/KOH → Soap + glycerol

⇒Soaps are fatty acid salts.

**Answer: A**

64. Which of the following is false about soaps?

A. They are alkaline
B. They are not used in hard water
C. They have no draw backs.
D. They are neutral.

**Solution**

➦ Soaps are alkaline because they are formed from weak acid and strong base. They have no draw backs and used in soft water.

**Answer: D**

65. Explain the difference between soaps and detergents.

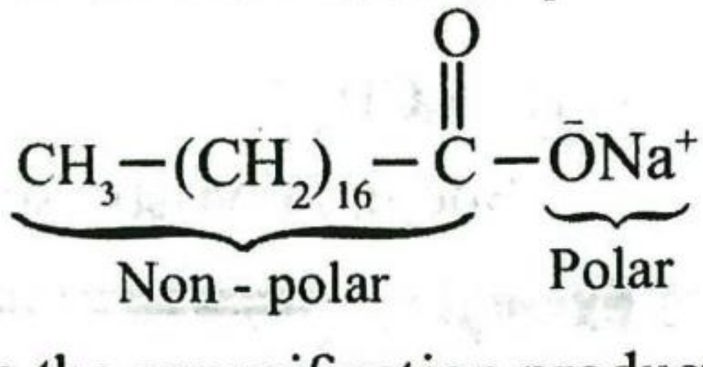
**Explanation**

| Soaps | Detergents |
|---|---|
| ➦ are basic | ➦ are neutral |
| ➦ used in soft water | ➦ used in soft and hard water |
| ➦ no draw backs | ➦ have draw backs but now a days improved |

66. Explain the cleaning actions of soaps.

**Explanation**

➦ Soaps have polar head and non - polar tail. Due to the like dissolves like principle's polar head removes polar dirt and non - polar tail removal non - polar dirt.

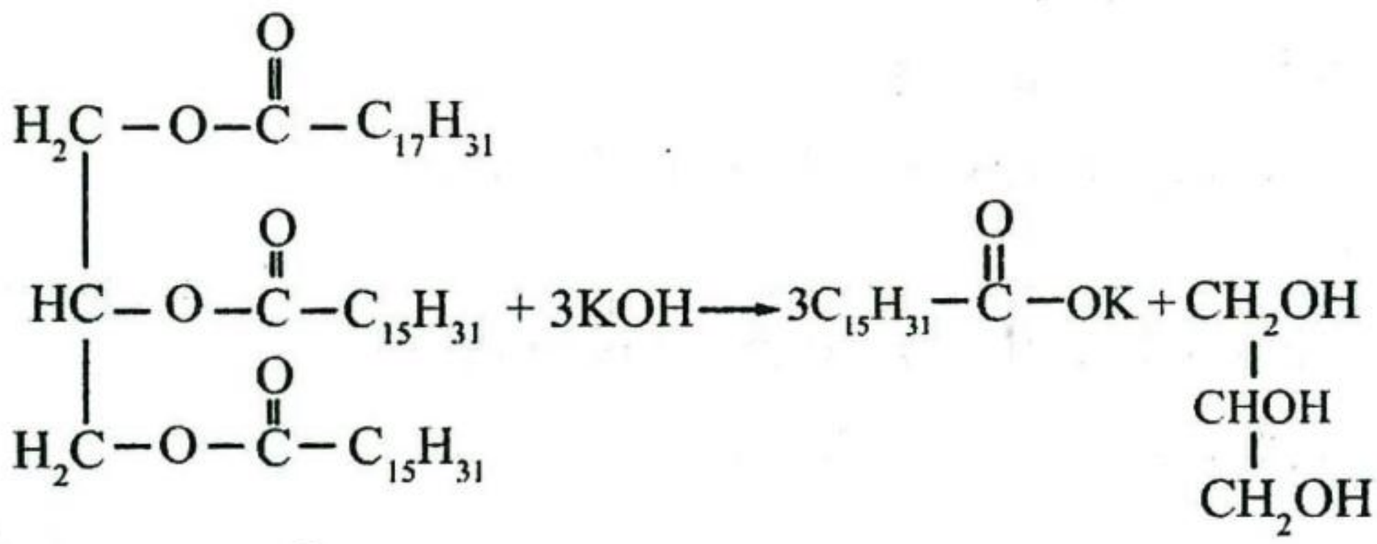



67. What are the saponification products obtained when the following fats are boiled with KOH?

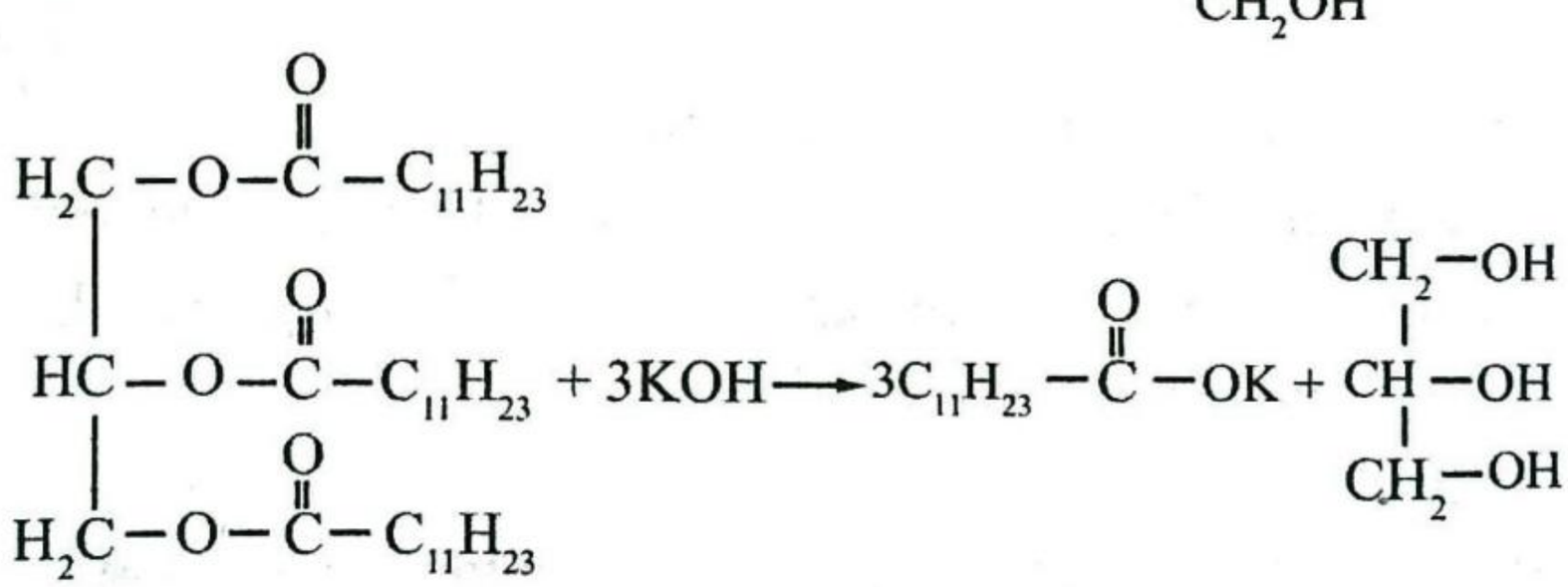
a. Glyceryltrilaurate b. Glyceryl triplamitate

**Solution**

a.



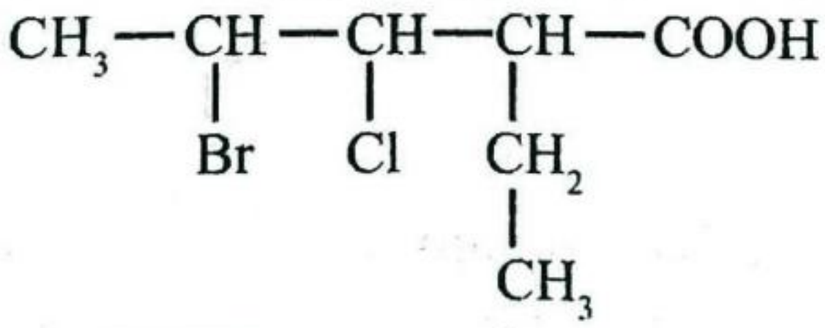

b.



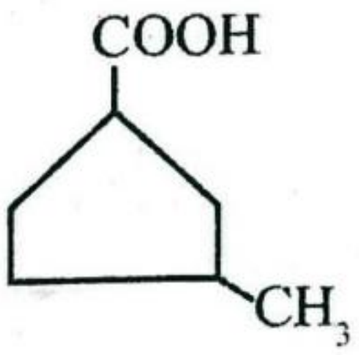

## Solved Problems

68. Write the IUPAC name for the following carboxylic acids.

a.

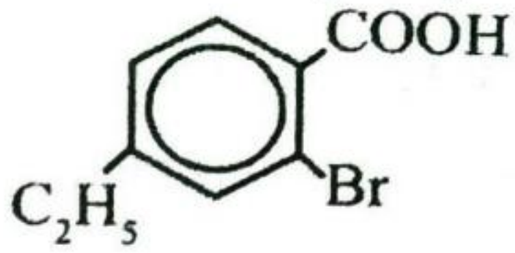



b.



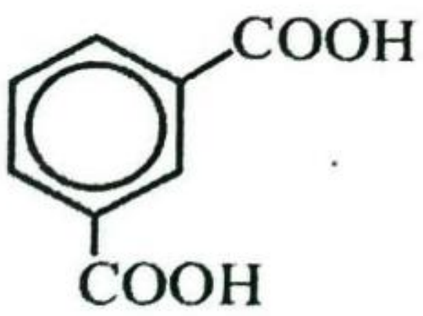
c. $CH_3-C\equiv C-CH_2-\overset{\overset{O}{\|}}{C}OH$

d. $HOOCCH_2CH_2COOH$

e. 

f. $CH_3(CH_2)_4COOH$

g. $CH_3(CH_2)_{14}COOH$

h. 

**Solution**

Follow the rules given to write the IUPAC name for each structure.

a. 4-Bromo-3-chloro-2-ethylpentanoic acid

b. 3-methylcyclopentane carboxylic acid

c. 3-pentynioic acid

d. Butanedioic acid

e. 2-Bromo-4-ethylbenzene carboxylic acid

f. Hexanoic acid

g. Hexadecanoic acid

h. 1, 3-benzene dicarboxylic acid.

69. Draw the structure of the following carboxylic acids.

a. 2,3-Dibromopropanoic acid

b. 2 – hydroxybenzene carboxylic acid

c. 5 – Bromopentanoic acid

d. 2- Heptenoic acid

e. 3-chlorocyclohexane carboxylic acid

f. Trichloro ethanoic acid

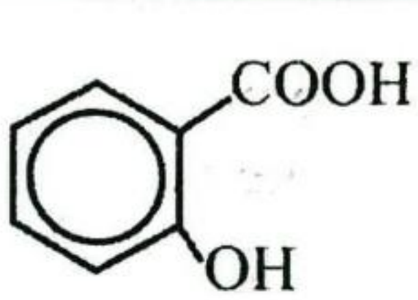
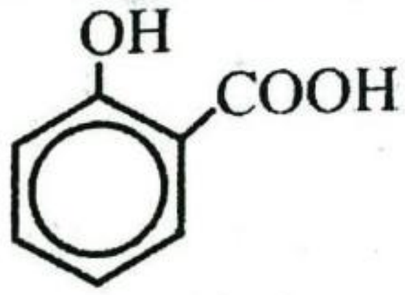
g. Tetradecanoic acid

h. Ethanedioic acid

**Solution**

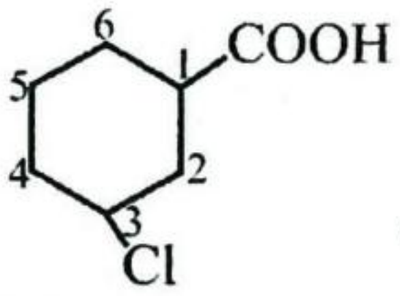
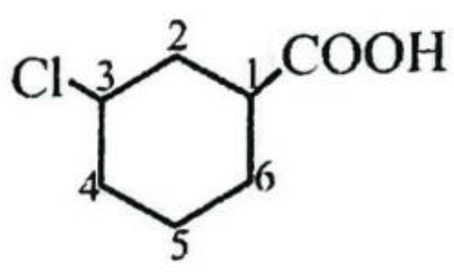
First determine the number of carbon atoms from the parent name and place the side chains (structures) on their respective (given) position.

a. $\begin{matrix} CH_2 & CHCOOH \\ | & | \\ Br & Br \end{matrix}$

b.  or 

c. $CH_2(Br)CH_2CH_2CH_2COOH$ or $CH_2Br(CH_2)_3COOH$

d. There is double bond on the second carbon
$CH_3CH_2CH_2CH_2CH = CHCOOH$

e.  or 

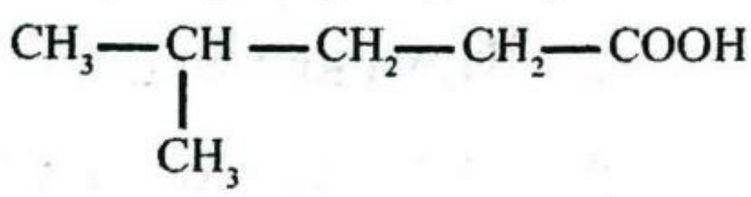
f. $CCl_3COOH$

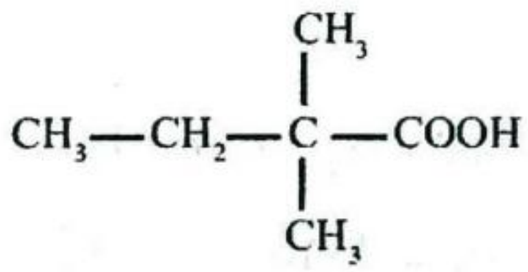
g. Has 14 carbonatoms
$CH_3(CH_2)_{12}COOH$

h. HOOCCOOH

70. Arrange the following in increasing order of boiling point

a. $CH_3CH_2CH_2CH_2CH_2COOH$

b. 

c. 

**Solution**

The three structure have the same carbon number. They differ in structural formula. As branching increases boiling point decreases.

$\Rightarrow c < b < a$

71. Arrange the following aliphatic carboxylic acid in the decreasing order of solubility in water.

a. Methanoic acid
b. Heptanoic acid
c. decanoic acid
d. pentanoic acid

**Solution**

As carbon number increases solubility decreases.

$\Rightarrow a > d > b > c$

72. Arrange the following organic compounds in increasing order of boiling point.

a. Pentane ($CH_3CH_2CH_2CH_2CH_3$)
b. 1-pentanol ($CH_3CH_2CH_2CH_2CH_2OH$)
c. Pentanoic acid ($CH_3CH_2CH_2CH_2COOH$)
d. Pentanedioic acid ($HOOCCH_2CH_2CH_2COOH$)

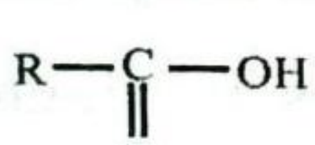
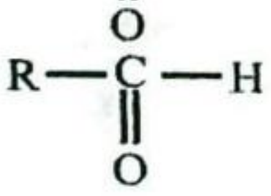
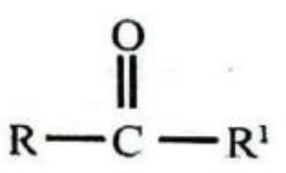
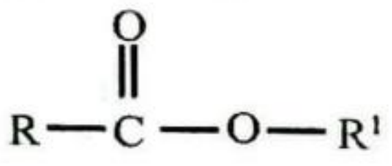
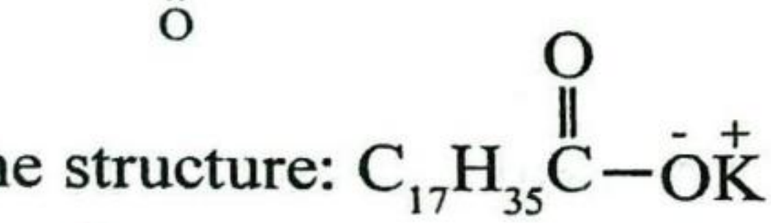
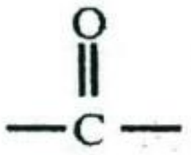
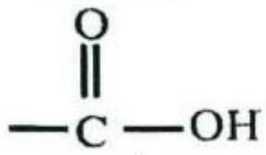
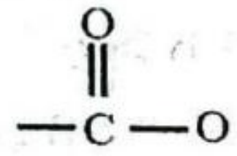
**Solution**

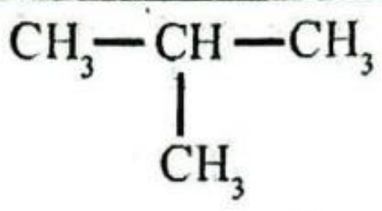
Alkanes have the lowest boiling point since their molecules is held together with the weakest intermolecular force called London force.

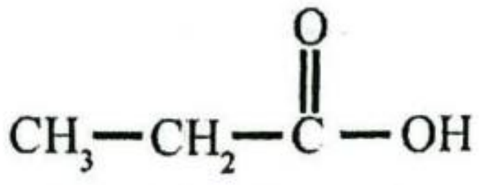
$\Rightarrow a < b < c < d$

## Supplementary Exercises

I. Multiple Choice questions

1. Which of the following does not contain oxygen?
   A. Methane
   B. Methanol
   C. Methanoic acid
   D. Methanal
2. Which of the following elements is necessarily found in all organic compounds?
   A. Hydrogen
   B. Carbon
   C. Oxygen
   D. nitrogen
3. Which of the following represents the general formula of an ester?
   A. 
   B. 
   C. 
   D. 
4. The structure:  represents
   A. detergent
   B. polymer
   C. soap
   D. Fast and oils
5. The conversion of ester into alcohol and carboxylic acid is
   A. Esterification
   B. Saponification
   C. dehydration
   D. Hydrolysis
6. The carbonyl group is represented by the structure
   A. 
   B. 
   C. 
   D. D. – C – OH
7. Conversion of propanpic acid into 1-propanol is an example of
   A. reduction
   B. oxidation
   C. hydrolysis
   D. dehydration

III 

IV 

A. II > IV > I > III > IV

B. V > II > I > III> IV

C. IV > II > I > III

D. II > I > IV > III

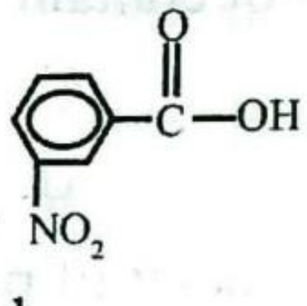
22. The structure for α-methyl hexanoic acid is

A. $CH_3CH(CH_3)CH_2CH_2CH_2COOH$

B. $CH_3CH_2CH(CH_3)CH_2CH_2COOH$

C. $CH_3CH_2CH_2CH(CH_3)CH_2COOH$

D. $CH_3CH_2CH_2CH_2CH(CH_3)COOH$

23. The common name for 

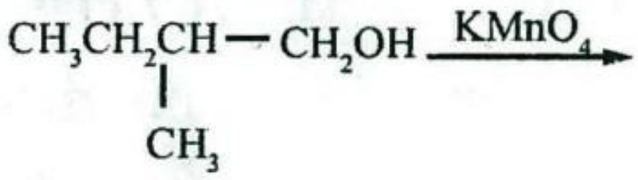
A. O-nitrobenzoic acid

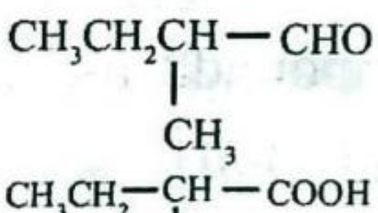
B. m-nitro benzoic acid

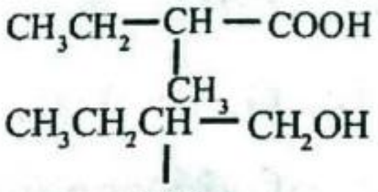
C. P-nitro benzoic acid

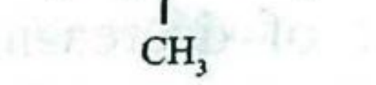
D. Nitrobenzoic acid

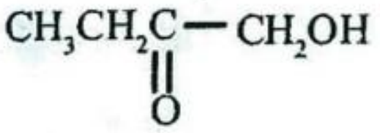
24. Consider the reaction below, the structure of the product is



A. 

B. 

C. 

D. 

25. Which of the following is not natural ester?

A. Bees wax

B. Animal fat

C. vegetable oil

D. Glycerol

II. Give short answer

26. Write the structure of the following
    a. O – Ethylbenzoic acid
    b. β – methylvaleric acid
    c. Isopropyl formate
    d. Adipic acid
    e. Pimelic acid
27. Write the chemical equations for the reaction between methanoic acid and
    a. NaOH
    b. $K_2CO_3$
    c. $NH_3$
28. The organic acid that is produced industrially from fermentation of ethanol by Quick vinegar fermentation process is _____
29. _____ are pleasant smelling organic compounds responsible for the odour of many fruits.
30. Classify the following as saturated or unsaturated fatty acids.
    a. Myristic acid
    b. Oleic acid
    c. Linolic acid
    d. lauric acid
    e. Palmitic acid
    f. Linolenic acid
    g. Stearic acid
31. Toluene (methyl benzene) can be converted in to benzoic acid by its reaction with _____
32. Given the following acids
    - Acetic acid
    - Oxalic acid
    - Caproic acid

    a. which acid can form two sodium salts?
    b. which acid is the least soluble in cold water
    c. which of the acids, when concentrated, is called 'glacial' because in cold water it crystallizes?

33. Complete the table by naming the ester and/or identifying the original alcohol and carboxylic acid it was made from.

| Ester | Alcohol | Carboxylic acid |
|---|---|---|
| $CH_3CH_2COOCH_3$ | a | b |
| c | $C_2H_5OH$ | HCOOH |
| $CH_3COOC_3H_7$ | d | e |
| f | $C_5H_{11}OH$ | $CH_3CH_2COOH$ |
| $C_2H_5COOC_4H_9$ | g | h |

34. The process of converting unsaturated esters of fatty acids into saturated esters by hydrogenation using a nickel catalyst is _____
35. Hydrolysis of n-propyl propanoate results _____.

አጋዥ መጽሐፍቶችን ለማግኘት በ
Telegram @ethiolibrary1 ላይ
ይቀላቀሉን
+251927273995
+251909282436
Email: jemal6454@gmail.com

አጋዥ መጽሐፍቶችን ለማግኘት በ Telegram @ethiolibrary1 ላይ ይቀላቀሉን
+251927273995
+251909282436
jemal6454@gmail.com

# Unit 7

## Solution

### Unit Content Outline:

7.1 Solution
7.2 The Solution Process
7.3 Rate of Dissolution
7.4 Solution formation and equilibrium
7.5 Ways of expressing the Concentration of Solution
7.6 Conversion of concentration units
7.7 Solution preparation
7.8 Solution Stoichiometry
7.9 Describing reactions in solution
7.10 Colligative Properties of solution

## Introduction

Substances are classified into two, pure substance and mixture. Mixture is further classified into two, these are homogenous and heterogeneous. mixtures.

**Heterogeneosu mixture** has two or more phases.

### Example

Blood, milk, rock salt, soil, a mixture of oil and water etc.

**Homogenous mixture** is a mixture that has one phase.

### Example

Sugar solution, salt solution, air, alloys, sea water, etc

**Suspension** is heterogenous mixture in which particles settle down after a given period of time.

### Example

- Milk is magnesia ($Mg(OH)_2(aq)$)
- A mixture of sand and water.
- A mixture of flour and water.

All suspensions are heterogenous mixture but all heterogenous mixture are not necessarily suspensions.

**Colloids** are mixtures containing small particles that are small enough not to settle down wards.

### Example

- Paint dissolved in water.
- Smoke
- Cloud
- Smog
- Fog

## Illustrative Examples

1. Classify the following as pure substances, homoegenous mixture or heterogenous mixture.
   a. Air
   b. Water
   c. Bloo
   d. Brass
   e. Mud
   f. Copper

   **Solution**

   - Pure substances are either compounds or elements.
   - b and f are pure substances.
   - a and d are homogeneous mixtures.
   - c and e are heterogeneous mixtures.

2. Classify the following as colloids or suspensions.
   a. Cell fluid
   b. Aluminum hydroxide in water.
   c. Sand and water
   d. Cloud

**Solution**

➦ a and d are colloids.

➦ b and c are suspensions.

## 7.1 Solution

**Solution** is homogeneous mixture. It has two components, solute and solvent.

**Solute** - present in small amount
- undergoes phase changes

**Solvent** - usually exists in large quantity.
- does not under go phase changes.
- affects the phase of a solution

**Example**

In sugar solution, sugar is solute while water is solvent.

### Types of Solutions

➦ Based on the states of solute and solvent there are six main types of solution. See table 7.1 below.

Table 7.1. Types of solutions and examples

| NO. | State of solute | State of solvent | State of solution | Types of solution | Example |
|---|---|---|---|---|---|
| 1 | Solid | Solid | Solid | Solid - solid | Brass, bronze |
| 2 | Solid | Liquid | Liquid | Solid - liquid | Salt solution<br>Sugar solution |
| 3 | Liquid | Solid | Solid | Liquid - solid | Amalgam |
| 4 | Liquid | Liquid | Liquid | Liquid - liquid | Ethanol in water |
| 5 | Gas | Liquid | Liquid | Gas - liquid | Pepsi, coca - cola |
| 6 | Gas | Gas | Gas | Gas - Gas | Air |

## Illustrative Examples

3. What is the solvent in 70% alcohol solution? (EUEE ... 2008)

A. Water
B. Alcohol
C. Sugar
D. Kerosene

**Solution**

Even though the amount of alcohol is higher than water, water is the solvent because it is universal solvent.

**Answer: A**

4. What kind of solution forms when gasoline evaporates in air? (EUEE)

A. Gas in gas solution.
B. Gas in liquid solution
C. Liquids in liquid solution.
D. Liquid in gas solution

**Solution**

Gasoline is converted into gas due to evaporation. Thus, it is gas - gas solution.

**Answer: A**

5. Which one is not a solution?

A. Milk
B. Brass
C. Whisky
D. Coca - cola drink

**Answer: A**

6. Which of the following types of solutions are possible? (EUEE ... 2002 E.C)

I. Solid dissolved in a liquid.
II. Gas dissolved in a liquid.
III. Gas dissolved in a gas.
IV. Solid dissolved in a solid.

A. I and II
B. I, II, III and IV
C. I, II and V
D. I

**Solution**

➦ All types of solution are possible.

**Answer: B**

7. What are the possible physical states of solute and solvent in

a. Solid solution
b. Liquid solution
c. Gaseous solution

1. Solvent - solvent interaction
2. Solute - solute interaction
3. Solute - solvent interaction.

➦ If 3 > 2 and 1, $\Delta H_{sol} < 0$ (exothermic mixing). It is favorable process.

➦ If 3 < 2 and 1, $\Delta H_{sol} > 0$ (endothermic mixing).

➦ If 3 = 2 and 1, $\Delta H_{sol} = 0$ (ideal solution)

If solute - solvent interaction is much less than solvent - solvent and solute - solute interactions, dissolution does not occur to any significant extent.

## Illustrative Examples

11. The process of solute particles being surrounded by solvent particles is known as (EUEE .. 2007 E.C.)

A. Saturation
B. Agglomeration
C. Solvation
D. Dehydration

**Answer: C**

12. When pure methanol is mixed with water, the solution becomes warm. Is this an ideal solution? (EUEE)

A. Yes, for ideal solution, $\Delta H_{sol} = 0$
B. No, for ideal solution, $\Delta H_{sol} = 0$
C. Yes, for ideal solution, $\Delta H_{sol} > 0$
D. No, for ideal solution, $\Delta H_{sol} < 0$

**Solution**

The solution is warm, meaning it is formed by exothermic mixing. For ideal solution, $\Delta H_{sol} = 0$.

**Answer: B**

13. What is the exothermic step in a solution of $CaCl_2$ in water?

A. Separation of the salt into ions in gaseous state.
B. Separation of the water molecule.
C. Hydration (solvation) of the ions from the salt.
D. Separation of the salt into ions in gaseous state and separation of the water molecules.

**Solution**

➦ Hydration or solvation may be the exothermic step.

**Answer: C**

14. For gaseous solute and solvent, write heat of solution interms of $\Delta H_1$, $\Delta H_2$ and $\Delta H_3$.

**Solution**

Gaseous molecules are already separated. Thus, $\Delta H_1$ and $\Delta H_2$ are zero.

$\Delta H_{sol} = \Delta H_3$

15. For ionic solute, what is the energy required to separate into gaseous ions? Write heat of solution.

**Solution**

Lattice energy ($\Delta H_{LE}$) is the energy that separates ionic solute into ions.

$\Delta H_{sol} = \Delta H + \Delta H_{LE} + \Delta H_{hyd}$

For ionic compounds, the solvent is water.

$\Delta H_{hyd}$ - enthalpy of hydration.

## 7.3 Rates of Dissolution

➦ The speed at which the solute enters into a solution is called **rate of dissolution**.

➦ The rate of dissolution is mainly affected by interparticle force (intermolecular forces).

i **Ion - dipole forces** are forces involved when an ionic compound dissolves in water.

ii **Hydrogen bonding** is the principal force in solutions of polar, F-, O- and N- containing organic and biological compounds such as alcohols, amines and amino acids.

iii **Dipole - dipole forces,** in the absence of hydrogen bonding, allow polar molecules to dissolve in polar solvents.

iv **Ion - induced dipole forces**, one type of charge - induced dipole force, rely on polarizability. They arise when an ion's charge distorts the electron cloud of a nearby non - polar molecule. This type of force initiates the binding of the $Fe^{2+}$ ion in hemoglobin to an $O_2$ molecule entering a red blood cell.

v **Dipole - induced dipole forces**, also based on polarizability, arise when a polar molecule distorts the electron cloud of a non - polar molecule.

vi **Dispersion forces** contribute to the solubility of non - polar molecule in non - polar solvent.

➦ Rate of dissolution is also affected by:

- temperature
- surface area
- pressure of gaseous solute

## Solutions of liquids in liquids

➦ The solubility of solute in a solvent follows the '**like dissolves like principle**'. This means polar dissolves polar and non - polar dissolves non - polar.

### Example

➦ Water and ethanol are miscible due to hydrogen bonding.

➦ Benzene and oil are miscible due to London dispersion force.

## Solutions of Solids in Liquids

➦ Polar solids dissolve in polar solvent and non - polar solids dissolve in non - polar solvent.

### Example

➦ Iodine dissolve in benzene due London dispersion force.

➦ Sugar is soluble in water because of hydrogen bonding.

➦ Sodium chloride is soluble in water because of ion – dipole force.

## Solubility of Ionic Compounds

➦ The solubility of ionic compounds in water is affected by

1. Hydration energy
2. Lattice energy

1. **Hydration energy** is the energy released when water molecules surround the ions of the salt.

   ➦ The greater the hydration energy, the more likely that the ionic solid will dissolve in water.

2. **Lattice energy** is the energy required to break one mole of an ionic solid into isolated gaseous ions.

The lower the lattice energy, the more soluble is an ionic solid in water.

### Example

Consider NaCl

$NaCl(s) \longrightarrow Na^+(g) + Cl^-(g), \Delta H_{LE}$

$Na^+(g) \xrightarrow{H_2O} Na^+(aq), \Delta H_2 < 0$

$Cl^-(g) \xrightarrow{H_2O} Cl^-(aq), \Delta H_3 < 0$

$NaCl(s) \xrightarrow{H_2O} Na^+(aq) + Cl^-(aq), \Delta H_{sol} = \Delta H_1 + \Delta H_2 + \Delta H_3$

$\Delta H_{sol} \approx 5 kJ/mol$

- ➦ $\Delta H_{sol} > 0$, the solution is not formed easily.
- ➦ $\Delta H_{sol} < 0$, the solution is formed easily.

## Illustrative Examples

16. What is the most important type of solute - solvent interaction in a solution of ammonia in water? (EUEE ... 1996 E.C)

A. Ion - dipole
B. Dipole - dipole
C. Hydrogen bonding
D. London - dispersion force

**Solution**

Ammonia ($NH_3$) and water ($H_2O$) have hydrogen bonding between their molecules.

**Answer: C**

17. Which of the following is most likely to be miscible in carbon tetra chloride, $CCl_4$? (EUEE)

A. $Na_2CO_3$
B. $CH_3F$
C. $(CH_2)_2O_2$
D. $I_2$

**Solution**

- ➦ $CCl_4$ is non - polar solvent that dissolves non - polar solute.

**Answer: D**

18. The rate of dissolution is mainly depends on

A. Surface area
B. Pressure
C. Temperature
D. Interparticle force

**Answer: D**

19. Sodium carbonate ($Na_2CO_3$) is soluble in water but calcium carbonate ($CaCO_3$) is insoluble in water. Why?

**Solution**

Calcium carbonate has higher lattice energy than sodium carbonate.

20. Identify the solute - solvent interaction responsible for the solubility of
    a. Hydrogen fluoride in water.
    b. Bromine in benzene.
    c. Calcium chloride in water.
    d. Hydrogen chloride in water

**Solution**

a. Hydrogen bonding
b. London - dispersion force
c. Ion - dipole force
d. Dipole - dipole force

21. Arrange the following ionic compounds in increasing order of solubility. LiCl, NaCl, RbCl, KCl

**Solution**

$$LiCl < NaCl < KCl < RbCl$$

22. In the laboratory by using your sense organ how can you identify whether a given solution is formed by exothermic or endothermic process?

**Solution**

By touching the apparatus with our hands. If we feel hot, the given solution is formed by exothermic mixing but if we feel cold it is formed by endothermic mixing.

**Example**

Solutions of $NH_4Cl$ and $KNO_3$ are cold. Solutions of $Na_2CO_3$ and $H_2SO_4$ are hot.

23. Explain the difference between enthalpy of solution and enthalpy of solvation.

**Solution**

- Enthalpy of solution ($\Delta H_{sol}$) is the energy released or absorbed when one mole of solute enters into solution.
- Enthalpy of solvation ($\Delta H_{solvation}$) is the energy released or absorbed when solvent molecules surround solute particles.

## 7.4 Solution formation and equilibrium

Based on equilibrium, solutions are classified into three. These are

1. Saturated solution
2. Unsaturated solution
3. Supersaturated solution

1. **Saturated solution** is a solution at equilibrium that contains the maximum amount of dissolved solute at a given temperature.

➦ It does not dissolve additional solute at the given temperature.

2. **Unsaturated solution** contains less than the equilibrium concentration of dissolved solute.
   ➦ It dissolves additional solute at a given temperature.
3. **Supersaturated solution** contains more than the equilibrium concentration.
   ➦ It is unstable and develop a crystal easily.
   ➦ A super saturated solution can be prepared from saturated solution if the solute is more soluble at higher temperature.

## Effect of temperature on solubility

**Solubility** is the maximum amount of solute that will dissolve in a given amount of solvent at a given temperature to form stable solution.

➦ Most solids are more soluble at high temperature.
➦ Some solids have higher solubility up to a certain temperature and lower solubility at still higher temperature.

Example

Consider the following salts.

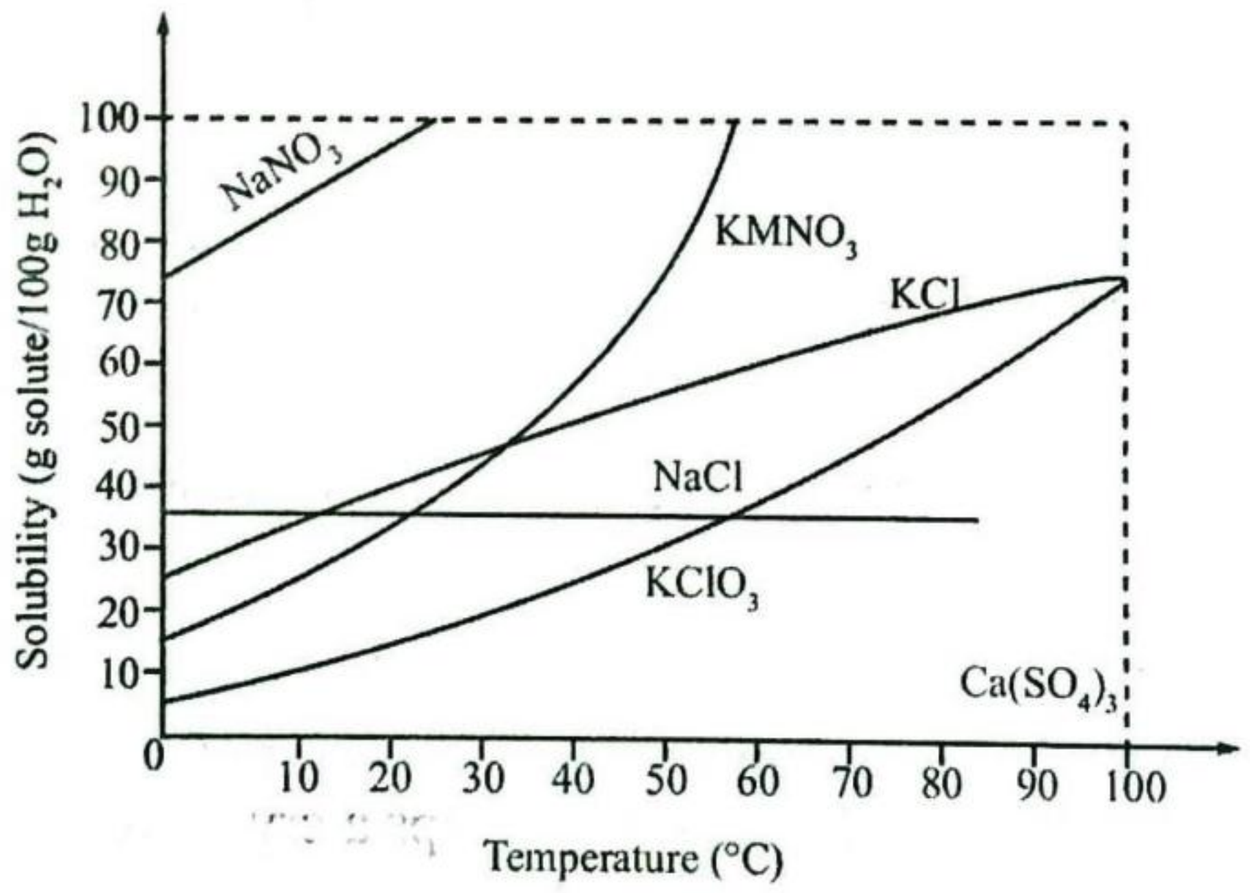



Figure 7.2 Relationship between solubility of some ionic compounds and temperature

**Solute (s) + water + heat ⇌ saturated solution**

➦ For gaseous solutes, increasing temperature decreases solubility because gas mix exothermically.

**Solute (g) + water ⇌ Saturated solution + heat**

## Effect of Pressure on solubility

- Pressure has little effect on the solubility of liquids and solids because they are almost incompressible.
- Pressure has effect on solubility of gaseous solute. As pressure increases, solubility of gas also increases. This effect was explained by **Henry's law**.

**Henry law** states that the solubility of a gas (Cg) is directly proportional to the partial pressure of the gas (Pg) above the solution.

$$Cg = K_H . P_g$$

$K_H$ - Henry's law constant

### Example

The partial pressure of carbondixoide gas inside a bottle of coca is 4 atm at 25°C. What is the solubility of $CO_2$? ($k_H$ for $CO_2$ = $3.3 \times 10^{-2}$ mol/L.atm)

**Solution**

$$C_{CO_2} = k_H . P_{CO_2}$$
$$= 3.3 \times 10^{-2} mol / L / atm \times 4atm$$
$$= 0.132 mol / L$$

## Illustrative Examples

24. When a small amount of crystal solute is added to the supersaturated solution, the solute crystal will: (EUEE ... 2008 E.C)

A. Grow bigger
B. Slightly dissolve
C. Remain unchanged
D. Dissolve completely

**Solution**

Supersaturated solution develops a crystal easily. As a result the size of the crystal increases.

**Answer: A**

25. The solubility of oxygen gas in water at 25°C and 1atm pressure of oxygen is 0.041g/L. The solubility of oxygen in water at 3atm and 25°C is _____ g/L. (EUEE ... 2007 E.C.)

A. 0.014 B. 0.31 C. 0.041 D. 0.123

**Solution**

➦ Solubility is directly proportional to partial pressure. If the partial pressure of the gas triples, the solubility of the gas also triples.

$C_{O_2} = 3 \times 0.041 g/L = 0.123 g/L$

**Answer: D**

26. A solution which has the capacity to dissolve more solute is called (EUEE ...1995 E.C)

A. Unsaturated
B. Saturated
C. Supersaturated
D. Concentrated

**Solution**

Unsaturated solution dissolves additional solute at a given temperature.

**Answer: A**

27. In which type of solution does an equilibrium always exist?

A. Supersaturated
B. Saturated
C. Unsaturated
D. Diluted

**Solution**

➦ Dynamic equilibrium is established when the rate of dissolution is equal to the rate of crystallization. The solution that always exist at equilibrium is saturated solution.

**Answer: B**

28. Ambo water is carbondioxide by injecting the liquid with carbondixoide gas. Under what conditions is the solubility of carobndixoide gas the greatest?

A. Low temperature, low pressure
B. Low temperature, high pressure
C. High temperature, high pressure
D. High temperature, low pressure

**Solution**

➦ The solubility of gas in liquid is the highest at low temperature and high pressure.

**Answer: B**

29. Which of the following is NOT correct?

A. Addition of a strongly solvated solute decreases the solubility of a gas in liquid.
B. For solid dissolved in a liquid that is exothermic, increase in temperature increases the solubility.

C. The solubility of a substance is its concentration in the saturated solution.

D. For solution to occur, solvent - solute attractions must overcome solute - solute and solvent - solvent attractions.

**Solution**

➦ For exothermic mixing, increasing temperature, decreases solubility.

**Answer: B**

30. The heat of solution of sodium iodide is -7500 J/mol. Is NaI more soluble at 10°C or 40°C?

**Solution**

➦ Since the heat of solution is below zero, NaI is more soluble at 10°C.

31. The solubility of potassiumdichromate at 20°C and 40°C is 10g and 40g per 100g of water respectively. What is the sign of heat of solution for the salt?

**Solution**

➦ Heat of solution is positive because as temperature increases solubility also increases.

32. You are given three beakers containing aqueous solution of solute A, one saturated, one unsaturated and one supersaturated. How can you identify which solution is which?

**Solution**

➦ By adding a crystal of A, it is possible to identify the type of solutions. If:

Size of A decreases - unsaturated solution.

Size of A increases - supersaturated solution.

Size of A remains same - saturated solution.

33. At 25°C, unknown gas X collected over water at a total pressure of 100KPa is soluble to the extent of 0.04g/L. What would its solubility be if its partial pressure over water were 106 KPa? (Vapor pressure of water is 3kPa at 25°C)

**Solution**

Pressure according to Dalton's law of partial pressure is the sum of partial pressure of X and vapor pressure of water.

$P_T = P_X + P_{H_2O}$

$$P_X = P_T - P_{H_2O} = 100 - 3 = 97KPa$$

$$K_H = \frac{C_X}{P_X} = \frac{0.04g/L}{97kPa} = 4.12\times10^{-4} g/L.KPa$$

$\Rightarrow$ The solubility of X when $P_X = 106KPa$

$C_X = K_H.P_X = 4.12 \times 10^{-4}g/L.kPa.106\ KPa$

$C_X = 0.044g/L$

34. The solubility of sodium selenate, $Na_2SeO_4$ is 84g/100g of water at 35°C. If a solution is obtained by dissolving 92g of $Na_2SeO_4$ in 200g of water at 35°C. What is this solution?

**Solution**

- 84g/100g - saturated solution
- 92g/200g = 46g/100g becomes unsaturated solution

35. How can you prepare supersaturated solution?

**Solution**

Supersaturated solution is formed by saturating a solution at high temperature.

36. If air contains 78% $N_2$ by volume, what is the solubility of $N_2$ in water at 25°C and 1 atm ($K_H$ for $N_2$ in $H_2O$ at 25°C = $7 \times 10^{-4}$mol/L.atm)

**Solution**

- In a mixture of gases, the volume percent of each gas times the total pressure equals its partial pressure according to Dalton's law of partial pressure.

$$\Rightarrow P_{N_2} = 78\%, P_T = \frac{78}{100}.P_T = 0.78\times1atm$$

$$P_{N_2} = 0.78atm$$

$$C_{N_2} = K_H.P_{N_2} = 7\times10^{-4} mol/L.atm \times 0.78atm$$

$$C_{N_2} = 5.46\times10^{-4} mol/L$$

## 7.5 Ways of expressing the concentration of solution

$$\text{Concentration} = \frac{\text{amount of solute}}{\text{amount of solution or solvent}}$$

- The concentration of a solution can be expressed qualitatively and quantitatively.
- **Qualitative expression** - is expressed interms of dilute or concentrated solution.
- **Quantitative expression** - can be expressed using five different ways.

1. Mass percent, ppm, PPb, volume percent
2. Mole fraction
3. Molarity
4. Normality
5. Molality

## 1. Mass percent, ppm, PPb, volume percent

$$\text{Mass percent of solute} = \frac{\text{Mass solute}}{\text{Mass solute + mass of sovlent}} \times 100$$

$$= \frac{\text{Mass solute}}{\text{Mass of solution}} \times 100$$

Mass percent solute + mass percent of solvent = 100

**% - parts per hundred**

**ppm - parts per milliion**

**PPb - parts per billion**

- $\text{PPm solute} = \frac{\text{mass solute}}{\text{mass solution}} \times 10^6 = \%.10^4$
- $\text{PPb solute} = \frac{\text{mass solute}}{\text{mass solution}} \times 10^9 = \text{PPm} \times 10^3 = \% \times 10^7$
- $\text{Volume percent (\% v/v)} = \frac{\text{Volume of solute}}{\text{Volume of solution}} \times 100$

### Example

The label on a 0.5L bottle says 11% alcohol by volume. How many liters of alcohol does the wine contain?

**Solution**

$$\text{Volume of alcohol} = 0.5\text{L} \times 11\% = 0.055\text{L}$$

## Illustrative Examples

37. A beaker filled to the 100mL mark with salt (the mass of the salt is 100g) and another beaker to the 100mL mark with water (the water has a mass of 100g) are mixed in a bigger beaker until the salt is completely dissolved. What will be the mass of the solution?(EUEE ... 2006 E.C)

A. It will be much more than 200g.
B. It will be much smaller than 200g.
C. It will be exactly 200g.
D. It will be slightly more than 200g.

**Solution**

Mass of solution = Mass of salt + Masss of water

= 100g + 100g = 200g

Mass is additive but volume is not necessarily additive due to expansion and contraction.

**Answer: C**

38. What is teh concentration of a solution in parts permillion, if 0.02g of $Na_3PO_4$ is dissolved.

A. 0.02PPm
B. 0.2PPm
C. 2PPm
D. 20PPm

**Solution**

$$PPm = \frac{\text{mass of } Na_3PO_4}{\text{Mass of } Na_3PO_4 + \text{Mass of } H_2O} \times 10^6$$

$$= \frac{0.02g}{(0.02+1000)g} \times 10^6 = 20PPm$$

**Answer: D**

39. A commercial bleaching solution ("berekina") contains mass percent of 5% sodium hypochlorite, NaOCl. What is the mass of NaOCl in a bottle containing 2500g of bleaching solution?

**Solution**

Mass of NaOCl = 5%. mass of solution

$$= \frac{5}{100} \times 2500g = 125g$$

40. A 500g toothpaste sample has 0.2g fluoride concentration. What is the concentration of fluoride interms of PPm?

**Solve by your own: Answer: 400ppm**

41. What percent is gold in 18 karat gold?

**Solution**

Karat is parts per 24

$$\Rightarrow \%\ \text{gold} = \frac{18}{24} \times 100 = 75\%$$

42. A certain vinegar is 6.02% acetic acid ($C_2H_4O_2$) by mass. How many grams of $C_2H_4O_2$ are contained in a 355mL bottle of vinegar. Assume a density of 1.01g/mL.

**Solution**

$$\text{Mass of vinegar} = \text{density} \times \text{volume} = 1.01\ \text{g/mL} \times 355\text{mL}$$

$$= 358.55\text{g}$$

$$\%C_2H_4O_2 = \frac{\text{mass of } C_2H_4O_2}{\text{mass of vinegar}} \times 100$$

$$\text{mass of } C_2H_4O_2 = \frac{6.02 \times 358.55\text{g}}{100} = 21.58\text{g}$$

43. If 150g of orange juice contains 120mg of ascorbic acid, what is the concentration of ascorbic acid expressed in

a. % by mass b. PPm c. PPb

**Solution**

a. $$\%\ \text{ascorbic acid} = \frac{\text{mass of ascorbic acid}}{\text{mass of orange juice}} \times 100$$

$$= \frac{0.12\text{g}}{150\text{g}} \times 100 = 0.08\%$$

b. $\text{PPm} = \% \times 10^4 = 0.08 \times 10^4 = 800\text{PPm}$

c. $\text{PPb} = \text{PPm} \times 10^3 = 8 \times 10^5\text{ppb}$

44. A 6 mol $H_2SO_4$ solution has a density of 1.338 g/mL. What is the percent by mass of Sulphuric acid in 1L of this solution?

**Solution**

$V = 1\text{L} = 1000\text{mL}.\ d = 1.338\text{g/mL}$

$m_{sol} = d \times v = 1.338\text{g/mL} \times 1000\text{mL} = 1338\text{g}$

$$m_{H_2SO_4} = n \times M = 6mol \times 98g/mol = 588g$$

$$\%H_2SO_4 = \frac{n_{H_2SO_4}}{m_{solution}} \times 100 = \frac{588g}{1338g} \times 100 = 44\%$$

45. A water sample is found to have 80PPb of chloroform, $CHCl_3$. How many grams of $CHCl_3$ would be found in a glassful (2500mL) of this water?

**Solution**

density of water = 1g/mL

Mass of water sample = V.d = 2500mL × 1 g/mL = 2500g

$$m_{CHCl_3} = ppm.M_{water\ sample} = \frac{80}{10^9}.2500g = 2 \times 10^{-4} g$$

46. In the laboratories of an oil company, a research chemist dissolved 1.0g of octane, $C_8H_{18}$ in 150g of benzene, $C_6H_6$. What is the mass percent of the solute in the solution?

**Answer: 0.64%**

## 2. Mole fraction (X)

- Mole fraction is the ratio of the moles of a certain substance to the total moles in a mixture of substances.
- Solutions that do not transmit electricity are called **non - electrolytic solution.**

### A. For non - electrolytes

$$X_{solute} = \frac{n_{solute}}{n_{solute} + n_{solvent}}$$

$$X_{solvent} = \frac{n_{solvent}}{n_{solute} + n_{solvent}}$$

The sum of mole fractions of component of a mixture is one.

$X_{solute} + X_{solvent} = 1$

Mole fraction has no unit

- Solutions that transmit electricity are called **electrolytic solutions**.

### B. For electrolytic solutions

- There are positive ions, negative ions and solvent in electrolytic solution.

$$\Rightarrow X_{cation} = \frac{n_{cation}}{n_{cation} + n_{anion} + n_{solvent}}$$

$$X_{anion} = \frac{n_{anion}}{n_{cation} + n_{anion} + n_{solvent}}$$

$$X_{solvent} = 1 - (X_{cation} + X_{anion})$$

## Illustrative Examples

47. What is the mole fraction of methanol in a mixture of 16g methanol ($CH_3OH$) and 27g water ($H_2O$)?

A. 0.75 B. 0.25 C. 0.65 D. 0.35

**Solution**

$$n_{meta} = \frac{m_{meth}}{M_{meth}} = \frac{16g}{32g/mol} = 0.5mol$$

$$n_{H_2O} = \frac{m_{H_2O}}{M_{H_2O}} = \frac{27g}{18g/mol} = 1.5mol$$

$$X_{meth} = \frac{n_{meth}}{n_{meth} + n_{H_2O}} = \frac{0.5mol}{(0.5+1.5)mol} = 0.25$$

**Answer: B**

48. 3 moles of sodium sulphide $Na_2S$, is dissolved in 2 moles of water. Calculate the mole fraction of $Na^+$, $S^{2-}$ and $H_2O$.

**Solution**

$$Na_2S(s) \xrightarrow{H_2O} 2Na^+(aq) + S^{2-}(aq)$$

1mol 2mol 1m

3mol ? ?

$$\Rightarrow n_{Na^+} = 6mol, n_{S^{2-}} = 3mol, n_{H_2O} = 1mol$$

$$X_{Na^+} = \frac{n_{Na^+}}{n_{Na^+} + n_{S^2} + n_{H_2O}} = \frac{6mol}{(6+3+1)mol} = 0.6$$

$$X_{S^{2-}} = \frac{n_{S^{2-}}}{n_{Na^+} + n_{S^{2-}} + n_{H_2O}} = \frac{3mol}{(6+3+1)mol} = 0.3$$

$$X_{H_2O} = 1 - (X_{Na^+} + X_{S^{2-}}) = 1 - (0.6 + 0.3) = 0.1$$

49. Calculate the mole fractions of $H_2O$ and $CH_3CH_2OH$ in a mixture of equal masses of water and ethanol.

**Solution**

Let the mass be m.

$$X_{eth} = \frac{\frac{m}{M_{eth}}}{\frac{m}{M_{eth}} + \frac{m}{M_{wat}}} = \frac{\frac{m}{46}}{\frac{m}{46} + \frac{m}{18}} = \frac{\frac{m}{46}}{\frac{18m + 46m}{46 \times 18}}$$

$$X_{eth} = \frac{m}{46} \cdot \frac{46.18}{64m} = \frac{18}{64} = 0.28$$

$$X_{water} = 1 - X_{eth} = 1 - 0.28 = 0.72$$

For equal masses of two substances A and B to determine the mole fraction of A and B use the formula below.

$$X_A = \frac{M_B}{M_A + M_B} \qquad M_A \text{ - molar mass of A}$$

$$X_B = \frac{M_A}{M_A + M_B} \qquad M_B \text{ - molar mass of B}$$

50. Calculate the mole fraction of all the species present in a solution prepared by dissolving 171g of aluminum $(Al_2(SO_4)_3)$ in 27g of water.

**Solution**

$$n_{Al_2(SO_4)_3} = \frac{m}{M} = \frac{171g}{342g/mol} = 0.5mol$$

$$n_{H_2O} = \frac{m}{M} = \frac{27g}{18g/mol} = 1.5mol$$

$$\underset{\substack{1\,mol \\ 0.5mol}}{Al_2(SO_4)_3} \longrightarrow \underset{\substack{2mol \\ ?}}{2Al^{3+}} + \underset{\substack{3mol \\ ?}}{3SO_4^{2-}}$$

$$n_{Al}^{3+} = 1mol, n_{SO_4^{2-}} = 1.5mol$$

$$X_{Al^{3+}} = \frac{1mol}{(1.5 + 1.5 + 1)mol} = 0.25$$

$$X_{SO_4^{2-}} = \frac{1.5mol}{(1.5 + 1.5 + 1)mol} = 0.375$$

$$X_{H_2O} = \frac{1.5}{(1.5 + 1.5 + 1)} = 0.375$$

## 3. Molarity (M)

- Molarity is the most important concentration unit in the laboratory.
- Molarity is defined as number of mole of solute divided by volume of solution in liter.

$$\text{Molarity} = \frac{\text{number of mole of solute}}{\text{volume of solution in liter}}$$

$$\boxed{M = \frac{n}{v}} \quad \boxed{n = M.V}$$

$$n = \frac{m}{M_{wt}}$$ m – given mass of solute; $M_{wt}$ - molecular weight

$$\boxed{M = \frac{m}{M_{wt}.V}}$$

## Illustrative Examples

51. How many moles of $H_2SO_4$ are needed to prepare 5.0 liters of a 2.0M solution of $H_2SO_4$? (EUEE ... 2008 E.C)

A. 2.5 B. 5.0 C. 20 D. 10

**Solution**

$$n = M.V = 2\frac{mol}{L} \times 5L = 10mol$$

**Answer: D**

52. What is the molarity of a 5g hydrogen peroxide ($H_2O_2$) in 100mL solution that is used for hair bleaching? (EUEE ... 2007 E.C)

A. 0.015M B. 0.15M C. 1.5M D. 3M

**Solution**

$$V = 100mL = 0.1L,\ M_{wt} = 34\ g/mol,\ m = 5g$$

$$M = \frac{m}{M_{wt}.V} = \frac{5g}{34g/mol \times 0.1L} = 1.5M$$

**Answer: C**

53. The concentration of nitrate ion in a solution that contains 0.900M aluminum nitrate is

A. 0.90 B. 0.45 C. 0.30 D. 2.70

**Solution**

➦ The formula of aluminum nitrate is $Al(NO_3)_3$

$$\underset{\substack{1M\\0.90M}}{Al(NO_3)_3(s)} \xrightarrow{H_2O} Al^{3+}(aq) + 3\underset{\substack{3M\\?}}{NO_3^-}$$

$$\Rightarrow [NO_3^-] = 3 \times 0.90M = 2.70M$$

**Answer: D**

54. Which one of the following solutions has the greatest concentration of $Na^+$ ions? (EUEE)

A. 0.026M $Na_2CO_3$
B. 0.014M NaCl
C. 0.013M $Na_3PO_4$
D. 0.032M $NaNO_3$

**Solution**

$[Na^+]$ = subscript of Na × [salt]

A. $[Na^+] = 2 \times 0.026M = 0.052M$
B. $[Na^+] = 1 \times 0.014M = 0.014M$
C. $[Na^+] = 3 \times 0.013M = 0.039M$
D. $[Na^+] = 1 \times 0.032M = 0.032M$

**Answer: A**

55. What is the mass of $CuCl_2.6H_2O$ required to prepare 300mL of 0.10M aqueous solution?

**Solution**

$$m = M.M_{wt}.V$$

$$m = 0.1\frac{mol}{L} \times 242.5\frac{g}{mol} \times 0.3L = 7.28g$$

56. How many chloride ions are there in 1 mole of $CaCl_2$? (EUEE)

**Solution**

$$n_{Cl^-} = 2 \times n_{CaCl_2} = 2 \times 1mol = 2mol$$

1 mol = $6.02 \times 10^{23}$ $Cl^-$ ions
2 mol = ?

$\Rightarrow$Number of $Cl^-$ ions = $2 \times 6.02 \times 10^{23} = 1.204 \times 10^{24}$

57. Calculate the number of grams of $H_2SO_4$ needed to make 200mL of 0.200M $H_2SO_4$?

**Answer:** $m_{H_2SO_4} = 3.92g$

58. What volume of 5.0M sulphuric acid contains 0.100mol $H_2SO_4$ in milliliters?

**Solution**

$$V = \frac{n}{M} = \frac{0.1mol}{5mol/L} = 0.02L = 20mL$$

59. What is the molarity of a solution obtained by dissolving 0.01mol of NaCl in 500mL of solution? (EUEE)

**Answer: 0.02M**

60. How many grams of $Ca(OH)_2$ are contained in 1500mL of 0.0250M $Ca(OH)_2$ solution?

**Solution**

$m = M.\ M_{wt}.\ V = 0.0250mol/L \times 74g/mol \times 1.5L = 2.78g$

## 4. Normality (N)

➦ Normality is defined as the number of chemical equivalents of solute per liter of solution.

$$\text{Normality} = \frac{\text{Number of equivalents of solute}}{\text{Volume of solution in liter}}$$

$$\boxed{N = \frac{No.eq}{V}}$$

$$No.eq = \frac{\text{given mass}}{\text{Equivalent mass}}$$

$$\boxed{No.eq = \frac{m}{E}} \qquad E - \text{equivalent mass}$$

$$\boxed{N = \frac{m}{E.V}}$$

**Equivalent mass (E)** - its unit is gram per equivalents.

**i For acids**

$$E_{acid} = \frac{M_{acid}}{\text{No. of } H^+ \text{ ionized}}$$

$M_{acid}$ - molar mass of the acid

**Example**

What is the equivalent mass of $H_2SO_4$? Assume complete ionization.

**Solution**

$$H_2SO_4 \longrightarrow 2H^+ + SO_4^{2-}$$

No. of $H^+$ ionized = 2 eq/mol

$M_{H_2SO_4} = 98g/mol$

$$E_{H_2SO_4} = \frac{M_{H_2SO_4}}{\text{No. of } H^+} = \frac{98g/mol}{2eq/mol} = 49g/eq$$

### ii For bases

$$E_{base} = \frac{M_{base}}{\text{No. of } OH^- \text{ ionized}}$$

### iii For salts

$$E = \frac{M_{salt}}{\text{total positive charge}}$$

**Example**

What is the equivalent mass of $Ca(OH)_2$?

$Ca(OH)_2 \rightarrow Ca^{2+} + 2OH^-$

$\Rightarrow$No. of $OH^-$ ionized = 2 eq/mol

$$E = \frac{M_{CA(OH)_2}}{\text{No. of } OH^-} = \frac{74 \text{ g/mol}}{2 \text{ eq/mol}} = 37 \text{ g/eq}$$

**Example**

What is the equivalent mass of $Ca(OH)_2$?

**Solution**

$Al_2(SO_4)_3 \longrightarrow 2Al^{3+} + 3SO_4^{2-}$

$\Rightarrow$Total positive charge $2 \times 3 = 6$ eq/mol

$$E_{Al_2(SO_4)_3} = \frac{M}{\text{total positive change}} = \frac{342 \text{ g/ mol}}{6 \text{ eq/mol}} = 57 \text{ g/eq}$$

### iv For oxidizing and reducing agents

$$E_{ox} = \frac{M_{ox}}{\text{electrons gained}}$$

$$E_{red} = \frac{M_{red}}{\text{electrons lost}}$$

$E_{ox}$ - equivalent mass of oxidizing agent.
$M_{ox}$ - molar mass of oxidizing agent
$E_{red}$ - equivalent mass of reducing agent
$M_{red}$ - molar mass of reducing agent

**Example**

Find the equivalent mass of $KMnO_4$ based on the reaction below.

$2KMnO_4 + 10KCl + 6H_2SO_4 \rightarrow 6K_2SO_4 + 2MnSO_4 + 5Cl_2 + 6H_2O$

**Solution**

Determine the oxidation number of Mn in $KMnO_4$ and $MnSO_4$.

$$2\overset{+7}{K}\overset{+sc}{Mn}O_4 \longrightarrow 2\overset{+2}{Mn}SO_4$$

$\Rightarrow$ 5 electrons are gained

$$E_{KMnO_4} = \frac{M_{kMnO_4}}{\text{electrons gained}} = \frac{158\ g/mol}{5\ eq/mol}$$

$$E_{KMnO_4} = 31.6\ g/eq$$

$$\boxed{E = \frac{M_{wt}}{Z}}$$

$M_{wt}$ - molar mass of acid, base, salt, oxidizing agent or reducing agent

Z = No. of $H^+$ for acids

- ➦ No. of $OH^-$ for bases
- ➦ Total positive change for salts.
- ➦ Electrons lost for reducing agent.
- ➦ Electrons gained for oxidizing agent.

$$N = \frac{m}{E.V} \quad \text{but} \quad E = \frac{M_{wt}}{Z}$$

$$N = \frac{m}{\frac{M_{wt}}{Z}.V}$$

$$N = \frac{m}{M_{wt}V}Z. \quad \text{but} \quad n = \frac{m}{M_{wt}}$$

$$N = \frac{n}{V}.Z, \quad \text{but } M = \frac{n}{V}$$

$$\boxed{N = M.\ Z,\ N \geq M,\ \text{No. of eq} = n \times Z}$$

The unit of normality is eq/L or normal (N).

## Illustrative Examples

61. What is the normality of a 2.3M sulphuric acid solution? (UEE ... 2001 E.C)

A. 1.15N B. 2.3N C. 4.6N D. 6.9N

**Solution**

$$H_2SO_4 \longrightarrow 2H^+ + SO_4^{2-}$$

$\Rightarrow Z = 2\ eq/mol$

$N = M.\ z = 2.3mol/L \times 2\ eq/mol = 4.6N$

**Answer: C**

62. What is the normality of 1.0M solution of $Na_2CO_3$? (EUEE...2002 E.C.)

A. 1N B. 2N C. 0.5N D. 3N

**Solution**

$$Na_2CO_3 \longrightarrow 2Na^+ + CO_3^{2-}$$

$\Rightarrow Z = 2 \times (+1) = 2eq/mol$

$N = M.\ Z = 1\ mol/L \times 2\ eq/mol = 2N$

**Answer: B**

63. Calculate the number of equivalent present in 1.5mol $H_3A$ if the acid is

a. Completely neutralized to give $A^{3-}$

b. Converted to $HA^{2-}$ c. Converted to $H_2A^-$

**Solution**

a. $H_3A \rightarrow 3H^+ + A^{3-}$

$\Rightarrow Z = 3\ eq/mol$

No. of eq = $n \times z = 1.5\ mol \times 3\ eg/mol = 4.5eq$

b. $H_3A \rightarrow 2H^+ + HA^{2-}$

$\Rightarrow Z = 2\ eq/mol$

No. of eq = $n \times z = 1.5\ mol \times 2\ eq/mol = 3\ eq$

c. $H_3A \rightarrow H^+ + H_2A^-$

$\Rightarrow z = 1\ eq/mol$

No. eq = $n \times z = 1.5mol \times 1\ eq/mol = 1.5eg$

64. Determine the normality of a solution prepared by dissolving 16g of $Fe_2(SO_4)_3$ in 200ml of solution.

**Solution**

$$Fe_2(SO_4)_3 \longrightarrow 2Fe^{3+} + 3SO_4^{2-}$$

$\Rightarrow Z = 2 \times 3 = 6\ eq/mol,\ M_{wt} = 400g/mol$

$V = 0.2L$

$$N = \frac{m.Z}{M_{wt}.V} = \frac{16\ g \times 6\ eq/mol}{400\ g/mol \times 0.2L} = 1.2N$$

65. Calculate the mass of $CuSO_4$ in 250mL of solution if the concentration is 0.65N.

**Solution**

$$CuSO_4 \longrightarrow Cu^{2+} + SO_4^{2-}$$

$\Rightarrow Z = 2$ eq/mol, $M_{wt} = 160$ g/mol

$$N = \frac{m.Z}{M_{wt}.V}$$

$$m = \frac{N.M_{wt}.V}{Z} = \frac{0.65\ \text{eq/L} \times 160\ \text{g/mol} \times 0.25\,\text{L}}{2\ \text{eq/mol}}$$

$m = 13\text{g}$

66. Determine the equivalent mass of $H_3PO_4$ based on the reactions below.
    a. $H_3PO_4 + NaOH \rightarrow NaH_2PO_4 + H_2O$
    b. $H_3PO_4 + 2NaOH \rightarrow Na_2HPO_4 + 2H_2O$
    c. $H_3PO_4 + 3NaOH \rightarrow Na_3PO_4 + 3H_2O$

**Solution**

a. One H is replaced with Na, $Z = 1$ eq/mol

$$E_{H_3PO_4} = \frac{98}{1} = 98\text{g/eq}$$

b. Two H are replaced, $Z = 2$ eq/mol

$$E_{H_3PO_4} = \frac{98}{2} = 49\ \text{g/eq}$$

c. Three H are replaced, $Z = 3$ eq/mol

$$E_{H_3PO_4} = \frac{98}{3} = 32.7\ \text{g/eq}$$

67. What volume of a 0.35N solution contains 4 meq of solute?

**Solution**

$$N = \frac{\text{No. of eq}}{V}$$

$$V = \frac{\text{No. eq}}{N} = \frac{4\ \text{meq}}{0.35\ \text{eq/L}} = 11.4\text{mL}$$

## 5. Molality

- The moality of a solution is the number of mole of solute per mass of solvent in kilogram.

$$m = \frac{n}{m(\text{kg})\text{solvent}}$$

- The unit of molality is mole per kilogram or molal.
- To prepare molal solution, we need to measure the mass of solvent not the amount of solution like molarity and normality.

**Example**

What is the molality of a solution obtained by dissolving 11.5g of ethanol is 400g of water?

**Solution**

$$n_{eth} = \frac{m}{M} = \frac{11.5g}{46\ g/mol} = 0.25mol$$

$$m_{H_2O} = 400g = 0.40kg$$

$$\Rightarrow m = \frac{n}{kgH_2O} = \frac{0.25mol}{0.40kg} = 0.625m$$

## Illustrative Examples

**68. Calculate the molality of a solution prepared by dissolving 47.5g of $MgCl_2$ in 200g of water.**

**Solution**

$$n = \frac{m}{M} = \frac{47.5\ g}{95\ g/mol} = 0.5mol$$

$$m = \frac{n}{kgH_2O} = \frac{0.5mol}{0.2kg} = 2.5m$$

**69. Find the mass of calcium chloride required to prepare 0.125m $CaCl_2$ in 800g of water.**

**Solution**

$$n = m.\ kg\ H_2O = 0.125mol/kg \times 0.80kg = 0.1mol$$

$$m_{CaCl_2} = n \times M_{CaCl_2} = 0.1mol \times 111g/mol = 11.1g$$

**70. What is the mass of water required to dissolve 14.5g of glucose, if the molality of the solution is 0.24m?**

**Solution**

$$m = \frac{m_{glucose}}{M_{glucose}.kgH_2O}$$

$$kgH_2O = \frac{14.5g}{18\ g/mol \times 0.24\ mol/kg} = 0.336kg$$

$\Rightarrow$ mass of $H_2O$ in grams in 336

# 7.6 Conversion of Concentration Units

➦ Mass percent and density are converted into molarity, normality and molality.

- Mass percent is converted into molality and mole fraction.
- Mole fraction is converted into molality.
- Molality is also converted into mole fraction.

## Illustrative Examples

71. What is the moality of a solution that contains 51.2g of naphthalene, $C_{10}H_{18}$ in 500mL of carbon tetra chloride? The density of $CCl_4$ is 1.60g/mL. (EUEE ... 2008 E.C.)

A. 0.750M B. 0.500M C. 0.840M D. 1.69M

**Solution**

$$m_{CCl_4} = V.d = 500mL \times 1.60g/mL = 800g = 0.80kg$$

$$n_{C_{10}H_8} = \frac{m}{M_{C_{10}H_8}} = \frac{51.2g}{128g/mol} = 0.4mol$$

$$m = \frac{n}{kgCCl_4} = \frac{0.4mol}{0.80kg} = 0.50m$$

**Answer: B**

72. An aqueous solution is 70% nitric acid ($HNO_3$) by mass. What is the concentration of $HNO_3$ expressed in molality? (EUEE ... 2006 E.C)

A. 0.55m B. 8.62m C. 11.1m D. 37m

**Solution**

70% by mass $HNO_3$ means 70g $HNO_3$ dissolved in 30g $H_2O$.

$$n_{HNO_3} = \frac{m}{M} = \frac{70g}{63g/mol} = 1.11mol$$

$$m = \frac{n}{kgH_2O} = \frac{1.11mol}{0.03kg} = 37.04m$$

**Answer: D**

73. What is the mole fraction of one molal aqueous solution? (EUEE)

A. 0.009 B. 0.018 C. 0.027 D. 0.036

**Solution**

$$1\ molal = \frac{1mol\ solute}{1kg\ H_2O}$$

$$\Rightarrow n_{solute} = 1mol, m_{H_2O} = 1kg = 1000g$$

$$n_{H_2O} = \frac{m_{H_2O}}{M_{H_2O}} = \frac{1000g}{18\ g/mol} = 55.6mol$$

$$\Rightarrow x_{solute} = \frac{n_{solute}}{n_{solute} + n_{H_2O}} = \frac{1mol}{(1+55.6)mol} = 0.0176$$

**Answer: B**

74. A solution contains equal masses of glycerol $C_3H_8O_3$ and water. The density of the solution is 1.10g/mL. What is the molality of the solution?

**Solution**

$$\Rightarrow m_{solution} = d \times v = 1.10g/mL \times 1000mL = 1100g$$

Volume of solution is assumed to be 1L

$$m_{water} = m_{glycerol} = \frac{1100g}{2} = 550g$$

$$m = \frac{m_{glycerol}}{M_{glycerol}.kgH_2O} = \frac{550g}{92g/mol \times 0.550kg}$$

$$m = 10.90m$$

75. A 2.45M sulphate acid solution has a density of 1.15 g/mL at 20°C. Calculate

a. Normality
b. Molality
c. Mass percent of $H_2SO_4$
d. Mole fraction of $H_2SO_4$

**Solution**

a. $N = M.Z = 2.45\ mol/L \times 2\ eq/mol = 4.90N$

b. $2.45M = 2.45mol/L$

$$\Rightarrow n_{H_2SO_4} = 2.45mol, V_{solution} = 1L = 1000mL$$

$$m_{H_2SO_4} = n \times M = 2.45mol \times 98g/mol = 240g$$

$$m_{solution} = d.v = 1.15\ g/mL \times 1000mL = 1150g$$

$$m_{H_2O} = m_{solution} - m_{H_2SO_4} = 1150 - 240 = 910g$$

$$\Rightarrow m = \frac{n}{kgH_2O} = \frac{2.45mol}{0.910kg} = 2.69m$$

c. $$\%H_2SO_4 = \frac{m_{H_2SO_4}}{m_{solution}} \times 100 = \frac{240g}{1150g} \times 100 = 21\%$$

d. $n_{H_2O} = \frac{m}{M} = \frac{910g}{18g/mol} = 50.6mol$

$$X_{H_2SO_4} = \frac{n_{H_2SO_4}}{n_{H_2SO_4} + n_{H_2O}} = \frac{2.45mol}{(2.45+50.6)mol} = 0.046$$

76. What is the molarity of potassium iodide in a solution that is 5% KI by mass and has a density of 1.038 g/mL?

**Solution**

**Case I:**

$$M = \frac{d \times \% \times 10}{M_{solute}} = \frac{1.038 \times 5 \times 10}{166} = 0.313M$$

**Case II:**

The volume of a solution is 1L or 1000mL.

$m_{solution} = d \times v = 1.038g/mL \times 1000mL = 1038g$

$m_{kI} = 5\%.M_{solution} = \frac{5}{100}.1038g = 51.9g$

$m_{kI} = \frac{m_{kI}}{M_{kI}} = \frac{51.9g}{166\ g/mol} = 0.313mol$

$M = \frac{n}{V} = \frac{0.313mol}{1L} = 0.313M$

77. What is the molarity of a 300mL solution which is 25% by mass $H_2SO_4$? The solution mass is 353.4g?

**Solution**

$$m_{H_2SO_4} = 25\%.m_{solution} = \frac{25}{100} \times 353.4g = 88.35g$$

$$M = \frac{m_{H_2SO_4}}{M_{H_2SO_4}.v} = \frac{88.35g}{98g/mol.0.3L} = 3.0M$$

78. An aqueous solution of ethylene glycol, $C_2H_6O_2$ has a mole fraction of ethylene glycol of 0.162. What is the molality of the solution?

**Solution**

Consider mole fraction as number of mole to determine molality.

$\Rightarrow X_{ethy} = 0.162 \Rightarrow n_{ethyl} = 0.162mol$

$$X_{water} = 1 - 0.162 = 0.838 \Rightarrow n_{H_2O} = 0.838mol$$

$$m_{H_2O} = n \times M = 0.838mol \times 18g/mol = 15.084g$$

$$m = \frac{n_{ethyl}}{kg\ H_2O} = \frac{0.162mol}{0.015084kg} = 10.74m$$

79. Calculate the molality of ethanol solution if mole fraction of ethanol is 0.25.

**Solve by your own: Answer: 18.5m**

80. A solution of lead (11) nitrate, $Pb(NO_3)_2$ contains 18% $Pb(NO_3)_2$ by mass. If the density of this solution is $1.18g/cm^3$, find

a. Molarity
b. Normality
c. Molality
d. Mole fraction of all the species in the solution

**Solution**

a. $$M = \frac{d \times \% \times 10}{M_{Pb(NO_3)_2}} = \frac{1.18 \times 18 \times 10}{207} = 1.03M$$

b. $N = M.Z = 1.03mol/L \times 2\ eq/mol = 2.06N$

c. $18\%\ Pb(NO_3)_2 \Rightarrow 18g\ Pb(NO_3)_2$ in $82g$ of $H_2O$

$$m = \frac{m_{Pb(NO_3)_2}}{M_{Pb(NO_3)_2}.kgH_2O} = \frac{18g}{207\ g/mol.0.082kg} = 1.06m$$

d. $$n_{Pb(NO_3)_2} = \frac{m}{M} = \frac{18g}{207\ g/mol} = 0.087mol$$

$$\Rightarrow nPb^{2+} = 0.087mol, n_{NO_3^-} = 2 \times 0.087 = 0.174mol$$

$$n_{H_2O} = \frac{m}{M} = \frac{82g}{18\ g/mol} = 4.56mol$$

$$X_{Pb^{2+}} = \frac{n_{Pb^{2+}}}{n_{Pb^{2+}} + n_{NO_3^-} + n_{H_2O}} = \frac{0.087mol}{(0.087 + 0.174 + 456)mol}$$

$$X_{Pb^{2+}} = 0.018$$

$$X_{NO_3^-} = \frac{n_{NO_3^-}}{n_{Pb^{2+}} + n_{NO_3^-} + n_{H_2O}} = \frac{0.174mol}{(0.087 + 0.174 + 4.56)\,mol}$$

$$X_{NO_3^-} = 0.036$$

$$X_{H_2O} = 1 - (0.018 + 0.036) = 0.946$$

81. The density of 5 molal solution of urea is 1.3 g.cm$^{-3}$. What could be the concentration of this solution interms of molarity?

**Solution**

$$5m = \frac{5mol}{1kgH_2O} \Rightarrow n_{urea} = 5mol$$

$$m_{H_2O} = 5kg = 1000g$$

$$V_{solution} = \frac{m}{d} = \frac{1000g}{1.3g/cm^3} = 769cm^3$$

$$M = \frac{n}{v} = \frac{5mol}{0.769L} = 6.5M$$

82. 10% aqueous solution of certain substance is isotonic with 5% aqueous solution of fructose ($C_6H_{12}O_6$). What is the molecular mass of the substance?

**Solution**

Isotonic solutions have the same concentration. For two solutes A and B for isotomic solutions.

$$\frac{\%A}{M_A} = \frac{\%B}{M_B}$$

$$\frac{10}{M_{sub}} = \frac{5}{M_{fructose}}$$

$$\frac{10}{M_{sub}} = \frac{5}{180}$$

$$M_{sub} = \frac{180 \times 10}{5} = 360g/mol$$

83. The molarity of 20% solution of sulphuric acid is 2.55M. What is the density of the solution?

**Solution**

$$M = \frac{d \times \% \times 10}{M_{solute}}$$

$$d = \frac{M.M_{H_2SO_4}}{\% \times 10} = \frac{2.55 \times 98}{20 \times 10}$$

$$d = 1.25g/cm^3$$

84. Gold occurs in a sample of sea water at a concentration of $1.1 \times 10^{-2}$ ppb. How many liters of sea water must be processed to recover 31.1g of gold? (Density of sea water = 1.025g/mL)

**Solution**

$$\text{PPb gold} = \frac{\text{mass gold}}{\text{mass sea warter}} \times 10^9$$

$$\text{mass sea water} = \frac{\text{mass gold}}{\text{PPb gold}} \times 10^9$$

$$= \frac{31.1\text{g}}{1.1\times10^{-2}} \times 10^9 = 2.83\times10^{12}\,\text{g}$$

$$V = \frac{m}{d} = \frac{2.83\times10^{12}\,\text{g}}{1.025\,\text{g/mL}} = 2.76\times10^{12}\,\text{mL}$$

$$V = 2.76\text{GL}$$

## 7.7 Solution Preparation

➦ The type of water used to prepare a solution is usually **distilled water.**

Steps used to prepare a solution in the laboratory.

1. Weigh the solute accurately.
2. Transfer the weighed solute into volumetric flask.
3. Add water through funnel.
4. Dissolve the solid gently.
5. Add more water to bring the level of solution exactly to the volumeric mark.

**Example**

A I molar $CaCl_2$ solution can be prepared.

- By dissolving 111g of $CaCl_2$ in water until the volume becomes one liter.
- By dissolving 55.5g of $CaCl_2$ in water until the volume becomes 500mL.
- By dissolving 27.75g $CaCl_2$ in water until the volume becomes 250mL.
- By dissolving 13.875g of $CaCl_2$ in water until the volume becomes 125mL.

## Diluting solution

➦ **Dilution** is the process of preparing less concentrated, solution from more concentrated solution by adding solvent.

➦ During dilution, concentration and volume change but number of mole of solute and equivalents of solute are not changed.

$$n_{\text{solute before dilution}} = n_{\text{solute after dilution}}$$

No. eq solute before dilution = No. eq. of solute after dilution

- $M_iV_i = M_fV_f$
- $N_iV_i = N_fV_f$
- $M_i$ - initial molarity
- $V_i$ - initial volume
- $M_f$ - final molarity
- $V_f$ - final volume
- $N_i$ - initial normality
- $V_i$ - initial volume
- $N_f$ - final normality
- $V_f$ - final volume

$$V_{H_2O} \text{ added} = V_f - V_i$$

- **Concentrating solution** is preparing more concentrated solution from less concentrated solution by removing water.

$$V_{H_2O} \text{ removed} = V_i - V_f$$

## Illuminative Examples

85. If a student wishes to prepare approximately 100mL of an aqueous solution of 6M HCl using 12M HCl. Which procedure is correct? (EUEE ... 2007 E.C)
A. Adding 50mL of 12M HCl to 50mL of water while stirring the mixture steadily.
B. Adding 25mL of 12M HCl to 50mL of water while stirring the mixture steadily.
C. Adding 50mL of water to 50mL of 12M HCl while stirring the mixture steadily.
D. Adding 25mL of water to 50mL of 12M HCl while stirring the mixture steadily.

**Solution**

$M_i = 12M$ $\qquad$ $M_f = 6M$

$V_i = ?$ $\qquad$ $V_f = 100mL$

$$V_i = \frac{M_f.V_f}{M_i} = \frac{6M \times 100mL}{12M} = 50mL$$

$$V_{H_2O} \text{ added} = V_f - V_i = 100 - 50 = 50mL$$

- During dilution of concentrated acid, acid is added to water not water to acid.

**Answer: A**

86. What is the final concentration of $Cl^-$ ion when 250mL of 0.20M $CaCl_2$ solution is mixed with 250mL of 0.40MNKCl solution? (EUEE)

A. 1.60M B. 0.40M C. 0.20M D. 0.60M

**Solution**

$$n_{CaCl_2} = M.V = 0.20mol/L \times 250mL = 50mmol$$

$$CaCl_2 \longrightarrow Ca^{2+} + 2Cl^-$$

$$n_{Cl^-} = 2 \times 50mmol = 100mmol$$

$$n_{kCl} = M.V = 0.40\ mol/L \times 250mL = 100mmol$$

$$KCl \rightarrow K^+ + Cl^-$$

$$n_{Cl^-} = 100mmol$$

$$V_T = (250 + 250)mL = 500mL$$

$$n_T = (100 + 100)mmol = 200mmol$$

$$[Cl^-] = \frac{n_T}{V_T} = \frac{200mmol}{500mL} = 0.40M$$

**Answer: B**

87. A laboratory instructor is preparing 5.0 liters of a 0.10M $Pb(NO_3)_2$ (molecular mass 331) solution, what is the mass required?

A. 165.5g of $Pb(NO_3)_2$ and add 5kg of $H_2O$

B. 165.6g of $Pb(NO_3)_2$ and add $H_2O$ until the solution has a volume of 5.0 liters.

C. 33.1 g of $Pb(NO_3)_2$ and add $H_2O$ until the solution has a volume of 5.0 liters.

D. 33.1g of $Pb(NO_3)_2$ and add 5.0 liters of $H_2O$.

**Solution**

➦ $m = M.\ M_{wt}.\ V = 0.10mol/L \times 331g/mol \times 5.0L = 165.6g$

➦ Thus, 165.5g of $Pb(NO_3)_2$ and add water until the solution has a volume of 5.0 liters.

**Answer: B**

88. How would you prepare 0.90L of $1.2 \times 10^{-2}$M NaCl aqueous solution?

**Solution**

$m = M.\ M_{wt}.\ V = 1.2 \times 10^{-2}mol/L \times 58.5g/mol \times 0.90L = 0.63g$

Thus, weigh 0.63g of NaCl, dissolve in a small amount of water and dilute to 0.90L.

89. How much water has to be evaporated from 250mL of 1M $Ca(OH)_2$ to make it 3M?

**Solution**

$V_i = 250m$ $\qquad$ $M_f = 3M$

$M_i = 1M$ $\qquad$ $V_f = ?$

$$V_f = \frac{M_i V_i}{M_f} = \frac{250mL \times 1M}{3M} = 83.3mL$$

$$V_{H_2O\ evaporated} = V_i - V_f = 250 - 83.3 = 166.7mL$$

90. How many mL of water is required to dilute 50mL of 3.5M $H_2SO_4$ to 2M $H_2SO_4$?

**Solution**

$$V_f = \frac{M_i V_i}{M_f} = \frac{50mL \times 3.5M}{2M} = 87.5mL$$

$$V_{H_2O\ added} = V_f - V_i = 87.5mL - 50mL = 37.5mL$$

91. How do you prepare 30mL of 4M $HNO_3$ from a stock solution of 16M $HNO_3$?

**Solution**

$$V_i = \frac{M_f V_f}{M_i} = \frac{4M \times 30mL}{16M} = 7.5mL$$

➦ You should dilute 7.5mL of $16HNO_3$ with sufficient water to give the final volume 30mL.

92. How could you prepare a 5% by mass KI solution is 40g of water?

**Solution**

Let the mass of KI be m

$$\%\ KI = \frac{m}{m + m_{H_2O}} \times 100$$

$$5 = \frac{m}{m + 40} \times 100$$

$\frac{5}{100} = \frac{m}{m+40}$, solving for m, $m = 2.1g$

$\Rightarrow$ 5% by mass KI solution is prepared by dissolving 2.1g of KI in 40g of water.

93. Calculate the final concentration of a solute when 2L of 3M sugar solution and 3L of 2.5M sugar solution are mixed and then diluted to 10L with water.

**Solution**

Solution 1: $n_1 = M_1 . V_1 = 3M \times 2L = 6mol$

Solution 2: $n_2 = M_2 . V_2 = 2.5M \times 3L = 7.5mol$

$n_T = n_1 + n_2 = 6 + 7.5 = 13.5mol$

$$M_f = \frac{n_T}{V_T} = \frac{13.5mol}{10L} = 1.35M$$

94. What is the volume water that must be added to a mixture of 200mL of 5M HCl and 800mL of 3M HCl to obtain a 2M solution,

**Solution**

$$M_1V_1 + M_2V_2 = M_3V_3$$

$$V_3 = \frac{M_1V_1 + M_2V_2}{M_3} = \frac{200 \times 5 + 800 \times 3}{2}$$

$V_3 = 1700mL$

$V_{H_2O\ added} = V_3 - (V_1 + V_2) = 1700mL - (800+200)mL = 700mL$

## 7.8 Solution Stoichiometry

**Stoichiometry** is the study of the quantitative relationship between amounts of reactants and reactants, reactants and products, or products and products in a balanced chemical equation.

➦ Solution stoichiometry differs from pure substances stoichiometry in that, the number of mole of pure substances is obtained when given mass is divided by molar mass. However, the number of mole of solute is the product of molarity and volume.

$$n_{pure\ substance} = \frac{m}{M_{wt}}$$

➦ There are three types of solution stoichiometry.

### 1. Mole - mass relationship

➦ From a given mole or volume and molarity it is possible to determine mass.

**Example**

How many grams of zinc react with 500mL of 1M HCl to form zinc chloride and hydrogen gas?

**Solution**

- First write the balanced chemical equation.

  $Zn\,(s) + 2HCl\,(aq) \rightarrow ZnCl_2(aq) + H_2(g)$

- Then determine the number of mole of HCl.

  $n_{HCl} = M.\,V = 1\ mol/L \times 0.5\ L = 0.5mol$

- By mole - mass - relationship, find the mass of Zn.

$$\underset{65g}{Zn(s)} + \underset{2mol}{2HCl(aq)} \longrightarrow ZnCl_2(aq) + H_2(g)$$

$$\frac{x}{65g} = \frac{0.5mol}{2mol}$$

$X$ = mass of Zn = 16.25g

## 2. Mole - Volume relationship

- From a given molarity and volume, find number of mole or vice versa.

**Example**

What volume of 0.5MKOH solution is needed to react with 50mL of $0.5MH_2SO_4$ solution?

**Solution**

- Write balanced chemical equation.

  $2KOH\,(aq) + H_2SO_4(aq) \rightarrow K_2SO_4(aq) + H_2O(l)$

- Find number of mole of $H_2SO_4$.

  $n_{H_2SO_4} = M_{H_2SO_4} \cdot V_{H_2SO_4} = 0.5\ mol/L \times 0.05L = 0.025mol$

- By mole - mole relationship find the number of mole of kOH

$$2\overset{x}{K}OH(aq) + \underset{1}{\overset{0.025mol}{H_2SO_4}}(aq) \longrightarrow K_2SO_4(aq) + H_2O(l)$$

$$\frac{x}{2} = \frac{0.025mol}{1},\ x = n_{kOH} = 0.05mol$$

- From mole and molarity, find the volume of KOH.

$$V_{KOH} = \frac{n_{KOH}}{M_{KOH}} = \frac{0.05mol}{0.5\ mol/L} = 0.1L = 100mL$$

## 3. Mole - number of particles relationship

- A mole is Avogadro's number of particles.

1 mole = $6.02 \times 10^{23}$ particles (atoms, ions, molecules, formula units)

## Example

What volume of 0.125M sucrose ($C_{12}H_{22}O_{22}$) contains $1.204 \times 10^{23}$ molecules of sucrose?

**Solution**

$$1 \text{ mol } C_{12}H_{22}O_{11} = 6.02 \times 10^{23} \text{ molecules} = 1.204 \times 10^{23} \text{ molecules}$$

$$n = \frac{1.204 \times 10^{23} \text{ molecules}}{6.02 \times 10^{23} \text{ molecules}} \times 1 \text{ mol}$$

$$n = 0.2\text{mol}$$

$$V = \frac{n}{M} = \frac{0.2\text{mol}}{0.125\text{mol/L}} = 1.6\text{L}$$

## Illustrative Examples

95. What volume of 0.500M NaOH is required to neutralize 25.0mL of 1.2M $H_2SO_4$? (Assume complete ionization of the acid) (EUEE ...2006 E.C.)

A. 60mL B. 90mL C. 100mL D. 120mL

**Solution**

$$n_{H_2SO_4} = M_{H_2SO_4} \cdot V_{H_2SO_4} = 25\text{mL} \times 1.2\text{mol/L} = 30\text{mmol}$$

$$2NaOH + H_2SO_4 \rightarrow Na_2SO_4 + 2H_2O$$

For 1 mol $H_2SO_4$, 2 mol NaOH is required. Thus,

$$n_{NaOH} = 2 \times 30\text{mmol} = 60 \text{ mmol}$$

$$V_{NaOH} = \frac{n_{NaOH}}{M_{NaOH}} = \frac{60\text{mmol}}{0.5\text{mol/L}} = 120\text{mL}$$

**Answer: D**

96. How many grams of $NaHCO_3$ (baking soda) would you need to neutralize 500mL of battery acid ($H_2SO_4$) that has spilled on your floor. Assume that the concentration of the battery acid is 12 molar. The balanced equation is

$$H_2SO_4(aq) + 2NaHCO_3(l) \rightarrow Na_2SO_4(aq) + 2CO_2(g) + 2H_2O(l)$$

A. $1.008 \times 10^3$kg C. 0.504kg

B. 1.008kg D. 0.252kg

**Solution**

$$n_{H_2SO_4} = 0.5\text{L} \times 12\text{mol/L} = 6\text{mol}$$

$\Rightarrow$ For 1 mol $H_2SO_4$ 2 mol $NaHCO_3$ is required. As a result,

$$n_{NaHCO_3} = 2 \times 6mol = 12mol$$

$$m_{NaHCO_3} = n.M_{wt} = 12mol \times 84g/mol = 1008g = 1.008kg$$

**Answer: B**

97. How many grams of $CaCO_3$ will be formed by passing excess $CO_2$ gas through 100mL of 0.25M $Ca(OH)_2$ solution?

$Ca(OH)_2 + CO_2 \rightarrow CaCO_3 + H_2O$

**Solution**

$$n_{Ca(OH)_2} = M.V = 0.25mol/L \times 0.1L = 0.025mol$$

For 1 mol $Ca(OH)_2$, we produce 1 mol $CaCO_3$.

$$\Rightarrow n_{CaCO_3} = 0.025mol$$

$$m_{CaCO_3} = n \times M_{wt} = 0.025mol \times 100g/mol = 2.5g$$

98. How many milliliters of 10M $H_3PO_4$ would be required to react with 200mL of 5M $Ca(OH)_2$? **Solve by your own: Answer 67 mL**

## 7.9 Describing Reactions in solution

➦ Chemists use three types of equations to represent aqueous ionic reaction. These are

1. Molecular equation
2. Total ionic equation
3. Net ionic equation

1. **Molecular equation** shows species that are actually in solution and no ions are present.

**Example**

$NaCl(aq) + AgNO_3(aq) \rightarrow AgCl(s) + NaNO_3(aq)$

2. **Total ionic equation** reveals all the soluble ionic substances as they actually exist in solution, where there are dissociated into ions.

$$Na^+(aq) + Cl^-(aq) + Ag^+(aq) + NO_3^-(aq) \longrightarrow AgCl(s) + Na^+(aq) + NO_3^-(aq)$$

3. **The net ionic equation** is very important because it eliminates the spectator ions and shows only the actual chemical change.

**Spectator ions** are ions present in the total ionic equation but not in the net ionic equation.

➦ They are not involved in actual chemical change but are only as part of the reactants or products.

- $Ag^+(aq) + Cl^-(aq) \rightarrow AgCl(s)$ ... net ionic equation
- $Na^+$ and $NO_3^-$ are spectator ions.

**Note**

Insoluble solids, gases and liquids do not appear in ionic form in the total ionic equation.

## Illustrative Examples

99. What is the balanced net ionic equation for the reaction of $CaCl_2$ and $AgNO_3$? (EUEE ... 2008 E.C.)

A. $CaCl_2(aq) + 2AgNO_3(aq) \rightarrow Ca(NO_3)_2(aq) + 2AgCl(s)$

B. $Ca^{2+}(aq) + 2Cl^-(aq) + 2Ag^+(aq) + 2NO_3^-(aq) \longrightarrow$
$Ca^{2+}(aq) + 2NO_3^-(aq) + 2AgCl(s)$

C. $Cl^-(aq) + Ag^+(aq) \rightarrow 2AgCl(s)$

D. $2Cl^-(aq) + 2Ag^+(aq) \rightarrow 2AgCl(s)$

**Solution**

➦ $CaCl_2(aq) + 2AgNO_3(aq) \rightarrow Ca(NO_3)_2(aq) + 2AgCl(s)$
⇒Molecular equation

➦ $Ca^{2+}(aq) + 2Cl^-(aq) + 2Ag^+(aq) + 2NO_3^-(aq) \longrightarrow Ca^{2+}(aq) + 2NO_3^-(aq) + 2AgCl(s)$ total ionic equation
$Ca^{2+}$ and $NO_3^-$ are spectator ions.

➦ $2Ag^+(aq) + 2Cl(aq) \rightarrow 2AgCl(s)$, net ionic equation

**Answer: D**

100. Consider the following balanced equation:
$3Ba(NO_3)_2(aq) + Fe_2(SO_4)_3(s) \rightarrow 2Fe(NO_3)_3(aq) + 3BaSO_4(s)$
The net ionic equation to describe this balanced equation is (EUEE ... 2004 E.C)

A. $3Ba^{2+}(aq) + SO_4^{2-}(aq) \longrightarrow 3BaSO_4(s)$

B. $3Ba^{2+}(aq) + 3SO_4^{2-}(aq) \longrightarrow 3BaSO_4(s)$

C. $3NO_3^-(aq) + 2Fe^{2+}(aq) \longrightarrow 2Fe(NO_3)_3(aq)$

D. $3Ba^{2+}(aq) + 2NO_3^-(aq) + 2Fe^{3+}(aq) + 3SO_4^{2-}(aq) \longrightarrow$
$3BaSO_4(s) + 3Fe^{3+}(aq) + 6NO_3^-(aq)$

**Solution**

$3Ba^{2+}(aq) + 6NO_3^-(aq) + 2Fe^{3+}(aq) + 3SO_4^{2-}(aq) \longrightarrow$

$3BaSO_4(s) + 2Fe^{3+}(aq) + 6NO_3^-(aq)$

$\Rightarrow NO_3^-$ and $Fe^{3+}$ are spectator ions.

$\Rightarrow 3Ba^{2+}(aq) + 3SO_4^{2-}(aq) \longrightarrow 3BaSO_4(s)$

**Answer: B**

101. Write the total ionic equation, net ionic equation and identify the spectator ion (s) for the following molecular equations.

a. $2HNO_3(aq) + Ca(HCO_3)(aq) \rightarrow Ca(NO_3)_2(aq) + 2H_2O(l) + 2CO_2(g)$

b. $5FeCl_2(aq) + KMnO_4(aq) + 8HCl(aq) \rightarrow 5FeCl_3(aq) + MnCl_2(aq) + KCl(aq) + 4H_2O(l)$

c. $La_2(CO_3)_3(s) + 6HCl(aq) \rightarrow 2LaCl_3(aq) + 2CO_2(g) + 3H_2O(l)$

**Solution**

a. $2H^+(aq) + 2NO_3^-(aq) + Ca^{2+}(aq) + 2CHO_3^-(aq) \longrightarrow$

$Ca^{2+}(aq) + 2NO_3^-(aq) + 2H_2O(l) + 2CO_2(g)$

➦ $2H^+(aq) + 2HCO_3^-(aq) \longrightarrow 2H_2O(l) + 2CO_2(g)$

➦ $Ca^{2+}$ and $NO_3^-$ are spectator ions.

b. $5Fe^{2+}(aq) + 10Cl^-(aq) + k^+(aq) + MnO_4^-(aq) + 8H^+(aq) + 8Cl^-(aq)$

$\Rightarrow 5Fe^{3+}(aq) + 15Cl^-(aq) + Mn^{2+}(aq) + 2Cl^-(aq) + K^+(aq) + Cl^-(aq) + 4H_2O$

➦ $5Fe^{2+}(aq) + MnO_4^-(aq) + 8H^+(aq) \longrightarrow$

$5Fe^{3+}(aq) + Mn^{2+}(aq) + 4H_2O(l)$

➦ Spectator ions = $K^+$, $Cl^-$

c. $La(CO_3)_3(s) + 6H^+(aq) + 6Cl^-(aq) \rightarrow 2La^{3+}(aq) + Cl^-(aq) + 2CO_2(aq) + 3H_2O(l)$

➦ $La_2(CO_3)_3(s) + 6H^+(aq) \rightarrow 2La^{3+}(aq) + 2CO_2(g) + 3H_2O(l)$

Spectator ion = $Cl^-$

## 7.10 Colligative Properties of Solutions

**Colligative (means 'collective') properties of solutions** are properties of solutions that depend only on the number of solute particles but not on their chemical identity. There are four colligative properties. These are:

1. Vapor pressure lowering ($\Delta P$)
2. Boiling point elevation ($\Delta T_b$)
3. Freezing point depression ($\Delta T_f$)
4. Osmotic Pressure ($\pi$)

## Colligative Properties of non - volatile non - electrolytic solutions

- **Non - electrolytes** are compounds such as sugar and alcohol that do not dissociate into ions. They are non - electrolytes because their solutions do not conduct a current.
- For non - electrolytes, one mole of compound yields one mole of particles when it dissolves in solution.

**Example**

0.5M sugar contains 0.5 mole of solute particles per liter.

### 1. Vapor pressure lowering ($\Delta P$)

- The vapor pressure of non - volatile non - electrolytic solution is always less than the vapour pressure of the pure solvent. The difference in vapor pressure is called **vapor pressure lowering**.

$$\Delta P = P^\circ - P \qquad \ldots 1$$

- $P^\circ$ - vapor pressure of pure solvent
- P - vapore pressure of solution

## Quantifying Vapor Pressure Lowering

- This was explained by Raoult's law. **Raoult's law** states that the vapor pressure of the solvent above a solution (P) equals the mole fraction of the solvent ($X_{solvent}$) times the vapor pressure of the pure solvent ($P^\circ$).

$$P = X_{solvent} . P^\circ \qquad \ldots (2)$$

$$\Delta P = P^\circ - P$$

$$\Delta P = P^\circ - (X_{solvent} . P^\circ)$$

$$\Delta P = P^\circ(1 - X_{solvent}),\ X_{solute} = 1 - X_{solvent}$$

$$\Delta P = X_{solute} . P^\circ$$

**Example**

Calculate the vapor pressure lowering of a solution of 2g of aspirin ($M_{wt}$ = 180g/mol) in 50g of methanol ($CH_3OH$) at 21°C. Methanol has a vapour pressure of 101 torr at 21°C.

**Solution**

⇒Solvent is methanol and the solute is aspirin.

$$n_{aspirin} = \frac{m}{M} = \frac{2g}{180\ g/mol} = 0.01mol$$

$$n_{methanol} = \frac{m}{M} = \frac{50g}{32\ g/mol} = 1.56mol$$

$$X_{aspirin} = \frac{n_{aspirin}}{n_{aspirin} + n_{methanol}} = \frac{0.011mol}{(0.011+1.56)mol} = 0.007$$

$$\Delta P = X_{solute}.\ P° = 0.007 \times 101torr = 0.71torr$$

$$P = P° - \Delta P = 101torr - 0.71\ torr = 100.29torr$$

➦ According to Dalton's law of partial pressure, for two gases A and B, the total pressure is the sum of the partial pressure of A and B.

$$P_T = P_A + P_B \qquad P_B = X_B P°_B$$

$$P_A = X_A.\ P°_A$$

$$\boxed{P_T = X_A P° + X_B P_B^{\circ}}$$

**Example**

1 mol of pure A and 3 mole of pure B are mixed, calculate the total pressure of the system $\left(P_B^{\circ} = 1140mmHg,\ P_B^{\circ} = 570mmHg\right)$

**Solution**

$$X_A = \frac{n_A}{n_A + n_B} = \frac{1}{(1+3)mol}0.25$$

$$X_B = 1 - X_A = 1 - 0.25 = 0.75$$

$$P_T = X_A P_A^{\circ} + X_B P_B^{\circ}$$

$$= (0.25 \times 1140 + 0.75 \times 570)mmHg = 712.5mmHg$$

## Ideal Solution

**Ideal solution** is a solution whose vapor pressure equals the mole fraction of the solvent times the vapor pressure of the pure solvent.

## Characteristics of ideal solution

- ➦ Ideal solution obeys Raout's law.
- ➦ The heat of ideal solution is always zero.

- The total pressure of the ideal solution is the sum of the partial pressures of the components of the solution.
- The total volume of the solution is the sum of the volume of the components.
- The components of the ideal solution have similarity in their chemical structure.

### Example

Benzene and toluene.

- The solute – solvent interaction force is equal to the solute – solute and solvent – solvent interaction force.

## Non - ideal (Real) Solutions

- Solutions that do not obey Raoult's law are called **non - ideal solutions**.
- Non - ideal solutions deviate from Raoult's law negatively and positively.

## Negative deviation

- If the solute - solvent interaction is greater than the solute - solute and solvent - solvent interaction, the deviation is negative and heat of solution becomes negative (exothermic mixing).
- The total vapor pressure is less than the sum of the vapor pressure as predicated by Raoult's law.

## Positive deviation

- If the solute - solvent interaction is less than the solute - solute and solvent - solvent interaction, the deviation is positive and heat of solution becomes positive (endothermic mixing).
- The total vapor pressure is greater than the sum of the vapor pressure predicated by Raoult's law.

## Illustrative Examples

102. If the solute - solvent interactions are greater than the solute - solute and solvent - solvent interactions, what will be the total vapor pressure of the solution? (EUEE ... 2006 E.C)

A. Greater than that calculated from Raoult's law.
B. Less than that of calculated from Raoult's law.
C. The same as calculated from Raoult's law.
D. Raoult's law cannot be applied for such type of calculations.

**Solution**

➦ If the solute - solvent interaction is greater than the solute - solute and solvent - solvent interactions, the total pressure is less than the sum of the vapor pressures.

**Answer: B**

103. A solution is prepared by mixing 1 mol of ethanol and 2 moles of water at 25°C. What is the total vapor pressure above the solution? (At 25°C, the vapor pressure of the ethanol and pure water are 44 torr and 23.8torr respectively). (EUEE - 1999 E.C)

A. 30.5 torr
C. 45 torr
B. 33.9 torr
D. 67.8 torr

**Solution**

$$P_T = P_{ethanol} + P_{water}$$

$$P_T = X_{eth.}P^{\circ}_{eth} + X_{water}.P^{\circ}_{water}$$

$$= \frac{1mol}{(1+2)mol}.44\ torr + \frac{2\ mol}{(1+2)mol}.23.8\ torr$$

$$= 30.5\ torr$$

**Answer: A**

104. Nitroglycerine is used in the manufacture of dynamite and in the treatment of certain coronary problems. If you add 0.454g of the compound to 100g of benzene at 25°C, the vapor pressure of the solution is 94.85mmHg. What is the molar mass of nitroglycerine? (Vapor pressure of pure benzene at 25°C is 95mmHg).

**Solution**

$$\Delta P = P^{\circ} - P = 95 - 94.85 = 0.15\ mmHg$$

$$\Delta P = X_{solute}.\ P^{\circ}$$

$$X_{solute} = \frac{\Delta P}{P^{\circ}} = \frac{0.15mmHg}{95mmHg} = 0.0016$$

$$n_{benzene} = \frac{m}{M} = \frac{100g}{78g/mol} = 1.28mol$$

$$X_{solute} = \frac{n_{solute}}{n_{solute} + n_{benzene}}$$

$$0.0016 = \frac{n_{solute}}{n_{solute} + 1.28mol}$$

$$n_{solute} = 0.002mol$$

$$M_{solute} = \frac{m}{n_{solute}} = \frac{0.454g}{0.002mol} = 227g/mol$$

105. Calculate the vapor pressure of benzene $(C_6H_6)$ and toluene $(C_7H_8)$ and the total pressure at 25°C above a solution with equal masses of the two liquids. $\left(P^{\circ}_{C_6H_6} = 94.1 \text{ torr}, P^{\circ}_{C_7H_8} = 28.4 \text{ torr}\right)$

**Solution**

Let the mass of benzene and toluene be m

$$X_{benzene} = \frac{M_{toluene}}{M_{benzene} + M_{toluene}} = \frac{92}{78+92}$$

$$X_{benzene} = 0.46$$

$$X_{toluene} = 1 - X_{benzene} = 0.45$$

$$P_{toluene} = X_{tol}.P^{\circ}_{tol} = 0.46 \times 28.4 \text{ tor} = 13.1 \text{ torr}$$

$$P_T = P_{benzene} + P_{totuene} = 64.5 \text{ torr}$$

106. Calculate $\Delta P$ when 10mL of glycerol $(C_3H_8O_3)$ is added to 500ml of water at 50°C.

$\left(P^{\circ}_{H_2O} = 92.5 \text{ torr}, \text{ density of } H_2O = 0.988 \text{g/mol}, \text{density of glycerol} = 1.26 \text{g/ml}\right)$

**Solution**

$$m_{H_2O} = d.v = 0.988 \text{ g/mol} \times 500 \text{mL} = 494 \text{g}$$

$$m_{gly} = d.\ v = 1.26 \text{ g/mL} \times 10 \text{mL} = 12.6 \text{g}$$

$$n_{H_2O} = \frac{m_{H_2O}}{M_{H_2O}} = \frac{494 \text{g}}{18 \text{g/ mol}} = 27.4 \text{mol}$$

$$n_{gly} = \frac{m_{gly}}{M_{gly}} = \frac{12.6 \text{g}}{18 \text{ g/mol}} = 0.14 \text{mol}$$

$$X_{gly} = \frac{n_{gly}}{n_{gly} + n_{H_2O}} = 0.005$$

$$\Delta P = X_{gly}.P^{\circ}_{H_2O} = 0.005 \times 92.5 = 0.46 \text{ torr}$$

107. How many grams of sucrose $(C_{12}H_{22}O_{11})$ must be dissolved in 550g of water to give a solution pure water at 20°C.

$\left(P^{\circ}_{H_2O} = 17.5 \text{mmHg}\right)$

**Solution**

$$\Delta P = 33 \text{mmHg}, n_{H_2O} = \frac{550 \text{g}}{18 \text{ g/mol}} = 30.6 \text{mol}$$

$$\Delta P = \frac{n_{solute}}{n_{sucrose} + n_{H_2O}}.P^{\circ}_{H_2O}$$

$$\frac{\Delta P}{P^{\circ}_{H_2O}} = \frac{n_{solute}}{n_{sucrose} + n_{H_2O}}$$

$$\frac{3 \text{mmHg}}{17.5 \text{mmHg}} = \frac{n_{sucrose}}{n_{sucrose} + 30.6 \text{mol}}$$

$\Rightarrow n_{sucrcose} = 6.27mol$

$m_{sucrose} = n.\ M = 6.27mol \times 342\ g/mol = 2144.3g$

108. The vapor pressure of a pure solvent A is 0.80 atm. When a non - volatile solute B is added to the solvent, its vapor pressure drops to 0.60atm, what is the mole fraction of component B?

**Solution**

$\Delta P = P° - P = 0.80atm - 0.60atm = 0.20\ atm$

$\Delta P = X_B, P°$

$X_B = \frac{\Delta P}{P°} = \frac{0.20atm}{0.80atm} = 0.25$

109. Arrange the following in increasing order of vapour pressure, $H_2O$, $CH_3OH$, $CH_3CH_2OH$, $C_6H_6$

**Solution**

$\Rightarrow$ Volatile substances have higher vapour pressure than non - volatile substances.

$H_2O < CH_3CH_2OH < CH_3OH < C_6H_6$

## 2. Boiling Point Elevation ($\Delta T_b$)

➦ A solution boils at a higher temperature than the pure solvent. The difference between the boiling points of the solution and the solvent is known as **boiling point elevation**.

$$\boxed{\Delta T_b = \Delta T_b - T_b°}$$

$T_b$ - boiling point of solution

$T_b°$ - boiling point of pure solvent

➦ Boiling point elevation is proportional to the molal concentration of a solute.

$$\boxed{\Delta T_b \ \alpha\ m \text{ or } \Delta T_b = k_b m}$$

➦ m - is molality of a solution

➦ $k_b$ - molal boiling point elevation constant in °C/m.

### Example

You add 500g of ethylene glycol ($C_2H_BO_2$) anti freeze to 4 kg of water in your car's radiator. Calculate (kb = 0.512°C/m)

a. $\Delta T_b$ b. $T_b$

**Solution**

a. $$m = \frac{m_{C_2H_6O_2}}{M_{C_2H_6O_2}.kgH_2O} = \frac{500g}{62g/mol \times 4kg} = 2m$$

$$\Delta T_b = mk_b = 2m \times 0.512°C/m = 1.024°C$$

b. $\Delta T_b = T_b - T_b^0$, boiling point of water is 100°C.

$T_b = \Delta T_b + T_b^0 = 1.024 + 100 = 101.024°C$

### 3. Freezing point depression ($\Delta T_f$)

➦ A solution freezes at a lower temperature than the pure solvent. The difference between the freezing point of pure solvent and solution is known as **freezing point depression**.

$$\Delta T_f = T_f^\circ - T_f$$

➦ Freezing point depression is proportional to the molal concentration of solute.

$$\Delta T_f \ \alpha \ m \text{ or } \Delta T_f = k_f m$$

➦ $k_f$ - is the molal freezing point depression constant in °C/m.

Table 7.2. Molal boiling point elevation and freezing point depression constant of several solvents

| Solvent | Boiling point in °C | kb (°C/m) | Freezing point in °C | $k_f$ (°C/m) |
|---|---|---|---|---|
| Acetic acid | 117.9 | 3.07 | 16.6 | 3.90 |
| Benzene | 80.1 | 2.53 | 5.5 | 4.90 |
| Carbondisulfide | 46.2 | 2.34 | -11.5 | 3.83 |
| Carbon tetrachloride | 76.5 | 5.03 | -23 | 30 |
| Chloroform | 61.7 | 3.63 | -63.5 | 4.70 |
| Diethylether | 34.5 | 2.02 | -116.2 | 1.79 |
| Ethanol | 78.5 | 1.22 | -117.3 | 1.99 |
| Water | 100.0 | 0.512 | 0.0 | 1.86 |

For the same solution

$$\frac{\Delta T_b}{\Delta T_f} = \frac{k_b}{k_f}$$

## Illustrative Examples

110. An aqueous solution freezes at -0.186°C $k_f$ = 1.86°C.kg.mol$^{-1}$, $k_b$ = 0.512°C.kg.mol$^{-1}$) what is the elevation in boiling point? (EUEE ...2003 E.C.)

A. 0.186°C
B. 0.512C
C. 0.512/1.86°C
D. 0.0512°C

**Solution**

$$\Delta T_f = T_f^\circ - T_f = 0°C - (-0.186°C) = 0.186°C$$

$$\frac{\Delta T_b}{\Delta T_f} = \frac{k_b}{k_f}$$

$$\Delta T_b = \frac{k_b}{k_f}.\Delta T_f = \frac{0.512}{1.86} \times 0.186°C = 0.0512°C$$

**Answer: D**

111. What would be the boiling point of 2.00m table sugar ($C_{12}H_{22}O_{11}$) in water at 1 atm? ($K_b$ for water = 0.51°C m$^{-1}$)

A. 100.50°C
B. 101.0°C
C. 101.5°C
D. 102.0°C

**Solution**

$$\Delta T_b = mk_b = T_b - T_b^\circ$$

$$T_b = mk_b + T_b^\circ = 2m \times 0.51°C.m^{-1} + 100°C$$

$$T_b = 101.012°C$$

**Answer: B**

112. An aqueous solution boils at 100.5°C. What is the freezing point of the solution? ($k_b$ = 0.512°C.m$^{-1}$, $k_f$ = 1.86°C.m$^{-1}$)

**Solution**

$$\Delta T_b = T_b - T_b^\circ = 100.5 - 100 = 0.5°C$$

$$\Delta T_f = \frac{k_f}{k_b}.\Delta T_b$$

$$\Delta T_f = \frac{1.86°C/m}{0.512°C/m}.0.5°C = 1.82°C$$

$$T_f = T_f^\circ - T_f = 0°C - 1.82°C = -1.82°C$$

113. Calculate the freezing point of a solution of 4g of $C_{12}H_{10}$ and 6g of $C_{10}H_8$ dissolved in 100g of benzene, $C_6H_6$. ($k_f$ for $C_6H_6 = 5.12°C.m^{-1}$, $T_f^\circ = 5.5°C$)

**Solution**

$$n_{C_{12}H_{10}} = \frac{4g}{154\ g/mol} = 0.026mol$$

$$n_{C_{12}H_{10}} = \frac{6g}{128\ g/mol} = 0.047mol$$

$$n_T = (0.026 + 0.047)mol = 0.073\,mol$$

$$\Delta T_f = mk_f = \frac{n_T}{kgC_6H_6}.k_f = \frac{0.073mol}{0.10kg} \times 5.12°C/m$$

$$\Delta T_f = 3.74°C$$

$$T_f = T_f^\circ - \Delta T_f = 5.5 - 3.74 = 1.76°C$$

114. When 10.6g of a non - volatile substance is dissolved in 740g of ether, its boiling point elevation is 0.28°C. ($k_b$ of ether = 2.11°C/m) What is the molar mass of the solute?

**Solution**

$$\Delta T_b = mk_b, m = \frac{\Delta T_b}{k_b} = \frac{0.28°C}{2.11°C/m} = 0.133m$$

$$m = \frac{m_{solute}}{M_{solute}.kg\ ether}$$

$$M_{solute} = \frac{m_{solute}}{m.kg\ ether} = \frac{10.6g}{0.133mol/kg \times 0.740kg}$$

$$M_{solute} = 108g/mol$$

115. Calculate the boiling point of a 20% by mass ethanol solution in water.

**Solution**

20% ethanol = 20g ethanol + 80g water

$$\Delta T_b = mk_b = \frac{m_{ethanol}}{M_{ethanol}.kgH_2O}.k_b$$

$$\Delta T_b = \frac{20g}{46g/mol.0.080kg} 0.512°C/m = 2.78°C$$

$$\Delta T_b = T_b^\circ + \Delta T_b = 100°C + 2.78°C$$

$$T_b = 102.78°C$$

## 4. Osmotic Pressure ($\pi$)

**Osmosis** is the net movement of solvent molecules through a semipermeable membrane from a pure solvent or from a dilute solution to a more concentrated solution.

**Osmotic pressure** is the pressure required to stop osmosis.

- The osmotic pressure is directly proportional to the number of mole of solute in a given solution volume, that is, to the molarity.

$$\pi \ \alpha \ \frac{n_{solute}}{V_{solution}} \text{ or } \pi \ \alpha \ M$$

$$\pi = \frac{n_{solute}}{V_{solution}} RT = MRT$$

From ideal gas law, we know that $PV = nRT$

$$P = \frac{n}{V} RT$$

$P = MRT$ if we replace P with $\pi$

$$\pi = MRT$$

- Osmotic pressure is used to determine the molecular mass of macro - molecules.
- To determine molecular mass of non - volatile solute from osmotic pressure of known mass (m) and know volume in liter (V).

$$M_{solute} = \frac{mRT}{V\pi}$$

- To determine density of solution from osmotic pressure, use the formula below.

$$\pi = \frac{nRT}{V} \text{ but } n = \frac{m}{M_{wt}}$$

$$\pi = \frac{mRT}{VM_{wt}} \text{ but } d = \frac{m}{V}$$

$$\pi = \frac{dRT}{M_{wt}}$$

d - desnity

## Illustrative Examples

116. Osmotic pressure of a solution increased by

A. Increasing volume

B. Decreasing temperature

C. Measuring the number of molecules of solute

D. None

**Solution**

➦ Osmotic pressure of a solution is increased by

- Increasing temperature
- Decreasing volume
- Increasing number of mole of solute

**Answer: C**

117. In the phenomenon of osmosis, the semipermeable membrane allow the passage of

A. Solute particles

B. Solvent molecules only

C. Both solute and solvent

D. None

**Answer: B**

118. What is the osmotic pressure at 0°C of an aqueous solution containing 46g of glycerol ($C_3H_8O_3$) per liter?

**Solution**

$T = 0°C = 0 + 273 = 273\ K$

$R = 0.082\ atm.L/mol.\ K$

$$\pi = MRT = \frac{nRT}{V} = \frac{mRT}{M_{wt}.V}$$

$$\pi = \frac{46g/mol \times 0.082atm.L/mol.K \times 273K}{92g/mol \times 1L}$$

$$\pi = 11.2atm$$

119. What is the molecular weight of solute A, if the osmotic pressure of a solution containing 10g/L is 10 torr at 27°C.

**Solution**

$$\pi = 10\text{torr} = \frac{1}{76}\text{atm}$$

$$T = 27°C = 300K$$

$$M_A = \frac{dRT}{\pi} = \frac{10\text{g/L} \times 0.082\text{atm.L / mol.K} \times 300K}{\frac{1}{76}\text{atm}}$$

$$M_A = 18696 \text{ g/mol}$$

120. Calculate the concentration of solute particles in human blood if the osmotic pressure is 7.53atm at 37°C, the temperature of a body.

**Solution**

$$T = 37°C = 310K,\ \pi = 7.53 \text{ atm}$$

$$\pi = MRT$$

$$M = \frac{\pi}{RT} = \frac{7.53\text{atm}}{0.082\text{atm.L/mol.K} \times 310K}$$

$$M = 0.30M$$

## Colligative Properties for electrolytic solutions

**Electrolytes** are solutions that conduct a current because the solutes separate into ions as they dissolve.

- Soluble salts, strong acids and strong bases are **strong electrolytes.**
- Weak acids, weak bases and sparingly soluble salts are **weak electrolytes**.
- For colligative properties of strong electrolyte solutions, the formula of the solute tells us the number of particles (ions). For example, the boiling point elevation ($\Delta T_b$) of 0.1m NaCl should be twice that of 0.1m glucose ($C_6H_{12}O_6$) because NaCl dissociates into two particles per formula unit. Thus, we use a multiplying factor called **the Van't Hoff factor (i)** named after the Dutch chemist Jacoubs Van't Hoff.

$$i = \frac{\text{measured value for electrolytic solution}}{\text{expected value for non electrolyte solution}}$$

- The four colligative properties for strong electrolyte solutions should have the formula below.

$$\Delta P = i\, X_{solute}.\, P°$$

$$\Delta T_b = imk_b$$

$$\Delta T_f = imk_f$$

$$\pi = i\, MRT$$

As i increases,

- $T_b$ increases
- $T_f$ decreases
- P decreases

## Note

- For non - electrolyte solutions, i = 1
- For electrolyte solutions, i > 1
- The value of i is almost equal to the number of moles of ions produced.

## Example

$CaCl_2 \rightarrow Ca^{2+} + 2Cl^-$ $\quad i \approx 1 + 2 = 3$

## Illustrative Examples

121. What will be the ratio of the boiling point elevation of a 0.1M solution of sucrose, sodium sulphate and potassium bromide? (EUEE ...2009 E.C)

A. 1: 3: 1 B. 1: 1: 2 C. 1: 3: 2 D. 1: 2: 2

**Solution**

For non - electrolyte, sucrose, i = 1

For sodium sulphate ($Na_2SO_4 \rightarrow 2Na^+ + SO_4^{2-}$), $i_2 = 3$

For potassium bromide ($KBr \rightarrow K^+ + Br$), $i_3 = 2$

Sucrose: $Na_2SO_4$: $KBr \Rightarrow i_1\Delta T_b$: $T_2{}^0\Delta T_b$: $i_3\Delta T_b$

$\Delta T_b : 3\Delta T_b : 2\Delta T_b \Rightarrow 1: 3: 2$

**Answer: C**

122. A solution is made by dissolving 250g of potassium chromate crystals ($K_2CrO_4$, molar mass 194.2g) in 1kg of water. What will be the freezing point of the solution? (kf for water is $1.86°C.m^{-1}$) (EUEE ... 2006 E.C)

A. -8.87°C B. -7.18°C C. -5.73°C D. -1.86°C

**Solution**

$$\Delta T_f = imk_f$$

$$K_2CrO_4 \longrightarrow 2K^+ + CrO_4^{2-} \quad i = 3$$

$$\Delta T_f = 3 \times \frac{250g}{194.2g/mol.1kg} \times 1.86°C.m^{-1} = 7.18°C$$

$$\Delta T_f = T_f^\circ - T_f$$

$$T_f = T_f^\circ - \Delta T_f = 0°C - 7.18°C = -7.18°C$$

**Answer: B**

123. Which of the following will produce the largest increase in boiling point upo the addition of 1kg of water? (EUEE ... 2003 E.C)

A. 1 mol $Ca(NO_3)_2$
B. 2 mol KCl
C. 3 mol ethylene glycol ($C_2H_6O_2$)
D. 1 mol of $CaCl_2$

**Solution**

$$\Delta T_b = imk_b = T_b - T_b^\circ$$

$$T_b = T_b^\circ + imkb$$

$\Rightarrow$ As the value of i increases, boiling point of solution also increases.

1 mol $Ca(NO_3)_2 \longrightarrow Ca^{2+} + 2NO_3^-, i = 3$

2 mol $KCl \longrightarrow 2K^+ + 2Cl^-, i = 4$

3 mol $C_2H_2O_2, i = 3$

1 mol $CaCl_2 \longrightarrow Ca^{2+} + 2Cl^-, i = 3$

**Answer: B**

124. Which aqueous solution has the lowest freezing point? (EUEE)

A. 0.20m NaCl
B. 0.20m $CaCl_2$
C. 0.20mKCl
D. 0.20m $CH_3CH_2OH$

**Solution**

$$\Delta T_f = imkf = T_f^\circ - T_f$$

$$\Rightarrow T_f = T_f^\circ - imkf$$

➦ As the value of i increases, freezing point of the solution decreases.

**Answer: B**

125. Arrange the following aqueous solution sin order of increasing freezing points.

a. 0.10m $C_2H_5OH$
b. 0.1m $Al_2(SO_4)_2$
c. 0.1m $K_2SO_4$
d. 0.1m KCl
e. 0.10m$FeCl_3$

**Solution**

b < e < c < d < a

126. Arrange the following in decreasing order of freezing point depression.

A. 1m $CaCl_2$
B. 1m$(NH_4)_3PO_4$
C. 1m $NH_4NO_3$
D. 1m $C_6H_{12}O_6$

**Solution**

➦ As the value of i increases freezing point depression also increase.

$\Rightarrow$b > a > c > d

127. The osmotic pressure of a 0.01M aqueous solution of $CaCl_2$ is found to be 0.67 atm at 25°C. Calculate the Van't Hoff factor, i, for the solution.

**Solution**

$\pi = iMRT$

$$i = \frac{\pi}{MRT} = \frac{0.67atm}{0.01mol/L \times 0.082atm.L/mol.K} \times 298K$$

## Solved Problems

128. Explain step by step how to attain a crystallized supersaturated solution.

**Solution**

➦ First a solute is added to a solvent, then more solute would be added and dissolved in the solvent until the solution is saturated. Then, the saturated solution is heated to dissolve more solute. The solution is left to cooled undisturbed. Once the solution is cooled, a 'seed' crystal of solute may be added, that will cause the solution to crystallize.

129. The Henry's law constant of gas A is 0.12mol/L.atm at 25°C. What is the solubility of A in water at 25°C at a partial pressure of 0.20atm?

**Solution**

$$C_A = K_H.P_A = 0.12\text{mol/L.atm} \times 0.20\text{ atm} = 0.024\text{ mol/L}$$

130. Two beakers contain 1L and 200mL of pure water respectively. What is the same in them?

**Solution**

➦ The molarity of pure water is the same in both beakers, it is 55.6M.

131. A sample of sea water was found to contain $5 \times 10^{-3}$g of dissolved oxygen per kilogram. What is the concentration of oxygen in ppm?

**Solution**

$$\text{PPm } O_2 = \frac{\text{mass } O_2}{\text{mass sea water}} \times 10^6 = \frac{5 \times 10^{-3}\text{g}}{1000\text{g}} \times 10^6 = 5\text{PPm}$$

132. How much water is required to dilute 10mL of 10N hydrochloric acid to deci normal (dN)?

**Solution**

$$N_1V_1 = N_2V_2,\ N_1 = 10\text{N},\ V_1 = 10\text{mL},\ V_2 = ?,\ N_2 = 1\text{dN} = 0.1\text{N}$$

$$V_2 = \frac{N_1V_1}{N_2} = \frac{10\text{N} \times 10\text{mL}}{0.1\text{N}} = 1000\text{mL}$$

$$V_{H_2O}\text{ added} = V_2 - V_1 = 1000\text{mL} - 10\text{mL} = 990\text{mL}$$

133. What is the molarity of $H_2SO_4$ solution that has a density of 1.84 g/cc at 35°C and contains 98% by weight?

**Solution**

$$M = \frac{d \times \% \times 10}{M_{wt}} = \frac{1.84 \times 98 \times 10}{98} = 18.4\text{M}$$

134. An aqueous solution of a non - electrolytic containing $3.01 \times 10^{23}$ molecules of solute in 250g of water will freeze at _____K.

**Solution**

1 mol solute = $6.02 \times 10^{23}$ molecules

? = $3.01 \times 10^{23}$ molecules

$\Rightarrow n_{solute} = 0.5\text{mol}$

$$m = \frac{n}{kg\ H_2O} = \frac{0.5mol}{0.250kg} = 2m$$

$$\Delta T_f = mk_f = 2m \times 1.86°C/m = 3.72°C$$

$$T_f = T_f^\circ - T_f = 0°C - 3.72°C = -3.72°C$$

$$k = °C + 273 = -3.72 + 273 = 269.28K$$

135. 10ml of 0.1M HCl is mixed with 5mL of 0.2M $HNO_3$ and 20ml of 0.3M $H_2SO_4$. What is the normality of the acids mixture?

**Solution**

$$n_{HCl} = M.\ V = 10mL \times 0.1mol/L = 1mmol = No.\ eq\ HCl$$

$$n_{HNO_3} = M.V = 5mL \times 0.2\ mol/L = 1\ mmol = No.\ eq\ HNO_3$$

$$n_{H_2SO_4} = M.V = 2mol \times 0.3mol/L = 6mmol$$

$$No.\ eq\ H_2SO_4 = 2 \times 6 = 12\ m\ eq$$

$$(No.\ eq)_{total} = 1m\ eq + 1\ m\ eq + 12\ meq = 14\ meq$$

$$N_T = \frac{(No.\ eq)_{total}}{V_{HCl} + V_{HNO_3} + V_{H_2SO_4}} = \frac{14meq}{(10+5+20)mL} = 0.4N$$

136. Write two concentration units that are affected by temperature.

**Solution**

Molarity, Normality and mass percent

137. The vapor Pressure at a given temperature of an ideal solution containing 0.2mol of a non - volatile solute and 0.6mol of solvent is 70mmHg. What is the vapor pressure of the pure solvent at the same temperature?

**Solution**

$$X_{solvent} = \frac{0.6\,mol}{(0.2+0.6)\,mol} = 0.75$$

$$P = X_{solvent} \cdot P^\circ$$

$$P^\circ = \frac{P}{X_{solvent}} = \frac{70mmHg}{0.75} = 93.3mmHg$$

138. An ideal solution is made by mixing 2 moles of benzene ($P^\circ = 266mmHg$) and 3 moles of another liquids ($P^\circ = 236\ mmHg$). What is the total vapor pressure of the solution?

**Solution**

Le the unknown liquid be A

$$P_T = P_{benzene} + P_A$$

$$= X_{benzene}.P^{\circ}_{benzene} + X_A.P^{\circ}_A$$

$$= \left(\frac{2}{2+3}\right).266 + \frac{3}{2+3}.236 = 248mmHg$$

139. Determine the normality of $H_3PO_4$ solution, 150mL of which is neutralized by 250mL of 0.5N NaOH.

**Solution**

One equivalent of acid always reacts with one equivalent of base.

$\Rightarrow N_A.V_A = N_B.V_B$, A - acid, B - base

$$N_A = \frac{N_B.V_B}{V_A} = \frac{0.5N \times 250mL}{150mL} = 0.83N$$

140. A solution of $KMnO_4$ is reduced to $MnO_2$ and has a normality of 2.75N. Calculate

a. the equivalent mass of $KMnO_4$

b. the molarity of the solution

Solve by your own

a. 52.67 g/eq b. 0.92M

141. The density of a 2M solution of weak acid HA is 1.02g/mL calculate the molality of the solution. ($M_{HA}$ = 48 g/mol)

**Solution**

$$m_{solution} = d \times v = 1.02g/mL \times 1000mL = 1020g$$

$$m_{HA} = n \times M_{HA} = 2mol \times 48g/mol = 96g$$

$$m_{solvent} = m_{solution} - m_{HA} = 1020 - 96 = 924g$$

$$m = \frac{n}{kg\ solvent} = \frac{2mol}{0.924\ kg} = 2.16m$$

Use the formula below to find the answer.

$$m = \frac{1000M}{1000d - M.M_{solute}}$$

d - density, M - molarity, $M_{solute}$ - molar mass of solute

$$m = \frac{1000 \times 2}{1000 \times 1.02 - 2 \times 48} = 2.16m$$

142. $H_2SO_4(aq) + 2KOH\ (aq) \rightarrow K_2SO_4(aq) + 2H_2O(l)$ If it takes 34.5mL of 0.1234MKOH to neutmline 20mL of $H_2SO_4$ solution, what is the concentration of $H_2SO_4$ solution?

**Answer: 0.106M**

143. Cow's milk typically contains 4.5% by mass of lactose ($C_{12}H_{22}O_{11}$), calculate the mass of lactose present in 175g of milk.

**Solution**

$$m_{lactose} = 4.5\%.\text{mass of milk} = \frac{4.5}{100} \times 175g = 7.875g$$

144. What mass of water must be added to 425 g of formaldehyde to prepare 40% by mass solution of formaldehyde (formalin)?

**Solution**

$m_{form}$ - mass of formaldehyde

$m_w$ - mass of water

$$\%\ form = \frac{m_{form}}{m_{form} + m_w} \times 100$$

$$40 = \frac{425}{425 + m_w} \times 100$$

$$\Rightarrow m_w = 637.5g$$

➦ Thus, 637.5g of water must be added to 425g of formaldehyde to prepare 40% by mass solution of formaldehyde. Or

Let the mass of solution be X

$40\%\ X = 425\ g$

$X = 1062.5g$

$m_w = X - m_{form} = 1062.5g - 425g = 637.5g$

145. What volume of a 0.25M NaOH solution is required to react exactly with 0.135L of 0.5M $H_3PO_4$?

**Solution**

$$H_3PO_4 + 3NaOH \rightarrow Na_3PO_4 + 3H_2O$$

$$n_{H_3PO_4} = M.V = 0.5mol/L \times 0.135L = 0.0675mol$$

For 1 mol $H_3PO_4$, 3 mol of NaOH is required.

Thus, for 0.0675 mol $H_3PO_4$, $3 \times 0.0675$ mol NaOH is required.

$$V_{NaOH} = \frac{n}{M} = \frac{3 \times 0.0675mol}{0.25mol/L} = 0.81L$$

146. The freezing point of a sample of naphthalene was found to be 80.6°C, when 0.512g of a substance is dissolved in 7.03g of naphthalene, the solution has freezing point of 75.2°C. What is the molecular weight of the solute? ($k_f = 6.80°C.m^{-1}$)

**Solution**

$$\Delta T_f = T_f^\circ - T_f = 80.6 - 75.2 = 5.4°C$$

$$\Delta T_f = \frac{m_{solute}}{M_{solute}.kg\ solvent}.k_f$$

$$M_{solute} = \frac{m_{solute}}{\Delta T_f.\ kg\ solvent}.k_f$$

$$= \frac{0.512g}{5.4 \times 0.00703} \times 6.80 = 92g/mol$$

147. What is the boiling point of a 10% by mass solution of ethanol ($C_2H_5OH$) in water? ($k_b$ for $H_2O = 0.512°C.m^{-1}$)

**Solution**

10% (m/m) ethanol = 10g ethanol + 90g $H_2O$

$$\Delta T_b = \frac{10g}{46g/mol \times 0.09kg} \times 0.512°C.m^{-1} = 1.24°C$$

$$T_b = T_b^\circ + \Delta T_b = 100 + 1.24 = 101.24°C$$

148. When 36g of a solute having empirical formula $CH_2O$ is dissolved in 1.20kg of water, the solution freezes at -0.93°C. What is the molecular formula of the solute?

**Solution**

$$\Delta T_f = T_f^\circ - T_f = 0°C - (-0.93°C) = 0.93°C$$

$$M_{solute} = \frac{m_{solute}}{\Delta T_f.kg\ solvent}.\ k_f$$

$$= \frac{36g}{0.93°C \times 1.20kg} \times 1.86°C.m^{-1} = 60g/mol$$

Molecular formula = (Empirical formula) x

$60 = (CH_2O)X$

$60 = (12 + 2 + 16)x$

$X = 2$

$\therefore$ Molecular formula = $(CH_2O)_2 = C_2H_4O_2$

149. A solution containing 8.3g of a non - volatile non - dissociating substances dissolved in 1 mole fo chloroform, $CHCl_3$, has a vapor pressure of 68.1kPa. The vapor pressure of pure $CHCl_3$, at the same temperature id 70.1KPa. Calcualte

a. The mole fraction of the solute

b. The number of mole of solute.

c. The molar mass of the solute.

**Solution**

a. $\Delta P = P° - P = 70.1 - 68.1 = 2KPa$

$$X_{solute} = \frac{\Delta P}{P°} = \frac{2KPa}{70.1KPa} = 0.03$$

b. $$X_{solute} = \frac{n_{solute}}{n_{solute} + n_{solvent}}$$

$$0.03 = \frac{n_{solute}}{n_{solute} + 1\,mol}$$

$$n_{solute} = 0.031mol$$

c. $$M_{solute} = \frac{m_{solute}}{n_{solute}} = \frac{8.3g}{0.031mol} = 268g/mol$$

150. When you open Ambo water or soft drinks, you hear a hissing sound and also observe bubbles escaping out of the water or drink. What does the escape of the gas bubbles indicate?

**Solution**

➦ When we open the bottle containing ambo water or soft drinks, the partial pressure of the gas decreases. Solubility of a gas in water decreases as partial pressure of the gas decreases.

151. What molarity of 7.5mL $Na_2SO_4$ solution is needed to precipitate barium as $BaSO_4$ from 12.5mL of 0.15M $Ba(NO_3)_2$ solution?

**Solution**

$$Na_2SO_4(aq) + Ba(NO_3)_2(aq) \rightarrow 2NaNO_3(aq) + BaSO_4(s)$$

$$n_{Ba(NO_3)_2} = 12.5mL \times 0.15mol/L = 1.875mmol$$

Since the coefficient ratio of $Ba(NO_3)_2$ to $Na_2SO_4$ is 1: 1,

$$n_{Na_2SO_4} = 1.875mmol$$

$$M_{Na_2SO_4} = \frac{n}{V} = \frac{1.875mmol}{7.5mL} = 0.25mol/L$$

152. If 0.454 kg of nitroglycerin explodes according to the following reaction. $4C_3H_5(NO_2)_3(l) + 5O_2(g) \rightarrow 12CO_2(g) + 6N_2(g) + 10H_2O(g)$ what is the total volume of gases in $dm^3$ produced at STP?

**Solution**

➦ The volume of one mole of any gas at STP is 22.4 $dm^3$.
Determine the number of mole of nitroglycerin (nitro).

$$n_{nitro} = \frac{m}{M_{wt}} = \frac{454g}{179\ g/mol} = 2.54\,mol$$

⇒ By mole - mole relationship find the number of moles of gases produced.

$$\overset{2.54mol}{4}C_3H_5(NO_2)_3(l) + 5O_2(g) \longrightarrow \overset{x}{12}CO_2(g) + \overset{y}{6}N_2(g) + \overset{z}{10}H_2O(g)$$

$$\Rightarrow \frac{2.54mol}{4} = \frac{x}{12}$$

$$x = n_{CO_2} = 7.62mol$$

$$\Rightarrow \frac{2.54mol}{4} = \frac{x}{6}$$

$$y = n_{N_2} = 3.81mol$$

$$\Rightarrow \frac{2.54mol}{4} = \frac{z}{10}$$

$$z = n_{H_2O} = 6.35mol$$

$$\therefore V_{CO_2} = n_{CO_2} \times 22.4dm^3 = 170.688dm^3$$

$$V_{N_2} = n_{N_2}, 22.4dm^3 = 85.344dm^3$$

$$V_{H_2O} = n_{H_2O}.22.4dm^3 = 142.24dm^3$$

$$V_T = V_{CO_2} + V_{N_2} + V_{H_2O}$$

$$= 170.688 + 85.344 + 142.24$$

$$= 398.272dm^3$$

153. When aqueous solution of $Na_2SO_4$ and $Pb(NO_3)_2$ are mixed, $PbSO_4$ precipitates. Calculate the mass of $PbSO_4$ formed when 1.25L of 0.05M $Pb(NO_3)_2$ and 2L of 0.025M $Na_2SO_4$ are mixed and identify the spectator ion (s).

**Solution**

$$n_{Pb(NO_3)_2} = M.V = 1.25L \times 0.05mol/L = 0.0625mol$$

$$n_{Na_2SO_4} = M.V = 2L \times 0.025mol/L = 0.05mol$$

$$Na_2SO_4(aq) + Pb(NO_3)_2(aq) \rightarrow 2NaNO_3(aq) + PbSO_4(s)$$

$$2Na^+ + SO_4^- + Pb^{2+} + 2NO_3^- \longrightarrow 2Na^+ + 2NO_3^- + PbSO_4(s)$$

$$Pb^{2+}(aq) + SO_4^{2-}(aq) \longrightarrow PbSO_4(s)$$

➦ Find number of moles of $Pb^{2+}$ and $SO_4^{2-}$ and identify the limiting reactant.

$$Pb(NO_3)_2(aq) \longrightarrow Pb^{2+}(aq) + 2NO_3^-(aq)$$

$$\Rightarrow n_{Pb(NO_3)_2} = n_{Pb^{2+}} = 0.0625mol$$

$$Na_2SO_4(aq) \longrightarrow 2Na^+(aq) + SO_4^{2-}(aq)$$

$$\Leftrightarrow n\,Na_2SO_4 = n_{SO_4^{2-}} = 0.05mol$$

$\therefore SO_4^{2-}$ is limiting reactant and determines the amount of $PbSO_4$.

$$n_{SO_4^{2-}} = n_{PbSO_4} = 0.05mol$$

$$\Rightarrow m_{PbSO_4} = n \times M_{wt} = 0.05mol \times 303g/mol = 15.2g$$

$\Rightarrow Na^+$ and $NO_3^-$ are spectator ions.

154. The osmotic pressure of 2.75g fo non - volatile solute in 100mL of solution at 27°C is 62 cmHg. Determine the molecular weight of the solute.

**Answer: 825 g/mol**

155. Classify the following substances as strong electrolytes, weak electrolytes or non - electrolytes

a. Sodium per manganase ($NaMnO_4$)

b. Glucose ($C_6H_{12}O_6$)

c. Hydrochloric acid (HCl)

d. Ammonia ($NH_3$)

e. Sodium hydroxide (NaOH)

f. Methanol ($CH_3OH$)

g. Acetic acid ($CH_3COOH$)

**Solution**

➦ a, c and e are strong electrolytes.
➦ b and f are non - electrolytes.
➦ d and g are weak electrolytes.

156. At 37°C, 0.30M sucrose has about the same osmotic pressure as blood. What is the osmotic pressure of blood?

**Solution**

$$\pi = MRT = 0.30mol/L \times 0.082\ atm.L/mol.k \times 310k$$
$$\pi = 7.63\ atm$$

157. Why does the solubility of any gas in water decrease with rising temperature?

**Solution**

Because gases mix with water exothermically.

158. Predict the products, write total ionic equation netionic equation and identify the spectator ions $K_3PO_4(aq) + Ca(NO_3)_2(aq) \rightarrow$

**Solution**

$$2K_3PO_4(aq) + 3Ca(NO_3)_2(aq) \rightarrow 6KNO_3(aq) + Ca_3(PO_4)_2(s)$$

$$6K^+(aq) + 2PO_4^{3-}(aq) + 3Ca^{2+}(aq) + 6NO_3^-(aq) \longrightarrow$$

$$6K^+(aq) + 6NO_3^-(aq) + Ca_3(PO_4)_2(s)$$

$$2PO_4^{3-}(aq) + 3Ca^{2+}(aq) \longrightarrow Ca_3(PO_4)_2(s)$$

Spectator ions = $K^+$ and $NO_3^-$

## Supplementary Exercises

### I. Multiple choices

1. Which one is homogeneous mixture

   A. Milk
   B. Blood
   C. Concrete
   D. Brass

2. Which of the following liquid pairs shows a negative deviation from Raoult's Law?

   A. $C_2H_5OH$ and $C_6H_6$
   B. $CCl_4$ and $H_2O$
   C. $CH_3OH$ and $H_2O$
   D. $C_{10}H_{22}$ and $CH_3OH$

3. The solubility of NaCl at 0°C is 34.2g per 100g of $H_2O$. What is the mole fraction of NaCl?

   A. 0.905 B. 0.095 C. 0.75 D. 0.075

4. Which of the following is solid solution?

   A. Air
   B. Sea water
   C. Brass
   D. Coca-Cola

5. Which of the following are immiscible with each other?

   A. $C_2H_5OH$ and $CH_3OH$
   B. $H_2O$ and $CH_3OH$
   C. $CCl_4$ and $C_6H_6$
   D. $C_6H_6$ and $H_2O$

6. Which of the following is the most soluble salt in water?
   A. KCl B. NaCl C. $MgCl_2$ D. $CaCl_2$

7. Which of the following is true?
   A. Boiling point is the temperature at which solid's vapour pressure equals the atmospheric pressure
   B. Boiling point increases if a non-volatile solute is added to a solvent.
   C. The boiling point elevation is directly proportional to the mole fraction of the solvent.
   D. Freezing point of a solvent increases when a non-volatile solute is added to it.

8. Which of the following liquids pairs shows a positive deviation from Raoult's law?
   A. Octane, chloroform
   B. Ethanol, water
   C. Benzene, methanol
   D. Water, nitric acid

9. Sulphuric acid reacts with potassium hydroxide to form calciumhydrogen sulphate and water. The equivalent mass of sulphuric acid in this reaction will be:
   A. 98g/eq B. 49g/eq C. 24.5g/eq D. 32.6g/eq

10. The solvent in aqueous solution is
   A. benzene
   B. water
   C. ethanol
   D. chloroform

11. A bleaching solution contains 5.2 mass percent NaCl. What is the mass of NaOCl in a bottle containing 2500g of bleaching solution?
   A. 90 B. 130 C. 480 D. 520

12. What is the normality of a 1M solution of $H_3PO_4$ given the following reaction?
$$H_3PO_4 + 2NaOH \rightarrow Na_2HPO_4 + 2H_2O$$
   A. 3N B. 2N C. 6N D. 1N

13. If 20mL of 0.5N salt solution is diluted to 1L, what is the new concentration?
   A. 0.01N B. 0.1N C. 0.001N D. 10N

14. Which of the following is the least soluble alcohol in water?
   A. $CH_3OH$
   B. $CH_3(CH_2)_4CH_2OH$
   C. $CH_3(CH_2)_{10}CH_2OH$
   D. $CH_3CH_2OH$

15. The average osmotic pressure, π, of blood is 12.3 atm at 27°C. What concentration of glucose, $C_6H_{12}O_6$, will be isotonic with blood?

A. 2.0M B. 1.0M C. 1.5M D. 0.5M

16. What type of solute-solvent attraction is responsible for the solubility of bromine in carbontetrachloride?

A. London force
B. Hydrogen bond
C. dipole-dipole force
D. ion- dipole force

17. All of the following are heterogeneous mixture except.

A. blood B. milk C. air D. concrete

18. During dilution, which of the following remain same?

A. Volume of solution
B. Concentration of solution
C. Volume of solvent
D. Mole of solute

19. Which of the following has the highest concentration of potassium ion?

A. 0.1MKCl
B. $0.1MKNO_3$
C. $0.1MK_2SO_4$
D. $0.1MK_3PO_4^-$

20. Which of the following is not an electrolyte?

A. HCl
B. NaOH
C. $CH_3OH$
D. $CaCl_2$

II. Given short answer

21. A solution that contains as much less solute as will dissolve at a given temperature is said to be _____

22. A solution that is 10% by mass sucrose contains 10g of sucrose in every 100 grams of solution.

23. Rank the following aqueous solutions, 0.10m $NaNO_3$, 0.20m glucose and 0.10m $CaCl_2$, in order of increasing.

a. Osmotic pressure
b. Boiling point
c. Freezing point
d. Vapor pressure

24. Write net ionic equation for

a. $Ba(NO_3)_2 + K_2SO_4 \rightarrow BaSO_4 + 2KNO_3$
b. $Pb(NO_3)_2 + CuSO_4 \rightarrow PbSO_4 + Cu(NO_3)_2$

25. If 75mL of a $AgNO_3$ solution reacts with enough copper to form 1.08g of Ag by single displacement reaction. What is the molarity of the initial $AgNO_3$

26. Which one has higher vapour pressure

    a. $H_2O$ or $CH_3OH$ b. $H_2SO_4$ or $C_6H_6$

27. What mass of potassium permanganate is needed to prepare a 0.125m $KMnO_4(aq)$ solution with 500g of water?

28. What volume of 0.50M $H_2SO_4$ is required to react exactly with 0.250L of 0.80N KOH?

29. _____ is a solution whose concentration is accurately known.

30. Give the concentration of each ion in the following

    a. 0.4M $Co(NO_3)_2$

    b. 1M$FeCl_3$

አጋዥ መጽሐፍቶችን ለማግኘት በ
Telegram @ethiolibrary1 ላይ
ይቀላቀሉን
+251927273995
+251909282436
Email: jemal6454@gmail.com

ETHIOPIAN

DIGITAL LIBRARY

@ETHIOLIBRARY1

# Ethiopian Digital library

public channel

## Description

✔ መጽሀፍት በPDF ያገኛሉ
✓Top books
✓Extreme books
✅ ለትምህርታዊ መፅሀፍቶች
✅ ቴክኖሎጂ መፃሀፍቶች
✅ የሳይንስ መጽሀፍቶ
✅ የመደበኛ ተማሪዎችን መጽሀፍትን፣
✅ የሶሻልና የናቹራል መጽሀፎች
✅ የጥያቄና መልስ
የሚያገኙበት
@ethiolibrary1
@ethiolibrary1
jemal6454@gmail.com

t.me/ethiolibrary1

Invite Link

Notifications

On